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**Statistical  
Abstract of  
Undergraduate  
Programs  
in the  
Mathematical  
Sciences  
in the United  
States**



AMERICAN  
MATHEMATICAL  
SOCIETY

Richelle Blair  
Ellen E. Kirkman

**Statistical Abstract  
of Undergraduate Programs  
in the Mathematical Sciences  
in the United States**

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# **Statistical Abstract of Undergraduate Programs in the Mathematical Sciences in the United States**

**Fall 2021 CBMS Survey**

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# Acknowledgments

The CBMS 2021 survey was originally scheduled to be conducted in fall 2020, following the usual five-year cycle. Because of the COVID pandemic, the CBMS 2020 survey was rescheduled for fall 2021, and changes were made to the CBMS 2020 survey instruments. A shorter survey on the impact of the pandemic was conducted in fall 2020 and published in March 2021. Development of the CBMS 2021 survey took place from 2016–2024 and involved an unusually large number of people, in multiple roles, who we want to thank:

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Richelle Blair  
Ellen E. Kirkman  
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# Foreword

Every five years since 1965, the Conference Board of the Mathematical Sciences (CBMS) has sponsored a national survey of undergraduate mathematical and statistical sciences in the nation's two- and four-year colleges and universities. The 2021 CBMS survey, conducted with National Science Foundation (NSF) support, is the eleventh report in this series of now over fifty years of data. The CBMS surveys study the mathematics departments at two-year colleges and the undergraduate programs of mathematics departments and statistics departments at four-year colleges and universities. Three different instruments were sent to a stratified random sample of these three populations. This report, CBMS2021, presents the estimates computed using the responses to these questionnaires.

CBMS2021 is organized as follows:

- Chapter 1 gives an overview of the most interesting findings of the 2021 CBMS survey: tables in this chapter are designated with the label S, for “summary.” Many of the tables in this chapter are presented in more detail in later chapters (e.g., all data from four-year mathematics departments is broken down later by level of department, data on course categories for two- and four-year departments are given later by individual courses, etc.). Topics covered include summary data on enrollments, number of degrees awarded, appointment type (tenured, tenure-eligible, other full-time, postdoc, or part-time) of course instructor, faculty demographics, and pedagogy in precalculus/elementary mathematics and statistics.
- Chapter 2 reports on the special projects of the 2021 survey: tables in this chapter are designated with the label SP, for “special project.” The special projects in 2021 for two- and four-year institutions are the mathematical education of pre-college mathematics/statistics teachers, practices in distance/remote learning courses, academic resources available to undergraduates, trends in dual enrollments, attitudes towards online learning after the COVID pandemic, and DEI (diversity, equity, and inclusion) activities. Special projects for four-year departments include interdisciplinary courses in four-year mathematics departments, requirements in the national majors in mathematics and statistics in four-year departments, availability of upper-level classes in four-year mathematics departments and statistics departments, estimates of the post-graduation plans of graduates of four-year mathematics departments and statistics departments, assessment in four-year mathematics departments and statistics departments, divisional

graduation credit for advanced placement courses in four-year mathematics and statistics departments, statistics majors and minors at four-year mathematics departments, and profiles of other full-time faculty at four-year mathematics and statistics departments. In addition to the special projects above, special projects for two-year colleges include coordination of online instruction of on-campus and remote learners, challenges of remote learning, and Pathways programs (Chapter 6).

- Chapter 3 focuses on course enrollments and the number of undergraduate degrees awarded by mathematics departments and statistics departments at four-year colleges and universities, including data on the appointment type of the course instructor: tables in this chapter are designated with the label E, for “enrollment.”
- Chapter 4 concerns the demographics of faculty in mathematics and statistics departments of four-year colleges and universities: tables in this chapter are designated with the label F, for “faculty.” As explained in this chapter, these data were obtained from the Annual Survey, conducted by the American Mathematical Society.
- Chapter 5 studies courses taught primarily to beginning students in mathematics and statistics departments at four-year colleges and universities: tables in this chapter are designated with the label FY, for “first year.”
- Chapter 6 focuses on enrollments, course offerings, and instructional practices, including Pathways courses, at two-year colleges: tables in this chapter are designated with the label TYE, for “two-year enrollment.”
- Chapter 7 presents faculty demographics and special topics at two-year colleges: tables in this chapter are designated with the label TYF, for “two-year faculty.”

Other important information is included in appendices:

- Appendix I contains the enrollments (both with, and without, distance learning enrollments) for each individual course listed on the four-year mathematics and statistics department questionnaires, along with past enrollments (with distance learning enrollments included). Standard errors for the 2021 course enrollments are also included.
- Appendix II contains details about the survey procedures.
- Appendix III gives the list of responders to the 2021 survey.

- Appendix IV contains the report on the impact of the COVID pandemic. As it seemed unreasonable to do the usual 2020 CBMS survey in fall 2020, a smaller survey on the impact of the COVID pandemic was conducted. The results of that survey are online at <https://www.ams.org/profession/data/cbms-survey/cbms2020> and are contained as an appendix to this report.
- Appendices V, VI, VII, and VIII give the actual questionnaires used in the 2021 CBMS survey. The instruments themselves can be useful in interpreting the results of the survey.
- Appendix IX gives the standard errors for each of the tables. It is important to remember that the survey is based on a sample, and the numbers provided in the tables are estimates that are subject to sampling error.

Throughout this report, the standard error (SE) is often included with the estimate, e.g., “52% (SE 2.2),” meaning that the estimate is 52% and the standard error in this estimate is 2.2. Data from the 2021 survey are usually compared to similar data from earlier CBMS surveys. The change in an estimate from the estimate in a previous survey is sometimes expressed both as percentage change and as the number of SEs this change represents (e.g., “grew by about 13% (2 SEs)” means that if  $X(2021)$  is the estimate in 2021 and  $X(2015)$  is the estimate in

2015, then  $(X(2021)-X(2015))/X(2015) = 0.13$  and  $(X(2021)-X(2015))/SE(X(2021)) = 2$ ).

Throughout this report, enrollments do not include dual enrollments, unless indicated by table captions. Depending on the caption of the table, enrollments may, or may not, include distance/remote learning enrollments. One can use Appendix I to find enrollments of courses at four-year departments for fall 2021 with, or without, distance/remote learning enrollments included (this is not the case for CBMS surveys before the 2010 survey, as past appendices give enrollments only with distance learning enrollments included). One can use the tables in Chapter 6 to find enrollments of courses at two-year departments for fall 2021 with, or without, distance/remote learning enrollments included, depending on the caption of the table. In the text of this report, whether the enrollments cited include, or do not include, distance/remote learning enrollments is generally determined by the comparable historical data available.

This report refers to earlier CBMS reports (called CBMS2015, CBMS2010, etc.). This report, and the preceding nine CBMS reports (beginning with the 1970 report), are available online at <http://www.ams.org/profession/data/cbms-survey/cbms-survey>. Other references can be found in the Bibliography at the end of this report.

## Chapter 1

# Summary of the CBMS2021 Report

In the text that follows, the standard error (SE) in many of the estimates is provided along with the estimate (e.g., “52% (SE 2.2)”); the SEs for all CBMS2021 tables can be found in Appendix IX. The change in an estimate from the estimate in a previous survey is often expressed both as percentage change and as the number of SEs of the 2021 estimate that change represents (e.g., “grew by about 13% (2 SEs)”).

## Highlights of Chapter 1

### A. Enrollments.

- From NCES data, four-year college and university undergraduate enrollments grew over each of the CBMS surveys 2005–21: up 23% from 2005 to 2010, up 1% from 2010 to 2015, and up 2% from 2015 to 2021. During these same time intervals, the CBMS survey estimated fall total enrollments in these four-year institutions’ mathematics and statistics departments combined grew 26% from 2005 to 2010 and 13% from 2010 to 2015 but declined 29% from 2015 to 2021, returning to the four-year college and university mathematics and statistics department estimated fall enrollment levels observed in the 1990–2005 CBMS surveys. See Table S.1 and Figure S.1.1.
- NCES data for public two-year colleges shows growth to 17% from 2005 to 2010, a decrease of 14% from 2010 to 2015, and another decrease of 28% in 2015 to 2021. Similar trends were seen in estimated two-year college mathematics department enrollments, with an increase in 2005 to 2010 of 19%, a 5% decrease in 2010 to 2015, and a 31% decrease in 2015 to 2021 to an estimate of 1,333,000 (SE 144,700) students. Prior to fall 2021, the lowest mathematics enrollment had been 1,386,000 students in 2000. See Table S.1 and Figure S.1.1.
- Estimated numbers of enrollments in all course categories (except for middle- and upper-level computer science courses), for all levels of four-year mathematics departments combined, are lower in fall 2021 than in fall 2015, and lower in fall 2021 than in fall 2010 for every course category except upper-level statistics and middle-level computer science courses. Estimated total fall enrollments in all mathematics courses combined, over all levels of four-year mathematics departments combined, were down 30% (5.4 SEs) from fall 2015 to fall 2021 and down 21% (3.5 SEs) from fall 2010 to fall 2021. From fall 2015 to fall 2021, in all levels of four-year mathematics departments combined, estimated pre-college course enrollments were down 60% (9.4 SEs), introductory-level course enrollments were down 33% (5 SEs), calculus-level enrollments were down 18% (2.4 SEs), and advanced-level mathematics course enrollments were down 21% (1.9 SEs). See Table S.2 and Figure S.2.1.
- Estimated number of enrollments in statistics courses taught in four-year mathematics departments declined by 33% (4.1 SEs) from fall 2015 to fall 2021, with most of the decline occurring in the introductory-level statistics courses. In four-year statistics departments, estimated enrollments were down 10% (1 SE) from fall 2015 to fall 2021, with most of the decline occurring in upper-level courses. The estimated fall 2021 total undergraduate enrollments in statistics courses taught in mathematics departments exceeded those enrollments in statistics departments. See Table S.2 and Figure S.2.3.
- Estimated fall enrollments in computer science courses taught in four-year mathematics departments declined by 9% (0.3 SEs) from 2015 to 2021, due to the decline occurring in the lower-level courses. Enrollments in middle-level and upper-level computer science courses taught in mathematics departments increased. See Table S.2.
- Estimated enrollments in every two-year college mathematics course category experienced a similar downward trend as institutional enrollments in 2015 to 2021. Of special note, pre-college course enrollments decreased 55% to 348,000 students (SE 47,100) from 2015 to 2021, compared to a 32% decrease from 2010–2015. Other course categories are presented in Table S.2 and Figure S.2.2.

### B. Bachelor’s degrees granted.

- The CBMS2021 estimated total number of all mathematical sciences bachelor’s degrees granted through four-year mathematics and statistics departments combined during the 2020–1 academic year was 22,562 (SE 1886), down from 26,234 in 2014–15 (a 14% (1.9 SEs) decrease), but above the 2009–10 estimate of 21,377. The 2014–15 estimate reversed the trend of decline in the estimated number of bachelor’s degrees awarded by four-year mathematics and statistics departments combined that had been observed in the 1985–2010 CBMS surveys. The CBMS 1985 survey estimated that 27,928 bachelor’s degrees were granted during 1984–85, but

this total included an estimated 8691 degrees in computer science awarded by four-year mathematics and statistics departments. When degrees in computer science awarded by mathematics or statistics departments are removed for the totals, the 2021 estimate of degrees awarded in the previous academic year is the second largest number observed in a CBMS survey. See Table S.3 and Figure S.3.1.

- The estimated number of bachelor's degrees in computer science awarded by four-year mathematics departments in 2020–21 declined 71% (7.2 SEs) over the relatively high number of computer science degrees awarded in 2014–15. See Table S.3 and Figures S.3.1 and S.3.2.
- The CBMS survey estimated percentage of bachelor's degrees awarded to women through U.S. mathematics and statistics departments combined in 2020–21 was estimated at 39%, lower than the estimates in recent CBMS surveys: 43% in 1999–2000, 40% in 2004–5, 43% in 2009–10, and 42% in 2014–15. When degrees in computer science degrees awarded by mathematics departments are excluded, then the estimated percentage of degrees awarded to women through U.S. mathematics and statistics departments in 2020–21 was 40%; it was 47% in 1999–2000, 43% in 2004–5, 45% in 2009–10, and 43% in 2014–15. See Table S.3.

### C. Appointment type of instructors of undergraduate mathematics and statistics course sections.

- The estimated percentage of sections of calculus-level courses taught in all levels of four-year college mathematics departments combined by tenured or tenure-eligible faculty has been decreasing from 61% in fall 2005, to 59% in fall 2010, to 52% in fall 2015; in fall 2021, it was slightly larger at 55% (SE 2.9). The estimated percentage of sections of calculus-level courses taught by other full-time faculty increased from 15% in fall 2010, to 24% in fall 2015, to 28% (SE 2.7) in fall 2021. Further data on the appointment type of the instructor, broken down by the type of class and the format of the class, are given for calculus classes, introductory statistics classes (broken down by courses taught in mathematics departments or statistics departments), and computer science classes. See Tables S.4–S.8.
- In public two-year colleges, the estimated percentage of mathematics and statistics sections taught by full-time faculty decreased by 31 percentage points to 33% (SE 7.9), compared to 64% (SE 64) in fall 2015. Part-time instruction increased thirty-one points to 67% (SE 7.9). In fall 2021, Calculus courses (Mainstream and Non-Mainstream) were taught by full-time faculty 75–78% of

the time. The estimated average section size of all Calculus sections decreased to 15–17 students, compared to 26 students in 2015, while the estimated section size of Statistics and Probability increased to 33 (SE 4.5) students. A notable decrease of 57 percentage points to 23% (SE 17.7) of sections were taught by full-time faculty in Elementary/Introductory Statistics and Probability courses, compared to 80% (SE 5) in 2015. See Tables S.4–S.7.

### D. Distance/remote learning enrollments.

- In fall 2021, distance/remote learning courses were defined to be “courses in which half or more of the instruction occurs with the instructor and the student separated by time and/or place facilitated by technology.” In four-year mathematics departments, distance/remote learning enrollments in precalculus, introductory-level courses, calculus I and II, differential equations, linear algebra, and introductory statistics courses combined in fall 2021 were estimated at 244,309, almost triple the fall 2015 estimate of 86,197. In fall 2021, distance/remote learning enrollments represented 17% (up from 4% in 2015) of the total enrollments in these courses. See Table S.9.
- In two-year mathematics departments, fall 2021 estimated distance/remote learning enrollments exceeded the non-distance/remote learning enrollments in each course category presented in Table S.9 and other courses listed in Table TYE.11. Distance/remote learning enrollment in precalculus, introductory-level courses, calculus I and II, differential equations, linear algebra, and introductory statistics courses combined was estimated at 646,669 students in fall 2021, three and a half times the estimate of 184,618 in 2015. See Table S.9. Two-year college enrollment in all mathematics and statistics grew to an estimated 823,000 (SE 108,000) students, up 266% compared to 225,000 (SE 25,000) students in fall 2015. Estimated distance/remote learning enrollment accounted for 62% of all two-year college mathematics estimated enrollment. See Table TYE.11 in Chapter 6.
- Two-year college elementary statistics distance/remote learning estimated enrollment increased 423% to an estimated total of 169,586 (SE 25,613) students compared to 30,608 (SE 4000) students in 2015.
- In four-year statistics departments, distance/remote learning enrollments in introductory statistics courses in fall 2021 were estimated at 28,218, almost seven times the fall 2015 estimate of 4291, and these enrollments represented 30% (up from 5% in 2015) of the estimated total enrollments in introductory statistics courses in fall 2021. See Table S.9.

### E. Pedagogical methods used in teaching undergraduate mathematics and statistics courses.

- An estimated 71% (SE 6.5) of two-year college mathematics departments and 22% (SE 4.5) of four-year mathematics departments have offered “Pathways sequences courses” in the last five years. The CBMS 2015 survey estimated that 58% of two-year mathematics departments had implemented a Pathways sequence. See Table S.10.
- Table S.10 also presents data on how often departments use various instructional strategies, such as focusing on conceptual understanding, integrating real-world applications, using student-centered active learning strategies, and using assessments such as regular graded homework and quizzes to inform teaching. See Table S.10 and Table TYE.11 in Chapter 6.
- Comparisons in instructional strategies for introductory statistics courses taught in fall 2021 at two-year college mathematics departments, four-year mathematics departments, and statistics departments are given in Table S.11, and comparisons in perceived success in introducing technology in introductory statistics courses are given in Table S.12. See Tables S.11 and S.12.

### F. Numbers of faculty.

- From fall 2015 to fall 2021, the estimated number of full-time faculty increased 5% at two-year mathematics departments, 41% at four-year mathematics departments, and almost tripled in doctoral-level statistics departments. See Table S.13 and Figure S.13.1.
- From fall 2015 to fall 2021, the estimated number of part-time faculty paid by two-year colleges decreased 23% at two-year mathematics departments to 13,827 persons (SE 1631). Part-time faculty paid by a third party doubled in 2021, compared with 2015. The estimated number of part-time faculty increased 31% at four-year mathematics departments and was almost seven times the 2015 estimate at doctoral-level statistics departments. See Table S.13 and Figures S.13.2, S.13.4, and S.13.5.
- Most of the increase from fall 2015 to fall 2021 in the estimated number of full-time faculty at four-year mathematics departments combined occurred in the category of “other full-time faculty” (faculty who are not tenured or tenure-eligible, including postdocs), which rose 80% (12 SEs) in fall 2021 from fall 2015. The increase in the estimated number of tenured plus tenure-eligible (TTE) faculty at all four-year mathematics departments combined increased 23% (10 SEs). See Table S.14.

- In the doctoral-level statistics departments, from fall 2015 to fall 2021, the estimated number of tenured plus tenure-eligible faculty more than doubled (an increase of 25 SEs), and the number of other full-time faculty more than tripled (an increase of 12 SEs). See Table S.14.
- In fall 2021, the total estimated number of full-time faculty (permanent, continuing, and other) in two-year colleges was 10,328 (SE 1117), a 5% increase compared to 9800 (SE 894) in 2015. The 2021 persons increase in faculty can be viewed in light of the 28% decrease in institutional enrollment in two-year colleges to 4,478,000 students and the 31% decrease of 585,000 students in mathematics and statistics enrollment from 2015 to 2021.
- It was estimated that there were 9126 (SE 919) full-time permanent faculty in public two-year college mathematics departments in the U.S. in fall 2021, compared with 8314 (SE 840) in 2015. This 10% increase of 812 persons can be compared to a 15% (2 SEs) decrease of 1476 persons in 2015. See Table S.14 in Chapter 1, Table TYE.2 in Chapter 6, and Table TYF.1 in Chapter 7.

### G. Gender and race/ethnicity in mathematical sciences faculty.

- In fall 2021, women comprised an estimated 37% of full-time mathematics faculty over all levels of four-year mathematics departments combined, up from the 31% observed in the CBMS 2015 survey. The estimated percentages of women among tenured faculty, tenure-eligible faculty, other full-time faculty, and postdocs over all levels of four-year mathematics departments combined each increased over the respective 2015 CBMS survey estimate. See Table S.15.
- In fall 2021, women comprised an estimated 38% of doctoral-level full-time statistics faculty, the same estimate as in fall 2015, though each individual category had a greater percentage of women in fall 2021 than in fall 2015. The 2021 CBMS survey estimated that the percentage of women among tenured doctoral-level statistics faculty in fall 2021 was 32% and among tenure-eligible faculty was 36%, percentages very similar to the percentages of tenured and tenure-eligible faculty over all levels of four-year mathematics departments combined, but higher than the estimated percentages in fall 2021 of women faculty at doctoral-level mathematics departments (where 19% of tenured faculty at doctoral-level mathematics departments were women and 29% of tenure-eligible faculty were women) (see Table F.1 in Chapter 4). See Table S.15.



- Tables on gender and race/ethnicity of faculty at two-year colleges appear in Tables TYF.8–TYF.12 in Chapter 7. In fall 2021, 49% (SE 5.3) of full-time permanent faculty were women and 51% (SE 5.4) men. With part-time faculty, an estimated 46% (SE 3.1) were women and 54% (SE 3.1) of men. See Table TYF.9 in Chapter 7.
- Slight changes in the estimated distribution of race/ethnicities among four-year mathematics and statistics faculty were observed in the 2021 CBMS survey. The estimated percentage of White, not Hispanic faculty among full-time male four-year mathematics faculty declined from 53% in fall 2015 to 43% in fall 2021, while the percentage of White, not Hispanic female faculty was estimated at 24% in both 2015 and 2021. Among four-year doctoral-level statistics faculty an estimated 45% of male faculty and 15% of female faculty were White, not Hispanic in fall 2015, while in 2021, these percentages were estimated at 31% and 16%, respectively. Distributions of race/ethnicities amounts for four-year mathematics and statistics part-time faculty are given in Chapter 4, Table F.6. See Tables S.18 and S.19.
- In fall 2021, the estimated percentage of racial/ethnic minorities among full-time permanent faculty members in mathematics departments in two-year colleges reflected an 8% decrease compared with 2015. The total number of racial/ethnic minority faculty was 501 fewer persons in 2021 compared with 2015. The majority of minority faculty represented (faculty other than White) was Asian/Pacific Islander, down 3 percentage points in 2021 and 245 persons compared with 2015. See Table TYF.10 in Chapter 7. The percentage of women in each ethnic group is displayed in Table TYF.12 in Chapter 7.

#### **H. Age distribution and changes in the mathematical sciences faculty due to deaths and retirements.**

- Estimated age distributions and average ages of tenured and tenure-eligible four-year mathematics faculty and permanent two-year college faculty in fall 2021 are given in Table S.16 and Figures S.16.1 and S.16.2. The analogous tables for faculty at doctoral-level statistics departments in fall 2021 are given in Table S.17 and Figure S.17.1.
- The total number of deaths and retirements in 2020–21 among four-year mathematics faculty was estimated to be about 6% of the fall 2021 estimated number of tenured/tenure-eligible four-year mathematics faculty. The analogous percentage for doctoral-level statistics faculty was estimated at 4%. In both cases, the percentage was 1% higher than that observed in the 2015 CBMS survey. See Table S.20.
- The estimated total number of two-year college full-time permanent faculty who were no longer part of the faculty in 2021–2022 was 256 persons (SE 63), approximately 3% of the estimated total full-time permanent faculty. See Table TYF.3 in Chapter 7.

Data on undergraduate course enrollments in mathematics departments and statistics departments in four-year colleges and universities and in two-year college mathematics departments is summarized here in Chapter 1. Data on enrollments in four-year departments is broken down further in Chapters 3 and 5; data on individual course enrollments, both with and without distance/remote learning enrollments, for four-year departments is given in Appendix I by course. Data on enrollments in two-year departments is discussed further in Chapter 6.

#### **Tables S.1 and S.2: An Overview of Enrollments**

Table S.1 gives an overall historical view of fall enrollments in courses taught in mathematics and statistics departments of four-year U.S. colleges and universities and in mathematics departments of public two-year colleges. The table also presents estimates of institutional undergraduate enrollments, so that one can compare changes in estimated mathematical sciences course enrollments with overall changes in institutional undergraduate enrollments. Tables S.1 and S.2 present data on enrollments (including distance/remote learning enrollments, but not dual enrollments) in four-year mathematics and statistics departments in fall 2005, 2010, 2015, and 2021 for mathematics, statistics, and computer science courses, with the fall 2021 estimated enrollment broken down into four-year college and university mathematics department enrollments and statistics department enrollments; enrollment data for mathematics departments in two-year colleges are also presented. The enrollment data in mathematical science courses in these tables was obtained from the CBMS surveys in those years. The total institutional undergraduate enrollments in four-year colleges and universities, and in two-year colleges, came from the National Center for Educational Statistics (NCES) and are based on data that post-secondary education institutions must submit to the Integrated Post-Secondary Educational Data System (IPEDS). The estimates are for full- and part-time students in public and private four-year colleges and universities and for full- and part-time students in public two-year colleges. Most national data cited in this report are drawn from the NCES

**TABLE S.1.** Enrollment in (1000s) in undergraduate mathematics, statistics, and computer science courses taught in mathematics departments and statistics departments of four-year colleges and universities, and in mathematics departments of two-year colleges. Also NCES data on total fall enrollments in two-year colleges and four-year colleges and universities in fall 2005, 2010, 2015, and 2021. NCES data include both public and private four-year colleges and universities and include only public two-year colleges. Enrollments include distance learning enrollments but not dual enrollments.

|                                                           | Four-Year College & University<br>Mathematics & Statistics Departments |                 |                 |       |              |        | Two-Year College<br>Mathematics Departments <sup>1</sup> |                 |                 |      |
|-----------------------------------------------------------|------------------------------------------------------------------------|-----------------|-----------------|-------|--------------|--------|----------------------------------------------------------|-----------------|-----------------|------|
|                                                           | Fall                                                                   |                 |                 |       | 2021 by Dept |        | Fall                                                     |                 |                 |      |
|                                                           | 2005                                                                   | 2010            | 2015            | 2021  | Math         | Stat   | 2005                                                     | 2010            | 2015            | 2021 |
| Mathematics                                               | 1607                                                                   | 1971            | 2213            | 1551  | 1551.2       | na     | 1580                                                     | 1887            | 1639            | 1076 |
| Statistics                                                | 260                                                                    | 371             | 457             | 340   | 210.64       | 128.95 | 117                                                      | 137             | 280             | 257  |
| Computer Science                                          | 59                                                                     | 77 <sup>2</sup> | 68 <sup>2</sup> | 62    | 61.698       | na     | -- <sup>2</sup>                                          | -- <sup>2</sup> | -- <sup>2</sup> | na   |
| Total                                                     | 1925                                                                   | 2419            | 2738            | 1952  | 1823.5       | 128.95 | 1697                                                     | 2024            | 1918            | 1333 |
| NCES Total Fall<br>Undergraduate Enrollments <sup>3</sup> | 8476                                                                   | 10399           | 10547           | 10765 |              |        | 6184                                                     | 7218            | 6224            | 4478 |

<sup>1</sup> Starting in 2005, data on mathematics, statistics, and computer science enrollments in two-year colleges include only public two-year colleges.

<sup>2</sup> Computer science totals in two-year colleges before 1995 included estimates of computer science courses taught outside of the mathematics department. In 1995 and 2000, only those computer science courses taught in the mathematics department were included. Starting in 2005, no computer science courses were included in the two-year mathematics survey, and starting in 2010, no computer science courses were included in the statistics survey.

<sup>3</sup> Data for 2005, 2010, 2015, and 2021 are from Table 303.70 from *Digest of Education Statistics: 2022*, NCES, [https://nces.ed.gov/programs/digest/d22/tables/dt22\\_303.70.asp](https://nces.ed.gov/programs/digest/d22/tables/dt22_303.70.asp).

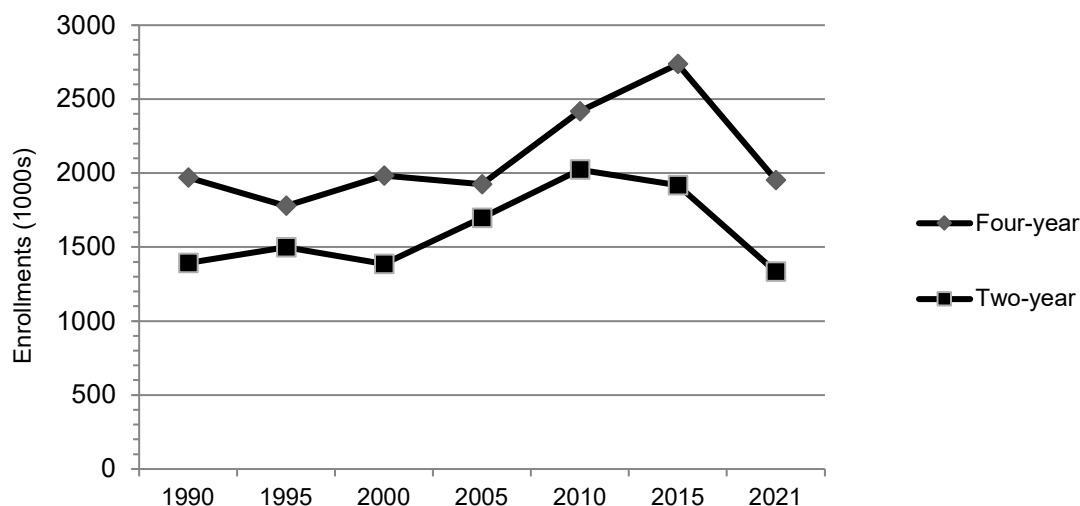
publication *Digest of Education Statistics: 2022*, which is available at [https://nces.ed.gov/programs/digest/d22/tables/dt22\\_303.70.asp](https://nces.ed.gov/programs/digest/d22/tables/dt22_303.70.asp).

We note that many of the NCES reports contain projections that are updated every two years, and, in updates, projections are replaced by actual enrollments. Therefore, enrollments from NCES data for a given year in CBMS Table S.1 may change in Table S.1 in subsequent CBMS reports, as we replace NCES projected enrollments with NCES actual enrollments.

Figure S.1.1 displays estimated enrollments in mathematical sciences undergraduate courses taught in mathematics and statistics departments of four-year colleges and universities and in two-year colleges, obtained from the CBMS surveys in fall 1990, 1995, 2000, 2005, 2010, 2015, and 2021. From Figure S.1.1, we see that four-year college mathematical sciences enrollments were relatively constant from 1990–2005, but the CBMS 2010 and CBMS 2015 surveys showed relatively large increases in mathematical sciences course enrollments. These increases were

likely due in part to the increases in four-year college institutional enrollments shown in Table S.1 (and in Table S.1 of CBMS2015); between fall 2000 and fall 2015, four-year college and university undergraduate enrollments grew by about 46%, and enrollments in those institutions' mathematics and statistics undergraduate courses grew by an estimated 38%. The CBMS2021 estimate shows fall mathematical science four-year college enrollments declining to the levels of the CBMS 2005 and previous surveys.

From 2010 to 2015, public two-year colleges experienced an overall total enrollment decrease of 14%, an estimated total of 6,224,000 students, based on National Center for Education Statistics (NCES) projections updated in 2016 for fall 2015. In NCES projections updated in 2022 for fall 2021, public two-year colleges experienced an overall total enrollment decrease of 28%, an estimated total of 4,478,000 in 2021 and a decrease of 1,746,000 students from 2015 to 2021. The 2021 two-year college decrease can be viewed in comparison with an overall increase at four-year colleges of 2%, an estimated total of 218,000 students. Enrollment in



**FIGURE S.1.1.** Combined enrollment (in 1000s) in undergraduate mathematics, statistics, and computer science courses at four-year colleges and universities within mathematics departments and statistics departments and within mathematics departments of two-year colleges: Fall 1990, 1995, 2000, 2005, 2010, 2015, and 2021. Data beginning in 2005 include only public two-year colleges.

Note: Before 1995, two-year enrollment totals included computer science enrollments taught outside of the mathematics program. In 1995 and 2000, only computer science courses taught within the mathematics department of two-year colleges were counted. Starting in 2005, no computer science courses were included in the CBMS survey of two-year mathematics programs, and starting in 2010, no computer science data were included in the survey of statistics departments.

two-year colleges in fall 2021 constituted about 30% of the total undergraduate enrollment in the U.S. See Table S.1 in Chapter 1 and Table TYE.1 in Chapter 6.

In the period 2000 to 2010, CBMS survey data presented an approximate growth of 46% in the mathematics and statistics enrollments in the mathematics departments of the nation's public two-year colleges, followed by a 5% decrease from fall 2010 to fall 2015 of an estimated 2,105,000 (SE 118,000) students. This total included 94,000 (SE 23,000) dually enrolled students and 225,000 (SE 25,000) distance learning enrollments. Enrollment in mathematics and statistics courses in mathematics programs at public two-year colleges received from the CBMS2021 survey was estimated to be 1,495,000 (SE 157,543) students. This total includes 162,000 (SE 6,386–20,634) dually enrolled students and 823,000 (SE 108,000) distance/remote learning enrollments. Enrollment in mathematics and statistics at two-year colleges in fall 2021 constituted approximately 41% of the total mathematics and statistics undergraduate enrollment in postsecondary institutions (not including dual enrollments).

Figure S.1.1 presents a graphical display of increases in estimated mathematical sciences course fall enrollments in two-year colleges from 1990 to 1995 and 2000 to 2010 and decreases from 1995 to 2000 and 2010 to 2021. Additional

information and discussion about trends in enrollments in mathematics courses and programs at two-year colleges can be found in Chapter 6, Tables TYE.1, TYE.2 (includes dual enrollments), and TYE.3. It should also be noted that the sample frame in 2005 and following years includes only public two-year colleges.

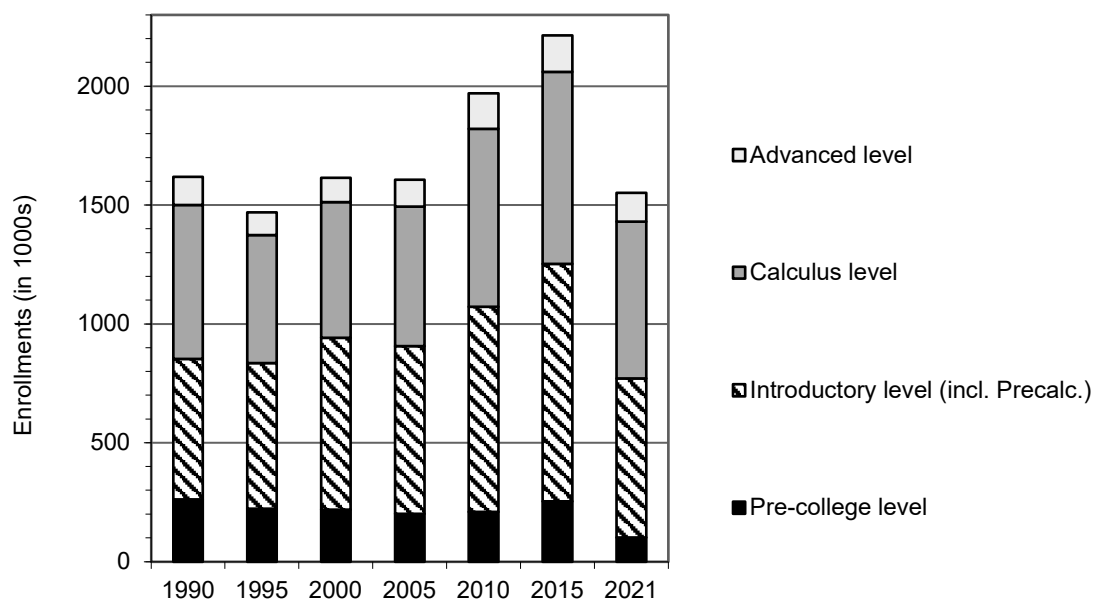
Table S.2 begins the process of breaking the total mathematical sciences course fall 2021 enrollments (including distance/remote learning enrollments, but not dual enrollments) down into its component parts. Among four-year college and university mathematics departments, the enrollment course categories used were pre-college-level courses, introductory-level courses, calculus-level courses, and advanced-level courses. Beginning with the 2010 CBMS survey, the pre-college courses (e.g., Arithmetic, Pre-Algebra, Elementary Algebra, and Intermediate Algebra) were treated as one block, and not itemized, as they had been in previous CBMS surveys. The intermediate-level course list was essentially unchanged from the previous 2015 CBMS survey, and included courses in liberal arts mathematics, mathematics for K–8 mathematics teachers, and a cluster of courses with names such as College Algebra, Precalculus, and Trigonometry. The calculus-level courses listed in the 2021 survey included, as in previous CBMS surveys, linear algebra, differential equations,

**TABLE S.2.** Total enrollment (in 1000s), including distance learning enrollment, by course level in undergraduate mathematics, statistics, and computer science courses taught in mathematics and statistics departments at four-year colleges and universities and in mathematics departments at two-year colleges in fall 2005, 2010, 2015, and 2021.

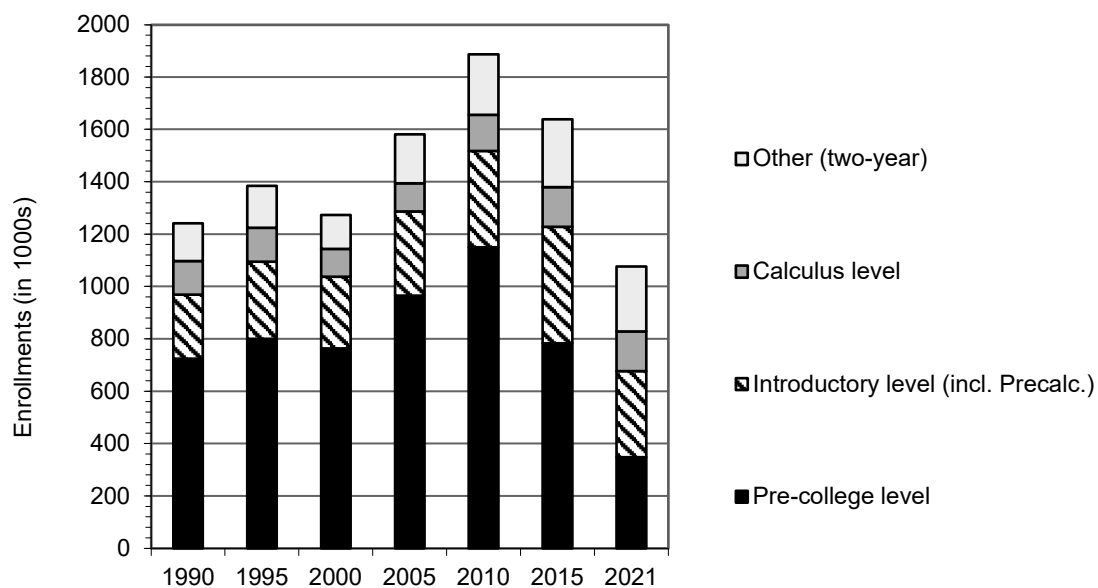
|                                               | Mathematics Departments |      |      |      | Statistics Departments |      |      |      | Two-Year College Mathematics Departments |      |      |      |
|-----------------------------------------------|-------------------------|------|------|------|------------------------|------|------|------|------------------------------------------|------|------|------|
| Course Level                                  | 2005                    | 2010 | 2015 | 2021 | 2005                   | 2010 | 2015 | 2021 | 2005                                     | 2010 | 2015 | 2021 |
| Mathematics courses                           |                         |      |      |      |                        |      |      |      |                                          |      |      |      |
| Pre-college level                             | 201                     | 209  | 253  | 101  | --                     | --   | --   | --   | 965                                      | 1150 | 782  | 348  |
| Introductory level<br>(including Precalculus) | 706                     | 863  | 1000 | 670  | --                     | --   | --   | --   | 321                                      | 368  | 445  | 329  |
| Calculus level                                | 587                     | 748  | 807  | 659  | --                     | --   | --   | --   | 108                                      | 138  | 152  | 151  |
| Advanced level                                | 112                     | 150  | 154  | 121  | --                     | --   | --   | --   | 0                                        | 0    | 0    | 0    |
| Other (two-year)                              | --                      | --   | --   | --   | --                     | --   | --   | --   | 187                                      | 231  | 259  | 248  |
| Total Mathematics courses                     | 1607                    | 1971 | 2213 | 1551 | --                     | --   | --   | --   | 1580                                     | 1887 | 1639 | 1076 |
| Probability and Statistics courses            |                         |      |      |      |                        |      |      |      |                                          |      |      |      |
| Introductory level                            | 148                     | 231  | 253  | 159  | 54                     | 81   | 94   | 93   | 117                                      | 137  | 280  | 257  |
| Upper level                                   | 34                      | 32   | 60   | 52   | 24                     | 27   | 50   | 36   | 0                                        | 0    | 0    | 0    |
| Total Probability and Statistics courses      | 182                     | 262  | 313  | 211  | 78                     | 108  | 144  | 129  | 117                                      | 137  | 280  | 257  |
| Computer Science courses <sup>1</sup>         |                         |      |      |      |                        |      |      |      |                                          |      |      |      |
| General education                             |                         |      |      | 7    |                        |      |      |      |                                          |      |      |      |
| Lower level                                   | 44                      | 56   | 45   | 25   | 2                      | --   | --   | --   | --                                       | --   | --   | --   |
| Middle level                                  | 8                       | 12   | 16   | 21   | 0                      | --   | --   | --   | --                                       | --   | --   | --   |
| Upper level                                   | 5                       | 10   | 6    | 8    | 0                      | --   | --   | --   | --                                       | --   | --   | --   |
| Total Computer Science courses <sup>1</sup>   | 57                      | 77   | 68   | 62   | 2                      | --   | --   | --   | --                                       | --   | --   | --   |
| Grand total                                   | 1845                    | 2310 | 2594 | 1823 | 80                     | 108  | 144  | 129  | 1697                                     | 2024 | 1918 | 1333 |

Note: Round-off may make column totals seem inaccurate.

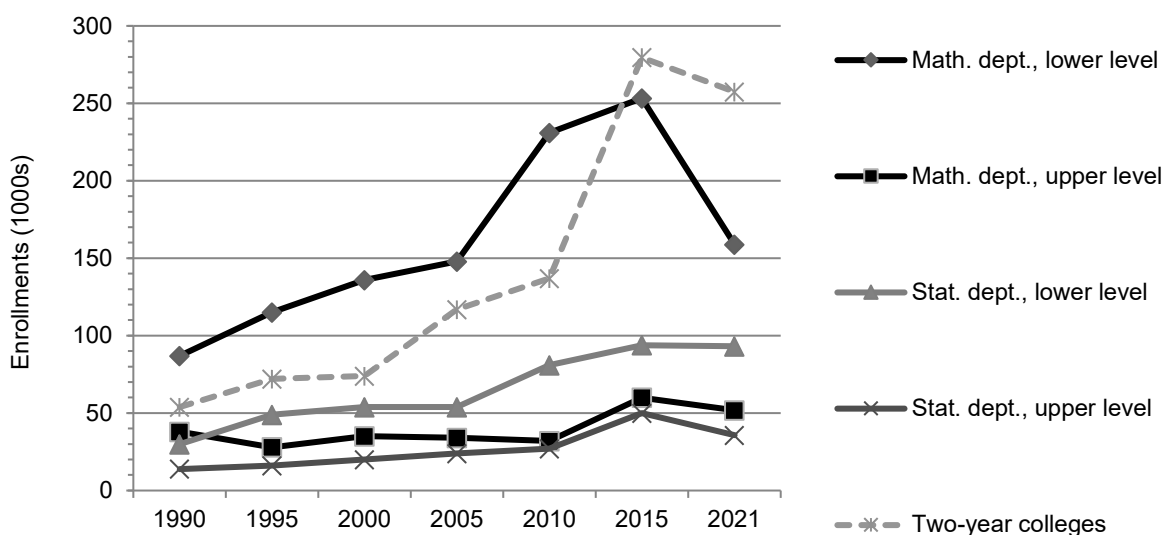
<sup>1</sup> Beginning in 1995, computer science enrollment included only courses taught in mathematics programs. Beginning in 2005, computer science courses were no longer included in the two-year college survey. Beginning in 2010, computer science courses were no longer included in the statistics survey.



**FIGURE S.2.1.** Enrollments (in 1000s) in undergraduate mathematics courses in mathematics departments of four-year colleges and universities by level of course in fall 1990, 1995, 2000, 2005, 2010, 2015, and 2021.



**FIGURE S.2.2.** Enrollments (in 1000s) in undergraduate mathematics courses (statistics not included) in two-year college mathematics departments by level of course in fall 1990, 1995, 2000, 2005, 2010, 2015, and 2021.



**FIGURE S.2.3.** Enrollments (in 1000s) in statistics courses in two-year college mathematics departments and in mathematics departments and in statistics departments of four-year colleges and universities in fall 1990, 1995, 2000, 2005, 2010, 2015, and 2021.

discrete mathematics, and various calculus courses; from the individual course enrollments, which are included in Appendix I, we see that calculus course enrollments accounted for 77% of the total calculus-level course enrollments (including distance/remote learning enrollments) and 76% of non-distance/remote learning enrollments. Four-year college statistics course enrollments, offered in either mathematics or statistics departments, were broken into introductory-level and upper-level enrollments, and computer science course enrollments offered in four-year mathematics departments were broken into the three previous levels and an additional new level, “General Education Computer Science Courses”; several changes were made in the list and titles of the statistics courses in the 2021 CBMS survey. The list of computer science courses was based on the course recommendations in the Association for Computer Machinery report, *Computer Science Curricular 2013*, available at [https://www.acm.org/binaries/content/assets/education/cs2013\\_web\\_final.pdf](https://www.acm.org/binaries/content/assets/education/cs2013_web_final.pdf).

Beginning in 2010, enrollments in computer science courses offered through statistics departments were not gathered in the CBMS survey, but they were gathered, as was done previously, from mathematics departments at four-year institutions in 2021.

In four-year college and university mathematics departments, the estimated total of all undergraduate enrollments in courses taught in mathematics departments fell from 2,594,000 (SE 157,000) in fall 2015 to 1,823,000 (SE 134,000) in fall 2021, according to Table S.2. All categories of courses taught in four-year mathematics departments

listed in Table S.2, except middle- and upper-level computer science courses, showed decreased estimated enrollments in fall 2021 over fall 2015, and all categories of courses, except upper-level statistics courses and middle-level computer science courses, had estimated enrollments in fall 2021 that were less than those in fall 2010. Enrollments in courses in mathematics (excluding statistics and computer science) taught in four-year mathematics departments decreased by 30% (5.4 SEs) from fall 2015 to fall 2021.

Table S.2 also shows estimated enrollments and trends in various course categories in public two-year college mathematics and statistics courses, including distance enrollments, but not dual enrollments.

In fall 2021, approximately 26% of all two-year college mathematics and statistics estimated enrollment was in pre-college (remedial/developmental) courses, compared with 41% in fall 2015 and 57% in fall 2010. Enrollment in pre-college mathematics courses (Arithmetic, Pre-Algebra, Elementary and Intermediate Algebra, and Geometry) at two-year colleges in 2021 was estimated to be 348,000 (SE 47,000), compared to 782,000 students (SE 65,000) in 2015. This represents a 55% decrease from 2015 to 2021, compared to a 32% decrease from 2010 to 2015 and a 19% increase from 2005 to 2010. All courses within the cohort of pre-college courses showed a decrease in estimated enrollment.

The trend of increasing estimated fall enrollments at two-year college mathematics departments in Precalculus/introductory-level courses (College Algebra, Trigonometry, College and Trig, Mathematical Modeling, and Elementary Functions) seen in 2010 and 2015 did not continue in 2021,

with an estimated decrease of 26% (SE 42,000) and 116,000 fewer students at two-year colleges. Enrollments in Precalculus/introductory courses represented 25% of all mathematics enrollments, a total of 329,000 (SE 42,000) students. All courses within the cohort of Precalculus/introductory-level courses showed a decrease in estimated enrollments, except for Mathematical Modeling with an increase of 1000 students.

Fall 2021 estimated enrollments at two-year-college mathematics departments in all calculus-level courses (Mainstream Calculus I, II, and III, Non-Mainstream Calculus I and II, and Differential Equations together) showed a 1% decrease to 151,000 (SE 26,000) students, compared with an 11% increase from 2010 to 2015 (152,000 students; SE 15,000). From 2015 to 2021, Mainstream Calculus I, II, and III experienced a 4% increase from 2015 to 2021, up to 124,000 students, compared to a 9% increase to a total of 119,000 students in 2015. Non-Mainstream Calculus I and II decreased 31% to 18,000 students (8000 fewer students) in 2021, compared to an increase of 18% to 26,000 students in 2015.

College-level, transferable statistics and probability courses in two-year colleges showed an 8% estimated decrease from 2015 to 2021, a total of 257,000 students (SE 24,000), 23,000 fewer students than in 2015. The 2021 decrease follows an increase of 104% in 2015.

For the first time, CBMS2021 surveyed the course Quantitative Literacy/Quantitative Reasoning in two-year college mathematics departments. Fall 2021 showed a total of 73,000 (SE 15,000) students enrolled in 47% (SE 6.65) of the colleges. Specific information can be found in Tables TYE.3–TYE.5 in Chapter 6.

Figure S.2.1 presents a bar graph showing the estimated fall enrollments in mathematics courses taught at four-year mathematics departments, broken down by course category, from 1990–2021. From fall 2015 to fall 2021, in all levels of four-year mathematics departments combined, estimated pre-college course enrollments were down 60% (9.3 SEs), introductory-level course enrollments were down 33% (5 SEs), calculus-level enrollments were down 18% (2.4 SEs), and advanced-level mathematics course enrollments were down 21% (1.9 SEs).

Figure S.2.2 presents a similar bar graph for estimated enrollments in mathematics departments at two-year colleges. Two-year college mathematics and statistics on-campus and distance/remote enrollment was an estimated 1,495,000 (SE 15,7543) in 2021, decreasing 26% from 2015 to 2021, compared with the decrease of 4% from 2010 to 2015. When dual-enrollment students are excluded, as shown in Figures S.2.1 and S.2.2, the enrollment was 1,333,000 (SE 2613) with a decrease of 31% in 2021 compared with 5% in 2015. The bar graph in Figure S.2.2 highlights the growth experienced in 2010 and then the decreases in 2015 and

2021. The 26% decrease in pre-college enrollments is particularly evident in the graph. It should be noted that Statistics and Probability enrollments are not included in Figure S.2.2. The changes in mathematics enrollments should be viewed considering the overall total institutional enrollment decrease of 28%, an estimated total of 1,746,000 students, from 2015 to 2021, at two-year colleges.

Between 1990 and 2015, the nation's undergraduate statistics courses continued a trend of enrollment growth in courses taught in mathematics departments of four- and two-year colleges, as well as in statistics departments of four-year institutions; Figure S.2.3 displays the enrollments in both introductory and upper-level statistics course enrollments for two-year colleges, four-year mathematics departments, and four-year statistics departments from 1990–2021. As Table S.2 and Figure S.2.3 show, estimated statistics enrollments were down across the board in fall 2021, though the upper-level enrollments in four-year statistics departments were only very slightly down. By Table S.2, of the estimated fall 2021 enrollments in introductory-level statistics courses at four- and two-year departments, 51% occurred at two-year mathematics departments, 31% at four-year mathematics departments, and 18% at statistics departments. Table S.2 shows that the estimated total fall 2021 enrollments in statistics departments were 129,000 (SE 14,200), below the fall 2015 estimated enrollment, but above the fall 2010 enrollment estimate. Most of the decline in the estimated statistics course enrollments in mathematics departments occurred in the introductory-level statistics courses, while most of the decline in statistics departments occurred in the upper-level statistics courses.

At two-year college mathematics departments, the Statistics and Probability course estimated fall 2021 enrollment showed the following trends. College-level, transferable statistics and probability courses at two-year colleges enrolled 257,000 (SE 33.8) students in 2015, an 8% decrease of 23,000 fewer students than in 2015, after an increase of 104% in 2015. See Tables S.2, TYE.3, and TYE.4 and Figure S.2.2.

Computer science enrollments within mathematics departments at four- and two-year institutions, as well as in statistics departments, have been generally declining as these courses move to departments of computer science. The 1990 CBMS survey [CBMS1990, Table S.1, p. 2] estimated that there were 180,000 enrollments in computer science courses taught in four-year mathematics departments, 311,000 enrollments in computer science courses taught in four-year computer science departments, and 98,000 enrollments taught in two-year colleges. The 2021 CBMS survey made estimates only for computer science courses taught in four-year mathematics departments and estimated there were 62,000 (SE 18.8) such enrollments in fall 2021, down from 68,000 in fall 2015. As will be noted in Chapter 3,

most of the computer science enrollments in mathematics departments occur in bachelor's-level departments, and they play a significant role in some four-year mathematics departments. Later in this chapter, we will consider the estimated number of undergraduate computer science bachelor's degrees awarded by mathematics departments in 2020–21.

### Table S.3: Bachelor's Degrees in the Mathematical Sciences

Table S.3 presents the estimated total number of bachelor's degrees awarded through the nation's four-year mathematics and statistics departments (combined) in the academic years 1994–95, 1999–2000, 2004–5, 2009–10, 2014–15, and 2020–1. The survey instructions specify that double majors should be included in the count of degrees awarded. The degrees awarded by four-year mathematics departments are categorized as degrees in mathematics, mathematics education, statistics, computer science, actuarial mathematics, joint majors, or “other.” The bachelor's degree categories awarded by four-year statistics departments were changed in the 2021 survey and were categorized as degrees in statistics, biostatistics, actuarial science, data science, joint statistics and mathematics, joint statistics and computer science, statistics education, or “other.” Surveys of four-year mathematics departments conducted before 2010 contained the additional option of a major in operations research, and the numbers of operations research majors from those previous years have been added to the “other” category in Table S.3; furthermore, prior surveys broke down the category of joint majors offered by mathematics departments into different subcategories, while beginning in 2010, the CBMS surveys considered all joint majors as one category. Beginning in 2010, computer science degrees are counted only in four-year mathematics departments. Table E.1.A in Chapter 3 gives the estimated numbers of bachelor's degrees awarded by mathematics departments in 2020–21, and Table E.1.B gives the estimated numbers of bachelor's degrees awarded by statistics departments in 2020–21; both tables give further breakdowns of the degrees awarded, including by the level (bachelor's, master's, or doctoral) of the department awarding the undergraduate degree.

Table S.3 shows that the estimated total number of mathematical sciences bachelor's degrees granted through the nation's four-year mathematics and statistics departments combined in the 2020–21 academic year was 22,562 (SE 1886), down from the estimated 26,234 degrees (SE 2587) awarded in the 2014–15 academic year. The six CBMS surveys preceding the 2015 survey (see Table S.3 for the surveys of 1995, 2000, 2005, and 2010 and CBMS2000, Table SE.4, p. 14 for 1985 and 1990) reported a declining trend in the total number of bachelor's degrees awarded

by four-year mathematics and statistics departments combined, and over the 25 years, 1985–2010, the estimated total number of bachelor's degrees awarded during the previous academic year had decreased by 23% (but when degrees in computer science are removed from the totals, the estimated numbers of degrees awarded in the previous academic year by four-year mathematics and statistics departments combined are relatively constant and are virtually the same in 1985 and 2010) (see Figure S.3.1 and CBMS2000, Table SE.4, p. 14). The CBMS 2021 survey estimate of 21,409 (SE 1829) degrees awarded by four-year mathematics and statistics departments combined in 2020–21, excluding degrees awarded in computer science, is the second highest such number observed in a CBMS survey (the highest being the 2015 estimate of 22,266 (SE 2008)).

Table S.3 and Figure S.3.2 show the breakdown of bachelor's degrees awarded into the different categories of majors, in 2020–21 and over the last five CBMS surveys. The estimated number of bachelor's degrees in mathematics education has been declining; the 2020–21 estimate is 60% (4.1 SEs) less than the 1999–2000 estimate and is the smallest estimate among the six CBMS surveys in Table S.3. The estimated number of bachelor's degrees awarded during 2020–21 in statistics by four-year mathematics and statistics departments combined is estimated at 3703 (SE 591), up from 1509 in 2014–15. The estimated number of bachelor's degrees awarded in computer science, while small, and mainly confined to bachelor's-level mathematics departments, is still a significant number, e.g., in 2020–21 it was larger than the estimated number of bachelor's degrees awarded in statistics by four-year mathematics departments (see Table E.1.A in Chapter 3). For the first time, the CBMS 2021 survey asked four-year statistics departments for the number of bachelor's degrees they awarded in data science.

The CBMS 2021 survey defined a “joint major” as “a student who completes a single major in your department that integrates courses from mathematics and some other program or department, and typically requires fewer credit hours than is the sum of the credit hours required by the separate majors.” “Double majors,” students who complete two separate majors, were counted in the CBMS survey according to the category of mathematics or statistics major they complete. Beginning in the CBMS 2010 survey, all joint mathematics majors were joined into one category of “joint majors,” rather than breaking them down into possible kinds of joint majors, which had been the CBMS survey practice before 2010. In 2020–21, the estimated number of degrees awarded in the category of joint majors awarded by four-year mathematics departments decreased 15% (0.5 SEs) from 2014–15.

Table S.3 also shows that the percentage of all bachelor's degrees awarded to women through four-year U.S. mathematics and statistics departments combined has



**TABLE S.3.** Combined total of all bachelor's degrees in mathematics and statistics departments at four-year colleges and universities between July 1 and June 30 in 1994–95, 1999–2000, 2004–2005, 2009–10, 2014–15, and 2020–21 by selected majors and gender.

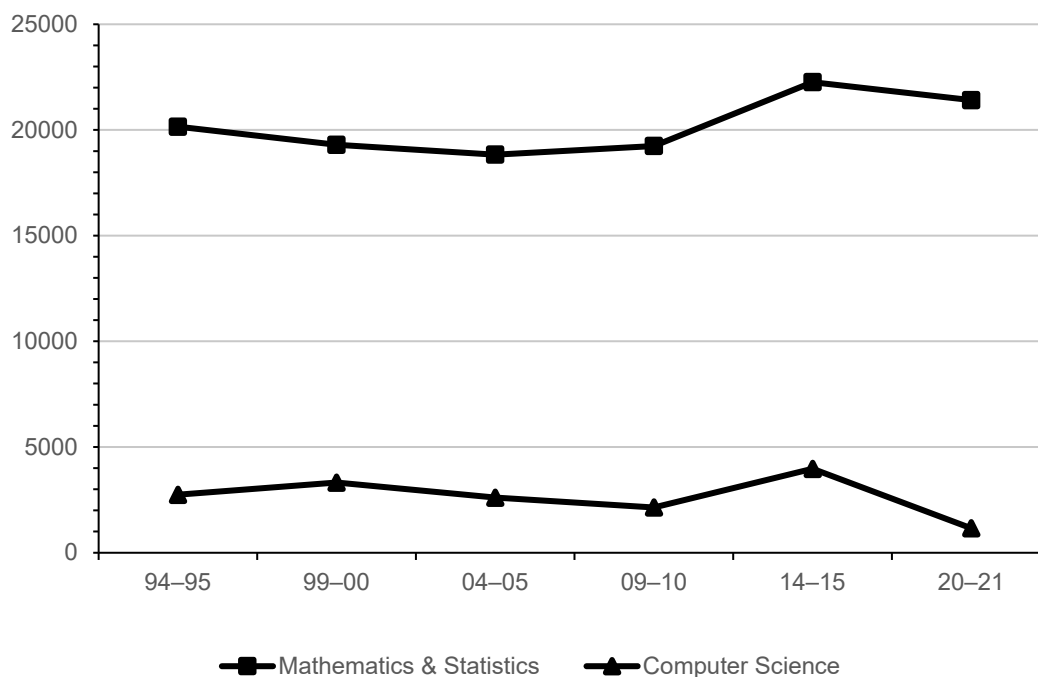
| Major                                                           | 94–95 | 99–00 | 04–05 | 09–10 | 14–15 | 20–21 |
|-----------------------------------------------------------------|-------|-------|-------|-------|-------|-------|
| Mathematics (except as reported below)                          | 12456 | 10759 | 12316 | 12468 | 12794 | 11701 |
| Mathematics Education                                           | 4829  | 4991  | 3369  | 3614  | 2880  | 1993  |
| Statistics Education                                            | na    | na    | na    | na    | na    | 0     |
| Statistics (except Actuarial Science)                           | 1031  | 502   | 527   | 856   | 1509  | 3703  |
| Biostatistics                                                   | na    | na    | na    | na    | na    | 0     |
| Actuarial Mathematics                                           | 620   | 425   | 499   | 849   | 2354  | 706   |
| Actuarial Science                                               | na    | na    | na    | na    | na    | 139   |
| Data Science                                                    | na    | na    | na    | na    | na    | 537   |
| All Joint Majors in Math Depts (combined) <sup>1</sup>          | --    | --    | --    | 1222  | 1821  | 1542  |
| Joint Statistics & Computer Science                             | na    | na    | na    | na    | na    | 14    |
| Joint Mathematics & Computer Science                            | 453   | 876   | 719   | --    | --    | na    |
| Joint Mathematics & Statistics <sup>2</sup>                     | 188   | 196   | 203   | --    | --    | 368   |
| Joint Math/Stat & Business or Economics                         | na    | na    | 214   | --    | --    | na    |
| Other (includes Operations Research prior to 2010) <sup>3</sup> | 577   | 1550  | 985   | 231   | 907   | 705   |
| Total Mathematics, Statistics & Joint degrees                   | 20154 | 19299 | 18833 | 19241 | 22266 | 21409 |
| Number of women                                                 | 9061  | 9017  | 8192  | 8692  | 9643  | 8641  |
| Computer Science degrees                                        | 2741  | 3315  | 2603  | 2137  | 3968  | 1154  |
| Number of women                                                 | 532   | 808   | 465   | 394   | 1302  | 251   |
| Total degrees                                                   | 22895 | 22614 | 21437 | 21377 | 26234 | 22562 |
| Number of women                                                 | 9593  | 9825  | 8656  | 9086  | 10946 | 8892  |

Note: Round-off may make column totals seem inaccurate.

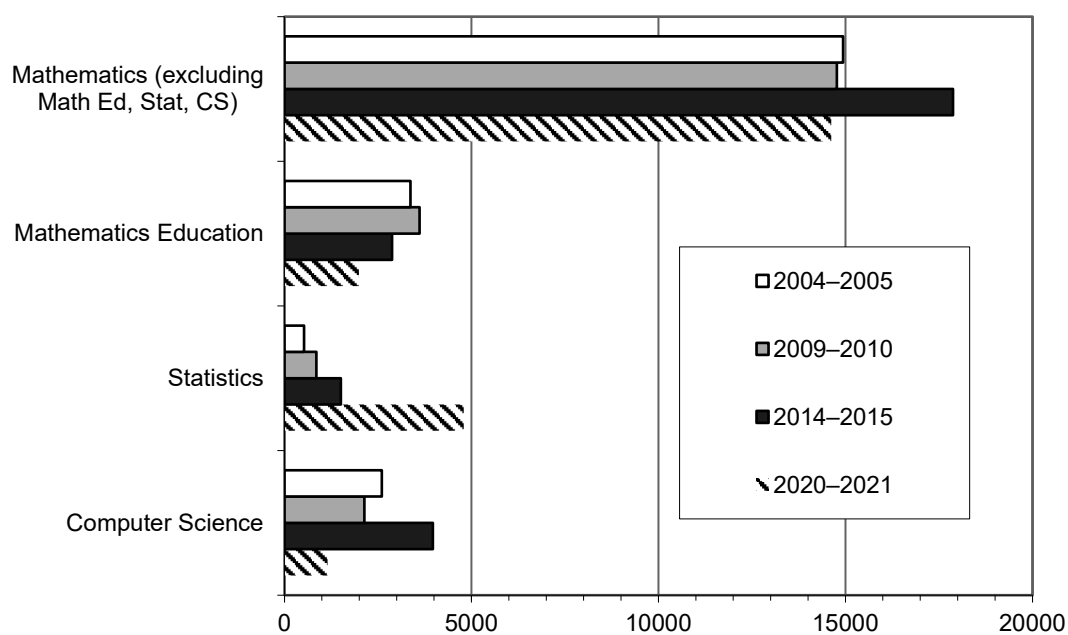
<sup>1</sup> Beginning in 2010, the survey asked for the total number of all joint majors.

<sup>2</sup> In statistics departments.

<sup>3</sup> Prior to 2010, Operations Research was a separate category. Beginning in 2010, Operations Research is included in Other Mathematics.



**FIGURE S.3.1.** Number of bachelor's degrees in mathematics and statistics and in computer science granted through mathematics and statistics departments in academic years 1994–1995, 1999–2000, 2004–2005, 2009–2010, 2014–2015, and 2020–2021.



**FIGURE S.3.2.** Number of bachelor's degrees awarded by mathematics and statistics departments (combined) at four-year colleges and universities between July 1 and June 30 in 2004–2005, 2009–2010, 2014–2015, and 2020–2021.

remained relatively constant; it was estimated at 43% in 1999–2000, 40% in 2004–5, 43% in 2009–10, 42% in 2014–15, and the 2020–21 estimate is slightly lower at 39%. When degrees in computer science degrees awarded by mathematics departments are excluded, then the estimated percentage of bachelor's degrees awarded to women through U.S. mathematics and statistics departments was 47% in 1990–2000, 44% in 2004–5, 45% in 2009–10, 43% in 2014–15, and the 2020–21 estimate is 40%. Tables E.1.A and E.1.B in Chapter 3 show that these percentages vary across levels of mathematics and statistics departments and categories of majors.

There are other estimates of numbers of bachelor's degrees awarded by four-year mathematical sciences departments. NCES also provides data on the numbers of bachelor's degrees awarded; these data come from the IPEDS data submitted by a university office. The Annual Survey administered by the American Mathematical Society gathers data from department chairs, including the number of bachelor's degrees awarded the previous academic year. The NCES, Annual Survey, and CBMS estimates of number of degrees awarded are not identical. The NCES estimate for number of bachelor's degrees awarded in mathematics and statistics in 2020–21 is 27,092, up from their estimate of 21,854 degrees awarded in 2014–15. The NCES data is not broken down into type of major. Unlike the Annual Survey and CBMS data, the NCES data do not always include double majors or mathematics education majors, and the NCES data may not include computer science majors given in a mathematics department in the totals of mathematics degrees awarded. NCES data is census data, the Annual Survey data comes from a census (but it is not required to be submitted, as the NCES data are), and the CBMS data are estimates based upon a stratified random sample of a sample space that is not identical to that of the IPEDS data.

### **Tables S.4–S.8: Appointment Type of Instructors in Undergraduate Mathematics and Statistics Sections**

CBMS2021 Tables S.4–S.8 provide information about the instructors in undergraduate mathematics and statistics sections of courses taught in mathematics and statistics departments of four- and two-year colleges and universities; information on instructors in computer science courses taught in four-year mathematics departments is also given. For the CBMS 2021 survey, the appointment types of the instructors of courses taught at four-year institutions were broken into four categories: tenured and tenure-eligible (TTE) faculty, other full-time (OFT) faculty who are full-time but not TTE, part-time (PT) faculty, and graduate teaching assistants (GTAs). A course was to be reported as being taught by a GTA if and only if the GTA was the “instructor of record”

for the course. GTAs who ran discussion or recitation sections as part of a lecture/recitation course were not included in this category. In CBMS2021, two-year college full-time faculty were broken into three categories: full-time permanent faculty (usually tenured), full-time continuing faculty (usually non-tenured), and other temporary full-time faculty. Part-time faculty were divided into two categories: part-time faculty paid by the college and others paid by a third party, such as a high school. Tables S.4–S.8 are broken down further, by courses and by the level of the department, in tables in Chapters 3, 5, 6, and 7.

There are several different class formats, and the size of a section depends on the class format. In the 2015 and 2021 CBMS surveys, three formats for sections of courses taught at four-year departments were considered: (1) lecture with separately scheduled recitation/problem/laboratory sessions, (2) individual sections (not included in (1)) that meet as a class with an instructor at a regularly scheduled time, and (3) other, not included in (1) or (2). For pre-college-level courses (combined), introductory-level courses (combined), calculus courses, other calculus-level courses (combined), lower-level statistics courses taught, respectively, in four-year mathematics and statistics departments, and lower-level computer science courses (combined) taught in four-year mathematics departments, the survey asked each department to give the total number of sections taught in each of these formats, and for each of these formats used, to give the breakdown into the number of sections taught by each of the instructor appointment types. For classes taught in the lecture/recitation format, the number of recitation sections was also requested.

The percentages of sections in mathematics and statistics courses at two-year colleges in fall 2021 taught by full- and part-time faculty are presented in Tables S.4–S.8. The number of sections in all courses and the percentage taught by part-time faculty at two-year colleges are presented in Table TYE.9 in Chapter 6. For the average section size for in-person and distance/remote courses in two-year colleges, see Tables TYE.7 and TYE.8 in Chapter 6.

In Tables S.4–S.8, the total number of faculty of each appointment type did not always match the total number of sections reported. When there were more sections than total faculty itemized, the difference was put in the category “unknown.” When there were more faculty given than there were total number of sections, each appointment type was weighted so that the percentages totaled 100%.

For courses taught at four-year mathematics or statistics departments, Table S.4 gives the estimated percentages of the various faculty appointment types for instructors of sections of pre-college-level courses (combined), introductory-level courses (combined), calculus-level courses (combined), lower-level statistics courses (combined), and lower-level computer science courses

**TABLE S.4.** Percentage of fall 2021 sections (excluding distance learning sections) in courses of various types taught in mathematics and statistics departments of colleges and universities by various types of instructors, and percentage of sections taught by full- and part-time faculty in mathematics departments at public two-year colleges in fall 2021, with data for fall 2015 from CBMS2015, Table S.4, p. 15, and with data for fall 2010 from CBMS2010, Table S.5, p. 15. Also total enrollments (in 1000s).

|                                             | Percentage of Sections Taught by                  |                         |                |                                         |              |                                 |
|---------------------------------------------|---------------------------------------------------|-------------------------|----------------|-----------------------------------------|--------------|---------------------------------|
|                                             | Tenured/<br>Tenure-<br>Eligible <sup>1</sup><br>% | Other<br>Full-Time<br>% | Part-Time<br>% | Graduate<br>Teaching<br>Assistants<br>% | Unknown<br>% | Total<br>Enrollment<br>in 1000s |
| Four-Year Colleges & Universities           |                                                   |                         |                |                                         |              |                                 |
| Mathematics Department courses              |                                                   |                         |                |                                         |              |                                 |
| Mathematics Courses                         |                                                   |                         |                |                                         |              |                                 |
| Pre-college level 2021                      | 15                                                | 46                      | 33             | 6                                       | 0            | 89                              |
| Pre-college level 2015                      | nc                                                | nc                      | nc             | nc                                      | nc           | 244                             |
| Pre-college level 2010                      | 18                                                | 20                      | 44             | 9                                       | 9            | 201                             |
| Introductory level 2021                     | 28                                                | 38                      | 24             | 8                                       | 2            | 537                             |
| Introductory level 2015                     | nc                                                | nc                      | nc             | nc                                      | nc           | 954                             |
| Introductory level 2010                     | 32                                                | 22                      | 27             | 8                                       | 10           | 834                             |
| Calculus level 2021                         | 55                                                | 28                      | 9              | 6                                       | 2            | 580                             |
| Calculus level 2015                         | 52                                                | 24                      | 10             | 7                                       | 7            | 790                             |
| Calculus level 2010                         | 59                                                | 15                      | 12             | 7                                       | 8            | 743                             |
| Advanced level 2021                         | 82                                                | 14                      | 1              | 1                                       | 3            | 108                             |
| Advanced level 2015                         | 70                                                |                         |                |                                         | 30           | 154                             |
| Statistics courses                          |                                                   |                         |                |                                         |              |                                 |
| Introductory level 2021                     | 45                                                | 28                      | 25             | 2                                       | 1            | 129                             |
| Introductory level 2015                     | 41                                                | 21                      | 25             | 4                                       | 8            | 235                             |
| Introductory level 2010                     | 48                                                | 14                      | 22             | 4                                       | 12           | 218                             |
| Upper level 2021 sections                   | 86                                                | 10                      | 2              | 1                                       | 2            | 47                              |
| Upper level 2015 sections                   | 53                                                |                         |                |                                         | 47           | 60                              |
| Computer Science courses                    |                                                   |                         |                |                                         |              |                                 |
| Lower level 2021                            | 58                                                | 22                      | 20             | 0                                       | 0            | 28                              |
| Lower level 2015                            | 46                                                | 20                      | 14             | 0                                       | 21           | 44                              |
| Lower level 2010                            | 50                                                | 17                      | 29             | 1                                       | 3            | 52                              |
| Statistics Department courses               |                                                   |                         |                |                                         |              |                                 |
| Introductory level 2021                     | 18                                                | 46                      | 12             | 24                                      | 0            | 65                              |
| Introductory level 2015                     | 14                                                | 25                      | 10             | 31                                      | 20           | 90                              |
| Introductory level 2010                     | 33                                                | 17                      | 12             | 15                                      | 23           | 77                              |
| Upper level 2021                            | 54                                                | 33                      | 6              | 7                                       | 0            | 28                              |
| Upper level 2015                            | 55                                                |                         |                |                                         | 45           | 50                              |
| Two-Year College<br>Mathematics Departments | Full-Time <sup>2</sup> Part-Time                  |                         |                |                                         |              |                                 |
| All 2021 sections                           | 33                                                |                         | 67             |                                         |              | 510                             |
| All 2015 sections                           | 64                                                |                         | 36             |                                         |              | 1693                            |
| All 2010 sections                           | 54                                                |                         | 46             |                                         |              | 1836                            |

Note: Some faculty sections do not add to 100% because certain institutions reported fewer sections by individual faculty type than the total.

<sup>1</sup> Before 2010, the category was “tenured/tenure-eligible”; the word “permanent” was added in 2010. In 2015, the word “permanent” was deleted.

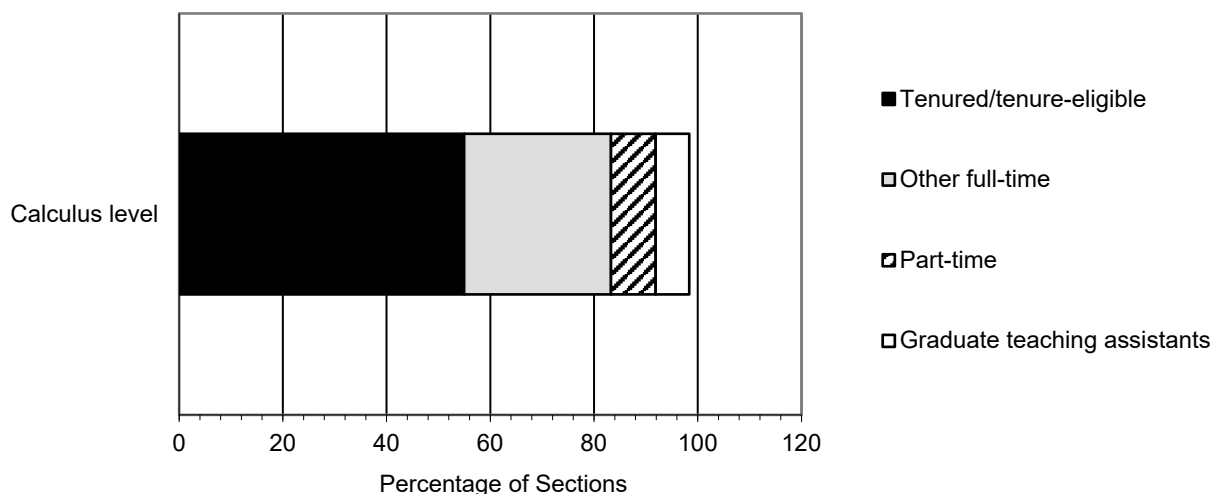
<sup>2</sup> “Full-time” includes full-time permanent, full-time continuing, and other full-time faculty at two-year colleges. For a detailed explanation of these terms, see Chapter 7.

\* Beginning in 2005, the CBMS survey asked departments to specify the number of upper-division sections and the number taught by tenured and tenure-eligible faculty. The deficit from 100% is reported as “unknown.”

Some rows do not sum to 100% due to round-off.

Note: 0 means less than one half of 1%.

nc = Not collected in 2015



**FIGURE S.4.1.** Percentage of sections in calculus-level mathematics courses in mathematics departments at four-year colleges and universities by type of instructor in fall 2021. Deficits from 100% represent unknown instructors.

(combined) in fall 2010, 2015, and 2021 (when the current data was gathered); the estimated total enrollments in these course categories is also given. The list of courses included in each of these categories can be seen in Appendix I. Table S.4 provides a macroscopic view of the appointment types of instructors at various levels of courses; in Chapter 3, Tables E.5 and E.6 break down some of the data on four-year departments in Table S.4 by the level (bachelor's, master's, or doctoral) of the mathematics and statistics department, revealing important trends in the data. Over the three surveys considered, from fall 2010 to fall 2021, there perhaps appears to be a somewhat greater use of OFT faculty generally, smaller use of TTE faculty in lower-level courses, and greater use of TTE faculty in upper-level courses, though some of the SEs are large. Figure S.4.1 contains a bar graph showing the distribution of estimated appointment types of instructors in calculus-level courses taught in four-year mathematics departments.

Table S.4 shows that an estimated 33% (SE 7.9) of sections in mathematics courses (excluding distance and dual enrollments) were taught by full-time faculty in two-year college mathematics departments in fall 2021. The number of sections taught by part-time faculty was 67% (SE 7.9),

an increase of 31 percentage points from 2015. Increases in part-time instruction were experienced in all courses and course categories. A detailed list of courses taught by part-time faculty is presented in Table TYE.9 in Chapter 6.

Calculus courses are important for the mathematics major, as well as for many other STEM majors, and hence CBMS surveys have paid particular attention to calculus courses. The 2021 survey made the same simplifying assumptions about calculus courses that were made in recent CBMS surveys. First, the CBMS survey divided all calculus courses into two pieces: “Mainstream Calculus” and “Non-Mainstream Calculus.” “Mainstream Calculus” consists of the calculus courses that are prerequisites for upper-level mathematics courses, as well as courses required in the physical sciences and in engineering, while “Non-Mainstream Calculus” consists of all the other calculus courses (often with titles such as “Calculus for Business and Social Science” or “Calculus for the Life Sciences”).

Table S.5 presents the estimated percentages of sections taught by faculty of the various appointment types for Mainstream Calculus I & II in fall 2021 and includes the comparable data from the CBMS surveys from 2010 and 2015 for courses offered in four-year mathematics

**TABLE S.5.** Percentage of fall 2021 sections in Mainstream Calculus I and II (not including distance learning and dual-enrollment sections) taught by various kinds of instructors in mathematics departments at four-year colleges and universities by size of sections with fall 2010 and 2015 data. Percentage of sections taught by full- and part-time faculty in mathematics programs at public two-year colleges in fall 2021, 2015, and 2010. Also total enrollments (in 1000s), and average section sizes.

|                                       | Percentage of Sections Taught by                  |                         |                |                                         |              |                        |                            |
|---------------------------------------|---------------------------------------------------|-------------------------|----------------|-----------------------------------------|--------------|------------------------|----------------------------|
|                                       | Tenured/<br>Tenure-<br>Eligible <sup>1</sup><br>% | Other<br>Full-Time<br>% | Part-Time<br>% | Graduate<br>Teaching<br>Assistants<br>% | Unknown<br>% | Enrollment<br>in 1000s | Average<br>Section<br>Size |
| Four-Year Colleges & Universities     |                                                   |                         |                |                                         |              |                        |                            |
| Mainstream Calculus I                 |                                                   |                         |                |                                         |              |                        |                            |
| Lecture with separate recitation      | 28                                                | 42                      | 4              | 24                                      | 0            | 91                     | 68                         |
| Sections that meet as a class         | 52                                                | 25                      | 16             | 5                                       | 2            | 118                    | 27                         |
| Other sections                        | 23                                                | 12                      | 0              | 65                                      | 0            | 0                      | 23                         |
| Course total 2021                     | 47                                                | 29                      | 13             | 10                                      | 2            | 209                    | 36                         |
| Course total 2015                     | 50                                                | 24                      | 12             | 7                                       | 8            | 255                    | 40                         |
| Course total 2010                     | 53                                                | 18                      | 15             | 7                                       | 8            | 234                    | 35                         |
| Mainstream Calculus II                |                                                   |                         |                |                                         |              |                        |                            |
| Lecture with separate recitation      | 38                                                | 38                      | 2              | 23                                      | 0            | 31                     | 52                         |
| Sections that meet as a class         | 56                                                | 31                      | 7              | 4                                       | 2            | 47                     | 22                         |
| Other sections                        | 40                                                | 60                      | 0              | 0                                       | 0            | 0                      | 1                          |
| Course total 2021                     | 52                                                | 32                      | 6              | 8                                       | 1            | 78                     | 29                         |
| Course total 2015                     | 54                                                | 26                      | 7              | 6                                       | 7            | 125                    | 39                         |
| Course total 2010                     | 59                                                | 14                      | 12             | 7                                       | 8            | 128                    | 36                         |
| Total Mainstream Calculus I & II 2021 | 48                                                | 30                      | 11             | 9                                       | 2            | 287                    | 34                         |
| Total Mainstream Calculus I & II 2015 | 51                                                | 6                       | 8              | 5                                       | 7            | 381                    | 40                         |
| Total Mainstream Calculus I & II 2010 | 55                                                | 16                      | 14             | 7                                       | 8            | 362                    | 35                         |
|                                       |                                                   |                         |                |                                         |              |                        |                            |
| Two-Year Colleges <sup>2</sup>        | Full-Time <sup>3</sup><br>%                       |                         |                |                                         |              |                        |                            |
|                                       | Part-Time<br>%                                    |                         |                |                                         |              |                        |                            |
| Total Mainstream Calculus I & II 2021 | 75                                                |                         |                |                                         |              | 40                     | 17                         |
| Total Mainstream Calculus I & II 2015 | 84                                                |                         |                |                                         |              | 94                     | 26                         |
| Total Mainstream Calculus I & II 2010 | 89                                                |                         |                |                                         |              | 93                     | 21                         |

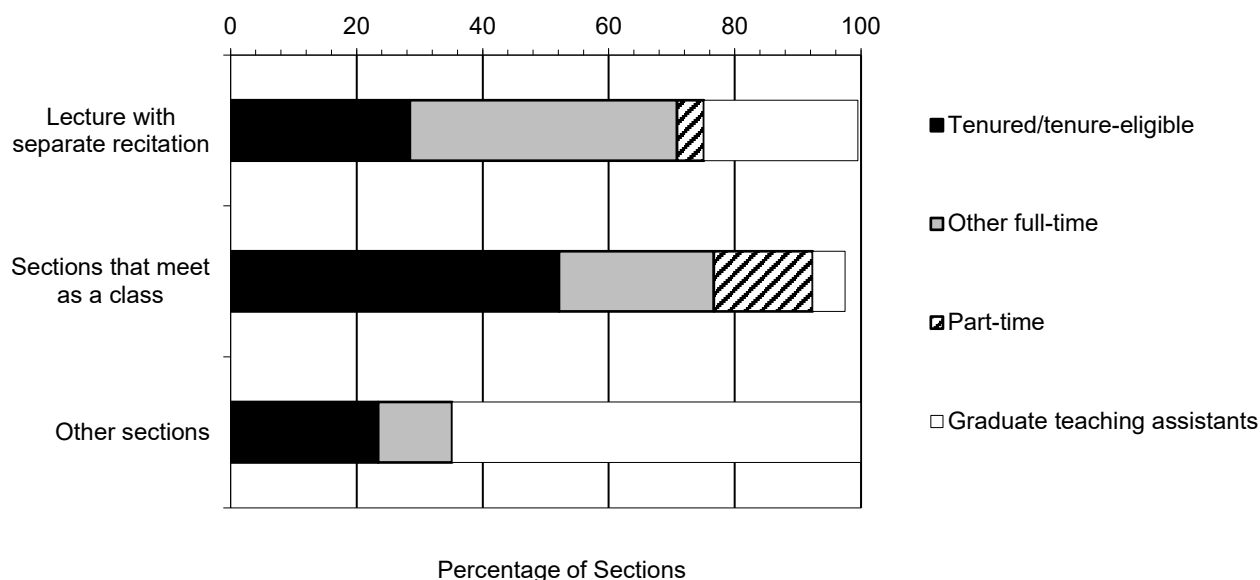
Note: Some faculty sections do not add to 100% because certain institutions reported fewer sections by individual faculty type than the total.

Percentage sums across rows may differ from 100% due to round-off.

<sup>1</sup> Before 2010, the category was “tenured/tenure-eligible”; the word “permanent” was added in 2010. In 2015, the word “permanent” was deleted.

<sup>2</sup> Due to the changes in the 2021 survey, statistics of total Mainstream Calculus are shown.

<sup>3</sup> “Full-time” includes full-time permanent, full-time continuing, and other full-time faculty at two-year colleges. For a detailed explanation of these terms, see Chapter 7.



**FIGURE S.5.1.** Percentage of sections in Mainstream Calculus I taught by tenured/tenure-eligible, other full-time, part-time, and graduate teaching assistants in mathematics departments at four-year colleges and universities by type of sections in fall 2021. Deficits from 100% represent unknown instructors.

departments and in public two-year college mathematics departments. The estimated total enrollment and estimated average section size in fall 2021 are also given. The fall 2021 estimates for four-year departments are broken down by the format type of the calculus course. Figure S.5.1 is a bar graph that displays the estimated fall 2021 percentages of the various appointment types broken down by the format of the class for Mainstream Calculus I classes in four-year mathematics departments. Table S.6 provides this same data for Non-Mainstream Calculus.

Table S.7 provides the data for introductory statistics courses, broken down by course offered in four- and two-year college mathematics departments and statistics departments, and Table S.8 and Figure S.8.1 provide data on the appointment type of the instructor for introductory-level statistics courses in statistics departments. Tables S.5–S.8 also present total (non-distance learning) enrollments and average section size. Data on computer science courses is provided in Chapter 3, Tables E.7 and E.8. Further detail on the appointment type of sections of courses taken by beginning students at four-year colleges and universities is given in Chapter 5, Tables FY.1, FY.2, FY.3, and FY.4.

There has been concern expressed at the decreasing percentages of TTE faculty, and increasing percentages of OFT and PT faculty, teaching lower-level mathematics and statistics courses. The small changes observed in the CBMS 2021 survey are consistent with these trends.

### Table S.9: Distance/Remote Learning Enrollments

Among the largest changes from past surveys observed in the CBMS 2021 survey data is the increase in estimated distance/remote learning enrollments. The COVID-19 pandemic and consequent pivoting to courses delivered online developed faculty skills and student interest in online education (see CBMS COVID study in Appendix IV). Even before the pandemic, two-year college mathematics departments had already begun to report increased online enrollments in 2015, and the 2021 CBMS survey showed that an estimated 62% of two-year college enrollments occurred online (see Table TYE.11 in Chapter 6). At four-year mathematics departments in fall 2021, there was a small

**TABLE S.6.** Percentage of sections in Non-Mainstream Calculus I and Non-Mainstream Calculus II, III, etc. taught by various kinds of instructors in mathematics departments at four-year colleges and universities by size of sections, and percentage of sections taught by full- and part-time faculty in mathematics departments at public two-year colleges in fall 2021. Also total enrollments (in 1000s), and average section sizes. Distance learning and dual-enrollment sections are not included. (Data in parentheses show percentage of sections in 2010 and 2015.)

|                                                    | Percentage of Sections Taught by                  |                         |                |                                         |              |                        |                            |
|----------------------------------------------------|---------------------------------------------------|-------------------------|----------------|-----------------------------------------|--------------|------------------------|----------------------------|
|                                                    | Tenured/<br>Tenure-<br>Eligible <sup>1</sup><br>% | Other<br>Full-Time<br>% | Part-Time<br>% | Graduate<br>Teaching<br>Assistants<br>% | Unknown<br>% | Enrollment<br>in 1000s | Average<br>Section<br>Size |
| Four-Year Colleges & Universities                  |                                                   |                         |                |                                         |              |                        |                            |
| Non-Mainstream Calculus I                          |                                                   |                         |                |                                         |              |                        |                            |
| Lecture with separate recitation                   | 11                                                | 76                      | 6              | 8                                       | 0            | 10                     | 57                         |
| Sections that meet as a class                      | 30                                                | 45                      | 16             | 9                                       | 0            | 60                     | 41                         |
| Other sections                                     | 0                                                 | 0                       | 100            | 0                                       | 0            | 0                      | 41                         |
| Course total 2021                                  | 28                                                | 49                      | 15             | 8                                       | 0            | 71                     | 43                         |
| (2010, 2015)                                       | (31,28)                                           | (24,29)                 | (23,19)        | (12,17)                                 | (11,7)       | (99,91)                | (42,42)                    |
| Non-Mainstream Calculus II, III, etc. <sup>2</sup> |                                                   |                         |                |                                         |              |                        |                            |
| Course total 2021                                  | 41                                                | 47                      | 12             | 0                                       | 0            | 7                      | 55                         |
| (2010, 2015)                                       | (34,32)                                           | (15,19)                 | (17,36)        | (11,6)                                  | (22,7)       | (22,16)                | (29,37)                    |
| Total Non-Mnstrm Calculus I & II, III, etc.        | 29                                                | 48                      | 15             | 8                                       | 0            | 78                     | 43                         |
| (2010, 2015)                                       | (31,29)                                           | (22,27)                 | (21,22)        | (12,15)                                 | (14,7)       | (121,106)              | (39,42)                    |
| Two-Year Colleges <sup>3</sup>                     |                                                   |                         |                |                                         |              |                        |                            |
|                                                    | Full-Time <sup>4</sup><br>%                       |                         | Part-Time<br>% |                                         |              |                        |                            |
| Total Non-Mnstrm Calculus I & II                   | 78                                                |                         | 22             |                                         |              | 5                      | 15                         |
| (2010, 2015)                                       | (73,71)                                           |                         | (27,29)        |                                         |              | (21,23)                | (21,26)                    |

Note: Some faculty sections do not add to 100% because certain institutions reported fewer sections by individual faculty type than the total.

<sup>1</sup> Before 2010, the category was “tenured/tenure-eligible”; the word “permanent” was added in 2010. In 2015, the word “permanent” was deleted.

<sup>2</sup> The 2010 survey asked for “Non-Mainstream Cal I, II, and III, etc.”—the data here are our best estimate for Calculus II, III, etc. Previous surveys asked only for Non-Mainstream Calculus II.

<sup>3</sup> Due to the changes in the 2021 survey, statistics of total Non-Mainstream Calculus are shown.

<sup>4</sup> “Full-time” includes full-time permanent, full-time continuing, and other full-time faculty at two-year colleges. For a detailed explanation of these terms, see Chapter 7.

Sums of percentages across rows may differ from 100% due to round-off.



**TABLE S.7.** Percentage of sections in introductory probability and statistics courses taught by various types of instructors in mathematics departments at four-year colleges and universities by size of sections, and percentage of sections taught by full- and part-time faculty in mathematics departments at public two-year colleges in fall 2021; comparable data for (2010, 2015) when available. Also total enrollments (in 1000s), and average section sizes. Distance learning and dual-enrollments are not included. (Data in parentheses show percentage of sections in 2010 and 2015.)

|                                                                                   | Percentage of Sections Taught by                  |                         |                |                                         |              |                        |                            |
|-----------------------------------------------------------------------------------|---------------------------------------------------|-------------------------|----------------|-----------------------------------------|--------------|------------------------|----------------------------|
| Four-Year Colleges & Universities<br>Mathematics Departments                      | Tenured/<br>Tenure-<br>Eligible <sup>1</sup><br>% | Other<br>Full-Time<br>% | Part-Time<br>% | Graduate<br>Teaching<br>Assistants<br>% | Unknown<br>% | Enrollment<br>in 1000s | Average<br>Section<br>Size |
| Introductory Statistics (F1) <sup>2</sup> (no calculus prerequisite) <sup>3</sup> |                                                   |                         |                |                                         |              |                        |                            |
| Lecture with separate recitation                                                  | 34                                                | 57                      | 9              | 0                                       | 0            | 6                      | 31                         |
| Sections that meet as a class                                                     | 44                                                | 29                      | 23             | 3                                       | 1            | 100                    | 27                         |
| Other sections                                                                    | 0                                                 | 25                      | 75             | 0                                       | 0            | 3                      | 22                         |
| Course total (F1)<br>                                                             |                                                   |                         |                |                                         |              |                        |                            |

Note: Some faculty sections do not add to 100% because certain institutions reported fewer sections by individual faculty type than the total.

<sup>1</sup> Before 2010, the category was “tenured/tenure-eligible”; the word “permanent” was added in 2010. In 2015, the word “permanent” was deleted.

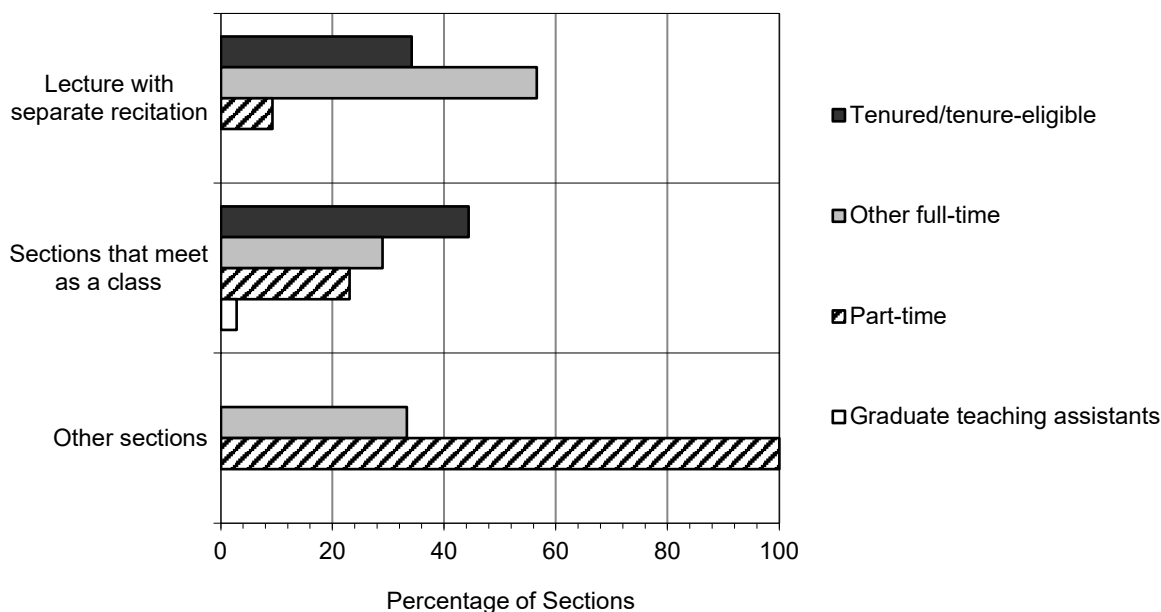
<sup>2</sup> F1 is the statistics course number on the four-year mathematics survey form.

<sup>3</sup> This course was called “Elementary Statistics” in previous CBMS surveys.

<sup>4</sup> “Full-time” includes full-time permanent, full-time continuing, and other full-time faculty at two-year colleges. For a detailed explanation of these terms, see Chapter 7.

Sums of percentages across rows may differ from 100% due to round-off.

Note: 0 means less than one half of 1%.



**FIGURE S.7.1.** Percentage of sections in Introductory Statistics (no Calculus prerequisite) taught by tenured/tenure-eligible/permanent, other full-time, part-time, and graduate teaching assistants in mathematics departments at four-year colleges and universities by type of sections in fall 2021. Deficits from 100% represent unknown instructors.

Note: Some faculty sections do not add to 100% because certain institutions reported fewer sections by individual faculty type than the total.

percentage (estimated 17%) of estimated enrollments in the courses listed in Table S.9, compared to fall 2015 distance/remote enrollments which were 4% of the estimated enrollments in these courses. At four-year statistics departments, distance/remote learning enrollments were estimated at 30% of fall 2021 Introductory Statistics enrollments, compared to 5% in fall 2015. More discussion of distance/remote learning enrollments in four-year departments occurs in Chapter 3 (where Table S.9 is repeated as Table E.4) and for two-year departments in Chapter 6 in Table TYE.11. Practices in distance/remote education are discussed in Chapter 2 (see Tables SP.8.A–SP.11.B).

### Table S.10: Pathways Course Sequences

The 2021 CBMS survey asked four- and two-year college mathematics departments if they had offered “a non-traditional ‘Pathways’ course sequence” in the last five years;

“a Pathways course sequence” was defined to be “a single course or course sequence that enables students to complete a college-level gateway mathematics or statistics course that is aligned to students’ academic and/or career goals within one academic year.” Table S.10 shows that in fall 2021 an estimated 71% (SE 6.48) of two-year college mathematics departments and an estimated 22% (SE 4.5) of all levels of mathematics departments combined have offered a Pathways sequence in the past five years. Eighty-two percent, 82% (SE 4), of four-year mathematics departments combined in fall 2021 offered some pre-college or introductory-level course (including precalculus) for credit. Table S.10 breaks this data down by level of the four-year mathematics department and shows that these Pathways sequences are more likely to be found in bachelor’s- and master’s-level departments than in doctoral-level departments (in fall 2021, only an estimated 6% (SE 4.5) of doctoral-level departments indicated they had offered a Pathways

**TABLE S.8.** Percentage of sections in Introductory Statistics for non-majors/minors taught by various kinds of instructors in statistics departments at four-year colleges and universities by size of sections in fall 2021. Also, total enrollments (in 1000s), and average section sizes. Distance learning enrollments are not included.

|                                                                                   | Percentage of Sections Taught by                  |                         |                |                                         |              | Enrollment<br>in 1000s | Average<br>Section<br>Size |
|-----------------------------------------------------------------------------------|---------------------------------------------------|-------------------------|----------------|-----------------------------------------|--------------|------------------------|----------------------------|
|                                                                                   | Tenured/<br>Tenure-<br>Eligible <sup>1</sup><br>% | Other<br>Full-Time<br>% | Part-Time<br>% | Graduate<br>Teaching<br>Assistants<br>% | Unknown<br>% |                        |                            |
| <b>Statistics Departments</b>                                                     |                                                   |                         |                |                                         |              |                        |                            |
| Introductory Statistics (no calculus prerequisite) <sup>2</sup> (E1) <sup>3</sup> |                                                   |                         |                |                                         |              |                        |                            |
| Lecture with separate recitation                                                  | 10                                                | 60                      | 10             | 20                                      | 0            | 22                     | 83                         |
| Sections that meet as a class                                                     | 18                                                | 33                      | 17             | 32                                      | 0            | 27                     | 58                         |
| Other sections                                                                    | 50                                                | 50                      | 0              | 0                                       | 0            | 0                      | 30                         |
| Course total                                                                      | 15                                                | 43                      | 15             | 27                                      | 0            | 49                     | 67                         |
| (2010, 2015) <sup>4</sup>                                                         | (29,13)                                           | (18,23)                 | (14,10)        | (16,33)                                 | (24,21)      | (56,66)                | (47,59)                    |
| Introductory Statistics (calculus prerequisite) (for non-majors) (E2)             |                                                   |                         |                |                                         |              |                        |                            |
| Lecture with separate recitation                                                  | 13                                                | 81                      | 0              | 0                                       | 6            | 3                      | 106                        |
| Sections that meet as a class                                                     | 31                                                | 54                      | 5              | 10                                      | 0            | 11                     | 68                         |
| Other sections                                                                    |                                                   |                         |                |                                         |              | 0                      |                            |
| Course total                                                                      | 28                                                | 58                      | 4              | 9                                       | 1            | 14                     | 73                         |
| (2015)                                                                            | (20)                                              | (33)                    | (8)            | (24)                                    | (15)         | (20)                   | (60)                       |
| Statistics for Pre-service Teachers (E3,E4)                                       |                                                   |                         |                |                                         |              |                        |                            |
| Course total (E3, E4)                                                             |                                                   |                         |                |                                         |              |                        |                            |
| Other Introductory-Level Statistics courses (E5)                                  |                                                   |                         |                |                                         |              |                        |                            |
| Course total (E5)                                                                 | 18                                                | 9                       | 18             | 55                                      | 0            | 1                      | 61                         |
| Total All Introductory courses                                                    |                                                   |                         |                |                                         |              |                        |                            |
| Course total (E1+E2+E3+E4+E5)                                                     | <b>18</b>                                         | <b>46</b>               | <b>13</b>      | <b>24</b>                               | <b>0</b>     | <b>65</b>              | <b>68</b>                  |

Note: Some faculty sections do not add to 100% because certain institutions reported fewer sections by individual faculty type than the total.

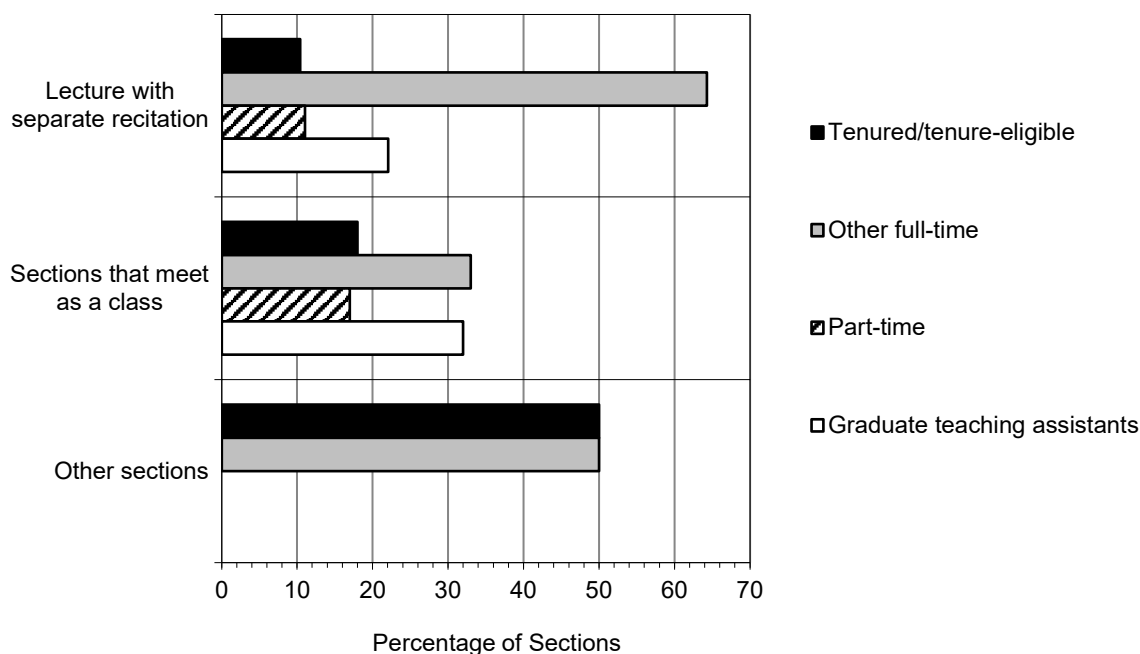
<sup>1</sup> Beginning in 2010, the CBMS survey added the word “permanent” to the description “tenured/tenure eligible” that was used previously. In 2015, the word “permanent” was deleted.

<sup>2</sup> In previous CBMS surveys, this course was called “Elementary Statistics.”

<sup>3</sup> E1 is the statistics course number on the four-year statistics survey form.

<sup>4</sup> Previous CBMS surveys gathered data for a course described as Probability and Statistics (no calculus prerequisite). Beginning in 2010, this description was replaced with Introductory Statistics (calculus prerequisite) (for non-majors).

Sums of percentages across rows may differ from 100% due to round-off.



**FIGURE S.8.1.** Percentage of sections in Introductory Statistics (no calculus prerequisite) taught by tenured/tenure-eligible faculty, other full-time faculty, part-time faculty, and graduate teaching assistants in statistics departments at four-year colleges and universities by type of sections in fall 2021.

sequence, and only 67% (SE 9.1) offered some pre-college or introductory-level mathematics course (including pre-calculus) for credit).

The CBMS 2015 survey found that 58% (SE 5.1) of two-year college mathematics departments had implemented a Pathways sequence (under a slightly different definition of Pathways); four-year mathematics departments were not asked about Pathways courses in the CBMS survey in 2015. At two-year colleges, the fall 2021 estimate of 71% (SE 6.48) of colleges implementing Pathways, mentioned above, was a thirteen-point increase compared with 2015. Pathways courses in Foundations increased 37 percentage points, Quantitative Reasoning/Literacy increased 28 percentage points, and Statistics increased 11 percentage points in fall 2021. See Table TYE.10 in Chapter 6.

Table S.10 also presents data on how often various pedagogical strategies were used in teaching pre-college and introductory classes in fall 2021, again providing a breakdown of this data by level of the four-year mathematics

department and including two-year college data. The strategy of “using assessments such as regular graded homework and quizzes to inform teaching” was used at least once a week in fall 2021 at an estimated 91% (SE 3.6) of two-year college mathematics departments, 92% (SE 3.3) of four-year mathematics departments combined, and only 78% (SE 10.4) of doctoral level mathematics departments. “Focusing on conceptual understanding” was used at least once a week at an estimated 87% (SE 4.5) of two-year college mathematics departments and 71% (SE 5.9) of four-year mathematics departments combined. “Integrating real-world applications” was used at an estimated 68% (SE 6.2) of two-year college mathematics departments and 56% (SE 6) of four-year mathematics departments combined. “Using student-centered active learning strategies” was used at an estimated 45% (SE 6.6) of two-year college mathematics departments and 47% (SE 6.5) of four-year mathematics departments combined.

**TABLE S.9.** Enrollments in distance/remote learning courses (meaning courses offered for credit in which half or more of the instruction occurs with the instructor and the students separated by time and/or place [e.g., courses in which half or more of the course is taught online either synchronously or asynchronously, by computer software, or by other technologies]), and other sections for various freshman and sophomore courses, by type of department, in fall 2021. Includes only distance/remote courses offered in normal practice, not courses that became distance/remote due to COVID-19 pandemic. (Fall 2015 data in parentheses.)

|                                                | Four-year Mathematics Departments |                    | Two-year Mathematics Departments |                    | Statistics Departments        |                   |
|------------------------------------------------|-----------------------------------|--------------------|----------------------------------|--------------------|-------------------------------|-------------------|
|                                                | Distance-Learning Enrollments     | Other Enrollments  | Distance-Learning Enrollments    | Other Enrollments  | Distance-Learning Enrollments | Other Enrollments |
| Pre-college Level                              | 12284<br>(8405)                   | 88747<br>(244475)  | 208883<br>(89035)                | 138952<br>(693252) |                               |                   |
| Introductory Level                             | 133444<br>(45226)                 | 536604<br>(954356) | 196368<br>(55227)                | 132425<br>(390066) |                               |                   |
| Calculus I<br>(mainstream and non-mainstream)  | 44760<br>(8968)                   | 280151<br>(346343) | 60350<br>(7455)                  | 33473<br>(84537)   |                               |                   |
| Calculus II<br>(mainstream and non-mainstream) | 10541<br>(3410)                   | 84923<br>(125126)  | 19203<br>(1813)                  | 12227<br>(32523)   |                               |                   |
| Differential Equations & Linear Algebra        | 13123<br>(1492)                   | 102737<br>(137567) | 11482<br>(480)                   | 5064<br>(13559)    |                               |                   |
| Introductory Statistics                        | 30157<br>(18696)                  | 128699<br>(234558) | 169586<br>(30608)                | 87832<br>(220671)  | 28218<br>(4291)               | 64955<br>(89620)  |

Note: For some enrollments and distance learning enrollments in this table, the SE was very large. See SEs in Appendix IX.

**TABLE S.10.** Instructional strategies used in mathematics departments of four- and two-year colleges in fall 2021. A “Pathways course sequence” is “a single course or course sequence that enables students to complete a college-level gateway mathematics or statistics course that is aligned to students’ academic and/or career goals within one academic year.”

|                                                                                                 | Mathematics Departments |           |              |       | Two-Year Colleges |  |
|-------------------------------------------------------------------------------------------------|-------------------------|-----------|--------------|-------|-------------------|--|
|                                                                                                 | Univ (PhD)              | Univ (MA) | College (BA) | Total | Total             |  |
| Department offered a "Pathways" course sequence in the last five years)                         | 6                       | 36        | 23           | 22    | 71                |  |
| Departments offer some Precollege of Introductory Level course for credit                       | 67                      | 81        | 85           | 82    |                   |  |
| Of those departments offering a course, how often is each of the instructional strategies used? |                         |           |              |       |                   |  |
| Focusing on conceptual understanding                                                            |                         |           |              |       |                   |  |
| At least once a week                                                                            | 69                      | 74        | 71           | 71    | 87                |  |
| Occasionally                                                                                    | 31                      | 19        | 29           | 28    | 13                |  |
| Almost never                                                                                    | 0                       | 7         | 29           | 1     |                   |  |
| Integrating real-world applications                                                             |                         |           |              |       |                   |  |
| At least once a week                                                                            | 57                      | 65        | 54           | 56    | 68                |  |
| Occasionally                                                                                    | 43                      | 28        | 46           | 43    | 31                |  |
| Almost never                                                                                    | 0                       | 7         | 0            | 1     | 1                 |  |
| Using student-centered active learning strategies                                               |                         |           |              |       |                   |  |
| At least once a week                                                                            | 32                      | 54        | 48           | 47    | 45                |  |
| Occasionally                                                                                    | 68                      | 34        | 46           | 47    | 54                |  |
| Almost never                                                                                    | 0                       | 12        | 6            | 6     | 2                 |  |
| Using assessments such as regular graded homework and quizzes to inform teaching                |                         |           |              |       |                   |  |
| At least once a week                                                                            | 78                      | 91        | 93           | 92    | 91                |  |
| Occasionally                                                                                    | 22                      | 4         | 7            | 8     | 9                 |  |
| Almost never                                                                                    | 0                       | 5         | 0            | 1     |                   |  |

Note: Blank entries represent strategies with no responses.

### Tables S.11 and S.12: Pedagogy in Introductory Statistics Course Instruction

The CMBS 2021 survey asked mathematics departments at two-year colleges and four-year colleges and at four-year college statistics departments questions about how the course “Introductory Statistics with no calculus prerequisite” was taught in fall 2021; this data for four-year departments is also discussed in Chapter 5.

Table S.11 shows the estimated percentages of four-year mathematics and statistics departments that offer Introductory Statistics with no calculus prerequisite in fall 2021 and the number of different kinds of courses offered. The Consortium for the Advancement of Undergraduate Statistics Education (CAUSE), <https://www.causeweb.org/cause/>, is among the organizations promoting changes to how Introductory Statistics courses are taught, and the CBMS 2021 survey probed the extent to which such changes had been implemented in mathematics and statistics departments. Table S.11 presents data on strategies used in teaching the course, while Table S.12 considers the use of technology in the course. The CBMS 2015 survey asked a slightly different set of questions of four-year mathematics and statistics departments regarding instruction in introductory statistics courses; that data can be found in CBMS2015, Table S.12 (p. 29), Table FY.5 (p. 145), and FY.6 (p. 146).

In fall 2021, ninety-three percent (93%, SE 3.8) of two-year mathematics departments offered Introductory Statistics. Another twelve percent (12%, SE 4.59) of the departments offered a probability course. See Table TYE.5 in Chapter 6.

Table S.11 shows that very similar use of these pedagogical methods was reported among two-year mathematics, four-year mathematics, and statistics departments in fall 2021. At least once a week, nearly all two-year mathematics, four-year mathematics, and statistics departments used assessments such as regular graded homework or quizzes to inform teaching, they integrated real world applications, and they emphasized conceptual understanding over formulas and procedures. About half of the departments indicated that they used student-centered active learning strategies at least once a week. About a quarter of the four-year departments said that students collected, organized, and analyzed real data at least once a week, but about 65% of these departments said that students did this occasionally, rather than at least once a week. The distribution of the percentages was quite similar over all levels of four-year mathematics departments, except for the use of regular graded homework or quizzes, which was said to be used less frequently at the doctoral-level departments.

Table S.12 records departments’ assessment of their level of success in implementing technology into courses in introductory statistics with no prerequisite. Again, the responses were relatively similar over the two-year mathematics departments, four-year mathematics departments, and statistics departments. All the departments indicated the greatest sense of success with the instructors’ use of technology to demonstrate concepts; somewhat less success was indicated in students’ abilities to use technology to explore data, analyze data, or solve problems.

### Demographics of the Mathematical Sciences Faculty

The remaining tables in this chapter present a snapshot of faculty demographics in mathematics and statistics departments of four-year colleges and universities as well as in the mathematics departments of two-year colleges during fall 2021. Further details about faculty in mathematics and statistics departments of four-year colleges and universities appear in Chapter 4, while additional information about faculty in mathematics departments of public two-year colleges is given in Chapter 7.

**Source of demographic data.** The demographic data on mathematics and statistics department faculty in four-year colleges and universities contained in the CBMS2021 report were not collected using the same survey instrument as the other data, nor was a random sample of institutions used. The demographic data were collected as part of the Annual Survey, a census, conducted each year by the American Mathematical Society and overseen by the Joint Data Committee of five professional societies: the American Mathematical Society, the American Statistical Association, the Institute of Mathematical Statistics, the Mathematical Association of America, and the Society for Industrial and Applied Mathematics. Reports on the Annual Survey are published each year in several issues of the *Notices of the American Mathematical Society* and online at <http://www.ams.org/profession/data/annual-survey/annual-survey>. Beginning with the CBMS survey in 2005, the demographic data on faculty for the CBMS surveys were collected as part of the Annual Surveys; sampled departments were asked additional demographic questions that do not normally appear on the Annual Survey but are a part of the CBMS surveys. As with the CBMS survey, the response rates for the Annual Survey were lower than in the past (see Appendix II for details on the survey methods).

**TABLES.11.** Instructional strategies in Introductory Statistics instruction in mathematics departments of four- and two-year colleges and statistics departments of four-year colleges in fall 2021 (percentages of departments).

|                                                                                                                                          | Mathematics Departments |           |              |       | Statistics Departments | Two-Year Colleges |
|------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|-----------|--------------|-------|------------------------|-------------------|
|                                                                                                                                          | Univ (PhD)              | Univ (MA) | College (BA) | Total | Total                  | Total             |
| Department offered an Introductory Statistics course                                                                                     | 38                      | 80        | 81           | 76    | 84                     | 93                |
| How many different kinds of Introductory Statistics courses designed for non-majors with no calculus prerequisite does department offer? |                         |           |              |       |                        |                   |
| 1                                                                                                                                        | 61                      | 61        | 71           | 69    | 18                     |                   |
| 2                                                                                                                                        | 20                      | 39        | 22           | 25    | 33                     |                   |
| 3                                                                                                                                        | 20                      | 0         | 3            | 4     | 23                     |                   |
| More than 3                                                                                                                              | 0                       | 0         | 3            | 2     | 28                     |                   |
| Of those departments offering a course, how often is each of the instructional strategies used?                                          |                         |           |              |       |                        |                   |
| Focusing on conceptual understanding                                                                                                     |                         |           |              |       |                        |                   |
| At least once a week                                                                                                                     | 89                      | 90        | 75           | 78    | 87                     | 82                |
| Occasionally                                                                                                                             | 11                      | 10        | 23           | 20    | 13                     | 18                |
| Almost never                                                                                                                             | 0                       | 0         | 2            | 1     |                        |                   |
| Integrating real-world applications                                                                                                      |                         |           |              |       |                        |                   |
| At least once a week                                                                                                                     | 80                      | 85        | 85           | 85    | 87                     | 89                |
| Occasionally                                                                                                                             | 20                      | 15        | 13           | 14    | 13                     | 11                |
| Almost never                                                                                                                             | 0                       | 0         | 2            | 1     |                        |                   |
| Students collect, organize, and analyze real data                                                                                        |                         |           |              |       |                        |                   |
| At least once a week                                                                                                                     | 26                      | 33        | 17           | 20    | 26                     |                   |
| Occasionally                                                                                                                             | 65                      | 62        | 66           | 65    | 64                     |                   |
| Almost never                                                                                                                             | 9                       | 5         | 18           | 15    | 10                     |                   |
| Using student-centered active learning strategies                                                                                        |                         |           |              |       |                        |                   |
| At least once a week                                                                                                                     | 32                      | 44        | 43           | 42    | 49                     | 57                |
| Occasionally                                                                                                                             | 59                      | 56        | 46           | 48    | 49                     | 41                |
| Almost never                                                                                                                             | 9                       | 0         | 12           | 10    | 3                      | 2                 |
| Using assessments such as regular graded homework and quizzes to inform teaching                                                         |                         |           |              |       |                        |                   |
| At least once a week                                                                                                                     | 69                      | 95        | 91           | 90    | 85                     | 92                |
| Occasionally                                                                                                                             | 31                      | 0         | 7            | 8     | 15                     | 8                 |
| Almost never                                                                                                                             | 0                       | 5         | 2            | 2     |                        |                   |

Note: Blank entries represent strategies with no responses.



**TABLE S.12.** Use of technology in teaching Introductory Statistics in mathematics departments of four- and two-year colleges and statistics departments of four-year colleges in fall 2021. Departments were asked, "How successful is your program in adopting each of the following uses of technology to your Introductory Statistics (no calculus prerequisite) course taught in fall 2021?"

|                                                                   | Mathematics Departments |           |              |       | Statistics Departments | Two-Year Colleges |
|-------------------------------------------------------------------|-------------------------|-----------|--------------|-------|------------------------|-------------------|
|                                                                   | Univ (PhD)              | Univ (MA) | College (BA) | Total | Total                  | Total             |
| Students use technology to explore concepts                       |                         |           |              |       |                        |                   |
| Very successful                                                   | 41                      | 43        | 48           | 47    | 64                     | 51                |
| Somewhat successful                                               | 59                      | 57        | 50           | 52    | 31                     | 46                |
| Not successful                                                    | 0                       | 0         | 2            | 1     | 5                      | 3                 |
| Instructors use technology to demonstrate concepts                |                         |           |              |       |                        |                   |
| Very successful                                                   | 50                      | 66        | 77           | 74    | 74                     | 70                |
| Somewhat successful                                               | 50                      | 34        | 21           | 25    | 26                     | 29                |
| Not successful                                                    | 0                       | 0         | 2            | 1     |                        | 2                 |
| Students use technology to analyze data                           |                         |           |              |       |                        |                   |
| Very successful                                                   | 50                      | 49        | 66           | 62    | 67                     | 66                |
| Somewhat successful                                               | 50                      | 51        | 32           | 36    | 28                     | 30                |
| Not successful                                                    | 0                       | 0         | 2            | 2     | 5                      | 4                 |
| Students' ability to use technology to solve problems is assessed |                         |           |              |       |                        |                   |
| Very successful                                                   | 41                      | 36        | 55           | 51    | 51                     | 43                |
| Somewhat successful                                               | 59                      | 53        | 44           | 46    | 46                     | 49                |
| Not successful                                                    | 0                       | 10        | 2            | 3     | 3                      | 8                 |

Note: Blank entries represent uses of technology with no responses.

**TABLE S.13.** Number of full- and part-time faculty in mathematics departments at four-year colleges and universities, in doctoral statistics departments at universities, and in mathematics programs at two-year colleges in fall 2005, 2010, 2015, and 2021. (Two-year college data since 2005 include only public two-year colleges.)

|                                              | 2005  | 2010  | 2015  | 2021  |
|----------------------------------------------|-------|-------|-------|-------|
| <b>Four-Year Colleges &amp; Universities</b> |       |       |       |       |
| Mathematics Departments                      |       |       |       |       |
| Full-time faculty                            | 21885 | 22293 | 22532 | 31807 |
| Part-time faculty                            | 6536  | 6050  | 7682  | 10101 |
| Statistics Departments (PhD)                 |       |       |       |       |
| Full-time faculty                            | 946   | 1004  | 1237  | 3396  |
| Part-time faculty                            | 112   | 105   | 128   | 845   |
| <b>Two-Year College Mathematics Programs</b> |       |       |       |       |
| Full-time faculty                            | 9403  | 10873 | 9800  | 10328 |
| Part-time faculty <sup>1</sup>               | 18227 | 23453 | 17888 | 13827 |

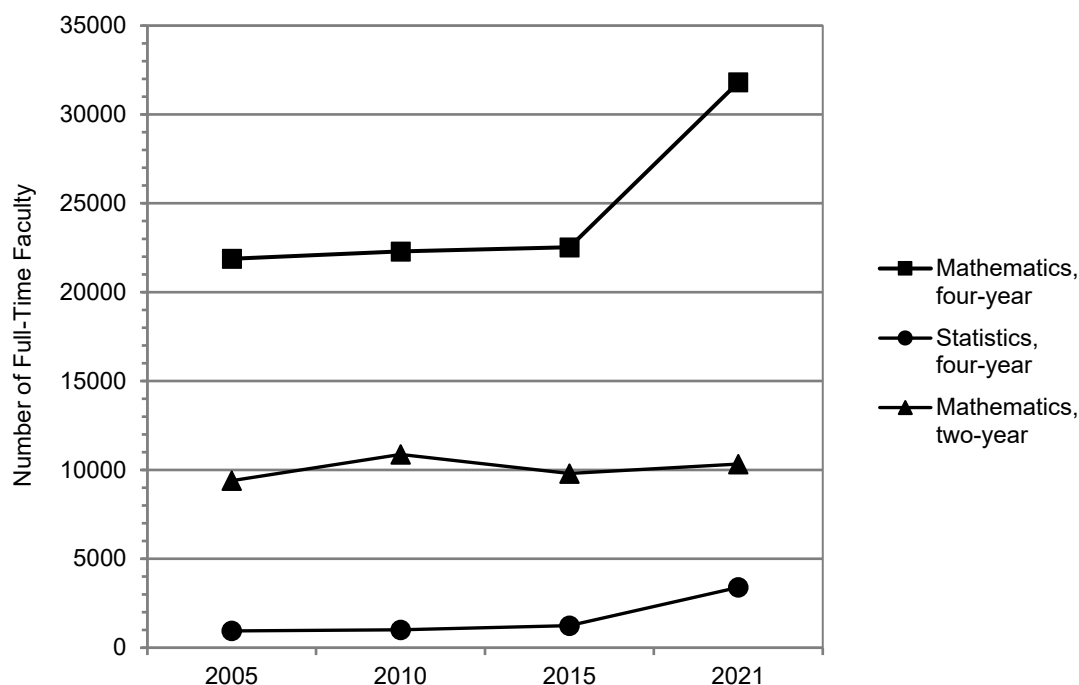
<sup>1</sup> Paid by two-year colleges. In fall 2000, there were an additional 776 part-time faculty in two-year colleges who were paid by a third party (e.g., by a school district for a dual-enrollment course). In 2005, the number paid by a third party was 1915; in 2010, the number paid by a third party was 2323; and in 2015, the number paid by a third party was 2359.

In comparing data from the CBMS surveys to the data published in the Annual Surveys, one must keep in mind several differences between the two surveys. The Annual Surveys do not include postdoctoral appointments as a part of “other full-time faculty” (OFT), while CBMS surveys do—i.e., CBMS survey tables list “other full-time faculty” (and these numbers include postdoctoral appointments), but they also break out the number of other full-time faculty who are postdoctoral appointments. The CBMS surveys of “statistics departments” include only statistics departments that offer an undergraduate program in statistics, while the Annual Surveys include all departments of statistics and biostatistics that award a PhD. However, the data for statistics departments that do not have an undergraduate program in statistics are not included in the tables that appear in this report. The 2005 and 2021 Annual Surveys did not include master’s-level statistics departments, while the 2010 and 2015 surveys included these departments; hence, comparisons to previous surveys remove the master’s-level statistics department data. The Annual Surveys use a census of bachelor’s-, master’s-, and doctoral-level programs, while the CBMS surveys use a stratified random sample. The demographic data for mathematics faculty at public two-year colleges were collected from the CBMS survey instruments and samples, as two-year college mathematics departments are not a part of the Annual Survey.

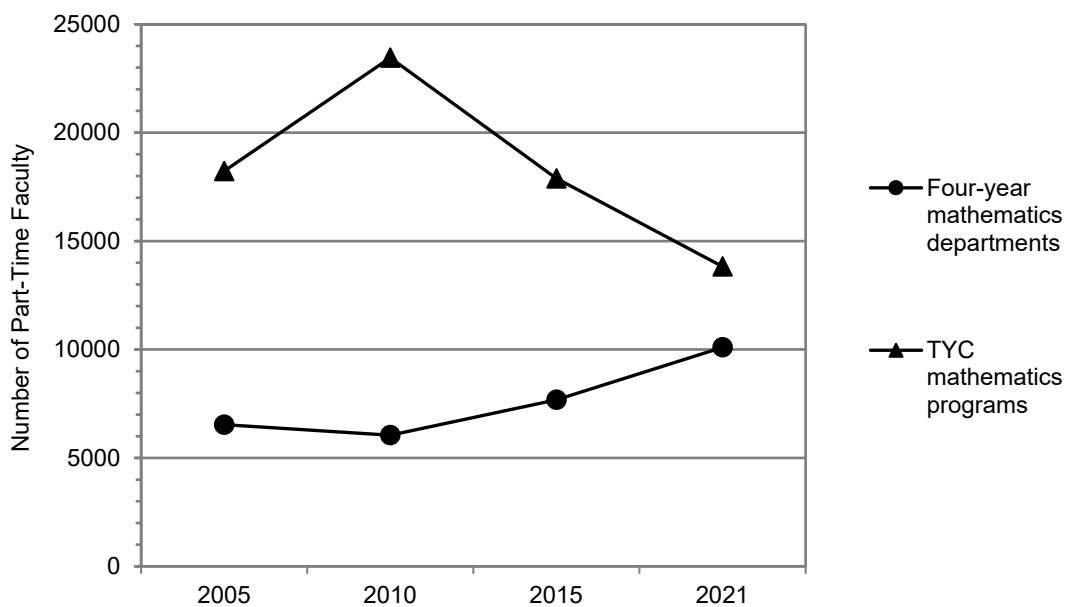
### Table S.13: Numbers of Faculty

Table S.13 shows the estimated numbers of full- and part-time mathematics faculty at two-year mathematics departments, all levels of four-year mathematics departments combined, and doctoral-level statistics departments in fall 2005, 2010, 2015, and 2021. Figure S.13.1 displays a graph of the numbers of full-time faculty at the three kinds of departments for each of the four years, while Figure S.13.2 shows the same information for the numbers of part-time mathematics faculty in two- and four-year institutions. Figures S.13.3, S.13.4, and S.13.5 display bar graphs of the numbers of full- and part-time faculty for mathematics departments at four-year institutions, mathematics departments at two-year colleges, and doctoral-level statistics departments, respectively. Further details on the estimated numbers of full- and part-time faculty for four-year colleges and universities are presented in Chapter 4, Table F.1, and for two-year colleges are presented in Chapter 7, Table TYF.1.

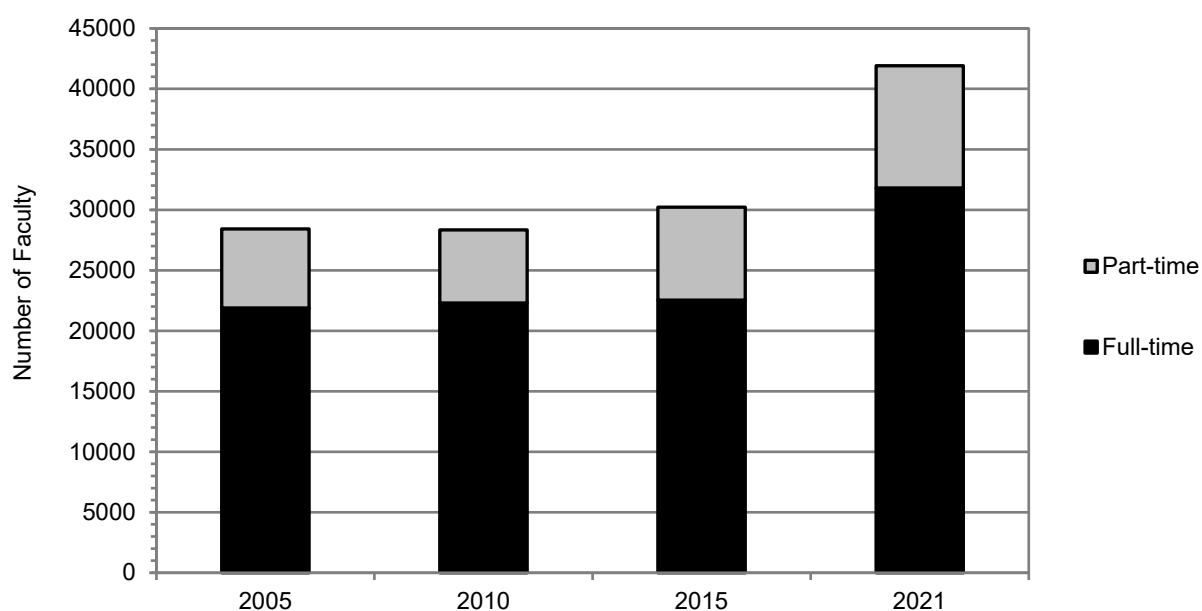
Despite the declines in enrollments in fall 2021 noted earlier, the estimated numbers of faculty increased from fall 2015 to fall 2021 for all categories of faculty in the table, except for part-time mathematics faculty at two-year colleges. According to Table S.13, the estimated number of full-time faculty from fall 2015 to fall 2021 increased 5% (0.5 SEs) in two-year mathematics departments, 41% (12 SEs) in



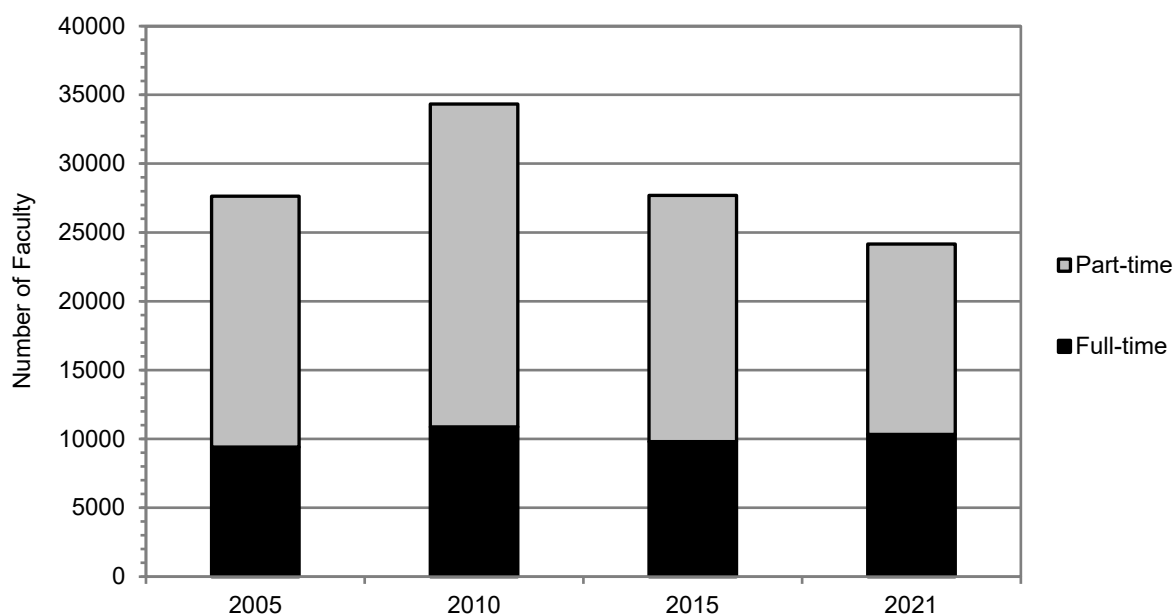
**FIGURE S.13.1.** Number of full-time faculty in mathematics departments of four year colleges and universities, in doctoral statistics departments, and in mathematics departments at public two-year colleges in fall 2005, 2010, 2015, and 2021. This figure can be compared to CBMS2015, Figure S.13.1, p. 33.



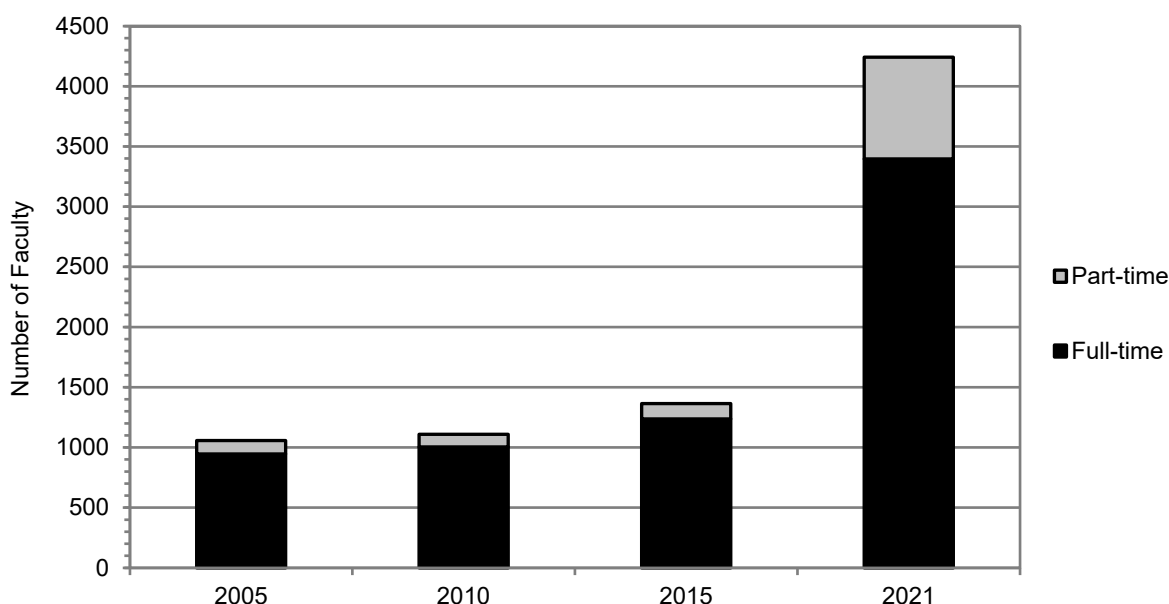
**FIGURE S.13.2.** Number of part-time faculty in mathematics departments at four-year colleges and universities and in mathematics departments at two-year colleges (TYCs) in fall 2005, 2010, 2015, and 2021. This figure can be compared to CBMS2015, Figure S.13.2, p. 33.



**FIGURE S.13.3.** Number of full- and part-time faculty in mathematics departments of four-year colleges and universities in fall 2005, 2010, 2015, and 2021. This figure can be compared to CBMS2015, Figure S.13.3, p. 34.



**FIGURE S.13.4.** Number of full- and part-time faculty in mathematics programs at public two-year colleges in fall 2005, 2010, 2015, and 2021. This figure can be compared to CBMS2015, Figure S.13.3, p. 34.



**FIGURE S.13.5.** Number of full- and part-time faculty in doctoral statistics departments in fall 2005, 2010, 2015, and 2021. This figure can be compared to CBMS2015, Figure S.13.5, p. 35.

four-year mathematics departments, and almost tripled (increased by 22 SEs) in doctoral statistics departments. Previous CBMS surveys showed that from fall 2000 to fall 2015, the estimated number of full-time mathematics faculty in four-year departments grew by 14%, while four-year mathematics departments' estimated total course enrollments grew by 36% (by 42% when computer science enrollments are removed) (see CBMS2015, Table S.2, p. 7), so the current increases in numbers of faculty at four-year mathematics departments may result from previous enrollment increases. Estimated doctoral-level statistics department enrollments have doubled since 2000 (see CBMS2015, Table S.2, p. 7).

When considering the 5% increase of full-time faculty at two-year mathematics departments (a total of 10,328 persons), it is important to examine the categories of full-time faculty. CBMS2015 and CBMS2021 collected data on full-time faculty in the three categories of faculty: permanent, continuing, and other faculty. In fall 2021, the total estimated number of faculty reported as "full-time permanent" faculty was 9126 (SE 918), a 10% increase of 812 persons from 2015. In 2021, full-time continuing and other faculty together totaled 1202 (SE 589), a decrease of 19% and 285 persons from fall 2015. See Table TYF.1 in Chapter 7.

Estimated numbers of part-time faculty also increased from fall 2015 to fall 2021: 31% (10 SEs) at four-year mathematics departments and almost 7 times the 2015 estimate at statistics departments. Two-year mathematics depart-

ments experienced a 23% decrease in part-time faculty paid by the two-year colleges to a total of 13,287 persons (SE 1630.9). Part-time faculty in two-year colleges fell into two categories: those paid by two-year colleges and others paid by a third party, usually high school teachers in a school with which the college had a dual-enrollment agreement. When both categories are combined, the estimated number of part-time faculty was 18,543 in fall 2021, an 8% decrease compared with 20,247 in fall 2015. The number of part-time faculty paid by a third party increased almost 100% from 2015 to 2021 to a total of 4716 (SE 963) persons in 2021, compared with 2359 (SE 528) persons in 2015. See Table TYF.1 in Chapter 7.

#### **Table S.14: Appointment Type and Degree Status of Full-Time Faculty**

Table S.14 gives the estimated numbers of full-time faculty in the mathematics and doctoral-level statistics departments of four-year colleges and universities in fall 2015 and fall 2021, broken down by their appointment type (tenured or tenure-eligible (TTE), other full-time (OFT), or post-doc) and the highest degree obtained by the faculty member (doctoral degree or other degree), along with two-year college mathematics faculty estimates, broken down by full-time permanent and other full-time. In this table (as in the other faculty tables in this, and the past, CBMS

**TABLE S.14.** Number of full-time faculty who are tenured and tenure-eligible (TTE), postdocs, and other full-time (OFT) in mathematics and doctoral-level statistics departments of four-year colleges and universities and in mathematics departments at two-year colleges in fall 2015 and fall 2021. (Postdocs are included in the other full-time category.)

| Four-Year Colleges and Universities | Fall 2015    |              |                 |             |  | Fall 2021    |              |                 |             |  |
|-------------------------------------|--------------|--------------|-----------------|-------------|--|--------------|--------------|-----------------|-------------|--|
|                                     | Total        | TTE          | Other Full-Time | Postdoc     |  | Total        | TTE          | Other Full-Time | Postdoc     |  |
| Mathematics Departments             |              |              |                 |             |  |              |              |                 |             |  |
| Full-time faculty                   | 22532        | 15270        | 7261            | 1317        |  | 31807        | 18753        | 13054           | 1379        |  |
| Having doctoral degree              | 18743        | 14869        | 3874            | 1317        |  | 25132        | 18271        | 6861            | 1379        |  |
| Having other degree                 | 3789         | 401          | 3387            |             |  | 6675         | 482          | 6193            |             |  |
| Statistics Departments <sup>1</sup> |              |              |                 |             |  |              |              |                 |             |  |
| Full-time faculty                   | 1238         | 869          | 369             | 113         |  | 3396         | 2030         | 1366            | 178         |  |
| Having doctoral degree              | 1198         | 869          | 329             | 113         |  | 3138         | 2023         | 1115            | 178         |  |
| Having other degree                 | 40           | 0            | 40              |             |  | 258          | 7            | 251             |             |  |
| Total Math & Stats Depts            | 23770        | 16139        | 7630            | 1430        |  | 35203        | 20783        | 14420           | 1557        |  |
| Two-Year College Mathematics        |              |              |                 |             |  |              |              |                 |             |  |
| Full-time faculty                   | 9800         | 8314         | 1487            |             |  | 10328        | 9126         | 1202            |             |  |
| <b>Grand Total</b>                  | <b>33570</b> | <b>24453</b> | <b>9117</b>     | <b>1430</b> |  | <b>45531</b> | <b>29909</b> | <b>15622</b>    | <b>1557</b> |  |

Note: Round-off may make marginal totals seem inaccurate.

<sup>1</sup> This table does not include master's-level statistics departments. The comparable table in CBMS2015, Table S.14, p. 36 does.

<sup>2</sup> Other full-time in this table in 2015 includes full-time continuing faculty and other full-time faculty from Table TYF.1.

surveys), the category of other full-time four-year faculty includes postdoctoral appointments, but the number of postdocs is also broken out of the number of other full-time faculty so that trends in the growing category of postdoc faculty can be observed. In this table, the category of “other full-time” for two-year colleges includes full-time continuing faculty and other full-time faculty discussed above and in Chapter 7.

In fall 2021, as in fall 2015, a master’s degree was the terminal degree for 80% (SE 3) of full-time permanent mathematics faculty at two-year colleges. The percentage of faculty with a doctorate increased four points to 19% (SE 3) in 2021. The percentage of full-time faculty whose terminal degree was a bachelor’s degree decreased four points to 1% (SE less than 0.5) in 2021. Table TYF.4 and Figure TYF.4.1 in Chapter 7 present historical data from 2000 to 2021.

Most of the increase in the estimated numbers of full-time faculty at four-year mathematics departments from fall 2015 to fall 2021 occurred in the category of “other full-time faculty” (faculty who are not tenured or tenure-eligible, including postdocs), which by Table S.14, rose 80% (12 SEs) in fall 2021 from fall 2015; see also Figure F.1.1 in Chapter 4. Previous CBMS surveys have noted the growth in this non-tenure track category of faculty. In both fall 2015 and fall 2021, an estimated 53% of other full-time four-year mathematics faculty had a PhD. The estimated number of tenured plus tenure-eligible (TTE) faculty at four-year mathematics departments rose by 23% (10 SEs) during this period. The estimated number of postdocs in four-year mathematics departments combined increased 5% from fall 2015 to fall 2021. The percentage of full-time four-year mathematics faculty possessing a doctoral degree went from 83% in fall 2015 to 79% in fall 2021.

In the doctoral-level statistics departments, from fall 2015 to fall 2021, the estimated number of tenured plus tenure-eligible faculty more than doubled (an increase of 25 SEs), and the number of other full-time faculty was more than three times the 2015 estimate (an increase of 12 SEs). The estimated number of postdocs increased 58% from fall 2015 to fall 2021. In fall 2015, an estimated 97% of full-time doctoral statistics faculty had a doctoral degree, and in fall 2021, an estimated 92% had a doctoral degree.

### **Tables S.15–S.20: Gender, Age, and Race/ Ethnicity Among the Mathematical Science Faculty**

Table S.15 breaks Table S.14 down by gender and divides the category of TTE four-year faculty into tenured and tenure-eligible faculty. The table also provides information on the gender of full-time permanent two-year college mathematics faculty, broken down by whether the faculty are

younger than 40. Figure S.15.1 displays comparisons of the estimated percentages of women in tenured and in tenure-eligible appointment types in all levels of mathematics departments of four-year colleges and universities combined and in doctoral statistics departments in fall 2015 and fall 2021. Table S.15 is broken down further by level of four-year mathematics department in Chapter 4, Tables F.1, F.2, and F.3 and Figure F.1.1.

As noted in Table S.15, according to data published in the NCES *Digest of Education Statistics*, the percentage of women among doctorates in mathematics and statistics (combined) awarded over the period July 1, 2015–June 30, 2021 was 28%; this percentage is lower than the 31% reported for the previous five years. Table S.15 shows that, in fall 2021, women comprise an estimated 37% of full-time mathematics faculty over all levels of four-year mathematics departments combined, up from the estimated 31% observed in the CBMS 2015 survey. The percentage of women varies by appointment type, and by Table S.15, in fall 2021, women were an estimated 30% of tenured four-year mathematics faculty, 37% of tenure-eligible faculty, 45% of other full-time faculty, and 24% of postdocs. In fall 2015, the CBMS survey estimated 22% of tenured faculty were women and 22% of postdocs were women.

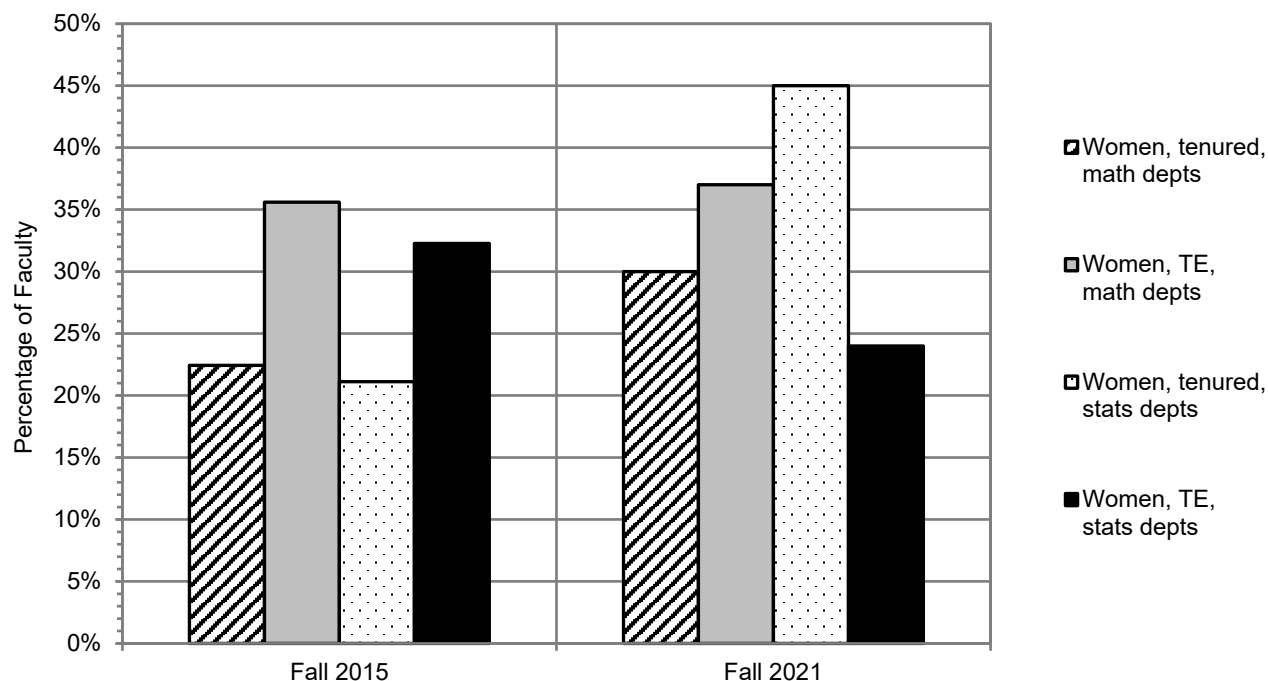
The estimated percentages of women full-time faculty vary among the levels of the department. Chapter 4, Tables F.1, F.2, and F.3 and Figure F.1.1, provide more detail on the estimated numbers of women faculty at four-year departments. From CBMS2015, Chapter 4, Table F.1, p. 117, we see that, in fall 2010, women comprised an estimated 11% of the tenured faculty at doctoral-level mathematics departments, and by 2015, this percentage had risen to 14%. By Table F.1 in Chapter 4 of the current report, we see that in fall 2021 this percentage is estimated at 19%. At bachelor’s-level mathematics departments, in 2010, women comprised an estimated 28% of the tenured faculty; in 2015, this percentage had risen to 29%; and in fall 2021, the estimated percentage of women among tenured faculty was 36%, almost double the percentage of tenured women at doctoral-level departments. The percentage of women among tenure-eligible faculty at bachelor’s-level mathematics departments in fall 2021 was estimated at 42% and at doctoral-level departments in fall 2021 was estimated at 29%, suggesting that the percentage of women among tenured faculty will continue to rise.

Annual surveys conducted by the American Mathematical Society have generally shown the percentage of women among new PhDs in statistics is greater than the percentage of women among new PhDs in mathematics. The CBMS 2010 survey showed the estimated percentage of women faculty at doctoral-level statistics departments was 26%; in 2015, it was 28%; and Table S.15 shows that women comprised an estimated 38% of total doctoral

**TABLE S.15.** Gender among full-time faculty in mathematics and statistics departments of four-year colleges and universities by type of appointment and among permanent full-time faculty in mathematics departments at two-year colleges in fall 2015 and fall 2021. Also gender among doctoral- and master's-degree recipients. (Postdocs are included in the other full-time category.)

| Four-Year Colleges and Universities                   | Fall 2015     |               |                 |                 |              | Fall 2021                    |               |                 |                 |              |
|-------------------------------------------------------|---------------|---------------|-----------------|-----------------|--------------|------------------------------|---------------|-----------------|-----------------|--------------|
| Mathematics Departments                               | Total         | Tenured       | Tenure-Eligible | Other Full-Time | Postdoc      | Total                        | Tenured       | Tenure-Eligible | Other Full-Time | Postdoc      |
| Full-time faculty                                     | 22532         | 11979         | 3291            | 7261            | 1317         | 31807                        | 14713         | 4040            | 13054           | 1379         |
| Number of women                                       | 6981<br>(31%) | 2688<br>(22%) | 1171<br>(36%)   | 3122<br>(43%)   | 288<br>(22%) | 11710<br>(37%)               | 4371<br>(30%) | 1508<br>(37%)   | 5831<br>(45%)   | 326<br>(24%) |
| Statistics Departments <sup>1</sup>                   |               |               |                 |                 |              |                              |               |                 |                 |              |
| Full-time faculty                                     | 896           | 649           | 220             | 369             | 113          | 3396                         | 1456          | 574             | 1366            | 178          |
| Number of women                                       | 341<br>(38%)  | 137<br>(21%)  | 71<br>(32%)     | 133<br>(36%)    | 22<br>(19%)  | 1295<br>(38%)                | 459<br>(32%)  | 205<br>(36%)    | 631<br>(46%)    | 64<br>(36%)  |
| Number of PhDs from US Math & Stat Depts <sup>2</sup> |               |               |                 |                 |              | July 1, 2015 – June 30, 2021 |               |                 |                 |              |
| Number of women among new PhDs <sup>2</sup>           |               |               |                 |                 |              | 8432                         |               |                 |                 |              |
| Number of women among new PhDs <sup>2</sup>           |               |               |                 |                 |              | 3340 (28%)                   |               |                 |                 |              |
| Number of women among new PhDs <sup>2</sup>           |               |               |                 |                 |              | 8432                         |               |                 |                 |              |
| Number of women among new PhDs <sup>2</sup>           |               |               |                 |                 |              | 3340 (28%)                   |               |                 |                 |              |
| Number of women among new PhDs <sup>2</sup>           |               |               |                 |                 |              | 8432                         |               |                 |                 |              |
| Number of women among new PhDs <sup>2</sup>           |               |               |                 |                 |              | 3340 (28%)                   |               |                 |                 |              |
| Number of women among new PhDs <sup>2</sup>           |               |               |                 |                 |              | 8432                         |               |                 |                 |              |
| Number of women among new PhDs <sup>2</sup>           |               |               |                 |                 |              | 3340 (28%)                   |               |                 |                 |              |
| Number of women among new PhDs <sup>2</sup>           |               |               |                 |                 |              | 8432                         |               |                 |                 |              |
| Number of women among new PhDs <sup>2</sup>           |               |               |                 |                 |              | 3340 (28%)                   |               |                 |                 |              |
| Number of women among new PhDs <sup>2</sup>           |               |               |                 |                 |              | 8432                         |               |                 |                 |              |
| Number of women among new PhDs <sup>2</sup>           |               |               |                 |                 |              | 3340 (28%)                   |               |                 |                 |              |
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| Number of women among new PhDs <sup>2</sup>           |               |               |                 |                 |              | 3340 (28%)                   |               |                 |                 |              |
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| Number of women among new PhDs <sup>2</sup>           |               |               |                 |                 |              | 3340 (28%)                   |               |                 |                 |              |
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| Number of women among new PhDs <sup>2</sup>           |               |               |                 |                 |              | 3340 (28%)                   |               |                 |                 |              |
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| Number of women among new PhDs <sup>2</sup>           |               |               |                 |                 |              | 3340 (28%)                   |               |                 |                 |              |
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| Number of women among new PhDs <sup>2</sup>           |               |               |                 |                 |              | 3340 (28%)                   |               |                 |                 |              |
| Number of women among new PhDs <sup>2</sup>           |               |               |                 |                 |              | 8432                         |               |                 |                 |              |
|                                                       |               |               |                 |                 |              |                              |               |                 |                 |              |





**FIGURE S.15.1.** Percentage of women in tenured and in tenure-eligible (TE) categories in mathematics departments of four-year colleges and universities and statistics departments in fall 2015 and 2021.

statistics faculty in fall 2021. The 2021 CBMS survey estimated that in fall 2021 the percentage of women among tenured doctoral-level statistics faculty was 32% and among tenure-eligible faculty was 36%; these are higher percentages than the analogous percentages for women in doctoral-level mathematics departments.

Table S.15 shows that in public two-year college mathematics departments, a decrease of 3 percentage points in the estimated percentage of women among full-time permanent mathematics faculty was reported in 2021 for the first time since 1975. In fall 2021, there were 49% (SE 2.8%) of full-time permanent faculty reported as women, compared with 52% in 2015 and 50% in 2010. Also see Table TYF.8 and Figure TYF.8.1 in Chapter 7. According to NCES data in Table TYF.9, the percentage of women mathematics master's-degree recipients among U.S. citizens/resident aliens was 35% in 2019–2020, compared with 36% in 2014–2015. Table TYF.17 in Chapter 7 presents the estimated percentage of full-time faculty in mathematics by age and gender and the estimated percentage of women by age.

In fall 2021, as in fall 2015, a master's degree was the terminal degree for an estimated 80% (SE 3) of full-time permanent mathematics faculty at two-year colleges. The estimated percentage of faculty with a doctorate increased four points to 19% (SE 3) in 2021. The estimated percentage of full-time faculty whose terminal degree was a bachelor's degree decreased four points to 1% (SE less than 0.5) in 2021. In 2021, the estimated total percentage of faculty whose most advanced degree (doctorate, master's, or bachelor's) was in the field of mathematics was 73% (SE 3). The estimated percentage of the faculty whose most advanced degree was in mathematics education increased two points to 15% (SE 2), compared to 2015. The estimated percentage of faculty whose degree was in statistics increased one point to 6% (SE 2). See Tables TYF.4 and TYF.5 and Figure TYF.4.1 in Chapter 7.

Table S.16 gives the estimated distribution of ages among full-time mathematics faculty at four-year colleges and universities in fall 2021, broken down by tenured or tenure-eligible status and by gender. The estimated average

age of tenured men in four-year mathematics departments had been rising but has remained about the same in the past three CBMS surveys; it was 52.4 in 2000 [CBMS2000, Table SF.9, p. 23], and by Table S.16, 53.7 in 2005, 54.6 in 2010, and 54.9 in 2015; the CBMS 2021 survey estimated the average age of tenured men as 54.6 (SE 0.3). The estimated average age of tenured women has also been rising slightly; it was 49.6 in 2000, 50.2 in 2005, 50.7 in 2010, and 51.0 in 2015; the CBMS 2021 survey estimated the average age of tenured women as 51.9 (SE 0.1). These data are broken down further by level of department in Chapter 4 in Table F.4.

In two-year college mathematics departments, the estimated percentages and numbers of full-time permanent faculty in eight age groups in 2021 are presented in Table S.16 and TYF.16 in Chapter 7. The estimated percentage of full-time permanent faculty under age 40 years dropped nine points to 16% compared to 2015. The estimated number of full-time permanent faculty under the age of 40 in 2021, 2015, 2010, and 2005 was 1424, 2045, 2914, and 2231, respectively. The estimated percentage of full-time permanent faculty between the ages of 50–59 years decreased seven points to 22% in 2021, representing a total decrease of 361 persons. The estimated percentage of full-time faculty over age 59 was down two points from 2015 to 13% (SE 2) in 2021, a decrease of 48 persons. The estimated total number of full-time permanent faculty over the age of 49 decreased by 409 persons from 2015 to 2021. The overall estimated average age of full-time permanent faculty at two-year college mathematics departments was 49.7 years (SE 0.8).

Figure S.16.1 displays the estimated age distribution of full-time faculty in four-year mathematics departments in fall 2021, broken down by gender. Figure S.16.2 displays the estimated age distribution for full-time faculty in two-year mathematics departments. These distributions can be compared to those in previous surveys.

Table S.17 gives the analogous estimated distribution of ages of tenured and tenure-eligible faculty at doctoral-level statistics departments in fall 2021, and the distribution, broken down by gender, is displayed in Figure S.17.1. The estimated average age of tenured men decreased slightly over the fall 2015 estimate, but the other average ages in the table increased; most notably, the estimated average age of tenure-eligible women was 43.1 in fall 2021, compared with the 2015 estimate of 34.4.

Table S.18 gives the estimated fall 2021 distribution of race/ethnicities of faculty, broken down by gender among full-time faculty at four-year mathematics departments. This table is broken down further by level of department in Chapter 4, Table F.5, and given for part-time faculty in

Table F.6. Table S.19 gives a similar table for full-time faculty in doctoral-level statistics departments. Slight changes in the estimated distribution of race/ethnicities among four-year mathematics and statistics faculty were observed in the 2021 CBMS survey. The estimated percentage of White, not Hispanic faculty among full-time male four-year mathematics faculty declined from 53% in 2015 to 43% in 2021, while the percentage of White, not Hispanic full-time female faculty was estimated at 24% in both 2015 and 2021. From Table S.19 in CBMS 2015, we see that among four-year doctoral-level statistics faculty, an estimated 45% of full-time male faculty and 15% of full-time female faculty were White, not Hispanic in fall 2015, while in 2021, from Table S.19, we see that these percentages were estimated at 31% and 16%, respectively.

In fall 2021, the estimated percentage of racial/ethnic minorities among full-time permanent faculty members in mathematics departments in two-year colleges was 15% (SE 2.6%), compared with 23% (SE 2) in 2015. The total number of racial/ethnic minority faculty was 1375 (SE 262.4) in 2021, down 501 persons from 1876 (SE 289) in 2015. Within specific racial/ethnic groups other than White (4785 persons, SE 613.2), the majority of faculty represented was Asian/Pacific Islander (509 persons; SE 113.2), down 3 percentage points and 245 persons from 2015. See Table TYF.10 in Chapter 7. The percentage of women in each racial/ethnic group is displayed in Table TYF.12 in Chapter 7.

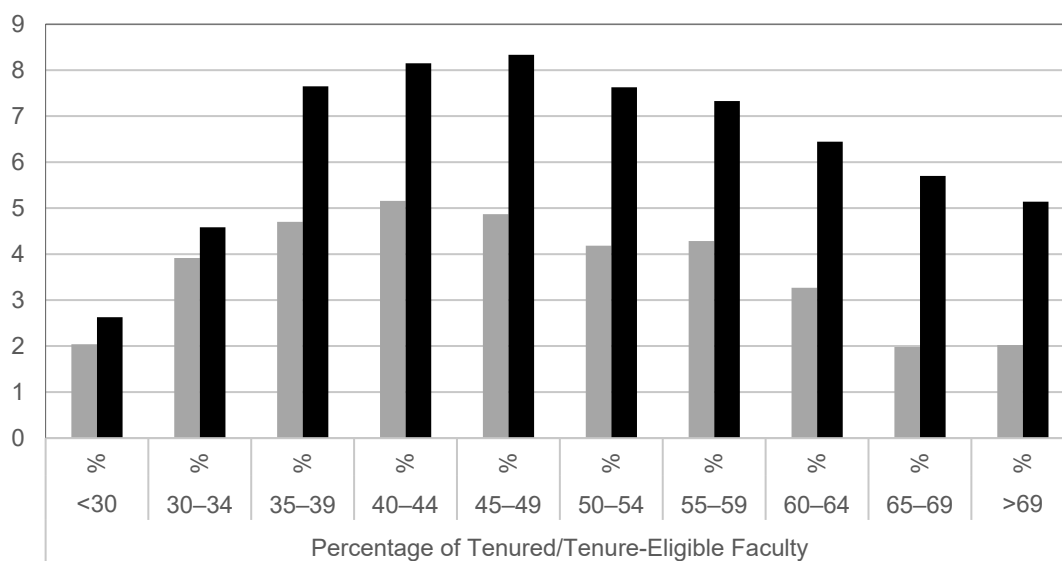
Table S.20 gives the estimated number of deaths and retirements in four-year mathematics and doctoral-level statistics departments, broken down by level of department for the academic years 2004–5, 2009–10, 2014–15, and 2020–21.

According to Table S.20, the total number of deaths and retirements among four-year mathematics faculty was estimated at 561 in 2014–15 and 1161 (SE 28) in 2020–21. Table S.20 also provides the estimated number of tenured/tenure-eligible four-year mathematics and doctoral-level statistics faculty in fall 2021, from which we see that the estimated number of deaths/retirements in 2020–21 was about 6% of estimated fall 2021 tenured/tenure-eligible mathematics faculty (the analogous percentage was 5% in 2014–15) and was about the same percentage over each level of mathematics department. Among doctoral-level statistics faculty, there were an estimated 29 deaths and retirements in 2014–15 and 92 (SE 7) in 2020–21; the number of deaths/retirements in 2020–21 represents about 5% of the fall 2021 estimated number of tenured/tenure-eligible doctoral-level statistics faculty (compared to 3% in 2014–15).

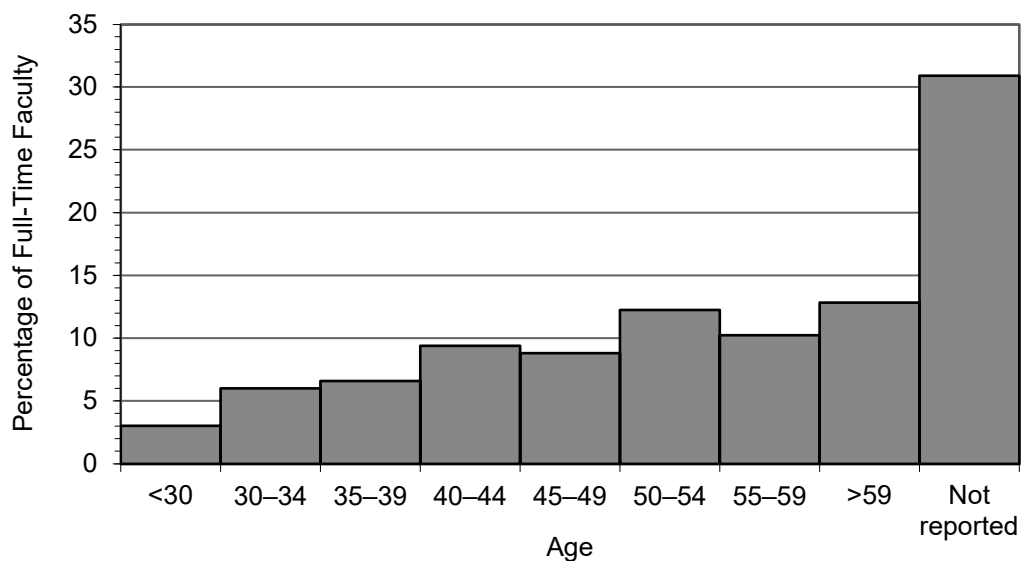
**TABLE S.16.** Percentage of all tenured and tenure-eligible faculty in mathematics departments of four-year colleges and universities in various age groups, and average age, by gender in fall 2021. Percentage of full-time permanent faculty in mathematics programs at public two-year colleges, by age, and average ages in fall 2021. This table can be compared to CBMS2015, Table S.16, p. 39.

| Four-Year College & University Mathematics Departments | Percentage of Tenured/Tenure-Eligible Faculty |       |       |       |       |       |       |       |       |     |  |  |  | Average Age 2010 | Average Age 2015 | Average Age 2021 |
|--------------------------------------------------------|-----------------------------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-----|--|--|--|------------------|------------------|------------------|
|                                                        | <30                                           | 30-34 | 35-39 | 40-44 | 45-49 | 50-54 | 55-59 | 60-64 | 65-69 | >69 |  |  |  |                  |                  |                  |
|                                                        | %                                             | %     | %     | %     | %     | %     | %     | %     | %     | %   |  |  |  |                  |                  |                  |
| Tenured men                                            | 1                                             | 1     | 3     | 5     | 5     | 5     | 5     | 5     | 5     | 4   |  |  |  |                  |                  | 54.6             |
| Tenured women                                          | 0                                             | 1     | 2     | 2     | 3     | 3     | 3     | 2     | 1     | 1   |  |  |  |                  |                  | 51.9             |
| Tenure-eligible men                                    | 2                                             | 4     | 5     | 3     | 3     | 2     | 2     | 2     | 1     | 1   |  |  |  |                  |                  | 44.6             |
| Tenure-eligible women                                  | 2                                             | 3     | 2     | 3     | 2     | 1     | 1     | 1     | 1     | 1   |  |  |  |                  |                  | 44.4             |
| Total tenured & tenure-eligible faculty                | 5                                             | 8     | 12    | 13    | 13    | 12    | 12    | 10    | 8     | 7   |  |  |  |                  |                  |                  |
| Two-Year College Mathematics Programs                  | Percentage of Permanent Full-Time Faculty     |       |       |       |       |       |       |       |       |     |  |  |  |                  |                  |                  |
|                                                        | <30                                           | 30-34 | 35-39 | 40-44 | 45-49 | 50-54 | 55-59 | >59   |       |     |  |  |  |                  |                  |                  |
| Full-time permanent faculty                            | 3                                             | 5     | 6     | 8     | 8     | 11    | 9     | 11    |       |     |  |  |  | 46.8             | 47.7             | 49.7             |

Note: 0 means less than one half of 1%. Round-off may cause some marginal totals to appear inaccurate.



**FIGURE S.16.1.** Percentage of all tenured and tenure-eligible (TTE) faculty in mathematics departments at four-year colleges and universities belonging to various age groups, by gender, in fall 2015. This figure can be compared to CBMS2015, Figure S.16.1, p. 40.



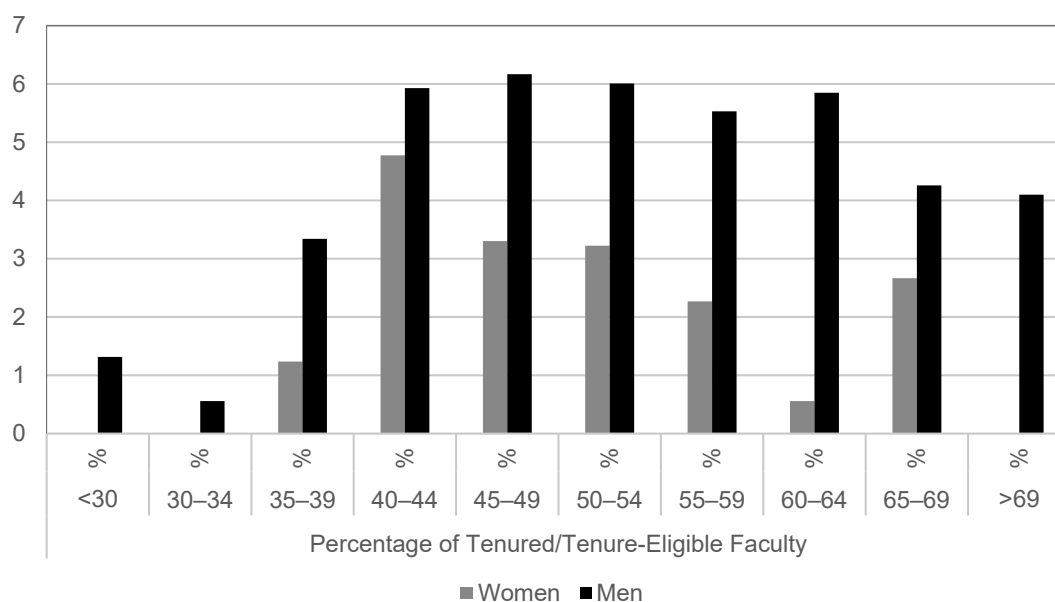
**FIGURE S.16.2.** Percentage of permanent full-time faculty in various age groups in mathematics departments at public two-year colleges in fall 2021.

**TABLE S.17.** Percentage of all tenured and tenure-eligible faculty belonging to various age groups in doctoral and master's statistics departments (combined) at universities by average age and by gender in fall 2021. Also average ages for doctoral statistics departments in fall 2010 and 2015. This can be compared to CBMS2015, Table S.17, p. 40.

| All Statistics<br>Departments           | Percentage of Tenured/Tenure-Eligible Faculty |            |            |            |            |            |            |            |            |          |  |  | Average<br>Age<br>2010 <sup>1</sup> | Average<br>Age<br>2015 | Average<br>Age<br>2021 |
|-----------------------------------------|-----------------------------------------------|------------|------------|------------|------------|------------|------------|------------|------------|----------|--|--|-------------------------------------|------------------------|------------------------|
|                                         | <30<br>%                                      | 30-34<br>% | 35-39<br>% | 40-44<br>% | 45-49<br>% | 50-54<br>% | 55-59<br>% | 60-64<br>% | 65-69<br>% | >69<br>% |  |  |                                     |                        |                        |
| Tenured men                             | 1                                             | 1          | 3          | 6          | 6          | 6          | 6          | 6          | 4          | 4        |  |  |                                     |                        |                        |
| Tenured women                           | 0                                             | 0          | 1          | 5          | 3          | 3          | 2          | 1          | 3          | 0        |  |  |                                     |                        |                        |
| Tenure-eligible men                     | 2                                             | 9          | 4          | 4          | 3          | 2          | 0          | 0          | 1          | 0        |  |  |                                     |                        |                        |
| Tenure-eligible women                   | 1                                             | 2          | 3          | 2          | 2          | 1          | 1          | 1          | 0          | 0        |  |  |                                     |                        |                        |
| Total tenured & tenure-eligible faculty | 4                                             | 12         | 12         | 16         | 15         | 13         | 9          | 7          | 8          | 4        |  |  |                                     |                        |                        |

Note: 0 means less than one half of 1%. Round-off may cause some marginal totals to appear inaccurate.

<sup>1</sup> Average ages for fall 2010 and 2015 from CBMS2015, Table S.17, p. 42.



**FIGURE S.17.1.** Percentage of tenured and tenure-eligible faculty in various age groups, by gender, in doctoral and master's statistics departments (combined) in fall 2021. This figure can be compared to CBMS2015, Figure S.17.1, p. 43.

**TABLE S.18.** Percentage of gender and of racial/ethnic groups among all tenured, tenure-eligible, postdoctoral, and other full-time faculty in mathematics departments of four-year colleges and universities in fall 2021.

| Mathematics Departments            | Asian<br>% | Black,<br>Not<br>Hispanic<br>% | Mexican<br>American/<br>Puerto Rican/<br>Other Hispanic<br>% | White,<br>Not<br>Hispanic<br>% | AIAN &<br>NHPI <sup>1</sup><br>% | Unknown<br>% |
|------------------------------------|------------|--------------------------------|--------------------------------------------------------------|--------------------------------|----------------------------------|--------------|
| Tenured men                        | 6          | 1                              | 1                                                            | 24                             | 0                                | 2            |
| Tenured women                      | 3          | 0                              | 1                                                            | 9                              | 0                                | 1            |
| Tenure eligible men                | 3          | 1                              | 1                                                            | 6                              | 0                                | 1            |
| Tenure eligible women              | 1          | 0                              | 0                                                            | 4                              | 0                                | 0            |
| Postdoctoral men                   | 1          | 0                              | 0                                                            | 3                              | 0                                | 0            |
| Postdoctoral women                 | 0          | 0                              | 0                                                            | 1                              | 0                                | 0            |
| Full-time men not included above   | 2          | 1                              | 1                                                            | 10                             | 0                                | 1            |
| Full-time women not included above | 2          | 0                              | 0                                                            | 10                             | 0                                | 1            |
| Total full-time men                | 12         | 2                              | 3                                                            | 43                             | 0                                | 4            |
| Total full-time women              | 6          | 1                              | 2                                                            | 24                             | 0                                | 2            |

<sup>1</sup> Includes the federal categories *American Indian or Alaska Native* (AIAN) and *Native Hawaiian or Other Pacific Islander* (NHPI).

Note: 0 means less than one half of 1%, and this may cause column sum inconsistencies.

**TABLE S.19.** Percentage of gender and of racial/ethnic groups among all tenured, tenure-eligible, postdoctoral, and other full-time faculty in doctoral statistics departments at universities in fall 2021.

| <b>All Statistics Departments</b>  | Asian<br>% | Black,<br>Not<br>Hispanic<br>% | Mexican<br>American/<br>Puerto Rican/<br>Other Hispanic<br>% | White,<br>Not<br>Hispanic<br>% | AIAN &<br>NHPI <sup>1</sup><br>% | Unknown<br>% |
|------------------------------------|------------|--------------------------------|--------------------------------------------------------------|--------------------------------|----------------------------------|--------------|
| Tenured men                        | 11         | 1                              | 1                                                            | 17                             | 0                                | 0            |
| Tenured women                      | 6          | 0                              | 1                                                            | 7                              | 0                                | 0            |
| Tenure eligible men                | 9          | 0                              | 1                                                            | 4                              | 0                                | 1            |
| Tenure eligible women              | 3          | 0                              | 1                                                            | 3                              | 0                                | 1            |
| Postdoctoral men                   | 4          | 0                              | 0                                                            | 3                              | 0                                | 0            |
| Postdoctoral women                 | 2          | 0                              | 0                                                            | 0                              | 0                                | 0            |
| Full-time men not included above   | 3          | 0                              | 1                                                            | 7                              | 0                                | 1            |
| Full-time women not included above | 4          | 1                              | 0                                                            | 6                              | 0                                | 1            |
| Total full-time men                | 26         | 2                              | 3                                                            | 31                             | 0                                | 2            |
| Total full-time women              | 15         | 2                              | 2                                                            | 16                             | 0                                | 2            |

<sup>1</sup> Includes the federal categories *American Indian or Alaska Native* (AIAN) and *Native Hawaiian or Other Pacific Islander* (NHPI).

Note: 0 means less than one half of 1%, and this may cause column sum inconsistencies.

**TABLE S.20.** Number of deaths and retirements of full-time faculty from mathematics departments and from doctoral statistics departments by type of department. This table can be compared to CBMS2015, Table S.21, p. 46.

| Four-Year College & University                                   | 2004–2005 | 2009–2010 | 2014–2015 | 2020–2021 | Number of<br>Tenured/<br>Tenure-Eligible<br>Faculty 2021 |
|------------------------------------------------------------------|-----------|-----------|-----------|-----------|----------------------------------------------------------|
| Mathematics Departments                                          |           |           |           |           |                                                          |
| Univ (PhD)                                                       | 139       | 146       | 182       | 369       | 6170                                                     |
| Univ (MA)                                                        | 140       | 91        | 128       | 198       | 3303                                                     |
| Coll (BA)                                                        | 219       | 123       | 251       | 594       | 9280                                                     |
| Total deaths and retirements in<br>all Mathematics Departments   | 499       | 360       | 561       | 1161      | 18753                                                    |
| Doctoral Statistics Departments:<br>Total deaths and retirements | 14        | 15        | 29        | 92        | 2030                                                     |

## Chapter 2

# CBMS2021 Special Projects

Each CBMS survey accepts proposals for special projects from various professional society committees. Special projects chosen for one CBMS survey might, or might not, be continued in the next CBMS survey. This chapter presents data from the special projects of CBMS2021 for two- and four-year mathematics departments:

- The mathematical education of pre-college mathematics teachers (Tables SP.1–SP.7)
- Percentage of departments offering distance/remote learning courses, and practices in distance learning courses (Tables SP.8.A–SP.11.B)
- Academic resources and special opportunities available to undergraduates (Tables SP.12–SP.14)
- Interdisciplinary courses in four-year mathematics departments (Tables SP.15)
- Dual-enrollment courses in mathematics and statistics (Tables SP.16 and SP.17)
- Requirements in the majors in mathematics and statistics in four-year departments (Tables SP.18–SP.19.B)
- Availability of upper-level classes in four-year mathematics and statistics departments (Tables SP.20 and SP.21)
- Estimates of post-graduation plans of graduates of four-year mathematics departments and statistics departments (Table SP.22)
- Assessment in four-year mathematics departments and statistics departments (Table SP.23)
- Divisional graduation credit for advanced placement courses in four-year mathematics and statistics departments (Table SP.24)
- Statistics majors and minors at four-year mathematics departments (Table SP.25)
- Departmental activities in response to diversity, equity, and inclusion issues (Table SP.26)
- Profiles of other full-time faculty at four-year mathematics and statistics departments (Tables SP.27 and SP.28)

When there is comparable data in CBMS2015, the appropriate comparison table will be given in the caption if the 2015 data is not included in the table. Also note that further discussion of selected special project issues at two-year colleges is given in the Topics of Special Interest for Mathematics Departments at Two-Year Colleges located in Chapter 6.

**Terminology:** Recall that in CBMS2021, the term “mathematics department” includes departments of mathematics, applied mathematics, mathematical sciences, and departments of mathematics and statistics. These departments may offer a broad spectrum of courses in mathematics education, actuarial science, and operations research, as well as mathematics, applied mathematics, and statistics. Computer science courses are sometimes also offered by mathematics departments. The term “statistics department” refers to graduate departments of statistics or biostatistics that offer undergraduate statistics courses. Courses and majors from separate departments of computer science, actuarial science, operations research, etc. are not included in CBMS2021. Departments are classified by the highest degree they offered; for example, “master’s-level department” refers to a department that offers a master’s degree, but not a doctoral degree.

In the text that follows, the standard error (SE) in many of the estimates is provided along with the estimate (e.g., “estimated 77% (SE 3.5)”); the SEs for all CBMS2021 tables can be found in Appendix IX. The change in an estimate in 2021 from the estimate in the 2015 survey is often expressed both as percentage change and as the number of SEs in the 2021 estimate that change represents (e.g., “increased 22% (1.2 SEs)”).

### Tables SP.1–SP.7: The Mathematical Education of Teachers of Pre-College Mathematics and Statistics

**Percentages of four-year mathematics departments whose institutions have elementary, middle school, and secondary mathematics teacher certification programs (Table SP.1).** Table SP.1 displays the estimated percentages of four-year mathematics departments whose institutions have a certification program that leads to teaching mathematics in grades K–5, grades 6–8, and grades 9–12 in fall 2015 and 2021, with some data for fall 2005 and 2010. These percentages are broken down by the level of mathematics department.



**TABLE SP.1.** Percentage of four-year mathematics departments whose institutions offer certification programs for some or all grades K–8, and also for secondary teachers, by type of department in fall 2015 and 2021. (Data for K–8 from fall 2005 and 2010, when available, in parentheses.)

|                         | Percentage Whose Institutions Have a Certification Program for: |              |              |              |                    |                            |              |
|-------------------------|-----------------------------------------------------------------|--------------|--------------|--------------|--------------------|----------------------------|--------------|
|                         | Grades K–5                                                      |              | Grades 6–8   |              | Grades K–8*        | Secondary<br>(Grades 9–12) |              |
|                         | Fall<br>2021                                                    | Fall<br>2015 | Fall<br>2021 | Fall<br>2015 | Fall<br>2005, 2010 | Fall<br>2021               | Fall<br>2015 |
| Mathematics Departments |                                                                 |              |              |              |                    |                            |              |
| Univ (PhD)              | 64                                                              | 52           | 69           | 47           | (78,62)            | 81                         | 75           |
| Univ (MA)               | 86                                                              | 63           | 95           | 64           | (92,90)            | 100                        | 92           |
| Coll (BA)               | 63                                                              | 52           | 80           | 50           | (88,70)            | 91                         | 75           |
| Total Math Depts        | 66                                                              | 53           | 81           | 51           | (87,72)            | 91                         | 77           |

\* Prior to 2015, CBMS asked about certification for pre-service K–8 teachers, while CBMS 2015 separated K–5 from 6–8. If the results for the two questions on CBMS 2015 are combined, then 63 percent of total mathematics departments responded that they had a program for certification for K–5 and/or 6–8 teachers.

The table indicates increased availability of mathematics teacher certification programs for all three grade levels, reversing the pattern of decreasing availability that was observed in previous CBMS surveys.

It follows from Table SP.1 that in fall 2021, an estimated 66% (SE 6) of all levels of four-year mathematics departments combined belonged to an institution that offered a mathematics teacher certification program for grades K–5 and an estimated 81% (SE 4.9) belonged to an institution that offered a mathematics teacher certification program for grades 6–8; this compares to an estimated 53% for grades K–5 and 51% for grades 6–8 reported in the 2015 CBMS survey. These results suggests an increase in the availability of these elementary/middle school mathematics teacher certification programs. The CBMS surveys before 2015 asked about the existence of a certification program for teaching mathematics in grades K–8, and the estimates of percentages of four-year mathematics departments combined that had such a program were 72% in fall 2010, 87% in fall 2005, and 84% in fall 2000. Hence, before the 2021 CBMS survey, the percentages of the availability of these programs seemed to be decreasing.

The availability of secondary (grades 9–12) mathematics teacher certification also appeared to increase from fall 2015, when it was estimated that 77% of all levels of four-year mathematics departments combined belonged to an institution with a secondary mathematics teacher certification program, to fall 2021, when an estimated 91% of mathematics departments indicated their institution had such a program.

When broken down by level of department, in each case, the largest percentage of four-year mathematics departments with mathematics teacher certification programs were the master’s-level departments, and in each of the cases, the estimated percentage of availability in fall 2021 was greater than the percentage estimated in fall 2015.

**Teacher preparation programs at two-year colleges (Tables SP.2 and SP.3).** Table SP.2 reports on organized programs at two-year colleges in which students can obtain their entire mathematics course requirement for teacher licensure. CBMS2021 and CBMS2015 data present decreasing numbers in elementary and secondary school programs, in contrast to the increasing trend at four-year colleges. The single largest component is the program for pre-service elementary school teachers reported by 22% (SE 5.7) in 2021 and 28% of two-year colleges in 2015. Pre-service middle school licensure-oriented programs reported a nine-point decrease to 5% (SE 2.3) of colleges. Between 2% and 15% of two-year colleges reported programs at the elementary or middle school levels for retraining by career switchers moving into teaching. Compared with 2015 in Table SP.2, all categories, except career-switchers aiming for elementary teaching, showed decreases in percentages of responding colleges in 2021.

The decreasing data in Table SP.2 is interesting given that Tables TYE.5 and TYE.6 in Chapter 6 show that 54% (SE 6.89), up thirteen points from 2015 (41%; SE 5), of two-year colleges offered the course Mathematics for Elementary School Teachers I in fall 2021. For the five-year CBMS

**TABLE SP.2.** Percentage of mathematical departments at public two-year colleges (TYCs) having organized programs that allow various types of pre- and in-service teachers to complete their entire mathematics course or licensure requirements in fall 2021. (Fall 2010, 2015 data in parentheses.)

|                                                    | Percentage of TYCs with an Organized Program<br>in Which Students Can Complete Their Entire<br>Mathematics Course or Licensure Requirements |         |
|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|---------|
| Pre-service elementary teachers                    | 22                                                                                                                                          | (41,28) |
| Pre-service middle school teachers                 | 5                                                                                                                                           | (24,14) |
| Pre-service secondary teachers                     | 2                                                                                                                                           | (13,7)  |
| In-service elementary teachers                     | 14                                                                                                                                          | (25,12) |
| In-service middle school teachers                  | 3                                                                                                                                           | (12,6)  |
| In-service secondary teachers                      | 2                                                                                                                                           | (10,4)  |
| Career-switchers aiming for elementary teaching    | 15                                                                                                                                          | (30,16) |
| Career-switchers aiming for middle school teaching | 4                                                                                                                                           | (17,13) |
| Career-switchers aiming for secondary teaching     | 2                                                                                                                                           | (13,5)  |

**TABLE SP.3.** Percentage of public two-year colleges (TYCs) that are involved with teacher preparation in various ways in fall 2021. (Data from fall 2010, 2015 in parentheses when available.)

|                                                                                        | Percentage of TYCs |
|----------------------------------------------------------------------------------------|--------------------|
| Assign a mathematics faculty member to coordinate K–8 teacher education in mathematics | 33 (36,35)         |
| Offer a special mathematics course for preservice K–8 teachers <sup>1</sup>            | 22 (7,55)          |
| Offer a special mathematics course for preservice secondary teachers <sup>2</sup>      | 2 (na,19)          |
| Offer mathematics pedagogy courses in the mathematics department                       | 9 (5,9)            |
| Offer mathematics pedagogy courses outside of the mathematics department               | 3 (9,6)            |

<sup>1</sup> In 2010, this question specifically excluded four courses listed in the detailed course matrix.

<sup>2</sup> Did not collect in 2010.

intervals beginning in 1990 through 2021, the percentages of two-year colleges teaching the course Mathematics for Elementary School Teachers I are successively 32%, 43%, 49%, 59%, 55%, 44%, and 54%. Of special note is the increase in departments offering Mathematics for Elementary School Teachers II, up 28 points to 45% (SE 6.79), from 2015 to 2021.

Table SP.3 reports on other involvements two-year college mathematics departments have with teacher education. Thirty-three percent (33%; SE 9) report that a faculty member is assigned to coordinate mathematics education for future K–8 teachers, compared with 35% in 2015. Twenty-two percent (22%; SE 5.7) of the reporting colleges designate special mathematics courses for future (pre-service) K–8 teachers, down 33 points from 2015. Fall 2021 saw a decrease of 17 points from 2015, down to 2% (SE 1.7), in offering a special mathematics course for pre-service secondary teachers. Nine percent (9%; SE 3.4) of mathematics departments offered mathematics pedagogy courses for future teachers within the department, and 3% (SE 1.9) of colleges offer such pedagogy courses outside the mathematics department.

**Four-year mathematics departments: Numbers of mathematics and statistics credits required for certification of pre-service mathematics K–8 teachers (Tables SP.4 and SP.5).** Tables SP.4 and SP.5 give the estimated percentages in fall 2021 of four-year mathematics departments belonging to a college or university having a program of study that leads to credentials to teach mathematics in grades K–5 (6–8) public schools that require various numbers of semester hours in specified courses for certification; the percentages are broken down by the level of the mathematics department. The courses considered are courses taught in the mathematics department, courses specifically in “fundamental ideas in mathematics appropriate for elementary (middle) school teachers,” and courses in statistics. Comparing Tables SP.4 and SP.5 shows that in fall 2021, certification for teaching mathematics in grades 6–8 generally required more semester hours in mathematics and statistics than did certification to teach mathematics in grades K–5. Tables SP.4 and SP.5 may be compared to CBMS2015, Tables SP.4 and SP.5, p. 51 and 52, respectively, which display comparable data for fall 2015, except that hours in statistics courses were not considered in the CBMS 2015 survey. The CBMS 2010 survey asked slightly different questions, and CBMS2010, Table SP.5, p. 51 gave the distribution of the number of mathematics courses (rather than semester hours) required for “early” grade (K–5) certification (if the institution made a distinction between kinds of K–8 certification, or all K–8 certification if no distinction was made) among the various levels of departments. The data from fall 2021

(Table SP.4) indicate a general trend toward fewer required semester hours in mathematics department courses and in courses in fundamental mathematical ideas for grades K–5 mathematics teacher certification than in fall 2015. The estimated distribution of the analogous percentages for required hours for grades 6–8 mathematics teacher certification in fall 2021 (Table SP.5) was roughly the same as in fall 2015.

Considering certification for teaching mathematics in grades K–5, as shown in Table SP.4, in fall 2021, an estimated 14% (SE 6.4) of programs at all levels of four-year institutions combined required zero hours of course work in the mathematics department for grades K–5 mathematics teacher certification; the analogous estimate in fall 2015 was 2%. At the high end of required hours, in fall 2015, an estimated 34% of programs at all levels of four-year mathematics departments combined required more than 12 hours of course work in the mathematics department, and in fall 2021, the estimate was 24% (SE 6.2). From Table SP.4, we see that in fall 2021, the interval of required semester hours taken in the mathematics department with the largest percentage of institutions was 4–6 required semester hours for each of the three levels of mathematics departments. The estimated distribution of institutions requiring various numbers of semester hours in “fundamental ideas in mathematics appropriate for elementary school teachers” also generally trended to be more concentrated at the lower end of required numbers of semester hours in fall 2021 than it was in fall 2015. A new question on the CBMS 2021 survey asked about required semester hours in statistics, and across all levels of four-year institutions combined, 61% (SE 7.2) required no hours in statistics. Each of the distributions in Table SP.4 is broken down by the level of the department.

Table SP.5 displays fall 2021 percentages for the analogous numbers of required semester hours of courses taught in the mathematics department for grades 6–8 mathematics teacher certification and shows substantially higher percentages of institutions requiring more hours of courses in the mathematics department for grades 6–8 mathematics teacher certification than for grades K–5 mathematics teacher certification. Table SP.5 shows that, in fall 2021, an estimated 90% (SE 6) of all levels of four-year mathematics departments combined belonged to an institution with a grades 6–8 mathematics teacher certification program that required more than 12 semester hours in the mathematics department, compared to an estimated 83% in fall 2015, and 9% (SE 4.7) required more than 12 semester hours in “fundamental ideas in mathematics appropriate for middle school teachers,” compared to an estimated 29% in fall 2015. The interval of required semester hours in mathematics courses on fundamental mathematical ideas having the largest percentage of institutions for grades 6–8 mathematics teacher certification with all levels

**TABLE SP.4.** Among all four-year colleges and universities with a grades K–5 certification program, the percentage of mathematics departments requiring various numbers of mathematics semester hours for certification, by type of department, in fall 2021.

|                                                                                                   | Percentage of Departments<br>with K–5 Certification Programs<br>that Require Various Numbers<br>of Mathematics Courses for Certification |             |             |            |
|---------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------------|------------|
| Number of Semester Hours<br>in Mathematics Department<br>Required for K–5 Certification           | Univ (PhD) %                                                                                                                             | Univ (MA) % | Coll (BA) % | All Math % |
| 0 required                                                                                        | 12                                                                                                                                       | 15          | 14          | 14         |
| 1–3 required                                                                                      |                                                                                                                                          | 4           | 3           | 3          |
| 4–6 required                                                                                      | 25                                                                                                                                       | 30          | 28          | 28         |
| 7–9 required                                                                                      | 17                                                                                                                                       | 29          | 17          | 19         |
| 10–12 required                                                                                    | 24                                                                                                                                       | 5           | 10          | 11         |
| More than 12 required                                                                             | 22                                                                                                                                       | 15          | 27          | 24         |
| Number of Semester Hours<br>in Fundamental Ideas of Mathematics<br>Required for K–5 Certification | Univ (PhD) %                                                                                                                             | Univ (MA) % | Coll (BA) % | All Math % |
| 0 required                                                                                        | 27                                                                                                                                       | 15          | 16          | 17         |
| 1–3 required                                                                                      | 6                                                                                                                                        | 20          | 17          | 16         |
| 4–6 required                                                                                      | 33                                                                                                                                       | 45          | 38          | 39         |
| 7–9 required                                                                                      | 23                                                                                                                                       | 15          | 17          | 18         |
| 10–12 required                                                                                    | 6                                                                                                                                        |             | 7           | 5          |
| More than 12 required                                                                             | 5                                                                                                                                        | 4           | 5           | 5          |
| Number of Semester Hours<br>in Statistics<br>Required for 6–8 Certification                       | Univ (PhD) %                                                                                                                             | Univ (MA) % | Coll (BA) % | All Math % |
| 0 required                                                                                        | 69                                                                                                                                       | 66          | 58          | 61         |
| 1–3 required                                                                                      | 24                                                                                                                                       | 11          | 19          | 19         |
| 4–6 required                                                                                      | 2                                                                                                                                        | 5           | 15          | 11         |
| 7–9 required                                                                                      |                                                                                                                                          | 13          |             | 2          |
| 10–12 required                                                                                    |                                                                                                                                          |             | 3           | 2          |
| More than 12 required                                                                             | 5                                                                                                                                        | 4           | 5           | 5          |

Note: Blank entries represent semester hours with no responses. Some percentages do not total 100% due to round-off.

**TABLE SP.5.** Among all four-year colleges and universities with a grades 6–8 certification program, the percentage of mathematics departments requiring various numbers of mathematics and statistics semester hours for certification, by type of department, in fall 2021.

|                                                                                                   | Percentage of Departments<br>with Grades 6–8 Certification Programs<br>that Require Various Numbers<br>of Mathematics Courses for Certification |             |             |            |
|---------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------------|------------|
| Number of Semester Hours<br>in Mathematics Department<br>Required for 6–8 Certification           | Univ (PhD) %                                                                                                                                    | Univ (MA) % | Coll (BA) % | All Math % |
| 0 required                                                                                        |                                                                                                                                                 |             |             |            |
| 1–3 required                                                                                      |                                                                                                                                                 |             | 5           | 4          |
| 4–6 required                                                                                      |                                                                                                                                                 | 8           |             | 2          |
| 7–9 required                                                                                      |                                                                                                                                                 |             |             |            |
| 10–12 required                                                                                    |                                                                                                                                                 |             | 5           | 4          |
| More than 12 required                                                                             | 100                                                                                                                                             | 92          | 89          | 90         |
| Number of Semester Hours in<br>Fundamental Ideas of Mathematics<br>Required for 6–8 Certification | Univ (PhD) %                                                                                                                                    | Univ (MA) % | Coll (BA) % | All Math % |
| 0 required                                                                                        |                                                                                                                                                 |             | 26          | 19         |
| 1–3 required                                                                                      |                                                                                                                                                 |             | 28          | 20         |
| 4–6 required                                                                                      | 28                                                                                                                                              | 20          | 25          | 24         |
| 7–9 required                                                                                      | 9                                                                                                                                               | 23          | 17          | 18         |
| 10–12 required                                                                                    | 54                                                                                                                                              | 17          | 4           | 9          |
| More than 12 required                                                                             | 9                                                                                                                                               | 39          |             | 9          |
| Number of Semester Hours in<br>Statistics Required for 6–8<br>Certification                       | Univ (PhD) %                                                                                                                                    | Univ (MA) % | Coll (BA) % | All Math % |
| 0 required                                                                                        |                                                                                                                                                 | 8           | 16          | 13         |
| 1–3 required                                                                                      | 37                                                                                                                                              | 61          | 78          | 72         |
| 4–6 required                                                                                      | 54                                                                                                                                              | 31          | 6           | 14         |
| 7–9 required                                                                                      |                                                                                                                                                 |             |             |            |
| 10–12 required                                                                                    |                                                                                                                                                 |             |             |            |
| More than 12 required                                                                             | 9                                                                                                                                               |             |             | 0          |

Note: Blank entries represent semester hours with no responses. Some percentages do not total 100% due to round-off.

of departments combined was 4–6 required hours in fundamental mathematical ideas (though it was 10–12 hours at doctoral-level mathematics departments), as it was for grades K–5 mathematics teacher certification. More hours in statistics appeared to be required for grades 6–8 mathematics teacher certification than for grades K–5 certification; Table SP.5 shows that, in fall 2021, an estimated 13% (SE 8.6) of all levels of four-year mathematics departments combined required no semester hours in statistics for grades 6–8 certification and 72% (SE 9.3) required 1–3 hours in statistics; Table SP.4 shows that an estimated 61% (SE 7.2) of all levels of four-year institutions combined required no semester hours in statistics for grades K–5 mathematics teacher certification and 19% (SE 5.5) required 1–3 hours in statistics. All the distributions in Table SP.5 are broken down by the level of the department.

#### **Four-year mathematics departments: Courses in secondary certification programs (Table SP.6).**

Table SP.6 gives the estimated percentages in fall 2021 of four-year mathematics departments belonging to a college or university having a program of study that leads to credentials to teach mathematics in grades 9–12 public schools where various specific courses are required or are generally taken. The distribution of the percentages of four-year institution programs, all levels of institutions combined, requiring the courses in fall 2021 was quite like the distribution observed in the 2015 CBMS survey, but the percentages indicating that the course was generally taken was, in most cases, higher than those observed in the 2015 CBMS survey. Departments were also asked if they offer courses specially designed for pre-service mathematics secondary teachers in each of the subjects; these percentages were also generally similar to those observed in the 2015 CBMS survey. The percentages in Table SP.6 are broken down by the level of the mathematics department. CBMS2015, Table SP.6, p. 53 and CBMS2010, Table SP.9, p. 54 present comparable data.

In fall 2021, as in fall 2015, across all four-year level institutions combined, geometry was the course most likely to be required, the course mostly likely for the mathematics department to offer a specially designed course for pre-service teachers, and the course, if not required, most likely generally to be taken (in fall 2015, number theory was the course most likely not required, but generally taken). Table SP.6 shows that in fall 2021, across all levels of mathematics departments combined, an estimated 84% (SE 4.9) of secondary certification programs at four-year institutions required a course in geometry, 81% (SE 6.5) required a course in modern algebra, 61% (SE 6.1) required a course in discrete mathematics, 57% (SE 6.3) required a course in probability, and 54% (SE 6.7) required a course in advanced calculus/analysis; in addition, 70% required some course in statistics (the 2021 CBMS survey broke down the possible

courses in statistics into four possible courses: an introductory statistics course with a simulation-based approach to inference, or without a simulation-based approach to inference, a courses with an introductory course prerequisite, and a statistical modeling course). Table SP.6 shows that in fall 2021, specially designed statistics courses for pre-service teachers were offered mainly at the bachelor's-level mathematics departments. Details about modern algebra and advanced calculus/analysis requirements in all, some, or none of other mathematics major requirements are given later in this chapter in Table SP.18. Also note that from Table SP.15, in fall 2021, an estimated 46% (SE 5.5) of four-year mathematics departments combined offered an interdisciplinary course in mathematics and education.

#### **Statistics departments: Courses for pre-service mathematics teachers (Table SP.7).**

Beginning with the 2015 CBMS survey, recognizing that statistics has become a basic part of pre-college curricula, CBMS surveys asked statistics departments about statistics courses taken in programs within their institutions that lead to certification to teach mathematics at public schools in grades K–5, grades 6–8, or grades 9–12. Table SP.7 summarizes data on the number of semester hours in statistics required for certification to teach grades K–5 and grades 6–8 mathematics. It also asks about statistics courses required, or not required but generally taken, as part of secondary certification to teach mathematics in grades 9–12, as well as statistics courses statistics departments have specially designed for pre-service secondary mathematics teachers. Table SP.7 can be compared to Tables SP.4, SP.5, and SP.6 that summarize the comparable data obtained from four-year mathematics departments. Table SP.7 is broken down by the level of the statistics department. Table SP.7 can be compared to CBMS2015, Table SP.7, p. 54.

Table SP.7 shows that in fall 2021, an estimated 79% (SE 11.7) of both levels of statistics departments combined whose institution had a program leading to certification to teach mathematics in grades K–5 public schools indicated that their institutions' program required no statistics courses (Table SP.4 shows that an estimated 61% (SE 7.2) of all levels of mathematics departments combined indicated that the grades K–5 mathematics certification program at their institution required no statistics courses). For certification to teach mathematics in grades 6–8, by Table SP.7, an estimated 43% (SE 21.7) of statistics departments indicated their institutions' grades 6–8 mathematics certification program required no statistics courses, and 43% (SE 21.7) indicated 1–3 hours of statistics courses were required (by Table SP.5, an estimated 72% (SE 9.3) of all levels of mathematics departments combined indicated that their institutions' programs required 1–3 hours of statistics).

**TABLE SP.6.** Among four-year colleges and universities with secondary pre-service teaching certification programs, for various courses, the percentage of mathematics departments whose program requires the course, or whose students generally take the course, or who offer a special course in the given subject that is designed for secondary teachers, by type of department, in fall 2021.

| Course                                                                     | Percentage of Departments with Secondary Certification Program Where: |             |             |            |  |                                             |             |             |            |                                                                                   |
|----------------------------------------------------------------------------|-----------------------------------------------------------------------|-------------|-------------|------------|--|---------------------------------------------|-------------|-------------|------------|-----------------------------------------------------------------------------------|
|                                                                            | Course Is Required                                                    |             |             |            |  | Course Is Generally Taken, but Not Required |             |             |            |                                                                                   |
|                                                                            | Univ (PhD) %                                                          | Univ (MA) % | Coll (BA) % | All Math % |  | Univ (PhD) %                                | Univ (MA) % | Coll (BA) % | All Math % | Math Dept Offers Special Course in the Subject for Secondary Pre-Service Teachers |
| Advanced Calculus/Analysis                                                 | 66                                                                    | 22          | 59          | 54         |  | 49                                          | 40          | 33          | 36         | Univ (PhD) % 6, Univ (MA) % 11, Coll (BA) % 9, All Math % 9                       |
| Modern Algebra                                                             | 88                                                                    | 90          | 79          | 81         |  | 69                                          | 100         | 61          | 67         | Univ (PhD) % 6, Coll (BA) % 0, All Math % 1                                       |
| Number Theory                                                              | 29                                                                    | 26          | 10          | 14         |  | 34                                          | 32          | 20          | 23         | Univ (PhD) % 6, Coll (BA) % 9, All Math % 8                                       |
| Geometry                                                                   | 84                                                                    | 90          | 82          | 84         |  | 71                                          | 100         | 81          | 83         | Univ (PhD) % 30, Univ (MA) % 13, Coll (BA) % 21, All Math % 21                    |
| Discrete Mathematics                                                       | 52                                                                    | 54          | 64          | 61         |  | 57                                          | 40          | 35          | 39         | Univ (PhD) % 6, Coll (BA) % 7, All Math % 6                                       |
| Introductory Statistics including a simulation-based approach to inference | 13                                                                    | 26          | 28          | 26         |  | 9                                           | 17          | 26          | 23         | Univ (PhD) % 11, Coll (BA) % 11, All Math % 8                                     |
| Introductory Statistics excluding a simulation-based approach to inference | 19                                                                    | 35          | 36          | 34         |  | 12                                          | 30          | 22          | 22         | Univ (PhD) % 5, Coll (BA) % 5, All Math % 4                                       |
| Statistical methods with an introductory course as a prerequisite          | 4                                                                     | 12          | 7           | 8          |  | 5                                           | 12          | 5           | 6          | Univ (PhD) % 4, Coll (BA) % 4, All Math % 3                                       |
| Statistical Modeling                                                       | 9                                                                     | 4           | 0           | 2          |  | 5                                           | 7           | 3           | 4          | Univ (PhD) % 2, Coll (BA) % 2, All Math % 1                                       |
| Probability                                                                | 65                                                                    | 52          | 57          | 57         |  | 56                                          | 43          | 60          | 58         | Univ (PhD) % 8, Coll (BA) % 8, All Math % 6                                       |
| History of Math                                                            | 63                                                                    | 41          | 38          | 42         |  | 72                                          | 71          | 20          | 31         | Univ (PhD) % 11, Univ (MA) % 18, Coll (BA) % 13, All Math % 14                    |
| Other                                                                      | 33                                                                    | 54          | 42          | 43         |  | 18                                          | 35          | 18          | 21         | Univ (PhD) % 13, Coll (BA) % 21, All Math % 4                                     |

Note: Some totals are less than 100% due to round-off.

**TABLE SP.7.** Among statistics departments at four-year colleges and universities with secondary pre-service teaching certification programs, for various courses, the percentage of statistics departments whose program requires the course, whose students generally take the course, or who offer a special course in the given subject that is designed for secondary teachers, and the number or semester hours required for certification in grades K–5 and 6–8, by type of department, in fall 2021.

| Course                                                                     | Percentage of Statistics Departments with Secondary Certification Program Where: |             |            |  | Stat Dept Offers Special Course in the Subject for Secondary Pre-Service Teachers |             |            |   |
|----------------------------------------------------------------------------|----------------------------------------------------------------------------------|-------------|------------|--|-----------------------------------------------------------------------------------|-------------|------------|---|
|                                                                            | Course Is Required                                                               |             |            |  | Course Is Generally Taken, but Not Required                                       |             |            |   |
|                                                                            | Univ (PhD) %                                                                     | Univ (MA) % | All Stat % |  | Univ (PhD) %                                                                      | Univ (MA) % | All Stat % |   |
| Introductory statistics including a simulation based approach to inference | 13                                                                               | 25          | 16         |  |                                                                                   |             |            | 5 |
| Introductory statistics excluding a simulation based approach to inference | 53                                                                               | 25          | 47         |  |                                                                                   |             |            |   |
| Statistical methods with an introductory course as a prerequisite          | 20                                                                               |             | 16         |  | 8                                                                                 |             | 6          |   |
| Statistical modeling                                                       | 7                                                                                |             | 5          |  | 7                                                                                 |             | 6          |   |
| Probability and/or statistics with calculus prerequisite                   | 40                                                                               | 50          | 42         |  | 33                                                                                |             | 27         |   |
| Other                                                                      | 14                                                                               | 25          | 17         |  | 8                                                                                 |             | 7          | 5 |
| Number of Semester Hours Required for K–5 Grade Teachers                   |                                                                                  |             |            |  |                                                                                   |             |            |   |
| None                                                                       | 73                                                                               | 100         | 79         |  |                                                                                   |             |            |   |
| 1–3 hours                                                                  | 27                                                                               |             | 21         |  |                                                                                   |             |            |   |
| 4–6 hours                                                                  |                                                                                  |             |            |  |                                                                                   |             |            |   |
| More than 6 hours                                                          |                                                                                  |             |            |  |                                                                                   |             |            |   |
| Number of Semester Hours Required for 6–8 Grade Teachers                   |                                                                                  |             |            |  |                                                                                   |             |            |   |
| None                                                                       | 25                                                                               | 67          | 43         |  |                                                                                   |             |            |   |
| 1–3 hours                                                                  | 50                                                                               | 33          | 43         |  |                                                                                   |             |            |   |
| 4–6 hours                                                                  | 25                                                                               |             | 14         |  |                                                                                   |             |            |   |
| More than 6 hours                                                          |                                                                                  |             |            |  |                                                                                   |             |            |   |

Note: Blank entries represent courses or semester hours with no responses. Some percentages do not total 100% due to round-off.



Table SP.7 gives the estimated percentages in fall 2021 of statistics departments belonging to a college or university having a program of study that leads to credentials to teach mathematics in grades 9–12 public schools where various specific statistics courses are required or are generally taken. It shows that an estimated 47% (SE 12) of these institutions require an introductory course in statistics that excludes a simulation-based approach to inference and 42% (SE 11.9) require a probability and/or statistics course with a calculus prerequisite; the options in the 2015 survey were different, but 41% required an introductory course and 42% required a probability or statistics courses with a calculus prerequisite. The 2015 CBMS survey showed much larger percentages of statistics courses generally taken as well as statistics courses specially designed for pre-service teachers than reported in the 2021 survey.

### Tables SP.8.A–SP.10: Practices in Distance/Remote Learning Courses

In the 2021 CBMS survey, the phrase “distance/remote learning” replaced “distance learning,” and a slightly modified definition of distance/remote learning was used. The 2021 CBMS survey used the definition: “Distance/remote learning courses are those courses offered by your institution for credit, in which half or more of the instruction occurs with the instructor and the students separated by time and/or place, and facilitated by technology (e.g., courses in which half or more of the course is taught online either synchronously or asynchronously, by computer software, or by other technologies). Include only distance/remote learning courses offered in normal practice, not courses that became distance/remote due to the COVID-19 pandemic.”

Past CBMS surveys have gathered data on distance learning courses. CBMS2015, Table E.4, p. 101 gives distance enrollments by course categories for four-year mathematics departments, two-year mathematics departments, and four-year statistics departments; the comparable data for the CBMS 2021 survey are given in Tables S.9 and E.4. These data show large increases in the number of enrollments in distance/remote learning courses, particularly at two-year mathematics departments.

As noted in Chapter 1, according to Table S.9, in four-year mathematics departments, distance/remote learning enrollments in precalculus, introductory-level courses, calculus I, calculus II, differential equations, linear algebra, and introductory statistics courses combined in fall 2021 were estimated at 244,309, almost triple the fall 2015 estimate of 86,197. In fall 2021, distance/remote learning enrollments represented an estimated 17% (up from 4% in 2015) of the total enrollments in these courses. Note that for four-year mathematics and statistics departments, Appendix I provides enrollment by course both with and with-

out distance/remote learning enrollments included (except for advanced-level mathematics course enrollments, where distance/remote learning enrollments are small and only enrollments with distance learning are provided).

In fall 2021, the estimated distance/remote learning enrollments at two-year mathematics departments exceeded the non-distance/remote learning enrollments in each of the course categories presented in Table S.9 and in every other course offered at two-year colleges as shown in Table TYE.11. Overall, distance/remote learning showed two-year college enrollment in mathematics and statistics growing to an estimated 823,000 (SE 108,000), up 266% compared to 225,000 (SE 25,000) students in fall 2015. Distance learning enrollment accounted for 62% of all two-year college mathematics enrollment. See Table S.9 in Chapter 1 and Tables TYE.2 and TYE.11 in Chapter 6 for individual course enrollments.

This section considers practices in distance/remote learning course instruction. The 2021 CBMS survey, delayed one year due to the COVID-19 pandemic, made changes to the distance learning section of the CBMS questionnaires that had been designed for the 2020 survey before the pandemic, including introducing some new questions. In fall 2020, the CBMS survey team also performed a modified survey of practices used during the pandemic; the results of this survey, The COVID Impact Study, are given in Appendix IV, and further information on this study is available at <https://www.ams.org/profession/data/cbms-survey/cbms2020>.

In the COVID Impact Study, departments were asked to respond to the statement regarding future course delivery plans: “We are considering offering a greater number of online courses.” Figure 6 of that study shows that in fall 2020, an estimated 35% of four-year mathematics departments, 54% of statistics departments, and 61% of two-year college mathematics departments indicated that they agreed or strongly agreed with that statement.

**Post-pandemic changes in faculty and student interest in online learning (Table SP.8.A).** Table SP.8.A contains data from questions that were asked of all four-year mathematics, statistics, and two-year college mathematics departments in fall 2021. It captures changes in attitudes and practices in distance/remote learning resulting from the experience of teaching during the COVID-19 pandemic. Table SP.8.A also records changes in faculty and student interest in online instruction and in the use of online tools, and it shows that generally both faculty and students had more favorable attitudes toward online learning in fall 2021 than before the pandemic.

Table SP.8.A shows that in fall 2021, the estimated percentage of departments whose faculty were generally more favorable toward online instruction than before the

**TABLE SP.8.A.** Overall, in fall 2021, how have attitudes toward online learning in four-year mathematics, statistics, and two-year mathematics departments changed as a result of the COVID-19 experience?

|                                     |                        | Mathematics Depts |           |              |       | Statistics Depts | Two-Year Colleges |
|-------------------------------------|------------------------|-------------------|-----------|--------------|-------|------------------|-------------------|
|                                     |                        | Univ (PhD)        | Univ (MA) | College (BA) | Total | Total            | Total             |
| Faculty interest in online teaching | Percent more favorable | 56                | 58        | 43           | 46    | 74               | 64                |
|                                     | Percent no change      | 33                | 24        | 32           | 31    | 16               | 23                |
|                                     | Percent less favorable | 12                | 18        | 25           | 23    | 9                | 13                |
| Faculty use of online tools         |                        |                   |           |              |       |                  |                   |
|                                     | Percent more favorable | 79                | 77        | 79           | 79    | 86               | 91                |
|                                     | Percent no change      | 21                | 23        | 21           | 21    | 14               | 8                 |
|                                     | Percent less favorable |                   |           |              |       |                  | 1                 |
| Student interest in online teaching |                        |                   |           |              |       |                  |                   |
|                                     | Percent more favorable | 49                | 40        | 28           | 32    | 63               | 69                |
|                                     | Percent no change      | 28                | 29        | 39           | 36    | 26               | 24                |
|                                     | Percent less favorable | 23                | 32        | 33           | 31    | 12               | 7                 |
| Student use of online tools         |                        |                   |           |              |       |                  |                   |
|                                     | Percent more favorable | 72                | 73        | 48           | 55    | 84               | 74                |
|                                     | Percent no change      | 25                | 27        | 47           | 41    | 14               | 23                |
|                                     | Percent less favorable | 3                 | 0         | 5            | 4     | 2                | 3                 |

Note: Blank entries represent attitude changes with no responses.

pandemic was 46% (SE 5.2) of all four-year mathematics departments combined, 74% (SE 6.8) of all statistics departments combined, and 64% (SE 6.4) of two-year mathematics departments. The estimated percentage of departments whose faculty were generally less favorable was 23% (SE 5) of all four-year mathematics departments combined, 9% (SE 4.5) of all statistics departments combined, and 13% (SE 4.6) of two-year mathematics departments; these percentages were relatively consistent across all levels of four-year mathematics departments, with the unfavorable percentages slightly higher at the bachelor's-level mathematics departments. Faculty use of online tools was generally categorized as either more favorable or unchanged after the pandemic across four-year mathematics departments, statistics departments, and two-year mathematics departments with almost no departments reporting less favorable attitudes.

These data can be compared to the data found in the COVID Impact Study, which showed generally less favorable attitudes toward online education. For example, Figure 4.A in the COVID Impact Study shows that in fall 2020, the estimated percentages of departments that disagreed with the statement that learning outcomes are better with face-to-face experience was 7% at four-year mathematics departments, 8% at statistics departments, and 15% at two-year mathematics departments. Figure 5 in the COVID Impact Study shows the increase in faculty preparedness to teach online courses before and after the pandemic.

Table SP.8.A also shows that in fall 2021, the estimated percentage of departments whose students were generally more favorable toward online instruction in fall 2021 than before the pandemic was 32% (SE 4.7) of all four-year mathematics departments combined, 63% (SE 7.5) of all statistics departments combined, and 69% (SE 6.4) of two-year mathematics departments. The estimated percentage of departments whose students were generally less favorable was 31% (SE 5.4) of all four-year mathematics departments combined, 12% (SE 5) of all statistics departments combined, and 7% (SE 3.2) of two-year mathematics departments. The student use of online tools was either more favorable or unchanged across most four-year mathematics departments, statistics departments, and two-year mathematics departments.

**Formats of courses offered pre-pandemic, during the pandemic, and in fall 2021 (Tables SP.8.B(a)–SP.8.B(c)).** In the CBMS 2021 survey, all four-year mathematics and statistics departments and two-year college mathematics departments were surveyed about their course formats before the pandemic, during the pandemic (spring 2020–summer 2021), and after the pandemic (fall 2021). The departments were asked about when they offered some sections of classes that were only for on-campus (face-to-face) learners, some sections of courses that were only for remote learners, and some “hybrid” courses that

**TABLE SP.8.B(a).** Percentage of four-year mathematics departments with the following forms of coordination of online instruction between on-campus learners and remote learners in the following time periods. (Percentage answering “yes”)

|                                                                                                        | Prior to Pandemic |              |                 |       | During Pandemic |              |                 |       | Fall 2021     |              |                 |       |
|--------------------------------------------------------------------------------------------------------|-------------------|--------------|-----------------|-------|-----------------|--------------|-----------------|-------|---------------|--------------|-----------------|-------|
|                                                                                                        | Univ<br>(PhD)     | Univ<br>(MA) | College<br>(BA) | Total | Univ<br>(PhD)   | Univ<br>(MA) | College<br>(BA) | Total | Univ<br>(PhD) | Univ<br>(MA) | College<br>(BA) | Total |
| Some sections offered only to on-campus (face-to-face) learners <sup>1</sup>                           | 71                | 89           | 68              | 71    | 53              | 47           | 42              | 44    | 74            | 85           | 62              | 67    |
| Some sections offered only to remote learners <sup>2</sup>                                             | 49                | 67           | 54              | 55    | 71              | 65           | 70              | 69    | 61            | 73           | 55              | 59    |
| Some sections offered to on-campus learners and remote learners in the same section ("hybrid courses") | 7                 | 11           | 10              | 9     | 66              | 67           | 70              | 69    | 41            | 44           | 31              | 34    |

<sup>1</sup> On-campus (face-to-face) learners: Students who would be expected to attend most classes in person.

<sup>2</sup> Remote learners: Students who would be expected to attend most classes remotely.

**TABLE SP.8.B(b).** Percentage of statistics departments with the following forms of coordination of online instruction between on-campus learners and remote learners in the following time periods. (Percentage answering “yes”)

|                                                                                                        | Prior to Pandemic | During Pandemic | Fall 2021 |
|--------------------------------------------------------------------------------------------------------|-------------------|-----------------|-----------|
| Some sections offered only to on-campus (face-to-face) learners <sup>1</sup>                           | 74                | 44              | 81        |
| Some sections offered only to remote learners <sup>2</sup>                                             | 53                | 77              | 65        |
| Some sections offered to on-campus learners and remote learners in the same section (“hybrid courses”) | 21                | 56              | 40        |

<sup>1</sup> On-campus (face-to-face) learners: Students who would be expected to attend most classes in person.

<sup>2</sup> Remote learners: Students who would be expected to attend most classes remotely.

**TABLE SP.8.B(c).** Percentage of two-year college departments with the following forms of coordination of on-line instruction between on-campus learners and remote learners in the following time periods. (Percentage answering “yes”)

|                                                                                                        | Prior to Pandemic | During Pandemic | Fall 2021 |
|--------------------------------------------------------------------------------------------------------|-------------------|-----------------|-----------|
| Some sections offered only to on-campus (face-to-face) learners <sup>1</sup>                           | 93                | 41              | 80        |
| Some sections offered only to remote learners <sup>2</sup>                                             | 72                | 78              | 74        |
| Some sections offered to on-campus learners and remote learners in the same section (“hybrid courses”) | 30                | 37              | 60        |

<sup>1</sup> On-campus (face-to-face) learners: Students who would be expected to attend most classes in person.

<sup>2</sup> Remote learners: Students who would be expected to attend most classes remotely.

had on-campus learners and remote learners in the same section. “On-campus learners” were defined to be students who would be expected to attend most classes in person, while “remote learners” were defined to be students who would be expected to attend most classes remotely. “Hybrid courses” were courses that contained on-campus and remote learners in the same section. Responses to these questions are summarized in Table SP.8.B(a) for four-year mathematics departments, broken down by level of department, Table SP.8.B(b) for statistics departments, and Table SP.8.B(c) for two-year mathematics departments. The tables generally show that the fall 2021 practice was not too different than the pre-pandemic practice, except for increased use of hybrid courses.

From Tables SP.8.B(a)–SP.8.B(c), we see that in fall 2021, departments reported that during the pandemic, less than half of each of the types of departments combined offered some courses only in face-to-face format. Furthermore, the distribution of responses on practices before the pandemic and in fall 2021 are quite similar; the largest change appears to be in the offering of hybrid courses. Pre-pandemic, an estimated 9% (SE 3.1) of four-year mathematics departments combined offered some hybrid course; during the pandemic, 69% (5.3 SE) offered some hybrid course; and in fall 2021, 34% (SE 5.1) offered some hybrid course. Pre-pandemic, an estimated 21% (SE 6.4) of all levels of statistics departments combined offered some hybrid course; during the pandemic, 56% (SE 7.7) offered some hybrid

course; and in fall 2021, 40% (SE 7.6) offered some hybrid course. Pre-pandemic, an estimated 30% (SE 6) of two-year mathematics departments offered some hybrid course; during the pandemic, 37% (SE 6.8) of two-year departments offered some hybrid course; and in fall 2021, 60% (SE 6.6) offered some hybrid course. In fall 2021, 59% (SE 5.3) of four-year mathematics departments, 65% (SE 7.4) of statistics departments, and 74% (SE 6.2) of two-year mathematics departments offered some course in a remote-only format.

**Structure of distance/remote learning courses and administration of tests (Table SP.9).** The CBMS 2021 survey asked four-year mathematics, statistics, and two-year mathematics departments if they had offered any distance/remote learning courses during 2018–2021 (other than courses that were moved online in response to the COVID-19 pandemic); if so, they were asked the structure that best characterized the majority of their distance/remote learning courses and how the majority of tests were administered. The departments' responses are summarized in Table SP.9. This data can be compared to parts of CBMS2015, Table SP.8, p. 55 as well as to the COVID Impact Report.

According to Table SP.9, the estimated percentage of departments that have offered a distance/remote learning course (other than courses moved online in response to the COVID-19 pandemic) during 2018–2021 was 75% (SE 4.8) at all levels of four-year mathematics departments combined (compared to 52% in fall 2015), 79% (6.3 SE) at all levels of statistics departments combined (compared to 64% in fall 2015), and 89% (SE 4.8) at two-year mathematics departments (compared to 87% in fall 2015).

The 2021 CBMS survey asked departments to choose the best characterization of the format/structure of the majority of their distance/remote courses from these options: (1) only online and only asynchronously, (2) only online with an opportunity to meet synchronously online, (3) a mixture of online and face-to-face sessions, or (4) other; in the 2015 survey, the options were (1) completely online, (2) hybrid, or (3) other. In the COVID Impact report from fall 2020, Figure 1 shows that the four-year mathematics departments, statistics departments, and two-year mathematics departments all indicated that the largest percentage of courses were offered online synchronously. The CBMS 2021 survey responses were relatively consistent across all the types of departments and indicate growing use of asynchronous online course delivery. Table SP.9 shows that only online and only asynchronously was the most popular response with an estimated 57% (SE 6.8) of all levels of four-year mathematics departments combined, 56% (SE 8.7) of all levels of statistics departments combined, and 68% (SE 6.4) of all two-year mathematics departments. Only online

with an opportunity to meet synchronously was the next most popular response with an estimated 21% (SE 5.2) of all levels of four-year mathematics departments combined, 32% (SE 8.2) of all levels of statistics departments combined, and 25% (SE 6.3) of all two-year mathematics departments. In the 2015 CBMS survey, an estimated 69% of four-year mathematics departments, 66% of statistics departments, and 69% of two-year mathematics departments chose “completely online.” Combining the data from Tables SP.8.A–SP.8.B(c) and SP.9, it appears that while more departments are offering some hybrid options, in fall 2021, for 78% of four-year mathematics departments, 88% of statistics departments, and 93% of two-year mathematics departments, most of their distance/remote offerings are only online.

The final section of Table SP.9 summarizes how the majority of tests were monitored in distance/remote courses—from among the options: online and not monitored, online using some kind of monitoring technology, at a monitored testing site, or a combination of monitoring methods. Some of the responses differed by the type of department. In fall 2021, an estimated 24% (5.6 SE) of four-year mathematics departments combined, 21% (7.1 SE) of statistics departments combined, and 24% (SE 6.3) of two-year mathematics departments chose online and not monitored, while 40% (7 SE) of four-year mathematics departments combined, 41% (8.6 SE) of statistics departments combined, and 21% (SE 5.6) of two-year mathematics departments chose online using some kind of monitoring technology. A combination of methods was used by 28% (5.9 SE) of four-year mathematics departments combined, 38% (8.5 SE) of statistics departments combined, and 37% (SE 6.6) of two-year mathematics departments. In the CBMS 2015 survey, a proctored testing site was the option chosen by the majority of all three types of departments (see CBMS2015, Table SP.8, p. 55).

**Challenges faced in offering distance/remote learning courses (Table SP.10).** In the CBMS 2021 survey, four-year mathematics, statistics, and two-year mathematics departments that had offered any distance/remote learning courses during 2018–2021 (other than courses moved online in response to the COVID-19 pandemic) were asked to rate various issues as not a challenge, somewhat of a challenge, or a very significant challenge. The responses are summarized in Table SP.10, broken down by the level of the department. In the COVID Impact Study, departments were asked to list challenges and opportunities that arose from the teaching during the pandemic; a summary of those free response comments is listed in the final section of that report.

According to Table SP.10, the issue that the largest percentage of each of the three types of departments listed as a very significant challenge in fall 2021 was maintaining

**TABLE SP.9.** Percentage of mathematics, statistics, and public two-year college departments offering distance/remote learning courses,<sup>1</sup> and the use of various practices with regard to distance learning in fall 2021.

|                                                                                 | Mathematics Depts |           |              |       | Statistics Depts |           |       | Two-Year Depts |
|---------------------------------------------------------------------------------|-------------------|-----------|--------------|-------|------------------|-----------|-------|----------------|
|                                                                                 | Univ (PhD)        | Univ (MA) | College (BA) | Total | Univ (PhD)       | Univ (MA) | Total |                |
| Taught any distance/remote learning course within the calendar years 2018–2021: |                   |           |              |       |                  |           |       |                |
| Yes                                                                             | 70                | 78        | 75           | 75    | 77               | 88        | 79    | 89             |
| No                                                                              | 30                | 22        | 25           | 25    | 23               | 13        | 21    | 11             |
| Format of majority of distance/remote learning:                                 |                   |           |              |       |                  |           |       |                |
| Online without an opportunity to meet synchronously online                      | 59                | 57        | 57           | 57    | 56               | 57        | 56    | 68             |
| Online with an opportunity to meet synchronously online                         | 23                | 15        | 23           | 21    | 37               | 14        | 32    | 25             |
| Hybrid                                                                          | 14                | 18        | 15           | 15    | 7                | 29        | 12    | 7              |
| Other                                                                           | 4                 | 11        | 5            | 6     |                  |           |       |                |
| How distance/remote learning students take majority of tests:                   |                   |           |              |       |                  |           |       |                |
| Not at a monitored testing site                                                 | 13                | 10        | 28           | 24    | 15               | 43        | 21    | 24             |
| Online, using monitoring technology                                             | 34                | 37        | 41           | 40    | 41               | 43        | 41    | 21             |
| At proctored testing site                                                       | 15                | 9         | 8            | 9     |                  |           |       | 18             |
| Combination of both                                                             | 39                | 44        | 23           | 28    | 44               | 14        | 38    | 37             |

<sup>1</sup> Distance/remote learning courses are those courses offered by your institution for credit, in which half or more of the instruction occurs with the instructor and the students separated by time and/or place, and facilitated by technology (e.g., courses in which half or more of the course is taught online either synchronously or asynchronously, by computer software, or by other technologies). Only distance/remote learning courses offered in normal practice, not courses that became distance/remote due to the COVID-19 pandemic, were included.

**TABLE SP.10.** Percentage of four-year mathematics and statistics and two-year mathematics departments rating challenges faced when creating/offering distance/remote learning courses.<sup>1</sup>

|                                                                      | Mathematics Depts |           |              |       | Statistics Depts |       | Two-Year Colleges |       |
|----------------------------------------------------------------------|-------------------|-----------|--------------|-------|------------------|-------|-------------------|-------|
|                                                                      | Univ (PhD)        | Univ (MA) | College (BA) | Total | Total            | Total | Total             | Total |
| Designing assessments of student learning                            |                   |           |              |       |                  |       |                   |       |
| Not a challenge                                                      | 8                 | 9         | 15           | 13    | 18               | 19    | 19                | 19    |
| Somewhat of a challenge                                              | 53                | 75        | 65           | 65    | 59               | 54    | 54                | 54    |
| Very significant challenge                                           | 39                | 17        | 20           | 22    | 24               | 27    | 27                | 27    |
| Maintaining academic integrity on assessments                        |                   |           |              |       |                  |       |                   |       |
| Not a challenge                                                      | 4                 | 9         | 5            | 6     | 9                | 8     | 8                 | 8     |
| Somewhat of a challenge                                              | 22                | 36        | 24           | 26    | 38               | 43    | 43                | 43    |
| Very significant challenge                                           | 74                | 56        | 70           | 69    | 53               | 49    | 49                | 49    |
| Grade inflation                                                      |                   |           |              |       |                  |       |                   |       |
| Not a challenge                                                      | 20                | 46        | 34           | 34    | 44               | 31    | 31                | 31    |
| Somewhat of a challenge                                              | 64                | 45        | 52           | 52    | 47               | 54    | 54                | 54    |
| Very significant challenge                                           | 15                | 9         | 15           | 14    | 9                | 15    | 15                | 15    |
| Building/maintaining community among faculty and students            |                   |           |              |       |                  |       |                   |       |
| Not a challenge                                                      | 13                | 0         | 2            | 3     |                  |       |                   |       |
| Somewhat of a challenge                                              | 36                | 42        | 39           | 39    | 50               | 61    | 61                | 61    |
| Very significant challenge                                           | 51                | 58        | 59           | 58    | 50               | 39    | 39                | 39    |
| Maintaining quality academic instruction                             |                   |           |              |       |                  |       |                   |       |
| Not a challenge                                                      | 25                | 11        | 13           | 14    | 35               | 24    | 24                | 24    |
| Somewhat of a challenge                                              | 56                | 75        | 62           | 63    | 59               | 59    | 59                | 59    |
| Very significant challenge                                           | 19                | 14        | 25           | 23    | 6                | 17    | 17                | 17    |
| Engaging students online                                             |                   |           |              |       |                  |       |                   |       |
| Not a challenge                                                      | 9                 | 0         | 2            | 2     | 3                | 1     | 1                 | 1     |
| Somewhat of a challenge                                              | 44                | 54        | 34           | 38    | 62               | 52    | 52                | 52    |
| Very significant challenge                                           | 47                | 46        | 64           | 60    | 35               | 47    | 47                | 47    |
| Replicating active learning in a virtual environment                 |                   |           |              |       |                  |       |                   |       |
| Not a challenge                                                      | 4                 | 0         | 2            | 2     | 12               | 2     | 2                 | 2     |
| Somewhat of a challenge                                              | 36                | 33        | 34           | 34    | 50               | 45    | 45                | 45    |
| Very significant challenge                                           | 59                | 67        | 64           | 64    | 38               | 53    | 53                | 53    |
| Availability of equipment and technical support for faculty/students |                   |           |              |       |                  |       |                   |       |
| Not a challenge                                                      | 46                | 32        | 27           | 30    | 32               | 20    | 20                | 20    |
| Somewhat of a challenge                                              | 54                | 51        | 59           | 57    | 59               | 63    | 63                | 63    |
| Very significant challenge                                           | 0                 | 17        | 15           | 13    | 9                | 17    | 17                | 17    |

Note: Blank entries represent challenges with no responses.

<sup>1</sup> Distance/remote learning courses are those courses offered by your institution for credit, in which half or more of the instruction occurs with the instructor and the students separated by time and/or place, and facilitated by technology (e.g., courses in which half or more of the course is taught online either synchronously or asynchronously, by computer software, or by other technologies). Only distance/remote learning courses offered in normal practice, not courses that became distance/remote due to the COVID-19 pandemic, were included.

**TABLE SP.11.A.** Percentage of four-year mathematics departments offering various upper-level mathematics courses by distance/remote learning, by department type, in fall 2021.

|                                                                 | Mathematics Departments |              |                 |       |
|-----------------------------------------------------------------|-------------------------|--------------|-----------------|-------|
|                                                                 | Univ<br>(PhD)           | Univ<br>(MA) | College<br>(BA) | Total |
| E23. Introduction to Proofs                                     | 1                       | 11           | 5               | 5     |
| E24-1. Modern Algebra I                                         | 3                       | 3            | 5               | 5     |
| E24-2. Modern Algebra II                                        | 7                       |              | 0               | 1     |
| E25. Number Theory                                              |                         | 7            |                 | 1     |
| E26. Combinatorics                                              |                         |              | 2               | 2     |
| E27. Actuarial Mathematics                                      | 3                       | 4            |                 | 1     |
| E28. Logic/Foundations (not E23)                                |                         |              |                 |       |
| E29. Discrete Structures                                        | 1                       |              |                 | 0     |
| E30. History of Mathematics                                     | 6                       | 4            | 2               | 2     |
| E31. Geometry                                                   | 6                       | 3            | 3               | 4     |
| E32-1. Advanced Calculus I and/or Real Analysis I               | 4                       | 4            | 2               | 2     |
| E32-2. Advanced Calculus II and/or Real Analysis II             | 4                       |              | 2               | 2     |
| E33. Advanced Mathematics for Engineering and Physical Sciences | 3                       | 4            |                 | 1     |
| E34. Advanced Linear Algebra (beyond E18, E20)                  | 7                       |              |                 | 1     |
| E35. Vector Analysis                                            | 3                       |              |                 | 0     |
| E36. Advanced Differential Equations (beyond E19)               |                         |              |                 |       |
| E37. Partial Differential Equations                             | 4                       |              |                 | 0     |
| E38. Numerical Analysis I and II                                | 4                       | 4            | 1               | 2     |
| E39. Applied Mathematics (Modeling)                             |                         |              | 3               | 2     |
| E40. Complex Variables                                          | 4                       |              |                 | 0     |
| E41. Topology                                                   | 4                       |              |                 | 0     |
| E42. Mathematics of Finance (not E28, E40)                      |                         |              |                 |       |
| E43. Codes and Cryptology                                       |                         | 3            |                 | 0     |
| E44. Biomathematics                                             |                         |              |                 |       |
| E45. Operations Research (all courses)                          |                         |              |                 |       |
| E46. Senior Seminar/ Independent Study in Mathematics           | 3                       | 4            | 3               | 3     |
| E47. Other advanced-level mathematics                           |                         |              | 5               | 4     |
| E48. Mathematics for Secondary School Teachers                  |                         | 4            |                 | 1     |

Note: These estimates are based on small numbers and have large SE. Blank entries represent courses with no responses, while 0 entries indicate percentages that round to 0%.



**TABLE SP.11.B.** Percentage of four-year mathematics and statistics departments offering upper-level statistics courses by distance/remote learning, by department type, in fall 2021.

|                                                               | Mathematics Departments |           |              |       | Statistics Departments |           |       |
|---------------------------------------------------------------|-------------------------|-----------|--------------|-------|------------------------|-----------|-------|
|                                                               | Univ (PhD)              | Univ (MA) | College (BA) | Total | Univ (PhD)             | Univ (MA) | Total |
| E7. Combined Probability & Statistics (calculus prerequisite) | 30                      | 38        | 22           | 25    | 49                     | 63        | 51    |
| E8. Probability (calculus prerequisite)                       | 70                      | 47        | 20           | 29    | 54                     | 13        | 47    |
| E9. Mathematical Statistics (calculus prerequisite)           | 38                      | 22        | 18           | 21    | 54                     | 38        | 51    |
| E10. Stochastic Processes                                     | 11                      | 3         |              | 2     | 20                     |           | 16    |
| E11. Data Science/Analytics                                   | 13                      | 22        | 12           | 14    | 54                     | 88        | 60    |
| E12. Design & Analysis of Experiments                         | 4                       |           |              | 0     | 6                      |           | 5     |
| E13. Applied Regression                                       | 28                      | 18        | 8            | 11    | 60                     | 25        | 53    |
| E14. Linear Models I                                          |                         | 5         |              | 1     | 23                     | 13        | 21    |
| E15. Linear Models II                                         |                         |           |              |       | 6                      |           | 5     |
| F16. Biostatistics                                            |                         | 3         | 4            | 4     | 6                      | 38        | 12    |
| E17. Nonparametric Statistics                                 | 4                       |           | 1            | 1     | 20                     | 13        | 19    |
| E18. Categorical Data Analysis                                |                         | 8         | 2            | 2     | 26                     | 50        | 30    |
| E19. Sample Survey Design & Analysis                          | 13                      |           | 0            | 2     | 23                     | 50        | 28    |
| E20. Statistical Computing and/or Software                    | 8                       | 9         | 2            | 4     | 51                     | 75        | 56    |
| E21. Bayesian Statistics                                      |                         |           |              |       | 31                     | 38        | 33    |
| E22. Statistical Consulting                                   | 4                       |           | 1            | 1     | 11                     | 13        | 12    |
| E23. Senior Seminar/ Independent Studies                      | 4                       |           | 7            | 6     | 26                     | 13        | 23    |
| E24. All other upper-level Probability & Statistics           | 15                      | 18        | 11           | 12    | 49                     | 38        | 47    |

Note: These estimates are based on small numbers and have large SE. Blank entries represent courses with no responses, while 0 entries indicate percentages that round to 0%.

academic integrity on assessments; an estimated 69% (5.7 SE) of four-year mathematics departments combined, 53% (8.8 SE) of statistics departments combined, and 49% (SE 6.9) of two-year departments rated it a very significant challenge. Building and maintaining community among faculty and students was cited as a very significant challenge by an estimated 58% (SE 6.8) of four-year mathematics departments combined, 50% (SE 8.8) of statistics departments combined, and 39% (SE 6.9) of two-year mathematics departments. Replicating active learning in a virtual environment was cited as a very significant challenge by an estimated 64% (SE 5.8) of four-year mathematics departments combined, 38% (SE 8.5) of statistics departments combined, and 53% (SE 6.9) of two-year mathematics departments, with similar percentages for the issue of engaging students online.

Generally, in fall 2021, the issues of grade inflation, designing assessments of student learning, and maintaining quality instruction were not seen as significant challenges.

### **Tables SP.11.A and SP.11.B: Upper-Level Courses Offered in Distance/Remote Format at Four-Year Mathematics and Statistics Departments**

Beginning in 2010, the CBMS survey asked four-year departments to check each upper-level course offered in distance/remote learning format. The numbers of departments reporting such courses were small in both 2010 and 2015 but have increased in 2021. The data in Tables SP.11.A (four-year upper-level mathematics courses) and SP.11.B (four-year upper-level statistics courses) may be compared to the data reported in CBMS2015, Tables SP.11.A and SP.11.B, pp. 58–9 and CBMS2010, Tables 13.A and 13.B, pp. 58–9. Beginning in the CBMS 2021 survey, data on distance/remote learning enrollments in upper-level courses were gathered. Appendix I presents four-year mathematics and statistics department enrollments both with and without distance/remote learning enrollments. Distance/remote learning enrollments in four-year advanced mathematics courses were small enough that they were not separated from non-distance/remote learning enrollments in Appendix I; for advanced mathematics courses, only enrollments with distance/remote learning are reported in Appendix I.

### **Tables SP.12–SP.14: Academic Resources Available to Undergraduates**

Tables SP.12 and SP.13 present the estimated percentages of mathematics and statistics departments in four-year colleges and universities, and of mathematics departments at public two-year colleges, that offer various kinds of special opportunities for undergraduates, by level of department,

in fall 2021; the tables also include the percentages observed in the 2015 CBMS survey. The biggest changes reported in Table SP.12 from the 2015 CBMS survey include the increase in percentages of departments with a mathematics or statistics club, which was up from an estimated 67% in 2015 to 82% (SE 5.6) in fall 2021 in all levels of four-year mathematics combined and up from 46% in 2015 to 68% (SE 7.5) in fall 2021 in four-year statistics departments combined. Both levels of statistics departments combined reported increases in special programs for women (14% in 2015 and 37% (SE 8) in 2021), for minorities (12% in 2015 and 26% (SE 7.3) in 2021), and in mathematical or statistical contests (54% in 2015 and 76% (SE 7) in 2021).

According to Table SP.13, the estimated percentage of all levels of four-year mathematics departments combined with opportunities for students to tutor, TA, or grade papers increased from 17% in fall 2015 to 85% (SE 4.2) in fall 2021, while the percentage with opportunities to consult with clients dropped from 83% in fall 2015 to 21% (SE 5.1) in fall 2021. Opportunities to participate in undergraduate research at all four-year mathematics departments combined increased from 77% in fall 2015 to 91% (SE 3.6) in fall 2021.

At two-year college mathematics departments, noting the decreases in enrollment in 2021, Tables SP.12 and SP.13 show mostly decreases in offerings of various opportunities to students. A decrease of 10 percentage points to 22% (SE 4.9) was seen in mathematics clubs. Programs to encourage women were down 5 percentage points to 10% (SE 3.7). Outreach in K–12 schools was down 6 percentage points to 40% (SE 7.1). Independent studies in mathematics were down 16 percentage points to 24% (SE 5.5). Special programs for minorities in mathematical studies experienced a one percent increase to 16% (SE 4.5). Opportunities for undergraduate student research remained constant at 17% (SE 4.8). See Tables SP.12 and SP.13 and Table TYE.13 in Chapter 6.

Table SP.14 presents estimated totals of undergraduate students from four-year mathematics and statistics departments who were involved in undergraduate research projects, mathematical sciences internships, or mathematical or statistical consulting opportunities from September 1, 2020 through August 31, 2021, broken down by the level of the department. These data can be compared to CBMS2015, Table SP.14, p. 62, though in both surveys the SEs are large. Numbers of participants in undergraduate research projects reported by four-year mathematics departments in the 2021 CBMS survey were relatively comparable to the numbers observed in the 2015 CBMS survey. The estimated number of students in all levels of four-year mathematics departments combined involved in internships in the mathematical sciences decreased from 6031 in fall 2015 to 3910 (SE 874) in fall 2021; it seems likely that concerns about the pandemic contributed to the decrease in internships.

**TABLE SP.12.** Percentage of mathematics and statistics departments in four-year colleges and universities, and of mathematics departments at public two-year colleges, that offer various kinds of special opportunities for undergraduates, by type of department, in fall 2021. (Fall 2015 data in parentheses.)

| Percentage with Special Opportunities for Undergraduates | Honors Sections of Courses for Majors % | Math or Stat Club % | Special Programs for Women % | Special Programs for Minorities % | Math or Stat Contests % | Special Math or Stat Colloquia for Undergrads % | Outreach in K-12 Schools % |
|----------------------------------------------------------|-----------------------------------------|---------------------|------------------------------|-----------------------------------|-------------------------|-------------------------------------------------|----------------------------|
| Mathematics Departments                                  |                                         |                     |                              |                                   |                         |                                                 |                            |
| Univ (PhD)                                               | 56 (69)                                 | 100 (94)            | 67 (41)                      | 27 (25)                           | 93 (91)                 | 52 (77)                                         | 68 (61)                    |
| Univ (MA)                                                | 47 (39)                                 | 100 (91)            | 26 (37)                      | 24 (31)                           | 85 (78)                 | 79 (87)                                         | 71 (77)                    |
| Coll (BA)                                                | 15 (28)                                 | 75 (56)             | 19 (16)                      | 13 (8)                            | 73 (64)                 | 44 (53)                                         | 50 (43)                    |
| Total Mathematics Depts                                  | 25 (35)                                 | 82 (67)             | 26 (22)                      | 16 (14)                           | 77 (70)                 | 50 (61)                                         | 55 (50)                    |
| Statistics Departments                                   |                                         |                     |                              |                                   |                         |                                                 |                            |
| Univ (PhD)                                               | 47 (38)                                 | 72 (55)             | 35 (18)                      | 26 (13)                           | 77 (56)                 | 68 (70)                                         | 23 (18)                    |
| Univ (MA)                                                | 43 (50)                                 | 50 (18)             | 43                           | 29 (8)                            | 71 (45)                 | 71 (42)                                         | (42)                       |
| Total Statistics Depts                                   | 46 (41)                                 | 68 (46)             | 37 (14)                      | 26 (12)                           | 76 (54)                 | 68 (63)                                         | 19 (24)                    |
| Two-Year College Mathematics Departments                 | na (28)                                 | 22 (32)             | 10 (15)                      | 16 (15)                           | na (40)                 | 14 (21)                                         | 40 (46)                    |

Note: 0 means less than one half of 1 %. Blank entries represent opportunities with no responses.

**TABLE SP.13.** Percentage of mathematics and statistics departments in four-year colleges and universities, and of mathematics departments in public two-year colleges, that offer various additional special opportunities for undergraduates, by type of department, in fall 2021. (Fall 2015 data, where available, in parentheses.)

[illegible]

**TABLE SP.14.** Total number of majors (best estimate) who participated in various activities over September 1, 2020, through August 31, 2021.

| Activity                                                    | All Math<br>Depts | PhD<br>Math | MA<br>Math | BA<br>Math | All Stat<br>Depts | PhD<br>Stat | MA<br>Stat |
|-------------------------------------------------------------|-------------------|-------------|------------|------------|-------------------|-------------|------------|
| Undergraduate research project in the mathematical sciences | 11428             | 2749        | 1310       | 7369       | 490               | 408         | 83         |
| Internship in mathematical sciences                         | 3910              | 806         | 299        | 2805       | 738               | 516         | 222        |
| Mathematical or statistical consulting to client            | 1780              | 81          | 174        | 1525       | 339               | 181         | 158        |

**TABLE SP.15.** Percentage of all four-year mathematics departments offering interdisciplinary courses, by type of department, in fall 2021.

|                                                      | Univ (PhD)             | Univ (MA)              | Coll (BA)              | All<br>Departments     |
|------------------------------------------------------|------------------------|------------------------|------------------------|------------------------|
| Offered Course in:                                   | Offered<br>Course<br>% | Offered<br>Course<br>% | Offered<br>Course<br>% | Offered<br>Course<br>% |
| Mathematics and finance or business                  | 48                     | 32                     | 30                     | 33                     |
| Mathematics and biology                              | 49                     | 15                     | 12                     | 17                     |
| Mathematics and the study of the environment         | 6                      |                        | 4                      | 3                      |
| Mathematics and engineering or the physical sciences | 36                     | 19                     | 16                     | 19                     |
| Mathematics and economics                            | 26                     | 8                      | 3                      | 7                      |
| Mathematics and social sciences other than economics | 10                     | 4                      | 8                      | 8                      |
| Mathematics and education                            | 34                     | 60                     | 45                     | 46                     |
| Mathematics and the humanities                       |                        | 4                      | 2                      | 2                      |
| Mathematics and computer science                     | 36                     | 35                     | 41                     | 40                     |
| Mathematics and social justice science               | 6                      |                        | 5                      | 5                      |
| Other                                                | 7                      | 17                     | 10                     | 11                     |

Note: Blank entries represent courses with no responses, while 0 entries indicate percentages that round to 0%.

The estimated numbers obtained from the four-year statistics departments reported in the 2021 CBMS survey were comparable to the numbers reported in the 2015 survey.

### **Table SP.15: Interdisciplinary Courses in Four-Year Mathematics Departments**

Each CBMS survey collects enrollments in specific courses that occur regularly in most departments, including the category “other,” but the surveys have also collected data on the existence of a changing list of possible interdisciplinary courses that are offered at four-year mathematics departments. An interdisciplinary course is defined as one in which mathematics is taught with relation to another field (calculus courses are not included). The possible interdisciplinary courses listed in the CBMS 2021 survey are the same as in the CBMS 2015 survey, with the addition of the course “mathematics and social justice.” Table SP.15 gives the estimated percentages of four-year mathematics departments that offer each of the listed interdisciplinary courses, including “other.” This table can be compared to CBMS2015, Table SP.15, p. 63, which has a similar distribution. As in the CBMS 2015 survey, in fall 2021, the interdisciplinary course offered by the largest percentage of departments was “mathematics and education,” which was offered by an estimated 46% (SE 5.5) of four-year mathematics departments; the interdisciplinary course offered by the next largest percentage of four-year mathematics departments combined was “mathematics and computer science,” which was offered by an estimated 40% (SE 6.1) four-year mathematics departments combined, followed by “mathematics and finance or business,” which was offered by an estimated 33% (SE 5) of four-year mathematics departments combined (at 48% (SE 9.5) of doctoral-level mathematics departments). Generally, the percentage of departments offering an interdisciplinary course was higher in the doctoral-level departments than in master’s- or bachelor’s-level mathematics departments. In fall 2021, the interdisciplinary course “mathematics and biology” was offered by an estimated 49% (SE 9.5) of doctoral-level mathematics departments but an estimated 17% (SE 4.8) of all four-year mathematics departments combined.

### **Tables SP.16 and SP.17: Dual Enrollments—College Credit for High School Courses**

Dual-enrollment courses were defined to be “courses conducted on a high school campus and taught by high school teachers, for which high school students may obtain high school credit and, simultaneously, college credit through your institution.” This arrangement is not the

same as obtaining college credit based on an AP or IB exam or as high school students enrolling in a course at a college. Dual enrollment is encouraged by many state governments as a way of utilizing state-wide educational resources efficiently.

Table SP.16 gives the percentages of four-year mathematics departments over all levels combined, statistics departments over all levels combined, and two-year colleges that offered dual-enrollment courses and gives the enrollments in various dual-enrollment courses in spring 2021 and fall 2021 compared to the total of all other enrollments in fall 2021 (including distance/remote learning enrollments) in fall 2021. The percentages of departments that offered dual-enrollment courses for fall 2015 are given, and the rest of the 2021 table can be compared to CBMS2015, Table SP.16, p. 65 and CBMS2010, Table SP.18, p. 65.

Originally, dual-enrollment courses were offered primarily through two-year colleges. According to the data obtained in the CBMS 2010 survey in fall 2010, an estimated 61% of mathematics departments at two-year colleges, 17% of mathematics departments at four-year colleges and universities, and 8% of statistics departments offered dual-enrollment courses. The CBMS 2015 survey found that in fall 2015, the estimated percentage of four-year mathematics departments offering dual-enrollment courses rose to 26% (SE 4.1). The percentages of two-year mathematics departments and statistics departments offering dual-enrollment courses in 2015 were about comparable to percentages in 2010. Moreover, the estimated number of enrollments in dual-enrollment courses offered in 2015 by four-year mathematics departments increased dramatically over the number of dual enrollments estimated in 2010. The estimated enrollment in dual-enrollment courses offered by mathematics departments in four-year colleges and universities in spring and fall (combined) in 2010 was 42,862, with slightly more than half of the enrollments occurring in fall 2010; in 2015, the estimated number of enrollments had risen to 117,399, and again, slightly more than half of the enrollments occurred in fall 2015.

Mathematics departments in two-year colleges had an estimated total of 170,970 enrollments in spring and fall (combined) 2015, compared to 158,097 enrollments in spring and fall (combined) 2010. In 2010, two-year mathematics departments had almost four times the estimated dual enrollments of mathematics departments at four-year colleges and universities, while in 2015, the estimated enrollments in four-year college dual-enrollment courses were about two-thirds of the estimated enrollments in dual-enrollment courses offered by two-year colleges. Statistics departments had a much smaller estimated number of dual enrollments at 1478 in 2015, compared with 1573 dual enrollments in 2010.

**TABLE SP.16.** Percentage of departments offering dual-enrollment courses taught in high school by high school teachers and enrollments in various dual-enrollment courses in spring 2021 and fall 2021 compared to total of all other enrollments (including distance learning enrollments) in fall 2021 by type of department. (Fall 2015 data in parentheses.)

| Percentage of Departments with Dual-Enrollment Courses | Four-Year Mathematics |           |                   |           | Two-Year Mathematics |           |                   |           | Four-Year Statistics |           |                   |           |
|--------------------------------------------------------|-----------------------|-----------|-------------------|-----------|----------------------|-----------|-------------------|-----------|----------------------|-----------|-------------------|-----------|
|                                                        | Dual Enrollments      |           | Other Enrollments |           | Dual Enrollments     |           | Other Enrollments |           | Dual Enrollments     |           | Other Enrollments |           |
|                                                        | Spring 2021           | Fall 2021 | Spring 2021       | Fall 2021 | Spring 2021          | Fall 2021 | Spring 2021       | Fall 2021 | Spring 2021          | Fall 2021 | Spring 2021       | Fall 2021 |
| College algebra                                        | 13815                 | 21116     |                   | 191596    | 40751                | 65799     |                   | 192625    | na                   | na        |                   | na        |
| Precalculus <sup>1</sup>                               | 6764                  | 4041      |                   | 122177    | 42566                | 36985     |                   | 80504     | na                   | na        |                   | na        |
| Calculus I <sup>2</sup>                                | 4162                  | 8083      |                   | 324911    | 12673                | 15700     |                   | 93823     | na                   | na        |                   | na        |
| Statistics                                             | 6038                  | 4420      |                   | 150756    | 22835                | 22204     |                   | 257418    | 38                   | 1058      |                   | 91366     |
| Other courses                                          | 5165                  | 2529      |                   | na        | 22078                | 21193     |                   | na        | na                   | na        |                   | na        |

<sup>1</sup> Precalculus, Elementary Functions, and Analytic Geometry are included in the "Two-Year Mathematics" column.

<sup>2</sup> The question on dual enrollments did not differentiate between Mainstream and Non-Mainstream Calculus. To provide comparable data, the column for "Other Enrollments" also combines Mainstream and Non-Mainstream calculus even though separate statistics are shown elsewhere in this report.

**TABLE SP.17.** Percentage of departments in four-year colleges and universities and in public two-year colleges that assign their own full-time or part-time faculty members to teach (in high school courses that award both high school and college credit) and number of students enrolled in fall 2021. (Fall 2015 data in parentheses.)

|                                                              | Four-Year<br>Mathematics<br>Departments | Two-Year<br>Mathematics<br>Departments | Statistics<br>Departments |
|--------------------------------------------------------------|-----------------------------------------|----------------------------------------|---------------------------|
| Assign their own members to teach<br>dual-enrollment courses | 6<br>(6)                                | 34<br>(44)                             | 0                         |
| Number of students enrolled                                  | 3717<br>(4,014)                         | 9687<br>(*)                            | (0)                       |

\* The estimate of 36,368 from the data shows very large SEs. The only clear finding is that there has been a large increase in this practice, but not necessarily as large as the estimate indicates.

The 2021 CBMS survey showed that the estimated percentages of four-year mathematics departments and statistics departments offering dual-enrollment courses remained about the same. In fall 2021, two-year colleges experienced an increase to an estimate of 76% (SE 6) to a total of 161,881 students, up from 63% in fall 2015. Dual-enrollment increases were noted in every course category in 2021. Spring and fall 2021 (combined) enrolled a total of 302,784 students in two-year college mathematics dual-enrollment courses. This represented a 77% increase from spring and fall 2015 and an increase of 131,816 students. See Tables SP.16 and SP.17 and Table TYE.2 in Chapter 6.

According to Table SP.16, the estimated number of dual enrollments offered through statistics departments in fall 2021 was about the same as the estimated number of dual enrollments in fall 2021, but the estimated number of spring 2021 dual enrollments offered through statistics departments was lower than the estimated number of dual enrollments in spring 2015. Except for dual enrollments in statistics courses, the estimated numbers of dual enrollments offered through four-year mathematics departments in 2021 were much lower than the numbers observed in 2015; e.g., dual enrollments in “Precalculus” (which, for four-year mathematics departments, also includes Analytic Geometry and Elementary Functions) were estimated at 15,702 in fall 2015 and 4040 (SE 2082) in fall 2021, though the SEs are large, making comparisons difficult.

Table SP.17 examines the practice of colleges and universities sending their own faculty members into high schools to teach courses that grant both high school and college credit through their institution; this practice differs from dual-enrollment courses, where the instructor is a high school teacher. Table SP.17 gives the percentages of four-year mathematics departments, statistics departments, and two-year mathematics departments who participated in this practice and the estimated number of students enrolled in fall 2021 with comparative data for fall 2015.

The estimated numbers of students involved in these courses have been smaller than the numbers of enrollments in dual-enrollment courses. Table SP.17 shows that an estimated less than 1% of statistics departments and only 6% (SE 2.3) of four-year mathematics departments participated in this practice. According to data from CBMS2010 and CBMS2015, the percentage of two-year mathematics departments that assigned their own faculty to teach in a high school for dual credit was 22% in fall 2010 and 44% in fall 2015. The 2021 estimate is 34% (SE 7) and accounts for almost 10,000 enrollments with an SE of almost 3000. Table TYF.1 in Chapter 7 shows that 4716 (SE 969) part-time faculty were paid by a third party (e.g., a high school faculty member teaching a dual-enrollment course for college credit), a 99.9% increase from 2015 to 2021.



### Tables SP.18–SP.19.B: Curricular Requirements of Mathematics and Statistics Majors in Four-Year Departments

Requirements for a major in mathematics have become more flexible, as can be seen, for example, in the MAA's Committee on Undergraduate Programs in Mathematics (CUPM) recommendations on requirements for the mathematics major: <https://maa.org/resource/cupm-guide/>.

Many departments have several tracks (sets of graduation requirements) and more flexible requirements for mathematics majors than in the past. The CBMS 2005 and 2010 surveys asked about these requirements, and some of these questions were repeated in the 2015 and 2021 surveys. Table SP.18 summarizes data from four-year mathematics departments, broken down by level of department, on whether each course option was required in all the department's majors, required in some, but not all, of their majors, or required in none of their majors; these numbers are broken down by the level of the department. The SEs on these percentages are rather large, indicating there is variability in the requirements, even when broken down by the level of the department. Table SP.18 can be compared to CBMS2010, Table SP.20, p. 67 and CBMS2015, Table SP.18, p. 67.

Table SP.18 shows that in fall 2021, the three courses listed that are required in all majors by the highest estimated percentages of doctoral-level mathematics departments were Real Analysis I (required by 53% (SE 10.2) of doctoral-level departments), at least one computer science course (required by 52% (SE 10.2) of doctoral-level departments), and Modern Algebra I (required by 51% (SE 10.2) of doctoral-level departments); in the 2015 CBMS survey, the analogous estimates were 55% for at least one computer science course, 34% for Modern Algebra, and 31% for Real Analysis I. In 2015, an estimated 48% of doctoral-level mathematics departments required a one-year upper-level sequence, and in the 2021 survey, this percentage was estimated as 32% (SE 9.4).

Master's- and bachelor's-level mathematics departments seem to favor more flexible requirements. The three courses listed that were required in all majors by the highest estimated percentages of master's-level mathematics departments in fall 2021, as shown in Table SP.18, were at least one statistics course (required by 80% (SE 9.9) of master's-level departments), a capstone experience (required by 63% (SE 12.8) of master's-level departments), and at least one computer science course (required by 54% (SE 14.3) of master's-level departments); in the 2015 CBMS survey, the analogous estimates were a capstone experience at 68%, at least one computer science course at 67%, Real Analysis I at

49%, and at least one statistics course at 46%. In fall 2021, Modern Algebra and Real Analysis were required in no major at an estimated 18% (SE 13.8) and 28% (SE 14.2), respectively, of master's-level mathematics departments.

For bachelor's-level mathematics departments, the three courses listed that were required in all majors by the highest estimated percentages of bachelor's-level mathematics departments in fall 2021 were a capstone experience (required by 80% (SE 6.3) of bachelor's-level departments), at least one computer science course (required by 64% (SE 8.7) of bachelor's-level departments), and at least one statistics course (required by 51% (SE 7.6) of bachelor's-level departments); in the 2015 CBMS survey, the analogous estimates were a capstone experience at 76%, at least one computer science course at 69%, Real Analysis I at 49%, and at least one statistics course at 59%. In fall 2021, Modern Algebra and Real Analysis were required in no major at an estimated 19% (SE 6.6) and 31% (SE 7.8), respectively, of bachelor's-level mathematics departments.

Tables SP.19.A and SP.19.B consider the requirements in fall 2021 for a major in statistics, awarded through a mathematics department (Table SP.19.A) and through a statistics department (Table SP.19.B), in both cases broken down by the level of the department. The SEs in Table SP.19.A for four-year mathematics departments and in Table SP.19.B for master's-level statistics departments are large, and hence indicating there is a large variety of requirements. The comparable data from the 2015 CBMS survey can be found in CBMS2015, Tables SP.19A and SP.19.B, pp. 68–9.

In terms of the mathematics courses required for statistics majors, according to Table SP.19.A in fall 2021, all doctoral-level mathematics departments required Calculus I and II, Multivariable Calculus, and Linear Algebra for all their statistics majors. Bachelor's-level mathematics departments almost entirely required Calculus I and II and Linear Algebra for all their statistics majors, and an estimated 69% (SE 24.6) of bachelor's-level mathematics departments required Multivariable Calculus. All master's-level mathematics departments required Calculus I and II, 64% (SE 43.3) required Linear Algebra, and 64% (SE 43.3) required Multivariable Calculus for all their majors. Table SP.19.B, summarizing requirements for statistics majors in statistics departments, shows that Calculus I and II, Linear Algebra, and Multivariable Calculus were required for all their majors by 91% (SE 5.1), 88% (SE 5.8), 79% (SE 7.3), and 69% (SE 8.4), respectively, of doctoral-level statistics departments.

Beyond mathematics requirements for a statistic major, according to Table SP.19.B, in doctoral-level statistics departments in fall 2021, the estimated percentages of departments requiring the following courses for all majors were: a computer science course at 81% (SE 7.1),

**TABLE SP.18.** Percentage of four-year mathematics departments requiring certain courses (or exit exam) in all, some, or none of their majors, by type of department, in fall 2021.

|                                                                                           | Required in All Majors |                   |                      | Required in Some but Not All Majors |                   |                      | Not Required in Any Major |                   |                      |
|-------------------------------------------------------------------------------------------|------------------------|-------------------|----------------------|-------------------------------------|-------------------|----------------------|---------------------------|-------------------|----------------------|
|                                                                                           | Univ<br>(PhD)<br>%     | Univ<br>(MA)<br>% | College<br>(BA)<br>% | Univ<br>(PhD)<br>%                  | Univ<br>(MA)<br>% | College<br>(BA)<br>% | Univ<br>(PhD)<br>%        | Univ<br>(MA)<br>% | College<br>(BA)<br>% |
| Mathematics Department Requirements                                                       |                        |                   |                      |                                     |                   |                      |                           |                   |                      |
| Modern Algebra I                                                                          | 51                     | 46                | 42                   | 45                                  | 36                | 39                   | 4                         | 18                | 19                   |
| Real Analysis I                                                                           | 53                     | 33                | 28                   | 43                                  | 39                | 41                   | 4                         | 28                | 31                   |
| Modern Algebra I or Real Analysis I (major may choose either to fulfill this requirement) | 28                     | 15                | 21                   | 18                                  | 32                | 20                   | 54                        | 53                | 59                   |
| A one-year upper-level sequence                                                           | 32                     | 24                | 38                   | 29                                  | 4                 | 27                   | 39                        | 71                | 36                   |
| At least one computer science course                                                      | 52                     | 54                | 64                   | 24                                  | 13                | 12                   | 24                        | 33                | 24                   |
| At least one statistics course                                                            | 32                     | 80                | 51                   | 31                                  | 14                | 38                   | 37                        | 6                 | 11                   |
| At least one applied mathematics course beyond course E21                                 | 17                     | 30                | 44                   | 42                                  | 40                | 28                   | 41                        | 30                | 28                   |
| A capstone experience (senior project, thesis, seminar, internship)                       | 36                     | 63                | 80                   | 23                                  | 15                | 5                    | 41                        | 21                | 15                   |
| An exit exam (written or oral)                                                            | 4                      | 4                 | 21                   | 13                                  | 11                | 5                    | 84                        | 84                | 74                   |

**TABLE SP.19.A.** Percentage of mathematics departments that offer a major in statistics requiring certain courses (or exit exam) in all, some, or none of their majors, by type of department, in fall 2021.

| Percentage of Mathematics Departments that Require:                                  | Required in All Majors |             |                | Required in Some but Not All Majors |             |                | Not Required in Any Major |             |                |
|--------------------------------------------------------------------------------------|------------------------|-------------|----------------|-------------------------------------|-------------|----------------|---------------------------|-------------|----------------|
|                                                                                      | Univ (PhD) %           | Univ (MA) % | College (BA) % | Univ (PhD) %                        | Univ (MA) % | College (BA) % | Univ (PhD) %              | Univ (MA) % | College (BA) % |
| (a) Calculus I                                                                       | 100                    | 100         | 100            |                                     |             |                |                           |             |                |
| (b) Calculus II                                                                      | 100                    | 100         | 91             |                                     |             | 9              |                           |             |                |
| (c) Multivariable Calculus                                                           | 100                    | 64          | 69             |                                     |             | 9              |                           | 36          | 22             |
| (d) Linear Algebra/Matrix Theory                                                     | 100                    | 64          | 91             |                                     | 36          | 9              |                           |             |                |
| (e) At least one computer science course                                             | 72                     | 36          | 70             | 28                                  | 36          |                |                           | 29          | 30             |
| (f) At least one applied mathematics course, not incl. (a), (b), (c), (d)            | 22                     | 36          | 52             |                                     |             |                | 78                        | 64          | 48             |
| (g) A capstone experience (e.g., a senior thesis or project, seminar, or internship) | 28                     | 64          | 88             | 28                                  |             |                | 44                        | 36          | 12             |
| (h) An exit exam (oral or written)                                                   |                        |             | 22             |                                     |             |                | 100                       | 100         | 78             |
| (i) One Probability course                                                           | 72                     | 71          | 75             | 28                                  | 29          | 3              |                           |             | 22             |
| (j) One Mathematical Statistics course                                               | 72                     | 100         | 69             | 28                                  |             | 9              |                           |             | 22             |
| (k) One applied statistics course                                                    | 72                     | 100         | 100            | 28                                  |             |                |                           |             |                |
| (l) One Linear Models course                                                         | 28                     | 100         | 39             | 22                                  |             | 3              | 50                        |             | 58             |
| (m) One Bayesian Inference course                                                    | 50                     |             |                |                                     | 29          | 12             | 50                        | 71          | 88             |

**TABLE SP.19.B.** Percentage of statistics departments requiring certain courses (or exit exam) in all, some, or none of their majors, by type of department, in fall 2021.

| Percentage of Statistics Departments that Require:                                   | Required in All Majors |             | Required in Some but Not All Majors |             | Not Required in Any Major |             |
|--------------------------------------------------------------------------------------|------------------------|-------------|-------------------------------------|-------------|---------------------------|-------------|
|                                                                                      | Univ (PhD) %           | Univ (MA) % | Univ (PhD) %                        | Univ (MA) % | Univ (PhD) %              | Univ (MA) % |
| (a) Calculus I                                                                       | 91                     | 67          | 3                                   | 33          | 6                         |             |
| (b) Calculus II                                                                      | 88                     | 67          | 6                                   | 33          | 6                         |             |
| (c) Multivariable Calculus                                                           | 69                     | 40          | 13                                  | 20          | 19                        | 40          |
| (d) Linear Algebra/Matrix Theory                                                     | 79                     | 60          | 9                                   |             | 12                        | 40          |
| (e) At least one computer science course                                             | 81                     | 80          | 9                                   | 20          | 9                         |             |
| (f) At least one applied mathematics course, not incl. (a), (b), (c), (d)            | 16                     | 20          | 22                                  |             | 63                        | 80          |
| (g) A capstone experience (e.g., a senior thesis or project, seminar, or internship) | 42                     | 40          | 12                                  | 20          | 45                        | 40          |
| (h) An exit exam (oral or written)                                                   |                        | 20          |                                     | 20          | 100                       | 60          |
| (i) One Probability course                                                           | 76                     | 75          | 9                                   |             | 15                        | 25          |
| (j) One Mathematical Statistics course                                               | 64                     | 80          | 21                                  |             | 15                        | 20          |
| (k) One applied statistics course                                                    | 76                     | 100         | 12                                  |             | 12                        |             |
| (l) One Linear Models course                                                         | 58                     | 25          | 15                                  |             | 27                        | 75          |
| (m) One Bayesian Inference course                                                    | 10                     | 20          | 13                                  |             | 77                        | 80          |

Note: Blank entries represent courses with no responses, while 0 entries indicate percentages that round to 0%.

a probability course at 76% (SE 7.6), an applied statistics course at 76% (SE 7.6), a mathematical statistics course at 64% (SE 8.6), a linear models course at 58% (SE 8.8), and a capstone experience at 42% (SE 8.8). These percentages observed in the CBMS 2021 survey are close to the estimates observed in the CBMS 2015 survey, except for a required mathematical statistics course, which in fall 2015, was required for all their statistics majors by an estimated 89% of doctoral-level statistics departments.

According to Table SP.19.A, in addition to the mathematics courses noted above, 72% (SE 32) of doctoral-level mathematics departments and all master's- and bachelor's-level mathematics departments required one applied statistics course for all their statistics majors. One mathematical statistics course was required for all majors by 72% (34 SE) of doctoral-level mathematics departments, by 100% of master's-level mathematics departments, and by 69% (22.6 SE) of bachelor's-level mathematics departments. One probability course was required for all majors by 72% (SE 32.0) of doctoral-level mathematics departments, by 71% (SE 36.3) of master's-level mathematics departments, and by 75% (SE 23.8) of bachelor's-level mathematics departments. These percentages are comparable to those of doctoral-level statistics departments. According to Table SP.19.B, doctoral-level statistics departments were more likely to require a course in linear models or a Bayesian statistics course than mathematics departments.

### **Tables SP.20 and SP.21: Availability of Upper-level Courses in Mathematics and Statistics at Four-Year Departments**

Table SP.20 presents the estimated percentages of four-year mathematics departments that offered specific mathematics courses at least once during the two academic years 2020–21 and 2021–22; the 2020–22 data is broken down by level of mathematics department and can be compared to data from CBMS2015, Table SP.20, p. 70–71 and CBMS2010, Table SP.23, p. 70–71. Generally, there was less availability of upper-level mathematics courses in 2020–22 than in 2014–16, and the availability in 2014–16 was slightly less than in 2009–11. We note that 2020–22 involved semesters where some of the lack of availability could be due to disruptions caused by the COVID-19 pandemic. According to Table S.2 in Chapter 1, enrollments in upper-level mathematics courses across all four-year mathematics departments combined dropped from an estimated 154,000 in fall 2015 to 121,000 in fall 2021 (for more information on four-year department enrollments, see Chapter 3 and Appendix I).

According to Table SP.20, across all four-year mathematics departments combined during 2020–22, an estimated 61% (SE 5.5) of departments offered Geometry (compared to 71% in 2014–2016), 55% (SE 6) offered Introduction to Proofs (compared to 56% in 2014–2016), 53% (SE 5.6) offered Modern Algebra I (compared to 78% in 2014–16), and 50% (SE 5.4) offered Advanced Calculus/Real Analysis 1 (compared to 72% in 2014–16). The declining availability of Modern Algebra 1 and Advanced Calculus/Real Analysis 1 occurred primarily at the bachelor's-level mathematics departments. The availability in doctoral-level mathematics departments was generally higher than in master's-level mathematics departments, and the availability in master's-level mathematics departments was generally higher than in bachelor's-level mathematics departments.

Table SP.21 presents the estimated percentages of four-year mathematics departments combined and four-year statistics departments combined that offered specific statistics courses at least once during the two academic years 2020–21 and 2021–22; the 2020–22 data is broken down by level of mathematics department and level of statistics department and can be compared to data from CBMS2015, Table SP.21, p. 73 and CBMS2010, Table SP.24, p. 72, though the list of specific statistics courses has experienced some changes, making certain comparisons difficult.

The availability of upper-level statistics courses in mathematics departments was slightly less in 2020–22 than in 2014–16, and generally comparable in 2014–16 and 2009–11. The availability of upper-level statistics courses in statistics departments was generally less in 2020–22 than in 2014–16 but was generally greater than in 2014–16 and in 2009–11. According to Table S.2, enrollments in upper-level statistics courses across all four-year mathematics departments combined dropped from an estimated 60,000 in fall 2015 to 52,000 in fall 2021; enrollments in upper-level statistics courses across all four-year statistics departments combined dropped from an estimated 50,000 in fall 2015 to 36,000 in fall 2021 (for more information on four-year department enrollments see, Chapter 3 and Appendix I).

According to Table SP.21, across all four-year mathematics departments combined during 2020–22, an estimated 38% (SE 5.6) of mathematics departments offered mathematical statistics, 30% (SE 4.8) offered probability, 23% (SE 4.5) offered probability and statistics combined, and 17% (SE 3.8) offered data science (compared to 7% in 2014–16). Generally, there was greater availability of upper-level statistics courses at doctoral-level mathematics departments than at master's-level mathematics departments and greater availability at master's-level mathematics departments than at bachelor's-level mathematics departments.

**TABLE SP.20.** Percentage of four-year mathematics departments offering various upper-division mathematics courses at least once in the two academic years 2020–2021 and 2021–2022, plus historical data on the two-year period 2014–2016, by type of department.

| Upper-Level Mathematics Courses         | All Math Depts<br>2014–2016<br>% | Academic Years 2020–2021 & 2021–2022 |               |              |              |
|-----------------------------------------|----------------------------------|--------------------------------------|---------------|--------------|--------------|
|                                         |                                  | All Math Depts<br>2020–2022<br>%     | PhD Math<br>% | MA Math<br>% | BA Math<br>% |
| Modern Algebra I                        | 78                               | 53                                   | 78            | 65           | 47           |
| Modern Algebra II                       | 27                               | 21                                   | 60            | 28           | 14           |
| Number Theory                           | 37                               | 31                                   | 72            | 46           | 20           |
| Combinatorics                           | 22                               | 17                                   | 43            | 24           | 11           |
| Actuarial Mathematics                   | 21                               | 21                                   | 33            | 22           | 19           |
| Foundations/Logic                       | 12                               | 3                                    | 12            | 8            |              |
| Discrete Structures                     | 21                               | 7                                    | 19            | 7            | 5            |
| History of Mathematics                  | 47                               | 34                                   | 49            | 58           | 27           |
| Geometry                                | 71                               | 61                                   | 67            | 73           | 57           |
| Math for Secondary Teachers             | 33                               | 21                                   | 23            | 35           | 18           |
| Adv Calculus/ Real Analysis I           | 72                               | 50                                   | 74            | 71           | 42           |
| Adv Calculus/Real Analysis II           | 31                               | 25                                   | 65            | 30           | 17           |
| Adv Mathematics for Engineering/Physics | 12                               | 10                                   | 24            | 24           | 5            |
| Advanced Linear Algebra                 | 22                               | 24                                   | 54            | 32           | 17           |
| Introduction to Proofs                  | 56                               | 55                                   | 64            | 82           | 48           |

**TABLE SP.20 (continued).** Percentage of mathematics departments offering various upper-division mathematics courses at least once in the two academic years 2020–2021 and 2021–2022, plus historical data on the two-year period 2014–16, by type of department.

| Upper-Level Mathematics Courses, continued | All Math Depts 2014–2016 % | Academic Years 2020–2021 & 2021–2022 |            |           |           |
|--------------------------------------------|----------------------------|--------------------------------------|------------|-----------|-----------|
|                                            |                            | All Math Depts 2020–2022 %           | PhD Math % | MA Math % | BA Math % |
| Vector Analysis                            | 11                         | 5                                    | 15         | 10        | 2         |
| Advanced Differential Equations            | 16                         | 11                                   | 25         | 6         | 9         |
| Partial Differential Equations             | 29                         | 16                                   | 63         | 36        | 4         |
| Numerical Analysis I and II                | 43                         | 33                                   | 70         | 53        | 23        |
| Applied Math/Modeling                      | 36                         | 19                                   | 42         | 39        | 12        |
| Complex Variables                          | 43                         | 38                                   | 76         | 35        | 32        |
| Topology                                   | 28                         | 16                                   | 63         | 21        | 7         |
| Mathematics of Finance                     | 13                         | 9                                    | 28         | 14        | 4         |
| Codes & Cryptology                         | 11                         | 14                                   | 27         | 35        | 7         |
| Biomathematics                             | 8                          | 6                                    | 31         | 9         | 0         |
| Operations Research                        | 18                         | 10                                   | 27         | 13        | 7         |
| Math senior seminar/Ind study              | 66                         | 51                                   | 62         | 70        | 46        |
| All other advanced-level mathematics       | 25                         | 33                                   | 45         | 28        | 31        |

**TABLE SP.21.** Percentage of mathematics and statistics departments offering various undergraduate statistics courses at least once in the two academic years 2020–2021 and 2021–2022 and at least once in the two academic years 2014–2016, by type of department.

| Upper-Level Statistics Courses             | All Math Depts 2014–2016 % | Academic Years 2020–2021 & 2021–2022 |            |           |           | All Stat Depts 2014–2016 % | Academic Years 2020–2021 & 2021–2022 |            |           |
|--------------------------------------------|----------------------------|--------------------------------------|------------|-----------|-----------|----------------------------|--------------------------------------|------------|-----------|
|                                            |                            | All Math Depts %                     | PhD Math % | MA Math % | BA Math % |                            | All Stat Depts %                     | PhD Stat % | MA Stat % |
| Introductory Probability and/or Statistics | 18                         | na                                   | na         | na        | na        | 48                         | na                                   | na         | na        |
| Mathematical Statistics                    | 34                         | 38                                   | 56         | 48        | 33        | 73                         | 63                                   | 69         | 38        |
| Probability                                | 37                         | 30                                   | 60         | 39        | 24        | 70                         | 40                                   | 46         | 13        |
| Combined Probability and Statistics        | 32                         | 23                                   | 30         | 28        | 22        | 48                         | 47                                   | 46         | 50        |
| Stochastic Processes                       | 12                         | 6                                    | 35         | 13        | 0         | 49                         | 21                                   | 20         | 25        |
| Applied Statistical Analysis               | 12                         | na                                   | na         | na        | na        | 46                         | na                                   | na         | na        |
| Experimental Design                        | 9                          | 6                                    | 25         | 16        | 2         | 59                         | 49                                   | 43         | 75        |
| Applied Regression                         |                            | 9                                    | 19         | 14        | 6         |                            | 56                                   | 49         | 88        |
| Linear Models I                            |                            | 1                                    | 4          | 3         |           |                            | 21                                   | 23         | 13        |
| Linear Models II                           |                            | 1                                    |            | 5         |           |                            | 16                                   | 17         | 13        |
| Regression & Correlation                   | 15                         | na                                   | na         | na        | na        | 78                         | na                                   | na         | na        |
| Biostatistics                              | 7                          | 4                                    | 4          |           | 4         | 36                         | 12                                   | 9          | 25        |
| Nonparametric Statistics                   | 6                          | 1                                    | 4          |           | 1         | 44                         | 23                                   | 26         | 13        |
| Categorical Data Analysis                  | 4                          | 3                                    | 4          | 8         | 3         | 30                         | 23                                   | 20         | 38        |
| Sample Survey Design                       | 4                          | 2                                    | 4          | 3         | 1         | 50                         | 23                                   | 23         | 25        |
| Stat Software & Computing                  | 11                         | 4                                    | 19         | 13        | 1         | 62                         | 60                                   | 63         | 50        |
| Data Science                               | 7                          | 17                                   | 23         | 39        | 13        | 36                         | 53                                   | 51         | 63        |
| Bayesian Statistics                        | na                         | 3                                    |            | 4         | 3         | 47                         | 37                                   | 40         | 25        |
| Statistical Consulting                     | na                         | 1                                    |            |           | 1         | 34                         | 21                                   | 20         | 25        |
| Senior Seminar/Independent Study           | 9                          | 11                                   | 11         | 8         | 12        | 56                         | 33                                   | 34         | 25        |

Note: 0 means less than one half of 1%. Blank entries represent courses with no responses.

\* In 2010, this appeared as two separate items in the statistics questionnaire, with 41% reporting courses in statistical computing and 35% reporting courses in statistical software.



**TABLE SP.22.** Departmental estimates of the percentage of graduating mathematics or statistics majors from academic year 2020–2021 who had various post-graduation plans, by type of department, in fall 2021. (Fall 2015 data in parentheses.)

| Departmental Estimates of Post-College Plans                                           | Mathematics Departments |                |                   | Statistics Departments |                |
|----------------------------------------------------------------------------------------|-------------------------|----------------|-------------------|------------------------|----------------|
|                                                                                        | Univ (PhD)<br>%         | Univ (MA)<br>% | College (BA)<br>% | Univ (PhD)<br>%        | Univ (MA)<br>% |
| Students who went into pre-college teaching                                            | 11<br>(12)              | 21<br>(25)     | 23<br>(26)        | 1<br>(1)               | 1<br>(1)       |
| Students who went to graduate school in the mathematical or statistical sciences       | 14<br>(11)              | 18<br>(13)     | 15<br>(12)        | 10<br>(17)             | 7<br>(10)      |
| Students who went to graduate or professional school outside of mathematics/statistics | 9<br>(8)                | 1<br>(4)       | 9<br>(7)          | 7<br>(10)              | 6<br>(1)       |
| Students who took jobs in business, government, etc.                                   | 22<br>(27)              | 37<br>(19)     | 29<br>(34)        | 32<br>(34)             | 44<br>(20)     |
| Students who had other plans known to the department                                   | 2<br>(3)                | 0<br>(3)       | 2<br>(4)          | 1<br>(3)               | 2<br>(0)       |
| Students whose plans are not known to the department                                   | 42<br>(40)              | 23<br>(36)     | 21<br>(16)        | 48<br>(36)             | 41<br>(68)     |

According to Table SP.21, across all four-year statistics departments combined during 2020–22, an estimated 63% (SE 7.5) of statistics departments offered mathematical statistics, 60% (SE 7.6) offered statistical software and computing, 56% (SE 7.7) offered applied regression analysis, and 53% (SE 7.7) offered data science (compared to 36% in 2014–16). The percentages of departments offering upper-level courses at doctoral-level statistics departments and master’s-level statistics departments were about the same. As would be expected, the upper-level statistics courses were more likely to be offered in statistics departments than in mathematics departments.

### Table SP.22: Estimates of Post-Graduation Plans of Graduates of Four-Year Mathematics Departments and Statistics Departments

Table SP.22 presents estimates from four-year mathematics departments and statistics departments of the post-graduation plans of their 2020–21 graduating undergradu-

ate majors, broken down by the level of department. Many departments do not know the post-graduation plans of all their majors; among doctoral-level mathematics departments, an estimated 42% (SE 7.4) of graduates had plans unknown to the department. Of the plans of majors that are known to doctoral-level mathematics departments, the largest estimated percentage of 22% (SE 4.2) took jobs in business and industry, followed by 14% (SE 2.8) of their majors who attended graduate school in the mathematical or statistical sciences; the next largest estimated percent of 11% (SE 3) began careers in pre-college teaching, and finally the next largest percentage of 9% (SE 2.4) attended graduate or professional school outside of the mathematical or statistical sciences.

Master’s- and bachelor’s-level mathematics departments indicated that they knew more of the majors’ plans; according to Table SP.22, an estimated 23% (SE 9.0) of graduates from master’s-level departments had plans unknown to the department, and this percentage was estimated at 21% (SE 5.1) at bachelor’s-level departments.

The distributions of percentages of various plans of graduates of master's- and bachelor's-level mathematics departments that were known to the department are slightly different than the doctoral-level departments. According to Table SP.22, the largest estimated percentage of master's- and bachelor's-level mathematics department majors took jobs in business and industry, followed by majors who began careers in pre-college teaching; the next largest percent attended graduate school in the mathematical or statistical sciences, and finally the next largest percentage attended graduate or professional school outside of the mathematical or statistical sciences. These patterns are like those observed in the 2015 CBMS survey, except in the 2015 CBMS survey in the master's-level departments, the largest percent of their majors whose plans were known to the department began pre-college teaching careers.

Most of the majors from statistics departments whose plans were known to the department took jobs in business or government. The next largest percentage attended graduate school in the mathematical or statistical sciences, followed by attending graduate or professional school outside of the mathematical or statistical sciences; an estimated 1% began careers in pre-college teaching. The percentages reported in doctoral- and master's-level statistics departments were similar.

### **Table SP.23: Assessment Activities in Four-Year Mathematics Departments and Statistics Departments**

Table SP.23 gives the percentages of four-year mathematics and statistics departments that have used various methods to assess their programs in the last six years. The percentages are broken down by the level of the department; the SEs in the master's-level statistics program are large enough to make these estimates unreliable. Among all levels of four-year mathematics and statistics departments, the two assessment activities cited by the largest percentages of departments were assessment of teaching objectives and the development of student learning outcomes. Generally, the next highest percentages were for surveying program graduates and studying student progress in successive courses. Evaluating the use of placement tests was not used widely in statistics departments. Outside consultants were consulted in 51% (SE 9.7) of doctoral-level mathematics departments, 34% (SE 11) of master's-level mathematics departments, 37% (SE 6.8) of bachelor's-level mathematics departments, 29% (SE 8.4) of doctoral-level statistics departments, and 43% (SE 21.7) of master's-level statistics departments. Regardless of the methods of assessment used, an estimated 74% (SE 8.3) of four-year doctoral-level mathematics departments, 88% (SE 6.8) of four-year master's-

level mathematics departments, 88% (SE 5) of four-year bachelor's-level mathematics departments, 77% (SE 7.7) of four-year doctoral-level statistics departments, and 57% (SE 21.7) of four-year master's-level statistics departments indicated that assessment methods led to changes in their program.

### **Table SP.24: Institutional or Divisional Graduation Requirements Satisfied by Advanced Placement Courses in Four-Year Mathematics and Statistics Departments**

Table SP.24 summarizes the responses of four-year mathematics and statistics departments to the question of whether advanced placement courses (taken when in high school) could be used to meet their institutions' mathematical sciences divisional graduation requirements. An estimated 95% (SE 2.5) of all levels of mathematics departments combined indicated that in fall 2021, advanced placement (AP) courses could satisfy a divisional requirement, and as did an estimated 90% (SE 4.6) of all levels of statistics departments combined. These results are slightly higher than observed in the CBMS 2015 survey (see CBMS2015, Table SP.24, p. 77).

### **Table SP.25: Statistics Minors and Majors in Four-Year Mathematics Departments**

Table SP.25 gives the estimated percentages in fall 2021 of four-year mathematics departments offering a minor in statistics and gives an estimate of the number of students graduating between July 1, 2020–June 30, 2021 with a statistics minor awarded by a mathematics department. It also presents an estimate of the percentages of mathematics departments offering a major in statistics. All the data are broken down by level of the department. The SEs on these estimates are large enough to make the estimates unreliable. The comparable data from the CBMS 2015 survey is CBMS2015, Table SP.28, p. 79.

Table SP.25 gives an estimate of 26% (SE 4.7) of all four-year mathematics departments combined that, in fall 2021, offered a minor in statistics and of 12% (SE 3.8) that offered a major in statistics; the estimates in the 2015 survey were that 16% offered a minor and 10% offered a major. The estimated number of students graduating with a minor in statistics awarded by a mathematics department during July 1, 2020–June 30, 2021 was 4132 (SE 2409). By Table E.1.A in Chapter 3, an estimated 915 bachelor's degrees in statistics were awarded by mathematics departments in the period of July 1, 2020–June 30, 2021.

**TABLE SP.23.** Percentage of four-year mathematics and statistics departments undertaking various assessment activities during the last six years, by type of department, in fall 2021. (Fall 2015 data, when known, in parentheses.)

| Percentage Using Various Assessment Tools         | Four-Year Mathematics Departments |                |                   | Statistics Departments |                |
|---------------------------------------------------|-----------------------------------|----------------|-------------------|------------------------|----------------|
|                                                   | Univ (PhD)<br>%                   | Univ (MA)<br>% | College (BA)<br>% | Univ (PhD)<br>%        | Univ (MA)<br>% |
| Consult outside reviewers                         | 51<br>(36)                        | 34<br>(57)     | 37<br>(40)        | 29<br>(44)             | 43<br>(42)     |
| Survey program graduates                          | 70<br>(67)                        | 58<br>(83)     | 72<br>(59)        | 71<br>(70)             | 43<br>(67)     |
| Consult other departments                         | 33<br>(44)                        | 20<br>(42)     | 45<br>(38)        | 43<br>(46)             | 29<br>(17)     |
| Study data on students' progress in later courses | 61<br>(63)                        | 80<br>(77)     | 79<br>(62)        | 58<br>(21)             | 50<br>(33)     |
| Developed a set of student learning outcomes      | 74<br>na                          | 100<br>na      | 94<br>na          | 91<br>na               | 86<br>na       |
| Assessed teaching objectives                      | 80<br>(78)                        | 100<br>(81)    | 91<br>(85)        | 81<br>(98)             | 71<br>(67)     |
| Evaluate placement system                         | 52<br>(72)                        | 82<br>(52)     | 72<br>(57)        | 16<br>(18)             | 17<br>(25)     |
| Change undergraduate program due to assessment    | 74<br>(80)                        | 88<br>(76)     | 88<br>(70)        | 77<br>(76)             | 57<br>(75)     |

**TABLE SP.24.** Percentage of mathematics and statistics departments that allow a student to meet an institutional or divisional graduation requirement using an advanced placement course in fall 2021.

| Meets Requirement | All Math<br>Depts | PhD<br>Math | MA<br>Math | BA<br>Math | All Stat<br>Depts | PhD<br>Stat | MA<br>Stat |
|-------------------|-------------------|-------------|------------|------------|-------------------|-------------|------------|
| Yes (%)           | 95                | 89          | 96         | 96         | 90                | 97          | 63         |
| No (%)            | 5                 | 11          | 4          | 4          | 10                | 3           | 38         |

**TABLE SP.25.** Percentage of four-year mathematics departments offering a minor in statistics, the number of students graduating with such a minor between July 1, 2020 and June 30, 2021, and the percentage of four-year mathematics departments offering a major in statistics.

| Number of Tracks                | Mathematics Departments |           |              |       |
|---------------------------------|-------------------------|-----------|--------------|-------|
|                                 | Univ (PhD)              | Univ (MA) | College (BA) | Total |
| Offer a minor in statistics (%) | 29                      | 58        | 20           | 26    |
| Number of graduates             | 645                     | 3052      | 435          | 4132  |
| Offer a major in statistics (%) | 15                      | 13        | 12           | 12    |

Some totals are less than 100% due to round-off.

### Table SP.26: Departmental Activities in Response to Diversity, Equity, and Inclusion Issues

Table SP.26 summarizes responses from all three types of departments about the extent to which six possible departmental activities related to diversity, equity, and inclusion (DEI) issues had taken place in the department during the previous year; the possible responses for each activity were “none,” “some,” and “a lot” (and in the case of the activity involving graduate students, “not applicable”). The results are broken down by level of four-year mathematics department, both levels of statistics departments combined, and two-year mathematics departments.

The activity “faculty discussion designed to create awareness of DEI issues” appears to be the activity that was most likely to occur. Table SP.26 shows that it is the activity with generally the largest percentages of responses of “a lot”: the response of 23% (SE 5.4) of all levels of four-year mathematics departments combined, of 32% (SE 7.9) of both levels of statistics departments combined, and of 33% (SE 6.7) of two-year mathematics departments. This was also the activity with the lowest percentage of answers of “none.” The estimated percentage of departments answering “none” was 15% (SE 4.5) of all levels of four-year mathematics departments combined, 16% (SE 6.2) of both levels of statistics departments combined, and 14% (SE 5.2) of two-year mathematics departments.

The activity with generally the next largest responses of “a lot” and the next smallest percentages of responses of “none” was “consideration of existing or new programs to assist underrepresented groups and/or at-risk students in the mathematical sciences,” where the percentage of de-

partments answering “c lot” was 11% (SE 3.8) of all levels of four-year mathematics departments combined, 32% (SE 7.9) of both levels of statistics departments combined, and 17% (SE 5.1) of two-year mathematics departments. The estimated percentage of departments answering “none” was 35% (SE 6.3) of all levels of four-year mathematics departments combined, 16% (SE 6.2) of both levels of statistics departments combined, and 38% (SE 7.2) of two-year mathematics departments.

From the distributions of percentages in Table SP.26, the other activities listed, “student discussion designed to increase awareness of DEI issues,” “program or policy changes to affect the demographic balance of faculty in the mathematical sciences,” “program or policy changes to affect the demographic balance of undergraduate students in the mathematical sciences,” and “program or policy changes intended to affect the demographic balance of graduate students in the mathematical sciences” seemed to have received less attention in departments.

The percentage of departments responding “a lot” was higher for both levels of statistics departments combined than for all levels of four-year mathematics departments combined and for two-year mathematics departments for most of the issues.

### Tables SP.27 and SP.28: Profiles of Other Full-Time Faculty in Four-Year Mathematics and Statistics Departments

Concern has been voiced about the early-career profiles of individuals with PhDs in the mathematical sciences. Data on numbers of faculty obtained from the CBMS survey in

**TABLE SP.26.** Response to the question in fall 2021: Which of the following activities have taken place in the past year in your department in response to increased national attention to diversity, equity, and inclusion issues (DEI)? (Percentage of departments.)

|                                                                                                                                  |                | Mathematics Depts |           |              |       | Statistics Depts | Two-Year Colleges |
|----------------------------------------------------------------------------------------------------------------------------------|----------------|-------------------|-----------|--------------|-------|------------------|-------------------|
|                                                                                                                                  |                | Univ (PhD)        | Univ (MA) | College (BA) | Total | Total            | Total             |
| Faculty discussion designed to increase awareness of DEI issues                                                                  |                |                   |           |              |       |                  |                   |
|                                                                                                                                  | None           | 8                 | 8         | 17           | 15    | 16               | 14                |
|                                                                                                                                  | Some           | 63                | 73        | 61           | 63    | 51               | 53                |
|                                                                                                                                  | A lot          | 29                | 19        | 22           | 23    | 32               | 33                |
| Student discussion designed to increase awareness of DEI issues                                                                  |                |                   |           |              |       |                  |                   |
|                                                                                                                                  | None           | 33                | 23        | 41           | 38    | 28               | 38                |
|                                                                                                                                  | Some           | 61                | 65        | 56           | 58    | 44               | 58                |
|                                                                                                                                  | A lot          | 6                 | 12        | 2            | 4     | 28               | 4                 |
| Program or policy changes to affect the demographic balance of faculty in the mathematical sciences                              |                |                   |           |              |       |                  |                   |
|                                                                                                                                  | None           | 40                | 67        | 60           | 58    | 42               | 58                |
|                                                                                                                                  | Some           | 55                | 33        | 38           | 40    | 44               | 37                |
|                                                                                                                                  | A lot          | 4                 | 0         | 2            | 2     | 14               | 5                 |
| Program or policy changes to affect the demographic balance of undergraduate students in the mathematical sciences               |                |                   |           |              |       |                  |                   |
|                                                                                                                                  | None           | 51                | 65        | 50           | 52    | 42               | 58                |
|                                                                                                                                  | Some           | 49                | 35        | 37           | 38    | 44               | 39                |
|                                                                                                                                  | A lot          | 0                 | 0         | 13           | 10    | 14               | 3                 |
| Program or policy changes intended to affect the demographic balance of graduate students in the mathematical sciences           |                |                   |           |              |       |                  |                   |
|                                                                                                                                  | None           | 37                | 60        | 30           | 35    | 26               |                   |
|                                                                                                                                  | Some           | 59                | 25        | 7            | 17    | 43               |                   |
|                                                                                                                                  | A lot          | 4                 | 0         | 0            | 1     | 26               |                   |
|                                                                                                                                  | Not applicable | 0                 | 15        | 63           | 48    | 6                |                   |
| Consideration of existing or new programs to assist underrepresented groups and/or at-risk students in the mathematical sciences |                |                   |           |              |       |                  |                   |
|                                                                                                                                  | None           | 24                | 30        | 38           | 35    | 16               | 38                |
|                                                                                                                                  | Some           | 65                | 65        | 49           | 53    | 51               | 44                |
|                                                                                                                                  | A lot          | 11                | 4         | 13           | 11    | 32               | 17                |

fall 2021 are contained in Table S.15 in Chapter 1 and in the Chapter 4 tables (which are broken down by the level of the department). In the CBMS survey reports, full-time four-year faculty are classified as being tenured, tenure-eligible, or other full-time (a category that includes postdocs). By Table S.15, in the fall 2015 CBMS survey, the number of other full-time faculty at all levels of four-year mathematics departments combined was estimated at 32% of the number of full-time faculty, while in the fall 2021 survey, other full-time faculty represented an estimated 41% of full-time faculty. Indeed, the number of other full-time faculty in all levels of four-year mathematics departments combined in-

creased 80% from 2015 to 2021. At the same time, the percentage of full-time faculty who were tenure-eligible faculty (untentured faculty in tenure-track positions) dropped from 15% in 2015 to 13% in 2021.

Table S.15 also shows that at doctoral-level statistics departments, the number of other full-time faculty was estimated at 364 in fall 2015 and at 1366 in fall 2021, but in both years, other full-time faculty represented about 40% of the full-time faculty. The percentage of full-time faculty at doctoral-level statistics departments who were tenure-eligible was estimated at 25% in fall 2015 and 17% in fall 2015.

**TABLE SP.27.** Profile of non-tenure-track faculty with renewable appointments. This table can be compared with CBMS2015, Table SP.30, p. 82.

| Section B                                                   | Doctoral<br>Math | Master's<br>Math | Bachelor's<br>Math | All Math | Doctoral<br>Stat |
|-------------------------------------------------------------|------------------|------------------|--------------------|----------|------------------|
| Renewable positions filled for 2020–2021                    | 2218             | 1661             | 2965               | 6845     | 681              |
| Number that left renewable position for 2021                | 158              | 69               | 194                | 421      | 15               |
| Percent that left renewable position for 2021               | 7%               | 4%               | 7%                 | 6%       | 2%               |
| Renewable positions filled for 2021–2022                    | 2243             | 1723             | 2930               | 6896     | 700              |
| Number active in teaching                                   | 2243             | 1533             | 2669               | 6509     | 610              |
| Percent active in teaching                                  | 100%             | 89%              | 91%                | 94%      | 87%              |
| Number active in research                                   | 645              | 249              | 802                | 1696     | 409              |
| Percent active in research                                  | 29%              | 14%              | 27%                | 25%      | 58%              |
| Number that will attend a research conference with support  | 363              | 182              | 604                | 1150     | 334              |
| Percent that will attend a research conference with support | 16%              | 11%              | 21%                | 17%      | 48%              |
| Number that will attend a teaching conference with support  | 578              | 352              | 1258               | 2189     | 245              |
| Percent that will attend a teaching conference with support | 26%              | 20%              | 43%                | 32%      | 35%              |
| Number that will serve in department committees             | 1191             | 863              | 1904               | 3958     | 401              |
| Percent that will serve in department committees            | 53%              | 50%              | 65%                | 57%      | 57%              |
| Number that advise undergrad. research projects             | 305              | 121              | 705                | 1130     | 145              |
| Percent that advise undergrad. research projects            | 14%              | 7%               | 24%                | 16%      | 21%              |
| Number that serve as academic advisor                       | 602              | 420              | 1456               | 2478     | 267              |
| Percent that serve as academic advisor                      | 27%              | 24%              | 50%                | 36%      | 38%              |
| Number that serve on univ. committees                       | 525              | 416              | 1436               | 2377     | 227              |
| Percent that serve on univ. committees                      | 23%              | 24%              | 49%                | 34%      | 32%              |
| Number that serve as course coordinator                     | 871              | 404              | 1002               | 2277     | 108              |
| Percent that serve as course coordinator                    | 39%              | 23%              | 34%                | 33%      | 15%              |

There also has been interest in the responsibilities of both non-tenure-track faculty with renewable positions and those with fixed-term (non-renewable) positions. These issues are explored in Tables SP.27 and SP.28 using data obtained from the Annual Survey (administered by the American Mathematical Society), which contained two sets of questions that are included in the 2021 and 2015 CBMS surveys (see Appendix VI) to gain more information about other full-time faculty. These questions were part of the Department Profile Survey sent to all mathematics and statistics departments. Unlike in 2015 when these questions were sent as a stand-alone document, in 2021, these questions were placed at the end of a lengthy survey, and they were completed by only 23% of the respondents to the survey. This survey also asked for the same data on postdocs

that was gathered in the 2015 survey, but there were not enough responses to that portion of the survey to include in the 2021 report; hence a table comparable to CBMS2015, Table SP.29, p. 80 is not included in this report. Data from the questions related to the profiles of non-tenure-eligible faculty with renewable appointments are contained in Table SP.27, and data on the questions related to non-tenure-eligible faculty with fixed-term (non-renewable) appointments are reported in Table SP.28. The comparable tables in the 2015 CBMS survey are CBMS2015, Table SP.30, p. 82 and CBMS2015, Table SP.31, p. 84. We note that the response rate to these questions in 2021 was much less than in 2015, making comparisons between 2015 and 2021 somewhat problematic.

**TABLE SP.28.** Profile of non-tenure-track faculty. This table can be compared to CBMS2015, Table SP.31, p. 84.

|                                                             | Doctoral<br>Math | Master's<br>Math | Bachelor's<br>Math | All Math | Doctoral<br>Stat |
|-------------------------------------------------------------|------------------|------------------|--------------------|----------|------------------|
| Number of Fixed-term positions filled for 2019–2020         | 970              | 494              | 765                | 2229     | 198              |
| Number that left renewable position for 2020                | 98               | 49               | 107                | 254      | 8                |
| Percent that left renewable position for 2020               | 10%              | 10%              | 14%                | 11%      | 4%               |
| Number of Fixed-term positions filled for 2020–2021         | 984              | 551              | 849                | 2384     | 189              |
| Number active in teaching                                   | 525              | 484              | 759                | 1768     | 93               |
| Percent active in teaching                                  | 53%              | 88%              | 89%                | 74%      | 49%              |
| Number active in research                                   | 276              | 75               | 259                | 610      | 63               |
| Percent active in research                                  | 28%              | 14%              | 31%                | 26%      | 33%              |
| Number that will attend a research conference with support  | 163              | 67               | 239                | 469      | 70               |
| Percent that will attend a research conference with support | 17%              | 12%              | 28%                | 20%      | 37%              |
| Number that will attend a teaching conference with support  | 108              | 81               | 339                | 529      | 42               |
| Percent that will attend a teaching conference with support | 11%              | 15%              | 40%                | 22%      | 22%              |
| Number that will serve in department committees             | 166              | 196              | 432                | 794      | 48               |
| Percent that will serve in department committees            | 17%              | 36%              | 51%                | 33%      | 25%              |
| Number that advise undergrad. research projects             | 95               | 0                | 158                | 250      | 17               |
| Percent that advise undergrad. research projects            | 10%              | 0%               | 19%                | 10%      | 9%               |
| Number that serve as academic advisor                       | 122              | 79               | 285                | 485      | 25               |
| Percent that serve as academic advisor                      | 12%              | 14%              | 34%                | 20%      | 13%              |
| Number that serve on univ. committees                       | 75               | 84               | 255                | 415      | 21               |
| Percent that serve on univ. committees                      | 8%               | 15%              | 30%                | 17%      | 11%              |
| Number that serve as course coordinator                     | 108              | 66               | 159                | 334      | 13               |
| Percent that serve as course coordinator                    | 11%              | 12%              | 19%                | 14%      | 7%               |

The set of questions related to the career profile of faculty with renewable, but not tenure-eligible (and not postdoc), appointments refer to positions such as Lecturer, Teaching Professional, Professor of the Practice, Instructor, etc. at four-year mathematics departments and at doctoral-level statistics departments. Data was collected on the number of such positions in the academic years 2020–21 and 2021–22, the number of faculty in such positions in 2020–21 who left those positions after 2020–21, and the typical responsibilities of faculty in these positions. These data are given in Table SP.27 and are broken down by the level of the department; they may be compared to the data from the CBMS 2015 survey that are given in CBMS2015, Table SP.30, p. 82.

From Table SP.27, we see the numbers of faculty in non-tenure-track renewable appointments in the 2020–2021 academic year and, of those, the numbers of those faculty who were no longer in that position in fall 2021. By comparing Table SP.27 in this report to CBMS2015, Table SP.30, p. 82, we see that across all levels of four-year mathematics departments combined, the number of faculty in non-tenure-track renewable appointments increased from an estimated 4269 in fall 2014 to an estimated 6845 (SE 194) in fall 2020, a 60% increase; in doctoral statistics departments, the estimated number of faculty in non-tenure-track renewable appointments increased from an estimated 214 in fall 2014 to 681 (SE 33) in fall 2020, more than tripling. There were slight increases in the estimated number of

non-tenure-track renewable appointments from fall 2020 to fall 2021. Across all levels of mathematics departments combined, an estimated 6% of faculty in non-tenure-track renewable positions in fall 2020 left these positions by fall 2021 (compared to 17% of faculty in a non-tenure-track renewable position in 2014–15 who left the position by fall 2015). These percentages were about the same over all levels of mathematics and slightly lower (2%) in doctoral-level statistics departments.

The next set of questions concerned the typical responsibilities of non-tenure-track faculty in renewable positions; the responses to these questions are also summarized in Table SP.27 (with the 2015 responses in CBMS2015, Table SP.30, p. 82) and are broken down by level of the department. The estimated percentage of faculty in non-tenure-track renewable appointments involved in each of the activities listed in the 2021 CBMS survey was generally higher than the estimated percentage reported in the 2015 CBMS survey, perhaps indicating increased responsibilities generally expected of faculty in non-tenure-track renewable appointments. The percentage of non-tenure-track renewable faculty active in research across all levels of four-year mathematics departments combined in fall 2021 was estimated at 25% and in fall 2015 was estimated at 16%; in doctoral statistics departments, an estimated 58% of faculty in non-tenure-track renewable positions in fall 2021 were active in research, up from 36% in fall 2015. In fall 2021, the percentage of faculty with non-tenure-track renewable appointments who would attend a teaching conference with support was estimated at 32% across all levels of four-year mathematics departments combined and at 35% in doctoral-level statistics departments (but at 26% in doctoral-level mathematics departments). The percentage of faculty with non-tenure-track renewable appointments who would attend a research conference with support in 2021–22 was estimated at 17% across all levels of four-year mathematics departments combined and at 48% at doctoral-level statistics departments. The percentage of faculty in non-tenure-track renewable positions who would serve on department committees, advise undergraduate research projects, serve as an academic advisor, serve on university committees, or serve as a course coordinator were all higher at bachelor's-level mathematics departments than at doctoral-level mathematics departments.

The final set of questions dealt with the profile of faculty in fixed-term (non-renewable) appointments, and the same questions were asked about this group of faculty as were asked about faculty with non-tenure-track renewable appointments. The responses to these questions are summarized in Table SP.28, which is broken down by level of mathematics department and given for doctoral-level

statistics departments. The comparable data from the 2015 CBMS survey is given in CBMS2015, Table SP.31, p. 84.

From Table SP.28, we see that the number of full-time, fixed-term (non-renewable) appointments for 2020–21 was estimated at 2229 (SE 134) across all levels of four-year mathematics departments combined and at 198 (SE 26) at doctoral-level statistics departments; these numbers are much lower than, less than a third of, the numbers reported in Table SP.27 for renewable (but non-tenure-track) faculty. The estimated number of faculty across all levels of four-year mathematics departments combined with fixed-term appointments in 2020–21 reported in Table SP.28 represents a 52% increase over the fall 2015 estimates reported in CBMS2015, Table SP.31, p. 84, and the fall 2021 estimates were more than four times the 2015 estimates for numbers of faculty with fixed-term appointments in doctoral-level statistics departments. Even though these positions were non-renewable, the percentage of faculty leaving such a position in fall 2021 was estimated at 11% for all levels of four-year mathematics departments combined and at 4% for doctoral-level statistics departments; the comparable percentages observed in the 2015 CBMS survey were 30% for mathematics departments and 54% for doctoral-level statistics departments.

Comparing the responsibilities of faculty in fixed-term (non-renewable) positions to those in renewable (but non-tenure-track) positions, described in Tables SP.27 and SP.28, we see that in fall 2021, the percentage of faculty at all levels of four-year mathematics departments combined in fixed-term positions who were active in research was almost the same as that for faculty in renewable positions, but the percentage for doctoral-level statistics faculty was lower for faculty in fixed-term positions than for faculty in renewable positions. For each of the responsibilities mentioned in Tables SP.27 and SP.28, the percentage of faculty in fixed-term positions involved in the activity was less than the percentage of faculty with renewable-term positions; for example, the percentage of four-year mathematics faculty at all levels of departments combined in renewable-term positions who serve on a departmental committee in 2020–21 was 57%, and the percentage was 33% for faculty in non-renewable positions; similarly for doctoral statistics faculty, 57% of faculty in renewable-term positions serve on a committee, while 25% of faculty in non-renewable-term positions serve on a department committee.

The 2021 CBMS data shows that the use of other full-time faculty in both four-year mathematics and doctoral-level statistics departments continues to grow for non-tenure-track faculty in both renewable positions and in non-renewable positions, and these faculty seem to be taking on increasing responsibilities within departments.





## Chapter 3

# Mathematical Sciences Bachelor's Degrees and Enrollments in Four-Year Colleges and Universities

Mathematics and statistics departments in the nation's four-year colleges and universities offer a wide spectrum of undergraduate mathematical sciences courses and majors, sometimes including mathematics education, actuarial science, operations research, and computer science, as well as mathematics and statistics. The tables in this chapter describe:

- the number of bachelor's degrees awarded through the nation's mathematics and statistics departments (Tables E.1.A–E.1.B),
- enrollments in mathematical sciences courses and the numbers of mathematical sciences course sections (Tables E.2–E.3),
- distance/remote learning enrollments (Table E.4),
- the appointment type of instructors who teach undergraduate courses in mathematics and statistics departments (Table E.5–E.9), and
- the average sizes of sections of categories of courses taught in mathematics and statistics department and average sizes of recitation sections used in lecture/recitation classes for calculus and introductory statistics courses (Tables E.10–E.11).

These tables are broken down by the level of department based on the highest degree offered. The tables in this chapter expand upon Tables S.1–S.8 from Chapter 1, while Chapter 5 provides additional details about enrollments in first-year courses in mathematics and statistics. The enrollments in each course listed on the four-year mathematics and statistics questionnaires (both with, and without, distance learning enrollments) are given in Appendix I; in making comparisons to previous CBMS surveys, one should note that the appendix enrollments in CBMS reports prior to 2010 include distance learning enrollments. Enrollment data from two-year colleges appears in Chapter 6.

In the text that follows, the standard error (SE) in many of the estimates is provided along with the estimate (e.g., enrollment of 255,000 (SE 23,000)); the SEs for all CBMS2015 tables can be found in Appendix VII. The change in an estimate from an estimate in a previous survey is often expressed both as percentage change and as the number of SEs that change represents (e.g., “increased 21% (1.7 SEs)”).

## Highlights of Chapter 3

### A. Number of bachelor's degrees awarded.

- The estimated total number of mathematical sciences bachelor's degrees granted through four-year mathematics and statistics departments combined during the 2020–21 academic year was 22,562 (SE 1886), down from 26,234 in 2014–15 (14% (1.9 SEs) decrease) but above the 2009–10 estimate of 21,377. The 2014–15 estimate reversed the trend of decline in the estimated number of bachelor's degrees awarded by mathematics and statistics departments combined observed in the 1985–2010 CBMS surveys. The CBMS estimate was 27,928 bachelor's degrees granted during 1984–85. See Table S.3 in Chapter 1.
- The estimated total number of mathematical sciences bachelor's degrees granted through four-year mathematics departments during the 2020–21 academic year was 18,682 (SE 2472), down from 24,387 in 2014–15, a decrease of 23% (2.3 SEs). Most of the decline in the number of degrees awarded occurred in the master's- and bachelor's-level departments. See Table E.1.A and Figures E.1.1, E.1.3, and E.1.4.
- The estimated total number of mathematical sciences bachelor's degrees granted through statistics departments during the 2020–21 academic year was 3881 (SE 890), more than doubling the 2014–15 estimate of 1847. See Table E.1.B. The estimated total number of bachelor's degrees in statistics granted from mathematics departments during 2020–21 was 915 (with a large SE 424), up from 416 in 2014–15. See Table E.1.A.
- The estimated number of bachelor's degrees in each of mathematics education, actuarial mathematics, and computer science granted by mathematics departments declined from 2014–15 to 2020–21. See Table E.1.A and Figure E.1.2.
- The estimated percentage of the total number of bachelor's degrees awarded to women by mathematics departments declined from 42% in 2014–15 to 39% (SE 2%) in 2020–21. See Table E.1.A. The estimated percentage of the total number of bachelor's degrees awarded

to women by statistics departments decreased from 43% in 2014–15 to 40% (SE 2%) in 2020–21. See Table E.1.B.

## B. Enrollments and number of sections.

- The 2021 CBMS survey data show that the estimated four-year mathematics department enrollments in fall 2021 have fallen to levels comparable to the estimated enrollment levels observed in the CBMS surveys of 1990–2005. Estimated numbers of enrollments in every course category in Table E.2, for all levels of mathematics departments combined, are lower in fall 2021 than in fall 2015 and are lower in fall 2021 than in fall 2010 for every course category except upper-level statistics. Estimated total enrollments in all mathematics courses combined, over all levels of mathematics departments combined, were down 30% (5.8 SEs) from fall 2015 to fall 2021 and down 21% (3.7 SEs) from fall 2010 to fall 2021. From fall 2015 to fall 2021 in all levels of departments combined, estimated pre-college course enrollments were down 60% (9.5 SEs), introductory-level course enrollments were down 33% (5 SEs), calculus-level course enrollments were down 18% (2.3 SEs), and advanced-level mathematics course enrollments were down 21% (1.9 SEs). See Table S.2 in Chapter 1 or Table E.2.
- Estimated total fall 2021 enrollments in statistics courses taught in all levels of four-year mathematics departments combined were down 33% (4 SEs) over fall 2015. The decline in enrollments occurred mostly in lower-level courses. See Table S.2 in Chapter 1 or Table E.2.
- Estimated total fall 2021 computer science course enrollments taught in all levels of four-year mathematics departments combined were down 9% (0.3 SEs) from fall 2015 and increased 140% (0.9 SEs) in computer science courses taught in doctoral-level mathematics departments. See Table E.2.
- Estimated total fall 2021 enrollments in statistics courses taught in statistics departments were down 10% (1 SE) over fall 2015 estimated enrollments. The decline in enrollments occurred mostly in upper-level courses. See Table S.2 in Chapter 1 or Table E.2.
- The decline in estimated total enrollments from fall 2015 to fall 2021 at four-year mathematics departments occurred over each of the three levels of mathematics departments: total enrollments were down 26% (3 SEs) in bachelor's-level departments, 39% (3.6 SEs) in master's-level departments, and 28% (3.3 SEs) in doctoral-level departments. See Table E.2 and Figure E.2.3.
- The estimated number of sections of courses offered in mathematics and statistics departments in fall 2021 declined for each group of mathematics and statistics courses and each level of department (except lower-level statistics courses taught at master's-level statistics departments). The percent decline from fall 2015 to fall 2021 in the estimated number of sections for all levels of mathematics departments combined was smallest (18% (2.9 SEs)) for calculus-level courses and largest (56% (7.8 SEs)) in pre-college-level courses. There were some increases from fall 2015 to fall 2021 in estimated numbers of sections of computer science courses taught in mathematics departments; the SEs in these estimates were also large. See Table E.3.

## C. Distance/remote learning enrollments.

- In four-year mathematics departments, total distance/remote learning enrollments in precalculus, introductory-level courses, calculus I and II, differential equations, linear algebra, and introductory statistics courses in fall 2021 were estimated at 244,309, almost triple the fall 2015 estimate of 86,197. In fall 2021, distance/remote learning enrollments represented 17% (up from 4% in 2015) of the total enrollments in these courses. At two-year mathematics departments in fall 2021, the estimated distance/remote learning enrollments exceeded the non-distance/remote learning enrollments in each of these course categories. See Table E.4.
- In statistics departments, distance/remote learning enrollments in introductory statistics courses in fall 2021 were estimated at 28,218, almost seven times the fall 2015 estimate of 4291, and these enrollments represented 30% (up from 5% in 2015) of the estimated total enrollments in introductory statistics courses in fall 2021. See Table E.4.

## D. Appointment type of section instructor.

- In fall 2021, over all levels of mathematics departments combined, 55% of the estimated calculus-level course sections were taught by tenure-track/tenure-eligible faculty, 28% by other full-time faculty, 9% by part-time faculty, and 7% by graduate teaching assistants (in 2015, these percentages were 52%, 24%, 10%, and 7%, respectively). See Table E.5 and Figure E.5.1.
- In fall 2021, over all levels of mathematics departments combined, 43% of the estimated introductory statistics course sections were taught by tenure-track/tenure-eligible faculty, 28% by other full-time faculty, 25% by part-time faculty, and 2% by graduate teaching assistants (in 2015, these percentages were 41%, 21%, 25%,

and 4%, respectively, with 20% unknown appointment type). See Table E.6 and Figure E.6.1.

- In fall 2021, over all levels of statistics departments combined, 18% of the estimated introductory statistics course sections were taught by tenure-track/tenure-eligible faculty, 46% by other full-time faculty, 13% by part-time faculty, and 24% by graduate teaching assistants (in 2015, these percentages were 14%, 25%, 10%, and 31%, respectively, with 20% unknown appointment type). See Table E.6 and Figure E.6.1.
- In fall 2021, over all levels of mathematics departments combined, 82% of the estimated advanced-level mathematics course sections and 86% of all advanced-level statistics course sections were taught by tenured or tenure-eligible faculty, up from 70% and 53%, respectively, in fall 2015. See Table E.7.
- In fall 2021, over all levels of statistics departments combined, 56% of the estimated advanced-level statistics course sections were taught by tenured or tenure-eligible faculty, about the same percentage as in fall 2015. See Table E.7.
- In fall 2021, over all levels of mathematics departments combined, 66% of the estimated sections of lower-level computer science courses taught in mathematics departments were taught by tenured or tenure-eligible faculty, up from 46% in fall 2015. See Tables E.8 and E.9 and Figure E.9.1.

### E. Average section size.

- The estimated average section size of mathematics and statistics courses has remained relatively stable over the past 16 years but differs by the level of institution. In fall 2021, the estimated average section size of calculus-level courses was 54 (SE 6) at doctoral-level departments and 19 (SE 1) at bachelor's-level departments, and the estimated average section size of advanced courses was 26 (SE 4) at doctoral-level departments and 9 (SE 1) at bachelor's-level departments. See Table E.10.
- The estimated average section size of calculus and introductory statistics course recitation sections was generally higher, particularly at doctoral-level departments, in fall 2021 than in fall 2015. See Table E.11.

In the CBMS 2021 survey, the term “mathematics department” includes departments of mathematics, applied mathematics, mathematical sciences, and departments of mathematics and statistics. The term “statistics department” refers to departments of statistics that offer undergraduate statistics courses. The term “mathematical sciences courses” covers all courses that are taught in mathematics or statistics departments in the U.S.; it includes

courses in mathematics education, actuarial sciences, and operations research taught in a mathematics or statistics department as well as courses in mathematics, applied mathematics, and statistics. Computer science courses (and majors) are included in CBMS2021 totals when the courses (and majors) are taught (granted through) a mathematics department (previous CBMS surveys gathered data on computer science courses/majors offered through statistics departments, but these data were not collected beginning in 2010). CBMS2021 data do not include any courses or majors that are taught in, or granted through, separate departments of computer science, actuarial science, operations research, etc. Departments are classified by the highest degree offered. For example, the term “bachelor's-level department” refers to one that does not offer master's or doctoral degrees.

### Tables E.1.A and E.1.B: Bachelor's Degrees Granted Between July 1, 2020 and June 30, 2021

**Total numbers of degrees awarded by mathematics and statistics departments.** The CBMS survey estimates of the total numbers of degrees awarded by four-year mathematics and statistics departments combined over the past academic year are given in Table S.3 of Chapter 1, where the data are broken down by type of major for the CBMS surveys of 1995, 2000, 2005, 2010, 2015, and 2021; the estimated number of women mathematics and statistics majors combined, of computer science majors and of total majors, are also presented. Tables E.1.A and E.1.B in this chapter give the estimated total numbers of degrees awarded by mathematics and statistics departments, respectively, from July 1, 2020–June 30, 2021, broken down by the highest degree granted by the department; the percentage of women majors for each type of major is also given.

The CBMS 2021 survey (see Table S.3 and Figure S.3.1 of Chapter 1) estimated that the total number of mathematical sciences bachelor's degrees granted through the nation's four-year mathematics and statistics departments in the 2020–21 academic year was 22,562 (SE 1886), down from 26,234 in 2014–15 (14% (1.9 SEs) decrease) but above the 2009–10 estimate of 21,377. The 2014–15 estimate reversed the trend of decline in the estimated number of bachelor's degrees awarded by mathematics and statistics departments combined observed in the 1985–2010 CBMS surveys. The CBMS survey of 1985 estimated that 27,928 bachelor's degrees were granted during 1984–85. When computer science degrees are removed from the degree count, the estimated number of degrees awarded by mathematics and statistics departments appeared relatively constant in

**TABLE E.1.A.** Bachelor's degrees in mathematics, mathematics education, statistics, and computer science in mathematics departments awarded between July 1, 2020 and June 30, 2021, by gender of degree recipient and type of department.

| Bachelor's Degrees in Math Depts              | Mathematics Departments |              |              | Total Math Depts |
|-----------------------------------------------|-------------------------|--------------|--------------|------------------|
|                                               | Univ<br>(PhD)           | Univ<br>(MA) | Coll<br>(BA) |                  |
| <b>Mathematics Majors (including applied)</b> |                         |              |              |                  |
| Men                                           | 3502                    | 1249         | 2201         | 6952             |
| Women                                         | 1808                    | 684          | 1857         | 4349             |
| Nonbinary                                     | 339                     | 15           | 46           | 400              |
| <i>Percentage of women</i>                    | 32%                     | 35%          | 45%          | 37%              |
| <b>Total Math Degrees</b>                     | <b>5649</b>             | <b>1948</b>  | <b>4104</b>  | <b>11701</b>     |
| <b>Mathematics Education Majors</b>           |                         |              |              |                  |
| Men                                           | 457                     | 51           | 269          | 777              |
| Women                                         | 592                     | 117          | 507          | 1216             |
| Nonbinary                                     | 0                       | 0            | 0            | 0                |
| <i>Percentage of women</i>                    | 56%                     | 70%          | 65%          | 61%              |
| <b>Total Math Ed Degrees</b>                  | <b>1049</b>             | <b>168</b>   | <b>776</b>   | <b>1993</b>      |
| <b>Statistics Majors</b>                      |                         |              |              |                  |
| Men                                           | 50                      | 57           | 243          | 350              |
| Women                                         | 39                      | 75           | 298          | 412              |
| Nonbinary                                     | 154                     | 0            | 0            | 154              |
| <i>Percentage of women</i>                    | 16%                     | 57%          | 55%          | 45%              |
| <b>Total Stat Degrees</b>                     | <b>243</b>              | <b>132</b>   | <b>541</b>   | <b>915</b>       |
| <b>Computer Science Majors</b>                |                         |              |              |                  |
| Men                                           | 11                      | 167          | 725          | 903              |
| Women                                         | 6                       | 61           | 184          | 251              |
| Nonbinary                                     | 0                       | 0            | 0            | 0                |
| <i>Percentage of women</i>                    | 33%                     | 27%          | 20%          | 22%              |
| <b>Total CS Degrees</b>                       | <b>17</b>               | <b>228</b>   | <b>909</b>   | <b>1154</b>      |
| <b>Actuarial Mathematics Majors</b>           |                         |              |              |                  |
| Men                                           | 288                     | 16           | 153          | 456              |
| Women                                         | 177                     | 24           | 49           | 250              |
| Nonbinary                                     | 0                       | 0            | 0            | 0                |
| <i>Percentage of women</i>                    | 38%                     | 60%          | 24%          | 35%              |
| <b>Total Actuarial Math Degrees</b>           | <b>465</b>              | <b>39</b>    | <b>201</b>   | <b>706</b>       |
| <b>Joint Mathematics Majors</b>               |                         |              |              |                  |
| Men                                           | 427                     | 152          | 299          | 878              |
| Women                                         | 207                     | 73           | 343          | 623              |
| Nonbinary                                     | 42                      | 0            | 0            | 42               |
| <i>Percentage of women</i>                    | 38%                     | 60%          | 24%          | 35%              |
| <b>Total Joint Degrees</b>                    | <b>675</b>              | <b>225</b>   | <b>642</b>   | <b>1542</b>      |
| <b>Other Mathematics Majors</b>               |                         |              |              |                  |
| Men                                           | 93                      | 6            | 331          | 431              |
| Women                                         | 66                      | 13           | 161          | 240              |
| Nonbinary                                     | 0                       | 0            | 0            | 0                |
| <i>Percentage of women</i>                    | 41%                     | 67%          | 33%          | 36%              |
| <b>Total Other Math Degrees</b>               | <b>159</b>              | <b>19</b>    | <b>492</b>   | <b>670</b>       |
| <b>Total degrees - Men</b>                    | <b>4828</b>             | <b>1697</b>  | <b>4221</b>  | <b>10746</b>     |
| <b>Total degrees - Women</b>                  | <b>2895</b>             | <b>1046</b>  | <b>3398</b>  | <b>7340</b>      |
| <b>Nonbinary</b>                              | <b>534</b>              | <b>15</b>    | <b>46</b>    | <b>596</b>       |
| <i>Percentage of women</i>                    | 35%                     | 38%          | 44%          | 39%              |
| <b>Total All Degrees</b>                      | <b>8258</b>             | <b>2759</b>  | <b>7665</b>  | <b>18682</b>     |

Note: Round-off may make row and column sums seem inaccurate.

**TABLE E.1.B.** Bachelor's degrees in statistics departments awarded between July 1, 2020 and June 30, 2021, by gender of degree recipient and type of department.

| Bachelor's Degrees in Stat Depts                            | Statistics Departments |              |                  |
|-------------------------------------------------------------|------------------------|--------------|------------------|
|                                                             | Univ<br>(PhD)          | Univ<br>(MA) | Total Stat Depts |
| <b>Statistics Majors</b>                                    |                        |              |                  |
| Men                                                         | 1391                   | 299          | 1690             |
| Women                                                       | 983                    | 115          | 1098             |
| Nonbinary                                                   | 0                      | 0            | 0                |
| <i>Percentage of women</i>                                  | 41%                    | 28%          | 39%              |
| <b>Total Stat Degrees</b>                                   | <b>2373</b>            | <b>415</b>   | <b>2788</b>      |
| <b>Biostatistics Majors</b>                                 |                        |              |                  |
| Men                                                         | 0                      | 0            | 0                |
| Women                                                       | 0                      | 0            | 0                |
| Nonbinary                                                   | 0                      | 0            | 0                |
| <i>Percentage of women</i>                                  |                        |              |                  |
| <b>Total Biostat Degrees</b>                                | <b>0</b>               | <b>0</b>     | <b>0</b>         |
| <b>Actuarial Science Majors</b>                             |                        |              |                  |
| Men                                                         | 53                     | 26           | 79               |
| Women                                                       | 46                     | 14           | 60               |
| Nonbinary                                                   | 0                      | 0            | 0                |
| <i>Percentage of women</i>                                  | 47%                    | 35%          | 43%              |
| <b>Total Actuarial Science Degrees</b>                      | <b>100</b>             | <b>40</b>    | <b>139</b>       |
| <b>Data Science Majors</b>                                  |                        |              |                  |
| Men                                                         | 313                    | 3            | 317              |
| Women                                                       | 217                    | 2            | 219              |
| Nonbinary                                                   | 2                      | 0            | 2                |
| <i>Percentage of women</i>                                  | 41%                    | 33%          | 41%              |
| <b>Total Data Science Degrees</b>                           | <b>532</b>             | <b>5</b>     | <b>537</b>       |
| <b>Joint Statistics and Computer Science Majors</b>         |                        |              |                  |
| Men                                                         | 14                     | 0            | 14               |
| Women                                                       | 0                      | 0            | 0                |
| Nonbinary                                                   | 0                      | 0            | 0                |
| <i>Percentage of women</i>                                  | 0%                     |              | 0%               |
| <b>Total Joint Stat and CS degrees</b>                      | <b>14</b>              | <b>0</b>     | <b>14</b>        |
| <b>Joint Statistics and Mathematics Majors</b>              |                        |              |                  |
| Men                                                         | 210                    | 0            | 210              |
| Women                                                       | 158                    | 0            | 158              |
| Nonbinary                                                   | 0                      | 0            | 0                |
| <i>Percentage of women</i>                                  | 43%                    |              | 43%              |
| <b>Total Joint Stat and Math Degrees</b>                    | <b>368</b>             | <b>0</b>     | <b>368</b>       |
| <b>Joint Statistics and (Business or Economics) Majors</b>  |                        |              |                  |
| Men                                                         | 0                      | 0            | 0                |
| Women                                                       | 0                      | 0            | 0                |
| Nonbinary                                                   | 0                      | 0            | 0                |
| <i>Percentage of women</i>                                  |                        |              |                  |
| <b>Total Joint Stat and (Business or Economics) Degrees</b> | <b>0</b>               | <b>0</b>     | <b>0</b>         |
| <b>Statistics Education Majors</b>                          |                        |              |                  |
| Men                                                         | 0                      | 0            | 0                |
| Women                                                       | 0                      | 0            | 0                |
| Nonbinary                                                   | 0                      | 0            | 0                |
| <i>Percentage of women</i>                                  |                        |              |                  |
| <b>Total Stat Ed Degrees</b>                                | <b>0</b>               | <b>0</b>     | <b>0</b>         |
| <b>Other Statistics Majors</b>                              |                        |              |                  |
| Men                                                         | 17                     | 0            | 17               |
| Women                                                       | 17                     | 0            | 17               |
| Nonbinary                                                   | 0                      | 0            | 0                |
| <i>Percentage of women</i>                                  | 50%                    |              | 50%              |
| <b>Total Other Stat Degrees</b>                             | <b>34</b>              | <b>0</b>     | <b>34</b>        |
| <b>Total degrees - Men</b>                                  | <b>1998</b>            | <b>329</b>   | <b>2327</b>      |
| <b>Total degrees - Women</b>                                | <b>1421</b>            | <b>131</b>   | <b>1552</b>      |
| <b>Nonbinary</b>                                            | <b>2</b>               | <b>0</b>     | <b>2</b>         |
| <i>Percentage of women</i>                                  | 42%                    | 28%          | 40%              |
| <b>Total All Degrees</b>                                    | <b>3421</b>            | <b>459</b>   | <b>3881</b>      |

Note: Round-off may make row and column sums seem inaccurate. Blank entries represent courses with no responses.

past CBMS surveys: 19,237 in 1984–85 (the first year that computer science degrees were tabulated), 19,380 in 1989–90, 19,241 in 2009–10, 22,266 in 2014–15, and 20,408 in 2020–21 (see Table S.3 in the current report and CBMS2000, Table SE.4). As we will note below, the number of computer science degrees awarded by mathematics departments has continued to decrease (see Table S.3).

Table E.1.A presents the estimated number of bachelor's degrees awarded by mathematics departments from July 1, 2020–June 30, 2021, broken down by the level of the department and type of degree awarded (the subcategories of degree types are mathematics (including applied mathematics), mathematics education, statistics, actuarial mathematics, computer science, joint majors, and other degrees). Table E.1.B gives the estimated number of degrees awarded by statistics departments over that same period. The estimated total number of mathematical sciences bachelor's degrees granted through four-year mathematics departments during the 2020–21 academic year was 18,682 (SE 2472), down from 24,387 in 2014–15, a decrease of 23% (2.3 SEs). Most of the decline in the estimated number of degrees awarded occurred in the master's-level and bachelor's-level departments.

Table E.1.B shows that the estimated total number of mathematical sciences bachelor's degrees granted through statistics departments during the 2020–21 academic year was 3881 (SE 890), more than doubling the 2014–15 estimate of 1847. Table E.1.A shows that the estimated total number of bachelor's degrees in statistics granted from mathematics departments during 2020–21 was 915 (with a large SE 424), up from 416 in 2014–15.

**Degrees awarded by mathematics departments broken down by level of department.** Table E.1.A breaks down the estimated numbers of bachelor's degrees awarded in 2020–21 by the level of mathematics department awarding the degree. Comparing the total number of degrees awarded by each level of department, from Table E.1.A, we see that most of the decline in number of degrees awarded by mathematics departments in 2020–21 occurred at the bachelor's- and master's-level departments. In 2020–21, doctoral-level departments awarded an estimated 8258 degrees (SE 1487), while in 2014–15, the estimate was 8409 degrees. The master's-level departments awarded an estimated 2759 degrees (SE 638) in 2020–21 compared to 5383 degrees in 2014–15, and the bachelor's-level departments awarded an estimated 7665 degrees (SE 1818) in 2020–21 compared to 10,595 degrees in 2014–15.

In the 2005 and 2010 CBMS surveys, most of the growth in the number of bachelor's degrees awarded in mathematics occurred in the doctoral-level mathematics departments. In 2005, for the first time, the estimated number of bachelor's degrees in mathematics granted by doctoral-level

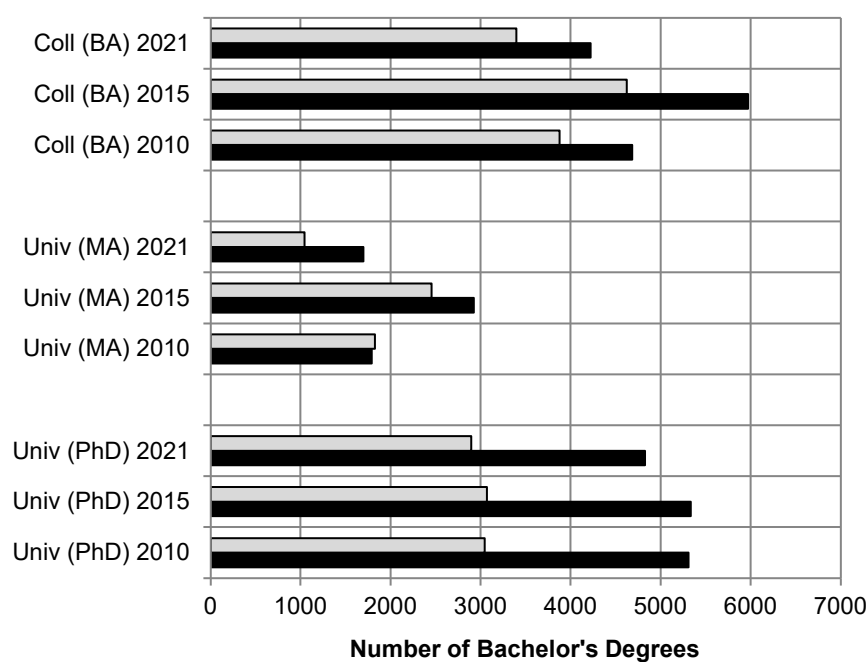
departments exceeded the number granted by bachelor's-level departments. In 2015, the largest growth in estimated degrees awarded occurred in the master's- and bachelor's-level departments, with bachelor's-level departments awarding more degrees total than doctoral-level departments, but when computer science degrees were removed, the situation was reversed. Figure E.1.1 displays the numbers of degrees, broken down by gender, awarded by each level of mathematics department in 2009–10, 2014–15, and 2020–21; Figures E.1.3 and E.1.4 display the percentage of mathematical science degrees awarded by each level of mathematics department and by statistics departments, with, and without, degrees in computer science awarded by mathematics departments included. In 2020–21, doctoral-level departments awarded 44% of all the estimated total degrees awarded by mathematics departments, and bachelor's-level departments awarded 41%; when computer science degrees awarded by mathematics departments were removed, doctoral-level departments awarded 47% of all the estimated degrees, and bachelor's-level departments awarded 39% of the degrees.

### Degrees awarded by mathematics departments broken down by type of major.

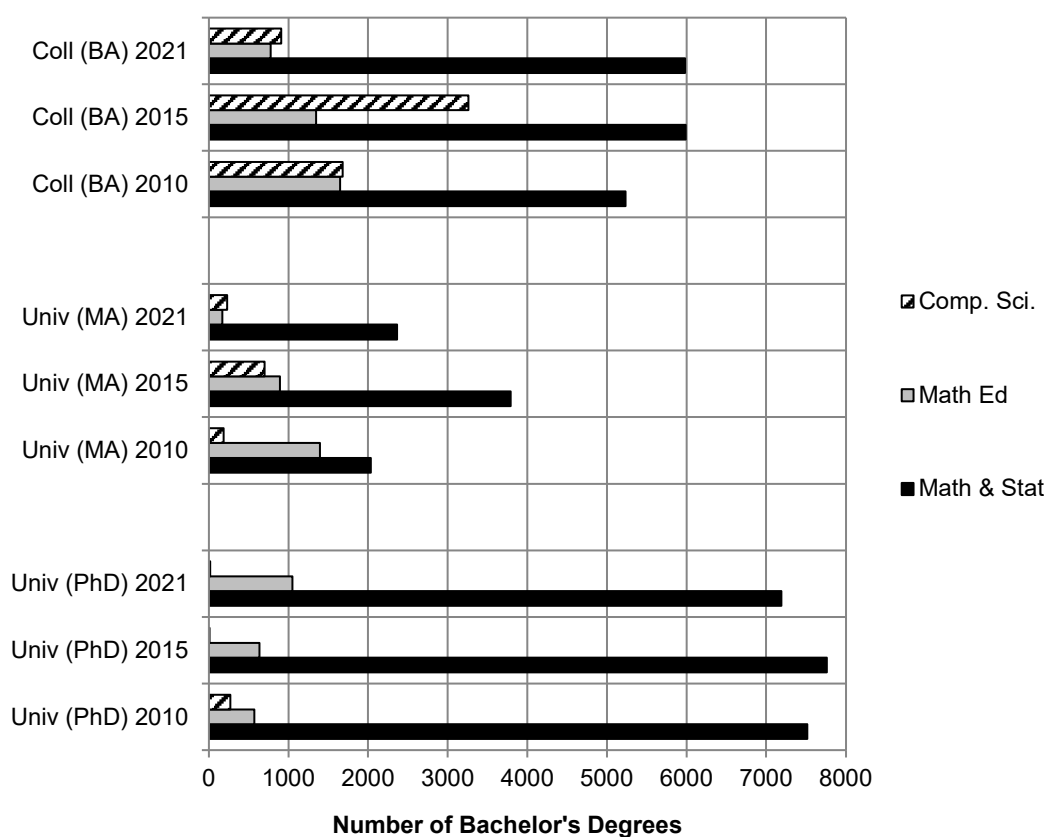
Table S.3 of Chapter 1 presents the estimated numbers of degrees awarded by mathematics and statistics departments combined, broken down by type of major for the CBMS surveys from 1994–95 through 2020–21. Table E.1.A shows the estimated number of degrees awarded by mathematics departments in 2020–21, broken down by the type of major and by the level of the department; Figure E.1.2 displays this breakdown for some of the types of majors awarded in 2009–10, 2014–15, and 2020–21, broken down by level of the department.

Table E.1.A shows that the estimated number of bachelor's degrees in the category “mathematics,” awarded in 2020–21 by all levels of mathematics departments combined, was 11,701 (SE 1581), a 9% (0.7 SE) decrease over 2014–15; Table S.3 of Chapter 1 shows that this is also a decrease over both the 2009–10 and 2004–05 estimates. Similarly, Table E.1.A shows that the estimated number of bachelor's degrees in the category “math education,” awarded in 2020–21 by all levels of mathematics departments combined, was 1993 (SE 726), a 31% (1.2 SE) decrease over 2014–15. Table S.3 of Chapter 1 shows that the number of degrees awarded in math education in 2020–21 is smaller than the estimated number of degrees awarded in each of the previous five CBMS surveys; the CBMS 1995 survey estimated that 4829 degrees were awarded in math education the previous year.

Table E.1.A shows that the estimated number of bachelor's degrees in the category of “statistics,” awarded in 2020–21 by all levels of mathematics departments combined, was 915 (SE 424), more than double the 2014–15 estimate. More

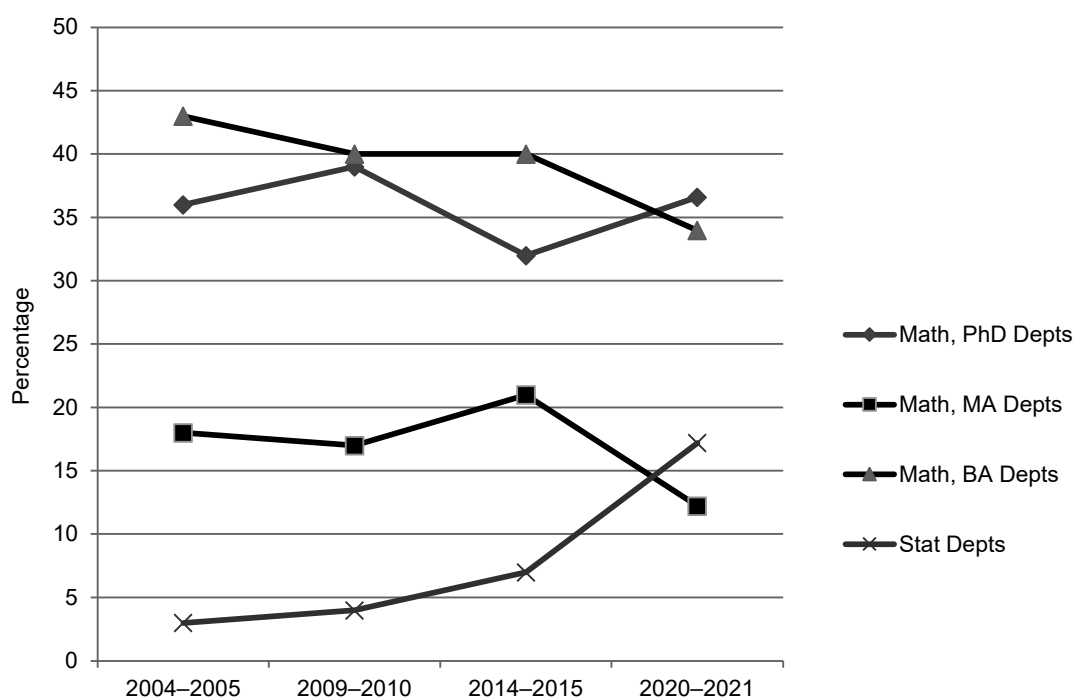


**FIGURE E.1.1.** Number of bachelor's degrees in mathematics departments awarded between July 1 and June 30 in the academic years 2009–2010, 2014–2015, and 2020–21, by gender and type of department.

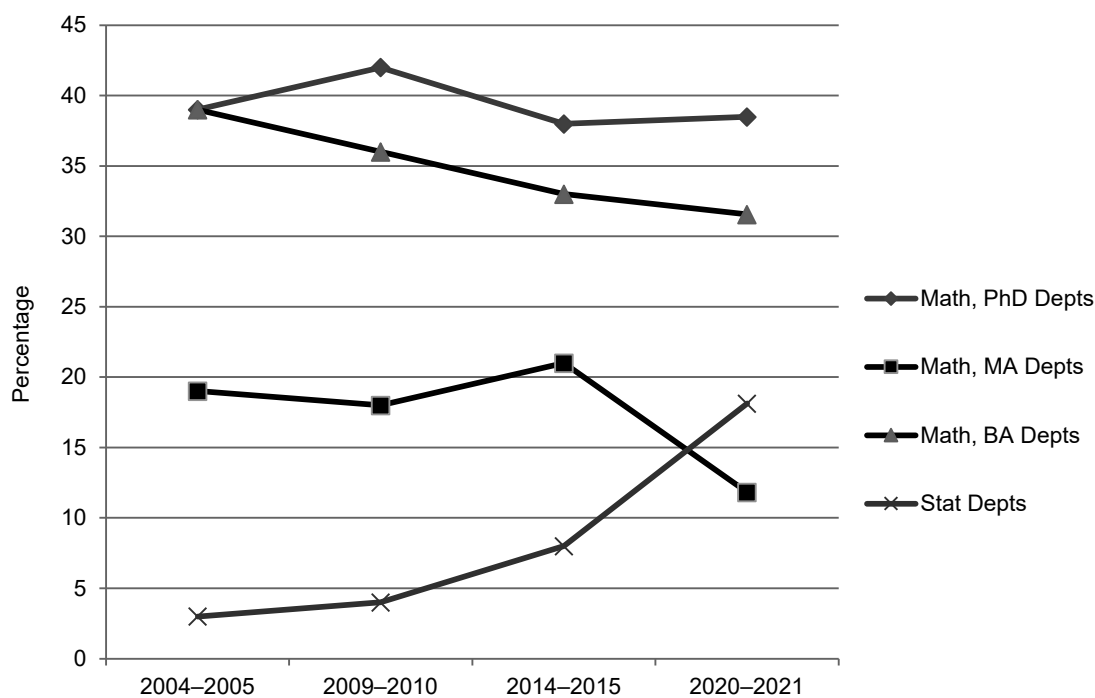


**FIGURE E.1.2.** Number of bachelor's degrees granted by mathematics departments in the academic years 2009–2010, 2014–2015, and 2020–2021, by type of major and type of department.





**FIGURE E.1.3.** Percentage of mathematical sciences bachelor's degrees (including computer science) awarded through mathematics and statistics departments of various kinds in the academic years 2004–2005, 2009–2010, 2014–2015, and 2020–2021.



**FIGURE E.1.4.** Percentage of mathematical sciences bachelor's degrees (excluding computer science) awarded through mathematics and statistics departments of various kinds in the academic years 2004–2005, 2009–2010, 2014–2015, and 2020–2021.

than half of these degrees were awarded at bachelor's-level colleges.

Table E.1.A shows that the estimated numbers of bachelor's degrees awarded by mathematics majors in the category of "actuarial mathematics" during 2020–21 decreased over the estimated number in 2014–15, while the estimated numbers of bachelor's degrees in joint majors increased over this period.

As was already observed, there was a decrease in the estimated number of bachelor's degrees awarded in computer science by mathematics departments. In 1994–95, the CBMS study estimated that mathematics departments awarded 2741 bachelor's degrees in computer science (Table S.3 of Chapter 1); in 2014–15, this number was estimated at 3968; and Table E.1.A shows that in 2020–21, this number was estimated at 1154 (SE 393). Most of the bachelor's degrees in computer science awarded during 2020–21 by mathematics departments were given by the bachelor's-level departments. As we will see later in this chapter, the 2021 report on enrollments shows that computer science estimated enrollments in mathematics departments declined from fall 2015 to fall 2021 (see Table E.3).

**Degrees awarded by statistics departments.** Table E.1.B breaks down the estimated 3881 degrees (SE 890) awarded by statistics departments in 2020–21 by the type of major and by the level of the department. For the first time, statistics departments were asked about the number of bachelor's degrees in "data science" that had been awarded by their department, and it is estimated that there were 537 (but with an SE 415) such degrees.

**Degrees awarded broken down by gender.** Tables E.1.A and E.1.B break down the estimated number of bachelor's degrees awarded by mathematics and statistics departments, respectively, by gender, and Figure E.1.1 displays the numbers of degrees awarded by mathematics departments, broken down by level of department and gender, for 2009–10, 2014–15, and 2020–21. Tables E.1.A and E.1.B show that the estimated percentages of mathematical sciences degrees awarded to women decreased from 2014–15 (see corresponding tables in CBMS2015) to 2020–21 at each level of mathematics and statistics departments, except at the bachelor's-level mathematics departments, where the percentage of degrees awarded to women remained the same (44% (SE 3)). Over the course of the last 30 years, the estimated percentage of bachelor's degrees awarded to women has decreased slightly in mathematics departments and remained relatively constant in statistics departments. Comparisons to previous CBMS surveys can be found at [CBMS1990, Table E.6, p. 30], [CBMS1995, Table E.1, p. 42], [CBMS2000, Table E.1, p. 71], [CBMS2005, Table E.1, p. 78], [CBMS2010, Table E.1, p. 78], and [CBMS2015, Tables E.1.A and E.1.B, pp. 88–89]. The total estimated percent-

age of undergraduate degrees awarded to women by all levels of mathematics departments combined in 2020–21 was 39% (SE 2). The comparable percentage in 2014–15 was 42% (SE 2) women, in 2009–10 was 43% women, and in 2004–5 was 40% women; in 1989–90, the estimated percentage of women was 46%. The estimated percentage of bachelor's degrees awarded to women by statistics departments in 2020–21 was 40% (SE 2) (Table E.1.B), in 2014–2015 was 43% (SE 0.5) (up from 40% in 2009–2010), in 2004–5 was 40%, in 1999–2000 was 43%, and in both 1989–90 and 1994–5 was 38%.

The percentage of degrees awarded to women varies by the level of department. The estimated percentage of all bachelor's degrees awarded to women by doctoral-level mathematics departments in 1989–90 was 37%, in 1994–5 was 43%, in 1999–2000 was 40%, in 2004–5 was 37%, in 2009–10 was 36%, in 2014–15 was 37% (SE 1.2), and in 2020–21 was 35% (SE 4) by Table E.1.A. The estimated percentage of bachelor's degrees awarded by master's-level mathematics departments to women decreased from 46% (SE 3.3) in 2014–15 to 38% (SE 3) in 2020–21; it was 50% in both 2009–10 and 1989–90. In the bachelor's-level departments, the percentage of bachelor's degrees awarded to women decreased from 45% in 2009–10 to 44% (SE 4.3) in 2014–15 and was 44% (SE 3) in 2020–21; it was 52% in 1989–90. See Figure E.1.1, which shows the estimated number of bachelor's degrees awarded, broken down by gender and level of department in 2009–10, 2014–15, and 2020–21.

Table E.1.A shows that the percentage of degrees awarded to women also varies by category of mathematics degree; it is highest in mathematics education (in 2020–21, it was 61% (SE 3); in 2014–15, it was 60% (SE 2.9); in 2009–10, it was 63%; and in 2004–5, it was 60%).

Table E.1.B breaks down the number of bachelor's degrees awarded by statistics departments into several categories. Though the numbers are small, the table shows that the percentage of bachelor's degrees in statistical education awarded to women was 50% (SE 8).

### Tables E.2 and E.3: Undergraduate Enrollments and Number of Sections Offered in Mathematics and Statistics Departments

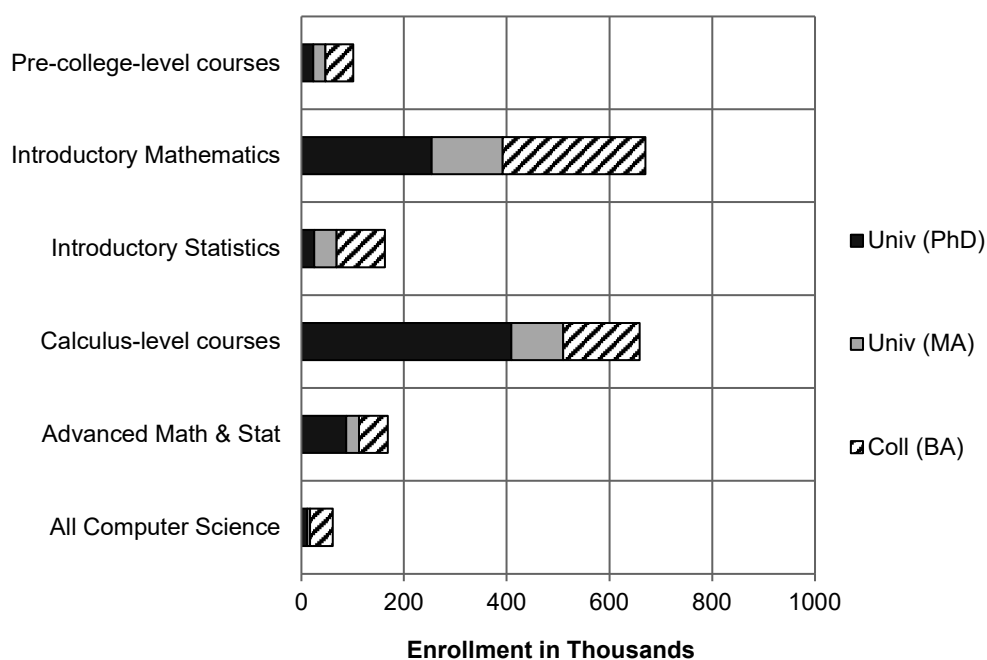
Table S.2 in Chapter 1 shows the estimated total enrollments in fall 2005, 2010, 2015, and 2021 for two-year mathematics departments, four-year mathematics departments, and statistics departments, broken down by the level of courses. Table E.2 presents estimated enrollments in fall 2010, 2015, and 2021 for four-year mathematics departments and statistics departments, broken down by the level of the department. Figure S.2.1 in Chapter 1 shows

**TABLE E.2.** Enrollment (in thousands) in undergraduate mathematics, statistics, and computer science courses (including distance learning enrollments) in mathematics and statistics departments by level of course and type of department in fall 2021. Numbers in parentheses are (2010, 2015) enrollments.

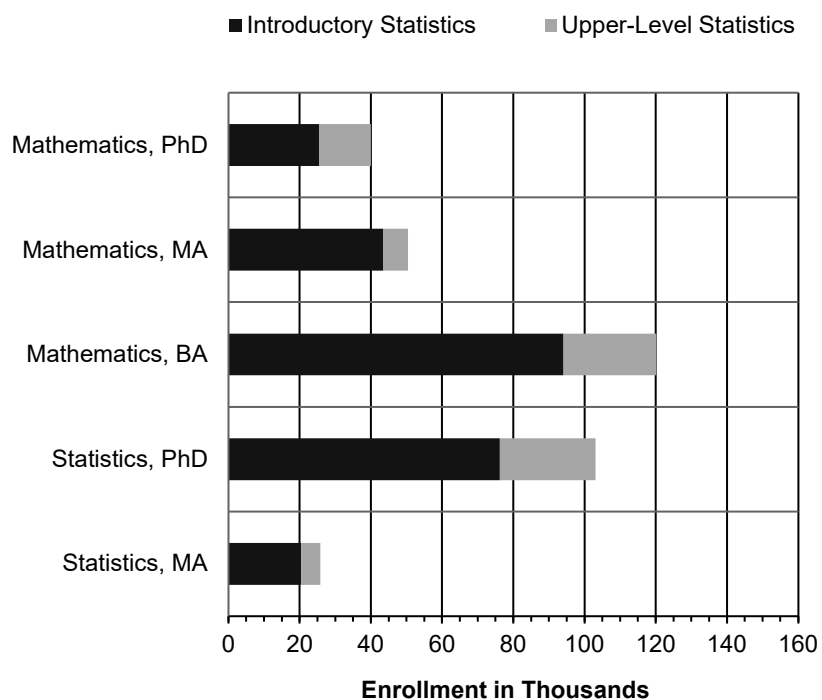
|                                 | Fall 2021 (2010,2015) Enrollments (in 1000s) |                  |                  |                     |                        |               |                     |
|---------------------------------|----------------------------------------------|------------------|------------------|---------------------|------------------------|---------------|---------------------|
|                                 | Mathematics Departments                      |                  |                  |                     | Statistics Departments |               |                     |
|                                 | Univ<br>(PhD)                                | Univ<br>(MA)     | Coll<br>(BA)     | Total Math<br>Depts | Univ<br>(PhD)          | Univ<br>(MA)  | Total Stat<br>Depts |
| <b>Mathematics Courses</b>      |                                              |                  |                  |                     |                        |               |                     |
| Pre-college level               | 24<br>(57,80)                                | 24<br>(64,48)    | 54<br>(88,125)   | 101<br>(209,253)    |                        |               |                     |
| Introductory (incl. Precalc)    | 254<br>(299,408)                             | 138<br>(214,226) | 278<br>(350,365) | 670<br>(863,1000)   |                        |               |                     |
| Calculus level                  | 409<br>(383,474)                             | 101<br>(145,157) | 149<br>(221,176) | 659<br>(748,807)    |                        |               |                     |
| Advanced Mathematics            | 73<br>(64,81)                                | 18<br>(39,30)    | 30<br>(47,43)    | 121<br>(150,154)    |                        |               |                     |
| <b>Total Math Courses</b>       | 760<br>(803,1043)                            | 281<br>(462,461) | 511<br>(706,709) | 1551<br>(1971,2213) |                        |               |                     |
| <b>Statistics Courses</b>       |                                              |                  |                  |                     |                        |               |                     |
| Lower Statistics                | 25<br>(51,57)                                | 43<br>(40,62)    | 91<br>(140,134)  | 159<br>(231,253)    | 73<br>(54,78)          | 20<br>(27,16) | 93<br>(81,94)       |
| Upper Statistics                | 15<br>(15,17)                                | 8<br>(6,24)      | 29<br>(11,20)    | 52<br>(32,60)       | 30<br>(15,45)          | 5<br>(12,5)   | 36<br>(28,50)       |
| <b>Total Stat Courses</b>       | 40<br>(66,74)                                | 50<br>(45,85)    | 120<br>(151,154) | 211<br>(262,313)    | 103<br>(70,124)        | 26<br>(39,20) | 129<br>(109,144)    |
| <b>Computer Science Courses</b> |                                              |                  |                  |                     |                        |               |                     |
| General Education courses       | 1                                            | 1                | 6                | 7                   |                        |               |                     |
| Lower Computer Science          | 5<br>(3,4)                                   | 1<br>(3,5)       | 19<br>(50,36)    | 25<br>(56,45)       |                        |               |                     |
| Middle Computer Science         | 6<br>(1,1)                                   | 2<br>(1,2)       | 14<br>(9,14)     | 21<br>(12,16)       |                        |               |                     |
| Upper Computer Science          | 1<br>(1,0)                                   | 1<br>(1,2)       | 7<br>(8,5)       | 8<br>(10,6)         |                        |               |                     |
| <b>Total CS Courses</b>         | 12<br>(5,5)                                  | 5<br>(6,8)       | 45<br>(67,55)    | 62<br>(77,68)       |                        |               |                     |
| <b>Total All Courses</b>        | 812<br>(874,1122)                            | 336<br>(513,554) | 675<br>(924,918) | 1823<br>(2310,2594) | 103<br>(70,124)        | 26<br>(39,20) | 129<br>(109,144)    |

Note: Beginning in 2010, the CBMS Survey did not include computer science courses taught in statistics departments.

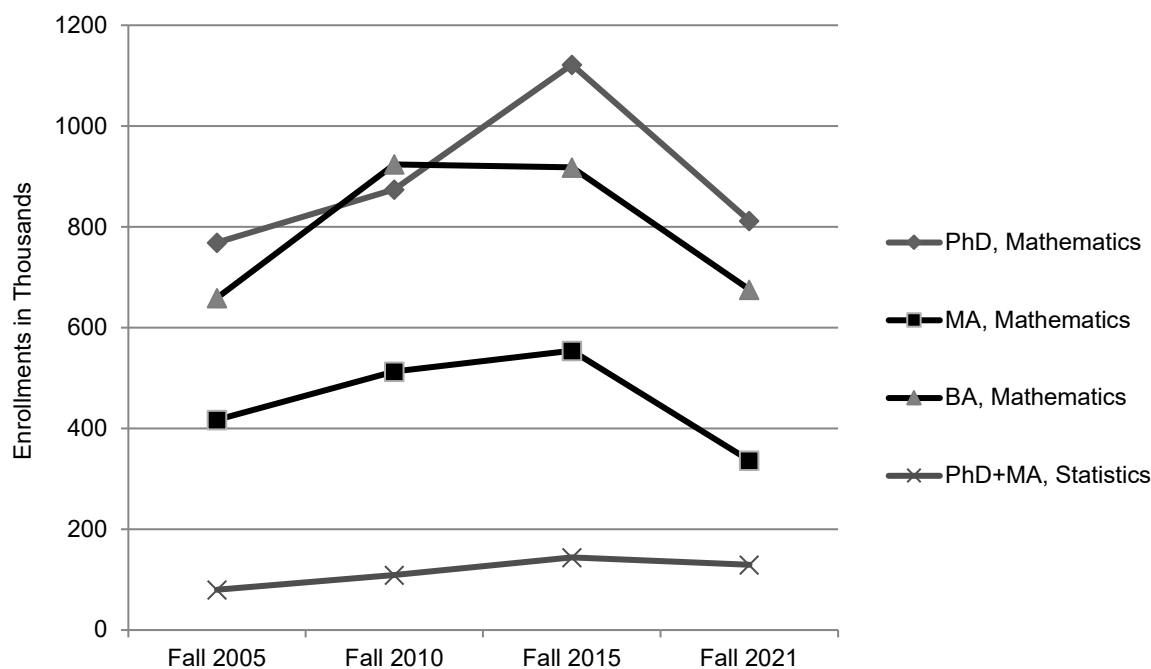
Note: Due to round-off, row and column sums may appear inaccurate.



**FIGURE E.2.1.** Enrollment (in thousands) in undergraduate mathematics, statistics, and computer science courses in four-year college and university mathematics departments by type of course and type of department in fall 2021.



**FIGURE E.2.2.** Enrollment (in thousands) in undergraduate statistics courses by level of course and type of department in fall 2021.



**FIGURE E.2.3.** Undergraduate enrollment (in thousands) by type of department in fall 2005, 2010, 2015, and 2021.

estimated mathematics enrollments in four-year mathematics departments since fall 1990, and Figure S.2.3 in Chapter 1 shows estimated statistics enrollments in two-year mathematics, four-year mathematics, and statistics departments over that 30-year period. The estimated enrollments in individual courses are given in Appendix I (where enrollments both with, and without, distance/remote learning enrollments can be found; in CBMS survey reports prior to 2010, Appendix I gives enrollments with distance/remote learning enrollments included). Enrollments in introductory-level mathematics, calculus-level, and lower-level statistics are considered in more detail in Chapter 5 (where tables generally do not include distance/remote learning enrollments). The specific courses considered in each of the course categories (introductory-level mathematics, calculus-level mathematics, advanced-level mathematics, lower-level statistics, upper-level statistics, and the various computer science course categories) are given in Appendix I. When a table in this report concerns sections of a course, the corresponding enrollments do not contain distance/remote learning enrollments; otherwise, distance/remote learning enrollments generally are included.

The CBMS2015 data showed that estimated enrollments in mathematical sciences courses were larger in fall 2015 than in fall 2010. The 2010 CBMS survey showed large growths in enrollments over 2005. The 2021 CBMS survey data show four-year mathematics department

estimated enrollments in fall 2021 have fallen to levels comparable to levels in fall 1990–2005.

**Enrollments in mathematics courses.** Estimated mathematics departments enrollments in every course category, except upper-level statistics courses, are lower in fall 2021 than in fall 2015 and fall 2010. From Table S.2 in Chapter 1 and Table E.2, we see that from fall 2015 to fall 2021 in all levels of mathematics departments combined, estimated pre-college course enrollments decreased 60% (9.5 SEs), introductory-level course enrollments decreased 33% (5 SEs), calculus-level course enrollments decreased 18% (2.3 SEs), and advanced-level mathematics course enrollments decreased 21% (1.9 SEs). From Table E.2, we see that these declines occurred in each level of department: mathematics department enrollments in mathematics courses were down 28% (3.1 SEs) in bachelor’s-level departments, 39% (3.5 SEs) in master’s-level departments, and 27% (3 SEs) in doctoral-level departments. Figure E.2.1 gives a bar graph that displays the total estimated fall 2021 enrollments in the various course categories, broken down by the type of four-year college (bachelor’s, master’s, or doctoral) where the enrollments occurred.

**Enrollments in statistics courses.** From Table E.2, we see that fall 2021 total estimated enrollments in statistics courses in all levels of mathematics departments

**TABLE E.3.** Number of sections (not including distance learning) of undergraduate mathematics, statistics, and computer science courses in mathematics and statistics departments by level of course and type of department in fall 2021 with fall 2015 figures in parentheses.

|                                 | Number of Sections: Fall 2021 (Fall 2015) |                  |                  |                     |                        |              |                     |
|---------------------------------|-------------------------------------------|------------------|------------------|---------------------|------------------------|--------------|---------------------|
|                                 | Mathematics Departments                   |                  |                  |                     | Statistics Departments |              |                     |
|                                 | Univ<br>(PhD)                             | Univ<br>(MA)     | Coll<br>(BA)     | Total Math<br>Depts | Univ<br>(PhD)          | Univ<br>(MA) | Total Stat<br>Depts |
| <b>Mathematics Courses</b>      |                                           |                  |                  |                     |                        |              |                     |
| Pre-college level               | 385<br>(2235)                             | 784<br>(1578)    | 2354<br>(4206)   | 3523<br>(8020)      |                        |              |                     |
| Introductory (incl. Precalc)    | 3974<br>(8245)                            | 4242<br>(6999)   | 10341<br>(16948) | 18557<br>(32192)    |                        |              |                     |
| Calculus                        | 6795<br>(8323)                            | 3084<br>(4579)   | 7388<br>(8285)   | 17267<br>(21186)    |                        |              |                     |
| Advanced Mathematics            | 2637<br>(3676)                            | 1258<br>(2633)   | 3098<br>(4461)   | 6993<br>(10771)     |                        |              |                     |
| <b>Total Math Courses</b>       | 13790<br>(22479)                          | 9369<br>(15788)  | 23181<br>(33901) | 46340<br>(72168)    |                        |              |                     |
| <b>Statistics Courses</b>       |                                           |                  |                  |                     |                        |              |                     |
| Introductory Statistics         | 477<br>(1319)                             | 1260<br>(1493)   | 3109<br>(4562)   | 4847<br>(7374)      | 663<br>(1256)          | 291<br>(238) | 953<br>(1494)       |
| Upper Statistics                | 486<br>(752)                              | 431<br>(1432)    | 1556<br>(1776)   | 2473<br>(3960)      | 571<br>(796)           | 120<br>(174) | 692<br>(970)        |
| <b>Total Stat Courses</b>       | 963<br>(2072)                             | 1692<br>(2925)   | 4665<br>(6338)   | 7320<br>(11334)     | 1234<br>(2052)         | 411<br>(412) | 1645<br>(2464)      |
| <b>Computer Science Courses</b> |                                           |                  |                  |                     |                        |              |                     |
| General Education courses       | 35                                        | 0                | 130              | 165                 |                        |              |                     |
| Lower Computer Science          | 204<br>(109)                              | 70<br>(186)      | 856<br>(1987)    | 1130<br>(2282)      |                        |              |                     |
| Middle Computer Science         | 202<br>(31)                               | 94<br>(69)       | 718<br>(1128)    | 1014<br>(1227)      |                        |              |                     |
| Upper Computer Science          | 21<br>(0)                                 | 39<br>(84)       | 543<br>(375)     | 603<br>(460)        |                        |              |                     |
| <b>Total CS Courses</b>         | 461<br>(140)                              | 203<br>(339)     | 2249<br>(3490)   | 2913<br>(3970)      |                        |              |                     |
| <b>Total All Courses</b>        | 15214<br>(24692)                          | 11263<br>(19053) | 30095<br>(43728) | 56572<br>(87472)    | 1234<br>(2052)         | 411<br>(412) | 1645<br>(2464)      |

Note: Due to round-off, row and column sums may appear inaccurate.

combined were down 33% (4 SEs) over fall 2015 enrollments. Fall 2021 estimated enrollments in introductory-level statistics courses in mathematics departments decreased 37% (4.7 SEs) over fall 2015 introductory-level statistics enrollments, and the enrollments in upper-level statistics courses in mathematics departments dropped 13% (1 SE) over fall 2015 upper-level statistics enrollments.

Total estimated fall 2021 enrollments in statistics courses taught in all levels of statistics departments combined were down 10% (1 SE) over fall 2015 enrollments. Fall 2021 enrollments in introductory-level statistics courses decreased slightly over 2015 enrollments, and fall 2021 enrollments in upper-level statistics courses in statistics departments decreased 28% (2.3 SEs) over 2015 enrollments in upper-level statistics courses. Figure E.2.2 gives a bar graph that displays the total estimated fall 2021 statistics enrollments, broken down by the type of four-year college (bachelor's-level mathematics, master's-level mathematics, doctoral-level mathematics, master's-level statistics, or doctoral-level statistics) where the enrollments occurred.

**Enrollments in computer science courses in mathematics departments.** Enrollments in computer science courses taught in mathematics departments have continued to decline as these enrollments have tended to move to computer science departments. From Table S.2 in Chapter 1 or Table E.2, we see there were an estimated 62,000 (SE 19,000) total enrollments in computer science courses taught within mathematics departments in fall 2021. From Table S.1 in Chapter 1, we see the past CBMS survey estimates were 123,000 enrollments in fall 2000, which dropped to 57,000 in fall 2005, rebounded to 77,000 in fall 2010, and was 68,000 in fall 2015. From Table E.2, we see that in fall 2021, 73% of the estimated mathematics department enrollments in computer science courses were from bachelor's-level departments, though the estimated total computer science enrollments from doctoral-level mathematics departments increased from 5000 in fall 2015 to 12,000 (SE 8000) in fall 2021. Computer science enrollments in mathematics departments, though small, are still significant in mathematics department enrollments; as one example, according to Table E.2, in fall 2021 (as in fall 2015 and fall 2010), the bachelor's-level mathematics departments had more total estimated enrollments in computer science courses than in advanced-level mathematics courses.

Figure E.2.3 presents a graph of estimated total enrollments in doctoral-level mathematics, master's-level mathematics, bachelor's-level mathematics, and all levels of statistics departments combined in fall 2005, 2010, 2015, and 2021. The figure shows that the general trend of increasing

enrollments from 2005 to 2015, with declining enrollments in fall 2021, occurred in all these departments' enrollments.

**Enrollments: Numbers of sections.** Table E.3 presents the estimated number of sections of courses (not offered in distance/remote learning format) for groups of courses at each level of department in fall 2021 and in fall 2015. Given that enrollments have decreased, it is not surprising that the estimated numbers of sections offered have also decreased. The estimated numbers of sections of courses offered in mathematics and statistics departments in fall 2021 declined over fall 2015 for each group of mathematics and statistics courses and each level of department in Table E.3; some enrollments in computer science course categories for some levels of departments increased, though the SEs of these estimates are large. The estimated total number of sections of all courses in mathematics offered in mathematics departments in fall 2021 at all levels of departments combined declined by 36% (7.6 SEs) over fall 2015: the number of pre-college course sections declined by 56% (7.8 SEs), the number of introductory-level course sections declined by 42%, (7.4 SEs), the number of calculus-level course sections declined by 18% (2.9 SEs), and the number of advanced-level course sections declined by 35% (6 SEs). The estimated total number of sections of all statistics courses offered in mathematics departments in fall 2021 declined by 35% (6 SEs) over fall 2015; the estimated total number of sections of all statistics courses in fall 2021 offered in statistics departments declined by 33% (4.1 SEs) over fall 2015.

#### Table E.4: Distance/Remote Education in Four-Year Colleges and Universities

The 2021 CBMS survey defined distance/remote learning courses as “those courses offered by your institution for credit, in which half or more of the instruction occurs with the instructor and the students separated by time and/or place, and facilitated by technology (e.g., courses in which half or more of the course is taught online either synchronously or asynchronously, by computer software, or by other technologies). Include only distance/remote learning courses offered in normal practice, not courses that became distance/remote due to the COVID-19 pandemic.” Data on various practices in distance/remote learning courses were discussed in Chapter 2 (see Tables SP.8.A–SP.11.B). Distance/remote learning enrollments in the mathematical sciences have grown following the COVID-19 pandemic. Particularly dramatic is the growth in distance/remote learning mathematical sciences enrollments at two-year colleges (more information regarding distance

**TABLE E.4.** Enrollments in distance/remote learning courses (meaning courses offered for credit in which half or more of the instruction occurs with the instructor and the students separated by time and/or place [e.g., courses in which half or more of the course is taught online either synchronously or asynchronously, by computer software, or by other technologies], and other sections for various freshman and sophomore courses, by type of department, in fall 2021. Includes only distance/remote courses offered in normal practice, not courses that became distance/remote due to COVID-19 pandemic. (Fall 2015 data in parentheses.)

|                                             | Four-Year Mathematics Departments |                    | Two-Year Mathematics Departments |                    | Statistics Departments        |                   |
|---------------------------------------------|-----------------------------------|--------------------|----------------------------------|--------------------|-------------------------------|-------------------|
|                                             | Distance-Learning Enrollments     | Other Enrollments  | Distance-Learning Enrollments    | Other Enrollments  | Distance-Learning Enrollments | Other Enrollments |
| Pre-college level                           | 12284<br>(8405)                   | 88747<br>(244475)  | 208883<br>(89035)                | 138952<br>(693252) |                               |                   |
| Introductory level                          | 133444<br>(45226)                 | 536604<br>(954356) | 196368<br>(55227)                | 132425<br>(390066) |                               |                   |
| Calculus I (mainstream and non-mainstream)  | 44760<br>(8968)                   | 280151<br>(346343) | 60350<br>(7455)                  | 33473<br>(84537)   |                               |                   |
| Calculus II (mainstream and non-mainstream) | 10541<br>(3410)                   | 84923<br>(125126)  | 19203<br>(1813)                  | 12227<br>(32523)   |                               |                   |
| Differential Equations & Linear Algebra     | 13123<br>(1492)                   | 102737<br>(137567) | 11482<br>(480)                   | 5064<br>(13559)    |                               |                   |
| Introductory Statistics                     | 30157<br>(18696)                  | 128699<br>(234558) | 169586<br>(30608)                | 87832<br>(220671)  | 28218<br>(4291)               | 64955<br>(89620)  |

Note: For some enrollments and distance learning enrollments in this table, the SE was very large. See Appendix IX.



learning enrollments at two-year colleges is contained in Chapter 6).

Table E.4 presents the estimated number of enrollments in distance/remote learning format and in non-distance/remote learning format for groups of courses offered at four-year mathematics departments, two-year mathematics departments, and four-year statistics departments in fall 2021 and in fall 2015. Appendix I gives estimated enrollments with, and without, distance/remote learning enrollments by course for all courses except advanced-level courses (where enrollments include distance learning enrollments) for four-year mathematics and statistics departments. Tables SP.11.A and SP.11.B in Chapter 2 present data on the advanced-level mathematics and statistics courses that were reported to be available in a distance/remote learning format in fall 2021.

Table E.4 shows that estimated enrollments in distance/remote learning courses listed in the table were up in fall 2021 over fall 2015 for each category of courses in the table. The total distance/remote learning enrollments for the courses (combined) that are listed in Table E.4 for four-year mathematics departments in fall 2021 was estimated at 244,309, almost triple the fall 2015 estimate of 86,197, which was more than double the fall 2010 estimate of 36,798. In fall 2021, the estimated distance/remote learning enrollments in the courses listed in Table E.4 were 17% of the total estimated enrollments in these courses; in fall 2015, distance/remote learning enrollments were 4% of the total estimated enrollments in these courses.

Table E.4 shows that in statistics departments, distance/remote learning enrollments in introductory statistics classes in fall 2021 were estimated at 28,218, almost seven times the fall 2015 estimate of 4291. In fall 2021, estimated distance/remote learning enrollments in introductory statistics courses were 30% of the enrollments in introductory statistics courses taught in statistics departments, while in fall 2015, they were 5%.

Table E.4 shows that in mathematics departments at two-year colleges, the estimated distance/remote learning enrollment in the total of all the courses listed in the table (combined) was 665,872 in fall 2021, up from 184,618 in fall 2015. In fall 2021, distance/remote learning estimated enrollments represented 62% of the total enrollments in these courses; in fall 2015, distance/learning estimated enrollments represented 11% of the total enrollments in these courses.

Many of the SEs for the data in Table E.4 are large, so the percentages of increase, as large as they appear, may be somewhat misleading; however, it does appear that distance learning enrollments were increasing in four-year mathematics and statistics departments and in two-year colleges, while as we have noted, total enrollments were decreasing.

### **Tables E.5–E.9: Appointment Type of Instructors in Mathematics and Statistics Courses at Four-Year Mathematics and Statistics Departments**

The 2021 survey continues the practice begun in 2005 of considering numbers of sections with various appointment type of instructors. Instructors were broken into the appointment type categories: tenured or tenure-eligible (TTE) faculty, other full-time (OFT) faculty (a category that includes, for example, postdocs, faculty with appointments that are renewable (but not tenure-eligible), and academic visitors), part-time (PT) faculty, graduate teaching assistant (GTA), and unknown (a category that was used when the response did not account for all sections of a course). Tables S.4–S.8 in Chapter 1 give data on appointment type of instructors for various categories of courses from the CBMS surveys of 2010, 2015, and 2021 over all levels of departments combined. Tables E.5–E.9 in this chapter give data from 2015 and 2021, broken down by level of department.

#### **Appointment type of calculus-level instructors.**

Table E.5 summarizes the estimated number of sections (excluding distance/remote learning sections) taught by each of the appointment types of calculus-level instructors in mathematics departments at four-year institutions in fall 2021 and fall 2015, broken down by level of department; the numbers in parentheses are from the 2015 CBMS survey. Figure E.5.1 presents a bar graph displaying, at each level of department, the estimated percentage of sections taught by each appointment type of faculty.

Over all levels of mathematics departments combined, the estimated percentage of calculus-level sections taught by TTE faculty had been decreasing; it was estimated at 61% in 2005, 59% in 2010, and 52% in 2015. In fall 2021, over all levels of departments combined, 55% of the estimated calculus sections were taught by TTE faculty, 28% by OFT faculty, 9% by PT faculty, and 6% by GTAs (in 2015, these percentages were 52%, 24%, 10%, and 7%, respectively). These percentages differ by the level of the department: for example in fall 2021, at doctoral-level departments, an estimated 38% of calculus sections were taught by TTE faculty; at master's-level departments, 57% of calculus sections were taught by TTE faculty; and at bachelor's-level departments, 69% sections were taught by TTE faculty.

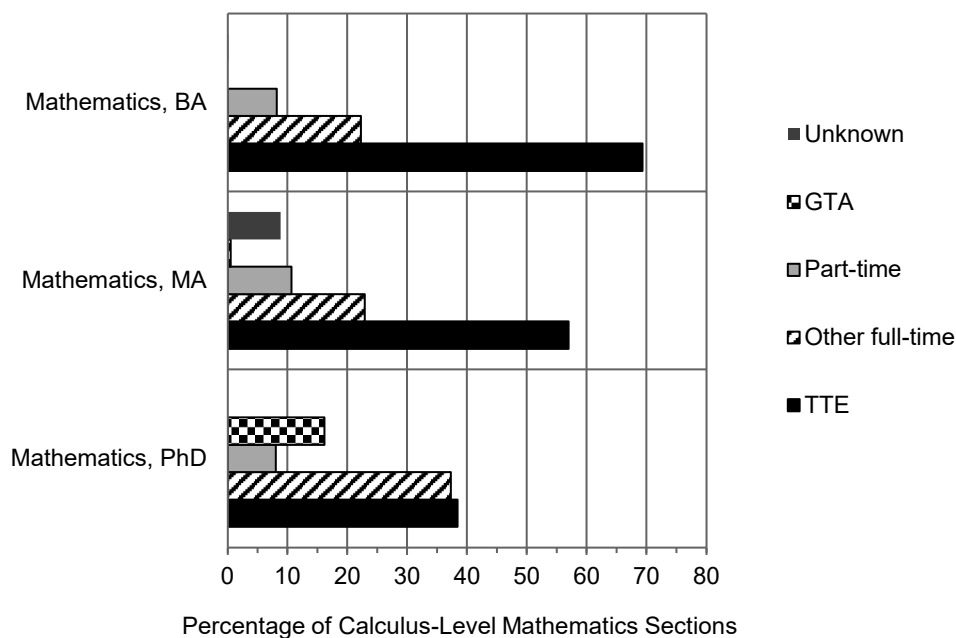
#### **Appointment type of introductory-level statistics instructors.**

Table E.6 and Figure E.6.1 give the estimated numbers of sections taught by each appointment type for introductory statistics courses in mathematics and statistics departments, by level of department, in fall 2021 and fall 2015. In fall 2021, over all levels of mathematics department combined, 43% of the estimated introductory statistics sections were taught by TTE faculty, 28% by OFT

**TABLE E.5.** Number of sections (excluding distance learning) of calculus-level courses in mathematics departments taught by various types of instructor, by type of department, in fall 2021 with fall 2015 figures in parentheses.

|                         | Number of Calculus-Level Sections Taught by |                    |                |                                   |               | Total Sections   |
|-------------------------|---------------------------------------------|--------------------|----------------|-----------------------------------|---------------|------------------|
|                         | Tenured/<br>Tenure-Eligible                 | Other<br>Full-Time | Part-Time      | Graduate<br>Teaching<br>Assistant | Unknown       |                  |
| Mathematics Departments |                                             |                    |                |                                   |               |                  |
| Univ (PhD)              | 2611<br>(2803)                              | 2536<br>(2962)     | 548<br>(733)   | 1101<br>(1370)                    | 0<br>(454)    | 6795<br>(8323)   |
| Univ (MA)               | 1758<br>(2365)                              | 708<br>(994)       | 330<br>(797)   | 16<br>(84)                        | 273<br>(339)  | 3084<br>(4579)   |
| Coll (BA)               | 5122<br>(5896)                              | 1647<br>(1078)     | 609<br>(585)   | 0<br>(0)                          | 10<br>(727)   | 7388<br>(8285)   |
| Total                   | 9490<br>(11064)                             | 4890<br>(5034)     | 1486<br>(2115) | 1117<br>(1454)                    | 283<br>(1520) | 17267<br>(21186) |

Note: Numbers of unknown sections were adjusted to match the reported total number of sections.

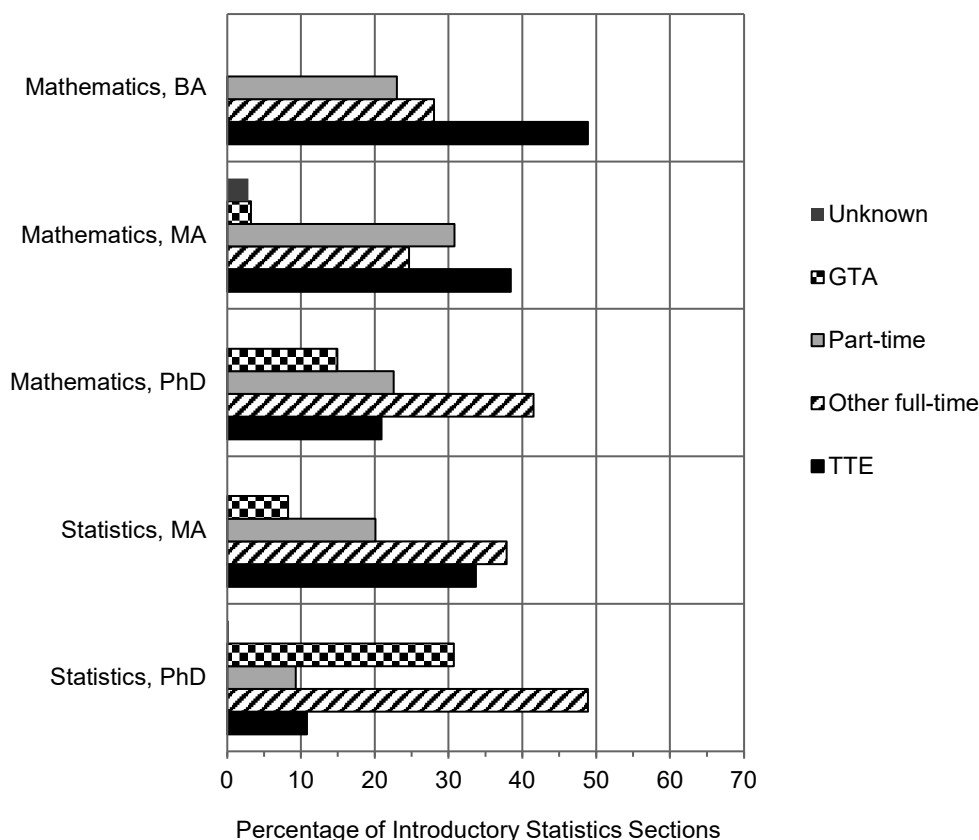


**FIGURE E.5.1.** Percentage of calculus-level mathematics sections in mathematics departments whose instructors were TTE faculty, OFT faculty, PT faculty, and GTAs, by type of department, in fall 2021. (Percentages may not sum to 100 due to “unknown” instructor percentages.)

**TABLE E.6.** Number of sections (excluding distance learning) of introductory statistics courses taught in mathematics departments and statistics departments by type of instructor and type of department in fall 2021 with fall 2015 figures in parentheses.

|                         | Number of Introductory Statistics Sections Taught by |                    |                |                                   |             | Total Sections |
|-------------------------|------------------------------------------------------|--------------------|----------------|-----------------------------------|-------------|----------------|
|                         | Tenured/<br>Tenure-Eligible                          | Other<br>Full-Time | Part-Time      | Graduate<br>Teaching<br>Assistant | Unknown     |                |
| Mathematics Departments |                                                      |                    |                |                                   |             |                |
| Univ (PhD)              | 100<br>(268)                                         | 198<br>(392)       | 108<br>(239)   | 71<br>(245)                       | 0<br>(175)  | 477<br>(1319)  |
| Univ (MA)               | 484<br>(781)                                         | 311<br>(467)       | 388<br>(216)   | 41<br>(0)                         | 36<br>(29)  | 1260<br>(1493) |
| Coll (BA)               | 1543<br>(2006)                                       | 885<br>(725)       | 726<br>(1389)  | 0<br>(30)                         | 3<br>(411)  | 3158<br>(4562) |
| Total                   | 2128<br>(3055)                                       | 1393<br>(1584)     | 1222<br>(1844) | 112<br>(275)                      | 40<br>(615) | 4895<br>(7374) |
| Statistics Departments  |                                                      |                    |                |                                   |             |                |
| Univ (PhD)              | 74<br>(136)                                          | 334<br>(281)       | 64<br>(111)    | 210<br>(466)                      | 2<br>(263)  | 683<br>(1256)  |
| Univ (MA)               | 98<br>(75)                                           | 110<br>(97)        | 59<br>(33)     | 24<br>(3)                         | 0<br>(31)   | 291<br>(238)   |
| Total                   | 172<br>(210)                                         | 444<br>(378)       | 122<br>(144)   | 234<br>(468)                      | 2<br>(295)  | 974<br>(1494)  |

Note: Total numbers of sections were adjusted to match the total number of sections reported taught by each faculty type.



**FIGURE E.6.1.** Percentage of introductory statistics sections in mathematics and statistics departments whose instructors were TTE faculty, OFT faculty, PT faculty, and GTAs, by type of department, in fall 2021. (Percentages may not sum to 100 due to “unknown” instructor percentages.)

faculty, 25% by PT faculty, and 2% of GTAs (in 2015, these percentages were 41%, 21%, 25%, and 4%, respectively). In fall 2021, over all levels of statistics department combined, 18% of the estimated introductory statistics sections were taught by TTE faculty, 46% by OFT faculty, 13% by PT faculty, and 24% of GTAs (in 2015, these percentages were 14%, 25%, 10%, and 31%, respectively, with 20% unknown).

**Appointment type of advanced-level course instructors.** Table E.7 gives the estimated numbers of sections of advanced-level (upper-level) courses in mathematics and statistics departments taught by each appointment type, by level of department, in fall 2021 and fall 2015. For all levels of mathematics departments combined, in fall 2021, an estimated 82% of the advanced-level sections of mathematics courses were taught by TTE faculty (up from 70% in fall 2015). This percentage varies by level of department: for example in fall 2021, for doctoral-level mathematics departments, it was estimated at 76%; at master’s-level mathematics departments, it was 80%; and at bachelor’s-

level mathematics departments, it was 87%. For all levels of mathematics departments combined, the estimated percentage of sections of upper-level statistics courses taught by TTE faculty in fall 2021 was estimated at 86% (up from 53% in fall 2015), while in all levels of statistics departments combined, in fall 2021, it was estimated at 56% (it was estimated at 55% in fall 2015).

**Appointment type of computer science instructors.** Tables E.8 and E.9 and Figure E.9.1 give the estimated percentages of sections taught by each appointment type for computer science courses in mathematics departments, by level of department, in fall 2021 and fall 2015. Table E.8 presents the data for lower-level computer science courses, Table E.9 presents the data for middle-level computer science courses, and Figure E.9.1 displays the data for all levels of computer science courses combined. As we have noted earlier, most of the computer science courses taught in mathematics departments are taught in bachelor’s-level departments. From Table E.8, we see that

**TABLE E.7.** Number of sections of advanced mathematics (including operations research) and statistics courses in mathematics departments and of advanced statistics courses in statistics departments taught by tenured/tenure-eligible (TTE) faculty and total number of advanced-level sections by type of department in fall 2021 with fall 2015 data in parentheses.

| Mathematics Departments      | Sections Taught by TTE | Total Sections  | Statistics Departments      | Sections Taught by TTE | Total Sections |
|------------------------------|------------------------|-----------------|-----------------------------|------------------------|----------------|
| Advanced Mathematics courses |                        |                 |                             |                        |                |
| Univ (PhD)                   | 1996<br>(2519)         | 2637<br>(3676)  |                             |                        |                |
| Univ (MA)                    | 1010<br>(1769)         | 1258<br>(2633)  |                             |                        |                |
| Coll (BA)                    | 2706<br>(3236)         | 3098<br>(4461)  |                             |                        |                |
| Total advanced mathematics   | 5712<br>(7525)         | 6993<br>(10771) |                             |                        |                |
| Advanced Statistics courses  |                        |                 | Advanced Statistics courses |                        |                |
| Univ (PhD)                   | 403<br>(452)           | 479<br>(752)    | Univ (PhD)                  | 281<br>(394)           | 539<br>(796)   |
| Univ (MA)                    | 336<br>(656)           | 391<br>(1432)   | Univ (MA)                   | 91<br>(140)            | 120<br>(174)   |
| Coll (BA)                    | 1232<br>(1010)         | 1411<br>(1776)  |                             |                        |                |
| Total advanced statistics    | 1971<br>(2118)         | 2281<br>(3960)  | Total advanced statistics   | 372<br>(533)           | 659<br>(970)   |
| Total all advanced courses   | 7683<br>(9643)         | 9274<br>(14731) | Total all advanced courses  | 372<br>(533)           | 659<br>(970)   |

Note: Round-off may make row and column sums seem inaccurate.

**TABLE E.8.** Number of sections (excluding distance learning) of lower-level computer science courses taught in mathematics departments by type of instructor and type of department in fall 2021 with fall 2015 figures in parentheses.

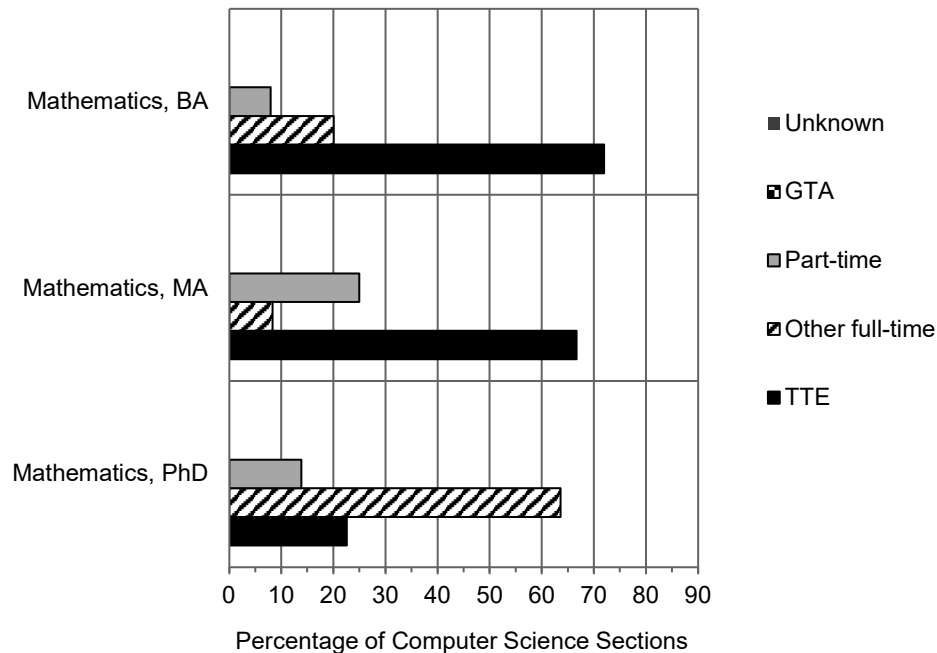
|                         | Number of Lower-Level Computer Science Sections Taught by |                    |              |                                   |            | Total Sections |
|-------------------------|-----------------------------------------------------------|--------------------|--------------|-----------------------------------|------------|----------------|
|                         | Tenured/<br>Tenure-Eligible/<br>Permanent                 | Other<br>Full-Time | Part-Time    | Graduate<br>Teaching<br>Assistant | Unknown    |                |
| Mathematics Departments |                                                           |                    |              |                                   |            |                |
| Univ (PhD)              | 58<br>(30)                                                | 104<br>(71)        | 42<br>(8)    | 0<br>(0)                          | 0<br>(0)   | 204<br>(109)   |
| Univ (MA)               | 23<br>(112)                                               | 23<br>(48)         | 23<br>(26)   | 0<br>(0)                          | 0<br>(0)   | 70<br>(186)    |
| Coll (BA)               | 664<br>(899)                                              | 130<br>(339)       | 63<br>(277)  | 0<br>(0)                          | 0<br>(472) | 856<br>(1987)  |
| Total                   | 745<br>(1042)                                             | 257<br>(458)       | 128<br>(311) | 0<br>(0)                          | 0<br>(472) | 1130<br>(2282) |

Note: Round-off may make row and column sums seem inaccurate.

**TABLE E.9.** Number of sections (excluding distance learning) of middle-level computer science courses taught in mathematics departments by type of instructor and type of department in fall 2021 with fall 2015 figures in parentheses.

|                         | Number of Middle-Level Computer Science Sections Taught by |                    |              |                                   |            | Total Sections |
|-------------------------|------------------------------------------------------------|--------------------|--------------|-----------------------------------|------------|----------------|
|                         | Tenured/<br>Tenure-Eligible                                | Other<br>Full-Time | Part-Time    | Graduate<br>Teaching<br>Assistant | Unknown    |                |
| Mathematics Departments |                                                            |                    |              |                                   |            |                |
| Univ (PhD)              | 46<br>(17)                                                 | 128<br>(0)         | 28<br>(5)    | 0<br>(0)                          | 0<br>(9)   | 202<br>(31)    |
| Univ (MA)               | 62<br>(55)                                                 | 8<br>(4)           | 23<br>(9)    | 0<br>(0)                          | 0<br>(0)   | 94<br>(69)     |
| Coll (BA)               | 529<br>(549)                                               | 148<br>(311)       | 59<br>(161)  | 0<br>(0)                          | 0<br>(107) | 736<br>(1128)  |
| Total                   | 637<br>(621)                                               | 284<br>(316)       | 110<br>(174) | 0<br>(0)                          | 0<br>(116) | 1031<br>(1227) |

Note: Total numbers of sections were adjusted to match the total number of sections reported taught by each faculty type.



**FIGURE E.9.1.** Percentage of computer science sections (all levels) in mathematics departments whose instructors were TTE faculty, OFT faculty, PT faculty, and GTAs, by type of department, in fall 2021. (Percentages may not sum to 100 due to “unknown” instructor percentages.)

**TABLE E.10.** Average section size (excluding distance learning) for undergraduate mathematics, statistics, and computer science courses in mathematics and statistics departments by level of course and type of department in fall 2021 with fall 2015 data, when available, in parentheses. Also, all departments' average section sizes from previous CBMS surveys.

|                              | Average Section Size Fall 2021 (2015) |            |            |                  |            |            |              |
|------------------------------|---------------------------------------|------------|------------|------------------|------------|------------|--------------|
|                              | Mathematics Depts                     |            |            | Statistics Depts |            |            | Overall Stat |
|                              | Univ (PhD)                            | Univ (MA)  | Coll (BA)  | Overall Math     | Univ (PhD) | Univ (MA)  |              |
| Mathematics Courses          |                                       |            |            |                  |            |            |              |
| Pre-college                  | 55<br>(34)                            | 26<br>(30) | 20<br>(29) | 25<br>(30)       |            |            |              |
| Introductory (incl. Precalc) | 51<br>(47)                            | 26<br>(31) | 22<br>(20) | 29<br>(30)       |            |            |              |
| Calculus level               | 54<br>(55)                            | 24<br>(34) | 19<br>(21) | 34<br>(37)       |            |            |              |
| Advanced Mathematics         | 26<br>(22)                            | 9<br>(11)  | 9<br>(10)  | 15<br>(14)       |            |            |              |
| Statistics Courses           |                                       |            |            |                  |            |            |              |
| Introductory Statistics      | 46<br>(40)                            | 25<br>(39) | 24<br>(27) | 26<br>(32)       | 76<br>(59) | 50<br>(65) | 68<br>(60)   |
| Upper Statistics             | 27<br>(23)                            | 15<br>(16) | 17<br>(11) | 19<br>(15)       | 40<br>(57) | 35<br>(27) | 39<br>(52)   |
| Computer Science Courses     |                                       |            |            |                  |            |            |              |
| General education            | 25                                    |            | 24         | 24               |            |            |              |
| Lower Computer Science       | 24<br>(38)                            | 18<br>(24) | 20<br>(18) | 21<br>(19)       |            |            |              |
| Middle Computer Science      | 27<br>(20)                            | 23<br>(22) | 16<br>(13) | 19<br>(13)       |            |            |              |
| Upper Computer Science       | 25<br>NA                              | 22<br>(19) | 12<br>(13) | 13<br>(14)       |            |            |              |
|                              |                                       |            |            |                  |            |            | 24           |
|                              |                                       |            |            |                  |            |            | 19 21 19 21  |
|                              |                                       |            |            |                  |            |            | 9 12 13 19   |
|                              |                                       |            |            |                  |            |            | 8 11 14 13   |

Note: NA = Not applicable (there were no upper-division computer science courses at doctorate-granting institutions). Blank entries represent courses with no responses.

**TABLE E.11.** Average recitation size in Mainstream Calculus I and II and other Calculus I courses and in introductory statistics courses that are taught using lecture/recitation method, by type of department in fall 2021, with fall 2015 data in parentheses. Distance learning sections are not included. (A calculus course is “Mainstream” if it leads to the usual upper-division mathematical sciences courses.)

| For Lecture/Recitation Courses | Average Recitation Section Size |              |                 |
|--------------------------------|---------------------------------|--------------|-----------------|
|                                | Univ<br>(PhD)                   | Univ<br>(MA) | College<br>(BA) |
| Calculus Courses               |                                 |              |                 |
| Mainstream Calculus I          | 82<br>(31)                      | 31<br>(34)   | 13<br>(17)      |
| Mainstream Calculus II         | 55<br>(29)                      | 39<br>(14)   | 17<br>(9)       |
| Mainstream Calculus III        | 73                              | 11           | 8               |
| Other Calculus I               | 56<br>(36)                      | 59<br>(16)   | (9)             |
| Introductory Statistics        |                                 |              |                 |
| In Mathematics Departments     | 53<br>(33)                      | 51<br>(19)   | 27<br>(26)      |
| In Statistics Departments      | 89<br>(25)                      | 72<br>(28)   | na<br>na        |

Note: Blank entries represent courses with no responses.

in fall 2021, an estimated 66% of the sections of lower-level computer science courses were taught by TTE faculty (up from 46% in fall 2015).

### Tables E.10 and E.11: Average Section Size

Table E.10 presents the average section size (excluding distance/remote learning sections) for mathematics, statistics, and computer science courses in four-year mathematics and statistics departments, by category of course and level of department, in fall 2021 and fall 2015. The estimated average section size for each category of course, across all levels of departments combined, is given for fall 2005, 2010, 2015, and 2021. The table shows that the estimated average section size has remained relatively stable over the

past 16 years. The section size differs by level of institution. For example, in fall 2021, the estimated average section size of calculus-level courses was 54 (SE 6) at doctoral-level departments and 19 (SE 1) at bachelor's-level departments, and the estimated average section size of advanced courses was 26 (SE 4) at doctoral-level departments and 9 (SE 1) at bachelor's-level departments.

Table E.11 gives the estimated average section size of recitation sections of calculus classes in mathematics departments and introductory statistics courses in both mathematics and statistics departments. The size of recitation sections in calculus courses (particularly at doctoral-level mathematics departments) and in statistics courses generally increased from fall 2015 to fall 2021, but many of the SEs in Table E.11 are large.





## Chapter 4

# Faculty Demographics in Mathematical Sciences Departments of Four-Year Colleges and Universities

### Introduction

In this chapter, we consider data on the number, gender, age, and ethnicity of mathematical science faculty in doctoral-, master's-, and bachelor's-level four-year mathematics departments, and in doctoral-level statistics departments possessing an undergraduate program in statistics. The same topics were presented in Chapter 1 tables for the four-year mathematics and statistics departments with all levels of departments combined. In this chapter, we will consider differences in numbers of faculty across departments grouped according to the highest degree offered ("level of department"), by "type of appointment" (tenured, tenure-eligible, other full-time, postdoc, and part-time), by highest degree obtained by the faculty ("doctoral" and "non-doctoral"), and by gender (male, female, and non-binary). The category of "other full-time faculty" includes full-time faculty who are not tenured or tenure-eligible and often contains faculty with titles such as visiting professor, teaching professional, professor of the practice, as well as postdoc. Note that in the tables throughout, as in past CBMS surveys, the numbers of postdoc faculty are included in the number of other full-time faculty but are also broken out of other full-time faculty for further information.

- From Table F.1.1 (and S.13 in Chapter 1), we see that the total number of full-time faculty in fall 2021 for all levels (bachelor's, master's, and doctoral) of four-year mathematics departments combined was estimated at 31,807 (SE 756), a 41% (12 SEs) increase over fall 2015. Most of this increase occurred in the category of "other full-time faculty" (faculty who are not tenured or tenure-eligible, including postdocs), which from Table F.1.1, we see rose 80% (12 SEs) from fall 2015; see Figure F.1.1. The numbers of faculty are broken down further by gender, non-doctoral faculty, and level of department in Tables F.1 and F.2.
- From Table F.1.1 (and S.15 in Chapter 1), we see that in all levels of four-year mathematics departments combined, the estimated number of tenured faculty and the number of tenure-eligible faculty each rose by 23% from fall 2015 to fall 2021; most of this increase occurred in the bachelor's-level departments.
- From Table F.1.1 (and S.13 in Chapter 1), we see that the estimated number of part-time faculty for all levels of four-year mathematics departments combined in fall 2021 increased 31% (10 SEs) from fall 2015.
- From Table F.1.1 (and S.13 in Chapter 1), we see that the total number of full-time faculty at doctoral-level statistics departments possessing an undergraduate program in statistics in fall 2021 was estimated at 3396 (SE 100), more than double the fall 2015 estimated total of 1238. The estimated number of other full-time statistics faculty rose from 369 in fall 2015 to 1366 (SE 87), the estimated number of tenured statistics faculty increased from 649 in fall 2015 to 1457 (SE 41) in fall 2021, the estimated number of tenure-eligible statistics faculty rose from 220 in fall 2015 to 573 (SE 21) in fall 2021, and the estimated number of part-time statistics faculty increased from 128 in fall 2015 to 845 (SE 74) in fall 2021.
- Table F.1 or F.2 displays the estimated number of faculty in each appointment type of faculty (tenured, tenure-eligible, other full-time, postdoc, and part-time) in fall 2021 broken down by the highest degree offered by the department. Note again that the numbers of postdocs are included in the numbers of other full-time faculty but also are broken out for information. From fall 2015 to fall 2021, the percentage increase in the estimated number of full-time faculty in all appointment types combined at mathematics departments offering PhDs was 32%, at those offering master's degrees was 47%, and at those offering bachelor's degrees was 47%.
- From Table F.1, we see that from fall 2015 to fall 2021 at mathematics departments offering a PhD, the estimated number of tenured faculty increased 9%, while at statistics departments offering a PhD, the number of tenured faculty increased 125%. At mathematics departments offering a PhD in fall 2021, there was an estimated total of 1274 postdocs and 1167 tenure-eligible faculty. Doctoral statistics departments had an estimated 178 postdocs and 573 tenure-eligible faculty. The analogous numbers for other levels of four-year departments can be seen in Table F.2.

- From Table F.1.1, we see that the estimated total number of full-time faculty who are women, over all levels of mathematics departments and all appointment types combined, increased from 31% in 2015 to 37% in 2021. In fall 2021, the estimated percentage of women among all tenured mathematics faculty, across all levels of mathematics departments combined, was 30%, up from 22% in fall 2015.
- Table F.2 (and Figure F.3.1) break the total estimated number of full-time fall 2021 and fall 2015 mathematics faculty in each appointment type (tenured, tenure-eligible, other full-time, and postdoc) down further by each of the three levels of mathematics department (doctoral, master's, and bachelor's). In fall 2021, the percentage of women in all appointment types combined was 27% at the doctoral-level mathematics departments (up from 23% in fall 2015), 42% at master's-level departments (up from 39% in fall 2015), and 43% at bachelor's-level departments (up from 31% in fall 2015). The percentage of women among tenured mathematics faculty in fall 2021 was 19% at doctoral-level departments (up from 14% in fall 2015), 33% at master's-level departments (up from 26% in fall 2015), and 36% at bachelor's-level departments (up from 29% in fall 2015). At doctoral-level mathematics departments, an estimated 59% of the women are in other full-time appointment types.
- Table F.3 and Figure F.3.1 give the estimated breakdown of the number of full-time doctoral-level statistics faculty for each of the four appointment types (tenured, tenure-eligible, other full-time, and postdoc) in fall 2021 by gender. In fall 2021, women were estimated at 38% of total full-time statistics faculty (up from 28% in fall 2015) and 31% of tenured statistics faculty (up from 21% in fall 2015).
- Table F.4 presents the estimated distribution of ages of tenured and tenure-eligible mathematics faculty and of doctoral-level statistics faculty in fall 2021, broken down by level of department and by gender. Figures F.4.1, F.4.2, and F.4.3 display the age distribution for tenured and tenure-eligible faculty combined at doctoral-, master's-, and bachelor's-level mathematics departments, respectively. An estimated 27.5% (8.4% women, 19.1% men) of tenured or tenure-eligible faculty at doctoral-level mathematics departments were under 40 in fall 2021, while the corresponding percentage was 21% (9% women, 12% men) in fall 2015. Furthermore, an estimated 28.4% of tenured or tenure-eligible faculty at doctoral-level mathematics departments were 60 or above in fall 2021, compared to 31%

in fall 2015. An estimated average age is given, and generally, it is higher in fall 2021 than it was in fall 2015.

- Table F.5 gives the estimated percentages of various ethnic groups of full-time faculty in fall 2021 broken down by gender, and Table F.6 presents the same data for part-time faculty. The percentage of White male faculty continues to decline in fall 2021 slightly from fall 2015 at most levels of departments.

## Data Sources and Notes on the Tables

Each fall, the AMS conducts national surveys of mathematical sciences departments at four-year institutions, titled the Annual Survey of the Mathematical Sciences (or just the Annual Survey when the context is clear). This work is sponsored by the AMS, ASA, IMS, MAA, and SIAM with oversight provided by the Joint Data Committee (JDC), whose members are appointed by the sponsoring societies. Reports on these surveys are published in the *Notices of the American Mathematical Society* each year and online at <http://www.ams.org/profession/data/annual-survey/annual-survey>. Beginning with the CBMS survey in 2005, the demographic data for the CBMS survey was collected as part of the Annual Survey; departments were asked additional demographic questions that do not normally appear on the Annual Survey.

In comparing data from the CBMS surveys to data published in the Annual Surveys, one must keep in mind several differences between the surveys. The Annual Surveys do not include postdoctoral appointments as a part of “other full-time faculty” (OFT), while CBMS surveys do—i.e., CBMS surveys list “other full-time faculty” (which includes postdoctoral appointments) and also breaks out the number of other full-time faculty who have postdoctoral appointments. The CBMS surveys of “statistics” include only statistics departments that offer an undergraduate program in statistics, while the Annual Surveys include all departments of statistics and biostatistics that award a PhD. The 2005 and 2021 Annual Surveys did not include master's-level statistics departments, and the 2010 and 2015 surveys did include these departments. The Annual Surveys use a census of bachelor's-, master's-, and doctoral-level programs. Hence, the demographic data on faculty in this chapter were obtained in a different manner than the other data in the 2021 CBMS survey report. Details on the survey methodology and response rates of both surveys can be found in Appendix II.

Table entries are rounded to the nearest integer, and the sum of rounded numbers is not always equal to the rounded sum.

In Chapter 1, Tables S.13, S.14, S.15, and S.16 and related figures present data on estimated numbers of faculty in four-year mathematics and statistics departments in fall 2021. In this chapter, these tables are broken down further.

Table F.1 gives a complete breakdown of estimated numbers of fall 2021 (fall 2015) faculty at four-year mathematics and doctoral-level statistics departments, broken down by the level of the mathematics department (doctoral, master's, and bachelor's), the type of appointment (tenured, tenure-eligible, other full-time (including postdoc), postdoc, and part-time), gender (total and the number of women), and whether the faculty possesses a doctoral degree. Tables F.1.1, F.2, and F.3 are derived from Table F.1. Table F.1.1 gives the estimated total numbers of faculty in fall 2021 (2015) for all levels of departments combined, broken down by the type of appointment (tenured, tenure-eligible, other full-time (including postdoc), postdoc, and part-time), gender (total number and number of women), and whether the faculty possesses a doctoral degree. Table F.2 gives a complete breakdown of numbers of full-time fall 2021 (fall 2015) faculty at four-year mathematics departments, broken down by the level of the department (doctoral, master's, and bachelor's), the type of appointment (tenured, tenure-eligible, other full-time (including postdoc), and postdoc), and gender (men, women, and nonbinary), but not by whether the faculty possesses a doctoral degree; Table F.3 presents the same information for doctoral-level statistics departments as appears in Table F.2 for mathematics departments.

## Total Numbers of Mathematics and Statistics Faculty

Table S.13 and Figure S.13.1 in Chapter 1 show a pattern of increases in the estimated number of full-time fall faculty in all levels of mathematics departments combined, and in statistics departments, in the CBMS surveys of 2005, 2010, 2015, and 2021. Figure S.13.2 displays the estimated number of part-time mathematics faculty, which generally increased over these four surveys, though it decreased in the 2010 survey. Table S.13 and Figure S.13.2 showed that the estimated total number of full-time mathematics faculty plus part-time mathematics faculty for all levels of four-year mathematics departments combined increased by 39% from 2015 to 2021; in the doctoral statistics departments, this number increased by 210% (see Figure S.13.5). From Table S.13 and Figures S.13.1 and S.13.3, we see that the estimated total number of full-time mathematics faculty in four-year colleges and universities across all types of departments was 31,807, up from the fall 2015 estimate of 22,532, an increase of 41% (12 SEs). The fall 2005 estimate was 21,885 faculty, and the fall 2000 estimate was 19,799 faculty. From fall 2015 to fall 2021, the estimated num-

ber of full-time plus part-time faculty in mathematics departments was up 30% at doctoral-level departments, 41% at master's-level departments, and 44% at bachelor's-level departments according to Table F.1.

From Table S.13 and Figures S.13.1 and S.13.5, we see that the estimated number of full-time faculty in doctoral statistics departments was 3396 in fall 2021, up from the fall 2015 estimate of 1237, and the number of part-time faculty in doctoral statistics departments was 845 in fall 2021, up from the fall 2015 estimate of 128.

As Figure F.1.1 makes clear, most of the growth in the numbers of four-year mathematics faculty is the result of growth in the numbers of part-time faculty and other full-time faculty.

## Numbers of Tenured and Tenure-Eligible Mathematics Faculty

From Table F.1.1, we see that across all levels of mathematics departments combined, from fall 2015 to fall 2021, the estimated number of tenured faculty increased 23% (9.6 SEs), and the estimated number of tenure-eligible faculty increased 23% (10.9 SEs). We see from Table F.2 that, broken down by level of the department, from fall 2015 to fall 2021, doctoral-level departments had a 9% (2.9 SEs) increase in the number of tenured faculty and a 17% (4.2 SEs) increase in the number of tenure-eligible faculty, master's-level departments had a 10% (2.9 SEs) increase in the number of tenured faculty and a 14% (3.5 SEs) increase in the number of tenure-eligible faculty, and bachelor's-level departments had a 42% (21.3 SEs) increase in the number of tenured faculty and a 30% (18.4 SEs) increase in the number of tenure-eligible faculty.

## Numbers of Tenured and Tenure-Eligible Statistics Faculty

From Table F.1.1 or F.3, we see that in the doctoral-level statistics departments, from fall 2015 to fall 2021, the estimated number of tenured faculty increased 125% (20 SEs), and the estimated number of tenure-eligible faculty increased 160% (17 SEs).

## Numbers of Other Full-Time Mathematics and Statistics Faculty

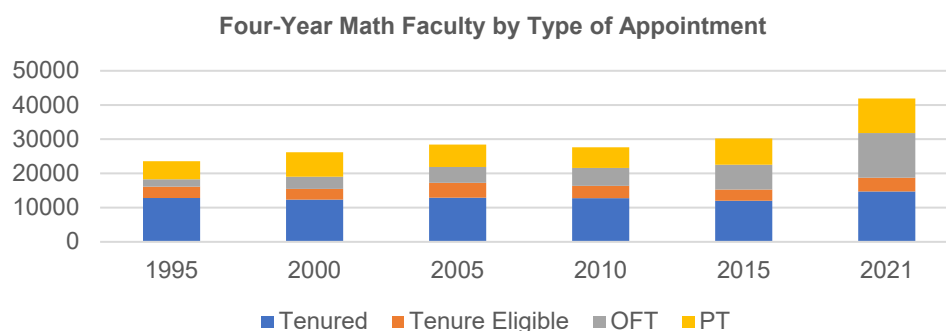
The category of "other full-time faculty" includes full-time faculty who are not tenured or tenure-eligible and contains faculty who often have titles such as visiting professor, teaching professional, professor of the practice, as well as postdoc. From Table F.1.1, across all levels of mathematics departments combined, from fall 2015 to fall 2021,

**TABLE F.1.** Number of faculty, and of female faculty (F), in various types of mathematics departments and PhD statistics departments by highest degree and type of department in fall 2021. (Fall 2015 figures are in parentheses, and postdocs are included in other full-time (OFT) faculty totals.)

|                          | Univ (PhD)     |                 |                            |                | Univ (MA)      |                |                 |                            | Coll (BA)  |                |                |                 |
|--------------------------|----------------|-----------------|----------------------------|----------------|----------------|----------------|-----------------|----------------------------|------------|----------------|----------------|-----------------|
|                          | Tenured        | Tenure-Eligible | OFT                        | Post-Docs      | Part-Time      | Tenured        | Tenure-Eligible | OFT                        | Post-Docs  | Part-Time      | Tenured        | Tenure-Eligible |
| <b>Mathematics Depts</b> |                |                 |                            |                |                |                |                 |                            |            |                |                |                 |
| Doctoral faculty         | 4970<br>(4591) | 1167<br>(998)   | 4329<br>(2336)             | 1274<br>(1150) | 745<br>(588)   | 2543<br>(2309) | 672<br>(608)    | 846<br>(398)               | 26<br>(31) | 802<br>(441)   | 6829<br>(4780) | 2090<br>(1582)  |
| Doctoral (F)             | 926<br>(635)   | 338<br>(260)    | 1175<br>(652)              | 291<br>(234)   | 177<br>(151)   | 838<br>(587)   | 238<br>(244)    | 311<br>(146) <sup>2</sup>  | 8<br>(3)   | 380<br>(148)   | 2465<br>(1346) | 895<br>(614)    |
| Non-doctoral faculty     | 34<br>(5)      | 0<br>(0)        | 1056<br>(833)              | 0<br>(0)       | 1015<br>(857)  | 59<br>(56)     | 28<br>(10)      | 2201<br>(942)              | 0<br>(0)   | 1640<br>(1469) | 278<br>(238)   | 82<br>(93)      |
| Non-doctoral (F)         | 16<br>(2)      | 0<br>(0)        | 651<br>(480)               | 0<br>(0)       | 507<br>(361)   | 22<br>(18)     | 16<br>(9)       | 1,216<br>(540)             | 0<br>(0)   | 727<br>(686)   | 103<br>(99)    | 21<br>(45)      |
| Total Mathematics        | 5004<br>(4596) | 1167<br>(998)   | 5385<br>(3170)             | 1274<br>(1150) | 1760<br>(1445) | 2602<br>(2365) | 700<br>(618)    | 3047<br>(1339)             | 26<br>(31) | 2442<br>(1911) | 7107<br>(5018) | 2172<br>(1675)  |
| Total Mathematics (F)    | 942<br>(637)   | 338<br>(260)    | 1826<br>(1133)             | 291<br>(234)   | 684<br>(512)   | 860<br>(605)   | 254<br>(252)    | 1527<br>(686) <sup>2</sup> | 8<br>(3)   | 1107<br>(835)  | 2568<br>(1445) | 916<br>(659)    |
| <b>Statistics</b>        |                |                 |                            |                |                |                |                 |                            |            |                |                |                 |
| Doctoral faculty         | 1454<br>(649)  | 569<br>(220)    | 1115<br>(329) <sup>2</sup> | 178<br>(113)   | 443<br>(91)    |                |                 |                            |            |                |                |                 |
| Doctoral (F)             | 459<br>(137)   | 201<br>(71)     | 463<br>(107)               | 64<br>(22)     | 112<br>(19)    |                |                 |                            |            |                |                |                 |
| Non-doctoral faculty     | 3<br>(0)       | 4<br>(0)        | 251<br>(40) <sup>2</sup>   | 0<br>(0)       | 402<br>(37)    |                |                 |                            |            |                |                |                 |
| Non-doctoral (F)         | 0<br>(0)       | 4<br>(0)        | 168<br>(26) <sup>2</sup>   | 0<br>(0)       | 220<br>(19)    |                |                 |                            |            |                |                |                 |
| Total Statistics         | 1457<br>(649)  | 573<br>(220)    | 1366<br>(369)              | 178<br>(113)   | 845<br>(128)   |                |                 |                            |            |                |                |                 |
| Total Statistics (F)     | 459<br>(137)   | 205<br>(71)     | 631<br>(133) <sup>2</sup>  | 64<br>(22)     | 332<br>(38)    |                |                 |                            |            |                |                |                 |

<sup>1</sup> These 2015 values are updates made April 18, 2017. These changes are not large.

<sup>2</sup> These 2015 values are updated and changed from the CBMS report after April 18, 2017. These changes are not large.



**FIGURE F.1.1.** Number of mathematics faculty by type of appointment at four-year institutions from 1995 to 2021.

**TABLE F.1.1.** Number of faculty, and of female faculty (F), in mathematics departments combined and of doctoral-level statistics departments in fall 2021. (Fall 2015 figures are in parentheses.)

|                          | Tenured                            | Tenure-Eligible | OFT                         | Post-Docs      | Part-Time       |
|--------------------------|------------------------------------|-----------------|-----------------------------|----------------|-----------------|
| <b>Mathematics Depts</b> | Univ (PhD) + Univ (MA) + Coll (BA) |                 |                             |                |                 |
| Doctoral faculty         | 14341<br>(11681)                   | 3930<br>(3188)  | 6861<br>(3874) <sup>1</sup> | 1379<br>(1317) | 3633<br>(1940)  |
| Doctoral (F)             | 4230<br>(2568)                     | 1471<br>(1118)  | 2252<br>(1219) <sup>1</sup> | 326<br>(288)   | 1429<br>(588)   |
| Non-doctoral faculty     | 372<br>(298)                       | 110<br>(103)    | 6193<br>(3780)              | 0<br>(0)       | 6467<br>(5742)  |
| Non-doctoral (F)         | 141<br>(120)                       | 38<br>(54)      | 3580<br>(1903)              | 0<br>(0)       | 3225<br>(2659)  |
| Total Mathematics        | 14713<br>(11979)                   | 4040<br>(3291)  | 13054<br>(7261)             | 1379<br>(1317) | 10100<br>(7682) |
| Total Mathematics (F)    | 4371<br>(2688)                     | 1509<br>(1171)  | 5832<br>(3122) <sup>1</sup> | 326<br>(288)   | 4654<br>(3248)  |
| <b>Statistics Depts</b>  | Univ (PhD)                         |                 |                             |                |                 |
| Doctoral faculty         | 1454<br>(649)                      | 569<br>(220)    | 1115<br>(329) <sup>2</sup>  | 178<br>(113)   | 443<br>(91)     |
| Doctoral (F)             | 459<br>(137)                       | 201<br>(71)     | 463<br>(107)                | 64<br>(22)     | 112<br>(19)     |
| Non-doctoral faculty     | 3<br>(0)                           | 4<br>(0)        | 251<br>(40) <sup>2</sup>    | 0<br>(0)       | 402<br>(37)     |
| Non-doctoral (F)         | 0<br>(0)                           | 4<br>(0)        | 168<br>(26) <sup>2</sup>    | 0<br>(0)       | 220<br>(19)     |
| Total Statistics         | 1457<br>(649)                      | 573<br>(220)    | 1366<br>(369)               | 178<br>(113)   | 845<br>(128)    |
| Total Statistics (F)     | 459<br>(137)                       | 205<br>(71)     | 631<br>(133) <sup>2</sup>   | 64<br>(22)     | 332<br>(38)     |

<sup>1</sup> These 2015 values are updates made April 18, 2017. These changes are not large.

**TABLE F.2.** Number of tenured, tenure-eligible, postdoctoral, and other full-time (OFT) faculty in mathematics departments at four-year colleges and universities by gender and type of department in fall 2021. (Note: Postdoctoral faculty are included in other full-time totals.)

|                              | Univ (PhD) |                 |      |                        | Univ (MA) |                 |                  |                        | Coll (BA) |                 |      |                        | Total   |                 |                   |                        |
|------------------------------|------------|-----------------|------|------------------------|-----------|-----------------|------------------|------------------------|-----------|-----------------|------|------------------------|---------|-----------------|-------------------|------------------------|
|                              | Tenured    | Tenure-Eligible | OFT  | Post-Docs <sup>1</sup> | Tenured   | Tenure-Eligible | OFT              | Post-Docs <sup>1</sup> | Tenured   | Tenure-Eligible | OFT  | Post-Docs <sup>1</sup> | Tenured | Tenure-Eligible | OFT               | Post-Docs <sup>1</sup> |
| Men, 2021                    | 4056       | 824             | 3547 | 976                    | 1733      | 444             | 1518             | 18                     | 4525      | 1246            | 2117 | 53                     | 10314   | 2514            | 7182              | 1047                   |
| Women, 2021                  | 942        | 338             | 1825 | 291                    | 861       | 255             | 1527             | 8                      | 2568      | 916             | 2479 | 27                     | 4371    | 1508            | 5831              | 326                    |
| Nonbinary, 2021              | 5          | 5               | 12   | 7                      | 8         | 3               | 1                | 0                      | 14        | 10              | 27   | 0                      | 28      | 18              | 40                | 7                      |
| Total, 2021                  | 5003       | 1167            | 5384 | 1274                   | 2602      | 702             | 3046             | 26                     | 7107      | 2172            | 4623 | 80                     | 14713   | 4040            | 13053             | 1380                   |
| Men, 2015                    | 3958       | 739             | 2037 | 916                    | 1760      | 366             | 653 <sup>2</sup> | 28                     | 3573      | 1015            | 1450 | 85                     | 9292    | 2120            | 4140 <sup>2</sup> | 1030                   |
| Women, 2015                  | 637        | 260             | 1133 | 234                    | 605       | 252             | 686 <sup>2</sup> | 3                      | 1445      | 659             | 1303 | 51                     | 2688    | 1171            | 3122 <sup>2</sup> | 288                    |
| Nonbinary, 2015 <sup>3</sup> | N/A        | N/A             | N/A  | N/A                    | N/A       | N/A             | N/A              | N/A                    | N/A       | N/A             | N/A  | N/A                    | N/A     | N/A             | N/A               | N/A                    |
| Total, 2015                  | 4596       | 998             | 3170 | 1150                   | 2365      | 618             | 1339             | 31                     | 5018      | 1675            | 2752 | 137                    | 11979   | 3291            | 7261              | 1317                   |

<sup>1</sup> A postdoctoral appointment is a temporary position primarily intended to provide an opportunity to extend graduate training or to further research experience. Postdoctoral faculty are included in the other full-time faculty totals throughout CBMS2015. This contrasts with publications of the AMS-ASA-IMS-MAA-SIAM Annual Survey since 2003, which list postdoctoral faculty as a category separate from other full-time faculty. Before 2003, separate counts of postdoctoral faculty were not collected by the Annual Survey.

<sup>2</sup> These 2015 values are updated and changed from the CBMS report after April 18, 2017. These changes are not large.

<sup>3</sup> Nonbinary data was not collected in 2015, and thus is represented as N/A.

**TABLE F.3.** Number of tenured, tenure-eligible, other full-time, and postdoctoral faculty in doctoral-level statistics departments by gender in fall 2021 and 2015. (Postdoctoral faculty are included in other full-time faculty totals.)

|                              | Doctoral Statistics Departments |                 |                  |                        |
|------------------------------|---------------------------------|-----------------|------------------|------------------------|
|                              | Tenured                         | Tenure-Eligible | Other Full-Time  | Post-Docs <sup>1</sup> |
| Men, 2021                    | 1000                            | 365             | 737              | 114                    |
| Women, 2021                  | 459                             | 205             | 631              | 64                     |
| Nonbinary, 2021              | 0                               | 3               | 0                | 0                      |
| Total, 2021                  | 1459                            | 573             | 1368             | 178                    |
| Men, 2015                    | 512                             | 148             | 236 <sup>2</sup> | 91                     |
| Women, 2015                  | 137                             | 71              | 133 <sup>2</sup> | 22                     |
| Nonbinary, 2015 <sup>3</sup> | N/A                             | N/A             | N/A              | N/A                    |
| Total, 2015                  | 649                             | 220             | 369              | 113                    |

<sup>1</sup> A postdoctoral appointment is a temporary position primarily intended to provide an opportunity to extend graduate training or to further research experience. Postdoctoral faculty are included in the other full-time faculty totals throughout CBMS2010. This contrasts with publications of the AMS-ASA-IMS-MAA-SIAM Annual Survey since 2003, which list postdoctoral faculty as a category separate from other full-time faculty. Before 2003, separate counts of postdoctoral faculty were not collected by the Annual Survey.

<sup>2</sup> These 2015 values are updated and changed from the CBMS report after April 18, 2017. These changes are not large.

<sup>3</sup> Nonbinary data was not collected in 2015, and thus is represented as N/A.

the estimated number of other full-time faculty increased 80%. From Table F.2, from fall 2015 to fall 2021, the estimated number of other full-time faculty at doctoral-level mathematics departments increased 70%, at master's-level mathematics departments increased 127%, at bachelor's-level mathematics departments increased 68%, and, from Tables F.1.1 or F.3, at doctoral-level statistics departments increased 270%.

### Number of Postdoc Mathematics and Statistics Faculty

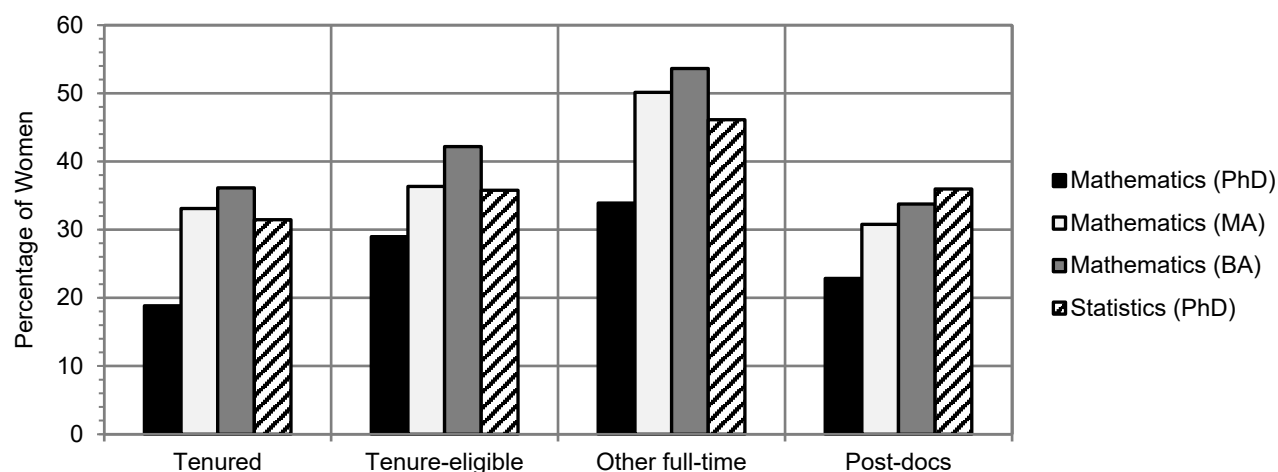
The estimated total number of postdocs across all levels of mathematics departments combined was 1317 in fall 2015 and 1379 in fall 2021 as seen in Table F.1.1. From Table F.2, we see that the estimated number of postdocs at bachelor's-level departments decreased from 2015 to 2021, while doctoral-level departments (where most postdocs reside) added an estimated 124 new postdocs. Doctoral-level statistics departments added an estimated 65 new postdocs from fall 2015 to fall 2021 according to Table F.3. In fall

2021, from Table F.2, we see that in doctoral-level mathematics departments, postdocs represented an estimated 24% of all other full-time positions (this percentage was estimated at 24% in fall 2015 also).

### Numbers of Part-Time Mathematics and Statistics Faculty

Across all levels of mathematics departments combined, by Table F.1.1, the estimated number of part-time faculty increased 31% (10 SEs) from fall 2015 to fall 2021. Broken down by level of department, by Table F.1, from fall 2015 to fall 2021, the estimated number of part-time faculty in doctoral-level mathematics departments increased 22% (2 SEs), in master's-level mathematics department increased 28% (4.6 SEs), and in bachelor's-level mathematics departments increased 36% (13 SEs). By Table F.1, in doctoral-level statistics departments, the estimated number of part-time faculty increased 560% (10 SEs) from fall 2015 to fall 2021.





**FIGURE F.3.1.** Percentage of women in various faculty categories by type of department in fall 2021.

## Non-Doctoral Faculty

According to Table F.1.1, most of the non-doctoral faculty are other full-time faculty or part-time faculty. From fall 2015 to fall 2021, across all levels of mathematics departments combined, the estimated number of non-doctoral full-time faculty increased by 60%, non-doctoral tenured faculty increased by 25% (3 SEs), non-doctoral tenure-eligible faculty increased by 7% (0.9 SEs), non-doctoral other full-time faculty increased by 64% (11 SEs), and non-doctoral part-time faculty increased by 13% (4 SEs). These numbers are broken down by level of department in Table F.1. From Table F.1, in statistics departments across all appointment types, from fall 2015 to fall 2021, the estimated total number of non-doctoral full-time faculty increased from 40 to 258; almost all the non-doctoral full-time statistics faculty are other full-time faculty. The number of non-doctoral statistics part-time faculty increased from 37 to 402.

## Gender

Table F.2 breaks down the fall 2021 total estimated numbers of full-time mathematics faculty in each of the four appointment types (tenured, tenure-eligible, other full-time, and postdoc) in all levels of mathematics departments combined by gender. The total percentage of full-time faculty who are women, over all levels of mathematics departments and all appointment types combined, increased from 31% in 2015 to 37% in 2021. In fall 2021, the estimated percentage of women among all tenured mathematics faculty, across all levels of mathematics departments combined, was estimated at 30%, up from 22% in fall 2015, and the percentage of women among all tenured-eligible mathemat-

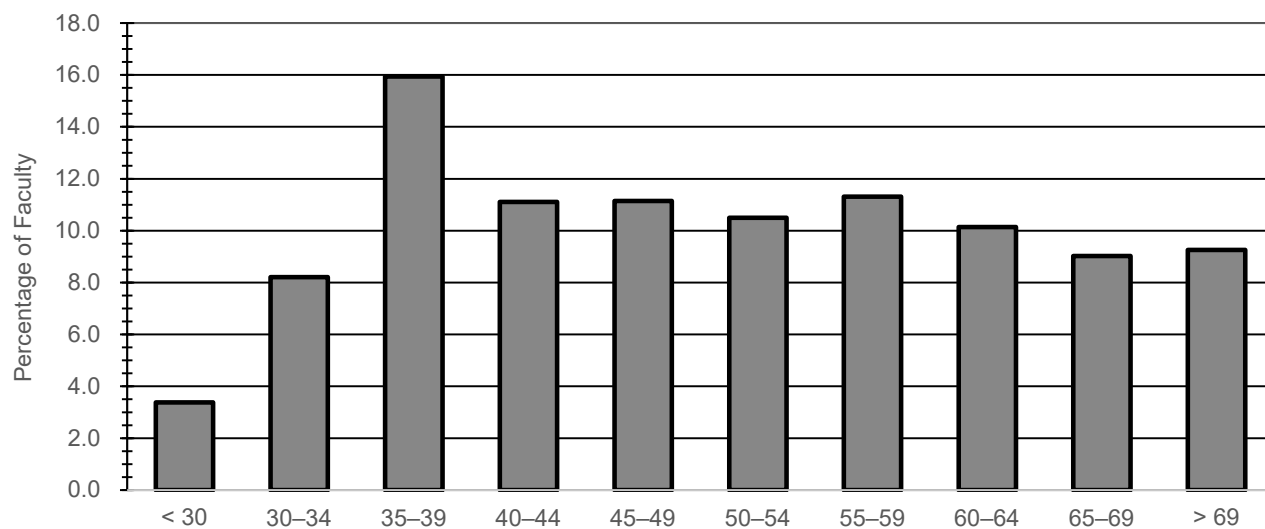
ics faculty, across all levels of mathematics departments combined, was estimated at 37%, slightly up from 36% in fall 2015.

Table F.2 provides estimated numbers of full-time faculty in mathematics departments, and Table F.3 provides estimated numbers of full-time faculty in doctoral-level statistics departments. Figure F.3.1, computed from these tables, displays data on both mathematics and statistics department faculty, giving the percentages of women in each of the full-time faculty appointment types (tenured, tenure-eligible, other full-time, and postdoc), broken down by the level of mathematics department (doctoral, master's, and bachelor's) and doctoral-level statistics departments. Women continue to represent a higher percentage of full-time faculty in doctoral statistics departments than in doctoral-level mathematics departments for each appointment type. In fall 2021, the percentage of women among all full-time appointment types combined was 27% at doctoral-level mathematics departments (up from 23% in fall 2015), 42% at master's-level departments (up from 39% in fall 2015), 43% at bachelor's-level departments (up from 36% in fall 2015), and, from Table F.3, 38% at doctoral-level statistics departments (up from 28% in fall 2015). From Table F.2, the percentage of women among tenured mathematics faculty in fall 2021 was 19% at doctoral-level mathematics departments (up from 14% in fall 2015), 33% at master's-level mathematics departments (up from 26% in fall 2015), 36% at bachelor's-level mathematics departments (up from 29% in fall 2015), and 31% at doctoral-level statistics departments (up from 21% in fall 2015). The percentage of women among tenure-eligible faculty is larger than the percentage of women among tenured faculty (29% at doctoral-level mathematics departments,

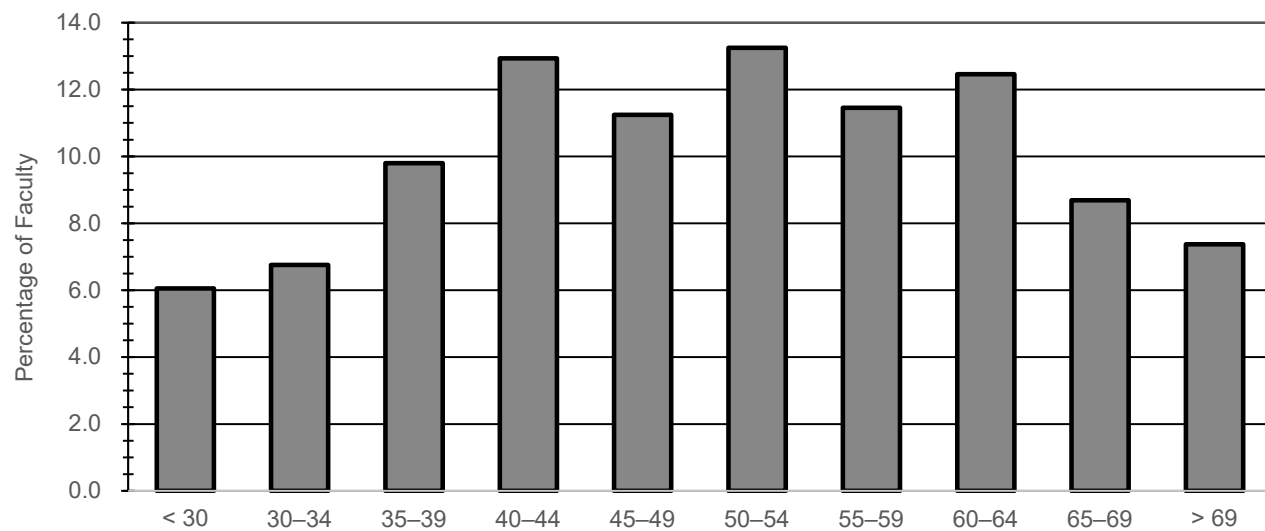
**TABLE F.4.** Percentage of tenured and tenure-eligible mathematics department faculty and statistics faculty at four-year colleges and universities belonging to various age groups by type of department and gender in fall 2021.

|                           | <30<br>% | 30–34<br>% | 35–39<br>% | 40–44<br>% | 45–49<br>% | 50–54<br>% | 55–59<br>% | 60–64<br>% | 65–69<br>% | >69<br>% | Average<br>Age 2010 | Average<br>Age 2015 | Average<br>Age 2021 |
|---------------------------|----------|------------|------------|------------|------------|------------|------------|------------|------------|----------|---------------------|---------------------|---------------------|
| <b>Mathematics Depts.</b> |          |            |            |            |            |            |            |            |            |          |                     |                     |                     |
| Univ (PhD)                |          |            |            |            |            |            |            |            |            |          |                     |                     |                     |
| Tenured men               | 0.0      | 0.9        | 4.1        | 5.5        | 6.0        | 7.3        | 7.4        | 6.5        | 7.3        | 7.1      | 55.4                | 55.9                | 56.1                |
| Tenured women             | 0.0      | 0.0        | 1.4        | 1.0        | 2.4        | 2.1        | 1.5        | 1.9        | 1.0        | 1.4      | 50.5                | 51.1                | 54.5                |
| Tenure-eligible men       | 2.4      | 4.5        | 7.2        | 3.2        | 1.8        | 0.8        | 1.2        | 1.4        | 0.5        | 0.5      | 36.3                | 36.0                | 41.2                |
| Tenure-eligible women     | 0.9      | 2.8        | 3.3        | 1.4        | 1.0        | 0.3        | 1.3        | 0.4        | 0.3        | 0.3      | 36.8                | 36.3                | 41.8                |
| Total Univ (PhD)          | 3.4      | 8.2        | 15.9       | 11.1       | 11.1       | 10.5       | 11.3       | 10.1       | 9.0        | 9.3      |                     |                     |                     |
| Univ (MA)                 |          |            |            |            |            |            |            |            |            |          |                     |                     |                     |
| Tenured men               | 0.0      | 0.3        | 2.3        | 4.4        | 4.2        | 6.5        | 5.2        | 7.3        | 3.5        | 5.3      | 54.1                | 55.1                | 56.3                |
| Tenured women             | 0.0      | 0.0        | 1.4        | 2.1        | 2.5        | 3.6        | 3.1        | 3.2        | 3.0        | 1.2      | 50.7                | 51.6                | 55.5                |
| Tenure-eligible men       | 4.1      | 4.7        | 3.8        | 2.9        | 2.8        | 2.2        | 1.6        | 0.6        | 1.8        | 0.6      | 37.3                | 36.1                | 42.8                |
| Tenure-eligible women     | 2.0      | 1.7        | 2.4        | 3.5        | 1.7        | 1.0        | 1.6        | 1.3        | 0.4        | 0.3      | 39.1                | 37.7                | 44.2                |
| Total Univ (MA)           | 6.1      | 6.8        | 9.8        | 12.9       | 11.2       | 13.2       | 11.5       | 12.5       | 8.7        | 7.4      |                     |                     |                     |
| Coll (BA)                 |          |            |            |            |            |            |            |            |            |          |                     |                     |                     |
| Tenured men               | 0.4      | 1.7        | 3.4        | 4.8        | 5.7        | 4.9        | 4.9        | 5.4        | 3.1        | 3.8      | 54.0                | 53.6                | 53.3                |
| Tenured women             | 0.0      | 1.4        | 2.8        | 2.8        | 2.8        | 2.6        | 3.1        | 1.8        | 1.6        | 0.8      | 50.9                | 50.8                | 50.6                |
| Tenure-eligible men       | 1.1      | 3.7        | 4.5        | 3.3        | 3.2        | 2.4        | 2.1        | 1.3        | 0.9        | 0.6      | 37.2                | 36.6                | 45.3                |
| Tenure-eligible women     | 2.1      | 3.4        | 3.3        | 2.6        | 2.1        | 1.6        | 1.7        | 1.0        | 1.0        | 0.5      | 37.4                | 37.0                | 44.0                |
| Total Coll (BA)           | 3.6      | 10.2       | 13.9       | 13.5       | 13.7       | 11.5       | 11.8       | 9.5        | 6.6        | 5.7      |                     |                     |                     |
| <b>Statistics Depts.</b>  |          |            |            |            |            |            |            |            |            |          |                     |                     |                     |
| Univ (PhD)                |          |            |            |            |            |            |            |            |            |          |                     |                     |                     |
| Tenured men               | 1.3      | 0.6        | 3.3        | 5.9        | 6.2        | 6.0        | 5.5        | 5.8        | 4.3        | 4.1      | 54.2                | 55.2                | 53.6                |
| Tenured women             | 0.0      | 0.0        | 1.2        | 4.8        | 3.3        | 3.2        | 2.3        | 0.6        | 2.7        | 0.0      | 48.1                | 48.0                | 51.1                |
| Tenure-eligible men       | 1.8      | 8.8        | 3.9        | 3.9        | 2.9        | 2.4        | 0.1        | 0.1        | 1.0        | 0.0      | 34.9                | 34.5                | 39.8                |
| Tenure-eligible women     | 1.4      | 2.2        | 3.2        | 1.6        | 2.3        | 1.0        | 1.5        | 0.9        | 0.0        | 0.2      | 36.2                | 34.4                | 43.1                |
| Total Univ (PhD)          | 4.4      | 11.5       | 11.7       | 16.1       | 14.6       | 12.6       | 9.4        | 7.4        | 7.9        | 4.3      |                     |                     |                     |

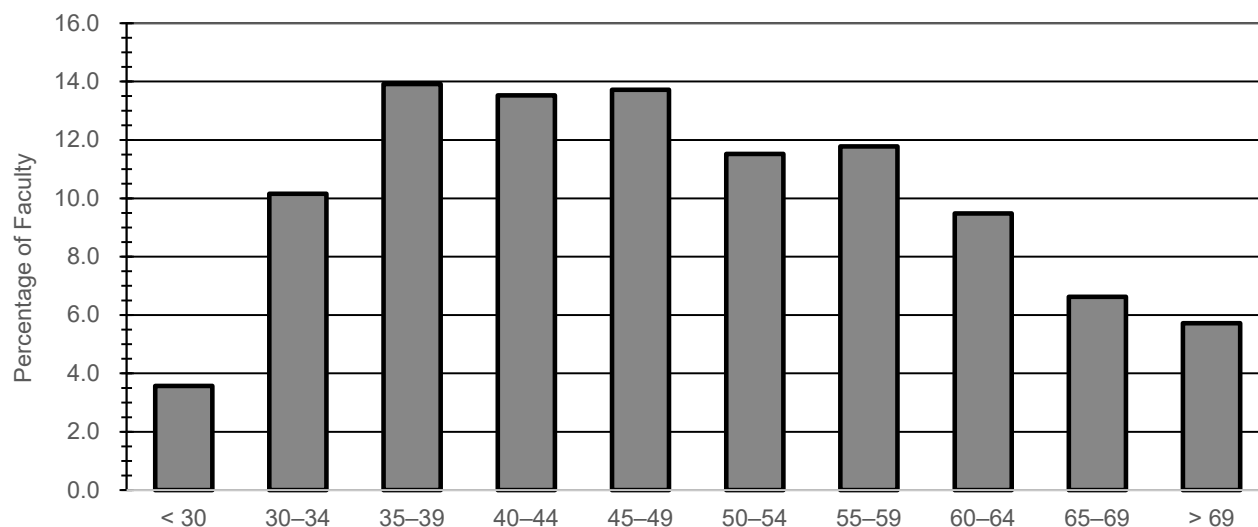
Note: 0 means less than one half of 1%.



**FIGURE F.4.1.** Percentage of tenured and tenure-eligible faculty in doctoral mathematics departments in various age groups in fall 2021.



**FIGURE F.4.2.** Percentage of tenured and tenure-eligible faculty in master's-level mathematics departments belonging to various age groups in fall 2021.



**FIGURE F.4.3.** Percentage of tenured and tenure-eligible faculty in bachelor's-level mathematics departments belonging to various age groups in fall 2021.

36% at master's-level mathematics departments, 37% at bachelor's-level mathematics departments, and 36% at doctoral-level statistics departments), suggesting that the percentage of tenured women will continue to rise.

### Age Distribution

Table S.16 in Chapter 1 presents the estimated age distribution of tenured, and of tenure-eligible, faculty broken down by gender for all levels of mathematics departments (combined) in fall 2021, and Table S.17 in Chapter 1 presents this same data for doctoral-level statistics departments. Tables S.16 and S.17 also show the average ages within each type of appointment (tenured or tenure-eligible) and each gender in fall 2010, 2015, and 2021 and, for each age group, the total percentages across all types of appointments in fall 2015, which are displayed in Figures S.16.1 and S.17.1 of Chapter 1. Table F.4 of this chapter presents the finer estimated breakdown of faculty ages by level of mathematics and statistics department, and Figures F.4.1, F.4.2, and F.4.3 display these distributions of ages, broken down by gender, for doctoral-, master's-, and bachelor's-level mathematics departments, respectively; these figures may be compared to those observed in the 2015 CBMS survey [CBMS2015, p. 124–5]. The percentages within each level of department total 100%, except for possible round-off errors. The standard errors (SEs) of the percentages in Tables S.16, S.17, F.4, and F.5 are relatively large, causing some uncertainty in the estimates.

From Table F.4 and Figure F.4.1, it appears that compared to fall 2015, in fall 2021, there are more younger and fewer older tenured or tenure-eligible faculty at doctoral-level mathematics departments. An estimated 27.5% (8.4% women, 19.1% men) of tenured or tenure-eligible faculty at doctoral-level mathematics departments were under 40 in fall 2021, while the corresponding percentage was 21% (9% women, 12% men) in 2015. Furthermore, 28.4% of tenured or tenure-eligible faculty at doctoral-level mathematics departments were 60 or above in fall 2021, compared to 31% in fall 2015. However, the estimated average age of doctoral-level faculty was slightly higher in fall 2021 than in fall 2015.

At master's-level mathematics departments, using Table F.4 and Figure F.4.2, it appears there are more older tenured or tenure-eligible faculty in fall 2021 than in fall 2015. An estimated 22.7% (7.5% women, 15.2% men) of tenured or tenure-eligible faculty were under 40 in fall 2021, while the corresponding percentage was 22% in 2015. Furthermore, an estimated 28.6% of tenured or tenure-eligible faculty at master's-level departments were 60 or above in fall 2021, compared to 26% in fall 2015. The estimated average age of tenured or tenure-eligible faculty at master's-level mathematics departments was higher in fall 2021 than in fall 2015.

At bachelor's-level mathematics departments, using Table F.4 and Figure F.4.3, it appears there are fewer older tenured or tenure-eligible faculty in fall 2021 than in fall 2015. An estimated 27.7% (13% women, 14.7% men) of tenured or tenure-eligible faculty were under 40 in fall 2021,

**TABLE F.5.** Percentages of full-time faculty belonging to various ethnic groups by gender and type of department in fall 2021. Except for round-off, the percentages within each departmental type sum to 100%.

|                             | Percentage of Full-Time Faculty |                             |                                                              |                             |                                   |              |
|-----------------------------|---------------------------------|-----------------------------|--------------------------------------------------------------|-----------------------------|-----------------------------------|--------------|
|                             | Asian<br>%                      | Black, Not<br>Hispanic<br>% | Mexican<br>American/<br>Puerto Rican/<br>Other Hispanic<br>% | White, Not<br>Hispanic<br>% | AIAN or<br>NHPI <sup>1</sup><br>% | Unknown<br>% |
| PhD Mathematics Departments |                                 |                             |                                                              |                             |                                   |              |
| All full-time men           | 16                              | 3                           | 3                                                            | 48                          | 0                                 | 3            |
| All full-time women         | 6                               | 0                           | 1                                                            | 18                          | 0                                 | 1            |
| MA Mathematics Departments  |                                 |                             |                                                              |                             |                                   |              |
| All full-time men           | 12                              | 3                           | 4                                                            | 37                          | 0                                 | 7            |
| All full-time women         | 7                               | 1                           | 2                                                            | 24                          | 0                                 | 3            |
| BA Mathematics Departments  |                                 |                             |                                                              |                             |                                   |              |
| All full-time men           | 12                              | 2                           | 3                                                            | 35                          | 1                                 | 5            |
| All full-time women         | 8                               | 2                           | 2                                                            | 26                          | 0                                 | 4            |
| PhD Statistics Departments  |                                 |                             |                                                              |                             |                                   |              |
| All full-time men           | 27                              | 1                           | 3                                                            | 30                          | 0                                 | 4            |
| All full-time women         | 13                              | 1                           | 3                                                            | 15                          | 0                                 | 2            |

<sup>1</sup> Includes the federal categories *American Indian or Alaskan Native* (AIAN) and *Native Hawaiian or Other Pacific* (NHPI).

Note: 0 means less than one half of 1%.

**TABLE F.6.** Percentages of part-time faculty belonging to various ethnic groups by gender and type of department in fall 2021. Except for round-off, the percentages within each departmental type sum to 100%.

|                             | Percentage of Part-Time Faculty |                             |                                                              |                             |                                   |              |
|-----------------------------|---------------------------------|-----------------------------|--------------------------------------------------------------|-----------------------------|-----------------------------------|--------------|
|                             | Asian<br>%                      | Black, Not<br>Hispanic<br>% | Mexican<br>American/<br>Puerto Rican/<br>Other Hispanic<br>% | White, Not<br>Hispanic<br>% | AIAN or<br>NHPI <sup>1</sup><br>% | Unknown<br>% |
| PhD Mathematics Departments |                                 |                             |                                                              |                             |                                   |              |
| All part-time men           | 6                               | 1                           | 0                                                            | 46                          | 1                                 | 2            |
| All part-time women         | 7                               | 0                           | 1                                                            | 36                          | 0                                 | 0            |
| MA Mathematics Departments  |                                 |                             |                                                              |                             |                                   |              |
| All part-time men           | 3                               | 4                           | 3                                                            | 46                          | 1                                 | 3            |
| All part-time women         | 3                               | 3                           | 3                                                            | 28                          | 0                                 | 2            |
| BA Mathematics Departments  |                                 |                             |                                                              |                             |                                   |              |
| All part-time men           | 8                               | 5                           | 5                                                            | 29                          | 1                                 | 7            |
| All part-time women         | 5                               | 4                           | 2                                                            | 29                          | 0                                 | 5            |
| PhD Statistics Departments  |                                 |                             |                                                              |                             |                                   |              |
| All part-time men           | 17                              | 3                           | 12                                                           | 34                          | 0                                 | 16           |
| All part-time women         | 7                               | 2                           | 1                                                            | 0                           | 0                                 | 8            |

<sup>1</sup> Includes the federal categories *American Indian or Alaskan Native* (AIAN) and *Native Hawaiian or Other Pacific Islander* (NHPI).

Note: 0 means less than one half of 1%.

while the corresponding percentage was 28% in fall 2015. Furthermore, an estimated 21.8% of tenured or tenure-eligible faculty at bachelor's-level departments were 60 or above in fall 2021, compared to 26% in fall 2015. The estimated average ages of tenure-eligible men and of tenure-eligible women were higher in fall 2021 than in 2015.

At doctoral-level statistics departments, it appears there are fewer younger and fewer older tenured or tenure-eligible faculty in fall 2021 than in fall 2015. Using Table F.4, it appears an estimated 27.6% (19.6% men, 8% women) of tenured or tenure-eligible faculty in doctoral-level statistics departments were under 40 in fall 2021, while the corresponding percentage was 33% (22% men, 11% women) in fall 2015. Furthermore, 19.6% of tenured or tenure-eligible faculty at doctoral-level statistics departments were 60 or above in fall 2021, compared to 22% in fall 2015. Except for tenured men, the estimated average ages were higher in fall 2021 than in 2015.

From Table S.16 (which contains the age distributions for all levels of mathematics departments combined) and Table S.17 (for doctoral-level statistics departments) in Chapter 1, we also note that the estimated percentage of tenured plus tenure-eligible faculty age 65 or over had been generally increasing. In four-year mathematics departments, this percentage was estimated at 5% in fall 2000, 8% in fall 2005, 12% in fall 2010, 13% in fall 2015, and 15% in fall 2021. In doctoral statistics departments, it was estimated at 7% in fall 2000, 8% in fall 2005, 11% in fall 2010, 13% in fall 2015, and 12% in fall 2021.

## Ethnicity and Gender

Table S.18 in Chapter 1 gives estimated percentages in various ethnic groups of all full-time faculty in all levels of mathematics departments combined, by gender and by type of appointment, in fall 2021; Table S.19 gives the same data for doctoral-level statistics departments. Table F.5 in this chapter presents these percentages broken down by the three levels of mathematics departments and doctoral statistics departments and by gender for all full-time types of appointment types combined. Table F.6 in this chapter presents the distribution of ethnic groups for part-time mathematics and statistics faculty. We note that in CBMS2015, data was collected for both doctoral- and master's-level statistics departments, and Tables S.19, F.5, and F.6 in CBMS2015 provide data on doctoral- and master's-level statistics departments combined, while in the current survey report, the data in these tables is for only doctoral-level statistics departments, making comparisons between 2015 and 2021 somewhat problematic for statistics departments. The SEs for percentages in Tables S.18, S.19, F.5, and F.6 are relatively large, causing some uncertainty in the estimates.

The Annual Surveys, from which these data are taken, follow the federal classification for racial and ethnic groups. However, in the text of this report, some of the more cumbersome federal classifications will be shortened. For example, "Mexican-American / Puerto Rican / other Hispanic" will be abbreviated to "Hispanic." Similarly, the federal classifications "Black, not Hispanic" and "White, not Hispanic" will be shortened to "Black" and "White," respectively, and "American Indian or Alaskan Native and Native Hawaiian / Pacific Islander" will be shortened to "AIAN & NHPI." For a small percentage of the faculty, race and ethnicity data were listed as "unknown" by the responding departments, and these faculty are listed as "unknown."

From Table F.5, we see some variation in the percentages of ethnic groups between mathematics departments and statistics departments, the most obvious difference being the percentage of Asian faculty, which in fall 2021, was estimated at 40% in doctoral-level statistics departments, 22% in doctoral-level mathematics departments, 19% in master's-level mathematics departments, and 20% in bachelor's-level mathematics departments. The estimated percentages of various racial/ethnic groups in each level of department in fall 2021 are not very different from those observed in fall 2015. The percentage of Black faculty is very slightly higher for all levels of departments, and the percentage of Hispanic faculty is the same or slightly higher for all levels of departments. Most notable is the continued decrease in the percentage of White men: at doctoral-level mathematics departments, it was estimated at 48% in fall 2021 and 55% in fall 2015; at master's-level departments, it was 37% in fall 2021 and 46% in 2015; at bachelor's-level mathematics departments, it was 35% in fall 2021 and 53% in fall 2015; and at doctoral-level statistics departments, it was 30% in fall 2021 and 45% in fall 2015.

From Table F.6, we see an increase in fall 2021 from fall 2015 in the estimated percentage of Black faculty among part-time faculty, except at the doctoral-level mathematics departments: doctoral-level mathematics departments (3% Black in 2015, and 1% Black in 2021); master's-level mathematics departments (4% Black in 2015, and 7% Black in 2021); bachelor's-level mathematics departments (2% Black in 2015, and 9% Black in 2021); statistics departments (3% Black in 2015 [all levels combined], and 5% Black in 2021 [doctoral-level]). Except for the doctoral-level departments, the estimated percentage of Black full-time faculty in fall 2021 reported in Table F.5 is less than the estimated percentage of Black part-time faculty in fall 2021 reported in Table F.6. The estimated percentage of White men part-time faculty reported in Table F.6 is smaller in fall 2021 than it was in fall 2015, except for the master's-level mathematics departments, where it is the same.



## Chapter 5

# First-Year Courses in Four-Year Colleges and Universities

The tables in this chapter explore the mathematics and statistics courses of four-year colleges and universities that are taught generally to beginning students. Tables S.5, S.6, S.7, S.8, and S.12 from Chapter 1 are broken down by the level of department in this chapter to provide more information about the following courses, which tend to be the focus of the early college experience:

- (1) Pre-college and Introductory-Level Mathematics (Appendix I)
- (2) Mainstream Calculus (Tables FY.1)
- (3) Non-Mainstream Calculus (Table FY.2)
- (4) Introductory Statistics (Tables FY.3–FY.8).

Beginning courses build the interest and skills that students need for further study of mathematics and statistics and for the many other disciplines that use mathematics or statistics. These courses constitute a substantial portion of four-year mathematics and statistics departments' course enrollments. Hence, these courses merit the careful consideration of the mathematical sciences community. The issues addressed in this chapter are the course enrollments, the format of the course (lecture/recitation, sections that meet as a class only, and other formats), the appointment type (tenured/tenure-eligible, other full-time, part-time, and graduate teaching assistant) of the course instructors, and pedagogy used in teaching Introductory Statistics. Note that issues involved in teaching beginning and introductory courses, and pathways through mathematics requirements that do not lead to calculus ("pathways sequences"), were addressed in Chapter 2.

**Enrollments:** Estimated enrollments for individual courses are given in Appendix I, but 2015 pre-college level course enrollments are gathered only as a group. Further information on enrollments in mathematics and statistics courses at four-year colleges and universities is given in Chapter 3. We note that the enrollments in Table E.2 of Chapter 3 include distance/remote learning enrollments, whereas the tables of this chapter and Chapter 1 generally do not. Tables A.1, A.2, and A.3 in Appendix I give the estimated enrollments (with distance/remote learning enrollments included) in fall 2005, 2010, 2015, and 2021 for each of the courses in the four-year mathematics and statistics questionnaires; they also present the non-distance/remote learning enrollments in fall 2015 and fall 2021 (except for

advanced-level mathematics courses where enrollments include distance/remote learning).

The Appendix I tables also give the estimated enrollments broken down by level of department (bachelor's, master's, or doctoral level) for enrollments in fall 2021; comparable breakdowns for fall 2010 and 2015 are given in the corresponding table of the CBMS 2015 report. In the discussion that follows, we present enrollments without distance/remote learning enrollments, as was done in the previous CBMS survey reports in this chapter, whenever these are available for some preceding years; we occasionally use enrollments with distance/remote learning included when necessary to compare to several previous years.

**Standard errors:** As the estimates produced from the survey data are broken down more finely, the estimates are made over smaller sets of departments, and the standard errors (SEs) typically increase, sometimes to magnitudes that make the estimates rather uncertain. This phenomenon occurs particularly in the master's-level mathematics and statistics departments, which are smaller in number and possibly less homogeneous, than the other levels of departments. In this chapter, data are broken down quite finely, and the SEs become an issue.

In the text that follows, the SE in many of the estimates is provided along with the estimate (e.g., enrollment of 255,000 (SE 23,000)); the SEs for all CBMS2015 tables can be found in Appendix VII. The change in an estimate from an estimate in a previous survey is often expressed both as percentage change and as the number of SEs that change represents (e.g., "increased 21% (1.7 SEs)").

## Highlights of Chapter 5

### A. Enrollments.

- As noted in Chapter 3, the estimated numbers of total enrollments in four-year mathematics department courses were generally significantly lower in fall 2021 than in fall 2015, even though enrollments in distance/remote learning courses were greater in fall 2021 than in fall 2015. (Chapter 3, Table E.2)
- The estimated number of total enrollments in pre-college level courses combined in all levels of mathematics departments combined in fall 2021 was less than half that in fall 2015 (and the estimated non-distance/remote learning enrollments in fall 2021



were about one-third of the non-distance/remote learning enrollments in fall 2015). (Appendix I and Chapter 3, Table E.4)

- Total estimated introductory-level course enrollments over all levels of departments combined also declined significantly in fall 2021 over fall 2015, even though distance/remote learning enrollments in introductory-level courses over that period tripled. (Appendix I and Chapter 3, Table E.4)
- Estimated enrollments in Mainstream Calculus I and II declined from fall 2015 to fall 2021, particularly at master's-level departments (Appendix I and Table FY.1), although distance/remote learning enrollments in these courses increased. The percentage of enrollments in lecture/recitation format, as opposed to classes that are not broken down into recitation sections, declined slightly from fall 2015 to fall 2021. (Table FY.1)
- Estimated enrollments in Non-Mainstream Calculus I continued the pattern of decline observed in the past three CBMS surveys, in fall 2021 declining 22% from fall 2015. Estimated fall 2021 Non-Mainstream Calculus II, III, etc. enrollment was roughly half the 2015 estimate. At doctoral-level departments in fall 2015, an estimated 46% of the Non-Mainstream Calculus I enrollment was in lecture/recitation format; in fall 2021, the estimate was 16%. (Table FY.2)
- Estimated enrollments in Introductory Statistics course enrollments in fall 2021 are down from fall 2015 in both mathematics and statistics departments, with bigger decreases in mathematics departments than in statistics departments. In both mathematics and statistics departments, a smaller percentage of the enrollments were delivered in lecture/recitation format in fall 2021 than in fall 2015. (Tables FY.3 and FY.4)
- Estimated total enrollment in Introductory Statistics courses across all levels of mathematics departments combined and all formats was 128,000 and across all levels of statistics departments was 67,000. (Tables FY.3 and FY.4)
- Estimated enrollments in Introductory Statistics courses and in data science courses taught outside of the surveyed departments were estimated. (Table FY.5)

## **B. Appointment type of instructors.**

- At doctoral-level departments, an estimated 22% (SE 4) of the Mainstream Calculus I instructors were tenured or tenure-eligible, while at the master's-level departments, the corresponding percentage was 48% (SE 11), and at the bachelor's-level departments, it was 64% (SE

6), comparable to the percentages observed in the 2015 CBMS survey. Generally, a larger percentage of sections of Mainstream Calculus II was taught by tenured or tenure-eligible faculty than percentages for Mainstream Calculus I. (Table FY.1)

- Across all Introductory Statistics courses taught in mathematics departments in fall 2021, the percentage of instructors who were tenured or tenure-eligible was 20% (SE 8) at doctoral-level mathematics departments, 37% (SE 8) at master's-level departments, and 54% (SE 7) at bachelor's-level departments. In statistics departments, the comparable percentages were 12% (SE 2) in doctoral-level statistics departments and 34% (SE 14) in master's-level statistics departments. These percentages are similar to those observed in the CBMS 2015 survey. (Tables FY.3 and FY.4)

## **C. Average section size.**

- The estimated average section sizes of Mainstream Calculus I and II classes generally declined slightly from fall 2015 to fall 2021 across all formats of sections and all levels of departments. (Table FY.1)
- In doctoral-level statistics departments, the estimated average section size of Introductory Statistics courses increased in fall 2021 over fall 2015. In doctoral-level mathematics departments, the estimated average section size of Introductory Statistics (non-calculus) in lecture/recitation format decreased, while the estimated average section size of the course in the individual course format increased. (Tables FY.3 and FY.4)

## **D. Pedagogy in Introductory Statistics.**

- Statistics departments typically offer more different kinds of Introductory Statistics with no calculus prerequisite courses than do mathematics departments. (Table FY.6)
- The reported pedagogy and use of technology in teaching Introductory Statistics with no calculus prerequisite across the various levels of four-year mathematics and statistics departments was quite similar. Many of the strategies advocated by statistical education groups such as CAUSE were reported to have been implemented. (Tables FY.7 and FY.8)

## **A. Appendix I and Chapter 1, Table S.10: Pre-College and Introductory-Level Courses**

As noted in Chapter 1, in recent years, there has been an increased interest in offering course options for beginning students other than those course sequences that lead

to calculus; these course sequences are often described as “pathways” course sequences. In the 2021 CBMS survey, four- and two-year mathematics departments were asked if they had offered “a non-traditional ‘pathways’ course sequence” in the last five years; “a pathways course sequence” was defined to be “a single course or course sequence that enables students to complete a college-level gateway mathematics or statistics course that is aligned to students’ academic and/or career goals within one academic year.” Table S.10 of Chapter 1 shows that in fall 2021, an estimated 71% (SE 6.5) of two-year mathematics departments and an estimated 22% (SE 4.5) of all levels of mathematics departments combined have offered a pathways sequence in the past five years, though 82% (SE 4) of four-year mathematics departments combined in fall 2021 offered some pre-college- or introductory-level course for credit. Table S.10 breaks this data down by level of the department and shows that these courses are more likely to be found in bachelor’s-level and master’s-level departments than in doctoral-level departments (in fall 2021, only an estimated 6% (SE 4.5) of doctoral-level departments indicated they had offered a pathways sequence). The CBMS 2015 survey found that 58% of two-year mathematics departments had implemented a pathways sequence (under a slightly different definition of pathways); four-year mathematics departments were not asked about pathways courses in 2015. Table S.10 also presents data on pedagogical strategies used in teaching pre-college and introductory classes, again providing a breakdown of this data by level of the four-year mathematics department.

The 2021 CBMS survey indicated that the estimated number of enrollments in pre-college-level courses in fall 2021 were about half of the number of enrollments observed in recent CBMS surveys; according to Appendix I, including distance/remote learning enrollments, the estimated numbers of enrollments in pre-college-level courses were 201,000 in fall 2005, 209,000 in fall 2010, 244,000 in fall 2015, and 101,000 (SE 16,200) in fall 2021. The non-distance/remote learning enrollments in pre-college-level courses were 89,000 in fall 2021 and 244,000 in fall 2015 (see also Table E.4 in Chapter 4). Comparing Table A.1 in the fall 2021 survey report to the corresponding table in the 2015 report, we see that this decline was about the same across each level of department. Comparing Table A.1 in 2021 and 2015, we see that all three levels of mathematics departments reported gains in estimated enrollments in distance/remote learning pre-college courses, especially at the bachelor’s-level departments.

The estimated number of enrollments in introductory-level courses also declined in fall 2021 over the numbers observed in recent CBMS surveys: according to Appendix I, including distance/remote learning enrollments, the estimated numbers of enrollments in all introductory-level

courses combined over all levels of departments combined were 706,000 in fall 2005, 863,000 in fall 2010, 1,000,000 in fall 2015, and 670,000 (SE 66,300) in fall 2021. According to Table E.4 in Chapter 4, distance/remote learning enrollments in all introductory-level courses combined over all levels of departments combined in fall 2021 were about three times those estimated enrollments in fall 2015. Appendix I breaks the introductory-level course enrollments down by course and by level of department, including distance/remote learning enrollments. The decline was larger at the doctoral- and master’s-level departments than at bachelor’s-level mathematics departments. Of the courses, Elementary Functions (which includes various precalculus courses) estimated enrollment in fall 2021 was comparable to the number of enrollments observed in previous surveys, and the estimated number of the enrollments in Introduction to Math Modeling in fall 2021 increased over fall 2015, as it has been increasing over the past few CBMS surveys. Estimated enrollments in “Math for the Liberal Arts” and “Finite Math” were both down significantly in fall 2021 over fall 2015.

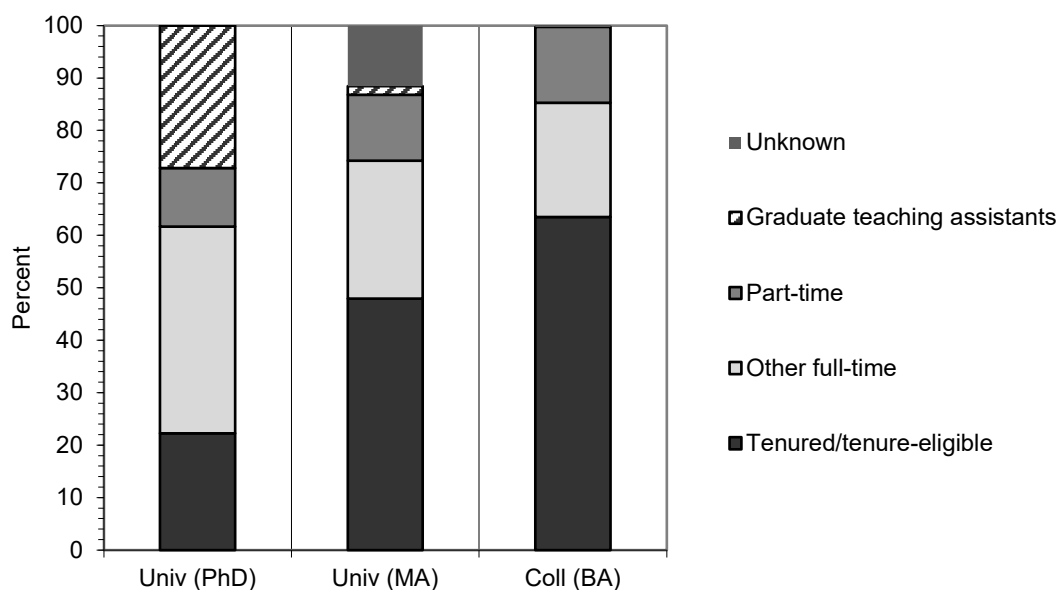
## **B. Table FY.1 and Figure FY.1.1: Mainstream Calculus**

“Mainstream Calculus” courses are the calculus courses needed for the mathematics major or for applications in the physical sciences or engineering. Other calculus courses, which tend to be for business, social science, or life science majors, are labeled “Non-Mainstream Calculus.”

Table FY.1 presents information on the appointment type (tenured/tenure-eligible, other full-time, part-time, or graduate teaching assistant) of the course instructors, the average section sizes, and the enrollments (excluding distance/remote learning sections) in Mainstream Calculus I and Mainstream Calculus II in fall 2021, broken down by the format of course sections and level of the department. This table containing data on Mainstream Calculus in fall 2021 can be compared to CBMS2015, Table FY.1, p. 136 and CBMS2015, Figure FY.1.1, p. 137.

Comparing Table FY.1 in the CBMS 2021 report and the corresponding table in the 2015 survey report, we see that estimated enrollments in Mainstream Calculus I courses declined from fall 2015 to fall 2021, particularly at master’s-level departments; in fall 2021, the estimated number of enrollments in Mainstream Calculus I (not including distance/remote learning sections) was estimated at 129,000 (SE 22,000) at doctoral-level departments, 23,000 (SE 5000) at master’s-level departments, and 57,000 (SE 9000) at bachelor’s-level departments; the comparable estimates in fall 2015 were 134,000 at doctoral-level departments, 58,000 at master’s-level departments, and 63,000 at bachelor’s-level departments. From the Appendix I tables, we can see





**FIGURE FY.1.1.** Percentage of sections (excluding distance learning) in Mainstream Calculus I in four-year mathematics departments by type of instructor and by type of department in fall 2021.

Note: Faculty sections do not add to 100% because some institutions reported fewer sections by individual faculty type than the total.

that although total enrollments declined, distance/remote learning enrollments increased from fall 2015 to fall 2021; in fall 2015, in the bachelor's- and master's-level departments, the distance/remote learning enrollments in Mainstream Calculus I were under 1000, while in fall 2021, they were approximately 4000 and 9000, respectively.

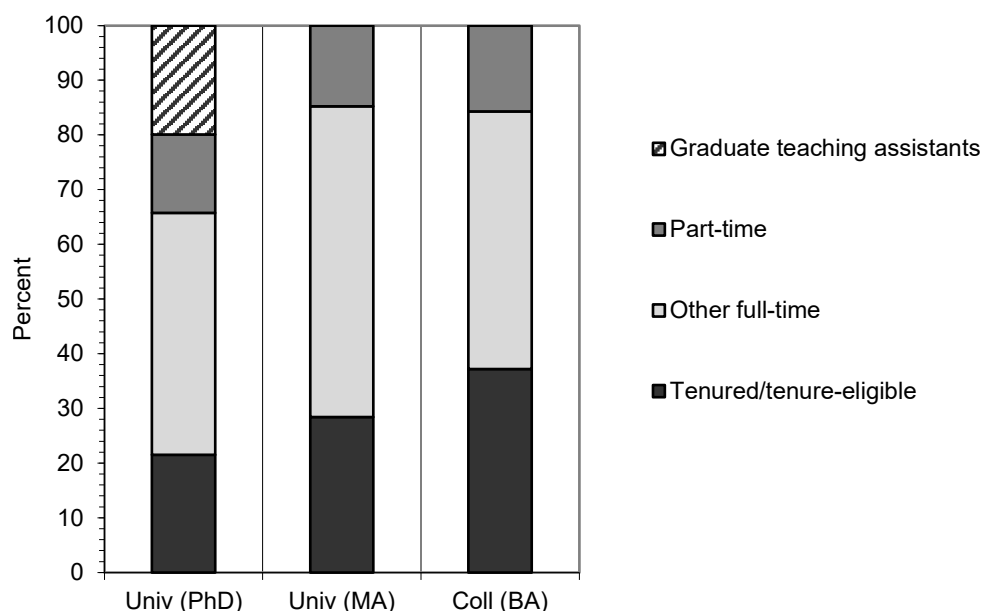
Similarly, the estimated enrollments in Mainstream Calculus II courses also declined; in fall 2021, the estimated number of enrollments in Mainstream Calculus II (not including distance/remote learning sections) was estimated at 46,000 (SE 7000) at doctoral-level departments, 10,000 (SE 2000) at master's-level departments, and 22,000 (SE 5000) at bachelor's-level departments; the comparable estimates in fall 2015 were 76,000 at doctoral-level departments, 21,000 at master's-level departments, and 29,000 at bachelor's-level departments, and there were increased distance/remote learning enrollments in fall 2021 as well.

Table FY.1 breaks down Mainstream Calculus I and II courses by the type of format: lecture/recitation, sections that meet as a class (with no breakdown into recitation sections), and other sections. Comparing Table FY.1 in 2021 and Table FY.1 in 2015, it appears that enrollments in lecture/recitation sections decreased, and more enrollments occurred in the individual class format, across all levels of departments. Table FY.1 shows that about 66% of the Main-

stream Calculus I enrollment at doctoral-level mathematics departments is taught in the lecture/recitation format (compared with 71% in fall 2015), while almost all the Mainstream Calculus enrollment at master's- and bachelor's-level departments is in the sections that meet as a class format. The average section sizes of Mainstream Calculus I and II classes also generally decreased slightly from 2015 to 2021 across the three formats and three levels of departments; as one example, the average section size of a lecture/recitation Mainstream Calculus I class at a doctoral-level mathematics department was estimated at 98 in fall 2015 and 85 (SE 17) in fall 2021.

Table FY.1 gives information on the appointment types of the Mainstream Calculus I and II instructors, indicating a similar distribution of appointment types of instructors in 2015 and 2021. At doctoral-level departments, 22% (SE 4) of the Mainstream Calculus I instructors were tenured or tenure-eligible, while at the master's-level departments, the corresponding percentage was 48% (SE 11), and at the bachelor's-level departments, it was 64% (SE 6). Generally, the percentage of faculty teaching Mainstream Calculus II who are tenured or tenure-track is greater than the percentage of faculty teaching Mainstream Calculus I who are tenured or tenure-track.





**FIGURE FY.2.1.** Percentage of sections (excluding distance learning sections) in Non-Mainstream Calculus I in four-year mathematics departments taught by various kinds of instructors in fall 2021 by type of department.

### C. Table FY.2 and Figure FY.2.1: Non-Mainstream Calculus

Non-Mainstream Calculus is the flavor of calculus that is not a part of the calculus sequence for mathematical and physical science majors, and tends to be for business, social science, or life science majors.

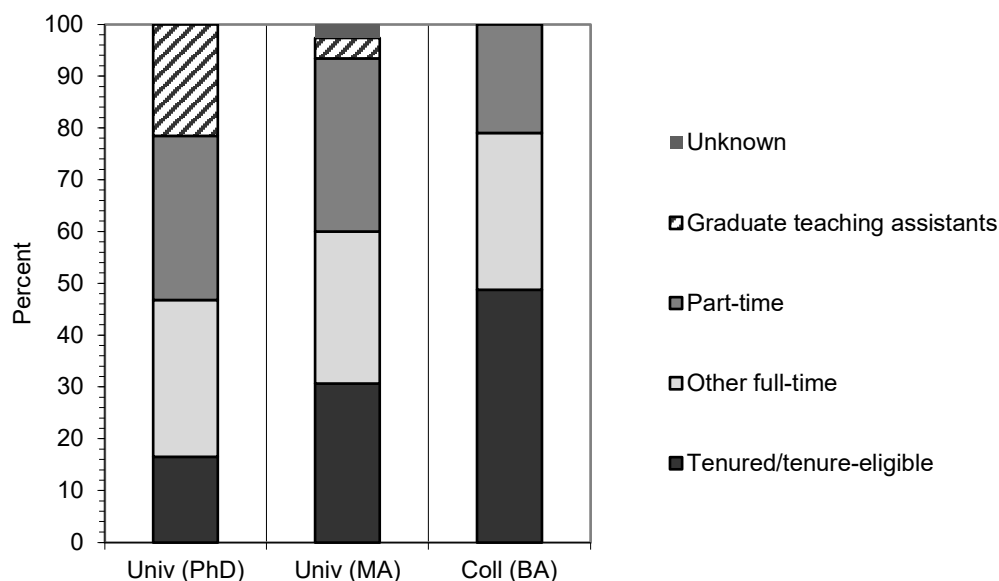
Table FY.2 presents information on the appointment type (tenured/tenure-eligible, other full-time, part-time, or graduate teaching assistant) of the course instructors, the average section sizes, and the enrollments (excluding distance/remote learning sections) in Non-Mainstream Calculus I and Non-Mainstream Calculus II, III etc. in fall 2021, broken down by the format of course sections and level of the department. This table containing data on Non-Mainstream Calculus in fall 2021 can be compared to CBMS2015, Table FY.2, p. 139 and CBMS2015, Figure FY.2.1, p. 140.

Enrollments in Non-Mainstream Calculus I have been declining over the past three CBMS surveys. According to Table FY.2, Non-Mainstream Calculus I had an estimated (non-distance/remote learning) enrollment in fall 2021 of roughly 71,000 (SE 12,200), compared to the fall 2015 estimate 91,000, the fall 2010 estimate 99,000 (CBMS2010,

Chapter 1, Table S.6), and the fall 2005 estimate 108,000; the fall 1995 estimate was 97,000 (CBMS2005, Chapter 1, Table S.8, p. 19). Non-Mainstream Calculus II, III etc. enrollment was estimated at 7000 (SE 3700), roughly half of the fall 2015 estimate.

Table FY.2 breaks down Non-Mainstream Calculus I and II courses by the type of format: lecture/recitation, sections that meet as a class (with no breakdown into recitation sections), and other sections. At doctoral-level departments in fall 2015, an estimated 46% of the Non-Mainstream Calculus I enrollment was in lecture/recitation format; in fall 2021, the estimate was 16%. However, the estimated average class size of the Non-Mainstream Calculus I classes at doctoral-level departments increased in fall 2021 over fall 2015. At master's- and bachelor's-level departments, most of the Non-Mainstream Calculus I and II enrollment continued to be the format of sections that meet as a class. The average section sizes of Non-Mainstream Calculus classes decreased slightly from 2015 to 2021 across bachelor's- and master's-levels of mathematics departments. The estimated percentages of tenured/tenure-eligible instructors in Non-Mainstream Calculus in fall 2021 are roughly the same over all levels of departments.





**FIGURE FY.3.1.** Percentage of sections (excluding distance learning sections) in Introductory Statistics (non-Calculus) in four-year mathematics departments by type of instructor and type of department in fall 2021.

Note: Faculty sections do not add to the total because some institutions reported a greater number of sections taught by each faculty type than total sections.

#### D. Table FY.3, FY.4, and FY.5: Introductory Statistics

Table FY.3 considers data on Introductory Statistics courses taught within a mathematics department in fall 2021, while Table FY.4 has data for these same courses when taught within a statistics department. The introductory courses that are considered are an introductory course with no calculus prerequisite, one with a calculus prerequisite that is for non-majors/minors, a statistics course for pre-service teachers, an intermediate non-calculus-based course, and any other introductory-level statistics courses. The first two introductory courses are broken down by the format of the course (lecture/recitation, meets as a class, and other), and all the data is broken down by level of department. Table FY.3, Figure FY.3.1, Table FY.4, and Figure FY.4.1 can be compared to the CBMS 2015 tables with the same numbers on p. 141, 142, 143, and 144, respectively, of the CBMS 2015 report. Table FY.5 gives mathematics and statistics departments fall 2021 estimates of the enrollment in Introductory Statistics courses taught outside the given departments; this table can be compared to CBMS2015, Table FY.9, p. 151.

The majority of estimated Introductory Statistics enrollments are in mathematics departments; estimated total enrollment in Introductory Statistics courses in fall 2021 across all levels of mathematics departments combined and all formats, by Table FY.3, was 128,000 and across all levels of statistics departments combined, by Table FY.4, was 67,000. In mathematics departments, every estimated number of enrollments in Table FY.3 for fall 2021 is below the number estimated in fall 2015. Combining all Introductory Statistics course enrollments, comparing Table FY.3 in 2021 and its corresponding table in 2015, we see that estimated total Introductory Statistics enrollments are down 58% at doctoral-level mathematics departments, 45% at master's-level mathematics departments, and 40% at bachelor's-level mathematics departments. Comparing Table FY.4 in 2021 and its corresponding table in 2015, we see that estimated total Introductory Statistics enrollments are down 28% at doctoral-level statistics departments and 7% at master's-level statistics departments. In both mathematics and statistics departments, a smaller percentage of the Introductory Statistics (both with and without calculus) enrollments in fall 2021 are in the lecture/recitation

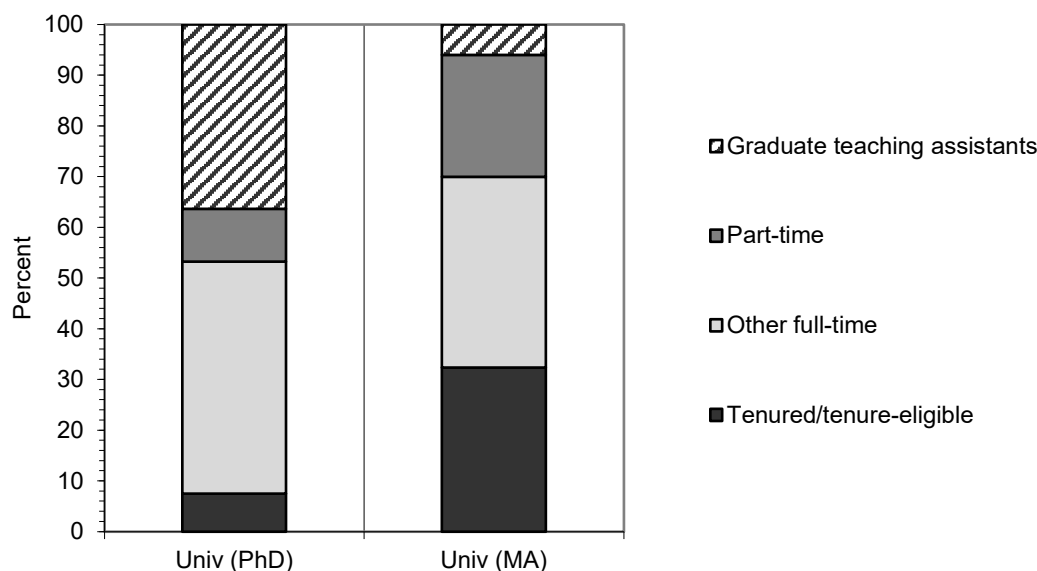


**TABLE FY.4.** Percentage of sections (excluding distance learning sections) in Introductory Statistics courses (for non-majors) taught by various types of instructors in statistics departments in fall 2021, by size of sections and type of department. Also average section size and total (non-distance learning) enrollments.

| Percentage of Sections Taught by                                           |                                  |     |                                       |     |                |     |                                         |     |              |    |                            |    |                       |    |
|----------------------------------------------------------------------------|----------------------------------|-----|---------------------------------------|-----|----------------|-----|-----------------------------------------|-----|--------------|----|----------------------------|----|-----------------------|----|
|                                                                            | Tenured/<br>Tenure-Eligible<br>% |     | Other<br>Full-Time<br>(with PhD)<br>% |     | Part-Time<br>% |     | Graduate<br>Teaching<br>Assistants<br>% |     | Unknown<br>% |    | Average<br>Section<br>Size |    | Enrollment<br>(1000s) |    |
|                                                                            | PhD                              | MA  | PhD                                   | MA  | PhD            | MA  | PhD                                     | MA  | PhD          | MA | PhD                        | MA | PhD                   | MA |
| Course & Statistics Department Type                                        |                                  |     |                                       |     |                |     |                                         |     |              |    |                            |    |                       |    |
| Introductory Statistics (Non-Calculus for Non-Majors/Minors) (E1)          |                                  |     |                                       |     |                |     |                                         |     |              |    |                            |    |                       |    |
| Lecture with separate recitation                                           | 9%                               | 11% | 57%                                   | 69% | 8%             | 20% | 26%                                     | 0%  | 0%           | 0% | 87                         | 72 | 18                    | 4  |
| Sections that meet as a class                                              | 6%                               | 40% | 37%                                   | 27% | 13%            | 26% | 45%                                     | 8%  | 0%           | 0% | 67                         | 43 | 20                    | 7  |
| Other sections                                                             | 50%                              |     | 50%                                   |     | 0%             |     | 0%                                      |     | 0%           |    | 30                         |    | 0                     |    |
| Total Introductory Statistics (Non-Calculus)                               | 7%                               | 32% | 46%                                   | 38% | 10%            | 24% | 36%                                     | 6%  | 0%           | 0% | 74                         | 50 | 38                    | 12 |
| Introductory Statistics (Calculus Prerequisite for Non-Majors/Minors) (E2) |                                  |     |                                       |     |                |     |                                         |     |              |    |                            |    |                       |    |
| Lecture with separate recitation                                           | 13%                              |     | 81%                                   |     | 0%             |     | 0%                                      |     | 6%           |    | 106                        |    | 3                     |    |
| Sections that meet as a class                                              | 25%                              | 45% | 57%                                   | 48% | 4%             | 7%  | 14%                                     | 0%  | 0%           | 0% | 78                         | 42 | 9                     | 2  |
| Other sections                                                             |                                  |     |                                       |     |                |     |                                         |     |              |    |                            |    | 0                     |    |
| Total Introductory Statistics (Calculus)                                   | 22%                              | 45% | 61%                                   | 48% | 4%             | 7%  | 12%                                     | 0%  | 1%           | 0% | 84                         | 42 | 12                    | 2  |
| Statistics for Pre-Service Teachers (E3,E4)                                |                                  |     |                                       |     |                |     |                                         |     |              |    |                            |    |                       |    |
| Intermediate Statistics (Non-Calculus) (E5)                                | 21%                              |     | 54%                                   |     | 4%             |     | 21%                                     |     | 0%           |    | 70                         |    | 2                     |    |
| Other Introductory-Level Statistics                                        | 25%                              | 14% | 25%                                   | 0%  | 50%            | 0%  | 0%                                      | 86% | 0%           | 0% | 49                         | 68 | 0                     | 1  |
| Total, All Introductory Probability & Statistics Courses                   | 11%                              | 34% | 49%                                   | 38% | 9%             | 20% | 30%                                     | 8%  | 0%           | 0% | 76                         | 50 | 53                    | 14 |

Note: 0 means less than one half of 1%. 0 enrollment means under 500. Row and column sums may appear inconsistent due to round-off. Blank entries represent sections with no responses.

Note: Faculty sections do not add to the total because some institutions reported a greater number of sections taught by each faculty type than total sections.



**FIGURE FY.4.1.** Percentage of sections (excluding distance learning sections) in Introductory Statistics (non-Calculus) taught in statistics departments by type of instructor and type of department in fall 2021.

Note: Faculty sections do not add to the total because some institutions reported a greater number of sections taught by each faculty type than total sections.

format than was observed in fall 2015. For example, in fall 2021, doctoral-level mathematics department enrollments in the lecture/recitation format of Introductory Statistics (non-calculus) represented 6% of this courses' enrollments, while in fall 2015, the lecture/recitation format was 37% of this courses' enrollments; in doctoral-level statistics departments, this percentage was 48% in fall 2021 and 65% in fall 2015.

Comparing Table FY.4 in 2021 and its corresponding table in 2015, we see that in doctoral-level statistics departments, the estimated average section size generally increased in fall 2021 over fall 2015, while at master's-level statistics departments, the average section size decreased over this time-period. For example, across all doctoral-level statistics departments combined, by Table FY.4, the estimated average section size of all Introductory Statistics sections combined in fall 2021 was 76 (SE 8), while in fall 2015, it was 59. Comparing Table FY.3 in 2021 and its corresponding table in 2015, we see that in doctoral-level mathematics departments, the estimated average section size in the lecture/recitation format sections of Introductory Statistics (non-calculus) declined from 141 in fall 2015 to 58 (SE 9) in fall 2021, but the average section size of the sections that meet as a class increased from 30 in fall 2015 to 54 (SE 6) in fall 2021.

The distribution of appointment types of instructors in all Introductory Statistics courses combined in fall 2021 was like that observed in fall 2015 in both mathematics and statistics departments. According to Table FY.3, over all formats of all Introductory Statistics courses combined, the percentage of instructors in fall 2021 who were tenured or tenure-eligible was 20% (SE 8) in doctoral-level mathematics departments, 37% (SE 8) in master's-level mathematics departments, and 54% (SE 7) in bachelor's-level mathematics departments. In statistics departments, from Table FY.4, the comparable percentages were 12% (SE 2) in doctoral-level statistics departments and 34% (SE 14) in master's-level statistics departments.

In Table FY.5, mathematics and statistics departments were asked to estimate the enrollment in Introductory Statistics courses taught at their institution, but not in their departments. This table can be compared to CBMS2015, Table FY.9, p. 151. In the 2021 CBMS survey, departments were also asked to estimate data science enrollments taught at their institution, but not in their department. From these numbers, weighted averages were computed to estimate the national total. These estimates are broken down by level of departments.

**TABLE FY.5.** Percentage of mathematics departments using various practices in the teaching of Introductory Statistics (no calculus prerequisite) in fall 2021 by type of department.

|                                                                                                           | Mathematics Departments |              |                 |                        | Statistics Departments |              |                        |
|-----------------------------------------------------------------------------------------------------------|-------------------------|--------------|-----------------|------------------------|------------------------|--------------|------------------------|
|                                                                                                           | Univ<br>(PhD)           | Univ<br>(MA) | College<br>(BA) | All Depts.<br>Combined | Univ<br>(PhD)          | Univ<br>(MA) | All Depts.<br>Combined |
| Percentage of departments that offer Introductory Statistics with no calculus prerequisite                | 38                      | 80           | 81              | 76                     | 83                     | 88           | 84                     |
| Number of different kinds of Introductory Statistics courses for non-majors with no calculus prerequisite |                         |              |                 |                        |                        |              |                        |
| 1                                                                                                         | 61                      | 61           | 71              | 69                     | 13                     | 38           | 18                     |
| 2                                                                                                         | 20                      | 39           | 22              | 25                     | 31                     | 38           | 33                     |
| 3                                                                                                         | 20                      |              | 3               | 4                      | 28                     |              | 23                     |
| More than 3                                                                                               |                         |              | 3               | 2                      | 28                     | 25           | 28                     |

Note: Blank entries represent practices with no responses.

**TABLE FY.6.** Instructional strategies used in Introductory Statistics courses.

|                                                                                         | Mathematics Departments    |              |                 | Statistics Departments     |              |                 |
|-----------------------------------------------------------------------------------------|----------------------------|--------------|-----------------|----------------------------|--------------|-----------------|
|                                                                                         | At Least<br>Once a<br>Week | Occasionally | Almost<br>Never | At Least<br>Once a<br>Week | Occasionally | Almost<br>Never |
| a. Focusing on conceptual understanding over formulas and procedures                    | 78                         | 20           | 1               | 87                         | 13           |                 |
| b. Integrating real world applications                                                  | 85                         | 14           | 1               | 87                         | 13           |                 |
| c. Students collect, organize, and analyze real data                                    | 20                         | 65           | 15              | 26                         | 64           | 10              |
| d. Using student-centered active learning strategies                                    | 42                         | 48           | 10              | 49                         | 49           | 3               |
| e. Using assessments such as regular graded homework or quizzes used to inform teaching | 90                         | 8            | 2               | 85                         | 15           |                 |

Note: Blank entries represent strategies with no responses.

**TABLE FY.7.** Extent of success in the use of technology in Introductory Statistics.

|                                                                      | Mathematics Departments |                     |                | Statistics Departments |                     |                |
|----------------------------------------------------------------------|-------------------------|---------------------|----------------|------------------------|---------------------|----------------|
|                                                                      | Very Successful         | Somewhat Successful | Not Successful | Very Successful        | Somewhat Successful | Not Successful |
| a. Students use technology to explore concepts                       | 47                      | 52                  | 1              | 64                     | 31                  | 5              |
| b. Instructors use technology to demonstrate concepts                | 74                      | 25                  | 1              | 74                     | 26                  |                |
| c. Students use technology to analyze data                           | 62                      | 36                  | 2              | 67                     | 28                  | 5              |
| d. Students' ability to use technology to solve problems is assessed | 51                      | 46                  | 3              | 51                     | 46                  | 3              |

Note: Blank entries represent uses with no responses.

**TABLE FY.8.** Of departments that offered Introductory Statistics (no calculus prerequisite) in fall 2021 and where a similar course is offered outside the mathematical sciences departments, the average estimated fall 2021 enrollment of all similar courses and an estimate of the total national enrollment.

|                                                                   | Mathematics Depts |           |              |        | Statistics Depts |           |       |
|-------------------------------------------------------------------|-------------------|-----------|--------------|--------|------------------|-----------|-------|
|                                                                   | Univ (PhD)        | Univ (MA) | College (BA) | Total  | Univ (PhD)       | Univ (MA) | Total |
| Average estimated outside enrollment in Introductory Statistics   | 655               | 158       | 76           | 149    | 160              | 185       | 162   |
| Estimated outside national enrollment in Introductory Statistics  | 60091             | 15677     | 49378        | 125146 | 2750             | 318       | 3068  |
| Average estimated outside enrollment in Introductory Data Science | 200               | 0         | 26           | 78     | 118              |           | 118   |
| Estimated outside national enrollment in Data Science             | 16052             | 0         | 4376         | 20428  | 2848             |           | 2848  |

Note: The estimates for statistics departments are for colleges with separate statistics departments. Since such colleges would be expected to also have mathematics departments, adding statistics for both types of departments together would result in duplicating the counts of some students.

### E. Tables FY.6, FY.7, and FY.8: Pedagogy in Introductory Statistics

The CMBS 2021 survey asked mathematics departments at two-year colleges and four-year colleges and four-year college statistics departments questions about how Introductory Statistics with no calculus prerequisite was taught in fall 2021; this data also appeared in Chapter 1, Tables S.11 and S.12. The focus here is on the differences between levels of four-year mathematics departments and statistics departments. Table FY.6 shows the estimated percentages of departments that offer Introductory Statistics with no calculus prerequisite in fall 2021, and the number of different kinds of courses offered. Table FY.7 focuses on how the course is taught, while Table FY.8 considers the use of technology in the courses. The CBMS 2015 survey asked a slightly different set of questions of four-year mathematics and statistics departments regarding instruction in Introductory Statistics courses; that data can be found in CBMS2015 Tables S.12 (p. 29), FY.5 (p. 145), and FY.6 (p. 146).

Table FY.6 shows that in fall 2021, Introductory Statistics with no calculus prerequisite is taught primarily at master's- and bachelor's-level mathematics departments and at both levels of statistics departments; only an estimated 38% (SE 10) of doctoral-level mathematics departments indicated they offered such a course in fall 2021, and as a result, the SEs on their responses are often quite large. Of all levels of mathematics departments combined offering such a course in fall 2021, an estimated 69% (SE 6) offered only one such course; this percentage was nearly the same across all levels of mathematics departments. Among all levels of statistics departments combined, only 18% (SE 6) offered only one such course, and 51% offered three or more different Introductory Statistics courses with no calculus prerequisite.

The Consortium for the Advancement of Undergraduate Statistics Education (CAUSE) <https://www.causeweb.org/cause/> is among the organizations promoting changes

to how Introductory Statistics courses are taught. The CBMS 2021 survey probed the extent to which such changes had been implemented in mathematics and statistics departments. Table FY.7 shows that very similar pedagogical methods were reported among four-year mathematics and statistics departments in fall 2021. At least once a week, nearly all levels of four-year mathematics and statistics departments used assessments such as regular graded homework or quizzes to inform teaching, they integrated real world applications, and they emphasized conceptual understanding over formulas and procedures. About half of the departments indicated that they used student-centered active learning strategies at least once a week. About a quarter of the departments said that students collected, organized, and analyzed real data at least once a week, but about 65% of the departments said that students did this occasionally. The distribution of the percentages was quite similar over all levels of four-year mathematics departments, except for the use of regular graded homework or quizzes, which was said to be used less frequently at the doctoral-level departments.

Table FY.8 summarizes data on departments' responses to questions on how successful they viewed the use of technology in teaching Introductory Statistics with no calculus prerequisite in fall 2021. All the departments indicated the greatest sense of success with the instructors' use of technology to demonstrate concepts; somewhat less success was indicated in students' abilities to use technology to explore data, analyze data, or solve problems. The bachelor's-level departments' responses were the most like the statistics departments' responses, with the doctoral- and master's-level departments indicating a slightly lower sense of success; for example, Table FY.8 shows that 77% (SE 6.2) of bachelor's-level mathematics departments indicated the instructors' use of technology was very successful, while 50% (SE 18) of doctoral-level mathematics departments indicated this use of technology was very successful.

## Chapter 6

# Enrollment, Course Offerings, and Instructional Practices in Mathematics Departments at Two-Year Colleges

This chapter reports estimated enrollment, course offerings, and instructional practices in mathematics and statistics courses at public two-year colleges in the United States in fall 2021. The data in this chapter has been rounded. Also included are total enrollment for these two-year colleges, average mathematics class size, trends in availability of mathematics courses, enrollment in mathematics courses offered outside of the mathematics departments, and services available to mathematics students. Some tables contain data from previous CBMS surveys (1975, 1980, 1985, 1990, 1995, 2000, 2005, 2010, and 2015) and hence allow for historical comparisons. Further analysis of many of the items discussed in this chapter can be found in Chapters 1 and 2, where they are discussed from a comprehensive point of view in comparison to similar data for four-year colleges and universities.

The table display code in this chapter is TYE, for “Two-Year Enrollment,” because this chapter addresses issues related to enrollment. There are also instances where “na” will be displayed in a table, indicating that similar data was not collected or was not available, or that the data was incomplete.

CBMS2005, CBMS2010, CBMS2015, and CBMS2021 surveys include data about public two-year colleges only. The two-year college data in this report were estimated from a stratified random sample of 228 institutions chosen from a sample frame of 1008 public two-year colleges. The survey return rate was 33% of the sample (76 of 228 colleges), despite intense follow-up efforts by the Two-Year College Committee to increase this return rate. For comparison purposes, the survey return rate for two-year colleges for CBMS2015 was 54% (119 of 222 colleges), CBMS2010 was 51% (105 of 205 colleges), CBMS2005 was 54% (130 of 241 colleges), CBMS2000 was 60% (179 of 300 colleges), and CBMS1995 was 65% (163 of 250 colleges). The return rate for four-year institutions was 37% (117 of 314 institutions). The completion rate for all institutions, two-year and four-year, in CBMS2021 was 34% (183 of 542 institutions). For more information on the sampling and projection procedures used in this survey, see Appendix II. A copy of the two-year college survey questionnaire for CBMS2021 may be found in Appendix VII.

The terms “full-time permanent,” “full-time continuing,” and “other full-time” faculty occasionally are used in this chapter and other chapters. For a detailed explanation of these terms, see Chapter 7.

In the text that follows, the standard error (SE) in many of the estimates is provided along with the estimate. The SEs for all CBMS2021 tables can be found in Appendix IX. The change in an estimate from the estimate in a previous survey is often expressed both as percentage change and as the number of SEs that change represents (e.g., “increased 22% (1.2 SEs)”).

## Highlights of Chapter 6

### Enrollments, class size, and course offerings in mathematics departments.

- From 2015 to 2021, public two-year colleges experienced an overall total enrollment decrease of 28%, an estimated total of 4,478,000 and 1,746,000 fewer students, based on National Center for Education Statistics (NCES) projections updated in 2022 for fall 2021. The 2021 decrease can be viewed in comparison with an overall increase at four-year colleges of 2%, an estimated total of 218,000 more students. From 2010 to 2015, public two-year colleges experienced an overall total enrollment decrease of 14%, an estimated total of 6,224,000 students, based on NCES projections updated in 2016 for fall 2015. Enrollment in two-year colleges in fall 2021 constituted about 30% of the total undergraduate enrollment in the United States. For details, see the discussion before and after Table S.1 in Chapter 1 and Table TYE.1 in this chapter.
- The fall 2021 enrollment in mathematics and statistics courses in mathematics departments at public two-year colleges received from the CBMS2021 survey was estimated to be 1,495,000 (SE 157,543) students, a 26% decrease from 2015. This total includes 162,000 (SE 6000) dually enrolled students and 823,000 (SE 108,000) distance learning enrollments. The fall 2015 enrollment in mathematics and statistics courses in mathematics departments at public two-year colleges received from

the CBMS2015 survey was estimated to be 2,012,000 (SE 118,000) students. This total included 94,000 (SE 23,000) dually enrolled students and 225,000 (SE 25,000) distance learning enrollments. Enrollment in mathematics and statistics at two-year colleges in fall 2021 constituted approximately 41% of the total mathematics and statistics undergraduate enrollment in United States postsecondary institutions (not including dual enrollment). See Table S.1 in Chapter 1, Table SP.16 in Chapter 2, and Tables TYE.2 and TYE.11 in this chapter.

- Two-year college mathematics and statistics on-campus and distance enrollment (excluding dual enrollment) in 2015 to 2021 decreased 26% to a total of 1,333,000 (SE 145,000), compared with the decrease of 4% from 2010 to 2015. During the period 2015 to 2021, four-year institutions had an enrollment decrease in mathematics courses of 30% (5 SEs), compared with the growth of 13% (2 SEs, excluding dual enrollment) from 2010 to 2015. See Table S.1 in Chapter 1 and Tables TYE.2 and TYE.3 in this chapter.
- In CBMS2021, dual-enrollment courses are defined to be “courses conducted in a high school campus by high school teachers, for which students may obtain high school credit and, simultaneously, college credit.” Two-year college dual enrollment increased 72% from 2015 to 2021 for 162,000 students compared with 63% from 2010 to 2015 and an estimated 94,000 students. See Tables SP.16 and SP.17 in Chapter 2 and Table TYE.2 in this chapter. Dual enrollment increases were noted in every course category in 2021.
- In 2021, enrollment in Pre-college mathematics courses (Arithmetic, Pre-Algebra, Elementary and Intermediate Algebra, and Geometry) at two-year colleges was estimated to be 348,000 (SE 47,000) compared to 782,000 students (SE 65,000) in 2015. This represents a 55% decrease from 2015 to 2021 compared to a 32% decrease from 2010 to 2015 and a 19% increase from 2005 to 2010. Within the cohort of Pre-college courses at two-year colleges, all courses showed a decrease in enrollment. Four-year college Pre-college enrollment between 2015 to 2021 decreased 60% (9.3 SEs) to 101,000 students after with an increase of 21% (5 SEs) to an estimate of 253,000 (SE 26,000) students from 2010 to 2015. See Table S.2 in Chapter 1 and Tables TYE.3 and TYE.4 in this chapter.
- Approximately 26% of all mathematics enrollment in two-year colleges in 2021 was in Pre-college (remedial/developmental) courses, compared with 41% of all two-year college mathematics and statistics enrollment in fall 2015 and 57% in fall 2010. See Table TYE.4.
- The trend of an increasing enrollment in Precalculus/Introductory-level courses (College Algebra, Trigonometry, College Algebra and Trigonometry [combined], Mathematical Modeling, Precalculus/Elementary Functions/Analytic Geometry) that was seen in 2010 and 2015 did not continue in 2021 with a decrease of 26% and 116,000 fewer students representing 25% of all mathematics enrollments, a total of 329,000 (SE 42,000) students. Within the cohort of Precalculus/Introductory-level courses, all courses showed a decrease in enrollment except for Mathematical Modeling showing an increase of 1000 students. See Tables TYE.3 and TYE.4.
- Enrollment in all Calculus-level courses (Mainstream Calculus I, II, and III and Non-Mainstream Calculus I and II and Differential Equations together) showed a 1% decrease to 151,000 (SE 26,000) students compared with an 11% increase from 2010 to 2015 (total 152,000 students; SE 15,000). From 2015 to 2021, Mainstream Calculus I, II, and III experienced a 4% increase from 2015 to 2021 up to 124,000 students compared to a 9% increase to a total of 119,000 students in 2015. Non-Mainstream Calculus I and II decreased 31% to 18,000 students, 8000 fewer students in 2021, compared to an increase of 18% to 26,000 students in 2015. See Tables TYE.3 and TYE.4.
- Among college-level, transferable statistics courses, after an increase of 104% in 2015, the data showed an 8% decrease from 2015 to 2021, a total of 257,000 (SE 34,000) students, 23,000 fewer students than in 2015. See Tables TYE.3 and TYE.4.
- For the first time, the CBMS2021 survey included the course Quantitative Literacy/Quantitative Reasoning. In fall 2021, a total of 73,000 (SE 15,000) students enrolled in this course offered in 47% (SE 7) of two-year colleges. See TYE.3 and TYE.5.
- The average section size of classes taught on two-year college campuses was 16.5 (SE 1) in 2021, compared to 22 (SE 2) students in 2015 and 24 students in 2010. Every course category showed a decrease from 2015. The largest decrease occurred in Calculus-level courses where section size was 25.4 in 2015 and 16 (SE 1) in 2021, a decrease of 9.4 students. In fall 2021, no course category had a percentage of on-campus sections with an average size greater than 30 (SE 0) students, compared to 25% of colleges with section size above 30 in 2015. See Tables TYE.7 and TYE.8. For comparable four-year data, the section-size range in Pre-college, Introductory, Calculus, and Statistics is 25–34 and 15 in advanced courses. See Table E.10 in Chapter 3.

- The average section size of all distance learning courses at two-year colleges in 2021 was 20.9 (SE 1) with a range of 18.2–23.6 students, compared with fall 2015 of 21 (SE 1) students, with a range of 11–27 students. The percentage of distance learning courses with an average size greater than 30 was 1.8% (SE 0) in 2021, compared to 17% (SE 4) in 2015. See Table TYE.7.1. Table TYE.8 displays the average on-campus and distance learning section size for each course.
- In fall 2021, 67% (SE 0.08) of mathematics class sections were taught by part-time faculty, up 31 points from 2015 (36%; SE 4) in 2015. The percentage of sections taught by part-time faculty varied significantly by course type, with every course category increasing since 2015. Specifically, part-time faculty taught 81% (SE 0.11) of Pre-college courses, 56% (SE 0.09) of Precalculus courses, 20% (SE 0.03) of Mainstream Calculus courses, 63% (SE 0.18) of Non-Mainstream Calculus courses, and 77% (SE 0.17) of Statistics and Probability courses. Technical mathematics courses showed 101% (SE 0.31) being taught by part-time faculty. See Table TYE.9 and Figure TYE.9.1.

### Redesign of mathematics programs—Pathways.

- CBMS2015 and CBMS2021 surveys asked questions about implementation of mathematics “Pathways,” defined to be “a single course or course sequence that enables students to complete a college-level gateway mathematics or statistics course that is aligned to students’ academic and/or career goals within one academic year.” In fall 2021, 71% (SE 6) of colleges reported having implemented a Pathways course sequence, compared with 2015 when 58% (SE 5) of colleges implemented Pathways. Departments oftentimes implemented multiple Pathways courses named Foundations (88%; SE 4.55), Quantitative Reasoning/Literacy (87%; SE 4.54), Statistics (74%; SE 6.63), and Other (64%; SE 12.9). An Intermediate Algebra prerequisite was required in 23% (SE 6.83) of the Foundations courses, 17% (SE 5.77) of the Quantitative Reasoning/Literacy courses, and 62% (SE 7.88) of Statistics courses. See Table TYE.10.

### Distance/remote learning courses and practices.

- Distance/remote learning, defined to be “courses in which half or more of the instruction occurs with the instructor and the student separated by time and/or place facilitated by technology,” showed enrollment in two-year college mathematics and statistics departments growing to an estimated 823,000 (SE 108,000) students, up 266% compared to 225,000 (SE 25,000) students in fall 2015. Distance/remote learning enroll-

ment accounted for 62% of all two-year college mathematics enrollment. See Table S.9 in Chapter 1 and Tables TYE.2 and TYE.11 in this chapter.

- Distance/remote learning enrollments accounted for 60% of all two-year college Pre-college course enrollments in fall 2021. The number of distance/remote learning students in the category of Pre-college courses was an estimated 208,883 (SE 22,745) students in 2021, compared with 89,000 (SE 16,000) students in fall 2015. See Table S.9 in Chapter 1 and Table TYE.11 for individual course enrollment.
- Distance/remote learning enrollment increased 263% in the category of Precalculus/Introductory courses (College Algebra, Trigonometry, and Precalculus) with an estimate of 196,368 (SE 21,815) students, compared with 54,000 (SE 7400) students in 2015. Distance/remote learning enrollment in Precalculus/Introductory courses represented 72% of all Precalculus enrollment. See Tables S.9 in Chapter 1 and Table TYE.11 in this chapter for individual course enrollment.
- Distance/remote learning enrollment in Elementary Statistics increased significantly to a total of 169,586 (SE 25,613) students in fall 2021, compared to 31,000 (SE 4000) students in 2015, which is an increase of 448%. See Table S.9 in Chapter 1 and Table TYE.11 for individual course enrollment.
- In 2021, distance/remote learning enrollment in all Calculus courses also increased significantly, representing 64% of all Calculus enrollments. 96,000 students enrolled in Mainstream Calculus, Non-Mainstream Calculus, and Differential Equations via distance, compared with 10,000 students in 2015, an 860% increase. See Table TYE.11.
- Table TYE.12 presents data on various distance learning practices, including the format of the majority of distance learning courses and how distance learning students took tests. The 2021 data were similar to 2015 in that 68% (SE 6.4) of the sections were taught only online, and only asynchronously. Thirty-seven percent (37%, SE 6.6) used a combination of test-taking formats (not monitored, online using monitoring technology, or monitored at testing site). See Table TYE.12. A discussion about the use of distance learning practices by mathematics departments is included in Chapter 2, Table SP.9.
- Table TYE.12.1 presents data on distance learning delivery and course design. “Replicating active learning in a virtual environment” was a very significant challenge in 53% (SE 6.89) of two-year college departments. “Maintaining academic integrity on assessments” was



the second significant challenge for 49% (SE 6.93) of departments. The third very significant challenge was “engaging students online” for 48% (SE 6.82) of the departments. In 2015, “maintaining a level of rigor in distance learning mathematics courses equivalent to courses offered face-to-face” was reported to be “somewhat of a challenge” by 41% (SE 5) of two-year colleges and a “very significant challenge” by 17% (SE 5). See Table TYE.12.1 and also Table SP.10 in Chapter 2.

### Placement and opportunities available to students.

- Questions about placement testing were reconstructed in CBMS2021 to gather data on the kinds of placement tools used to assess first-time enrollees. Seventy-four percent (74%, SE 6) of colleges required a placement test of first-time enrollees. A variety of tools (placement test, self-placement, counseling, scores on state test, high school record, ACT/SAT, and national assessment) were used. Eighty-five percent (85%, SE 7) of colleges reported the high school record and ACT/SAT were the favored tools for placement. Seventy-nine percent (79%, SE 6) periodically assess the effectiveness of their placement tests. See Table TYE.13.
- With decreases in mathematics enrollments in 2021, Table TYE.13 presents decreases in offerings of various opportunities to students such as honors sections, mathematics clubs, programs to encourage women and minorities in mathematical studies, undergraduate student research, and independent studies in mathematics. Small increases were noted in faculty advising and K–12 outreach activities. These are described in Tables SP.12 and SP.13 in Chapter 2 and Table TYE.13 in this chapter.

### Topics of special interest in CBMS2021.

- In each CBMS survey cycle, certain topics of special interest are chosen for data collection and comprehensive analysis from both two- and four-year institutions. Special topics for two- and four-year institutions are discussed in Chapters 2 of this report and at the end of this chapter. In 2021, the same questions were added in the two-year, four-year, and statistics questionnaires regarding instructional strategies.

## Enrollment, Class Size, and Course Offerings in Mathematics Departments

**Number of two-year college students.** In fall 2021, approximately 4,478,000 students, including 68% part-time students, were enrolled in public two-year colleges, compared with 6,224,000 students in fall 2015 (65% attending

part-time). This estimate is based on an overall 2022 enrollment projection for public two-year colleges by the National Center for Educational Statistics (NCES). These enrollments constitute a 28% enrollment decrease from 2015–2021, compared with a 14% enrollment decrease from 2010–2015 for public two-year colleges. NCES projections indicated about a 2% increase in four-year college enrollments in the same time period and totaled 10,766,000 students. Enrollment in two-year colleges in fall 2021 constituted about 30% of the total undergraduate enrollment in the United States. See Table TYE.1 and Figure TYE.1.1 in this chapter and Table S.1 in Chapter 1.

### Enrollment trends in mathematics departments.

Enrollment in mathematics and statistics courses in mathematics departments at public two-year colleges was estimated to be 1,495,000 (SE 157,543) students in 2021, compared with 2,012,000 (SE 118,000) students in 2015. See Table TYE.2 and Figure TYE.2.1.

Of the 1,495,000 students enrolled in mathematics, approximately 823,000 (SE 108,000) were enrolled in distance/remote learning courses and 162,000 (SE 6000) were dual enrolled. Fall 2021 enrollment represents a decrease of 26% from 2015 to 2021, compared with the 4% decrease from 2010 to 2015 and 19% increase from 2005 and 2010. If dual-enrolled students are excluded, the decrease in mathematics department enrollment from 2015 to 2021 was 31% and a total of 1,333,000 students. The 26% enrollment decrease in mathematics and statistics courses from 2015 to 2021 is consistent with the 28% decrease in two-year institutional enrollment mentioned above.

In CBMS2021, dual-enrolled students are “high school students who take courses conducted at a high school campus taught by high school teachers, for which the student may obtain high school credit and, simultaneously, college credit through the two-year college institution.” The estimated 162,000 dual-enrolled students in 2021 represented 11% of total mathematics and statistics enrollments, compared to 94,000 dual-enrolled students representing almost 5% of total mathematics and statistics enrollments in fall 2015. See Table SP.16 in Chapter 2.

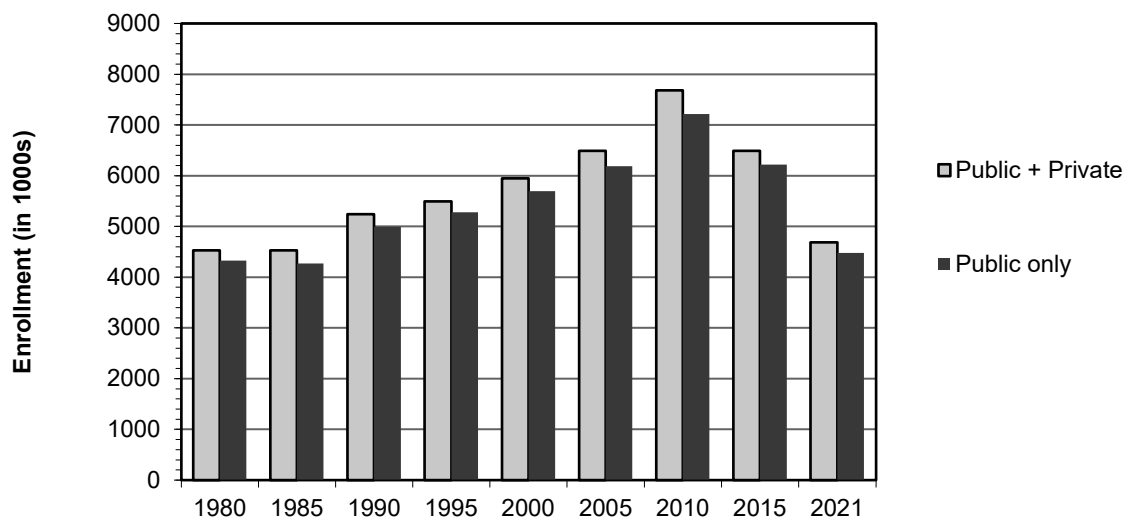
In fall 2021, the estimated 823,000 students in distance/remote learning mathematics and statistics courses represented 55% of the total mathematics and statistics enrollments in two-year colleges, compared to 225,000 students in distance learning mathematics courses who represented 12% of total mathematics and statistics enrollments in fall 2015. See Tables TYE.2 and TYE.11.

Table S.1 in Chapter 1 presents data on both two- and four-year institutions’ overall mathematics and statistics enrollments, excluding dual enrollments. The estimated total of 1,333,000 (SE 145,000) non-dual-enrolled students in 2021 represents a 31% decrease from 2015, compared

**TABLE TYE.1.** Total institutional enrollment (in thousands) and percentage of part-time enrollments in two-year colleges in fall for 1980 through 2015 and projected enrollments for fall 2021.<sup>1</sup> Enrollments include distance learning but not dual enrollments.

|                      | 1980 | 1985 | 1990 | 1995 | 2000 | 2005 | 2010 | 2015 | 2021 |
|----------------------|------|------|------|------|------|------|------|------|------|
| Public + Private     |      |      |      |      |      |      |      |      |      |
| Number of students   | 4525 | 4531 | 5240 | 5492 | 5948 | 6488 | 7684 | 6499 | 4683 |
| Percentage part-time | 61   | 63   | 64   | 64   | 63   | 59   | 56   | 61   | 65   |
| Public only          |      |      |      |      |      |      |      |      |      |
| Number of students   | 4328 | 4270 | 4996 | 5277 | 5697 | 6184 | 7218 | 6224 | 4478 |
| Percentage part-time | 63   | 65   | 66   | 65   | 65   | 61   | 59   | 65   | 68   |

<sup>1</sup> Data for the first three rows are from Table 303.70, *Digest of Educational Statistics: 2022*, NCES, [https://nces.ed.gov/programs/digest/d22/tables/dt22\\_303.70.asp](https://nces.ed.gov/programs/digest/d22/tables/dt22_303.70.asp). Data for the percentage part-time for public institutions are from Table 14, *Projections of Education Statistics to 2028*, <https://nces.ed.gov/pubs2020/2020024.pdf>.



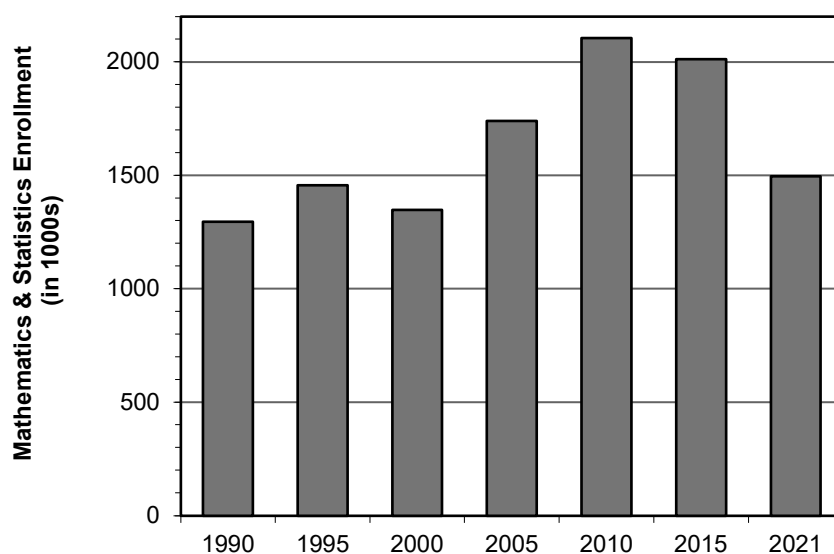
**FIGURE TYE.1.1.** Total enrollments (all disciplines) in public and private two-year colleges and in public-only two-year colleges in fall 1980 through fall 2021.

**TABLE TYE.2.** Enrollments in mathematics and statistics (no computer science) courses in mathematics programs at two-year colleges in fall 1985, 1990, 1995, 2000, 2005, 2010, 2015, and 2021.

|                                              | 1985    | 1990      | 1995      | 2000      | 2005 <sup>1</sup> | 2010 <sup>1</sup> | 2015 <sup>1</sup> | 2021      |
|----------------------------------------------|---------|-----------|-----------|-----------|-------------------|-------------------|-------------------|-----------|
| Mathematics & Statistics Enrollments in TYCs | 936,000 | 1,295,000 | 1,456,000 | 1,347,000 | 1,739,000         | 2,105,000         | 2,012,000         | 1,495,000 |

<sup>1</sup> Data for 2005, 2010, 2015, and 2021 include only public two-year colleges. 2021 data include 162,000 dual enrollments from Table SP.16 in Chapter 2 and 823,000 distance enrollments from Table TYE.12 in this chapter.

Note: Data for 1990, 1995, and 2000 in this table differ from corresponding data in Table S.1 of Chapter 1 because the totals in this table do not include any computer science courses, while the totals in Table S.1 do.



**FIGURE TYE.2.1.** Enrollments in mathematics and statistics courses (no computer science) in mathematics departments in two-year colleges in fall 1990, 1995, 2000, 2005, 2010, 2015, and 2021. (Data for 2005, 2010, 2015, and 2021 include only public two-year colleges. 2021 data includes 162,000 dual enrollments from Table SP.16 in Chapter 2 and 823,000 distance enrollments from Table TYE.12 in this chapter.)

with the 1,918,000 (SE 115,000) two-year college enrollment in 2015. Two-year college mathematics and statistics enrollment (excluding dual enrollment) comprised 41% of all postsecondary mathematics and statistics enrollments in fall 2021. See Table S.1 in Chapter 1, Tables SP.16 and SP.17 in Chapter 2, and Table TYE.2 in this chapter.

It is difficult to draw specific conclusions about the decrease in institutional and mathematics enrollment in two-year colleges in fall 2021. The influence of the COVID virus upon higher education, and lives in general, cannot be overstated. Two- and four-year colleges pivoted to offer courses to students under very stressed academic and physical conditions, leading to more remote course offerings and dual-enrollment arrangements.

The 2021 CBMS survey, delayed one year due to the COVID-19 pandemic, made changes to the distance learning section of the CBMS questionnaires that had been designed for the 2020 survey before the pandemic, including introducing some new questions. In fall 2020, the CBMS survey team also performed a modified survey of practices used during the pandemic. The COVID Impact Study, published in March 2021, is given in Appendix IV, and further information on this study is available at <https://www.ams.org/profession/data/cbms-survey/cbms2020>.

In the COVID Impact Study, departments were asked to respond to the statement regarding future course delivery plans: “We are considering offering a greater number of online courses.” Figure 6 of that study shows that in fall 2020, an estimated 35% of four-year mathematics departments, 54% of statistics departments, and 61% of two-year college mathematics departments indicated that they “strongly agree or agree” with this statement. Another highlight of the COVID survey was that during the pandemic, the format “only online synchronous” was the most frequently used option (50%), followed by “a mixture of online and face-to-face sessions” (26%).

Other factors that may have influenced mathematics enrollments include national degree completion and Pathways initiatives, changes in state legislation regarding decreased funding for developmental education and high school graduation requirements, and implementation of multiple placement measures/procedures. More discussion about trends in specific course enrollment and implementation of mathematics Pathways can be found later in this chapter.

Two-year college mathematics and statistics enrollment from 2015 to 2021 can be considered in light of the pattern in the nation’s four-year colleges and universities. Between 2015 and 2021, mathematics and statistics enrollment (excluding dual enrollments) at two-year colleges decreased 31%, and four-year mathematics and statistics enrollment decreased 30%. See Table S.1 in Chapter 1.

In addition to the tables that follow in this chapter, the reader should consult Chapters 1 and 2 in this report, which contain a detailed analysis of mathematics and statistics department enrollments at both two- and four-year colleges from 2015 to 2021.

**Enrollment trends in course groups and in specific courses.** Tables TYE.3 and TYE.4 report mathematics enrollments in two-year colleges. Table TYE.3 reports enrollment in individual mathematics courses. Table TYE.4 reports enrollment for categories of courses. Table TYE.4 is constructed from Table TYE.3 and reports headcounts and percentages from 2005 through 2021 for the following course groupings: Pre-College, Precalculus/Introductory, Calculus, Statistics, and Other Mathematics courses. Each category consists of five or more specific courses from Table TYE.3.

In fall 2021, approximately 348,000 (SE 47,000) students enrolled in Pre-college mathematics courses (Arithmetic, Pre-Algebra, Elementary Algebra, Intermediate Algebra, and Geometry), compared with 782,000 (SE 65,000) students in fall 2015. Enrollment in these courses comprised 26% of mathematics program enrollment in 2021, compared with 41% in 2015. These percentages are presented in Table TYE.4, which does not include the 162,000 students in dual-enrollment courses. Fall 2021 presented Pre-college course enrollment decreases of 55% at two-year colleges. In comparison, four-year college Pre-college enrollment was a total of 101,000 (SE 16,000) students and experienced a 60% decrease from 2015 to 2021. See Table S.2 in Chapter 1.

All two-year college Pre-college courses experienced a decrease in enrollment from 2015 to 2021: Arithmetic & Basic Mathematics at 42,000 (SE 8000) students, down 40%; Pre-Algebra at 49,000 (SE 8000) students, down 62%; Elementary Algebra at 89,000 (SE 13,000) students, down 68%; Intermediate Algebra at 145,000 (SE 29,000) students, down 52%; and Geometry at 1000 (SE 8000) students, down 88%. See Tables TYE.3 for enrollment in individual courses. For Precalculus/Introductory-level courses (College Algebra, Trigonometry, College Algebra & Trigonometry, Introduction to Mathematical Modeling, and Precalculus), 329,000 (SE 42,000) students accounted for 25% of all 2021 mathematics enrollment, compared with 445,000 (SE 39,000) students and 23% of 2015 enrollment. Precalculus courses, together with Pre-college courses, accounted for 51% of mathematics and statistics enrollment at public two-year colleges in fall 2021, a decrease of 13% from fall 2015. See Table TYE.4.

Within the cohort of Precalculus/Introductory-level courses, all courses experienced a decrease between 7–34%, except for Mathematical Modeling, which declined by 53%

**TABLE TYE.3.** Enrollment in thousands in mathematics and statistics courses (not including dual enrollments; including distance enrollments) in mathematics departments at two-year colleges in fall 2005, 2010, 2015, and 2021.

| Course Number                                  | Type of Course                                                 | 2005 | 2010 | 2015 | 2021 |
|------------------------------------------------|----------------------------------------------------------------|------|------|------|------|
|                                                | <b>Pre-College Level</b>                                       |      |      |      |      |
| 1                                              | Arithmetic & Basic Mathematics                                 | 104  | 146  | 71   | 42   |
| 2                                              | Pre-Algebra                                                    | 137  | 226  | 127  | 49   |
| 3                                              | Elementary Algebra (high school level)                         | 380  | 428  | 277  | 89   |
| 4                                              | Intermediate Algebra (high school level)                       | 336  | 344  | 299  | 145  |
| 5                                              | Geometry (high school level)                                   | 7    | 6    | 8    | 1    |
| 5a                                             | Other Developmental Mathematics Courses                        |      |      |      | 22   |
|                                                | <b>Precalculus Level</b>                                       |      |      |      |      |
| 6                                              | College Algebra (above Intermediate Algebra)                   | 206  | 230  | 292  | 193  |
| 7                                              | Trigonometry                                                   | 36   | 45   | 51   | 40   |
| 8                                              | College Algebra & Trigonometry (combined)                      | 14   | 11   | 13   | 12   |
| 9                                              | Introduction to Mathematical Modeling                          | 7    | 18   | 2    | 3    |
| 10                                             | Precalculus/Elem. Functions/Analytic Geometry                  | 58   | 64   | 87   | 81   |
|                                                | <b>Calculus Level <sup>1</sup></b>                             |      |      |      |      |
| 11                                             | Mainstream Calculus I                                          | 51   | 65   | 66   | 78   |
| 12                                             | Mainstream Calculus II                                         | 19   | 29   | 34   | 29   |
| 13                                             | Mainstream Calculus III                                        | 11   | 15   | 19   | 17   |
| 14                                             | Non-Mainstream Calculus I                                      | 21   | 20   | 26   | 15   |
| 15                                             | Non-Mainstream Calculus II                                     | 1    | 2    | 0    | 3    |
| 16                                             | Differential Equations                                         | 4    | 6    | 7    | 9    |
|                                                | <b>Other Mathematics Courses</b>                               |      |      |      |      |
| 17                                             | Linear Algebra                                                 | 3    | 5    | 7    | 8    |
| 18                                             | Discrete Mathematics                                           | 2    | 2    | 5    | 6    |
| 19                                             | Elementary Statistics (with or w/o Probability)                | 111  | 134  | 251  | 240  |
| 20                                             | Probability (with or w/o Statistics)                           | 7    | 3    | 28   | 17   |
| 21                                             | Finite Mathematics                                             | 22   | 18   | 40   | 11   |
| 22a                                            | Mathematics for Liberal Arts                                   | 59   | 91   | 97   | 83   |
| 22b                                            | Quantitative Literacy/Quantitative Reasoning <sup>2</sup>      |      |      |      | 73   |
| 23                                             | Mathematics for Elementary School Teachers I <sup>3</sup>      | 29   | 21   | 12   | 13   |
| 24                                             | Mathematics for Elementary School Teachers II <sup>4</sup>     | na   | 8    | 3    | 4    |
| 25                                             | Other Mathematics Courses for Teacher Preparation <sup>4</sup> | na   | 1    | 1    | 1    |
| 26a                                            | Business Mathematics                                           |      |      |      | 13   |
|                                                | Non-Calculus-Based Technical Mathematics                       |      |      |      |      |
|                                                | Calculus-Based Technical Mathematics                           |      |      |      |      |
| 26b                                            | Business Mathematics (not transferable)                        | 22   | 16   | 16   |      |
| 27                                             | Business Mathematics (transferable)                            | 17   | 4    | 10   |      |
| 28                                             | Technical Math (non-calculus-based)                            | 16   | 17   | 21   | 13   |
| 29                                             | Technical Math (calculus-based)                                | 1    | 1    | 3    | 6    |
| 30                                             | Other Mathematics Courses (not transferable) <sup>5</sup>      | 28   | 33   | 31   | 16   |
| 31                                             | Other Mathematics Courses (transferable) <sup>4</sup>          | na   | 14   | 12   | 3    |
| Total All Two-Year College Mathematics Courses |                                                                | 1696 | 2024 | 1918 | 1333 |

Note: 0 means fewer than 500 enrollments, and “na” means not available. Round-off may make column sums seem inaccurate.

<sup>1</sup> Mainstream Calculus is for mathematics, physics, science and engineering. Non-Mainstream Calculus is for biological, social, and management sciences.

<sup>2</sup> These courses were not listed before 2021.

<sup>3</sup> In 2005 and earlier surveys, there was a single course listed as Mathematics for Elementary School Teachers.

<sup>4</sup> This course was not listed in 2005 and earlier surveys.

<sup>5</sup> In 2005 and earlier surveys, there was a single course listed as Other Mathematics Courses.

**TABLE TYE.4.** Enrollment in 1000s (not including dual enrollments; including distance enrollments) and percentages of total enrollment in mathematics and statistics courses by type of course in mathematics departments at two-year colleges in fall 2000, 2005, 2010, 2015, and 2021.

| Course Numbers <sup>1</sup> | Type of Course          | 2000           | 2005           | 2010           | 2015           | 2021           |
|-----------------------------|-------------------------|----------------|----------------|----------------|----------------|----------------|
| 1–5a                        | Pre-college level       | 763<br>(57%)   | 964<br>(57%)   | 1150<br>(57%)  | 782<br>(41%)   | 348<br>(26%)   |
| 6–10                        | Precalculus level       | 274<br>(20%)   | 321<br>(19%)   | 368<br>(18%)   | 445<br>(23%)   | 329<br>(25%)   |
| 11–16                       | Calculus level          | 106<br>(8%)    | 107<br>(6%)    | 138<br>(7%)    | 152<br>(8%)    | 151<br>(11%)   |
| 19–20                       | Statistics, Probability | 74<br>(5%)     | 118<br>(7%)    | 137<br>(7%)    | 280<br>(15%)   | 257<br>(19%)   |
| 17–18 &<br>21–31            | Remaining courses       | 130<br>(10%)   | 186<br>(11%)   | 231<br>(11%)   | 259<br>(13%)   | 248<br>(19%)   |
| 1–31                        | Total, All Courses      | 1347<br>(100%) | 1696<br>(100%) | 2024<br>(100%) | 1918<br>(100%) | 1333<br>(100%) |

<sup>1</sup> For names of specific courses, see Table TYE.3.

Note: Round-off may make column totals seem inaccurate.

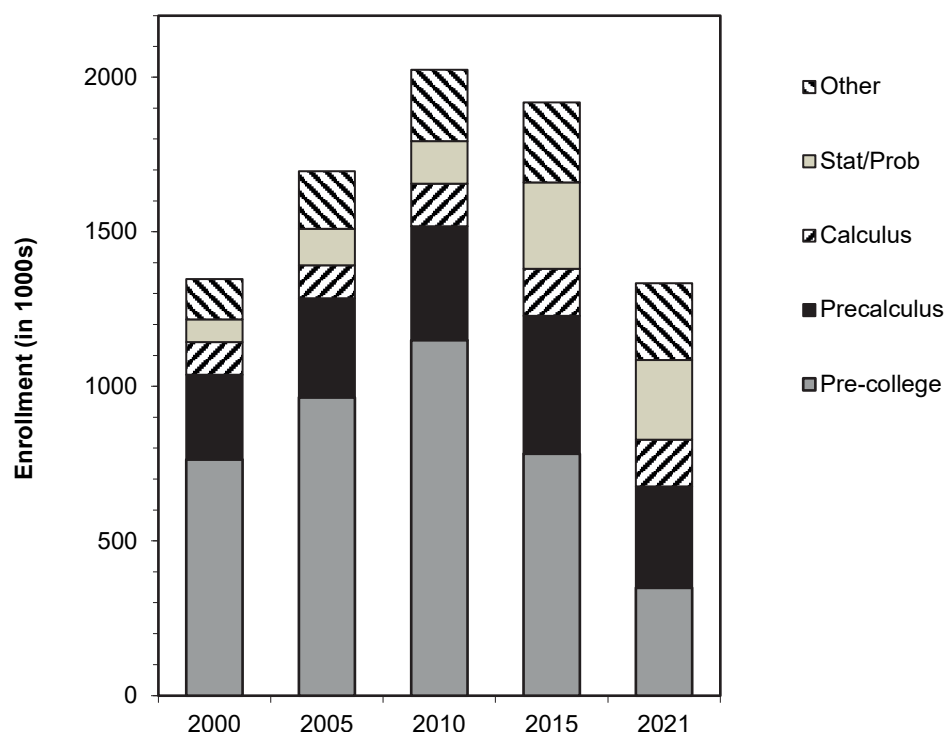
in enrollment. College Algebra enrollment showed the greatest decrease of 34% to 193,000 (SE 31,000) students, compared to an increase of 27% to 292,000 (SE 29,000) students in 2015. See Table TYE.3 for enrollment in individual courses.

In fall 2021, Mainstream Calculus I experienced a 20% increase totaling 78,000 (SE 14,000) students, while Mainstream Calculus II and III and Non-Mainstream Calculus experienced decreases of 15%, 11%, and 31%, respectively. Differential Equations increased by 29%, a total of 9000 (SE 2000) students. Overall, all Calculus courses and Differential Equations together showed an increase of 1000 students and represented 11% of all mathematics and statistics enrollments. See Table TYE.3 and TYE.4.

College-level, transferable statistics and probability courses, after an increase of 104% in 2015, showed an 8% decrease from 2015 to 2021 to a total of 257,000 (SE 43,000) students, 23,000 fewer students than in 2015. See Tables TYE.3 and TYE.4.

Summarizing the enrollment trend in mathematics course categories (see Tables TYE.3 and TYE.4) from fall 2015 to 2021, every category experienced a downward trend:

- Pre-college-level courses enrolled a total of 348,000 (SE 47,000) students, some 434,000 fewer students in 2021 than in 2015, representing a 55% decrease.
- Precalculus/Introductory courses enrolled a total of 329,000 (SE 42,000) students, some 116,000 fewer students in 2021 than in 2015, representing a 26% decrease.
- Mainstream and Non-Mainstream Calculus and Differential Equations courses enrolled a total of 151,000 (SE 26,000) students, some 1000 fewer students in 2021 than in 2015, representing a 1% decrease.



**FIGURE TYE.4.1.** Enrollment in 1000s (not including dual enrollments; including distance enrollments) in mathematics and statistics courses by type of course<sup>1</sup> in mathematics departments at two-year colleges in fall 2000, 2005, 2010, 2015, and 2021.

<sup>1</sup> For names of specific courses in each course grouping, see Table TYE.3.

- Elementary Statistics and Probability courses enrolled a total of 257,000 (SE 34,000) students, some 23,000 fewer students in 2021 than in 2015, representing an 8% decrease.
- Of special note is the collection of data for the first time of Quantitative Literacy/Quantitative Reasoning courses, with enrollment of 73,000 (SE 15,000) students in 2021.

In considering the factors listed above related to the decrease in total mathematics and statistics enrollment in 2021, the COVID virus impacted enrollment and instructional strategies in course categories and individual courses in two-year colleges. The 2021 CBMS COVID Impact Study is given in Appendix IV, and further information on this study is available at <https://www.ams.org/profession/data/cbms-survey/cbms2020>.

In addition, an increase in implementation of mathematics “Pathways,” defined as “a single course or course sequence that enables students to complete a college-level

gateway mathematics or statistics course that is aligned to students’ academic and/or career goals within one academic year,” also impacted course enrollments. Table TYE.10 shows that 71% (6 SEs) of responding colleges implemented a Pathways course sequence in fall 2021, compared with 58% in 2015.

**Trends in availability of courses in mathematics departments.** Tables TYE.5 and TYE.6 should be considered together and represent the availability of fall 2015 and 2021 course offerings and percentage of two-year college mathematics departments teaching individual courses.

Table TYE.5 reports that the percentage of two-year college mathematics departments offering a course titled Arithmetic/Basic Mathematics in 2021 was 46% (SE 6.9), compared to 36% (SE 5) in 2015. From 2015 to 2021, the percentage of mathematics departments offering a Pre-Algebra course, which generally included arithmetic and basic algebra skills, increased to 60% (SE 6.82), compared to 44% (SE 5) in 2015.

**TABLE TYE.5.** Percentage of two-year college mathematics departments teaching mathematics courses in fall 2015 and fall 2021.

| Course Number | Type of Course                                    | Fall 2015 | Fall 2021 |
|---------------|---------------------------------------------------|-----------|-----------|
| 1             | Arithmetic & Basic Mathematics                    | 36        | 46        |
| 2             | Pre-Algebra                                       | 44        | 60        |
| 3             | Elementary Algebra (high school level)            | 75        | 70        |
| 4             | Intermediate Algebra (high school level)          | 72        | 85        |
| 5             | Geometry (high school level)                      | 8         | 3         |
| 5a            | Other Developmental Mathematics Courses           | na        | 27        |
| 6             | College Algebra (above Intermediate Algebra)      | 79        | 81        |
| 7             | Trigonometry                                      | 57        | 55        |
| 8             | College Algebra & Trigonometry (combined)         | 20        | 18        |
| 9             | Introduction to Mathematical Modeling             | 5         | 6         |
| 10            | Precalculus/ Elem. Functions/ Analytic Geometry   | 54        | 58        |
| 11            | Mainstream Calculus I                             | 80        | 91        |
| 12            | Mainstream Calculus II                            | 65        | 75        |
| 13            | Mainstream Calculus III                           | 54        | 64        |
| 14            | Non-Mainstream Calculus I                         | 26        | 41        |
| 15            | Non-Mainstream Calculus II                        | 0         | 5         |
| 16            | Differential Equations                            | 25        | 45        |
| 17            | Linear Algebra                                    | 24        | 38        |
| 18            | Discrete Mathematics                              | 12        | 26        |
| 19            | Elementary Statistics (with or w/o Probability)   | 83        | 93        |
| 20            | Probability (with or w/o Statistics)              | 5         | 12        |
| 21            | Finite Mathematics                                | 23        | 24        |
| 22a           | Mathematics for Liberal Arts                      | 62        | 49        |
| 22b           | Quantitative Literacy/Quantitative Reasoning      | na        | 47        |
| 23            | Mathematics for Elementary School Teachers I      | 41        | 54        |
| 24            | Mathematics for Elementary School Teachers II     | 17        | 45        |
| 25            | Other Mathematics Courses for Teacher Preparation | 4         | 10        |
| 26            | Business Mathematics (not transferable)           | 25        | 32        |
| 27            | Business Mathematics (transferable)               | 12        | na        |
| 28            | Technical Mathematics (non-calculus-based)        | 38        | 30        |
| 29            | Technical Mathematics (calculus-based)            | 9         | 11        |
| 30            | Other Mathematics Courses (not transferable)      | 23        | 27        |
| 31            | Other Mathematics Courses (transferable)          | 10        | 13        |



**TABLE TYE.6.** Percentage of two-year college mathematics departments teaching selected mathematics courses in the fall terms of 2005, 2010, 2015, and 2021.

| Course Number | Type of Course                                            | Percentage of Two-Year Colleges Teaching Course |      |      |      |
|---------------|-----------------------------------------------------------|-------------------------------------------------|------|------|------|
|               |                                                           | 2005                                            | 2010 | 2015 | 2021 |
| 11            | Mainstream Calculus I                                     | 82                                              | 79   | 80   | 91   |
| 16            | Differential Equations                                    | 25                                              | 21   | 25   | 45   |
| 17            | Linear Algebra                                            | 19                                              | 19   | 24   | 38   |
| 18            | Discrete Mathematics                                      | 12                                              | 11   | 12   | 26   |
| 19            | Elementary Statistics (with or w/o Probability)           | 78                                              | 73   | 83   | 93   |
| 21            | Finite Mathematics                                        | 28                                              | 27   | 23   | 24   |
| 22a           | Mathematics for Liberal Arts                              | 56                                              | 44   | 62   | 49   |
| 23            | Mathematics for Elementary School Teachers I <sup>1</sup> | 59                                              | 55   | 41   | 54   |
| 28            | Technical Mathematics (non-calculus-based)                | 35                                              | 26   | 38   | 30   |
| 29            | Technical Mathematics (calculus-based)                    | 5                                               | 3    | 9    | 11   |

<sup>1</sup> In 2005 and earlier, there was a single course listed as Mathematics for Elementary School Teachers; the enrollment for that course is listed here.

The availability of Elementary Algebra within mathematics departments decreased in 2021 to 70% (SE 6.4), compared to 75% (SE 5) in 2015. Intermediate Algebra was offered in 85% (SE 4.75) of the sampled mathematics departments in 2021, compared with 72% (SE 5) in fall 2015. In fall 2021, 81% (SE 4.95) of two-year colleges offered a separate College Algebra course, compared with 79% (SE 4) in 2015. The percentage of colleges offering a separate Trigonometry course decreased two points in 2021 to 55% (SE 6.87).

Comparing fall 2015 to fall 2021, the percentage of colleges offering Mainstream Calculus I increased by 11 points to 91% (SE 4.28), up from 80% (SE 6) in 2015. The availability of Mainstream Calculus II grew by 10 points to 75% (SE 6.39). Mainstream Calculus III also grew by 10 points to 64% (SE 6.8). Non-Mainstream Calculus I was available at 41% (SE 6.44) of reporting colleges.

Quantitative Literacy/Quantitative Reasoning was first surveyed in 2021 and showed that 47% (SE 6.74) of two-year colleges offered those courses. These offerings mirror the increased offerings of Pathways courses as presented in Table TYE.10. Possibly related to the expanded offering of Quantitative Literacy is the decrease in the offering of the course “Mathematics for Liberal Arts,” which showed a decrease of thirteen points to 49% (SE 5.52) of colleges offering the course.

In general, availability of other courses not mentioned above increased in fall 2021. Availability of courses important to baccalaureate degrees in science, technology, engineering, mathematics, and computer science, such as Differential Equations, Discrete Mathematics, Elementary Statistics, and Finite Mathematics, is reported in Table TYE.6. The percentage of two-year college mathematics departments teaching each of those courses increased in 2021 between 10 to 20 percentage points.

**Trends in average section size.** The trend in the average number of students per on-campus class section in two-year college mathematics courses, exhibited in 1990 through 2005, shifted slightly upward in 2010 and downward again in 2015 and 2021. In fall 2021, the average on-campus section size was 16.5 (SE 0.54), down 5.2 students compared with fall 2015 with 21.7 students.

Pre-college courses exhibited the smallest section size of 13.9 (SE 1.04) students, Precalculus/Introductory and Statistics courses had a section size of 18.7 (SE 0.85), and Calculus-level courses had a section size of 16.0 (SE 1.09). See Table TYE.7. For a closer examination of individual course average section sizes in 2021, see Table TYE.8, displaying a range of 9 (line 25, Teacher Preparation) to 27 (line 5, Geometry) average section sizes of on-campus courses.

**TABLE TYE.7.** Average on-campus section size by type of course in mathematics departments at two-year colleges in fall 2010, 2015, and 2021. Also percentage of sections with enrollment above 30 in fall 2015 and 2021.

| Course Number <sup>1</sup> | Type of Course <sup>2</sup>   | 2010                 | 2015                 |                                       | 2021                 |                                       |
|----------------------------|-------------------------------|----------------------|----------------------|---------------------------------------|----------------------|---------------------------------------|
|                            |                               | Average Section Size | Average Section Size | Percentage of Sections with Size > 30 | Average Section Size | Percentage of Sections with Size > 30 |
| 1–5a                       | Pre-college level             | 24.0                 | 19.2                 | 19%                                   | 13.9                 | 0.0%                                  |
| 6–10                       | Precalculus level             | 26.0                 | 24.7                 | 31%                                   | 18.7                 | 0.3%                                  |
| 11–16                      | Calculus level                | 21.0                 | 25.4                 | 34%                                   | 16.0                 | 1.1%                                  |
| 19–20                      | Elem. Statistics, Probability | 28.0                 | 25.5                 | 33%                                   | 18.6                 | 0.3%                                  |
| 1–31                       | Total, All Courses            | 24.0                 | 21.7                 | 25%                                   | 16.5                 | 0.3%                                  |

<sup>1</sup> For names of specific courses, see Table TYE.3.<sup>2</sup> For specific course section size, see Table TYE.8.**TABLE TYE.7.1.** Average distance learning section size by type of course in mathematics departments at public two-year colleges in fall 2021. Also percentage of sections with enrollment above 30 in fall 2021.

| Course Number <sup>1</sup> | Type of Course <sup>2</sup> | 2021 Average Section Size | Percentage of 2021 Departments with Average Size > 30 |
|----------------------------|-----------------------------|---------------------------|-------------------------------------------------------|
| 1–5a                       | Pre-college level           | 18.2                      | 0.8%                                                  |
| 6–10                       | Precalculus level           | 22.6                      | 2.6%                                                  |
| 11–16                      | Calculus level              | 21.7                      | 2.9%                                                  |
| 19–20                      | Statistics, Probability     | 23.6                      | 1.7%                                                  |
| 1–31                       | Total, All Courses          | 20.9                      | 1.8%                                                  |

<sup>1</sup> For names of specific courses, see Table TYE.3.<sup>2</sup> For specific course section size, see Table TYE.8.

Fall 2021 data reflect no on-campus courses with section size greater than 30 students.

At four-year colleges and universities, the average section size for pre-college, Introductory, Calculus, Advanced level, and Statistics courses were 25, 29, 34, 15, and 26 students, respectively. See Tables E.10 in Chapter 3 for four-year institutional data. The average section size of distance learning classes in fall 2021 of 20.9 students (SE 0.89) was similar to 21 students in 2015, as reported in Table TYE.7.1. Fall 2021 data reflect no distance learning courses with section size greater than 30 students.

Pre-college courses exhibited the smallest distance section size of 18.2 (SE 1.89), Precalculus/Introductory and Statistics courses has a section size of 22.6 and 23.6, respectively (SE 0.85 and 0.99), and Calculus-level courses had a section size of 21.7 (SE 1.41).

Comparing the section sizes of distance learning by course category to on-campus section sizes, distance learning section size was generally slightly larger than on-campus courses. Table TYE.8 presents both on-campus and distance learning sections sizes for all two-year college mathematics courses.

**TABLE TYE.8.** Average on-campus and distance learning section size for public two-year college mathematics department courses in fall 2021.

| Course Number | Type of Course                                    | Average On-Campus Section Size | Average Distance Learning Section Size |
|---------------|---------------------------------------------------|--------------------------------|----------------------------------------|
| 1             | Arithmetic & Basic Mathematics                    | 13                             | 16                                     |
| 2             | Pre-Algebra                                       | 13                             | 14                                     |
| 3             | Elementary Algebra (high school level)            | 13                             | 16                                     |
| 4             | Intermediate Algebra (high school level)          | 16                             | 23                                     |
| 5             | Geometry (high school level)                      | 8                              | 27                                     |
| 5a            | Other Developmental Mathematics Courses           | 12                             | 17                                     |
| 6             | College Algebra (above Intermediate Algebra)      | 19                             | 23                                     |
| 7             | Trigonometry                                      | 17                             | 22                                     |
| 8             | College Algebra & Trigonometry (combined)         | 18                             | 21                                     |
| 9             | Introduction to Mathematical Modeling             | 17                             | 26                                     |
| 10            | Precalculus/Elem. Functions/Analytic Geometry     | 19                             | 23                                     |
| 11            | Mainstream Calculus I                             | 18                             | 24                                     |
| 12            | Mainstream Calculus II                            | 15                             | 20                                     |
| 13            | Mainstream Calculus III                           | 14                             | 22                                     |
| 14            | Non-Mainstream Calculus I                         | 14                             | 17                                     |
| 15            | Non-Mainstream Calculus II                        | 18                             | 21                                     |
| 16            | Differential Equations                            | 14                             | 20                                     |
| 17            | Linear Algebra                                    | 13                             | 24                                     |
| 18            | Discrete Mathematics                              | 16                             | 19                                     |
| 19            | Elementary Statistics (with or w/o Probability)   | 19                             | 24                                     |
| 20            | Probability (with or w/o Statistics)              | 15                             | 26                                     |
| 21            | Finite Mathematics                                | 23                             | 21                                     |
| 22a           | Mathematics for Liberal Arts                      | 18                             | 22                                     |
| 22b           | Quantitative Literacy/Quantitative Reasoning      | 20                             | 20                                     |
| 23            | Mathematics for Elementary School Teachers I      | 15                             | 17                                     |
| 24            | Mathematics for Elementary School Teachers II     | 14                             | 14                                     |
| 25            | Other Mathematics Courses for Teacher Preparation |                                | 9                                      |
| 26            | Business Mathematics (not transferable)           | 12                             | 20                                     |
| 27            | Business Mathematics (transferable)               |                                |                                        |
| 28            | Technical Mathematics (non-calculus-based)        | 16                             | 16                                     |
| 29            | Technical Mathematics (calculus-based)            | 18                             | 19                                     |
| 30            | Other Mathematics Courses (not transferable)      | 12                             | 22                                     |
| 31            | Other Mathematics Courses (transferable)          | 10                             | 14                                     |

Note: Blank entries represent courses with no responses.

**Trends in the use of part-time faculty.** Fall 2021 data shows that 18,543 (SE 2600) part-time faculty taught mathematics courses in two-year colleges (Table TYF.1 in Chapter 7). As seen in the faculty data in Table TYF.1 in Chapter 7, trends in part-time faculty fell into two categories: (1) paid by the two-year college or (2) paid by a third party. When faculty of every sort are included, full-time (permanent, continuing, and other) faculty, part-time faculty paid by a college, and part-time faculty paid by third parties made up 64% of the total mathematics faculty in fall 2021, as compared to 67% in 2015. If the 4716 (SE 969) third-party payee part-time faculty members are excluded, 57% of the faculty had part-time status in fall 2021; the comparable figure for 2015 was 65%. See Table TYF.1.

Part-time faculty represented about two-thirds (64%) of total faculty in 2021 by headcount and taught 67% (SE 0.08) of mathematics department class sections, compared with 36% (4 SEs) in fall 2015. See Table TYE.9. For historical reference, in fall 2000, 46% of class sections were taught by part-time faculty. In fall 1995, this figure was 38%.

Concerning the instructional issue of which types of courses are taught most often by part-time faculty, all courses showed more sections taught by part-time faculty in 2021 than in 2015. Eighty-one percent (81%; SE 0.11) of all pre-college-level sections were taught by part-time faculty, up 35 points compared with 2015 (46%; SE 10). Fifty-six percent (56%; SE 0.09) of part-time faculty taught Precalculus/Introductory-level sections, compared with 33% (SE 3) in 2015. Twenty percent (20%; SE 0.03) of Mainstream Calculus sections were taught by part-time faculty in 2021, compared with fifteen percent (15%; SE 2) in 2015. Sixty-three percent (63%; SE 0.18) of Non-Mainstream Calculus sections were taught by part-time faculty, compared with twenty-nine percent (29% with SE 10) in 2015. Seventy-seven percent (77%; SE 0.17) of statistics and probability courses were taught by part-time faculty, compared to 21% (SE 5) in 2015. See Tables TYE.9 and TYE.9.1.

**Redesign of mathematics programs and Pathways.** Strategies to improve success/completion rates and to update the curriculum were the focus of nationwide discussions starting in 2009 and continued into fall 2021. These efforts and discussions led to curricular programs called mathematics “Pathways.” In the CBMS2021 survey, “Pathways” is defined as “a single course or course sequence that enables students to complete a college-level gateway mathematics or statistics course that is aligned to students’ academic and/or career goals within one academic year.” These curricular changes resulted in new courses, revision of current courses, and (often) revisions of course prerequisites in those courses. Availability of Pathways courses and sequences may have impacted de-

creased enrollments in traditional Pre-college courses and increased enrollment in College Algebra, Quantitative Literacy, and Statistics courses. See Tables TYE.3, TYE.4, and TYE.10.

Table TYE.10 reports that 71% (SE 6.48) of responding two-year colleges implemented a Pathways course sequence in 2021, compared with 58% (SE 5) in fall 2015. The following Pathways courses were available in 2021 in the given percentage of mathematics departments, each showing an increase from 2015 availability: Foundations (88%; SE 5; compared to 51% in 2015), Quantitative Reasoning/Literacy (87%; SE 5; compared to 59% in 2015), Statistics (74%; SE 7; compared to 63% in 2015), and Other courses (64%; SE 13; compared to 32% in 2015).

CBMS2021 also asked department chairpersons if an Intermediate Algebra course was a prerequisite to Pathways courses. The responses showed that Intermediate Algebra was a prerequisite to Pathways courses called Foundations (23%; SE 6.83), Quantitative Literacy/Reasoning (17%; SE 5.66), and Statistics (62%; SE 13.72).

**Distance/remote learning courses.** In CBMS2021, distance/remote learning was defined to be “courses offered by your institution for credit, in which half or more of the instruction occurs with the instructor and the student separated by time and/or place facilitated by technology (e.g., courses in which the majority of the course is taught online, by computer software, or by other technologies).” Collecting data about distance/remote learning enrollments and section size in CBMS surveys began in 2005 through 2021. Chapter 2 discusses the survey questions in CBMS2021 about distance/remote learning and trends for both four- and two-year colleges, as well as below.

Enrollment data for two-year colleges (Table TYE.11 in this chapter and Table SP.9 in Chapter 2) reported that 62% (SE 0) of mathematics students enrolled via distance, which accounted for 823,000 (SE 107,700) students of the total 1,333,000 (SE 144,700) mathematics students, as compared to 12% (SE 1) which accounted for 225,000 students of the total 1,918,000 (SE 25,000) students in 2015. This represented a 266% increase and 598,000 more students in 2021 than in 2015. Comparing 2021 to 2015, two-year colleges had increases in the number of students enrolled in distance/remote learning courses in every course offering, except Mathematical Modeling, where distance/remote learning stayed constant. Four-year mathematics and statistics departments also saw increased distance enrollments in every course category.

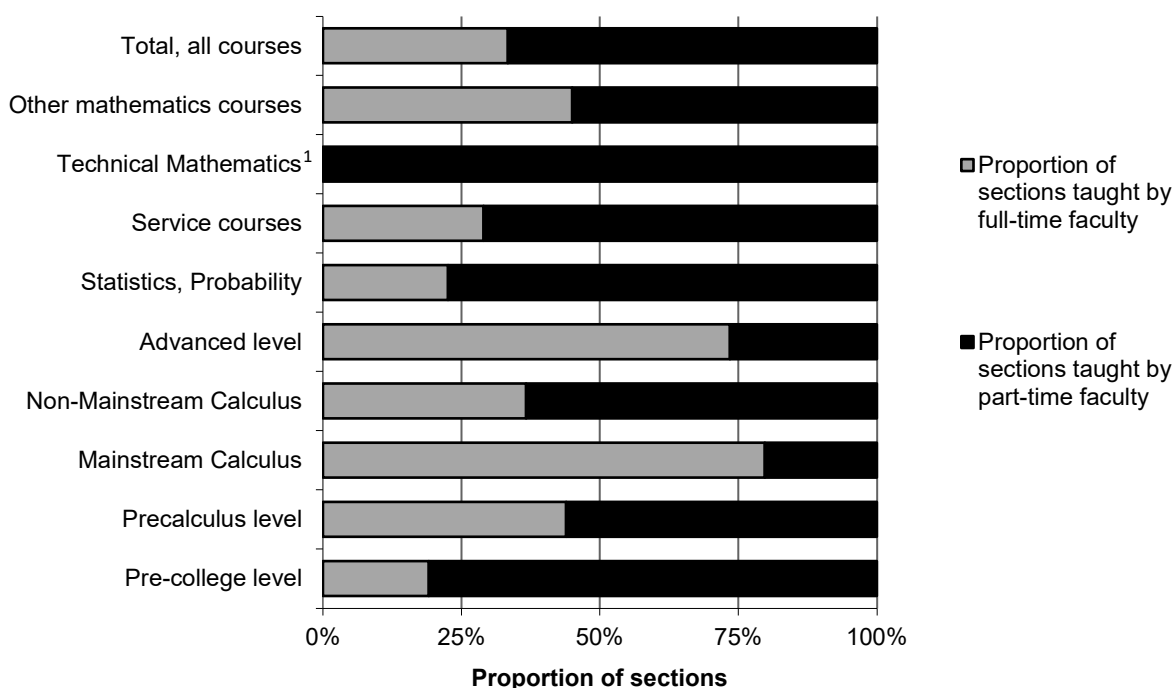
Distance/remote learning enrollment was more than 50% in most courses, except for Geometry, College Algebra & Trigonometry, Introduction of Mathematical Modeling,

**TABLE TYE.9.** Number of sections and number and percentage of sections taught by part-time faculty in mathematics departments at public two-year colleges by type of course in fall 2015 and 2021 (excluding distance learning and dual-enrollment sections).

| Course Number <sup>1</sup> | Type of Course                     | 2015               |                                                |                                                    | 2021                                           |                    |                                                    |
|----------------------------|------------------------------------|--------------------|------------------------------------------------|----------------------------------------------------|------------------------------------------------|--------------------|----------------------------------------------------|
|                            |                                    | Number of Sections | Number of Sections Taught by Part-Time Faculty | Percentage of Sections Taught by Part-Time Faculty | Number of Sections Taught by Part-Time Faculty | Number of Sections | Percentage of Sections Taught by Part-Time Faculty |
| 1-5a                       | Pre-college level                  | 36108              | 16515                                          | 46%                                                | 8110                                           | 10027              | 81%                                                |
| 6-10                       | Precalculus level                  | 15793              | 5173                                           | 33%                                                | 3968                                           | 7075               | 56%                                                |
| 11-13                      | Mainstream Calculus                | 4396               | 666                                            | 15%                                                | 577                                            | 2859               | 20%                                                |
| 14-15                      | Non-Mainstream Calculus            | 882                | 254                                            | 29%                                                | 235                                            | 371                | 63%                                                |
| 16-18                      | Advanced level                     | 761                | 62                                             | 8%                                                 | 129                                            | 485                | 27%                                                |
| 19-20                      | Statistics, Probability            | 9661               | 1977                                           | 21%                                                | 3651                                           | 4715               | 77%                                                |
| 21-27                      | Service courses                    | 7014               | 2053                                           | 29%                                                | 3088                                           | 4349               | 71%                                                |
| 28-29                      | Technical Mathematics <sup>2</sup> | 1433               | 501                                            | 35%                                                | 532                                            | 525                | 101%                                               |
| 30-31                      | Other mathematics courses          | 1845               | 813                                            | 44%                                                | 227                                            | 413                | 55%                                                |
| 1-31                       | Total, All Courses                 | 77893              | 28014                                          | 36%                                                | 20517                                          | 30819              | 67%                                                |

<sup>1</sup> For names of specific courses, see Table TYE.3.

<sup>2</sup> Nine institutions reported a greater number of sections taught by part-time faculty than total number of sections, resulting in a percentage over 100%.



**FIGURE TYE.9.1.** Proportion of sections of mathematics and statistics courses taught by full-time and by part-time faculty in mathematics departments at public two-year colleges by type of course<sup>2</sup> in fall 2015.

<sup>1</sup> Nine institutions reported a greater number of sections taught by part-time faculty than total number of sections, resulting in a percentage over 100%.

<sup>2</sup> For names of specific courses, see Table TYE.3.

and Probability. Elementary Statistics showed 67% (SE 0) of on-campus enrollments in distance/remote learning format. The 2021 percentage increased between 40–70% in every course from 2015. See Table TYE.11.

As stated earlier in the chapter, it is suggested that the distance/remote learning increases in enrollment for higher-education institutions were driven by the necessary change in instructional strategies due to the COVID virus pandemic. In order to assess the impact of the COVID virus on instruction, CBMS administered a specific COVID survey in March 2021, which is presented in Appendix IV.

CBMS2021 continued to collect data on characteristics of distance learning courses and programs in two-year colleges (see Table TYE.12 in this chapter and Table SP.8.B(c) in Chapter 2). Eighty-nine percent (89%; SE 4.8) of mathematics departments taught distance learning courses, a small change from 2015 (87%; SE 4). While 69% (SE 6) of those courses were taught only asynchronously online in 2015, there was a small decrease to 68% (SE 6.4) in 2021. In 2021, some sections were taught online, with an oppor-

tunity to meet synchronously online (25%; SE 6.3), while other sections used a mixture of online and face-to-face sessions (7%; SE 2.6).

Data was collected in 2015 and 2021 about how distance/remote learning students completed the majority of tests. A notable difference from 2015 is that in 2021, 18% (SE 5.3) completed tests at a monitored testing site, compared to 47% (SE 5) in 2015. In 2021, 37% (SE 6.6) of departments reported using a combination of non-monitored tests, online tests using monitoring technology, and monitoring at testing sites. See Table TYE.12.

A more detailed discussion about trends in distance learning enrollment in four-year institutions can be found in Table SP.9 in Chapter 2. In 2021, while the number of students enrolled in distance/remote learning in four-year mathematics departments was less than at two-year colleges, data showed increased distance/remote learning enrollments in all course categories.

Distance/remote learning delivery and course design presented unique challenges for departments. “Replicating

**TABLE TYE.10.** Percentage of mathematics departments at public two-year colleges which implemented a Pathways<sup>1</sup> course sequence in 2015 and 2021, an Intermediate Algebra prerequisite in 2021, and the types of courses implemented.

|                                        | Implemented in 2015? | Implemented in 2021? | Intermediate Algebra a Prerequisite in 2021? |
|----------------------------------------|----------------------|----------------------|----------------------------------------------|
| Pathways course                        | Yes                  | Yes                  | Yes                                          |
| Implemented a Pathways course sequence | 58                   | 71                   |                                              |
| Implemented Pathways course in:        |                      |                      |                                              |
| a. Foundations                         | 51                   | 88                   | 23                                           |
| b. Quantitative Reasoning/Literacy     | 59                   | 87                   | 17                                           |
| c. Statistics                          | 63                   | 74                   | 62                                           |
| d. Other                               | 32                   | 64                   | 29                                           |

Note: The percentage of mathematics departments at two-year colleges which implemented an Intermediate Algebra prerequisite was added to the table in 2021.

<sup>1</sup> Pathways is defined to be a redesign of a mathematics sequence that provides students with an alternative course or sequence to/through developmental mathematics and to/through a college-level mathematics or statistics course.

active learning in a virtual environment” was a very significant challenge in 53.3% (SE 6.89) of two-year college departments. “Maintaining academic integrity on assessments” was the second significant challenge for 49% (SE 6.93) of departments. The third very significant challenge was “engaging students online” for 47.5% (SE 6.82) of the departments. In 2015, “maintaining a level of rigor in distance learning mathematics courses equivalent to courses offered face-to-face” was reported to be “somewhat of a challenge” by 41% (SE 5) of two-year colleges and a “very significant challenge” by 17% (SE 5). See Table TYE.12.1 in this chapter and Table SP.10 in Chapter 2.

**Placement testing, math clubs, independent study, honors programs, programs for minorities, and programs for women.** Table TYE.13 reports that 74% (SE 6) of colleges usually required placement testing for first-time students, compared with 78% (SE 4) in 2015. Seventy-nine percent (79%; SE 6) of the colleges responding periodically assess the effectiveness of their placement tests, as reported in both 2021 and 2015. For the first time, CBMS2021 asked about how placement methods, such as testing alone, self-placement, counseling,

scores on state tests, high school record, and ACT/SAT and national assessments, were used for placing students into mathematics courses. Eighty-five percent (85%; SE 7 and 6, respectively) used high school record and ACT/SAT for placement.

Table TYE.13 in this chapter and Tables SP.12 and SP.13 in Chapter 2 report specific outside-of-class opportunities for two- and four-year college mathematics students. CBMS2021 results are similar to CBMS2015 or showed decreases for students’ participation in various activities: mathematics clubs (22% in 2021 (SE 5), compared with 32% in 2015 (SE 5)) and undergraduate research activities (17% in 2021 and 2015 (SE 5 and 3)). Independent studies in mathematics decreased 16% to 25% (SE 6) in 2021, compared with 41% (SE 6) in 2015. Special programs to encourage minorities in mathematics were reported in 16% (SE 4) of two-year colleges, up one percentage point from 2015. Special programs to encourage women in mathematics decreased five points to 10% (SE 4) in 2021, compared to 15% (SE 3) in 2015. The decreases in student activities at both two- and four-year colleges may be related to the COVID virus and difficulty in offering activities to students.

**TABLE TYE.11.** Enrollments in distance/remote learning (in 1000s) and percentage of distance/remote learning enrollments (distance learning courses are courses in which half or more of the instruction occurs with the instructor and the students separated by time and/or place facilitated by technology) among all enrollments (excluding dual enrollments) in mathematics departments at two-year colleges in fall 2010, 2015, and 2021.

| Course Number | Type of Course                                | 2010<br>Total<br>Enrollments <sup>1</sup><br>(1000s) | 2010<br>Distance<br>Enrollments<br>(1000s) | 2010<br>Percentage<br>Distance<br>Enrollments | 2015<br>Total<br>Enrollments <sup>1</sup><br>(1000s) | 2015<br>Distance<br>Enrollments<br>(1000s) | 2015<br>Percentage<br>Distance<br>Enrollments | 2021<br>Total<br>Enrollments <sup>1</sup><br>(1000s) | 2021<br>Distance<br>Enrollments<br>(1000s) | 2021<br>Percentage<br>Distance<br>Enrollments |
|---------------|-----------------------------------------------|------------------------------------------------------|--------------------------------------------|-----------------------------------------------|------------------------------------------------------|--------------------------------------------|-----------------------------------------------|------------------------------------------------------|--------------------------------------------|-----------------------------------------------|
| 1             | Arithmetic & Basic Mathematics                | 146                                                  | 11                                         | 7%                                            | 71                                                   | 9                                          | 13%                                           | 42                                                   | 25                                         | 59%                                           |
| 2             | Pre-Algebra                                   | 226                                                  | 14                                         | 6%                                            | 127                                                  | 9                                          | 7%                                            | 49                                                   | 27                                         | 55%                                           |
| 3             | Elementary Algebra (high school level)        | 428                                                  | 37                                         | 9%                                            | 277                                                  | 38                                         | 14%                                           | 89                                                   | 47                                         | 53%                                           |
| 4             | Intermediate Algebra (high school level)      | 344                                                  | 25                                         | 7%                                            | 299                                                  | 33                                         | 11%                                           | 145                                                  | 94                                         | 65%                                           |
| 5             | Geometry (high school level)                  | 6                                                    | 0                                          | 0%                                            | 8                                                    | 0                                          | 0%                                            | 1                                                    | 0                                          | 16%                                           |
| 5a            | Other Developmental Mathematics Courses       |                                                      |                                            |                                               |                                                      |                                            |                                               | 22                                                   | 15                                         | 71%                                           |
| 6             | College Algebra (above Intermed. Algebra)     | 230                                                  | 32                                         | 14%                                           | 292                                                  | 38                                         | 13%                                           | 193                                                  | 114                                        | 59%                                           |
| 7             | Trigonometry                                  | 45                                                   | 4                                          | 10%                                           | 51                                                   | 4                                          | 9%                                            | 40                                                   | 29                                         | 71%                                           |
| 8             | College Algebra & Trigonometry (combined)     | 11                                                   | 1                                          | 12%                                           | 13                                                   | 1                                          | 7%                                            | 12                                                   | 5                                          | 44%                                           |
| 9             | Introduction to Mathematical Modeling         | 18                                                   | 1                                          | 4%                                            | 2                                                    | 1                                          | 46%                                           | 3                                                    | 1                                          | 42%                                           |
| 10            | Precalculus/Elem. Functions/Analytic Geometry | 64                                                   | 3                                          | 5%                                            | 87                                                   | 10                                         | 12%                                           | 81                                                   | 47                                         | 59%                                           |
| 11            | Mainstream Calculus I                         | 65                                                   | 2                                          | 3%                                            | 66                                                   | 4                                          | 6%                                            | 78                                                   | 49                                         | 63%                                           |
| 12            | Mainstream Calculus II                        | 29                                                   | 0                                          | 1%                                            | 34                                                   | 2                                          | 5%                                            | 29                                                   | 18                                         | 61%                                           |
| 13            | Mainstream Calculus III                       | 15                                                   | 0                                          | 0%                                            | 19                                                   | 1                                          | 4%                                            | 17                                                   | 10                                         | 60%                                           |
| 14            | Non-Mainstream Calculus I                     | 20                                                   | 2                                          | 8%                                            | 26                                                   | 3                                          | 13%                                           | 15                                                   | 11                                         | 72%                                           |
| 15            | Non-Mainstream Calculus II                    | 2                                                    | 0                                          | 0%                                            | 0                                                    | 0                                          | 0%                                            | 3                                                    | 2                                          | 61%                                           |
| 16            | Differential Equations                        | 6                                                    | 0                                          | 2%                                            | 7                                                    | 0                                          | 1%                                            | 9                                                    | 6                                          | 70%                                           |
| 17            | Linear Algebra                                | 5                                                    | 0                                          | 4%                                            | 7                                                    | 0                                          | 6%                                            | 8                                                    | 5                                          | 69%                                           |
| 18            | Discrete Mathematics                          | 2                                                    | 0                                          | 12%                                           | 5                                                    | 1                                          | 13%                                           | 6                                                    | 4                                          | 72%                                           |

Note: 0% means less than one-half of one percent.

<sup>1</sup> Does not include dual enrollments.



**TABLE TYE.11 (continued).** Enrollments in distance/remote learning (in 1000s) and percentage of distance/remote learning enrollments (distance learning courses are courses in which half or more of the instruction occurs with the instructor and the students separated by time and/or place facilitated by technology) among all enrollments (excluding dual enrollments) in mathematics departments at two-year colleges in fall 2010, 2015 and 2021.

| Course Number | Type of Course                                            | 2010                                   | 2010                         | 2010                            | 2015                                   | 2015                         | 2015                            | 2021                                   | 2021                         | 2021                            |
|---------------|-----------------------------------------------------------|----------------------------------------|------------------------------|---------------------------------|----------------------------------------|------------------------------|---------------------------------|----------------------------------------|------------------------------|---------------------------------|
|               |                                                           | Total Enrollments <sup>1</sup> (1000s) | Distance Enrollments (1000s) | Percentage Distance Enrollments | Total Enrollments <sup>1</sup> (1000s) | Distance Enrollments (1000s) | Percentage Distance Enrollments | Total Enrollments <sup>1</sup> (1000s) | Distance Enrollments (1000s) | Percentage Distance Enrollments |
| 19            | Elementary Statistics (with or w/o Probability)           | 134                                    | 23                           | 17%                             | 251                                    | 31                           | 12%                             | 240                                    | 162                          | 67%                             |
| 20            | Probability (with or w/o Statistics)                      | 3                                      | 0                            | 7%                              | 28                                     | 2                            | 9%                              | 17                                     | 8                            | 46%                             |
| 21            | Finite Mathematics                                        | 18                                     | 2                            | 11%                             | 40                                     | 4                            | 11%                             | 11                                     | 7                            | 62%                             |
| 22a           | Math for Liberal Arts                                     | 91                                     | 15                           | 17%                             | 97                                     | 19                           | 19%                             | 83                                     | 50                           | 60%                             |
| 22b           | Quantitative Literacy/Quantitative Reasoning <sup>2</sup> |                                        |                              |                                 |                                        |                              |                                 | 73                                     | 43                           | 59%                             |
| 23            | Mathematics for Elementary School Teachers I              | 21                                     | 2                            | 11%                             | 12                                     | 2                            | 17%                             | 13                                     | 7                            | 56%                             |
| 24            | Mathematics for Elementary School Teachers II             | 8                                      | 2                            | 20%                             | 3                                      | 1                            | 32%                             | 4                                      | 3                            | 70%                             |
| 25            | Other Mathematics Courses for Teacher Preparation         | 1                                      | 0                            | 0%                              | 1                                      | 0                            | 0%                              | 1                                      | 1                            | 100%                            |
| 26a           | Business Mathematics                                      |                                        |                              |                                 |                                        |                              |                                 | 13                                     | 9                            | 71%                             |
|               | Non-Calculus-Based Technical Mathematics                  |                                        |                              |                                 |                                        |                              |                                 |                                        |                              |                                 |
|               | Calculus-Based Technical Mathematics                      |                                        |                              |                                 |                                        |                              |                                 |                                        |                              |                                 |
| 26b           | Business Mathematics (not transferable)                   | 16                                     | 3                            | 19%                             | 16                                     | 3                            | 21%                             |                                        |                              |                                 |
| 27            | Business Mathematics (transferable)                       | 4                                      | 0                            | 7%                              | 10                                     | 1                            | 11%                             |                                        |                              |                                 |
| 28            | Technical Mathematics (non-calculus-based)                | 17                                     | 1                            | 7%                              | 21                                     | 3                            | 12%                             | 13                                     | 7                            | 55%                             |
| 29            | Technical Mathematics (calculus-based)                    | 1                                      | 0                            | 37%                             | 3                                      | 0                            | 6%                              | 6                                      | 4                            | 59%                             |
| 30            | Other Mathematics Courses (not transferable)              | 33                                     | 2                            | 7%                              | 31                                     | 2                            | 7%                              | 16                                     | 12                           | 74%                             |
| 31            | Other Mathematics Courses (transferable)                  | 14                                     | 3                            | 19%                             | 12                                     | 1                            | 13%                             | 3                                      | 2                            | 69%                             |
|               | Total Enrollments                                         | 2024                                   | 188                          | 9%                              | 1918                                   | 225                          | 12%                             | 1333                                   | 823                          | 62%                             |

Note: 0% means less than one-half of one percent.

<sup>1</sup> Does not include dual enrollments.

<sup>2</sup> These courses were not listed before 2021.

**TABLE TYE.12.** Percentage of mathematics departments reporting use of distance learning in mathematics departments at two-year colleges in 2013–2021.

|                                                                                      | 2013–2015 | 2018–2021 |
|--------------------------------------------------------------------------------------|-----------|-----------|
| A. Department taught distance learning courses                                       |           |           |
| a. Yes                                                                               | 87        | 89        |
| b. No                                                                                | 13        | 11        |
| B. Format of majority of distance learning                                           |           |           |
| a. Sections are taught only online and only asynchronously                           | 69        | 68        |
| b. Sections are taught only online, with an opportunity to meet synchronously online | 22        | 25        |
| c. Sections use a mixture of online and face-to-face sessions                        | 8         | 7         |
| d. Other                                                                             |           |           |
| C. How distance learning students take majority of tests                             |           |           |
| a. Not monitored                                                                     | 11        | 24        |
| b. Online, but using monitoring technology                                           | 10        | 21        |
| c. At monitored testing site                                                         | 47        | 18        |
| d. Combination of above                                                              | 32        | 37        |

Note: Blank entries represent courses with no responses.

**TABLE TYE.12.1.** Percentage of departments with distance learning that described various factors as significant challenges or somewhat of a challenge in fall 2021.

|                                                                          | No Challenge | Somewhat of a Challenge | Very Significant Challenge |
|--------------------------------------------------------------------------|--------------|-------------------------|----------------------------|
| a. Designing appropriate assessments of student learning.                | 18.7         | 54.2                    | 27.1                       |
| b. Maintaining academic integrity on assessments.                        | 8.2          | 42.8                    | 49.0                       |
| c. Grade inflation.                                                      | 31.1         | 54.3                    | 14.6                       |
| d. Building/maintaining community among faculty and students.            |              | 60.5                    | 39.5                       |
| e. Maintaining academic quality instruction.                             | 24.3         | 59.0                    | 16.7                       |
| f. Engaging students online.                                             | 0.7          | 51.8                    | 47.5                       |
| g. Replicating active learning in a virtual environment.                 | 1.8          | 44.9                    | 53.2                       |
| h. Availability of equipment and technical support for faculty/students. | 20.0         | 63.0                    | 17.0                       |

Note: Blank entries represent courses with no responses. Factors were updated to match 2021 survey questions.

**TABLE TYE.13.** Percentage of two-year colleges offering various opportunities and services to mathematics students in fall 2010, 2015, and 2021.

| Opportunity/Service                                                             | 2010 | 2015 | 2021 |
|---------------------------------------------------------------------------------|------|------|------|
| A. Diagnostic or placement testing                                              | 90   | 94   | na   |
| a. Colleges that usually require placement tests of first-time enrollees        | 100  | 78   | 74   |
| Placement test alone                                                            |      |      | 52   |
| Self-placement                                                                  |      |      | 33   |
| Counseling/advisement                                                           |      |      | 47   |
| Scores on state test                                                            |      |      | 39   |
| High school record                                                              |      |      | 85   |
| ACT/SAT                                                                         |      |      | 85   |
| Scores on national assessment                                                   |      |      | 36   |
| b. Colleges that periodically assess the effectiveness of their placement tests | 75   | 79   | 79   |
| B. Advising by a member of the mathematics faculty                              | 42   | 49   | 33   |
| C. Opportunities to compete in mathematics contests                             | 41   | 40   | na   |
| D. Honors sections                                                              | 20   | 28   | na   |
| E. Mathematics club                                                             | 31   | 32   | 22   |
| F. Special mathematics programs to encourage minorities                         | 11   | 15   | 16   |
| G. Lectures/colloquia for students, not part of math club                       | 16   | 21   | 14   |
| H. Special mathematics programs to encourage women                              | 6    | 15   | 10   |
| I. K–12 outreach opportunities                                                  | 32   | 46   | 40   |
| J. Undergraduate research opportunities                                         | 14   | 17   | 17   |
| K. Independent mathematics studies                                              | 36   | 41   | 25   |
| L. Other                                                                        | 13   | 5    | 14   |

### Topics of Special Interest for Mathematics Departments at Two-Year Colleges

In each CBMS survey cycle, certain topics of special interest are chosen for data collection and comprehensive analysis across both two- and four-year colleges, or for two- or four-year institutions individually. Special topics for two- and four-year institutions are discussed in Chapters 1, 2, and 6 of this report. Of special note in CBMS2021 is an emphasis on collecting data regarding instructional strategies.

**Instructional practices, content, and delivery methods.** CBMS2021 included questions on all three questionnaires (two-year college, four-year college, and statistics) about content, delivery methods, and instructional strategies. Data regarding trends in content including students collecting and analyzing data, solving contextually

based problems, focus on quantitative reasoning, and less symbol manipulation are presented in Tables S.10, S.11, and S.12 in Chapter 1. For example, Table S.10 presents data on how often various pedagogical strategies were used in teaching pre-college and introductory classes in fall 2021, again providing a breakdown of this data by level of the four-year mathematics department and including two-year college data. The strategy of “using assessments such as regular graded homework and quizzes to inform teaching” was used at least once a week in fall 2021 at an estimated 91% (SE 3.6) of two-year college mathematics departments, 92% (SE 3.3) of four-year mathematics departments combined, and only 78% (SE 10.4) of doctoral-level mathematics departments. “Focusing on conceptual understanding” was used at least once a week in an estimated 87% (SE 4.5) of two-year college mathematics departments and

71% (SE 5.9) for four-year mathematics departments combined. “Integrating real-world applications” was used at an estimated 68% (SE 6.2) of two-year college mathematics departments and 56% (SE 6) of four-year mathematics departments combined. “Using student-centered active learning strategies” was used at an estimated 45% (SE 6.6) of two-year college mathematics departments and 47% (SE 6.5) of four-year mathematics departments combined.

Data regarding online instruction appears in Tables SP.8.A, SP.8.B(a)–SP.8.B(c), and SP.9 in Chapter 2. For example, Table SP.8.A shows that in fall 2021, the estimated percentage of departments whose faculty were generally more favorable toward online instruction than before the pandemic was 46% (SE 5.2) of all four-year mathematics departments combined, 74% (SE 6.8) of all statistics departments combined, and 64% (SE 6.4) of two-year mathematics departments. Table SP.8.B(c) presents that an estimated 30% (SE 6) of two-year mathematics departments offered some hybrid course pre-pandemic. Then during the pandemic, 37% (SE 6.8) offered some hybrid course, and in fall 2021, 60% (SE 6.6) offered some hybrid course. In fall 2021, 59% (SE 5.3) of four-year mathematics departments, 65% (SE 7.4) of statistics departments, and 74% (SE 6.2) of two-year mathematics departments offered some course in a remote-only format.

**A. Departmental activities in response to diversity, equity, and inclusion issues.** Table SP.26 in Chapter 2 summarizes responses from all three types of departments about the extent to which six possible departmental activities related to diversity, equity, and inclusion (DEI) issues had taken place in the department during the previous year; the possible responses were “none,” “some,” and “a lot” (and in the case of the activity involving graduate students, “not applicable”). The results are broken down by level of four-year mathematics department, both levels of statistics department combined, and two-year mathematics departments.

The activity “faculty discussion designed to create awareness of DEI issues” appears to be the activity that was most likely to occur. Table SP.26 shows that it is the activity with generally the largest percentages of responses of “a lot”: the response of 23% (SE 5.4) of all levels of four-year mathematics departments combined, of 32% (SE 7.9) of both levels of statistics departments combined, and of 33% (SE 6.7) of two-year mathematics departments. This was also the activity with the lowest percentage of answers of “none.” The estimated percentage of departments answering “none” was 15% (SE 4.5) of all levels of four-year mathematics departments combined, 16% (SE 6.2) of both levels of statistics departments combined, and 14% (SE 5.2) of two-year mathematics departments.

The activity with generally the next largest responses of “a lot” and the next smallest percentages of responses of “none” was “Consideration of existing or new programs to assist underrepresented groups and/or at-risk students in the mathematical sciences,” where the percentage of departments answering “a lot” was 11% (SE 3.8) of all levels of four-year mathematics departments combined, 32% (SE 7.9) of both levels of statistics departments combined, and 17% (SE 5.1) of two-year mathematics departments. The estimated percentage of departments answering “none” was 35% (SE 6.3) of all levels of four-year mathematics departments combined, 16% (SE 6.2) of both levels of statistics departments combined, and 38% (SE 7.2) of two-year mathematics departments.

From the distributions of percentages in Table SP.26, the other activities listed, “Student discussion designed to increase awareness of DEI issues,” “Program or policy changes to affect the demographic balance of faculty in the mathematical sciences,” “Program or policy changes to affect the demographic balance of undergraduate students in the mathematical sciences,” and “Program or policy changes intended to affect the demographic balance of graduate students in the mathematical sciences,” seemed to have received less attention in departments.

The percentage of departments responding “a lot” was higher for both levels of statistics departments combined than for all levels of four-year mathematics departments combined and for two-year mathematics departments for most of the issues.

**Scope and organization of pre-service mathematics education for K–16 teachers.** CBMS2021 continued an inquiry that began in 2000 about the level of involvement of two-year college mathematics departments in the mathematical education of future mathematics teachers. These data are reported primarily among the special topics in Chapter 2, Tables SP.2 and SP.3.

In the last two decades, involvement in teacher education at two-year colleges has been active as more students turned to them to take required mathematics and education courses. Fall 2021 saw student enrollment in Mathematics for Elementary School Teachers I increase to 13,000 (SE 2.1), as compared to the decrease reported in 2015 to 12,000 (SE 1.8). A similar increase of 1000 students (SE 0.94) in fall 2021 was seen in the second course, Mathematics for Elementary School Teachers II. See Table TYE.3.

Table TYE.5 shows that 54% (SE 6.65) of two-year colleges offered the course Mathematics for Elementary School Teachers I in fall 2021, up 13% from 2015 (41%; SE 5). For the five-year CBMS intervals beginning in 1990 through 2015, the percentages of two-year colleges teaching Mathematics for Elementary School Teachers I are successively

32%, 43%, 49%, 59%, 55%, 41%, and 54%. Of special note is the increase in departments offering Mathematics for Elementary School Teachers II in 2021, up 28 points to 45% (SE 6.79) since 2015. See Table TYE.6.

In contrast, Table SP.2 in Chapter 2 reports on “organized” programs at two-year colleges in which students can obtain their entire mathematics course requirement for teacher licensure. CBMS2021 and CBMS2015 data present decreasing numbers in elementary, middle, and secondary school programs. The single largest component is the program for pre-service elementary school teachers reported by 22% (SE 6) of two-year colleges in 2021 and 28% in 2015. Pre-service middle school licensure-oriented programs reported a 9% decrease to 5% (SE 2) of colleges. Between 2% and 15% (SE 2 and 5, respectively) of two-year colleges reported programs at the elementary- or middle-school levels for retraining by career switchers moving into teaching. Compared with 2015, all categories (except career switchers aiming for elementary teaching) showed decreases in percentages of responding colleges in 2021 in Table SP.2.

Table SP.3 in Chapter 2 reports on other involvements two-year college mathematics departments have with teacher education. Thirty-three percent (33%; SE 9) report that a faculty member is assigned to coordinate mathematics education for future K–8 teachers, compared with 35% in 2015. Twenty-two (22%; SE 6) of the reporting colleges designate special mathematics courses for future (pre-service) K–8 teachers, down 33 points from 2015. Fall 2021 saw a decrease of 17 points from 2015, down to 2% (SE 2), in offering a special mathematics course for pre-service secondary teachers. Showing increases in 2021, among mathematics departments, 81% (SE 16) offered mathematics pedagogy courses for future teachers within the department, and 28% (SE 18) of colleges offered such pedagogy courses outside the mathematics department.

**Dual enrollment.** In CBMS2021, dual-enrollment courses, defined to be “courses conducted in a high school campus by high school teachers, for which students may obtain high school credit and, simultaneously, college credit,” increased 72% (SE 6) from 2015 to 2021 for 162,000 students compared with 63% from 2010 to 2015 for an estimated 94,000 students. Dual-enrollment increases were noted in every course category in 2021. See Tables SP.16 and SP.17 in Chapter 2 and Table TYE.2 in this chapter.

In some cases, a faculty member teaching a dual-enrollment course was classified as a part-time faculty member at the two-year college that awarded college credit for the course, even though the salary was paid completely by a third party, e.g., the local school district. In 2021, two-year colleges assigned and paid their own faculty to teach courses in a high school that awards both high school and college credit in 34% (SE 7) of courses, compared to four-year colleges (6%; SE 2). See Table SP.17 in Chapter 2.

Increases in dual enrollment in 2021 accounted for 12% of all mathematics and statistics enrollment at two-year colleges, a total of 162,000 students comprising a 72% increase, compared with four-year college mathematics departments with 40,200 students, which accounts for a 41% decrease.

As reported in Tables TYF.24 and TYF.25 in Chapter 7, among all survey respondents (including respondents from colleges that do not have dual-enrollment arrangements), 5% (SE 3) of mathematics program heads in two-year colleges saw dual-enrollment courses as a “major problem,” down two points from 2015. Another 34% (SE 7) found dual-enrollment arrangements “somewhat of a problem,” down two points from 2015.

## Chapter 7

# Faculty, Administration, and Special Topics in Mathematics Departments at Two-Year Colleges

This chapter continues the presentation of data and analysis about mathematics faculty and departments in public two-year colleges. It reports the estimated number, teaching conditions, education, professional activities, age, gender, and ethnicity of the faculty in these mathematics departments in fall 2021. Additional analysis of some items discussed in this chapter can be found in Chapters 1 and 2, where they are discussed from a comprehensive point of view in comparison to similar data for four-year colleges and universities. In particular, Chapter 2 discusses issues related to dual enrollment, distance learning courses, and instructional strategies. CBMS survey data has been collected since 1965. However, unlike surveys prior to 1995, the surveys since 1995 do not include faculty who taught in computer science departments that were separate from mathematics departments. Also, CBMS2005, CBMS2010, CBMS2015, and CBMS2021 include data regarding public two-year colleges only. The estimated data in this chapter have not been rounded. Information on the sampling procedure used in the 2021 survey can be found in Appendix II. A copy of the CBMS2021 two-year college survey questionnaire can be found in Appendix VII.

The term “full-time permanent” faculty is used frequently in this document. Two-year college faculty in this category have an on-going, stable relationship with the college’s mathematics departments, and are tenured and tenure-eligible faculty, including those on leave or on sabbatical. They occupy a recurring position in the college’s budget and are subject to the college’s long-term evaluation and re-appointment policy. These faculty are responsible for teaching, curriculum development, student advising, committee appointments, and other forms of college service.

Full-time faculty who are employed on a non-tenure track, sometimes continuing, are called “full-time continuing” faculty in this document. Two-year colleges often have their own individual classification for a third category of “other” non-tenure-track full-time faculty. Data about this third classification of positions was collected for the first time in CBMS2015. This group is referred to as “other” full-time faculty in this document. Full-time “permanent” faculty are distinguished from “continuing” or “other” full-time faculty who are often meeting a short-

term institutional need. All three categories of full-time faculty teach full course assignments, distinguishing them from part-time or adjunct faculty.

The Table display code in this chapter is TYF, for “two-year faculty,” since the chapter discusses issues related to faculty. Some tables have entries of “not reported,” “unknown,” or “status not known or other” where data was not available or submitted. This is not surprising, given the low response rate and that questions regarding faculty appeared near the end of the questionnaire. A summary of the sample protocols, stratum, and number of completes can be found in Appendix II.

In the text that follows, the standard error (SE) in many of the estimates is provided along with the estimate (e.g., estimate 4596 (SE 58)). The SEs for all CBMS2021 tables can be found in Appendix IX. The change in an estimate from the estimate in a previous survey is often expressed both as percentage change and as the number of SEs that change represents (e.g., “increased 22% (1.2 SEs)”).

## Highlights of Chapter 7

### Number of full-time faculty and part-time faculty.

- In fall 2021, the total estimated number of full-time faculty (permanent, continuing, and other) in two-year colleges was 10,328 (SE 1117), a 5% increase compared to 9801 (SE 894) in 2015. The increase in faculty, 202 persons, can be viewed in light of the 28% decrease in institutional enrollment in two-year colleges to 4,478,000 students and the 26% decrease in mathematics and statistics enrollment to 1,495,170 (SE 157,543) students discussed in Chapter 6. See Table S.13 in Chapter 1, Table TYE.2 in Chapter 6, and TableTYF.1 in this chapter.
- It was estimated that there were 9126 (SE 918) full-time permanent faculty in public two-year college mathematics departments in the U.S. in fall 2021, compared with 8314 (SE 840) in 2015. This 10% increase of 812 persons can be compared to a 15% decrease of

1476 persons in 2015. See Table S.14 in Chapter 1 and Table TYF.1 in this chapter.

- In fall 2021, the estimated number of part-time faculty in two-year college mathematics departments was 18,543. This included 13,827 persons (SE 1631) paid by two-year colleges and 4716 persons (SE 969) paid by third parties such as school districts, an 8% decrease compared with 20,247 in 2015 (17,888, SE 1909, paid by two-year colleges and 2359, SE 528, paid by third parties such as school districts). Part-time faculty represented 64% of the total number of faculty in 2021, compared with 67% in 2015. When third party payees are omitted, part-time faculty represented 57% of the total number of faculty, down eight points from 2015. See Table TYF.1.
- In fall 2021, 67% (SE 7) of responding colleges reported the average teaching assignment for full-time permanent faculty to be 13–15 hours, down one point from 2015. The average weekly contact hours for full-time permanent faculty decreased to 15 hours (SE 0.5), compared to 18 (SE 2) hours in fall 2015. Table TYF.2 shows that 64% (SE 4) of full-time permanent faculty taught extra hours for extra pay at their own college in fall 2021, down ten points from 2015. Of those faculty who taught for extra pay, 33% (SE 4) taught 1–3 extra hours, 38% (SE 4) taught 4–6 hours, and 29% (SE 5) taught 7 or more hours. See Table TYF.2.
- In fall 2021, fifty-five percent (55%; SE 4) of part-time faculty taught 6 or more hours in 2021, down nine points from 2015. Sixty-seven percent (67%; SE 0) of all sections were taught by part-time faculty in fall 2021, a 31% increase from 2015. See Table TYE.9 in Chapter 6 and Table TYF.2 in this chapter.
- There were 256 (SE 63) faculty who were no longer part of the faculty in 2021–2022, compared to 612 (SE 132) who were no longer part of the faculty in 2015–2016. See Table TYF.3.

### **Educational credentials of faculty in mathematics departments.**

- In fall 2021, a master's degree was the terminal degree for 80% (SE 3) of the full-time permanent mathematics faculty at two-year colleges, the same percentage as in 2015. An additional 19% (SE 3) of full-time faculty held doctorates and 1% (SE 0) held bachelor's degrees. Of the total full-time permanent faculty, 73% (SE 3) held either a master's or doctoral degree in mathematics as their highest degree, 15% (SE 2) held either a master's or doctoral degree in mathematics education as their highest degree, and 6% (SE 2) held either a master's or doctoral degree in statistics as their highest degree. See Tables TYF.4 and TYF.5.
- Among part-time faculty in fall 2021, six percent (6%; SE 1) held a doctorate (down one point from 2015), 80% (SE 3) held a master's degree (up 4 points from 2015), and 14% (SE 3) held a bachelor's degree as their highest degree (down three points from 2015). A bachelor's degree may be considered an appropriate or terminal degree for those teaching pre-college courses or by accrediting agencies for faculty teaching highly specialized technical courses. See Table TYF.6.
- In fall 2021, fifty-seven percent (57%; SE 6) of part-time faculty held degrees in an academic major in the field of mathematics, 27% (SE 5) in the field of mathematics education, and 2% (SE 1) in the field of statistics. See Table TYF.7.

### **Gender, ethnic composition, and age of full-time permanent mathematics departments faculty.**

- In fall 2021, 49% (SE 3) of full-time permanent faculty were women and 51% (SE 3) were men. Forty-six percent (46%; SE 3) of the women were part-time faculty, and 54% (SE 3) of men were part-time faculty. See Tables TYF.8 and TYF.9.
- In fall 2021, the percentage of ethnic minorities among full-time permanent faculty in mathematics departments in two-year colleges was 15% (SE 3), compared with 23% (SE 2) in 2015. The total number of ethnic minority faculty was 1375 (SE 262) in 2021, down 501 persons from 1876 (SE 289) in 2015. Within specific ethnic groups other than White (4785 persons; SE 613), the majority of faculty represented was Asian/Pacific Islander (509 persons; SE 113), down 3 percentage points and 245 persons from 2015. The percentage of women in each ethnic group is displayed in Table TYF.12. See Tables TYF.10, TYF.11, and TYF.12.
- In 2021, the number of full-time permanent faculty under the age of 40 was 1424 (SE 67), down 621 persons compared with 2045 (SE 292) in 2015. Ethnic minorities accounted for 15% (SE 3) of all full-time permanent faculty and 20% (SE 9) of full-time faculty under the age of 40. The percentage of master's degrees awarded in the U.S. in 2020–21 to ethnic minorities increased to 36% in 2021, up from 29% in 2014–15. See Tables TYE.10 and TYF.13.
- Twenty-two percent (22%; SE 4), or 1798 persons, of the total 13,827 faculty in 2021 were part-time ethnic minority faculty paid by two-year colleges, the same as in 2015. The largest minority group, other than White, was Asian/Pacific Islander with 1196 (SE 192) persons, representing 9% of all part-time faculty. Forty-six percent (46%; SE 3) of all part-time faculty were women in

fall 2021, compared to 53% in 2015. See Tables TYF.14 and TYF.15.

- Distribution of faculty by age is displayed in Table TYF.16. While the total number of full-time permanent faculty increased compared with 2015, the data in Table TYF.16 shows a decrease in each age category possibly because of the large percentage, 31% and 2820 persons, of “not reported” on this question. In 2021, the category of faculty >59 years of age had the largest number of faculty with 1171 (SE 185) persons, representing 13% (SE 2) of all full-time faculty. The next largest was the category of 50–54 years, with 1117 (SE 191) persons (12%; SE 2). The largest percentage of full-time permanent women faculty was in age categories of 45–54 and >54 years (12% each; SE 2). The largest percentage of full-time permanent men faculty was in the category >54 years (12%; SE 2). See Table S.16 in Chapter 1 and Tables TYF.16 and TYF.17.

### **Demographics of full-time permanent faculty newly appointed by mathematics departments.**

- There were 255 (SE 70) newly appointed full-time permanent faculty in 2021, compared to 451 (SE 83) newly appointed full-time permanent faculty in fall 2015. The largest percentage of new full-time faculty was persons who had been part-time or full-time temporary faculty (42%; SE 16), compared with 26% (6 SE) in the same category in 2015. There was an eight-point increase of new faculty with a doctorate degree to 17% (SE 12), compared with 9% in 2015. The majority of new full-time faculty (75%; SE 13) held a master’s degree as their highest degree. See Tables TYF.18 and TYF.19.
- Newly appointed full-time permanent faculty in the ethnic minority groups of “Asian/Pacific Islander” and “Mexican American, Puerto Rican, or other Hispanic” made up 11% and 10% of all faculty, respectively (SE 6 and SE 5), in fall 2021. In 2021, forty-six percent (46%; SE 15) of all new appointments were women, down nine points from 2015. See Table TYF.20.

### **Teaching evaluations and professional development of mathematics program faculty.**

- The percentage of two-year colleges requiring periodic teaching evaluations for all full-time faculty was 100% in both fall 2021 and 2015 (SE 0). The percentages of colleges requiring evaluation of part-time faculty decreased to 93% (SE 4), down from 98% (SE 1) in 2015. Percentages of methods for evaluating full- and part-time faculty are reported in Table TYF.22. The most popular method of evaluation for both full- and part-time faculty was “forms completed by students” (97%; SE 2 and 3, respectively), compared with 95% (3 SEs) in

2015. The second most popular method for both full-time faculty and part-time faculty was “observation of classes by other faculty” (full-time: 66%; SE 7, and part-time: 78%; SE 7). See Tables TYF.21 and TYF.22.

- The percentage of two-year colleges requiring annual continuing education or professional development for full-time permanent faculty rose two points to 84% (SE 5) in 2021, compared to 82% (SE 4) in 2015. See Table TYF.23.
- Tables TYF.24 and TYF.25 present various problems experienced by mathematics departments in two-year colleges in 2021. The three items reported by mathematics program heads with the highest percentage as being a “major problem” in 2021 were: “low student motivation” (48%; SE 7), “students not understanding the demands of college work” (42%; SE 7), and “lack of student progress from developmental to advanced courses” (29%; SE 6).
- When considering issues reported as “somewhat of a problem,” the top three items and their percentages were: “lack of student progress from developmental to advanced courses” (55%; SE 7), “faculty salaries too low” (55%; SE 7), and “too few students who intend to transfer actually do” (54%; SE 7). See Tables TYF.24 and TYF.25.

### **Topics of special interest for mathematics departments.**

- Issues related to faculty involvement and instructional strategies in distance learning courses are discussed in Chapters 2 and 6. Distance learning, defined to be “courses in which half or more of the instruction occurs with the instructor and the student separated by time and/or place facilitated by technology,” showed enrollment in mathematics and statistics growing to an estimated 823,000 (SE 108,000) up 266% compared to 225,000 (SE 25,000) students in fall 2015. Distance learning enrollment accounted for 62% of all two-year college mathematics enrollment. Table TYE.12 presents data on various distance learning practices, including the format of the majority of distance learning courses and how distance learning students took tests. The 2021 data results were like 2015 in that 68% (SE 6) of the sections were taught only online, and only asynchronously. Thirty-seven percent (37%; SE 7) used a combination of test-taking formats (e.g., not monitored, online, monitored site). See Tables TYE.12 and TYE.12.1 in Chapter 6 and Tables SP.8.B(c) in Chapter 2.



- CBMS2015 and CBMS2021 asked questions about implementation of mathematics “Pathways,” defined to be “a single course or course sequence that enables students to complete a college-level gateway mathematics or statistics course that is aligned to students’ academic and/or career goals within one academic year.” In fall 2021, 71% (SE 7) of colleges reported having implemented a Pathways course sequence, compared with 2015 when 58% (SE 5) colleges implemented Pathways. Departments sometimes implemented multiple Pathways courses, including Foundations (88%; SE 5), Quantitative Reasoning/Literacy (87%; SE 5), Statistics (74%; SE 7), and Other (64%; SE 13). See Table TYE.10 and the discussion before Table TYE.10 in Chapter 6.
- In CBMS2021, dual-enrollment courses are defined to be “courses conducted in a high school campus by high school teachers, for which students may obtain high school credit and, simultaneously, college credit.” Dual-enrollment courses increased 72% from 2015 to 2021 for 162,000 students, compared with 63% from 2010 to 2015 for an estimated 94,000 students. See Tables SP.16 and SP.17 in Chapter 2 and Table TYE.2 Chapter 6.

## The Number and Teaching Assignments of Full-Time and Part-Time Mathematics Department Faculty

**Number of full-time permanent faculty and part-time faculty.** In fall 2021, the total estimated number of full-time faculty (permanent, continuing, and other) in two-year colleges was 10,328 (SE 1117), a 5% increase compared to 9801 (SE 894) in 2015. This increase is interesting in light of the 28% decrease in two-year college institutional enrollments and the 26% decrease in mathematics and statistics enrollment to 1,495,170 (SE 157,543). See tables in Chapter 1, 2, and 6.

In fall 2021, the total estimated number of faculty reported as “full-time permanent” faculty was 9126 (SE 918), compared with 8314 (SE 840), a 10% decrease of 812 persons from 2015. This data should be considered by examining data of categories of full-time faculty. CBMS2015 and CBMS2021 collected data on full-time faculty in the three categories of permanent, continuing, and other faculty, instead of the two categories, permanent and temporary, in CBMS2010. In 2021, full-time continuing and other faculty together totaled 1202, a decrease of 285 from fall 2015. See Chapter 6 for two-year college enrollment data, Chapter 1 for an overall enrollment data summary, and Table TYF.1.

For comparison, the total estimated number of all full-time faculty in four-year institutions, full-time (tenure-eligible), other full-time, and postdocs, was approximately

27,102 (SE 254) in fall 2021, compared with 22,532 (SE 313) in fall 2015, a 20% increase. The institutional enrollment at four-year institutions decreased 29% in fall 2021. See Table S.13 in Chapter 1.

Part-time faculty in two-year colleges fell into two categories: those paid by two-year colleges and others paid by a third party. The latter most often were high school teachers in a school with which the college had a dual-enrollment agreement. When both categories are included, the estimated number of part-time faculty was 18,543 in fall 2021, an 8% decrease compared with 20,247 in fall 2015. Part-time faculty represented 64% of the total two-year college teaching staff, down 3 percentage points from 2015. When the 4716 (SE 969) third party payees are excluded, the estimated number of part-time faculty was 13,827 (SE 1631), compared with 17,888 (SE 1909) in 2015, a decrease of 23% from 2015 to 2021. Part-time faculty represented 57% of total faculty, down 8 percentage points from 2015. The number of part-time faculty paid by a third party increased almost 100% from 2015 to 2021 to 4716 (SE 969) persons in 2021, compared with 2359 (SE 528) persons in 2015. See Table TYF.1.

Demographics and discussion of newly appointed full-time permanent faculty in fall 2021 are presented later in this chapter in Tables TYF.18, TYF.19, and TYF.20.

## Teaching assignments of full-time permanent and part-time faculty.

The average teaching assignment in weekly classroom contact hours for a full-time permanent mathematics faculty at a public two-year college in fall 2021 was 15 (SE 0.5) contact hours, compared to 18 (SE 2) weekly contact hours in 2015. This figure returned to the contact hours reported in 2010, where the average was also 15 hours. See Table TYF.2 and Figure TYF.2.1.

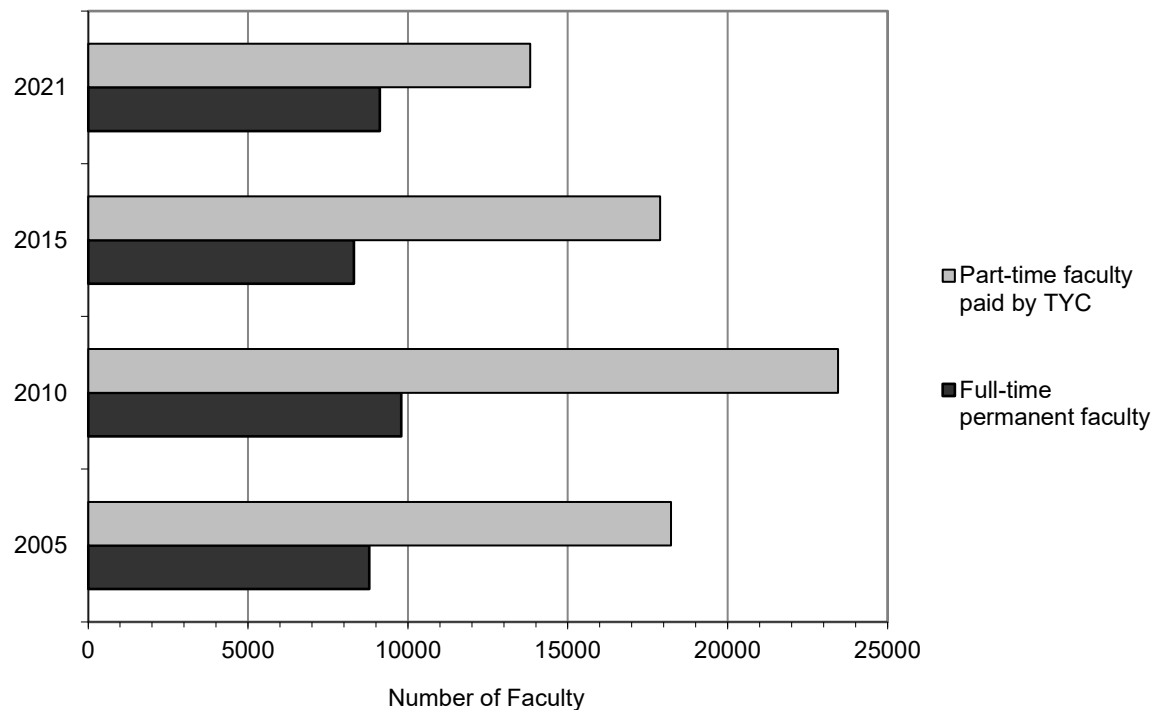
In 2021, most full-time faculty taught between 13 and 15 weekly contact hours in 67% (SE 7) of responding colleges, compared with 68% (SE 5) in 2015. Twenty-two percent (22%) of colleges reported weekly contact hour teaching assignments greater than 15 credit hours, up 3 points from 2015. This included 3% (SE 2) of colleges reporting that teaching assignments were more than 21 credit hours. Twelve percent (12%) had teaching assignments less than 13 weekly credit hours.

Table TYF.2 also shows that 64% (SE 4) of full-time permanent mathematics faculty at two-year colleges taught extra credit hours for extra pay at their own colleges in fall 2021, compared to 74% in 2015. Of those faculty who taught for extra pay in 2021, 33% (SE 4) of full-time permanent faculty taught 1–3 credit hours for extra pay, and 38% (SE 4) taught 4–6 credit hours. Twenty-nine percent (29%; SE 5) taught 7 or more extra credit hours for extra pay, an increase of 6 percentage points from 2015.

**TABLE TYF.1.** Number of full-time permanent faculty, full-time temporary faculty, other full-time faculty, and part-time faculty paid by two-year colleges (TYC) and by a third party (e.g., dual-enrollment instructors) in mathematics departments at two-year colleges in fall 2005, 2010, 2015, and 2021.

| Two-Year Colleges              | 2005  | 2010  | 2015  | 2021  |
|--------------------------------|-------|-------|-------|-------|
| Full-time permanent faculty    | 8793  | 9790  | 8314  | 9126  |
| Full-time continuing faculty   | 610   | 1083  | 1221  | 808   |
| Other full-time faculty        |       |       | 266   | 394   |
| Part-time faculty paid by TYC  | 18227 | 23453 | 17888 | 13827 |
| Part-time, paid by third party | 1915  | 2323  | 2359  | 4716  |

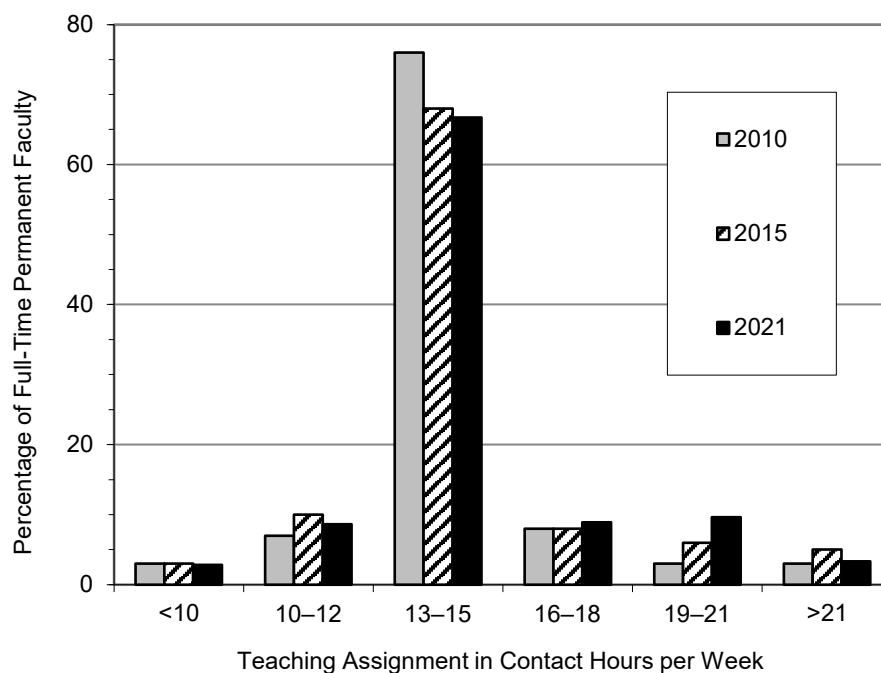
Note: Prior to 2015, there was no differentiation between full-time continuing faculty and other full-time faculty.



**FIGURE TYF.1.1.** Numbers of full-time permanent faculty and part-time faculty paid by TYC in mathematics departments in two-year colleges in fall 2005, 2010, 2015, and 2021.

**TABLE TYF.2.** Teaching assignment for full-time permanent faculty, and teaching and other duties of part-time faculty, in mathematics departments at two-year colleges in fall 2021 (2015 data in parentheses).

|                                                                                               | Teaching Assignment in Weekly Contact Hours |           |            |          |           |          |
|-----------------------------------------------------------------------------------------------|---------------------------------------------|-----------|------------|----------|-----------|----------|
|                                                                                               | <10                                         | 10 to 12  | 13 to 15   | 16 to 18 | 19 to 21  | >21      |
| Percentage of Two-Year Colleges                                                               | 3<br>(3)                                    | 9<br>(10) | 67<br>(68) | 9<br>(8) | 10<br>(6) | 3<br>(5) |
| <b>Full-Time Permanent Faculty</b>                                                            |                                             |           |            |          |           |          |
| A. Average weekly contact hours: 15 (18)                                                      |                                             |           |            |          |           |          |
| B. Percentage who teach extra hours for extra pay at their own two-year college: 64% (74%)    |                                             |           |            |          |           |          |
| C. Percentage teaching 1–3 extra hours for extra pay: 33% (38%)                               |                                             |           |            |          |           |          |
| D. Percentage teaching 4–6 extra hours for extra pay: 38% (39%)                               |                                             |           |            |          |           |          |
| E. Percentage teaching 7 or more extra hours for extra pay: 29% (23%)                         |                                             |           |            |          |           |          |
| <b>Part-Time Faculty</b>                                                                      |                                             |           |            |          |           |          |
| F. Percentage who teach 6 or more hours weekly: 55% (64%)                                     |                                             |           |            |          |           |          |
| G. Percentage of two-year colleges requiring part-time faculty to hold office hours: 37 (29%) |                                             |           |            |          |           |          |



**FIGURE TYF.2.1.** Percentage of full-time permanent faculty with various teaching assignments in mathematics departments at two-year colleges in fall 2010, 2015, and 2021.

**TABLE TYF.3.** Number of full-time permanent faculty in 2020–2021 who were no longer part of the faculty in 2025–2016 and 2021–2022.

|                                              | 2015 | 2021 |
|----------------------------------------------|------|------|
| Number no longer part of 2021–2022 faculty   | 612  | 256  |
| Total full-time permanent faculty, fall 2021 | 8314 | 9126 |

In fall 2021, sixty-seven percent (67%; SE 0) of all sections were taught by part-time faculty. Fifty-five percent (55%; SE 4) of part-time faculty in two-year college mathematics departments taught 6 credit hours or more, down 9 percentage points from 2015. Office hours were required of part-time faculty in 37% (SE 7) of two-year colleges, up 8 points from 2015. See Table TYF.2.

**Outflow of full-time permanent mathematics faculty.** Data about outflow of permanent faculty was collected in detail prior to CBMS2010, including specific information about faculty deaths, faculty retiring, and faculty taking positions at four-year institutions, other two-year institutions, high schools, or graduate schools. Because this detailed information is difficult to obtain, CBMS 2021, CBMS2015, and CBMS2010 collected only the total number of outflow of faculty. In 2021–2022, two hundred fifty-six persons (256; SE 63) and 3% of the total full-time permanent faculty were no longer a part of the faculty, compared with 612 (SE 132) persons in 2015–2016. The authors acknowledge that this data is difficult to collect and may not represent a true picture in the change in faculty numbers over time.

## Educational Credentials of Faculty in Mathematics Departments

### Highest degree of full-time permanent faculty.

In fall 2021, as in fall 2015, a master's degree was the terminal degree for 80% (SE 3) of full-time permanent mathematics faculty at two-year colleges. The percentage of faculty with a doctorate increased four points to 19% (SE 3) in 2021. The percentage of full-time faculty whose terminal degree was a bachelor's degree decreased four points to 1% (SE 0) in 2021. Table TYF.4 and Figure TYF.4.1 present historical data from 2000 to 2021. Data regarding the previous employment and degrees of newly appointed faculty in fall 2021 can be found in Tables TYF.18 and TYF.19.

The academic major and highest degree of full-time permanent two-year college mathematics faculty is shown in Table TYF.5. The total percentage of faculty whose most

advanced degree (doctorate, master's and bachelor's) was in the field of mathematics was 73% (SE 3), similar to 2015. In 2021, the percentage of the faculty whose most advanced degree was in the field of mathematics education increased two points to 15% (SE 2), compared to 2015. The percentage of degrees with majors in the field of statistics increased one point to 6% (SE 2).

**Highest degree of part-time faculty.** Table TYF.6, Figure TYF.6.1, and Table TYF.7 summarize data on the highest degrees held by part-time faculty, along with their fields of specialization. In fall 2021, a doctoral degree was the highest degree held by 6% (SE 1) of part-time faculty, down one point from fall 2015. A master's degree was the highest degree for 80% (SE 3) of part-time faculty, compared to 76% in 2015. A bachelor's degree was the highest degree for 14% (SE 3) of part-time faculty in 2021, a decrease of three points from 2015.

In 2021, the percentage of part-time faculty whose most advanced degree had mathematics or mathematics education as the major field of study was 84% (57%; SE 6 and 27%; SE 5, respectively), compared to the combined total of 77% in 2015. Two percent (2%; SE 1) of part-time faculty held degrees in statistics, down one point from 2015. A six-point decrease to 13% (3 SEs) was reported in "other fields." See Table TYF.7.

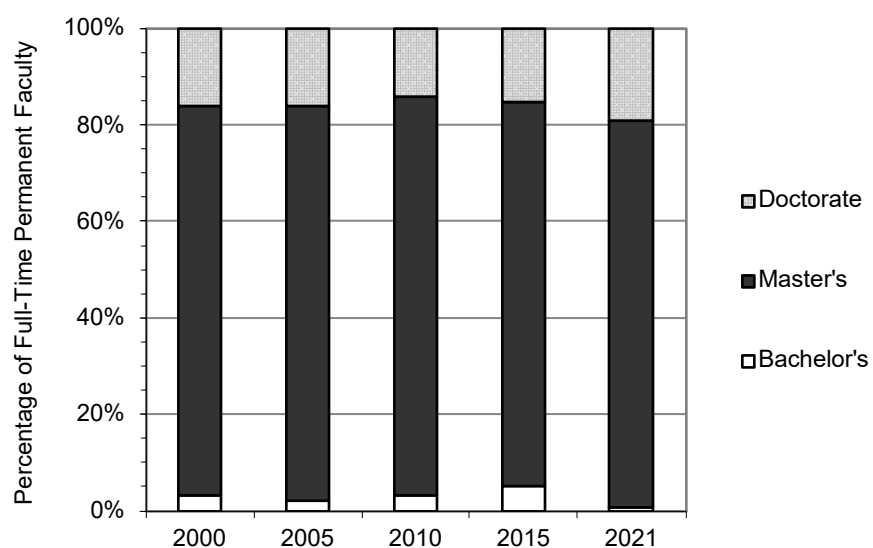
## Gender, Ethnic Composition, and Age of Full-Time Permanent Mathematics Program Faculty

### Gender of full-time permanent faculty and part-time faculty.

A decrease of 3 percentage points in the percentage of women among full-time permanent mathematics faculty at two-year colleges was reported in 2021 for the first time since 1975. In fall 2021, there were 49% (SE 3) of full-time permanent faculty reported as women, compared with 52% in 2015 and 50% in 2021. Therefore, 51% (SE 3) of full-time permanent faculty were men. Table TYF.9 reports that in fall 2021 the percentage of women among

**TABLE TYF.4.** Percentage of full-time permanent faculty in mathematics departments at two-year colleges by highest degree in fall 2000, 2005, 2010, 2015, and 2021.

| Highest Degree                        | Percentage of Full-Time Permanent Faculty |      |      |      |      |
|---------------------------------------|-------------------------------------------|------|------|------|------|
|                                       | 2000                                      | 2005 | 2010 | 2015 | 2021 |
| Doctorate                             | 16                                        | 16   | 14   | 15   | 19   |
| Master's                              | 81                                        | 82   | 83   | 80   | 80   |
| Bachelor's                            | 3                                         | 2    | 3    | 5    | 1    |
|                                       | 100%                                      | 100% | 100% | 100% | 100% |
| Number of full-time permanent faculty | 6960                                      | 8793 | 9790 | 8314 | 9126 |



**FIGURE TYF.4.1.** Percentage of full-time permanent faculty in mathematics departments at two-year colleges by highest degree in fall 2000, 2005, 2010, 2015, and 2021.

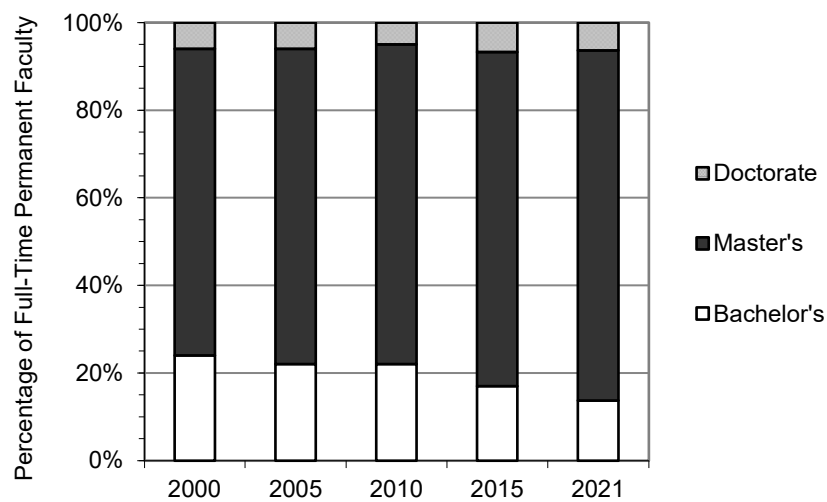
**TABLE TYF.5.** Percentage of full-time permanent faculty in mathematics programs at public two-year colleges by field and highest degree in fall 2021.

| Field of Degree                    | Percentage Having as Highest Degree |          |            | Total Percent in Field |
|------------------------------------|-------------------------------------|----------|------------|------------------------|
|                                    | Doctorate                           | Master's | Bachelor's |                        |
| Mathematics                        | 12                                  | 61       | 0          | 73                     |
| Statistics                         | 1                                   | 5        | 0          | 6                      |
| Mathematics Education              | 5                                   | 9        | 0          | 15                     |
| Other fields                       | 1                                   | 5        | 0          | 7                      |
| Total percentage by highest degree | 19                                  | 80       | 1          | 100                    |

Note: 0 means less than one half of 1%. Round-off may make column sums seem inaccurate.

**TABLE TYF.6.** Percentage of part-time faculty in mathematics departments at two-year colleges (including those paid by a third party, as in dual-enrollment courses) by highest degree in fall 2000, 2005, 2010, 2015, and 2021.

| Highest Degree              | Percentage of Part-Time Faculty |       |       |       |       |
|-----------------------------|---------------------------------|-------|-------|-------|-------|
|                             | 2000                            | 2005  | 2010  | 2015  | 2021  |
| Doctorate                   | 6                               | 6     | 5     | 7     | 6     |
| Master's                    | 70                              | 72    | 73    | 76    | 80    |
| Bachelor's                  | 24                              | 22    | 22    | 17    | 14    |
| Total                       | 100%                            | 100%  | 100%  | 100%  | 100%  |
| Number of part-time faculty | 14887                           | 20142 | 25775 | 20247 | 18543 |



**FIGURE TYF.6.1.** Percentage of part-time faculty in mathematics departments at two-year colleges (including those paid by a third party, as in dual-enrollment courses) by highest degree in fall 2000, 2005, 2010, 2015, and 2021.

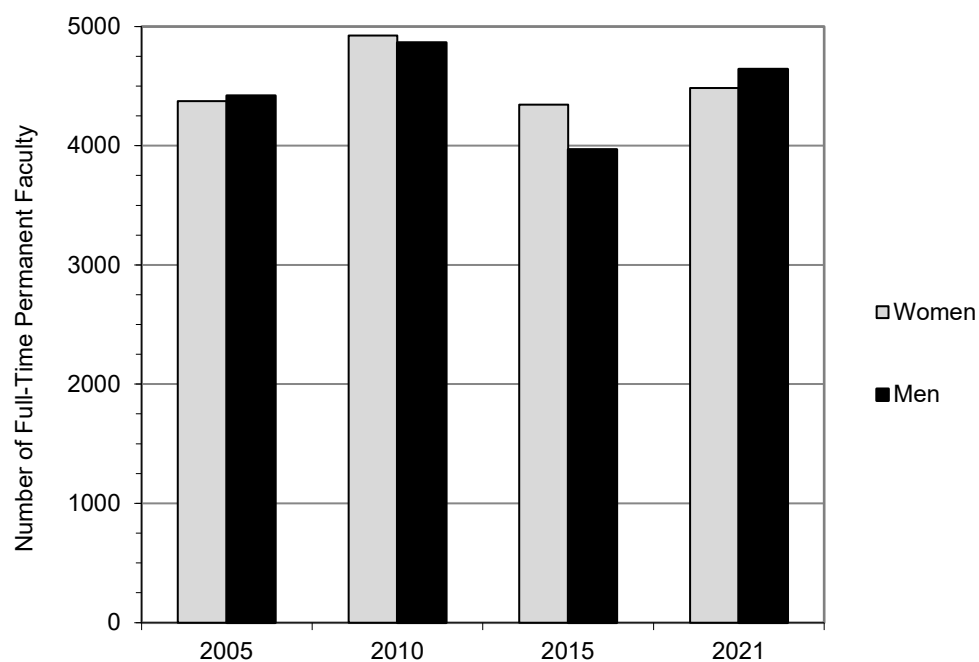
**TABLE TYF.7.** Percentage of part-time faculty in mathematics departments at two-year colleges (including those paid by a third party, as in dual-enrollment courses) by field and highest degree in fall 2021, with 2015 data in parentheses.

| Field of Degree                    | Percentage Having as Highest Degree |            |            | Total Percent in Field |
|------------------------------------|-------------------------------------|------------|------------|------------------------|
|                                    | Doctorate                           | Master's   | Bachelor's |                        |
| Mathematics                        | 3                                   | 45         | 9          | 57                     |
| Statistics                         | 0                                   | 2          | 0          | 2                      |
| Mathematics Education              | 1                                   | 23         | 3          | 27                     |
| Other fields                       | 2                                   | 9          | 2          | 13                     |
| Total percentage by highest degree | 6<br>(7)                            | 80<br>(76) | 14<br>(17) | 100%                   |

Note: 0 means less than one half of 1%. Round-off may make column sums seem inaccurate.

**TABLE TYF.8.** Number and percentage of total full-time permanent faculty in mathematics departments at two-year colleges by gender in fall 2005, 2010, 2015, and 2021.

|       | 2005         | 2010         | 2015         | 2021         |
|-------|--------------|--------------|--------------|--------------|
| Men   | 4420<br>50%  | 4866<br>50%  | 3969<br>48%  | 4643<br>51%  |
| Women | 4373<br>50%  | 4924<br>50%  | 4345<br>52%  | 4483<br>49%  |
| Total | 8793<br>100% | 9790<br>100% | 8314<br>100% | 9126<br>100% |



**FIGURE TYF.8.1.** Number of full-time permanent faculty in mathematics departments at two-year colleges by gender in fall 2005, 2010, 2015, and 2021.



**TABLE TYF.9.** Percentage of full-time permanent faculty and part-time faculty in mathematics departments at public two-year colleges by gender in fall 2021. Also master's degrees in mathematics and statistics granted in the U.S. to citizens and resident aliens, by gender, in 2019–20. Part-time faculty paid by a third party are not included.

|              | Percentage of               |                   |                                                                                                                          |
|--------------|-----------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------|
|              | Full-Time Permanent Faculty | Part-Time Faculty | Master's Degrees in Mathematics & Statistics Granted in the U.S. in 2019–20 to Citizens and Resident Aliens <sup>1</sup> |
| Men          | 51                          | 54                | 65                                                                                                                       |
| Women        | 49                          | 46                | 35                                                                                                                       |
| Total        | 100%                        | 100%              | 100%                                                                                                                     |
| Total number | 9126                        | 13827             | 5657                                                                                                                     |

<sup>1</sup> Report Tables 323.40 and 323.50 are from *Digest of Education Statistics: 2022*, NCES, [https://nces.ed.gov/programs/digest/current\\_tables.asp](https://nces.ed.gov/programs/digest/current_tables.asp).

part-time faculty was 46% (SE 3), down from 53% in fall 2015. See Tables TYF.8 and TYF.9. See Table TYF.8 and Figure TYF.8.1.

The percentage of women recipients of degrees in mathematics and statistics among U.S. citizens/resident aliens was 35% in 2019–2020, compared with 36% in 2014–2015. See Table TYF.9.

Table TYF.17 presents the percentage of full-time faculty in mathematics by age and gender and the percentage of women by age. Table TYF.20 presents data on the gender and ethnicity of newly appointed full-time permanent mathematics faculty in fall 2015 and 2010. In fall 2021, the percentage of women in this group was 46% (SE 15), down nine points from 2015.

**Ethnicity among full-time permanent and part-time faculty.** Demographics data about ethnic minority faculty among full-time permanent mathematics faculty at two-year colleges are given in Tables TYF.10–TYF.13 and Figure TYF.10.1. The minority groups referenced in the survey are listed in Table TYF.11. Tables TYF.10 and TYF.11 provide a historical perspective, while Tables TYF.12 and TYF.13 present more detailed information on the ethnic profile of the full-time permanent mathematics faculty in fall 2021, including information about both age and gender. Tables TYF.14 and TYF.15 present data on ethnicity of part-time faculty.

In fall 2021, ethnic minority faculty constituted 15% (SE 3) of the full-time permanent faculty and 1375 (SE 262) faculty. In fall 2015, 1876 (SE 289) full-time permanent ethnic minority faculty comprised 23% of total mathemat-

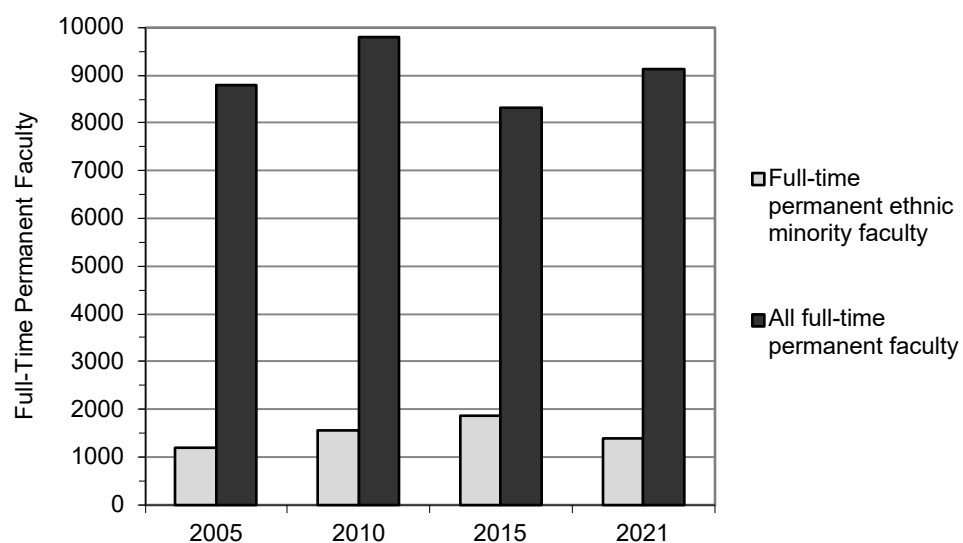
ics faculty. The decrease from 2015 to 2021 was 8 percentage points and 501 persons. See Table TYF.10 and Figure TYF.10.1.

The relative percentage of the full-time permanent minority faculty within individual ethnic groups decreased slightly between 2015 and 2021. Caution is suggested in reviewing the data, as 29% of colleges were “not reported” or “status unknown.” Asian/Pacific Islanders represented the largest ethnic minority groups in fall 2021, down three points from 2015, at 6% (SE 1) of full-time permanent faculty. The percentage of Black (non-Hispanic) faculty decreased 2 percentage points (4%; SE 1). The percentage of Mexican American / Puerto Rican / other Hispanic faculty was down 1 percentage point to 5% (SE 1). These changes impacted the percentage of White (non-Hispanic) full-time permanent faculty in 2021, down twenty-three points from 2015 to 52% (5 SEs). See Table TYF.11.

Table TYF.12 presents the number of full-time permanent faculty and the percentage of women within ethnic groups. Caution is again suggested in reviewing the data, as 29% of colleges were “not reported” or “status unknown.” In fall 2021, the largest percentage of women within a group occurred in White (non-Hispanic), of which 54% (SE 3) in that group or 2584 were women, of the total 4785 (SE 613) faculty in that group. This represented a two-point increase from 2015. The Asian/Pacific Islander group was the second highest percentage group among full-time faculty with 509 persons (6%; SE 1), of which 40% (SE 10), or 204 persons, were women. Third was the Mexican American, Puerto Rican, or other Hispanic group with 5% (SE 1) with

**TABLE TYF.10.** Percentage and number of ethnic minority full-time permanent faculty in mathematics departments at two-year colleges in fall 2005, 2010, 2015, and 2021.

|                                                                   | 2005 | 2010 | 2015 | 2021 |
|-------------------------------------------------------------------|------|------|------|------|
| Percentage of ethnic minorities among full-time permanent faculty | 14%  | 16%  | 23%  | 15%  |
| Number of full-time permanent ethnic minority faculty             | 1198 | 1566 | 1876 | 1375 |
| Number of full-time permanent faculty                             | 8793 | 9790 | 8314 | 9126 |



**FIGURE TYF.10.1.** Number of ethnic minority full-time permanent faculty and number of all full-time permanent faculty in mathematics departments at two-year colleges in fall 2005, 2010, 2015, and 2021.

**TABLE TYF.11.** Percentage of full-time permanent faculty in mathematics departments at two-year colleges by ethnicity in fall 2005, 2010, 2015, and 2021.

| Ethnic Group                                     | Percentage of Full-Time Permanent Faculty |      |      |      |
|--------------------------------------------------|-------------------------------------------|------|------|------|
|                                                  | 2005                                      | 2010 | 2015 | 2021 |
| American Indian / Eskimo / Aleut                 | 0                                         | 0    | 0    | 0    |
| Asian / Pacific Islander                         | 6                                         | 6    | 9    | 6    |
| Black (non-Hispanic)                             | 5                                         | 6    | 6    | 4    |
| Mexican American / Puerto Rican / other Hispanic | 3                                         | 4    | 6    | 5    |
| White (non-Hispanic)                             | 84                                        | 79   | 75   | 52   |
| Status unknown                                   | 2                                         | 5    | 3    | 4    |
| Not reported                                     | na                                        | na   | na   | 29   |
|                                                  | 100%                                      | 100% | 100% | 100% |
| Number of full-time permanent faculty            | 8793                                      | 9790 | 8314 | 9126 |

Note: 0 means less than one half of 1%.

Note: Percentages of full-time permanent faculty in mathematics programs at two-year colleges by ethnic group do not sum to 100% due to missing faculty reported per ethnic group relative to the total.

**TABLE TYF.12.** Number and percentage of full-time permanent faculty in mathematics departments at two-year colleges by ethnic group, and percentage of women within each ethnic group in fall 2021.

| Ethnic Group                                      | Number of Full-Time Permanent Faculty | Percentage of Ethnic Group in Full-Time Permanent Faculty | Percentage of Women in Ethnic Group |
|---------------------------------------------------|---------------------------------------|-----------------------------------------------------------|-------------------------------------|
| American Indian, Alaskan Native                   | 38                                    | 0                                                         | 26                                  |
| Asian / Pacific Islander                          | 509                                   | 6                                                         | 40                                  |
| Black or African American (non-Hispanic)          | 370                                   | 4                                                         | 20                                  |
| Mexican American, Puerto Rican, or other Hispanic | 458                                   | 5                                                         | 46                                  |
| White (non-Hispanic)                              | 4785                                  | 52                                                        | 54                                  |
| Status not known or other                         | 355                                   | 4                                                         | 32                                  |
| Not reported                                      | 2611                                  | 29                                                        | ---                                 |
| Total                                             | 9126                                  | 100%                                                      | 35                                  |

Note: 0 means less than one half of 1%.

Note: Percentages of full-time permanent faculty in mathematics programs at two-year colleges by ethnic group do not sum to 100% due to missing faculty reported per ethnic group relative to the total.

**TABLE TYF.13.** Percentage of full-time permanent faculty and of full-time permanent faculty under age 40 in mathematics departments at public two-year colleges by ethnic group in fall 2021. Also U.S. master's degrees in mathematics and statistics granted in the U.S. to citizens and resident aliens by ethnic group in 2020–21.

| Ethnic Group         | Percentage Among                |                                          |                                                                                                                          |
|----------------------|---------------------------------|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
|                      | All Full-Time Permanent Faculty | Full-Time Permanent Faculty Under Age 40 | Master's Degrees in Mathematics & Statistics Granted in the U.S. in 2020-21 to Citizens and Resident Aliens <sup>1</sup> |
| Ethnic minorities    | 15                              | 20                                       | 36                                                                                                                       |
| White (non-Hispanic) | 52                              | 76                                       | 64                                                                                                                       |
| Unknown              | 4                               | 4                                        |                                                                                                                          |
| Not reported         | 29                              | 0                                        |                                                                                                                          |
| Total                | 100%                            | 100%                                     | 100%                                                                                                                     |
| Number               | 9126                            | 1424                                     | 5657                                                                                                                     |

Note: Percentages of full-time permanent faculty in mathematics programs at two-year colleges by ethnic group do not sum to 100% due to missing faculty reported per ethnic group relative to the total.

Note: Number of full-time permanent faculty under age 40 in this table do not match with table TYF.16 due to missing faculty reported per ethnic group.

<sup>1</sup> Table 323.30 is from *Digest of Education Statistics: 2022*, NCES, [https://nces.ed.gov/programs/digest/d22/tables/dt22\\_323.30.asp](https://nces.ed.gov/programs/digest/d22/tables/dt22_323.30.asp).

These figures do not include a total of 6926 non-resident aliens.

**TABLE TYF.14.** Percentage of ethnic minority part-time faculty in mathematics departments at public two-year colleges in fall 2010, 2015, and 2021.

|                                                         | 2010  | 2015  | 2021  |
|---------------------------------------------------------|-------|-------|-------|
| Percentage of ethnic minorities among part-time faculty | 17    | 22    | 22    |
| Number of part-time faculty                             | 23453 | 17888 | 13827 |

**TABLE TYF.15.** Number and percentage of part-time faculty in mathematics departments at public two-year colleges by ethnic group, and percentage of women within each ethnic group in fall 2021.

| Ethnic Group                                      | Number of Part-Time Faculty | Percentage of                            |                           |
|---------------------------------------------------|-----------------------------|------------------------------------------|---------------------------|
|                                                   |                             | Ethnic Group Among All Part-Time Faculty | Women Within Ethnic Group |
| American Indian, Alaskan Native                   | 53                          | 0                                        | 19                        |
| Asian / Pacific Islander                          | 1196                        | 9                                        | 53                        |
| Black or African American (non-Hispanic)          | 890                         | 6                                        | 47                        |
| Mexican American, Puerto Rican, or other Hispanic | 867                         | 6                                        | 33                        |
| White (non-Hispanic)                              | 9202                        | 67                                       | 47                        |
| Status not known or other                         | 1619                        | 12                                       | 39                        |
| Total                                             | 13827                       | 100%                                     | 46                        |

458 full-time permanent faculty, of which 46% (SE 7) were women. The Black or African American (non-Hispanic) full-time faculty group consisted of 370 persons (4%; SE 1), including 74 women (20%; SE 7).

In fall 2021, Table TYF.13 shows that the total number of full-time permanent faculty under the age of 40 was 1424 (SE 67), compared with 2045 (SE 292) in 2015, a 30% decrease. These faculty under the age of 40 comprised 16% of all full-time permanent faculty, compared to 25% in 2015. The ethnic group “White” was 76% (SE 10) of full-time faculty under the age of 40, down 4 points from 2015. The percentage of ethnic minority full-time permanent mathematics faculty under the age of 40 decreased 6 percentage points to 20% (SE 9), or 285 persons, from 26% (3 SEs). Data on ethnicity of newly appointed full-time permanent faculty in fall 2015 are presented in Table TYF.20.

In fall 2021, twenty-two percent (22%; SE 4) of part-time faculty, or 3042 persons, were ethnic minorities (Asian/Pacific Islander, Black or African American, Mexican American, Puerto Rican, or other Hispanic), down 9 percentage points from 2015. Caution is suggested in reviewing the data, given “status unknown” with 1619 (SE 352) persons. Asian/Pacific Islanders comprised 9% (SE 0) of part-time faculty (1196 persons, SE 192) and Black or African American and Mexican American, Puerto Rican or other Hispanic each represented 6% (0 SE) of all part-time faculty for a total of 1757 persons. Women comprised 46% (SE 3) of all part-time faculty. See Tables TYF.14 and TYF.15.

**Number and age distribution of full-time permanent faculty.** As mentioned earlier in this chapter, it was estimated that there were 9126 (SE 919) full-time permanent faculty in public two-year college mathematics departments in the U.S. in fall 2021, compared with 8314 (SE 840) in 2015. This 10% increase of 812 persons can be compared to a 15% decrease of 1476 persons in 2015. When the 1202 continuing and other full-time faculty are included, the total was 10,328 (SE 1117) persons and represented an increase of 5% compared to 2015. See Table TYF.1.

The percentage and number of full-time permanent faculty in eight age groups in 2021 are presented in TYF.16. The percentage of full-time permanent faculty under age 40 years dropped eight points to 16% in 2021, a total of 1424 persons (SE 67) compared to 2045 persons (SE 292) in 2015 and a 30% decrease of 621 persons. The number of full-time permanent faculty under the age of 40 in 2021, 2015, 2010, and 2005 was 1424, 2045, 2914, and 2231, respectively. The percentage of full-time permanent faculty between the ages of 50–59 years decreased seven points to 22% in 2021, a decrease of 361 persons. The percentage of full-time faculty over age 59 was down two points from 2015 to 13% (SE 2) in 2021, a decrease of 48 persons. The total number of full-time permanent faculty over the age of 49 decreased by 409 persons from 2015 to 2021. See Table TYF.16.

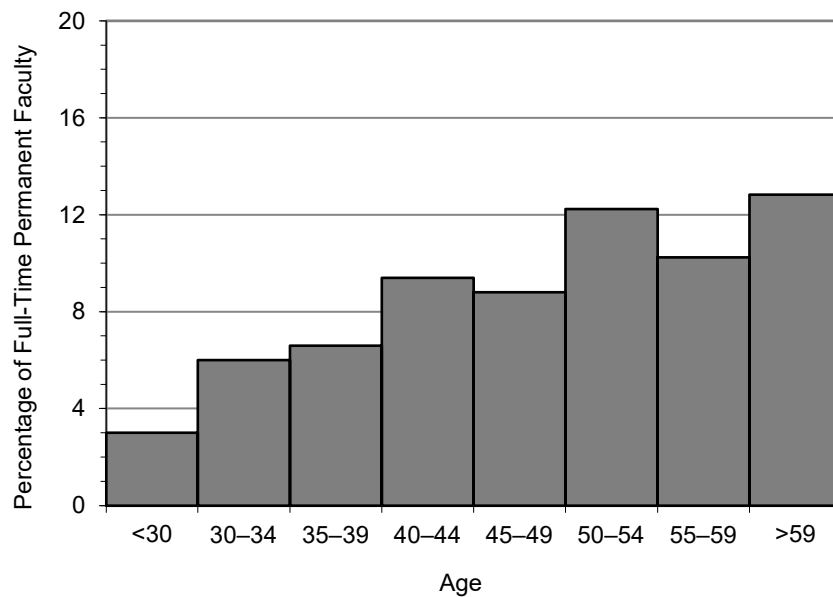
In 2021, full-time permanent women were again a slight minority with 49% (SE 5) overall. In the age group

**TABLE TYF.16.** Percentage and number of full-time permanent faculty in mathematics departments at two-year colleges by age in fall 2005, 2010, 2015, and 2021.

| Age          | Percentage of Full-Time Permanent Faculty |      |      |      | Number of Full-Time Permanent Faculty |      |      |      |
|--------------|-------------------------------------------|------|------|------|---------------------------------------|------|------|------|
|              | 2005                                      | 2010 | 2015 | 2021 | 2005                                  | 2010 | 2015 | 2021 |
| <30          | 5                                         | 8    | 4    | 3    | 478                                   | 832  | 363  | 274  |
| 30–34        | 8                                         | 9    | 6    | 6    | 716                                   | 893  | 529  | 548  |
| 35–39        | 12                                        | 12   | 14   | 7    | 1037                                  | 1189 | 1153 | 602  |
| 40–44        | 13                                        | 14   | 14   | 9    | 1163                                  | 1416 | 1159 | 857  |
| 45–49        | 15                                        | 15   | 18   | 9    | 1298                                  | 1475 | 1479 | 803  |
| 50–54        | 18                                        | 11   | 16   | 12   | 1574                                  | 1085 | 1357 | 1117 |
| 55–59        | 17                                        | 13   | 13   | 10   | 1528                                  | 1268 | 1055 | 934  |
| >59          | 11                                        | 17   | 15   | 13   | 999                                   | 1631 | 1219 | 1171 |
| Not reported | na                                        | na   | na   | 31   |                                       |      |      | 2820 |
| Total        | 100%                                      | 100% | 100% | 100% | 8793                                  | 9790 | 8314 | 9126 |

Note: Rounding may make column totals seem inaccurate.

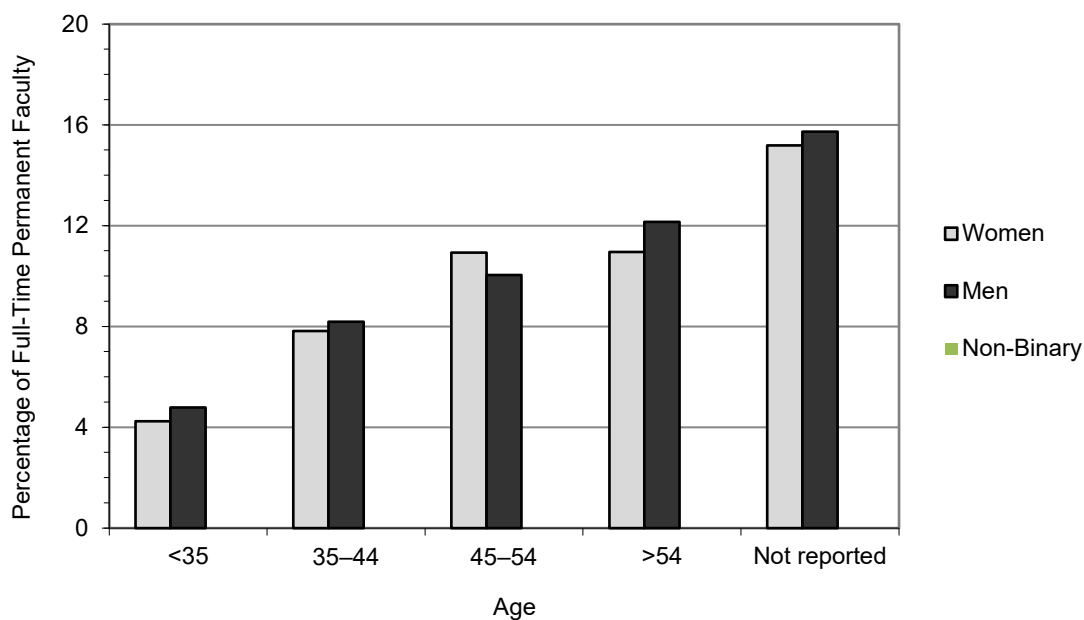
Note: Percentages of full-time permanent faculty in mathematics departments at two-year colleges by age group do not sum to 100% due to missing faculty reported per age group relative to the total.

**FIGURE TYF.16.1.** Percentage distribution of full-time permanent faculty in mathematics departments at public two-year colleges by age in fall 2021.

Note: Percentages of full-time permanent faculty in mathematics departments at two-year colleges by age group do not sum to 100% due to missing faculty reported per age group relative to the total.

**TABLE TYF.17.** Percentage of full-time permanent faculty in mathematics departments at public two-year colleges by age and by gender, and percentage of women by age in fall 2021.

| Age          | Percentage of Full-Time Permanent Faculty |     |            | Percentage of Women in Age Group |
|--------------|-------------------------------------------|-----|------------|----------------------------------|
|              | Women                                     | Men | Non-Binary |                                  |
| <35          | 4                                         | 5   | 0          | 47                               |
| 35–44        | 8                                         | 8   | 0          | 49                               |
| 45–54        | 11                                        | 10  | 0          | 52                               |
| >54          | 11                                        | 12  | 0          | 47                               |
| Not reported | 15                                        | 16  | 0          | 49                               |
| Total        | 49                                        | 51  | 0          |                                  |



**FIGURE TYF.17.1.** Percentage of full-time permanent faculty in mathematics departments at public two-year colleges by age and by gender in fall 2021.

less than 35 years, women represented 4% (SE 0.8) of full-time permanent faculty. Fifty-two percent (52%; SE 4.5) of the age group 45–54 were women, down four points from 2015. Forty-seven percent (47%; SE 4.6) of the age group >54 were women, up four points from 2015. See Table TYF.17 and Figure TYF.17.1.

### **Demographics of Full-time Permanent Faculty Newly Appointed by Mathematics Departments**

Two-year college mathematics departments hired 255 (SE 70) new full-time permanent faculty in fall 2021, down 196 persons and 43% from those appointed in 2015. See Table TYF.18.

Fall 2021 and earlier surveys presented sources of new appointments at two-year colleges. In 2005 and 2010, graduate school as a source remained steady at 23%. In fall 2015, that percentage from graduate school increased to 37% (SE 8) in 2015, 167 persons from a total of 451. In fall 2021, fourteen percent (14%; SE 8) and 36 persons of new appointments were from graduate school, a decrease of 23% from 2015. This was a decrease of 131 persons from 2015 to 2021. In contrast, the percentage of new hires who had been teaching at four-year institutions was 3% (SE 2) in 2021, compared to 4% (SE 2) in 2015, a decrease of 10 persons. In 2015, hiring from among part-time faculty at the same institution was 26% (SE 6), 116 persons. Fall 2021 saw a 16% increase to 42% (SE 16), however, that represented 107 faculty appointed from former part-time or full-time temporary faculty at the same college and a 10 person decrease from 2015. See Table TYF.18.

The master's degree was held by 75% (SE 13) of newly appointed full-time permanent faculty in fall 2021, down 12 percentage points from 2015. In 2021, the percentage of new faculty with a doctorate degree was 17% (SE 12), up 8 percentage points from 2015. See Table TYF.19.

The 2000, 2005, 2010, 2015, and 2021 CBMS data indicate a decrease of new appointments with a bachelor's degree from 19% to 5% to 2% to 0% to 0% (less than one percent and/or round-off may make 0% totals inaccurate), respectively.

In 2021, forty-six percent (46%; SE 15) of new mathematics faculty appointments were women, down 9 percentage points from fall 2015. The category of White (non-Hispanic) faculty comprised 63% (SE 12), compared to 82% (SE 5) of new appointments for 2015, down 19 percentage points and representing a decrease of 209 persons. Thirty-four percent (34%; SE 12), representing 55 persons of new appointments, were women in the White (non-Hispanic) group.

Twenty-three percent (23%; SE 6), or 59 persons, of the 255 new appointments in 2021 were ethnic minorities other than White. Fall 2021 was up fourteen points from 2015 and represented an increase of 17 persons. In 2021, nine percent (9%) of newly appointed faculty in ethnic minorities other than White were women, a total of 5 persons. Information about age of new appointments was not collected in CBMS2021, CBMS2015, and CBMS2010.

### **Teaching Evaluations and Professional Development of Mathematics Faculty and Concerns and Issues in Mathematics Departments**

In fall 2021, as in 2015, one hundred percent (100%; SE 0) of two-year colleges responding to the survey required periodic evaluation of the teaching of full-time permanent mathematics faculty. Periodic teaching evaluation was required for part-time faculty at 93% (SE 4) of colleges, compared to 98% reported in 2015. See Table TYF.21.

In 2021, 2015, 2010, and 2005, the most common method of evaluating full- and part-time teaching was the use of "evaluation instruments completed by students." In 2021, student evaluations were used for full-time faculty in 97% (SE 2) of reporting colleges and 97% (SE 3) of colleges for part-time faculty, increasing by 2 and 3 percentage points, respectively. The second most common method of evaluating both full- and part-time faculty was "observation of classes by other faculty." "Self-evaluation, such as teaching portfolios," was used as a component of the evaluation of full-time faculty by 49% (7 SEs) of colleges in 2021, up twenty-six points from 2015. In contrast, twenty percent (20%; SE 6) of responding colleges in 2021 used self-evaluation for part-time faculty, compared to 62% in 2015, a drop of 42 percentage points.

The percentage of colleges using "observation of classes by other faculty" decreased nine points to 66% (SE 7) for full-time faculty and increased fourteen points in 2021 to 78% (SE 7) for part-time faculty. The percentage of colleges that used "observation of classes by a division or head or other administrator" as a component of full-time faculty evaluation was 61% (SE 7), up sixteen points compared to 2015. In contrast, a decrease of 29 percentage points to 33% (SE 7) was reported in administrators observing part-time faculty in 2021. See Table TYF.22.

Other methods of evaluation, such as evaluation of written course materials such as lesson plans, self-evaluation/portfolios, and written peer evaluations are presented in Table TYF.22.



**TABLE TYF.18.** Percentage of newly appointed full-time permanent faculty in mathematics departments at two-year colleges coming from various sources in fall 2015 and 2021.

| Percentage of New Faculty from:                                    | 2015 | 2021 |
|--------------------------------------------------------------------|------|------|
| A. Graduate School                                                 | 37   | 14   |
| B. Teaching in a four-year college or university                   | 4    | 3    |
| C. Teaching in another two-year college                            | 19   | 18   |
| D. Teaching in a secondary school                                  | 1    | 8    |
| E. Part-time or full-time temporary employment at the same college | 26   | 42   |
| F. Nonacademic employment                                          | 1    | 0    |
| G. Unemployed                                                      | 4    | 0    |
| H. Unknown                                                         | 9    | 15   |
| Total                                                              | 100% | 100% |
| Total number appointed                                             | 451  | 255  |

Note: 0 means less than one half of 1%.

**TABLE TYF.19.** Percentage of full-time permanent faculty newly appointed in mathematics departments at two-year colleges by highest degree in fall 2015 and 2021.

| Highest Degree | Percentage of New Appointments |           |
|----------------|--------------------------------|-----------|
|                | 2015–2016                      | 2021–2022 |
| Doctorate      | 9                              | 17        |
| Master's       | 87                             | 75        |
| Bachelor's     | 0                              | 0         |
| Unknown        | 4                              | 8         |
| Total          | 100%                           | 100       |

Note: 0 means less than one half of 1%. Round-off may make column totals seem inaccurate.

**TABLE TYF.20.** Percentage of full-time permanent faculty newly appointed in mathematics departments at two-year colleges by ethnic group in fall 2015 and 2021. Also percentage of women within each ethnic group in fall 2021.

| Ethnic Group                                      | Percentage of New Appointments |           | Percentage of Women in Ethnic Group for 2021–2022 New Appointments |
|---------------------------------------------------|--------------------------------|-----------|--------------------------------------------------------------------|
|                                                   | 2015–2016                      | 2021–2022 |                                                                    |
| American Indian                                   | 0                              | 0         | 0                                                                  |
| Asian / Pacific Islander                          | 4                              | 11        | 7                                                                  |
| Black or African American (non-Hispanic)          | 2                              | 2         | 2                                                                  |
| Mexican American, Puerto Rican, or other Hispanic | 3                              | 10        | 0                                                                  |
| White (non-Hispanic)                              | 82                             | 63        | 34                                                                 |
| Other                                             | 3                              | 7         | 4                                                                  |
| Unknown                                           | 5                              | 8         | 0                                                                  |
| Percentage of women among all new hires           | 55                             | 46        |                                                                    |

Note: 0 means less than one half of 1%. Round-off may make column totals seem inaccurate.

### Professional development obligations and activities of full-time permanent faculty.

In fall 2021, some form of continuing education or professional development was required of full-time permanent faculty at 84% (SE 5) of two-year colleges, up two points from 2015. Data was collected in CBMS2015 about ways to meet professional development requirements but was not collected in 2021. See Table TYF.23.

### Concerns and issues in mathematics departments.

Table TYF.24 presents various problems experienced by mathematics departments in two-year colleges in 2021. The three items reported by mathematics program heads with the highest percentage as being a “major problem” in 2021 were: “low student motivation” (48%; SE 7), “students not understanding the demands of college work” (42%; SE 7), and “lack of student progress from developmental to advanced courses” (29%; SE 6).

Table TYF.25 includes additional data on the extent to which program heads thought items listed were a “major” problem, “somewhat” of a problem, or a “minor or no” problem. When considering issues reported as “somewhat of a problem,” the top three items and their percentages were: “lack of student progress from developmental to

advanced courses” (55%; SE 7), “faculty salaries too low” (55%; SE 7), and “too few students who intend to transfer actually do” (54%; SE 7). See Tables TYF.24 and TYF.25.

Obtaining travel funds for faculty professional development have historically been a department concern. “Lack of or reduced funds” available for faculty professional travel and other professional development activities was less of a concern to mathematics departments in 2021. In 2015, the concern about the level of travel funding for mathematics faculty by program heads was a “major concern” in 25% (SE 4) of reporting colleges and “somewhat of a problem” by 31% (SE 3) of reporting colleges; this decreased to 10% (SE 4) in each case in 2021.

In the CBMS surveys between 1985 and 2015, sixty percent or more of mathematics program heads classified “too many students needing remediation” as a “major” problem for their departments. This was the number one major problem in 2015, 2010, 2005, 2000, and 1995. In fall 2021, this figure decreased to 38% (SE 7), compared to 64% (SE 5) in 2015. A possible reason for this decrease could be the increase in the Pathways offerings (see Table TYE.10 and the discussion in Chapter 6). See Tables TYF.24 and TYF.25.

**TABLE TYF.21.** Percentage of two-year colleges that require periodic teaching evaluations for all full-time or all part-time faculty in fall 2015 and 2021.

|                                                                      | Percentage of Two-Year Colleges in Fall 2015 | Percentage of Two-Year Colleges in Fall 2021 |
|----------------------------------------------------------------------|----------------------------------------------|----------------------------------------------|
| Colleges that require teaching evaluations for all full-time faculty | 100                                          | 100                                          |
| Colleges that require teaching evaluations for all part-time faculty | 98                                           | 93                                           |

**TABLE TYF.22.** Percentage of mathematics departments at public two-year colleges using various methods of evaluating teaching of full- and part-time faculty in fall 2021.

| Method of Evaluating Teaching                                                               | Percentage of Departments Using Evaluation Method for |                   |
|---------------------------------------------------------------------------------------------|-------------------------------------------------------|-------------------|
|                                                                                             | Part-Time Faculty                                     | Full-Time Faculty |
| A. Observation of classes by other faculty                                                  | 78                                                    | 66                |
| B. Observation of classes by division head (if different from chair) or other administrator | 33                                                    | 61                |
| C. Evaluation forms completed by students                                                   | 97                                                    | 97                |
| D. Evaluation of written course material such as lesson plans, syllabus, or exams           | 53                                                    | 49                |
| E. Self-evaluation such as teaching portfolios                                              | 20                                                    | 49                |
| F. Written peer evaluations                                                                 | 16                                                    | 31                |
| G. Other methods                                                                            | 7                                                     | 7                 |

**TABLE TYF.23.** Percentage of two-year colleges that require some form of continuing education or professional development for full-time permanent faculty, and percentage of faculty using various methods to fulfill those requirements, in mathematics departments at two-year colleges in fall 2015 and 2021.

| Faculty Development                                                                                                   | Fall 2015 | Fall 2021 |
|-----------------------------------------------------------------------------------------------------------------------|-----------|-----------|
| Percentage of institutions requiring continuing education or professional development for full-time permanent faculty | 82        | 84        |

**TABLE TYF.24.** Percentage of program heads classifying various problems as “major” in mathematics departments at two-year colleges in fall 2005, 2010, 2015, and 2021.

| Problem                                                                                    | Percentage of Department Heads<br>Classifying Problem as Major |      |      |      |
|--------------------------------------------------------------------------------------------|----------------------------------------------------------------|------|------|------|
|                                                                                            | 2005                                                           | 2010 | 2015 | 2021 |
| A. Maintaining vitality of faculty                                                         | 2                                                              | 4    | 7    | 12   |
| B. Dual-enrollment courses                                                                 | 5                                                              | 11   | 7    | 5    |
| C. Staffing statistics courses                                                             | 3                                                              | 2    | 5    | 6    |
| D. Students don't understand demands of college work                                       | 55                                                             | 64   | 62   | 42   |
| E. Need to use part-time faculty for too many courses                                      | 30                                                             | 35   | 15   | 6    |
| F. Faculty salaries too low                                                                | 22                                                             | 21   | 39   | 32   |
| G. Class sizes too large                                                                   | 5                                                              | 3    | 5    | 6    |
| H. Low student motivation                                                                  | 50                                                             | 50   | 57   | 48   |
| I. Too many students needing remediation                                                   | 63                                                             | 67   | 64   | 38   |
| J. Lack of student progress from developmental to advanced courses                         | 34                                                             | 37   | 36   | 29   |
| K. Low success rate in transfer-level courses                                              | 7                                                              | 13   | 14   | 20   |
| L. Too few students who intend to transfer actually do                                     | 4                                                              | 11   | 8    | 4    |
| M. Inadequate travel funds for faculty                                                     | 22                                                             | 23   | 25   | 10   |
| N. Learning disabilities are not adequately supported                                      | na                                                             | na   | na   | 9    |
| O. Inadequate classroom facilities for use of technology                                   | 12                                                             | 10   | 4    | 8    |
| P. Inadequate classroom facilities for teaching with active learning                       | na                                                             | na   | na   | 8    |
| Q. Inadequate computer facilities for part-time faculty use                                | 9                                                              | 6    | 7    | 9    |
| R. Inadequate computer facilities for student services                                     | 1                                                              | 5    | 6    | 3    |
| S. Heavy classroom duties prevent personal & teaching enrichment by faculty                | 14                                                             | 11   | 13   | 11   |
| T. Coordinating mathematics courses with high schools                                      | 7                                                              | 14   | 21   | 13   |
| U. Lack of curricular flexibility because of transfer rules                                | 7                                                              | 5    | 2    | 5    |
| V. Other barriers that inhibit curricular changes <sup>1</sup>                             | na                                                             | na   | 7    | 4    |
| W. Maintaining high and consistent expectations across sections <sup>1</sup>               | na                                                             | na   | 8    | 16   |
| X. High cost of textbooks <sup>1</sup>                                                     | na                                                             | na   | 54   | 13   |
| Y. Lack of flexibility in curricular redesign <sup>1</sup>                                 | na                                                             | na   | 4    | 3    |
| Z. Maintaining common standards between distance learning and related courses <sup>1</sup> | na                                                             | na   | 2    | 23   |
| AA. Use of distance education <sup>1</sup>                                                 | 6                                                              | 6    | 4    | na   |

Note: 0 means less than one half of 1%.

<sup>1</sup> Data not collected before 2015.

**TABLE TYF.25.** Percentage of program heads of mathematics departments at public two-year colleges classifying various problems by severity in fall 2021.

| Problem                                                                       | Percentage of Department Heads<br>Classifying Problems as |                          |                  |
|-------------------------------------------------------------------------------|-----------------------------------------------------------|--------------------------|------------------|
|                                                                               | Minor or<br>No Problem                                    | Somewhat of<br>a Problem | Major<br>Problem |
| A. Maintaining vitality of faculty                                            | 48                                                        | 40                       | 12               |
| B. Dual-enrollment courses                                                    | 62                                                        | 34                       | 5                |
| C. Staffing statistics courses                                                | 58                                                        | 36                       | 6                |
| D. Students don't understand demands of college work                          | 6                                                         | 52                       | 42               |
| E. Need to use part-time faculty for too many courses                         | 69                                                        | 25                       | 6                |
| F. Faculty salaries too low                                                   | 14                                                        | 55                       | 32               |
| G. Class sizes too large                                                      | 62                                                        | 32                       | 6                |
| H. Low student motivation                                                     | 2                                                         | 51                       | 48               |
| I. Too many students needing remediation                                      | 8                                                         | 54                       | 38               |
| J. Lack of student progress from developmental to advanced courses            | 16                                                        | 55                       | 29               |
| K. Low success rate in transfer-level courses                                 | 21                                                        | 59                       | 20               |
| L. Too few students who intend to transfer actually do                        | 42                                                        | 54                       | 4                |
| M. Inadequate travel funds for faculty                                        | 66                                                        | 24                       | 10               |
| N. Learning disabilities are not adequately supported                         | 56                                                        | 35                       | 9                |
| O. Inadequate classroom facilities for use of technology                      | 66                                                        | 26                       | 8                |
| P. Inadequate classroom facilities for teaching with active learning          | 63                                                        | 29                       | 8                |
| Q. Inadequate computer facilities for part-time faculty use                   | 66                                                        | 25                       | 9                |
| R. Inadequate computer facilities for student services                        | 68                                                        | 29                       | 3                |
| S. Heavy classroom duties prevent personal & teaching enrichment by faculty   | 44                                                        | 46                       | 11               |
| T. Coordinating mathematics courses with high schools                         | 43                                                        | 45                       | 13               |
| U. Lack of curricular flexibility because of transfer rules                   | 56                                                        | 38                       | 5                |
| V. Other barriers that inhibit curricular changes                             | 54                                                        | 42                       | 4                |
| W. Maintaining high and consistent expectations across sections               | 30                                                        | 54                       | 16               |
| X. High cost of textbooks                                                     | 38                                                        | 49                       | 13               |
| Y. Lack of flexibility in curricular redesign                                 | 65                                                        | 32                       | 3                |
| Z. Maintaining common standards between distance learning and related courses | 34                                                        | 43                       | 23               |

## Appendix I

# Enrollments in Departmental Courses in Four-Year Colleges and Universities: 2005, 2010, 2015, and 2021

**TABLE A.1.A.** Enrollment (in 1000s) in mathematics courses in fall 2005, 2010, 2015, and 2021 [with standard error (SE) for 2005, 2010, 2015, and 2021 totals]. Round-off may cause marginal totals to appear incorrect.

| Courses                                        | 2005              | 2010<br>Total (Including<br>Distance Courses) | 2015<br>Total (Including<br>Distance Courses) | 2015<br>Total<br>(Non-Distance) | 2021<br>Total (Including<br>Distance Courses) | Fall 2021 Enrollment (in 1000s)    |              |               |                              |              |            |
|------------------------------------------------|-------------------|-----------------------------------------------|-----------------------------------------------|---------------------------------|-----------------------------------------------|------------------------------------|--------------|---------------|------------------------------|--------------|------------|
|                                                |                   |                                               |                                               |                                 |                                               | Mathematics Departments            |              |               |                              |              |            |
|                                                |                   |                                               |                                               |                                 |                                               | Total (Including Distance Courses) |              |               | Total (Non-Distance Courses) |              |            |
|                                                |                   | Univ<br>(PhD)                                 | Univ<br>(MA)                                  | Coll<br>(BA)                    | Univ<br>(PhD)                                 | Univ<br>(MA)                       | Coll<br>(BA) | Univ<br>(PhD) | Univ<br>(MA)                 | Coll<br>(BA) | Subtotal   |
| <b>Pre-College Level<sup>1</sup></b>           | <b>201</b> [18.8] | <b>209</b> [22.0]                             | <b>253</b> [26.5]                             | <b>244</b> [25.7]               | <b>101</b> [16.2]                             | <b>24</b>                          | <b>24</b>    | <b>54</b>     | <b>21</b>                    | <b>20</b>    | <b>47</b>  |
| <b>89</b>                                      |                   |                                               |                                               |                                 |                                               |                                    |              |               |                              |              |            |
| <b>Introductory (Including Pre-Calc) Level</b> |                   |                                               |                                               |                                 |                                               |                                    |              |               |                              |              |            |
| 6 College Algebra                              | 201 [17.2]        | 251 [15.9]                                    | 261 [24.7]                                    | 251 [24.1]                      | 192 [27.7]                                    | 81                                 | 44           | 68            | 63                           | 36           | 54         |
| 7 Trigonometry                                 | 30 [3.5]          | 42 [5.2]                                      | 47 [7.4]                                      | 46 [7.1]                        | 26 [5.1]                                      | 11                                 | 5            | 10            | 10                           | 5            | 8          |
| 8 Coll Alg & Trig Combined                     | 34 [6.8]          | 35 [7.6]                                      | 62 [15.6]                                     | 60 [15.1]                       | 4 [1.6]                                       | 1                                  | 1            | 2             | 1                            | 1            | 2          |
| 9 Elementary Functions <sup>2</sup>            | 93 [8.9]          | 114 [8.2]                                     | 128 [11.2]                                    | 125 [10.7]                      | 122 [10.6]                                    | 64                                 | 14           | 44            | 51                           | 8            | 37         |
| 10 Intro Math Modeling                         | 8 [3.1]           | 9 [2.2]                                       | 12 [4.2]                                      | 12 [4.0]                        | 19 [10.6]                                     | 4                                  | 4            | 10            | 2                            | 3            | 9          |
| 11 Math for Liberal Arts                       | 123 [11.7]        | 147 [14.4]                                    | 171 [21.9]                                    | 159 [19.5]                      | 72 [12.9]                                     | 17                                 | 12           | 43            | 12                           | 10           | 30         |
| 12 Finite Math                                 | 94 [16.1]         | 62 [6.7]                                      | 90 [11.1]                                     | 81 [10.2]                       | 41 [11.8]                                     | 23                                 | 1            | 16            | 18                           | 1            | 14         |
| 13 Business Math                               | 38 [5.8]          | 47 [7.7]                                      | 48 [8.8]                                      | 45 [7.9]                        | 48 [11.2]                                     | 25                                 | 8            | 15            | 23                           | 7            | 12         |
| 14 Math Elem Sch Tchrs                         | 72 [6.5]          | 85 [7.2]                                      | 73 [9.6]                                      | 72 [9.5]                        | 50 [7.2]                                      | 11                                 | 13           | 26            | 10                           | 10           | 18         |
| Quantitative Literacy/Reasoning                |                   |                                               |                                               |                                 | 33 [9.2]                                      | 9                                  | 14           | 10            | 8                            | 11           | 8          |
| 15 Other Intro-Level Math                      | 12 [2.5]          | 69 [10.5]                                     | 108 [24.2]                                    | 103 [23.5]                      | 64 [11.5]                                     | 7                                  | 21           | 36            | 4                            | 18           | 31         |
| <b>Subtotal Introductory Level</b>             | <b>706</b> [29.0] | <b>863</b> [35.0]                             | <b>1000</b> [79.6]                            | <b>954</b> [74.4]               | <b>670</b> [66.3]                             | <b>254</b>                         | <b>138</b>   | <b>278</b>    | <b>203</b>                   | <b>111</b>   | <b>223</b> |
| <b>537</b>                                     |                   |                                               |                                               |                                 |                                               |                                    |              |               |                              |              |            |

<sup>1</sup> Additional details on Pre-College-Level courses prior to 2010 are available in the CBMS Survey Fall 2010 report, [www.ams.org/cbms-survey](http://www.ams.org/cbms-survey).

<sup>2</sup> Elementary Functions, Precalculus, and Analytic Geometry.

**TABLE A.1.B.** Fall term mathematics course enrollment (in 1000s) [with SE for 2005, 2010, 2015, and 2021 totals].

| Fall 2021 Enrollment (in 1000s)           |            |                                               |                                               |                                 |                                               |                                       |              |              |                              |              |              |          |
|-------------------------------------------|------------|-----------------------------------------------|-----------------------------------------------|---------------------------------|-----------------------------------------------|---------------------------------------|--------------|--------------|------------------------------|--------------|--------------|----------|
| Mathematics Departments                   |            |                                               |                                               |                                 |                                               |                                       |              |              |                              |              |              |          |
| Courses                                   | 2005       | 2010<br>Total (Including<br>Distance Courses) | 2015<br>Total (Including<br>Distance Courses) | 2015<br>Total<br>(Non-Distance) | 2021<br>Total (Including<br>Distance Courses) | Total (Including Distance<br>Courses) |              |              | Total (Non-Distance Courses) |              |              |          |
|                                           |            |                                               |                                               |                                 |                                               | Univ<br>(PhD)                         | Univ<br>(MA) | Coll<br>(BA) | Univ<br>(PhD)                | Univ<br>(MA) | Coll<br>(BA) | Subtotal |
| Calculus Level                            |            |                                               |                                               |                                 |                                               |                                       |              |              |                              |              |              |          |
| 16 Mainstream Calc I                      | 201 [9.6]  | 235 [14.2]                                    | 261 [23.9]                                    | 255 [22.9]                      | 238 [26.6]                                    | 145                                   | 32           | 61           | 129                          | 23           | 57           | 209      |
| 17 Mainstream Calc II                     | 85 [4.9]   | 129 [13.7]                                    | 129 [11.7]                                    | 125 [10.7]                      | 87 [9.4]                                      | 51                                    | 13           | 23           | 46                           | 10           | 22           | 78       |
| 18 Mainstream Calc III, IV                | 74 [4.0]   | 104 [6.2]                                     | 117 [10.6]                                    | 115 [10.9]                      | 86 [11.2]                                     | 56                                    | 11           | 19           | 53                           | 7            | 18           | 78       |
| 19 Non-Mainstream Calc I                  | 108 [8.6]  | 99 [6.4]                                      | 94 [10.9]                                     | 91 [10.5]                       | 87 [14.3]                                     | 55                                    | 19           | 12           | 46                           | 14           | 11           | 71       |
| 20 Non-Mainstream Calc II                 | 11 [2.0]   |                                               |                                               |                                 |                                               |                                       |              |              |                              |              |              |          |
| 20.5 Non-Mainstream Calc II,<br>III, etc. |            | 22 [3.3]                                      | 16 [4.3]                                      | 16 [4.3]                        | 9 [4.5]                                       | 8                                     | 0            | 0            | 7                            | 0            | 0            | 7        |
| 21a Diff Eq & Lin Alg (Comb)              | 9 [2.2]    | 15 [2.6]                                      | 23 [3.4]                                      | 23 [3.4]                        | 5 [2.4]                                       | 3                                     | 1            | 2            | 2                            | 0            | 2            | 4        |
| 21b Differential Equations                | 36 [2.8]   | 56 [5.3]                                      | 61 [5.6]                                      | 60 [5.6]                        | 46 [6.1]                                      | 31                                    | 8            | 7            | 29                           | 5            | 7            | 42       |
| 22 Discrete Math                          | 17 [1.9]   | 25 [3.7]                                      | 25 [3.0]                                      | 25 [3.0]                        | 22 [4.4]                                      | 7                                     | 7            | 8            | 6                            | 6            | 8            | 21       |
| 23 Linear/Matrix Algebra                  | 37 [2.6]   | 46 [4.0]                                      | 55 [4.8]                                      | 54 [4.8]                        | 65 [8.1]                                      | 42                                    | 9            | 14           | 38                           | 7            | 13           | 57       |
| 21 Freshman Seminar                       |            |                                               | 5 [0.7]                                       | 5 [0.7]                         | 3 [1.2]                                       | 1                                     | 1            | 1            | 1                            | 1            | 1            | 3        |
| 24 Other Calculus Level                   | 9 [2.7]    | 17 [3.1]                                      | 21 [6.0]                                      | 20 [6.0]                        | 12 [7.4]                                      | 10                                    | 0            | 2            | 8                            | 0            | 2            | 10       |
| Subtotal Calculus Level                   | 586 [23.6] | 748 [35.2]                                    | 807 [62.3]                                    | 790 [60.7]                      | 659 [62.7]                                    | 409                                   | 101          | 149          | 366                          | 74           | 141          | 580      |

Note: 0 means less than 500 enrollments.

**TABLE A.1.C.** Fall term mathematics course enrollment (in 1000s) [with SE for 2005, 2010, 2015, and 2021 totals].

|                                                        |          |          |          |          | Fall 2021 Enrollment<br>(in 1000s) |              |              |
|--------------------------------------------------------|----------|----------|----------|----------|------------------------------------|--------------|--------------|
|                                                        |          |          |          |          | Math Departments                   |              |              |
| Courses                                                | 2005     | 2010     | 2015     | 2021     | Univ<br>(PhD)                      | Univ<br>(MA) | Coll<br>(BA) |
| <b>Advanced Level</b>                                  |          |          |          |          |                                    |              |              |
| 25 Intro to Proofs                                     | 12 [1.3] | 15 [1.2] | 18 [1.4] | 15 [2.6] | 8                                  | 3            | 4            |
| 26-1 Modern Algebra I                                  |          | 13 [1]   | 13 [1.7] | 11 [1.8] | 5                                  | 1            | 5            |
| 26-2 Modern Algebra II                                 |          | 1 [0.1]  | 1 [0.2]  | 1 [0.6]  | 1                                  | 0            | 0            |
| 26 Modern Algebra I & II                               | 11 [1.1] | 14 -     | 14 [1.7] |          |                                    |              |              |
| 27 Number Theory                                       | 3 [0.5]  | 4 [0.5]  | 4 [0.5]  | 4 [1.4]  | 2                                  | 1            | 2            |
| 28 Combinatorics                                       | 3 [0.5]  | 3 [0.5]  | 3 [0.8]  | 3 [0.7]  | 2                                  | 0            | 0            |
| 29 Actuarial Mathematics                               | 2 [0.5]  | 2 [0.3]  | 6 [1.1]  | 3 [0.7]  | 2                                  | 0            | 1            |
| 30 Logic/Foundations                                   | 1 [0.4]  | 1 [0.2]  | 1 [0.4]  | 0 [0.1]  | 0                                  | 0            | 0            |
| 31 Discrete Structures                                 | 3 [0.7]  | 4 [0.9]  | 4 [0.8]  | 3 [1.9]  | 2                                  | 1            |              |
| 32 History of Mathematics                              | 6 [1.0]  | 7 [1.4]  | 5 [0.9]  | 3 [0.9]  | 1                                  | 0            | 2            |
| 33 Geometry                                            | 8 [1.0]  | 10 [1]   | 8 [1.1]  | 5 [1.2]  | 3                                  | 1            | 1            |
| 34 Math for HS Teachers                                | 8 [2.2]  | 8 [1]    | 8 [1.4]  | 3 [1.0]  | 1                                  | 0            | 2            |
| 35-1 Advanced Calculus I, Real Analysis I              |          | 16 [1.6] | 15 [1.4] | 11 [2.0] | 7                                  | 1            | 3            |
| 35-2 Advanced Calculus II, Real Analysis II            |          | 2 [0.8]  | 1 [0.3]  | 1 [0.3]  | 1                                  | 0            |              |
| 35 Advanced Calculus I & II, Real Analysis I & II      | 15 [1.2] | 18 -     | 17 [1.6] |          |                                    |              |              |
| 36 Advanced Math for Engineering and Physical Sciences | 6 [1.1]  | 11 [5.3] | 6 [1.3]  | 5 [1.5]  | 3                                  | 1            | 1            |
| 37 Advanced Linear Algebra                             | 4 [0.7]  | 4 [0.5]  | 7 [0.9]  | 3 [1.3]  | 3                                  | 0            | 0            |
| 38 Vector Analysis                                     | 2 [0.8]  | 3 [0.5]  | 4 [1.3]  | 12 [7.9] | 12                                 | 0            | 0            |
| 39 Advanced Differential Equations                     | 1 [0.2]  | 3 [0.6]  | 3 [0.6]  | 2 [0.8]  | 2                                  | 0            |              |
| 40 Partial Differential Equations                      | 3 [0.5]  | 4 [0.5]  | 4 [1.0]  | 2 [0.7]  | 2                                  | 0            | 0            |
| 41 Numerical Analysis I & II                           | 5 [0.5]  | 7 [1.1]  | 8 [0.8]  | 7 [1.5]  | 5                                  | 2            | 0            |

Note: 0 means less than 500 enrollments. Blank entries represent courses with no responses.

Note: Totals for advanced-level course enrollments include distance courses. Unlike Tables A.1.A and A.1.B, totals for non-distance course enrollments were not generated due to the low enrollments by degree type.



**TABLE A.1.D.** Fall term mathematics course enrollment (in 1000s) [with SE for 2005, 2010, 2015, and 2021 totals].

|                                      |                    |                    |                     |                     | Fall 2021 Enrollment<br>(in 1000s) |              |              |
|--------------------------------------|--------------------|--------------------|---------------------|---------------------|------------------------------------|--------------|--------------|
|                                      |                    |                    |                     |                     | Math Departments                   |              |              |
| Courses                              | 2005               | 2010               | 2015                | 2021                | Univ<br>(PhD)                      | Univ<br>(MA) | Coll<br>(BA) |
| <b>Advanced Level (Cont.)</b>        |                    |                    |                     |                     |                                    |              |              |
| 42 Applied Math (Modeling)           | 2 [0.3]            | 3 [0.5]            | 4 [0.7]             | 3 [1.1]             | 2                                  | 0            | 1            |
| 43 Complex Variables                 | 3 [0.5]            | 3 [0.3]            | 3 [0.5]             | 2 [0.5]             | 1                                  | 0            | 0            |
| 44 Topology                          | 1 [0.3]            | 2 [0.2]            | 2 [0.3]             | 1 [0.3]             | 1                                  |              | 0            |
| 45 Math of Finance                   | 1 [0.4]            | 2 [0.4]            | 4 [1.1]             | 2 [0.7]             | 1                                  | 0            | 0            |
| 46 Codes & Cryptology                | 0 [0.2]            | 0 [0.1]            | 1 [0.3]             | 3 [1.0]             | 2                                  | 1            | 0            |
| 47 Biomathematics                    | 1 [0.2]            | 1 [0.2]            | 1 [0.2]             | 1 [0.5]             | 0                                  | 0            | 0            |
| 48 Senior Sem / Ind Study in Math    | 3 [0.5]            | 5 [0.5]            | 6 [0.6]             | 4 [1.0]             | 1                                  | 1            | 3            |
| 49 Other Adv-Level Courses           | 5 [0.7]            | 14 [3.8]           | 11 [2.6]            | 7 [2.2]             | 3                                  | 2            | 3            |
| <b>Operations Research</b>           |                    |                    |                     |                     |                                    |              |              |
| 58 Intro Oper Research               | 1 [0.2]            |                    |                     |                     |                                    |              |              |
| 59 Intro to Linear Programming       | 1 [0.4]            |                    |                     |                     |                                    |              |              |
| 60 Other Oper Research               | 0 [0.2]            |                    |                     |                     |                                    |              |              |
| 61 Operations Research (All Courses) |                    | 2 [0.4]            | 3 [1.2]             | 2 [0.9]             | 1                                  | 0            | 1            |
| <b>Subtotal Advanced Level</b>       | <b>112 [6.2]</b>   | <b>150 [6.6]</b>   | <b>154 [12.2]</b>   | <b>121 [17.1]</b>   | <b>73</b>                          | <b>18</b>    | <b>30</b>    |
| <b>Mathematics Total</b>             | <b>1606 [45.3]</b> | <b>1971 [72.5]</b> | <b>2213 [139.7]</b> | <b>1551 [121.7]</b> | <b>760</b>                         | <b>281</b>   | <b>511</b>   |

Note: 0 means less than 500 enrollments. Blank entries represent courses with no responses.

Note: Totals for advanced-level course enrollments include distance courses. Unlike Tables A.1.A and A.1.B, totals for non-distance course enrollments were not generated due to the low enrollments by degree type.

**TABLE A.2.A.** Enrollment (in 1000s) in statistics courses in fall 2005, 2010, 2015, and 2021 [with SE for 2005, 2015, and 2021 totals]. (SEs for 2010 Mathematics and Statistics department totals are in CBMS2010, Table A.2, p. 189.) Round-off may cause marginal totals to appear incorrect.

|                                                                | Fall 2021 Enrollment (in 1000s) |               |               |               |               |                           |              |               |               |              | Fall 2021 Enrollment (in 1000s) |            |               |              |           |                           |              |               |               |              |           |               |           |           |           |              |
|----------------------------------------------------------------|---------------------------------|---------------|---------------|---------------|---------------|---------------------------|--------------|---------------|---------------|--------------|---------------------------------|------------|---------------|--------------|-----------|---------------------------|--------------|---------------|---------------|--------------|-----------|---------------|-----------|-----------|-----------|--------------|
|                                                                | Mathematics Departments         |               |               |               |               |                           |              |               |               |              | Statistics Departments          |            |               |              |           |                           |              |               |               |              |           |               |           |           |           |              |
|                                                                | Total (Including Dist. Courses) |               |               |               |               | Total (Non-Dist. Courses) |              |               |               |              | Total (Inc. Dist. Courses)      |            |               |              |           | Total (Non-Dist. Courses) |              |               |               |              |           |               |           |           |           |              |
| Statistics Courses                                             | Total<br>2005                   | Total<br>2010 | Total<br>2015 | Total<br>2021 | Univ<br>(PhD) | Univ<br>(MA)              | Coll<br>(BA) | Subtotal      | Univ<br>(PhD) | Univ<br>(MA) | Coll<br>(BA)                    | Subtotal   | Univ<br>(PhD) | Univ<br>(MA) | Subtotal  | Univ<br>(PhD)             | Univ<br>(MA) | Subtotal      | Univ<br>(PhD) | Univ<br>(MA) | Subtotal  |               |           |           |           |              |
| <b>Lower-Level Statistics</b>                                  |                                 |               |               |               |               |                           |              |               |               |              |                                 |            |               |              |           |                           |              |               |               |              |           |               |           |           |           |              |
| Introductory Statistics (No Calc Prereq)                       | 167                             | [14.3]        | 243           | -             | 275           | [16.7]                    | 205          | [15.0]        | 19            | 36           | 81                              | 136        | [16.5]        | 17           | 27        | 66                        | 109          | [13.1]        | 54            | 16           | 70        | [9.5]         | 38        | 12        | 49        | [7.4]        |
| Introductory Statistics (Calc Prereq, for Non-Majors)          |                                 |               | 42            | -             | 56            | [6.2]                     | 37           | [3.8]         | 6             | 5            | 4                               | 15         | [3.3]         | 5            | 5         | 3                         | 13           | [3.0]         | 19            | 3            | 22        | [4.8]         | 12        | 2         | 14        | [3.8]        |
| Probability & Statistics (No Calc Prereq)                      | 21                              | [5.5]         | 19            | -             | NA            | NA                        |              |               |               |              |                                 |            |               |              |           |                           |              |               |               |              |           |               |           |           |           |              |
| Statistics for Pre-Service Elementary or Middle Grade Teachers |                                 |               | 66            | -             | 1             | [0.3]                     | 1            | [0.1]         |               |              | 1                               | 1          | [0.1]         |              |           | 1                         | 1            | [0.1]         |               |              |           |               |           |           |           |              |
| Statistics for Pre-Service High School Teachers                |                                 |               | 122           | -             | 0             | [0.2]                     | 0            | [0.0]         |               |              | 0                               | 0          | [0.0]         |              |           | 0                         | 0            | [0.0]         |               |              |           |               |           |           |           |              |
| Other Intro-Level Statistics                                   | 13                              | [2.5]         | 8             | -             | 15            | [2.9]                     | 9            | [4.2]         | 0             | 2            | 5                               | 7          | [4.8]         | 0            | 1         | 5                         | 6            | [4.3]         | 0             | 1            | 2         | [1.4]         | 0         | 1         | 1         | [0.9]        |
| <b>Subtotal Intro-Level Statistics</b>                         | <b>202</b>                      | <b>[14.9]</b> | <b>312</b>    | <b>-</b>      | <b>347</b>    | <b>[20.7]</b>             | <b>252</b>   | <b>[17.8]</b> | <b>25</b>     | <b>43</b>    | <b>91</b>                       | <b>159</b> | <b>[19.5]</b> | <b>22</b>    | <b>32</b> | <b>74</b>                 | <b>129</b>   | <b>[16.1]</b> | <b>73</b>     | <b>20</b>    | <b>93</b> | <b>[12.0]</b> | <b>50</b> | <b>14</b> | <b>65</b> | <b>[9.5]</b> |

Note: 0 means less than 500 enrollments. Blank entries represent courses with no responses. SEs for combined enrollments across mathematics and statistics departments were not calculated in 2010.

**TABLE A.2.B.** Enrollment (in 1000s) in statistics courses in 2005, 2010, 2015, and 2021 in mathematics and statistics departments [with SE for 2005, 2015, and 2021 totals]. (SEs for 2010 Mathematics and Statistics department totals are in CBMS2010, Table A.2, p. 190.) Round-off may cause marginal totals to appear incorrect.

|                                               |               |               |               |               | Fall 2021 Enrollment (in 1000s) |              |              |                           |               |              |                                 |          |               |                           |          |         |
|-----------------------------------------------|---------------|---------------|---------------|---------------|---------------------------------|--------------|--------------|---------------------------|---------------|--------------|---------------------------------|----------|---------------|---------------------------|----------|---------|
|                                               |               |               |               |               | Mathematics Departments         |              |              |                           |               |              | Statistics Departments          |          |               |                           |          |         |
|                                               |               |               |               |               | Total (Including Dist. Courses) |              |              | Total (Non-Dist. Courses) |               |              | Total (Including Dist. Courses) |          |               | Total (Non-Dist. Courses) |          |         |
| Statistics Courses                            | Total<br>2005 | Total<br>2010 | Total<br>2015 | Total<br>2021 | Univ<br>(PhD)                   | Univ<br>(MA) | Coll<br>(BA) | Subtotal                  | Univ<br>(PhD) | Univ<br>(MA) | Coll<br>(BA)                    | Subtotal | Univ<br>(PhD) | Univ<br>(MA)              | Subtotal |         |
| <b>Upper-Level Statistics</b>                 |               |               |               |               |                                 |              |              |                           |               |              |                                 |          |               |                           |          |         |
| Intermediate Statistics (Non-Calc)            |               |               |               | 8 [1.6]       | 0                               | 1            | 3            | 4 [1.7]                   | 0             | 1            | 3                               | 4 [1.7]  | 3             |                           | 3 [1.3]  | 2 [1.0] |
| Prob & Statistics for Majors (No Calc Prereq) |               |               | 38 [4.7]      |               |                                 |              |              |                           |               |              |                                 |          |               |                           |          |         |
| Math Statistics (Calc Prereq)                 | 12 [2.1]      | 8 -           | 8 [0.7]       | 8 [1.1]       | 2                               | 0            | 2            | 4 [1.0]                   | 1             | 0            | 2                               | 4 [1.0]  | 3             | 0                         | 3 [0.8]  |         |
| Probability (Calc Prereq)                     | 10 [1.0]      | 12 -          | 16 [1.2]      | 13 [1.7]      | 5                               | 2            | 4            | 11 [2.0]                  | 5             | 2            | 4                               | 10 [2.0] | 2             | 0                         | 3 [0.8]  | 2 [0.7] |
| Prob & Statistics Combined                    | 16 [2.0]      | 12 -          | 19 [1.8]      | 20 [2.4]      | 5                               | 3            | 4            | 11 [2.4]                  | 4             | 2            | 3                               | 9 [2.2]  | 7             | 2                         | 9 [2.4]  | 4 [1.5] |
| Stochastic Processes                          | 1 [0.2]       | 1 -           | 2 [0.3]       | 1 [0.3]       | 0                               | 0            |              | 0 [0.2]                   | 0             | 0            |                                 | 0 [0.2]  | 1             |                           | 1 [0.5]  | 1 [0.5] |
| Applied Statistical Analysis                  | 7 [1.2]       | 5 -           | 4 [0.9]       |               |                                 |              |              |                           |               |              |                                 |          |               |                           |          |         |
| Applied Regression                            |               |               |               | 5 [1.2]       | 1                               | 0            | 2            | 3 [1.3]                   | 1             | 0            | 2                               | 3 [1.3]  | 2             | 0                         | 2 [0.6]  | 2 [0.6] |
| Linear Models I                               |               |               |               | 1 [0.2]       |                                 | 0            |              | 0 [0.0]                   |               | 0            |                                 | 0 [0.0]  | 1             | 0                         | 1 [0.4]  | 1 [0.4] |
| Linear Models II                              |               |               |               | 0 [0.0]       |                                 |              |              |                           |               |              |                                 |          | 0             |                           | 0 [0.0]  | 0 [0.0] |
| Data Science/Analytics                        |               |               | 2 [0.2]       | 6 [2.4]       | 0                               | 0            | 6            | 6 [2.8]                   | 0             | 0            | 5                               | 6 [2.7]  | 3             | 1                         | 4 [1.1]  | 3 [0.9] |
| Design & Anal of Experiments                  | 1 [0.2]       | 2 -           | 2 [0.2]       | 1 [0.4]       | 0                               | 0            | 1            | 1 [0.5]                   | 0             | 0            | 1                               | 1 [0.4]  | 1             | 0                         | 1 [0.3]  | 1 [0.3] |
| Regression & Correlation                      | 3 [0.5]       | 4 -           | 5 [0.4]       |               |                                 |              |              |                           |               |              |                                 |          |               |                           |          |         |
| Biostatistics                                 | 2 [0.6]       | 1 -           | 2 [0.4]       | 6 [3.2]       |                                 | 0            | 5            | 5 [3.7]                   |               | 0            | 4                               | 4 [3.5]  | 0             | 1                         | 1 [0.4]  | 1 [0.4] |
| Nonparametric Statistics                      | 0 [0.1]       | 0 -           | 1 [0.0]       | 1 [0.2]       | 0                               |              | 0            | 0 [0.3]                   | 0             |              | 0                               | 0 [0.3]  | 0             |                           | 0 [0.1]  | 0 [0.1] |

Note: 0 means less than 500 enrollments. Blank entries represent courses with no responses. SEs for combined enrollments across mathematics and statistics departments were not calculated in 2010.

**TABLE A.2.C.** Fall term statistics enrollment (in 1000s) [with SE for 2005, 2015, and 2021 totals]. (SEs for 2010 Mathematics and Statistics department totals are in CBMS2010, Table A.2, p. 191.)

| Fall 2021 Enrollment (in 1000s)       |               |               |               |               |                                 |              |              |          |               |                           |              |                        |               |              |                                 |          |               |              |                           |               |              |          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |    |
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| Mathematics Departments               |               |               |               |               |                                 |              |              |          |               |                           |              | Statistics Departments |               |              |                                 |          |               |              |                           |               |              |          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |    |
| Statistics Courses                    | Total<br>2005 | Total<br>2010 | Total<br>2015 | Total<br>2021 | Total (Including Dist. Courses) |              |              |          |               | Total (Non-Dist. Courses) |              |                        |               |              | Total (Including Dist. Courses) |          |               |              | Total (Non-Dist. Courses) |               |              |          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |    |
|                                       |               |               |               |               | Univ<br>(PhD)                   | Univ<br>(MA) | Coll<br>(BA) | Subtotal | Univ<br>(PhD) | Univ<br>(MA)              | Coll<br>(BA) | Subtotal               | Univ<br>(PhD) | Univ<br>(MA) | Coll<br>(BA)                    | Subtotal | Univ<br>(PhD) | Univ<br>(MA) | Subtotal                  | Univ<br>(PhD) | Univ<br>(MA) | Subtotal |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |    |
| Upper-Level Statistics<br>(Continued) |               |               |               |               |                                 |              |              |          |               |                           |              |                        |               |              |                                 |          |               |              |                           |               |              |          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | </ |

Note: 0 means less than 500 enrollments. Blank entries represent courses with no responses. SEs for combined enrollments across mathematics and statistics departments were not calculated in 2010.

**TABLE A.3.** Enrollment (in 1000s) in computer science courses in fall 2005, 2010, 2015, and 2021 [with SE for 2005, 2010, 2015, and 2021 totals]. Round-off may cause marginal totals to appear incorrect.

| Fall 2021 Enrollment (in 1000s) |                                    |           |           |            |                              |           |            |           |           |          |  |
|---------------------------------|------------------------------------|-----------|-----------|------------|------------------------------|-----------|------------|-----------|-----------|----------|--|
| Mathematics Departments         |                                    |           |           |            |                              |           |            |           |           |          |  |
| CS Courses                      | Total (Including Distance Courses) |           |           |            | Total (Non-Distance Courses) |           |            |           |           |          |  |
|                                 | Univ (PhD)                         | Univ (MA) | Coll (BA) | Univ (PhD) | Univ (MA)                    | Coll (BA) | Univ (PhD) | Univ (MA) | Coll (BA) | Subtotal |  |
|                                 |                                    |           |           |            |                              |           |            |           |           |          |  |
|                                 |                                    |           |           |            |                              |           |            |           |           |          |  |
|                                 |                                    |           |           |            |                              |           |            |           |           |          |  |
|                                 |                                    |           |           |            |                              |           |            |           |           |          |  |
|                                 |                                    |           |           |            |                              |           |            |           |           |          |  |
|                                 |                                    |           |           |            |                              |           |            |           |           |          |  |
|                                 |                                    |           |           |            |                              |           |            |           |           |          |  |
|                                 |                                    |           |           |            |                              |           |            |           |           |          |  |
| Total Computer Science          |                                    |           |           |            |                              |           |            |           |           |          |  |

\* For 2005 and 2010, these courses were described in the 2001 ACM/IEEE report *Model Curricula for Computing*. For 2015 and 2021, these courses were described in the IEEE/ACM report *Computer Science Curricula 2013*.

# Sampling and Estimation Procedures

## Overview

A stratified simple random sample was employed in the CBMS 2021 survey, and strata were based on three variables: curriculum, highest degree level offered, and total institutional enrollment. Data were collected using an online survey with email and telephone followup.

## Sampling Approach

For CBMS 2021, the basic design was a stratified simple random sample of institutions. Neyman allocation based on a key outcome variable was used to determine targeted sample sizes for the 29 sampling strata. A two-phase sample design was applied to some of the strata to ease data collection workload when the sampling frame was imperfect.

## Target Population and Sampling Frames

The Integrated Postsecondary Education Data System (IPEDS), a database maintained by the National Center for Education Statistics within the U.S. Department of Education, was used as a basis for building a frame for this survey. For the academic year 2019–2020, there were approximately 2700 four-year colleges and universities across the country and 1300 two-year colleges, according to IPEDS. AMS conducts annual surveys of four-year institutions and thus has reasonably current information for four-year institutions; this information was used as a basis for updating the IPEDS frame. However, it was necessary to obtain updated information on two-year institutions, partly because they are surveyed only every five years and partly because of variations in how they are administered. Two-year institutions are sometimes centralized (with one institution having all required information, including for branch campuses) and sometimes decentralized (with each campus maintaining its own data, and there being no integrated database); the latter must be surveyed separately, so the sampling unit becomes the campus rather than the institution. Sometimes there is a mixture of centralization and decentralization at two-year colleges; for example, an administratively independent campus might have a satellite location that is not administratively separate from the campus. The sampling unit was that level that maintained administrative data on faculty and courses.

The target population of the CBMS 2021 survey consisted of undergraduate mathematics and statistics programs at two- and four-year colleges and universities in the U.S. Thus, the frame for the CBMS 2021 survey was divided into three parts: (A) 1524 institutions having four-year math programs, (B) 183 institutions having four-year statistics programs, and (C) 1008 institutions having two-year math programs. Note that parts A and B did not necessarily consist of mutually exclusive institutions since some institutions had both four-year math programs and four-year statistics programs. However, this was not problematic since the math and statistics programs within these institutions were the targets of interest, and the departments were sampled independently.

## Sampling Strata

The three parts of the frame were each stratified using the same two variables that were used in the previous three rounds of the CBMS survey, that is, “Highest Degree Granted by the Institution” (PhD, MA, or BA) and “Institutional Full-Time Equivalent (FTE) Undergraduate Fall Enrollment.” After an initial investigation on the population distributions of the two variables, it was determined that the strata from CBMS 2015 largely could be maintained with changes to the thresholds. The stratification for part A was similar to the design in CBMS 2015 except for a change in the boundaries. The stratification used in CBMS 2015 for part C was applied for this round except that the previous strata 9 was removed. The stratification for part B of the frame remained unchanged.

The final stratification can be seen in the first four columns of Table 1. The four-year mathematics programs were divided into sixteen strata, the four-year statistics programs were divided into five strata, and the two-year programs were divided into eight strata.

## Allocation Process

For the CBMS 2021 survey, a stratified simple random sample of 622 institutions was drawn from parts A, B, and C. The sampling rates were adjusted based on the response rates in CBMS2015. The sample for CBMS2021 consisted of 314 institutions sampled from part A, 228 institutions sampled from part C, and 75 institutions sampled

**TABLE 1.** Phase 1—Stratum designations and final allocation for the CBMS 2021 study (program types A, B, and C).

| Stratum | Program Type             | Highest Degree Granted | FTE Undergraduate Fall Enrollment | Universe (N) | Final Sample Allocation (n) | Sampling Rate (n/N) | Raw Sampling Weights (N/n) |
|---------|--------------------------|------------------------|-----------------------------------|--------------|-----------------------------|---------------------|----------------------------|
| 1       | Four-Year Math (A)       | PhD                    | 0–3499                            | 320          | 64                          | 0.19                | 5.16                       |
| 2       |                          |                        | 3500–8999                         | 222          | 59                          | 0.27                | 3.76                       |
| 3       |                          |                        | 9000–16,999                       | 106          | 37                          | 0.33                | 3.03                       |
| 4       |                          |                        | 17,000–27,999                     | 74           | 25                          | 0.34                | 2.96                       |
| 5       |                          |                        | 28,000–37,999                     | 36           | 13                          | 0.36                | 2.77                       |
| 6       |                          |                        | 38,000+                           | 8            | 8                           | 1                   | 1                          |
| 7       |                          | MA                     | 0–1499                            | 237          | 31                          | 0.13                | 7.65                       |
| 8       |                          |                        | 1500–3499                         | 167          | 27                          | 0.16                | 6.42                       |
| 9       |                          |                        | 3500–7499                         | 79           | 18                          | 0.23                | 4.39                       |
| 10      |                          |                        | 7500–14,999                       | 19           | 5                           | 0.26                | 3.8                        |
| 11      |                          |                        | 15,000+                           | 6            | 6                           | 1                   | 1                          |
| 12      |                          | BA                     | 0–499                             | 29           | 2                           | 0.07                | 14.5                       |
| 13      |                          |                        | 500–1499                          | 100          | 11                          | 0.11                | 9.09                       |
| 14      |                          |                        | 1500–2999                         | 66           | 9                           | 0.14                | 7.33                       |
| 15      |                          |                        | 3000–9999                         | 11           | 2                           | 0.18                | 5.5                        |
| 16      |                          |                        | 10,000+                           | 2            | 2                           | 1                   | 1                          |
| 17      | Four-Year Statistics (B) | PhD                    | 0–29                              | 48           | 16                          | 0.33                | 3                          |
| 18      |                          |                        | 30–59                             | 34           | 13                          | 0.38                | 2.61                       |
| 19      |                          |                        | 60–144                            | 50           | 22                          | 0.44                | 2.27                       |
| 20      |                          |                        | 145+                              | 28           | 11                          | 0.39                | 2.55                       |
| 21      |                          | MA / BA                | All                               | 13           | 13                          | 1                   | 1                          |
| 22      | Two-Year Schools (C)     | N/A                    | 0–1099                            | 174          | 18                          | 0.1                 | 9.67                       |
| 23      |                          |                        | 1100–2499                         | 275          | 40                          | 0.14                | 7.05                       |
| 24      |                          |                        | 2500–4499                         | 222          | 46                          | 0.2                 | 5.05                       |
| 25      |                          |                        | 4500–7499                         | 175          | 49                          | 0.3                 | 3.37                       |
| 26      |                          |                        | 7500–11,499                       | 87           | 35                          | 0.41                | 2.42                       |
| 27      |                          |                        | 11,500–16,999                     | 47           | 22                          | 0.49                | 2.04                       |
| 28      |                          |                        | 17,000–24,999                     | 20           | 11                          | 0.65                | 1.54                       |
| 29      |                          |                        | 25,000+                           | 7            | 7                           | 1                   | 1                          |

from part B. For part A, there were later 5 records from territories added with a sampling rate of 1 for each, bringing the total for part A to 319.

In order to allocate the sample optimally to each of the 29 strata, Neyman allocation was used. This form of allocation distributes sample to the strata proportionately to the overall number of institutions on the frame belonging to each stratum, while adjusting the allocation to give more sample to those strata with greater variability (larger standard deviations) with respect to key variables. The statistics of interest in this survey involve both the counts at the student level and the counts at the institution level. In the frame for the CBMS 2021 survey, the most reliable information for developing the design was the student enrollment, a count at the student level, so it was used as the key outcome variable to measure variability.

For part A, the standard deviation varied substantially, ranging from 153 in stratum 12 to 27,168 in stratum 16. To smooth out this broad range of variability, and not let it dominate the sample allocation, while balancing the precision of estimates at the institution level, a modified Neyman allocation, the square root of the standard deviation of the student FTE enrollment, was used to allocate the sample in strata 1 through 5, 7 through 10, and 12 through 15. Strata 6, 11, and 16 were selected with certainty.

For part C, the sample was distributed through strata 22 to 28 by Neyman allocation. Strata 29 was a certainty stratum. For part B, strata 17 and 18 were allocated using Neyman, and the remaining three strata were selected with certainty.

## Weighting Approach

Sampling weights that adjusted for nonresponding institutions were created for weighted data analysis. To facilitate the calculation of SEs, replicate weights were created using the unstratified jackknife method. Nonresponse adjustments were also applied to each set of replicate weights.

For the four-year math programs, a raking step was performed to adjust the weights to a variable with the corrected degree (PhD, MA, and BA) fields.

## Sampling Weights

For parts A, B, and C, the raw sampling weight in Table 1 serves as the base weight. The raw sampling weight in the  $h$ th stratum was computed as  $N_h/n_h$ , where  $N_h$  is the total number of institutions in the  $h$ th stratum and  $n_h$  is the number of selected institutions in the  $h$ th stratum. Among the sampled institutions,<sup>1</sup> a few were identified as ineligible for the following reasons:

- Institutions only offering math as part of a general studies requirement but that were classified as a four-year mathematics program based on the sampling frame;
- Institutions having math courses required for some other programs but that were classified as a two-year mathematics program;
- Institutions having statistics courses required for some other programs, i.e., business school, but that were classified as a four-year statistics program;
- A duplicate institution was found.

The ineligible institutions were out of scope of the population of interest, so they were excluded from the weighting adjustment. The rest of the sample was classified as either responding institutions or nonresponding institutions. To remove bias from the estimates and reduce variability of the estimates, the base weights were adjusted for nonresponse. Within stratum  $h$ , a nonresponse adjustment factor,  $f_h$  was calculated as

$$f_h = \frac{\sum_{\text{eligible}} W_h}{\sum_{\text{responding}} W_h},$$

where  $W_h$  is the base weight. Small cells in a stratum with less than 10 institutions or a large nonresponse adjustment exceeding 2.8 were collapsed with an adjacent cell within program type and highest degree granted. The analysis weight,  $W_h^*$  for any respondent in the  $h$ th stratum was computed as

$$W_h^* = W_h f_h.$$

See Tables 2, 3, and 4 for the weights used in the four-year mathematics, four-year statistics, and two-year mathematics categories, respectively. Two sets of weights were produced for two-year mathematics. One set of weights applied to all of the responding two-year institutions.

## Replicate Weights

Weighted estimates and SEs were calculated using the replication method JK1 (Jackknife method). The idea behind replication is to select subsamples (replicates) repeatedly from the whole sample, calculate the statistic of interest for each subsample, and then use these subsamples or replicate statistics to estimate the variance of the full-sample statistics. The JK1 method divides the sample into subsamples by excluding one unit at a time.

For the initial 2021 data collection, since all samples were set aside and all records on the frames were surveyed due to the COVID-19 pandemic, the base weights were reset to reflect this. This is why the JK1 method was used

<sup>1</sup>For the part C sample, the institutions refer to the campuses.



TABLE 2. Final sampling weights used in the four-year mathematics questionnaire.

| Stratum<br>( $h$ ) | Number of<br>Completes | Number of<br>Nonresponse | Number of<br>Ineligibles | Response<br>Rate | Base<br>Weight<br>( $W_h$ ) | Nonresponse<br>Adjusted Factor<br>( $f_h$ ) | Nonresponse<br>Adjusted Weight<br>( $W_h^*$ ) | Raking<br>Adjusted<br>Factors | Raking<br>Adjusted<br>Weight |
|--------------------|------------------------|--------------------------|--------------------------|------------------|-----------------------------|---------------------------------------------|-----------------------------------------------|-------------------------------|------------------------------|
| 1                  | 20                     | 42                       | 2                        | 0.323            | 1–5.194                     | 2.816                                       | 2.816–14.626                                  | 0.659–1.187                   | 3.341–17.354                 |
| 2                  | 23                     | 36                       | 0                        | 0.39             | 3.763                       | 2.816                                       | 10.596                                        | 0.659–1.187                   | 6.979–12.573                 |
| 3                  | 17                     | 20                       | 0                        | 0.459            | 1–3.029                     | 2.816                                       | 8.529                                         | 0.659–1.187                   | 5.617–10.12                  |
| 4                  | 14                     | 11                       | 0                        | 0.56             | 2.96                        | 2.816                                       | 8.336                                         | 0.659–0.736                   | 5.49–6.138                   |
| 5                  | 4                      | 9                        | 0                        | 0.308            | 2.769                       | 2.816                                       | 7.798                                         | 0.659                         | 5.136                        |
| 6                  | 3                      | 5                        | 0                        | 0.375            | 1                           | 2.816                                       | 2.816                                         | 0.659                         | 1.855                        |
| 7                  | 8                      | 21                       | 2                        | 0.276            | 7.71                        | 2.816                                       | 21.711                                        | 1.187                         | 25.761                       |
| 8                  | 10                     | 17                       | 0                        | 0.37             | 6.423                       | 2.816                                       | 18.088                                        | 0.736–1.187                   | 13.319–21.462                |
| 9                  | 5                      | 13                       | 0                        | 0.278            | 4.389                       | 2.816                                       | 12.36                                         | 0.736–1.187                   | 9.101–14.665                 |
| 10                 | 1                      | 4                        | 0                        | 0.2              | 3.8                         | 2.816                                       | 10.701                                        | 0.736                         | 7.88                         |
| 11                 | 2                      | 4                        | 0                        | 0.333            | 1                           | 2.816                                       | 2.816                                         | 1.187                         | 3.341                        |
| 12                 | 1                      | 1                        | 0                        | 0.5              | 14.5                        | 2.816                                       | 40.834                                        | 1.187                         | 48.45                        |
| 13                 | 3                      | 8                        | 0                        | 0.273            | 9.091                       | 2.816                                       | 25.601                                        | 0.736–1.187                   | 18.852–30.376                |
| 14                 | 4                      | 5                        | 0                        | 0.444            | 7.333                       | 2.816                                       | 20.652                                        | 1.187                         | 24.503                       |
| 15                 | 1                      | 1                        | 0                        | 0.5              | 5.5                         | 2.816                                       | 15.489                                        | 1.187                         | 18.378                       |
| 16                 | 1                      | 1                        | 0                        | 0.5              | 1                           | 2.816                                       | 2.816                                         | 1.187                         | 3.341                        |
| Total              | 117                    | 198                      | 4                        | 0.371            |                             |                                             |                                               |                               |                              |

**TABLE 3.** Final sampling weights used in the four-year statistics questionnaire.

|       | Number of<br>Completes | Number of<br>Nonresponse | Number of<br>Ineligibles | Response<br>Rate | Base<br>Weight<br>( $W_h$ ) | Nonresponse<br>Adjusted Factor<br>( $f_h$ ) | Final<br>Weight<br>( $W_h^*$ ) |
|-------|------------------------|--------------------------|--------------------------|------------------|-----------------------------|---------------------------------------------|--------------------------------|
| Total | 43                     | 31                       | 1                        | 0.581            | 1                           | 1.721                                       | 1.721                          |

**TABLE 4.** Final sampling weights used in the two-year mathematics questionnaire.

| Stratum<br>( $h$ ) | Number of<br>Completes | Number of<br>Nonresponse | Number of<br>Ineligibles | Response<br>Rate | Base<br>Weight<br>( $W_h$ ) | Nonresponse<br>Adjusted Factor<br>( $f_h$ ) | Final<br>Weight<br>( $W_h^*$ ) |
|--------------------|------------------------|--------------------------|--------------------------|------------------|-----------------------------|---------------------------------------------|--------------------------------|
| 22                 | 7                      | 10                       | 1                        | 0.412            | 9.667                       | 2.953                                       | 28.546                         |
| 23                 | 12                     | 28                       | 0                        | 0.3              | 7.051                       | 2.953                                       | 20.823                         |
| 24                 | 16                     | 29                       | 1                        | 0.356            | 5.045                       | 2.953                                       | 14.9                           |
| 25                 | 16                     | 32                       | 1                        | 0.333            | 3.365                       | 2.953                                       | 9.938                          |
| 26                 | 6                      | 29                       | 0                        | 0.171            | 2.417                       | 2.953                                       | 7.137                          |
| 27                 | 10                     | 11                       | 1                        | 0.476            | 2.043                       | 2.953                                       | 6.035                          |
| 28                 | 6                      | 5                        | 0                        | 0.545            | 1.538                       | 2.953                                       | 4.543                          |
| 29                 | 3                      | 4                        | 0                        | 0.429            | 1                           | 2.953                                       | 2.953                          |
| Total              | 76                     | 148                      | 4                        | 0.339            |                             |                                             |                                |

instead of the previous JK<sub>n</sub> method, which is preferred for stratified designs. Although the final math four-year sample was stratified, the JK<sub>1</sub> method was kept to stay in line with the other two samples.

For the CBMS 2021 survey, 200 replicates were created for the two- and four-year mathematics programs. For the four-year statistics program, 75 replicates were used due to the limited sample size. The same nonresponse adjustment used for the full sample was applied to each replicate.

### Variance Estimation

A change was made for CBMS2021 to use the JK<sub>1</sub> method instead of the JK<sub>n</sub> method for variance estimation.

Suppose that  $\hat{\theta}$  is the full-sample estimate of some population parameter  $\theta$ . The variance estimator using the JK<sub>1</sub> method,  $v(\hat{\theta})$ , is

$$v(\hat{\theta}) = \left( \frac{G-1}{G} \right) \sum_{g=1}^G h_g (\hat{\theta}_{(g)} - \bar{\theta})^2,$$

where:

- $\hat{\theta}_{(g)}$  is the estimate of  $\theta$  based on the observations included in the  $g$ th replicate;
- $G$  is the number of replicates formed; and
- $h_g$  is the JK<sub>1</sub> factor for replicate  $g$ .



## Appendix III

# List of Respondents to the Survey

### Two-Year Respondents

**Arizona Western College**

*Mathematics and Engineering Division*

**Austin Community College**

*Mathematics and Developmental Mathematics*

**Borough of Manhattan Community College**

*Mathematics*

**Caldwell Community College and Technical Institute**

*Mathematics Department*

**Cape Cod Community College**

*Mathematics*

**Catawba Valley Community College**

*Astronomy, Mathematics, and Physics*

**Chattahoochee Valley Community College**

*Mathematics Department*

**Chemeketa Community College Salem Campus**

*Mathematics Program*

**Chesapeake of Tidewater Community College**

*Science and Mathematics Pathway*

**Cincinnati State Technical and Community College**

*Humanities and Sciences Division*

**Cleveland Community College**

*Mathematics and Education*

**Clovis Community College**

*Math and Science Division*

**Coastal Bend College**

*Math, Science, and Kinesiology*

**Coconino Community College**

*Mathematics*

**College of the Muscogee Nation**

*Academic Affairs*

**Columbia Gorge Community College  
(Indian Creek Campus - Hood River)**

*Mathematics*

**Columbus State Community College**

*Mathematics*

**De Anza College**

*Mathematics*

**Des Moines Area Community College  
West Campus**

*Math and Engineering*

**Diablo Valley College**

*Mathematics*

**Dutchess Community College**

*Math and Computer Science Department*

**El Camino College**

*Mathematical Sciences*

**Estrella Mountain Community College**

*Mathematics*

**Florida State College at Jacksonville**

*Mathematics Department*

**Frederick Community College**

*Math*

**Georgia Northwestern Technical College**

*General Education - Mathematics*

**Glendale Community College**

*Mathematics and Computer Science*

**Glendale Community College (CA)**

*Mathematics*

**Hagerstown Community College**

*Mathematics*

**Harrisburg**

*Mathematics and Computer Science*

**Helena College**  
*General Education*

**Iowa Central Community College**  
*Mathematics*

**Ivy Tech Community College**  
*Mathematics*

**John Tyler Becoming Brightpoint Community College (Chester and Midlothian Campuses)**  
*Mathematics*

**Klamath Community College**  
*Mathematics*

**Lake Land Main Campus**  
*Math and Science*

**Lakeshore Technical College**  
*General Studies*

**Lamar State College Orange**  
*Academic Studies*

**Lewis & Clark Community College - Godfrey Campus**  
*Department of Mathematics*

**Luzerne County Community College**  
*Math, Computer Science, and Pre-Engineering*

**Main Campus**  
*Mathematics*

**Metropolitan Community College**  
*Mathematics Department*

**MiraCosta College**  
*Math*

**Mohawk Valley Community College**  
*School of STEM Transfer*

**Morton College**  
*Mathematics*

**Mount Wachusett Community College, Gardner**  
*Mathematics*

**Mountwest Community and Technical College**  
*Mathematics Department*

**Napa Valley College**  
*Math*

**Northampton Community College**  
*Mathematics*

**Northeast State Community College**  
*Mathematics*

**Oak Ridge Branch Campus**  
*Mathematics*

**Owens State Community College**  
*Department of Mathematics*

**Pasadena City College**  
*Mathematics*

**Pitt Community College**  
*Math and Physics*

**Reynolds Community College**  
*Science, Technology, Engineering, and Mathematics*

**Rio Salado College**  
*Mathematics*

**Rockland Community College**  
*Mathematics*

**Salt Lake Community College**  
*Mathematics*

**Scottsdale Community College**  
*Mathematics, Computer Science, and Engineering*

**Seminole State College of Florida**  
*Mathematics*

**Shoreline Community College**  
*Mathematics*

**South Plains College**  
*Mathematics, Engineering, and Computer Science*

**Southern Maine Community College**  
*Mathematics Department*

**Southwestern Indian Polytechnic Institute**  
*Liberal Arts*

**Spokane Falls Community College***Mathematics***St. Petersburg College - Clearwater Campus***Mathematics***Tarrant County College Northeast Campus***Mathematics***Tulsa Community College***Mathematics***University of Hawai'i Maui College***STEM/Mathematics***University of South Carolina Sumter***Division of Math, Science, and Engineering***University of South Carolina Union***Mathematics***Valencia College - West Campus***Mathematics***Victoria College***Mathematics Department***Virginia Highlands Community College***Arts and Sciences***Walla Walla Community College***Mathematics***Whatcom Community College***Mathematics Department***Four-Year Mathematics Respondents****Adelphi University***Mathematics and Computer Science***Andrews University***Department of Mathematics***Arizona State University***School of Mathematical and Statistical Sciences***Auburn University***Mathematics and Statistics***Barton College***Mathematics***Baylor University***Mathematics***Benedictine University***Mathematical and Computational Sciences***Brigham Young University***Mathematics***Brigham Young University-Idaho***Mathematics***Buena Vista University***Math***Cabrini University***Mathematics***California State University, Sacramento***Mathematics and Statistics***Case Western University***Mathematics, Applied Mathematics, and Statistics***Central Methodist University***Mathematics***Cleveland State University***Department of Mathematics and Statistics***Coastal Carolina University***Mathematics and Statistics***Colorado State University Pueblo***Mathematics***Concord University***Mathematics and Computer Science***Concordia University, St. Paul***Mathematics***Covenant College***Mathematics Department***Dakota Wesleyan University***Mathematics*

**Dalton State College***Department of Technology and Mathematics***Doane University***Mathematics and Data Analytics***Eastern New Mexico University***Mathematical Sciences***Emmanuel College***Mathematics Department***Fort Hays State University***Department of Mathematics***Harding University***Mathematics***Hofstra University***Mathematics***Hood College***Department of Mathematics***Huntington University***Mathematics and Computer Science***Idaho State University***Mathematics and Statistics***Illinois State University***Mathematics Department***Indiana University Southeast***Mathematics Program***Indiana Wesleyan University***Division of Mathematics and Computer  
Information Sciences***Inter American University San German Campus***Department of Sciences and Technology***Ithaca College***Mathematics***Jacksonville University***Mathematics***Kansas State University***Mathematics***Kean University***School of Mathematical Sciences***Lawrence University***Mathematics, Statistics, and Computer Science***Manhattan College***Mathematics***McDaniel College***Department of Mathematics and Computer Science***McMurry University***Mathematics***Metropolitan State University of Denver***Mathematics and Statistics***MidAmerica Nazarene University***Natural, Health, and Mathematical Sciences***Minnesota State University Moorhead***Mathematics***Missouri State University***Department of Mathematics***Montclair State University***Mathematics***Moravian University***Mathematics and Computer Science***Mount Mary University***Mathematics***Mount St. Joseph University***Mathematics***Murray State University***Department of Mathematics and Statistics***New England College***Science, Health, and Education Division***New York University***Department of Mathematics***North Carolina State University***Department of Mathematics*

**North Central College**  
*Mathematics and Actuarial Science*

**Northwest Missouri State University**  
*Mathematics and Statistics*

**Northwestern Oklahoma State University**  
*Math and Computer Science*

**Norwich University**  
*Department of Mathematics*

**Notre Dame College (Ohio)**  
*Mathematics*

**Oklahoma Baptist University**  
*Division of Mathematics*

**Piedmont University**  
*Mathematical Sciences*

**Quinnipiac University**  
*Department of Mathematics and Statistics*

**Regis University**  
*Mathematics*

**Rockford University**  
*Math, CSCI, and Physics*

**Rowan University**  
*Mathematics*

**Rutgers University Newark**  
*Mathematics and Computer Science*

**Samford University**  
*Mathematics and Computer Science*

**Scripps College**  
*Mathematics*

**Southeast Missouri State University**  
*Mathematics*

**Southwestern Oklahoma State University**  
*Mathematics*

**St. Cloud State University**  
*Mathematics and Statistics*

**St. Lawrence University**  
*Mathematics, Computer Science, and Statistics*

**St. Mary's University, San Antonio, Texas**  
*Mathematics*

**State University of New York at Fredonia**  
*Mathematical Sciences*

**Tennessee Tech University**  
*Mathematics*

**Texas A&M University**  
*Department of Mathematics*

**Texas A&M University-Commerce**  
*Mathematics*

**The Citadel**  
*Mathematical Sciences*

**The College of Wooster**  
*Department of Mathematical and Computational Sciences*

**The Master's University**  
*Mathematics*

**The University of Southern Mississippi**  
*School of Mathematics and Natural Sciences*

**Towson University**  
*Department of Mathematics*

**University at Albany - State University of New York**  
*Department of Mathematics and Statistics*

**University of Akron**  
*Mathematics*

**University of Alaska Fairbanks**  
*Mathematics and Statistics*

**University of Arkansas**  
*Mathematical Sciences*

**University of California, Santa Barbara**  
*Mathematics*

**University of Central Florida**  
*Department of Mathematics*

**University of Denver**  
*Department of Mathematics*



**University of Idaho***Mathematics and Statistical Science***University of Iowa***Mathematics***University of Miami***Mathematics***University of Michigan-Flint***Department of Mathematics and Applied Sciences***University of Nebraska-Lincoln***Mathematics***University of Nevada, Las Vegas***Department of Mathematical Sciences***University of North Carolina at Charlotte***Department of Mathematics and Statistics***University of North Carolina Wilmington***Mathematics and Statistics***University of North Dakota***Mathematics***University of Notre Dame***Mathematics***University of Oklahoma***Department of Mathematics***University of Pittsburgh***Mathematics***University of Pittsburgh at Johnstown***Mathematics***University of South Carolina Beaufort***Department of Computer Science and Mathematics***University of Texas at Arlington***Department of Mathematics***University of the Virgin Islands***Department of Mathematical Sciences***University of Virginia***Mathematics***Utah Valley University***Department of Mathematics***Valparaiso University***Mathematics and Statistics***Vanderbilt University***Department of Mathematics***Virginia Commonwealth University***Mathematics and Applied Mathematics***Wake Forest University***Mathematics and Statistics***Walla Walla University***Mathematics***Washington & Jefferson College***Department of Mathematics***Wesleyan University***Mathematics and Computer Science***Whitworth University***Mathematics and Computer Science***Four-Year Statistics Respondents****Baruch College, City University of New York***Paul H. Chook Department of Information Systems and Statistics***Brigham Young University***Statistics***California Polytechnic State University,  
San Luis Obispo (Cal Poly, SLO)***Statistics Department***California State University East Bay***Statistics and Biostatistics***Columbia University***Statistics***Cornell University***Statistics and Data Science***Duke University***Statistical Science***Florida State University***Statistics*

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|                                                                                                              |                                                                                                 |
|--------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| <b>George Mason University</b><br><i>Statistics</i>                                                          | <b>University of California, Santa Barbara</b><br><i>Statistics and Applied Probability</i>     |
| <b>George Washington University</b><br><i>Statistics</i>                                                     | <b>University of Central Florida</b><br><i>Statistics and Data Science</i>                      |
| <b>Indiana University</b><br><i>Statistics</i>                                                               | <b>University of Connecticut</b><br><i>Statistics</i>                                           |
| <b>Iowa State University</b><br><i>Department of Statistics</i>                                              | <b>University of Florida</b><br><i>Statistics</i>                                               |
| <b>Kennesaw State University</b><br><i>School of Data Science and Analytics</i>                              | <b>University of Georgia</b><br><i>Department of Statistics</i>                                 |
| <b>Louisiana State University</b><br><i>Experimental Statistics</i>                                          | <b>University of Michigan</b><br><i>Department of Statistics</i>                                |
| <b>Michigan State University</b><br><i>Statistics and Probability</i>                                        | <b>University of Minnesota</b><br><i>School of Statistics</i>                                   |
| <b>North Carolina State University</b><br><i>Statistics</i>                                                  | <b>University of Missouri</b><br><i>Statistics</i>                                              |
| <b>Rutgers University New Brunswick</b><br><i>Statistics</i>                                                 | <b>University of Nebraska-Lincoln</b><br><i>Department of Statistics</i>                        |
| <b>Southern Methodist University</b><br><i>Department of Statistical Science</i>                             | <b>University of North Carolina at Chapel Hill</b><br><i>Statistics and Operations Research</i> |
| <b>Temple University</b><br><i>Department of Statistics, Operations, and Data Science</i>                    | <b>University of Pennsylvania</b><br><i>Statistics and Data Science</i>                         |
| <b>Texas A&amp;M University</b><br><i>Department of Statistics</i>                                           | <b>University of South Carolina</b><br><i>Statistics</i>                                        |
| <b>The Ohio State University</b><br><i>Department of Statistics</i>                                          | <b>University of Virginia</b><br><i>Statistics</i>                                              |
| <b>The Pennsylvania State University</b><br><i>Department of Statistics</i>                                  | <b>University of Washington</b><br><i>Department of Statistics</i>                              |
| <b>University of Akron</b><br><i>Statistics</i>                                                              | <b>University of Wisconsin–Madison</b><br><i>Statistics</i>                                     |
| <b>University of Alabama</b><br><i>Department of Information Systems, Statistics, and Management Science</i> | <b>University of Wyoming</b><br><i>Department of Mathematics and Statistics</i>                 |
| <b>University of Arizona</b><br><i>Mathematics</i>                                                           | <b>Virginia Commonwealth University</b><br><i>Statistical Sciences and Operations Research</i>  |
| <b>University of California, Davis</b><br><i>Mathematics</i>                                                 | <b>Virginia Tech</b><br><i>Statistics</i>                                                       |





# The Impact of COVID-19 on Undergraduate Mathematical Sciences Education: Report on a CBMS Survey

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March 2021

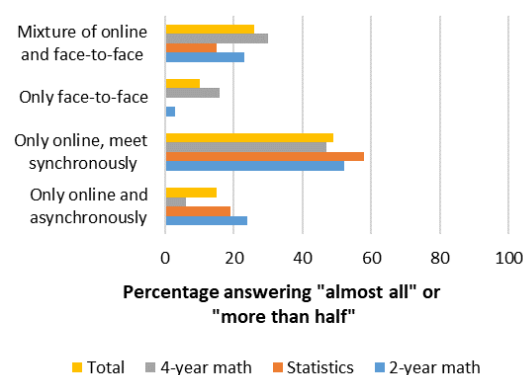
The following report is a summary of results from a survey of undergraduate mathematical and statistical sciences programs in two-year and four-year institutions in the US about how the COVID-19 pandemic has affected these programs, and what changes to future instruction might occur as a result of the experience. Administered on behalf of the Conference Board of the Mathematical Sciences (CBMS) in October and November 2020 by the American Mathematical Society in conjunction with Westat, Inc. and with funding from the National Science Foundation, the survey consists of six multiple-choice questions, a request for enrollments, and two free response questions. The survey instrument is located at <http://www.ams.org/profession/data/cbms-survey/cbms2020>, where response data is also available, broken down by department type, highest degree offered, institutional size, and institutional control (public/private).

In this report, the headings generally indicate take-away messages from the responses, and they generally follow the sequence of the survey instrument questions. Within the headings, the discussion, figures, and tables summarize or highlight the authors' findings. Information about methodology, response rates, and other statistical matters appear in the appendices along with tables giving fuller breakdowns on responses.

## "Online synchronous" was most commonly used format

The first question on the survey was designed to get at departments' choice of instructional delivery method under pandemic conditions. Figure 1 shows how various types of departments addressed class formats through the question, "Based on your current plans for the fall 2020 term, what proportion of your department's mathematical sciences sections are taught in the following formats?"

Figure 1. Distributions of class format, by department level



NOTE: Department chairs could choose more than one response. Percentages may add to more than 100.

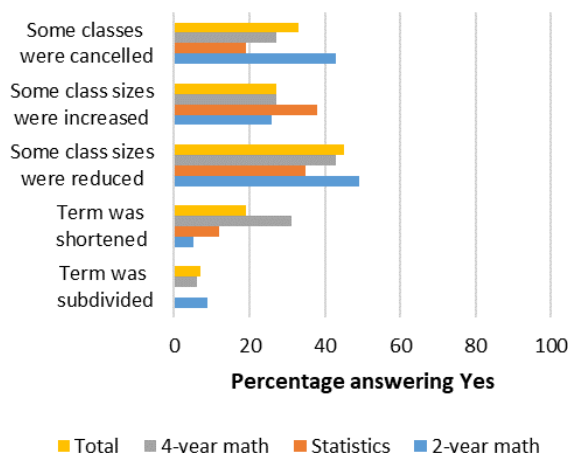


- Considering the total of all respondents and combining the two categories of “almost all” and “more than half of courses”, the format “only online synchronous” was the most frequently used option (50%), followed by “a mixture of online and face-to-face sessions” (26%; Figure 1).
- “Only online and asynchronous” and “only online synchronous” together accounted for the responses from 53% of the mathematics departments at four-year institutions, 77% of the statistics departments, and 76% of the mathematics departments at two-year colleges.
- The least used format in all the departments combined was “only face-to-face” (10%).
- There was more frequent use of the “only online only asynchronous” format at two-year colleges (24%) and in statistics departments (19%) than at four-year mathematics departments (7%).
- Private four-year mathematics departments reported offering “almost all” or “more than half” of their courses in “face-to-face” or a “mixture” format (60%) than departments at public universities (30%).

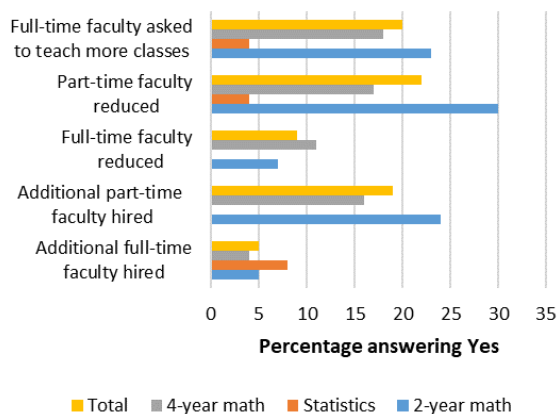
### Schedule and staff changes: Class sizes reduced, part-timers released, and full-timers teach more

In light of pandemic conditions, many departments may have changed their fall 2020 term length, number of sections offered, enrollment limits, and assignments to instructional staff, and the second question on the survey was designed to elicit information about these changes. Figure 2a gives insight to schedule changes that departments made. The percentages shown are of “yes” responses to the indicated changes. Note that these numbers are not meant to sum together. Figure 2b also shows “yes” responses to statements regarding staffing changes. The results indicate that the most frequently utilized personnel changes—except in the statistics group—were reductions in part-time faculty numbers and more sections assigned to full-time faculty. Table 2 in the Appendix gives further breakdowns for these two figures.

**Figure 2a. Changes in course schedule and size limits, by department type**



**Figure 2b. Instructional personnel changes, by department type**



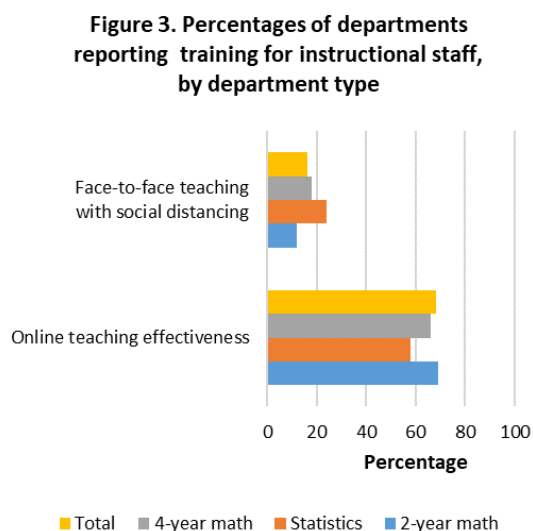
- Among the total of all responses, terms were subdivided at few (7%) departments, and the terms were shortened at 19% of departments (31% of four-year mathematics departments).
- Thirty-three percent of all departments cancelled some classes. Classes were more likely to be cancelled at two-year college mathematics departments (43%).



- Changes in the number of faculty hired or fired were most likely for part-time faculty (Figure 2b).
- Full-time faculty were asked to teach additional classes at 20% of all departments.
- The responses to these questions were quite similar for four-year and two-year college mathematics departments. In statistics departments there were few departments reporting changes in numbers of faculty hired, fired, or full-time faculty asked to teach additional courses.

### Training for teaching in the pandemic: Emphasis on online over in-person

Approximately two-thirds of respondents indicated that “more than half” or “almost all” of their staff received training in online instruction, and much smaller numbers cited similar training for in-person/socially-distanced teaching. Figure 3 shows these percentages across the three groups of departments.



- At all departments combined “almost all” plus “more than half” of their faculty received training in online instruction at 68% of departments and training in face-to-face instruction with social distancing at 16% of departments (Figure 3).

- The percentages in the table above were relatively consistent over all three types of institutions. More mathematics departments at four-year private colleges and universities provided instruction than at public institutions.

### Face-to-face is better, and not because students are ill-equipped

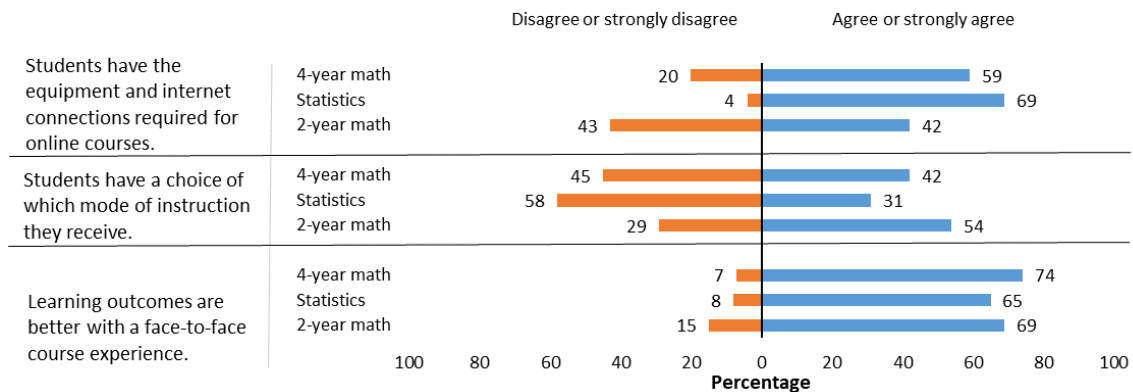
Department respondents voiced opinions on the effectiveness of face-to-face instruction, students’ ability to choose their mode of instruction, and students’ equipment. Figure 4a gives a breakdown of responses on these questions across the three main department types. Overwhelmingly, respondents felt that face-to-face outcomes are better. Their responses showed a mix of opinion regarding whether students have a choice of modality. While mathematics and statistics respondents felt that students were adequately equipped for online learning, there was disagreement among two-year respondents about how well students were equipped.

“Students’ learning outcomes from a face-to-face learning experience are better in an online experience”: 72% of all respondents agreed (i.e., “agreed” or “strongly agreed”), and this response was relatively consistent over the three types of institutions (Figure 4a).

- “Students have a choice of which mode of instruction they receive”: 47% of all respondents agreed and 38% disagreed, but this percentage was different across different types of institutions.
  - In four-year college mathematics departments, the percentage of those “(strongly) agreeing” was about the same as the percentage as those “(strongly) disagreeing”.
  - A larger percentage of statistics departments “(strongly) disagreed” than “(strongly) agreed”.
  - More two-year college mathematics departments “(strongly) agreed” than “(strongly) disagreed”.



Figure 4a. Percentages of department with various views on student experiences



- “Students taking courses online have the equipment and internet connections required for taking courses online”: Across the total of all respondents, 52% (strongly) agreed and 30% (strongly) disagreed.
  - In two-year college mathematics departments, there were about the same percentage of departments (strongly) agreeing (42%) as departments (strongly) disagreeing (43%).
  - In four-year college mathematics departments, more departments (strongly) agreed (59%) than (strongly) disagreed (20%).
  - In statistics departments, almost all departments (strongly) agreed (69%) as opposed to those that (strongly) disagreed (4%).

### Instructional staff have adequate technology, prefer to teach face-to-face, and have a choice of teaching mode

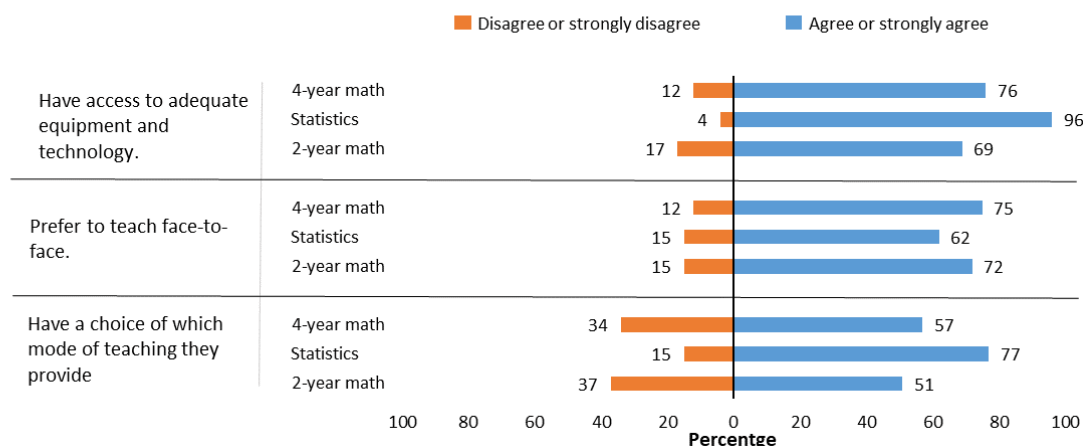
Figure 4b provides insight to faculty experiences teaching under pandemic conditions. Very broadly, respondents felt that instructional staff (1) have access

to adequate equipment for online teaching, (2) prefer face-to-face teaching, and (3) have choices about teaching modality.

- “Instructional staff teaching online have access to adequate equipment and technology for teaching online”: Ninety-six percent of statistics departments, 76% of four-year college mathematics departments, and 69% of two-year college mathematics departments (strongly) agreed.
- “Instructional staff prefer to teach face-to-face classes”: Across all respondents combined 73% (strongly) agreed and 13% (strongly) disagreed, and this percentage was relatively consistent across all three types of departments.
- “Instructional staff have a choice of which mode of teaching they provide”: Across all respondents combined 55% (strongly) agreed and 35% (strongly) disagreed. These percentages were about the same over four-year and two-year college mathematics departments, with the percentages in statistics departments showing more agreement (75% (strongly) agreeing and 15% (strongly) disagreeing). When data from



Figure 4b. Percentages of department with various views on faculty experiences



Tables 1 and 4 are combined, we see that at departments where almost all sections were in a particular format, there was less agreement with the statement that the instructor could choose their mode of instruction because choice was not possible.

### Training for online teaching increased dramatically with the pandemic

Figure 5 shows a dramatic shift in preparedness to teach online between the winter/spring 2020 and fall 2020 terms. Chairs were asked the question, “During the terms listed below, what proportion of your department’s instructional staff were/are adequately prepared to teach online?” and by fall, it was quite rare for instructors to be unprepared for online teaching.

- At each of the three types of department, the largest percentage was “less than half” of the faculty were adequately prepared to teach online pre-pandemic. Faculty were better prepared pre-pandemic at two-year college mathematics departments than at statistics departments or at

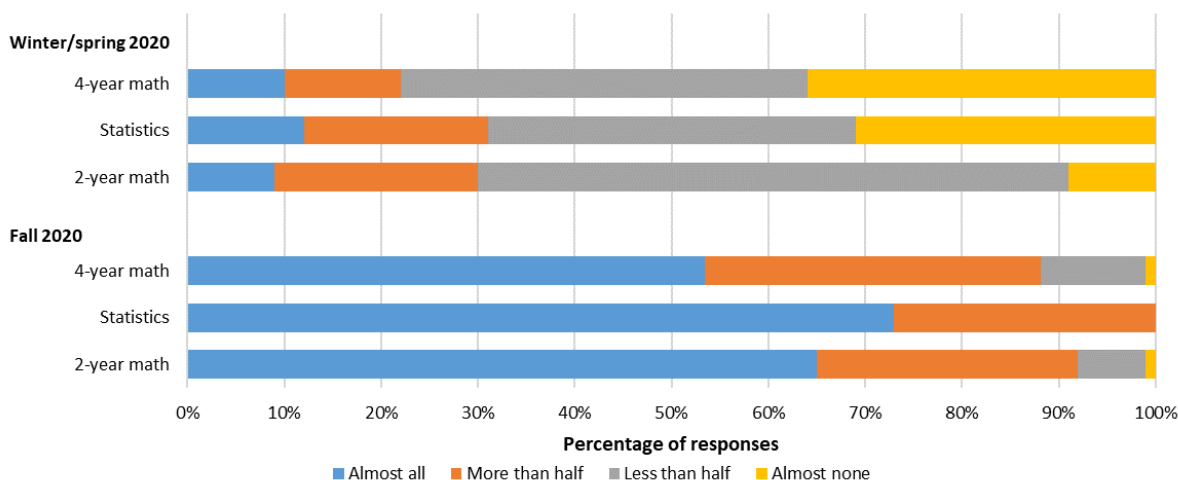
four-year college mathematics departments, as the percentage of “almost none” were adequately prepared was 9% at two-year college mathematics departments, 31% at statistics departments, and 36% at four-year college mathematics departments.

- Twenty-five percent of all departments combined had “almost all” or “more than half” of their faculty adequately prepared pre-pandemic. Ninety percent of all departments had “almost all” or “more than half” of their faculty adequately prepared after the pandemic—a dramatic change.
- The percentage of departments with “almost all” or “more than half” of their faculty adequately prepared in fall 2020 was relatively consistent across all types of institutions. In fall 2020 statistics departments had the largest percentage of “almost all” adequately prepared to teach online (73%), followed by two-year college mathematics departments (65%), and then by four-year college mathematics departments (54%).





Figure 5. Comparison of instructor preparedness for online teaching between winter/spring 2020 and fall 2020



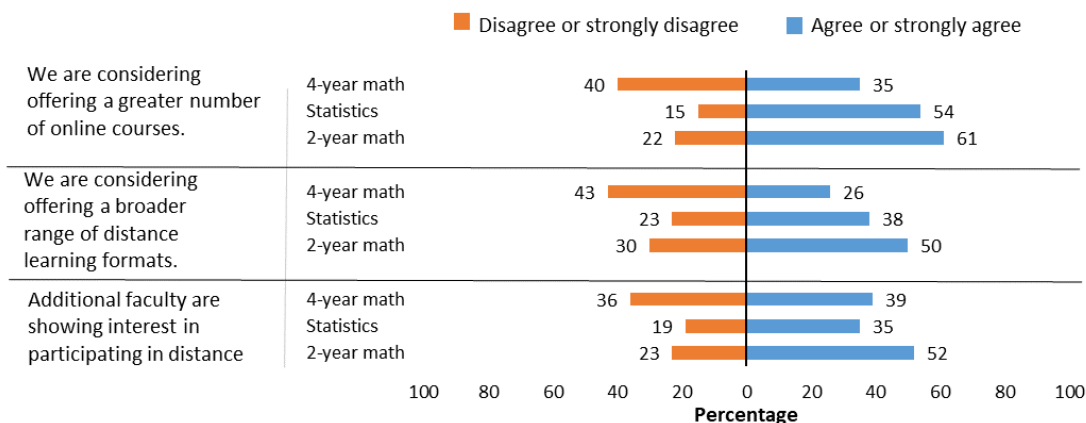
## Two-year departments most interested in expanded online offerings

Faculty and leaders in departments in two-year institutions overall show the strongest interest in more and broadened online course offerings, and smaller four-year mathematics and private institutions showed the least interest in these sorts of future changes. Overall, though, a majority of departments believe that more faculty are interested in teaching online courses and possibly more sections of online courses will be offered. Figure 6 summarizes.

- “We are considering offering a greater number of distance learning mathematical sciences classes.” Combining all respondents, 47% (strongly) agreed while 31% (strongly) disagreed. The percentage of two-year college mathematics departments (strongly) agreeing was 61%, while at four-year mathematics departments the percentage was 35%.
- Private mathematics departments at four-year colleges had the lowest (strongly) agreed percentage of departments considering offering more online courses in the future (25%).
- “We are considering offering a broader range of distance learning formats in mathematical sciences classes (e.g., more types of mathematics courses).” Overall, 37% of respondents (strongly) agreed and 37% (strongly) disagreed. Across all types of departments, a larger percentage (strongly) disagreed than (strongly) agreed, except for statistics departments and two-year college mathematics departments.
- “Additional faculty are showing interest in participating in distance learning.” Across all respondents 45% (strongly) agreed and 30% (strongly) disagreed, with 52% of two-year college mathematics departments (strongly) agreeing. The percentage of departments (strongly) disagreeing was greater at smaller four-year college mathematics departments and at private four-year college mathematics departments.



Figure 6: Departments' future course-delivery planning



### Falls 2019 to 2020, a third of departments experienced small enrollment changes, but almost as many had decreases of more than 10%

Departments were asked, “What are the total Fall enrollments in mathematics and statistics courses in your department for 2019 and 2020? If your Fall term has been split into shorter blocks, combine enrollments for all of the blocks in the term.” Overall, more departments reported essentially no change in enrollment between the two fall terms than any other size change. That said, nearly a third of departments reported enrollment decreases of more than 10%. The department group with the largest percentage citing enrollment increase was statistics. Figure 7 provides details.

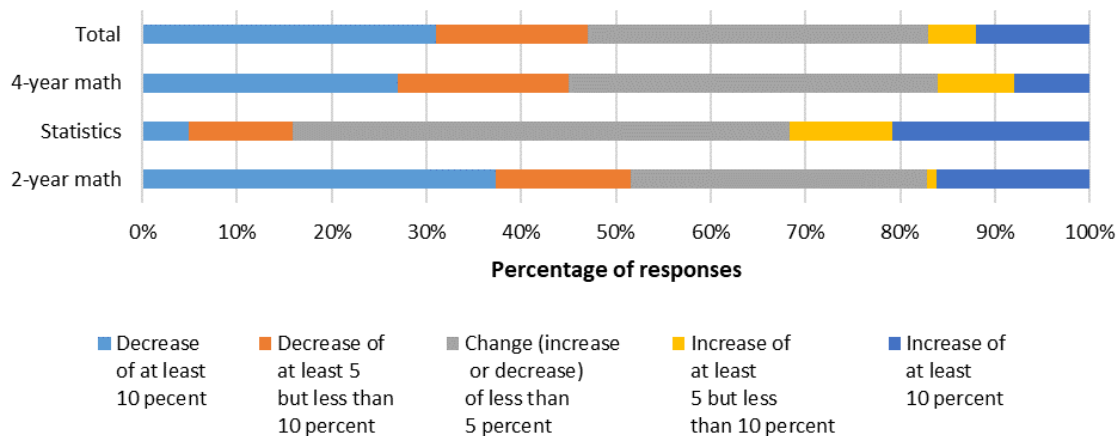
- The three types of departments each reported the largest percentage of departments had some decrease in enrollment (i.e., both categories of decrease added together). Generally, the next largest percentage of departments had little change in enrollment, and the lowest percentage of departments had some increase in enrollment (both categories of increase added together). Over

all departments combined these percentages were 31% of all departments reported a decrease at least 10%, 16% a decrease of 5 – 10%, 26% less than 5% change (either increase or decrease), 5% an increase of 5 – 10%, and 12% an increase of at least 10%.

- Statistics was the only type of department reporting a larger percentage of departments with some increase (both categories of increase added together) than some decrease (both categories of decrease added together).
- In most of the types of departments, the largest single category was a change of under 5% (i.e., little change) in enrollment—except at two-year college mathematics departments (where the largest percentage is a decrease of at least 10% in enrollment).
- Statistics departments are the type of department with the highest percentage of departments (21%) reporting at least 10% increase in enrollment—the next largest percentage occurs at two-year college mathematics departments, where 16% of departments reported at least a 10% increase in enrollment.



**Figure 7: Percentages of departments with various changes in fall enrollments from 2019 to 2020**



### Challenges: Student engagement, assessment, and integrity; institutional and faculty inertia

The survey incorporated two free-response questions, designed respectively to get at the respondents' sense of key challenges and opportunities presented by pandemic conditions. First, the challenge question was, "What stands out to you as the greatest difficulty your department has faced with relation to the COVID-19 pandemic? Describe only one."

In the free responses to this question from two-year college mathematics departments, **student assessment** and **making connections with students** emerged as frequently-cited challenges.

In connection with assessment, respondents mentioned both academic integrity and faculty time and expertise with online testing:

- Increase number of academic dishonesty incidents on exams. "Rampant cheating online, inability to stop it." "Continual struggle how to ensure academic integrity of exams."

- Proctoring exams: faculty felt online test proctoring software was an equity issue or an invasion of student privacy.
- Grades not a reflection of student learning and grade inflation.
- "Determining how to effectively do testing."
- Additional time required by instructor to enforce showing all workspace during tests to "ensure students are not using cell phone online solvers."
- "Providing students with annotate corrections on tests and quizzes."

Regarding the theme of making connections to students, respondents addressed the motivation of students, engaging students, and employing active learning strategies:

- "Interactions between students and faculty have reduced in quality and quantity to a point where students feel that they have issues with comprehension, mentorship, and competitiveness." Faculty and students feel isolated.
- "Online communication is a poor substitute for in-person."



- “Students are choosing to join class remotely out of convenience. It is difficult to connect with these students and they have various distractions at home.”
- Keeping students engaged and working on new materials and assignments.
- Replicating active learning in a virtual environment and how to do group activities online.

Other difficulties reported by two-year college respondents included:

- Ensuring the integrity and quality of instruction.
- Lack of bandwidth and technology equipment. Students not technically prepared for online classes. “Access to adequate equipment. Most technology in faculty hands are centered around face-to-face classes.”
- Converting to/from face-to-face instruction to remote instruction.
- Training faculty.

Among respondents in four-year mathematics and statistics programs, a wide variety of difficulties were reported. Indeed, one response was: “that there are SO MANY challenges at once.”

- Students and faculty rapidly pivoting to new modalities (including online, face-to-face with distancing, and mixed) for which they were initially unprepared, for which they had little training and without established departmental norms.
- Maintaining quality instruction: “the median quality of teaching is lower, and the variation is bigger”. Replicating active learning, group work, and office hours in online courses.
- Institutions using “HyFlex” modalities, where students could choose their modalities, and accommodating quarantined students, forced faculty to teach in different modalities in the same section.
- Supervising TAs and inexperienced faculty.

- Maintaining the usual course content.
- Engaging students in the new modalities and helping students who were struggling.
- Designing appropriate assessments and problems with student cheating was a frequently mentioned greatest difficulty in online courses.
- Building and maintaining community: among faculty, among students, and between students and faculty were frequently mentioned greatest difficulties.
- Finding adequate classrooms for face-to-face classes, given social distancing requirements; the inability to use computer labs compromised instruction.
- The administration’s uncertain and changing plans, cuts in budgets, poor communication, and lack of faculty involvement in decision-making.
- Cuts in numbers of faculty and increasing teaching loads. The new modalities required more faculty time resulting in low morale and burn-out.
- Concern that other responsibilities of faculty such as research were compromised
- A lack of equipment and technical support for faculty. Remote students having inadequate equipment, internet access, and good working environments; accommodating students abroad in different time zone.
- Addressing student and parent complaints.
- Pressure to offer face-to-face courses, and faculty reluctance to teach face-to-face.

### Opportunities: Learning new teaching methods and styles, greater faculty sympathy for students, and more inclusiveness

The second free-response question was, “What is the greatest benefit, if any, that you see as arising because of the COVID-19 pandemic? Describe only one.” Among responses from two-year college department chairpersons, the chief benefit cited was the



**opportunity to learn new methods and styles of teaching, learn new technology, and engage in professional development.**

- “Faculty have learned online teaching strategies and technology that will help them in their future classes.”
- “Potential development of web sections for courses that were not under consideration before.”
- “We are learning to use tools that we've had access to for a long time but sat dormant for lack of perceived need.”
- “Instructors have learned many things; using Zoom, using drawing tablets, e.g., to help the learning environment in different situations.”
- “More students and instructors were educated on the use of technology.”
- “More instructors are aware of technology that can be used to enhance their traditional classes.”

Two-year college chairs also mentioned the following:

- Faculty became more creative about instructional delivery.
- Meetings were streamlined and more convenient.
- Faculty realization that students are able to learn mathematics remotely.
- “More students are realizing they can succeed in online sections.”

Almost ten percent of respondents felt there was **no benefit**:

- “None. It has been an excruciating and problematic transition with no upside.”
- “None - too many changes and knee jerk reactions, enrollment down 13%.”

The following list is illustrative of the benefits cited by four-year mathematics and statistics program respondents:

- Faculty and students now are better equipped to teach and learn remotely (e.g., instead of cancelling class on hurricane or snow days classes might continue remotely, some office hours might

occur online), and some departments might offer online courses in the future.

- Faculty are now familiar with new technologies (e.g., making videos, using learning management systems, having students submit assignments electronically) that will be used in the return to normal instruction.
- Faculty found teaching techniques such as “flipped” classes, new assessments, additional materials for students, and new ways of capturing student attention (e.g., putting course content into modules) effective. These techniques are likely to be used in normal times.
- Faculty are now more open to trying new teaching methods and to rethinking course content.
- Learning Centers provided remote tutoring that may continue and serve more students.
- Faculty have become more sympathetic to the problems of students and colleagues, which may help inclusiveness and response to under-represented groups.
- Faculty discovered online instruction has some advantages over face-to-face instruction, including breakout groups that worked better than some face-to-face group work settings, that some students were more willing to write questions in the Zoom chat than to ask a question in class, that online classes solved some space problems on campus, that online courses may be more convenient for non-traditional students, and they can provide students from around the world expanding enrollments.
- Teaching remotely saved time that was usually lost commuting.
- Recordings of class sessions provided students the ability to watch class sessions again.
- Students have learned skills such as scanning and turning in assignments online.
- Some respondents stated that students spent more time on their classes due to lack of conflicting activities.



- Some chairs stated that videos of class sessions provided good ways of assessing teaching.
- Some departments were able to host and to participate in more seminars and colloquia with remote speakers, and faculty found it easier and cheaper to participate in some conferences.
- Some departments found enrollments increased due to the greater availability of courses, and that it was easier to find adjunct instructors for classes that were taught remotely.
- Technology resources in some departments increased because of the pandemic.

Virtual meetings were seen by some as more efficient and easier to schedule than face-to-face meetings, and virtual honor ceremonies and teas allowed alumni, donors, and parents to participate.

Similar to the two-year group, eight percent of four-year departments responded that there was no benefit. To some departments the pandemic provided confirmation that remote instruction is not effective, and it helped them understand the limitations of online instruction better. The pandemic experience made students and the public appreciate the privilege and value of face-to-face instruction. If nothing else, as one respondent noted, the pandemic has provided some good modeling problems.

#### CBMS surveys: Tracking the mathematical and statistical sciences in higher education since 1965

This survey was supported by the National Science Foundation under grant #DUE-1916764. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the authors and do not necessarily reflect the views of the National Science Foundation.

This COVID-focused survey has been possible through the flexibility of the NSF, which has been the underwriter of the CBMS Surveys of Undergraduate Mathematical Sciences Programs every five years since 1965. Like many regularly-occurring activities, the full 2020 Survey has been postponed to 2021, and this targeted COVID survey has been incorporated into the overall CBMS Survey project.

Examining programs at two- and four-year institutions, these national surveys are sponsored by the Conference Board of the Mathematical Sciences (CBMS), a consortium of nineteen professional associations. The project is administered by the American Mathematical Society, and survey reports can be downloaded from <http://www.ams.org/profession/data/cbms-survey/cbms-survey>. See <http://www.ams.org/profession/data/cbms-survey/cbms2020> for further information about the survey to be conducted in the fall of 2021.

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## Appendix 1: Methodology

Developed by the authors in consultation with Westat, Inc. and the CBMS Survey Steering Committee (listed below), this survey was sent to chairpersons of 855 two-year mathematics programs, 85 statistics programs, and 1,342 mathematics programs in the US under the management of Westat. These effectively constituted a census of four-year mathematics and statistics programs, and a random sample of two-year programs. For the purpose of this survey, a two-year mathematics program is one housed in an institution whose degrees awarded (undergraduate and graduate combined) consist of at least 51% associate recipients. Generally, the categories of level, institutional size, highest degree offered, and control were obtained from the Integrated Postsecondary Education Data System (IPEDS) operated

by the US Education Department's National Center for Education Statistics (NCES).

Usable responses were received from 81 two-year mathematics, 26 statistics, and 402 four-year mathematics programs. The data were weighted to adjust for the probability of selection and nonresponse. For consistency, statistics in this report are designed to match the appendix tables. When two different responses are summed, however, the actual statistics sometimes differ slightly because of rounding and other issues (such as missing data). The following table provides illustrative standard errors and confidence intervals for the survey statistics.

A special word of thanks goes to Dr. Brad Chaney of Westat. His insight and attention to detail have been invaluable in this COVID study and in the larger CBMS Survey project.

| CBMS Survey Steering Committee                                        |
|-----------------------------------------------------------------------|
| Ed Aboufadel, Grand Valley State University                           |
| Elizabeth Burroughs, Montana State University                         |
| Beth Chance, California Polytechnic State University, San Luis Obispo |
| Ted Coe, Northwest Evaluation Association                             |
| T. J. Hitchman, University of Northern Iowa                           |
| Dennis Pearl, Penn State University                                   |
| Emily Puckette, Sewanee: The University of the South                  |
| April Strom, Maricopa Community Colleges                              |
| Linda Zientek, Sam Houston State University                           |

| Sections are taught only online, with an opportunity to meet synchronously online (Q1b) |          |                           |                                   |         |
|-----------------------------------------------------------------------------------------|----------|---------------------------|-----------------------------------|---------|
| College type                                                                            | Percent  | Standard Error of Percent | 95% Confidence Limits for Percent |         |
| <b>Two-year colleges</b>                                                                |          |                           |                                   |         |
| Almost all                                                                              | 22.2063  | 4.7267                    | 12.8857                           | 31.5269 |
| More than half                                                                          | 29.4794  | 5.4213                    | 18.7892                           | 40.1696 |
| Less than half                                                                          | 48.3143  | 6.0481                    | 36.3881                           | 60.2404 |
| Total                                                                                   | 100.0000 |                           |                                   |         |
| <b>Statistics</b>                                                                       |          |                           |                                   |         |
| Almost all                                                                              | 23.0769  | 8.5427                    | 6.0918                            | 40.0621 |
| More than half                                                                          | 34.6154  | 9.6460                    | 15.4365                           | 53.7943 |
| Less than half                                                                          | 42.3077  | 10.0172                   | 22.3909                           | 62.2245 |
| Total                                                                                   | 100.0000 |                           |                                   |         |
| <b>Four-year colleges</b>                                                               |          |                           |                                   |         |
| Almost all                                                                              | 23.1738  | 2.2736                    | 18.6905                           | 27.6571 |
| More than half                                                                          | 23.9295  | 2.2562                    | 19.4805                           | 28.3784 |
| Less than half                                                                          | 52.8967  | 2.5279                    | 47.9119                           | 57.8815 |



## Appendix 2: Tables

| Table 1: Percentages of departments answering "more than half" or "almost all" to the question, "Based on your current plans for the fall 2020 term, what proportion of your department's mathematical sciences sections are taught in the following formats?" |                                |                                 |                   |                                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|---------------------------------|-------------------|------------------------------------|
|                                                                                                                                                                                                                                                                | Only online and asynchronously | Only online, meet synchronously | Only face-to-face | Mixture of online and face-to-face |
| <b>Total</b>                                                                                                                                                                                                                                                   | 15                             | 49                              | 10                | 26                                 |
| <b>4-year math</b>                                                                                                                                                                                                                                             | 6                              | 47                              | 16                | 30                                 |
| <b>Statistics</b>                                                                                                                                                                                                                                              | 19                             | 58                              | .                 | 15                                 |
| <b>2-year math</b>                                                                                                                                                                                                                                             | 24                             | 52                              | 3                 | 23                                 |
| <b>4-year math departments only</b>                                                                                                                                                                                                                            |                                |                                 |                   |                                    |
| <b>Highest degree offered</b>                                                                                                                                                                                                                                  |                                |                                 |                   |                                    |
| <b>Doctoral</b>                                                                                                                                                                                                                                                | 7                              | 52                              | 11                | 27                                 |
| <b>Master's</b>                                                                                                                                                                                                                                                | 7                              | 42                              | 24                | 31                                 |
| <b>Baccalaureate</b>                                                                                                                                                                                                                                           | 4                              | 38                              | 21                | 39                                 |
| <b>Institution size</b>                                                                                                                                                                                                                                        |                                |                                 |                   |                                    |
| <b>Under 5,000</b>                                                                                                                                                                                                                                             | 3                              | 34                              | 26                | 35                                 |
| <b>5,000 - 9,999</b>                                                                                                                                                                                                                                           | 10                             | 66                              | 8                 | 22                                 |
| <b>10,000 - 19,999</b>                                                                                                                                                                                                                                         | 10                             | 41                              | 7                 | 34                                 |
| <b>20,000 or more</b>                                                                                                                                                                                                                                          | 9                              | 75                              | 1                 | 16                                 |
| <b>Control of institution</b>                                                                                                                                                                                                                                  |                                |                                 |                   |                                    |
| <b>Public</b>                                                                                                                                                                                                                                                  | 10                             | 56                              | 7                 | 23                                 |
| <b>Private</b>                                                                                                                                                                                                                                                 | 3                              | 39                              | 24                | 36                                 |





Table 2. Percentages of departments answering "yes" to the question, "Which of these changes have occurred in your department in response to the COVID-19 pandemic?" Note that these percentages do not constitute distributions.

|                              | Term was subdivided | Term was shortened | Some class sizes were reduced | Some class sizes were increased | Some classes were cancelled | Additional full-time faculty hired | Additional part-time faculty hired | Full-time faculty reduced | Part-time faculty reduced | Full-time faculty asked to teach more classes |
|------------------------------|---------------------|--------------------|-------------------------------|---------------------------------|-----------------------------|------------------------------------|------------------------------------|---------------------------|---------------------------|-----------------------------------------------|
| Total                        | 7                   | 19                 | 45                            | 27                              | 33                          | 5                                  | 19                                 | 9                         | 22                        | 20                                            |
| 4-year math                  | 6                   | 31                 | 43                            | 27                              | 27                          | 4                                  | 16                                 | 11                        | 17                        | 18                                            |
| Statistics                   | .                   | 12                 | 35                            | 38                              | 19                          | 8                                  | .                                  | .                         | 4                         | 4                                             |
| 2-year math                  | 9                   | 5                  | 49                            | 26                              | 43                          | 5                                  | 24                                 | 7                         | 30                        | 23                                            |
| 4-year math departments only |                     |                    |                               |                                 |                             |                                    |                                    |                           |                           |                                               |
| Highest degree offered       |                     |                    |                               |                                 |                             |                                    |                                    |                           |                           |                                               |
| Doctoral                     | 4                   | 26                 | 36                            | 34                              | 28                          | 6                                  | 18                                 | 11                        | 18                        | 19                                            |
| Master's                     | 4                   | 36                 | 54                            | 16                              | 23                          | 2                                  | 9                                  | 11                        | 14                        | 16                                            |
| Baccalaureate                | 8                   | 13                 | 61                            | 26                              | 36                          | 1                                  | 12                                 | 10                        | 22                        | 10                                            |
| Institution size category    |                     |                    |                               |                                 |                             |                                    |                                    |                           |                           |                                               |
| Under 5,000                  | 7                   | 37                 | 52                            | 20                              | 23                          | 2                                  | 12                                 | 10                        | 13                        | 17                                            |
| 5,000 - 9,999                | 5                   | 29                 | 27                            | 26                              | 34                          | 3                                  | 13                                 | 13                        | 19                        | 19                                            |
| 10,000 - 19,999              | 3                   | 29                 | 43                            | 32                              | 32                          | 5                                  | 19                                 | 19                        | 25                        | 32                                            |
| 20,000 or more               | 3                   | 18                 | 28                            | 47                              | 25                          | 10                                 | 30                                 | 9                         | 21                        | 9                                             |
| Control of institution       |                     |                    |                               |                                 |                             |                                    |                                    |                           |                           |                                               |
| Public                       | 3                   | 28                 | 40                            | 36                              | 29                          | 6                                  | 19                                 | 14                        | 23                        | 21                                            |
| Private                      | 8                   | 35                 | 45                            | 20                              | 25                          | 3                                  | 13                                 | 9                         | 12                        | 16                                            |



**Table 3. Percentages of departments responding "almost all" or "more than half" to the question, "During the summer and/or fall of 2020, what proportion of instructional staff in your department received training in the following?"**

|                                     | Online teaching effectiveness | Face-to-face teaching with social distancing |
|-------------------------------------|-------------------------------|----------------------------------------------|
| <b>Total</b>                        | 68                            | 16                                           |
| <b>4-year math</b>                  | 66                            | 18                                           |
| <b>Statistics</b>                   | 58                            | 24                                           |
| <b>2-year math</b>                  | 70                            | 12                                           |
| <b>4-year math departments only</b> |                               |                                              |
| <b>Highest degree offered</b>       |                               |                                              |
| <b>Doctoral</b>                     | 65                            | 18                                           |
| <b>Master's</b>                     | 64                            | 18                                           |
| <b>Baccalaureate</b>                | 80                            | 22                                           |
| <b>Institution size category</b>    |                               |                                              |
| <b>Under 5,000</b>                  | 70                            | 24                                           |
| <b>5,000 - 9,999</b>                | 61                            | 13                                           |
| <b>10,000 - 19,999</b>              | 57                            | 17                                           |
| <b>20,000 or more</b>               | 68                            | 8                                            |
| <b>Control of institution</b>       |                               |                                              |
| <b>Public</b>                       | 57                            | 9                                            |
| <b>Private</b>                      | 74                            | 27                                           |



**Table 4a. Percentages of departments' responses to the question, "Please state how strongly you personally agree or disagree with the following statements about your department's experiences and plans relating to the COVID-19 pandemic in fall 2020."**

|                                                                                 | Learning outcomes are better with a face-to-face course experience |                               | Students have a choice of which mode of instruction they receive |                               | Students have the equipment and internet connections required for online courses |                               |
|---------------------------------------------------------------------------------|--------------------------------------------------------------------|-------------------------------|------------------------------------------------------------------|-------------------------------|----------------------------------------------------------------------------------|-------------------------------|
|                                                                                 | Strongly agree or agree                                            | Disagree or strongly disagree | Strongly agree or agree                                          | Disagree or strongly disagree | Strongly agree or agree                                                          | Disagree or strongly disagree |
| <b>Total</b>                                                                    | 72                                                                 | 11                            | 47                                                               | 38                            | 52                                                                               | 30                            |
| <b>level</b>                                                                    |                                                                    |                               |                                                                  |                               |                                                                                  |                               |
| 4-year math                                                                     | 74                                                                 | 7                             | 42                                                               | 45                            | 59                                                                               | 20                            |
| Statistics                                                                      | 65                                                                 | 8                             | 31                                                               | 58                            | 69                                                                               | 4                             |
| 2-year math                                                                     | 69                                                                 | 15                            | 54                                                               | 29                            | 42                                                                               | 43                            |
| <b>4-year math departments only</b>                                             |                                                                    |                               |                                                                  |                               |                                                                                  |                               |
| <b>Highest degree offered</b>                                                   |                                                                    |                               |                                                                  |                               |                                                                                  |                               |
| Doctoral                                                                        | 71                                                                 | 9                             | 42                                                               | 45                            | 56                                                                               | 20                            |
| Master's                                                                        | 76                                                                 | 4                             | 43                                                               | 44                            | 60                                                                               | 23                            |
| Baccalaureate                                                                   | 82                                                                 | 4                             | 41                                                               | 43                            | 67                                                                               | 12                            |
| <b>Institution size category</b>                                                |                                                                    |                               |                                                                  |                               |                                                                                  |                               |
| Under 5,000                                                                     | 78                                                                 | 6                             | 41                                                               | 45                            | 58                                                                               | 23                            |
| 5,000 - 9,999                                                                   | 68                                                                 | 8                             | 41                                                               | 48                            | 52                                                                               | 23                            |
| 10,000 - 19,999                                                                 | 69                                                                 | 11                            | 52                                                               | 41                            | 66                                                                               | 15                            |
| 20,000 or more                                                                  | 72                                                                 | 7                             | 37                                                               | 46                            | 62                                                                               | 15                            |
| <b>Control of institution</b>                                                   |                                                                    |                               |                                                                  |                               |                                                                                  |                               |
| Public                                                                          | 69                                                                 | 9                             | 46                                                               | 42                            | 61                                                                               | 19                            |
| Private                                                                         | 78                                                                 | 6                             | 38                                                               | 47                            | 56                                                                               | 21                            |
| Note: the percentages responding "undecided" or "not applicable" are not shown. |                                                                    |                               |                                                                  |                               |                                                                                  |                               |



| Table 4b. Percentages of departments' responses to the question, "Please state how strongly you personally agree or disagree with the following statements about your department's experiences and plans relating to the COVID-19 pandemic in fall 2020." |                                                                      |                               |                              |                               |                                                      |                               |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|-------------------------------|------------------------------|-------------------------------|------------------------------------------------------|-------------------------------|
|                                                                                                                                                                                                                                                           | Have access to adequate equipment and technology for teaching online |                               | Prefer to teach face-to-face |                               | Have a choice of which mode of teaching they provide |                               |
|                                                                                                                                                                                                                                                           | Strongly agree or agree                                              | Disagree or strongly disagree | Strongly agree or agree      | Disagree or strongly disagree | Strongly agree or agree                              | Disagree or strongly disagree |
| <b>Total</b>                                                                                                                                                                                                                                              | 74                                                                   | 14                            | 73                           | 13                            | 55                                                   | 35                            |
| <b>level</b>                                                                                                                                                                                                                                              |                                                                      |                               |                              |                               |                                                      |                               |
| 4-year math                                                                                                                                                                                                                                               | 76                                                                   | 12                            | 75                           | 12                            | 57                                                   | 34                            |
| Statistics                                                                                                                                                                                                                                                | 96                                                                   | 4                             | 62                           | 15                            | 77                                                   | 15                            |
| 2-year math                                                                                                                                                                                                                                               | 69                                                                   | 17                            | 72                           | 15                            | 51                                                   | 37                            |
| <b>4-year math departments only</b>                                                                                                                                                                                                                       |                                                                      |                               |                              |                               |                                                      |                               |
| <b>Highest degree offered</b>                                                                                                                                                                                                                             |                                                                      |                               |                              |                               |                                                      |                               |
| Doctoral                                                                                                                                                                                                                                                  | 78                                                                   | 13                            | 71                           | 14                            | 55                                                   | 37                            |
| Master's                                                                                                                                                                                                                                                  | 71                                                                   | 14                            | 82                           | 11                            | 59                                                   | 33                            |
| Baccalaureate                                                                                                                                                                                                                                             | 82                                                                   | 8                             | 80                           | 4                             | 65                                                   | 20                            |
| <b>Institution size category</b>                                                                                                                                                                                                                          |                                                                      |                               |                              |                               |                                                      |                               |
| Under 5,000                                                                                                                                                                                                                                               | 70                                                                   | 16                            | 81                           | 8                             | 58                                                   | 33                            |
| 5,000 - 9,999                                                                                                                                                                                                                                             | 77                                                                   | 13                            | 69                           | 16                            | 66                                                   | 29                            |
| 10,000 - 19,999                                                                                                                                                                                                                                           | 80                                                                   | 10                            | 69                           | 15                            | 57                                                   | 36                            |
| 20,000 or more                                                                                                                                                                                                                                            | 91                                                                   | 4                             | 69                           | 16                            | 47                                                   | 38                            |
| <b>Control of institution</b>                                                                                                                                                                                                                             |                                                                      |                               |                              |                               |                                                      |                               |
| Public                                                                                                                                                                                                                                                    | 82                                                                   | 10                            | 73                           | 11                            | 59                                                   | 33                            |
| Private                                                                                                                                                                                                                                                   | 71                                                                   | 15                            | 77                           | 12                            | 56                                                   | 34                            |
| Note: the percentages responding "undecided" or "not applicable" are not shown.                                                                                                                                                                           |                                                                      |                               |                              |                               |                                                      |                               |



**Table 5. Percentages of departments' responses to the question, "During the terms listed below, what proportion of your department's instructional staff were/are adequately prepared to teach online?"**

|                                     | Winter/spring 2020 |                |                |             | Fall 2020  |                |                |             |
|-------------------------------------|--------------------|----------------|----------------|-------------|------------|----------------|----------------|-------------|
|                                     | Almost all         | More than half | Less than half | Almost none | Almost all | More than half | Less than half | Almost none |
| <b>Total</b>                        | 9                  | 16             | 50             | 24          | 59         | 31             | 9              | 1           |
| <b>level</b>                        |                    |                |                |             |            |                |                |             |
| 4-year math                         | 10                 | 12             | 42             | 36          | 54         | 35             | 11             | 1           |
| Statistics                          | 12                 | 19             | 38             | 31          | 73         | 27             | .              | .           |
| 2-year math                         | 9                  | 21             | 61             | 9           | 65         | 27             | 7              | 1           |
| <b>4-year math departments only</b> |                    |                |                |             |            |                |                |             |
| <b>Highest degree offered</b>       |                    |                |                |             |            |                |                |             |
| Doctoral                            | 7                  | 14             | 42             | 37          | 54         | 34             | 12             | 0           |
| Master's                            | 14                 | 8              | 41             | 37          | 53         | 38             | 8              | 1           |
| Baccalaureate                       | 8                  | 12             | 45             | 35          | 58         | 33             | 8              | .           |
| <b>Institution size category</b>    |                    |                |                |             |            |                |                |             |
| Under 5,000                         | 13                 | 12             | 41             | 33          | 54         | 34             | 11             | 1           |
| 5,000 - 9,999                       | 5                  | 16             | 45             | 34          | 55         | 34             | 11             | .           |
| 10,000 - 19,999                     | 8                  | 10             | 34             | 48          | 50         | 33             | 17             | .           |
| 20,000 or more                      | 4                  | 9              | 49             | 38          | 57         | 38             | 4              | .           |
| <b>Control of institution</b>       |                    |                |                |             |            |                |                |             |
| Public                              | 8                  | 15             | 44             | 34          | 49         | 39             | 12             | .           |
| Private                             | 11                 | 10             | 41             | 38          | 59         | 31             | 9              | 1           |



**Table 6. Percentages of departments in agreement or disagreement with statements describing course delivery planning. The lead question was, "How has the COVID-19 experience influenced your department's plans for the future (after fall 2020)? Only report on changes that your department is considering because of your COVID-19 experience. Do not consider changes that you would have made anyway."**

|                                     | <b>We are considering offering a greater number of distance learning classes</b> |                                      | <b>We are considering offering a broader range of distance learning formats</b> |                                      | <b>Additional faculty are showing interest in participating in distance learning</b> |                                      |
|-------------------------------------|----------------------------------------------------------------------------------|--------------------------------------|---------------------------------------------------------------------------------|--------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------|
|                                     | <b>Strongly agree or agree</b>                                                   | <b>Disagree or strongly disagree</b> | <b>Strongly agree or agree</b>                                                  | <b>Disagree or strongly disagree</b> | <b>Strongly agree or agree</b>                                                       | <b>Disagree or strongly disagree</b> |
| <b>Total</b>                        | 47                                                                               | 31                                   | 37                                                                              | 37                                   | 45                                                                                   | 30                                   |
| <b>level</b>                        |                                                                                  |                                      |                                                                                 |                                      |                                                                                      |                                      |
| 4-year math                         | 35                                                                               | 40                                   | 26                                                                              | 43                                   | 39                                                                                   | 36                                   |
| Statistics                          | 54                                                                               | 15                                   | 38                                                                              | 23                                   | 35                                                                                   | 19                                   |
| 2-year math                         | 61                                                                               | 22                                   | 50                                                                              | 30                                   | 52                                                                                   | 23                                   |
| <b>4-year math departments only</b> |                                                                                  |                                      |                                                                                 |                                      |                                                                                      |                                      |
| <b>Highest degree offered</b>       |                                                                                  |                                      |                                                                                 |                                      |                                                                                      |                                      |
| Doctoral                            | 39                                                                               | 34                                   | 32                                                                              | 36                                   | 45                                                                                   | 28                                   |
| Master's                            | 34                                                                               | 44                                   | 19                                                                              | 50                                   | 36                                                                                   | 43                                   |
| Baccalaureate                       | 14                                                                               | 63                                   | 18                                                                              | 59                                   | 22                                                                                   | 55                                   |
| <b>Institution size category</b>    |                                                                                  |                                      |                                                                                 |                                      |                                                                                      |                                      |
| Under 5,000                         | 27                                                                               | 51                                   | 20                                                                              | 51                                   | 28                                                                                   | 49                                   |
| 5,000 - 9,999                       | 45                                                                               | 21                                   | 31                                                                              | 32                                   | 50                                                                                   | 23                                   |
| 10,000 - 19,999                     | 40                                                                               | 33                                   | 32                                                                              | 37                                   | 53                                                                                   | 18                                   |
| 20,000 or more                      | 46                                                                               | 31                                   | 37                                                                              | 35                                   | 51                                                                                   | 22                                   |
| <b>Control of institution</b>       |                                                                                  |                                      |                                                                                 |                                      |                                                                                      |                                      |
| Public                              | 46                                                                               | 28                                   | 38                                                                              | 31                                   | 52                                                                                   | 21                                   |
| Private                             | 25                                                                               | 51                                   | 17                                                                              | 53                                   | 28                                                                                   | 48                                   |



**Table 7. Percentage of departments experiencing various changes  
in fall enrollments from 2019 to 2020**

|                                     | Decrease<br>of at least<br>10 percent | Decrease of<br>at least 5<br>but less<br>than<br>10 percent | Change<br>(increase<br>or<br>decrease)<br>of less than<br>5 percent | Increase of<br>at least<br>5 but less<br>than 10<br>percent | Increase of<br>at least<br>10 percent |
|-------------------------------------|---------------------------------------|-------------------------------------------------------------|---------------------------------------------------------------------|-------------------------------------------------------------|---------------------------------------|
| <b>Total</b>                        | 31                                    | 16                                                          | 36                                                                  | 5                                                           | 12                                    |
| <b>level</b>                        |                                       |                                                             |                                                                     |                                                             |                                       |
| 4-year math                         | 27                                    | 18                                                          | 39                                                                  | 8                                                           | 8                                     |
| Statistics                          | 5                                     | 11                                                          | 53                                                                  | 11                                                          | 21                                    |
| 2-year math                         | 37                                    | 14                                                          | 31                                                                  | 1                                                           | 16                                    |
| <b>4-year math departments only</b> |                                       |                                                             |                                                                     |                                                             |                                       |
| <b>Highest degree offered</b>       |                                       |                                                             |                                                                     |                                                             |                                       |
| Doctoral                            | 25                                    | 18                                                          | 43                                                                  | 8                                                           | 6                                     |
| Master's                            | 27                                    | 19                                                          | 32                                                                  | 10                                                          | 12                                    |
| Baccalaureate                       | 38                                    | 13                                                          | 33                                                                  | 8                                                           | 10                                    |
| <b>Institution size category</b>    |                                       |                                                             |                                                                     |                                                             |                                       |
| Under 5,000                         | 33                                    | 15                                                          | 32                                                                  | 9                                                           | 12                                    |
| 5,000 - 9,999                       | 32                                    | 38                                                          | 21                                                                  | 6                                                           | 2                                     |
| 10,000 - 19,999                     | 26                                    | 11                                                          | 55                                                                  | 4                                                           | 4                                     |
| 20,000 or more                      | 9                                     | 14                                                          | 59                                                                  | 13                                                          | 5                                     |
| <b>Control of institution</b>       |                                       |                                                             |                                                                     |                                                             |                                       |
| Public                              | 25                                    | 20                                                          | 42                                                                  | 8                                                           | 5                                     |
| Private                             | 29                                    | 15                                                          | 36                                                                  | 9                                                           | 10                                    |

## **Four-Year Mathematics Questionnaire**



**SURVEY OF UNDERGRADUATE PROGRAMS IN THE MATHEMATICAL SCIENCES**



**General Information**

Four-Year College Mathematics Questionnaire

As part of a random sample, your department has been chosen to participate in the NSF-funded CBMS2021 National Survey of Undergraduate Mathematical Sciences Programs. The presidents of all U.S. mathematical sciences organizations have endorsed it and ask for your cooperation.

We assure you that no individual departmental data, except the names of responding departments, will be released.

This survey provides data about the nation's undergraduate mathematical and statistical effort that is available from no other source. You can see the results of a similar survey fielded six years ago by going to [www.ams.org/cbms](http://www.ams.org/cbms), where the CBMS 2015 report is available online.

All departments in this survey are in universities and colleges that offer at least a bachelor's degree. They may or may not offer a major in mathematics. Many of the departments in our random sample also offer higher degrees in mathematical sciences.

We have classified your department as belonging to a university or four-year college. If this is not correct, please contact Ellen Kirkman, Survey Director, at 336-758-5351 or at [Kirkman@wfu.edu](mailto:Kirkman@wfu.edu).

Please report on undergraduate programs in the broadly defined mathematical sciences (including applied mathematics, statistics, operations research, and computer science) that are under the direction of your department. Do not include data for other departments or for branches or campuses of your institution that are budgetarily separate from your own. Also, if your department is broader than just mathematics (e.g., Division of Mathematics and Sciences), please report only on the mathematics courses (as broadly defined here).

This survey may be completed either online or using a hard-copy questionnaire. We recommend using the online system because it will do some of the work for you; e.g., it will automatically skip those questions that are not applicable (based on the response you give), gray out portions of questions that do not apply, remind you of previous responses, and provide definitions when you let your cursor hover over certain highlighted words.

If you have any questions while filling out this survey form, please call the Survey Director, Ellen Kirkman, at 336-758-5351 or contact her by e-mail at [Kirkman@wfu.edu](mailto:Kirkman@wfu.edu). For help with the online questionnaire, call Westat at 855-770-0558 or send an email to [cbms2021@westat.com](mailto:cbms2021@westat.com).

**Please complete the questionnaire by October 29, 2021 online or by mailing a hard copy to:**

CBMS Survey  
Westat  
1600 Research Boulevard, RB 3103 Rockville,  
MD 20850-3129

**Please retain a copy of your responses to this questionnaire in case questions arise.**

**A. General Information**

Four-Year Mathematics Questionnaire

A1. Name of your institution: \_\_\_\_\_

A2. Name of your department: \_\_\_\_\_

A3. We have classified your department as being part of a university or four-year college. Do you agree?

Yes..... ☐ → If Yes, go to A4 below.No ..... ☐ → If No, please call Ellen Kirkman, Survey Director,  
at 336-758-5351A4. If your college or university does not recognize tenure, check this box. ☐

A5. Contact person in your department:

A6. Contact person's e-mail address:

A7. Contact person's phone number  
including area code:

A8. Contact person's mailing address:

a. Street .....

b. Street2.....

c. City .....

d. State .....

e. Zip code .....

**B. Dual-Enrollment Courses**

Four-Year Mathematics Questionnaire

**Definition:** We use the term dual (or concurrent) enrollment courses to refer to courses taught in a high school by high school teachers, for which high school students may obtain high school credit and, simultaneously, college credit through your institution.

B1. Does your department participate in any dual enrollment programs of this type?

Yes..... ☐ —————> If Yes, go to B2.

No ..... ☐ —————> If No, go to B3.

B2. Please complete the following table giving the number of students enrolled in your dual enrollment program (as defined above) for the previous term (spring 2021) and the current fall term of 2021. (Do not include these enrollments in subsequent sections)

| Course             | Total Dual Enrollments     |                          |
|--------------------|----------------------------|--------------------------|
|                    | Last Term =<br>Spring 2021 | This Term =<br>Fall 2021 |
| a. College Algebra |                            |                          |
| b. Precalculus     |                            |                          |
| c. Calculus I      |                            |                          |
| d. Statistics      |                            |                          |
| e. Other           |                            |                          |

B3. Does your department assign any of its own full-time or part-time faculty to teach courses in a high school for which high school students may receive both high school and college credit (through your institution)?

Yes..... ☐ —————> If Yes, go to B4.

No ..... ☐ —————> If No, go to C1.

B4. In fall 2021, how many students are enrolled in the courses taught in a high school by your full-time or part-time faculty and through which high school students may receive both high school and college credit (through your institution)? (Include these enrollments in subsequent sections)

Number of students.....

**C. Distance/Remote Learning**

Four-Year Mathematics Questionnaire

**Definition:** Distance/remote learning courses are those courses offered by your institution for credit, in which half or more of the instruction occurs with the instructor and the students separated by time and /or place, and facilitated by technology (e.g., courses in which half or more of the course is taught online either synchronously or asynchronously, or by computer software, or by other technologies). Include only distance/remote learning courses offered in normal practice, not courses that became distance/remote due to the COVID-19 pandemic.

- C1. Overall, how have attitudes towards online learning changed as a result of the COVID-19 experience?

|                                        | More favorable           | No change                | Less favorable           |
|----------------------------------------|--------------------------|--------------------------|--------------------------|
| a. Faculty interest in online teaching | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| b. Faculty use of online tools         | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| c. Student interest in online teaching | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| d. Student use of online tools         | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

C2. Many colleges have conducted online instruction as a way of addressing the COVID-19 pandemic, offering online instruction either as an alternative to face-to-face instruction or as a supplement to face-to-face instruction. How is that instruction coordinated with what has normally been called distance education? Please indicate which of the following applies to your department for each listed time period, using the following definitions.

- On-campus (face-to-face) learners—students who would be expected to attend most classes in person.
- Remote learners—students who would be expected to attend most classes remotely.

| Policy                                                                                                                                                                                    | Prior to pandemic (prior to spring 2020) |                          | During pandemic (spring 2020-summer 2021) |                          | Fall 2021                |                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|--------------------------|-------------------------------------------|--------------------------|--------------------------|--------------------------|
|                                                                                                                                                                                           |                                          |                          |                                           |                          |                          |                          |
|                                                                                                                                                                                           | Yes                                      | No                       | Yes                                       | No                       | Yes                      | No                       |
| a. We offer some course sections only to on-campus (face-to-face) learners; remote learners might be offered the same course, but the students would not normally be in the same section. | <input type="checkbox"/>                 | <input type="checkbox"/> | <input type="checkbox"/>                  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| b. We offer some course sections only to remote learners; on-campus (face-to-face) learners might be offered the same course, but the students would not normally be in the same section. | <input type="checkbox"/>                 | <input type="checkbox"/> | <input type="checkbox"/>                  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| c. We offer some "hybrid" courses to on-campus (face-to-face) learners and remote learners in the same section.                                                                           | <input type="checkbox"/>                 | <input type="checkbox"/> | <input type="checkbox"/>                  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

C3. Has your department taught any distance/remote learning course **(other than courses moved online in response to the COVID-19 pandemic)** within the calendar years 2018-2021?

Yes..... ☐ —————→ If Yes, go to C4.

No ..... ☐ —————→ If No, skip to section D.

- C4. Which best characterizes the format/structure of the majority of your distance/remote learning courses (not including courses moved online in response to the COVID-19 pandemic)? (*Choose one response.*)

Sections are taught only online, and only asynchronously ..... ☐

Sections are taught only online, with an opportunity to meet synchronously online ..... ☐

Sections use a mixture of online and face-to-face sessions ..... ☐

Other ..... ☐

- C5. In most of your distance/remote learning courses, how are the majority of the tests administered (not including courses moved online in response to the COVID-19 pandemic)? (*Choose one response.*)

Online and not monitored ..... ☐

Online, but using some kind of monitoring technology ..... ☐

At a monitored testing site ..... ☐

Combination of monitoring methods ..... ☐

- C6. Rate the following challenges that your department faces when creating and/or offering distance/remote learning mathematics courses. (Please check one box in each line.)

| Challenge                                                                | Not a challenge          | Somewhat of a challenge  | Very significant challenge |
|--------------------------------------------------------------------------|--------------------------|--------------------------|----------------------------|
| a. Designing appropriate assessments of student learning.                | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>   |
| b. Maintaining academic integrity on assessments.                        | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>   |
| c. Grade inflation.                                                      | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>   |
| d. Building/maintaining community among faculty and students.            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>   |
| e. Maintaining academic quality instruction.                             | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>   |
| f. Engaging students online.                                             | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>   |
| g. Replicating active learning in a virtual environment.                 | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>   |
| h. Availability of equipment and technical support for faculty/students. | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>   |

**D. Faculty Profile (Fall 2021)**

Four-Year Mathematics Questionnaire

Please indicate whether the following types of faculty are actively teaching one or more courses in fall 2021.

**Definitions**

- **Full-time faculty.** Faculty who are full-time employees in the institution and more than halftime in the department. For example, if a tenured physics professor with a joint appointment in your department teaches a total of two courses in fall 2021, with exactly one being in your department (i.e., mathematics is 50% of the fall teaching assignment), then that person would be counted as part-time in your department.

| Faculty Type                                                                                                                                                                                                                                |  | Teach in Fall 2021       |                          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--------------------------|--------------------------|
|                                                                                                                                                                                                                                             |  | Yes                      | No                       |
| D1. Full-time faculty who are more than halftime in your department                                                                                                                                                                         |  |                          |                          |
| a. Tenured or tenure-eligible, or permanent (if your institution does not recognize tenure) faculty.....                                                                                                                                    |  | <input type="checkbox"/> | <input type="checkbox"/> |
| b. Other full-time faculty (e.g., permanent faculty only if your institution also has tenured or tenure-eligible faculty, faculty with renewable positions such as teaching professionals, postdoctoral faculty, and visiting faculty)..... |  | <input type="checkbox"/> | <input type="checkbox"/> |
| D2. Part-time faculty (those who are halftime or less in your department) ....                                                                                                                                                              |  | <input type="checkbox"/> | <input type="checkbox"/> |
| D3. Graduate teaching assistant(s) who teach courses independently (not counting the teaching of recitation sessions) .....                                                                                                                 |  | <input type="checkbox"/> | <input type="checkbox"/> |

**E. Mathematics Courses (Fall 2021)**

Four-Year Mathematics Questionnaire

Which of the following courses are taught in your department in Fall 2021? You may use different titles for these courses, and may have multiple courses that match a particular course name.

- Enter an X in each applicable box.
- Do **not** include courses taught in other departments, learning centers, or developmental/remedial programs separate from your mathematics program or department.
- Include courses taught through distance/remote education.
- Please also indicate which catalog codes are used to identify those courses. This information will be used to generate a reduced course list suitable for your department so reporting on enrollments will be easier.
- Make sure that no course is reported in more than one row.

| Name of Course (or equivalent)                                                                 | Taught in fall 2021<br>(a) | Catalog code(s) used for courses offered FALL 2021 (not spring 2022 or previous academic year) (use comma to separate codes)<br>(e) |
|------------------------------------------------------------------------------------------------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>MATHEMATICS</b>                                                                             |                            |                                                                                                                                     |
| <b>PRECOLLEGE LEVEL</b>                                                                        | Enter X where applicable   |                                                                                                                                     |
| E1. Precollege level (e.g., arithmetic, pre-algebra, elementary algebra, intermediate algebra) | <input type="checkbox"/>   |                                                                                                                                     |
| <b>INTRODUCTORY LEVEL, INCLUDING PRE-CALCULUS</b>                                              |                            |                                                                                                                                     |
| E2. Mathematics for Liberal Arts                                                               | <input type="checkbox"/>   |                                                                                                                                     |
| E3. Finite Mathematics                                                                         | <input type="checkbox"/>   |                                                                                                                                     |
| E4. Business Mathematics (non-Calculus)                                                        | <input type="checkbox"/>   |                                                                                                                                     |
| E5. Mathematics for pre-service K-8 School Teachers (all courses)                              | <input type="checkbox"/>   |                                                                                                                                     |
| E6. College Algebra (not included in the Precollege E1 above)                                  | <input type="checkbox"/>   |                                                                                                                                     |
| E7. Trigonometry                                                                               | <input type="checkbox"/>   |                                                                                                                                     |
| E8. College Algebra & Trigonometry (combined)                                                  | <input type="checkbox"/>   |                                                                                                                                     |
| E9. Pre-Calculus                                                                               | <input type="checkbox"/>   |                                                                                                                                     |
| E10. Introduction to Mathematical Modeling                                                     | <input type="checkbox"/>   |                                                                                                                                     |
| E11. Quantitative Literacy/Reasoning                                                           | <input type="checkbox"/>   |                                                                                                                                     |
| E12. All other introductory-level non-Calculus courses                                         | <input type="checkbox"/>   |                                                                                                                                     |

<sup>1</sup>Distance/remote learning courses are those courses offered by your institution for credit, in which half or more of the instruction occurs with the instructor and the students separated by time and /or place, and facilitated by technology (e.g. courses in which the majority of the course is taught online either synchronously or asynchronously, or by computer software, or by other technologies).



**E. Mathematics Courses (Fall 2021) (cont.)**

Four-Year Mathematics Questionnaire

Which of the following courses are taught in your department in Fall 2021? You may use different titles for these courses, and may have multiple courses that match a particular course name.

- Enter an X in each applicable box.
- Do **not** include courses taught in other departments, learning centers, or developmental/remedial programs separate from your mathematics program or department.
- Include courses taught through distance/remote education.
- Please also indicate which catalog codes are used to identify those courses. This information will be used to generate a reduced course list suitable for your department so reporting on enrollments will be easier.
- Make sure that no course is reported in more than one row.
- **Calculus and Introductory Statistics classes.** You will be asked to list separately classes taught in a **large lecture format** (with recitation/problem/laboratory sections) and, sections that meet as a class with an instructor at a regularly scheduled time (and are not divided into recitation sections). Please treat any large class that is sometimes broken up into smaller units as a "lecture/recitation" class (even if there is no lecture); if neither the lecture/recitation or individual class format seems an appropriate description of the enrollment, enter the enrollment under "other."

| Name of Course (or equivalent)                                                                                  | Taught in fall 2021<br>(a) | Catalog code(s) used for courses offered FALL 2021 (not spring 2022 or previous academic year) (use comma to separate codes) <sup>1</sup><br>(e) |
|-----------------------------------------------------------------------------------------------------------------|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>MAINSTREAM<sup>2</sup> CALCULUS I</b>                                                                        |                            |                                                                                                                                                  |
| E13-1. Lecture with separately scheduled recitation/problem/laboratory sessions                                 | <input type="checkbox"/>   |                                                                                                                                                  |
| E13-2. Individual sections, not in E13-1, that meet as a class with an instructor at a regularly scheduled time | <input type="checkbox"/>   |                                                                                                                                                  |
| E13-3. Other sections, not listed above                                                                         | <input type="checkbox"/>   |                                                                                                                                                  |
| <b>MAINSTREAM<sup>2</sup> CALCULUS II</b>                                                                       |                            |                                                                                                                                                  |
| E14-1. Lecture with separately scheduled recitation/problem/laboratory sessions                                 | <input type="checkbox"/>   |                                                                                                                                                  |
| E14-2. Sections not in E14-1, that meet as a class with an instructor at a regularly scheduled time             | <input type="checkbox"/>   |                                                                                                                                                  |
| E14-3. Other sections not listed above                                                                          | <input type="checkbox"/>   |                                                                                                                                                  |
| <b>MAINSTREAM<sup>2</sup> CALCULUS III (and IV, etc.)</b>                                                       |                            |                                                                                                                                                  |
| E15-1. Lecture with separately scheduled recitation/problem/laboratory sessions                                 | <input type="checkbox"/>   |                                                                                                                                                  |
| E15-2. Individual sections not in E15-1 that meet as a class with an instructor at a regularly scheduled time   | <input type="checkbox"/>   |                                                                                                                                                  |

|                                        |                          |  |
|----------------------------------------|--------------------------|--|
| E15-3. Other sections not listed above | <input type="checkbox"/> |  |
|----------------------------------------|--------------------------|--|

<sup>1</sup> For E-13 through E-16, enter course identifiers that are sufficiently distinct to separate courses with recitation sessions, courses that meet as a class, and other sections.

<sup>2</sup> A calculus course is mainstream if it leads to the usual upper division mathematical sciences courses.

**E. Mathematics Courses (Fall 2021) (cont.)**

Four-Year Mathematics Questionnaire

| Name of Course (or equivalent)                                                                                | Taught in fall 2021<br>(a) | Catalog code(s) used for courses offered FALL 2021 (not spring 2022 or previous academic year) (use comma to separate codes) <sup>1</sup><br>(e) |
|---------------------------------------------------------------------------------------------------------------|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>NON-MAINSTREAM<sup>2</sup> CALCULUS</b>                                                                    |                            |                                                                                                                                                  |
| E16-1. Lecture with separately scheduled recitation/ problem/laboratory sessions <sup>6</sup>                 | <input type="checkbox"/>   |                                                                                                                                                  |
| E16-2. Individual sections not in E16-1 that meet as a class with an instructor at a regularly scheduled time | <input type="checkbox"/>   |                                                                                                                                                  |
| E16-3. Other sections not listed above                                                                        | <input type="checkbox"/>   |                                                                                                                                                  |
| E17. Non-mainstream <sup>5</sup> Calculus II, III, etc.                                                       | <input type="checkbox"/>   |                                                                                                                                                  |
| <b>OTHER CALCULUS LEVEL COURSES</b>                                                                           |                            |                                                                                                                                                  |
| E18. Differential Equations and Linear Algebra (combined)                                                     | <input type="checkbox"/>   |                                                                                                                                                  |
| E19. Differential Equations                                                                                   | <input type="checkbox"/>   |                                                                                                                                                  |
| E20. Linear Algebra or Matrix Theory                                                                          | <input type="checkbox"/>   |                                                                                                                                                  |
| E21. Discrete Mathematics (not Discrete Structures, which is E30)                                             | <input type="checkbox"/>   |                                                                                                                                                  |
| E22. Freshman seminar (Only count courses that are not included elsewhere)                                    | <input type="checkbox"/>   |                                                                                                                                                  |
| E23. Other calculus-level courses                                                                             | <input type="checkbox"/>   |                                                                                                                                                  |

Which of the following courses are taught in your department in Fall 2021, will be taught in Spring 2022, or were taught at any time in 2020-21? You may use different titles for these courses, and may have multiple courses that match a particular course name.

- Enter an X in each applicable box.
- Do **not** include courses taught in other departments, learning centers, or developmental/remedial programs separate from your mathematics program or department.
- Include courses taught through distance/remote education.
- Please also indicate which catalog codes are used to identify those courses. This information will be used to generate a reduced course list suitable for your department so reporting on enrollments will be easier.
- Make sure that no course is reported in more than one row.

| Name of Course (or equivalent)                                       | Taught in fall 2021<br>(a) | Will be taught in spring 2022<br>(b) | Taught during academic year 2020-21<br>(c) | Offer as distance/remote learning course<br>(d) | Catalog code(s) used for courses offered FALL 2021 (not spring 2022 or previous academic year) (use comma to separate codes) <sup>1</sup><br>(e) |
|----------------------------------------------------------------------|----------------------------|--------------------------------------|--------------------------------------------|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ADVANCED UNDERGRADUATE LEVEL</b>                                  |                            |                                      |                                            |                                                 |                                                                                                                                                  |
| E24. Introduction to Proofs                                          | <input type="checkbox"/>   | <input type="checkbox"/>             | <input type="checkbox"/>                   | <input type="checkbox"/>                        |                                                                                                                                                  |
| E25-1. Modern Algebra I                                              | <input type="checkbox"/>   | <input type="checkbox"/>             | <input type="checkbox"/>                   | <input type="checkbox"/>                        |                                                                                                                                                  |
| E25-2. Modern Algebra II                                             | <input type="checkbox"/>   | <input type="checkbox"/>             | <input type="checkbox"/>                   | <input type="checkbox"/>                        |                                                                                                                                                  |
| E26. Number Theory                                                   | <input type="checkbox"/>   | <input type="checkbox"/>             | <input type="checkbox"/>                   | <input type="checkbox"/>                        |                                                                                                                                                  |
| E27. Combinatorics                                                   | <input type="checkbox"/>   | <input type="checkbox"/>             | <input type="checkbox"/>                   | <input type="checkbox"/>                        |                                                                                                                                                  |
| E28. Actuarial Mathematics                                           | <input type="checkbox"/>   | <input type="checkbox"/>             | <input type="checkbox"/>                   | <input type="checkbox"/>                        |                                                                                                                                                  |
| E29. Logic/Foundations (not E24)                                     | <input type="checkbox"/>   | <input type="checkbox"/>             | <input type="checkbox"/>                   | <input type="checkbox"/>                        |                                                                                                                                                  |
| E30. Discrete Structures (beyond Discrete Mathematics, which is E21) | <input type="checkbox"/>   | <input type="checkbox"/>             | <input type="checkbox"/>                   | <input type="checkbox"/>                        |                                                                                                                                                  |
| E31. History of Mathematics                                          | <input type="checkbox"/>   | <input type="checkbox"/>             | <input type="checkbox"/>                   | <input type="checkbox"/>                        |                                                                                                                                                  |
| E32. Geometry                                                        | <input type="checkbox"/>   | <input type="checkbox"/>             | <input type="checkbox"/>                   | <input type="checkbox"/>                        |                                                                                                                                                  |

<sup>1</sup> For E-13 through E-16, enter course identifiers that are sufficiently distinct to separate courses with recitation sessions, courses that meet as a class, and other sections.

<sup>2</sup> A calculus course is mainstream if it leads to the usual upper division mathematical sciences courses.

**E. Mathematics Courses (Fall 2021) (cont.)**

Four-Year Mathematics Questionnaire

| Name of Course (or equivalent)                                                                                                          | Taught in fall 2021<br>(a) | Will be taught in spring 2022<br>(b) | Taught during academic year 2020-21<br>(c) | Offer as distance/remote learning course<br>(d) | Catalog code(s) used for courses offered FALL 2021 (not for Spring 2022 or previous academic year) (use comma to separate codes)<br>(e) |
|-----------------------------------------------------------------------------------------------------------------------------------------|----------------------------|--------------------------------------|--------------------------------------------|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>ADVANCED UNDERGRADUATE LEVEL (cont.)</b>                                                                                             |                            |                                      |                                            |                                                 |                                                                                                                                         |
| E33-1. Advanced Calculus I and/or Real Analysis I                                                                                       | <input type="checkbox"/>   | <input type="checkbox"/>             | <input type="checkbox"/>                   | <input type="checkbox"/>                        |                                                                                                                                         |
| E33-2. Advanced Calculus II and/or Real Analysis II                                                                                     | <input type="checkbox"/>   | <input type="checkbox"/>             | <input type="checkbox"/>                   | <input type="checkbox"/>                        |                                                                                                                                         |
| E34. Advanced Mathematics for Engineering and Physical Sciences (all courses)                                                           | <input type="checkbox"/>   | <input type="checkbox"/>             | <input type="checkbox"/>                   | <input type="checkbox"/>                        |                                                                                                                                         |
| E35. Advanced Linear Algebra (beyond Differential Equations and Linear Algebra (combined) and Linear Algebra or Matrix Theory E18, E20) | <input type="checkbox"/>   | <input type="checkbox"/>             | <input type="checkbox"/>                   | <input type="checkbox"/>                        |                                                                                                                                         |
| E36. Vector Analysis                                                                                                                    | <input type="checkbox"/>   | <input type="checkbox"/>             | <input type="checkbox"/>                   | <input type="checkbox"/>                        |                                                                                                                                         |
| E37. Advanced Differential Equations (beyond Differential Equations E19)                                                                | <input type="checkbox"/>   | <input type="checkbox"/>             | <input type="checkbox"/>                   | <input type="checkbox"/>                        |                                                                                                                                         |
| E38. Partial Differential Equations                                                                                                     | <input type="checkbox"/>   | <input type="checkbox"/>             | <input type="checkbox"/>                   | <input type="checkbox"/>                        |                                                                                                                                         |
| E39. Numerical Analysis (all courses)                                                                                                   | <input type="checkbox"/>   | <input type="checkbox"/>             | <input type="checkbox"/>                   | <input type="checkbox"/>                        |                                                                                                                                         |
| E40. Applied Mathematics (Modeling)                                                                                                     | <input type="checkbox"/>   | <input type="checkbox"/>             | <input type="checkbox"/>                   | <input type="checkbox"/>                        |                                                                                                                                         |
| E41. Complex Variables                                                                                                                  | <input type="checkbox"/>   | <input type="checkbox"/>             | <input type="checkbox"/>                   | <input type="checkbox"/>                        |                                                                                                                                         |
| E42. Topology                                                                                                                           | <input type="checkbox"/>   | <input type="checkbox"/>             | <input type="checkbox"/>                   | <input type="checkbox"/>                        |                                                                                                                                         |
| E43. Mathematics of Finance (not Actuarial Mathematics E28, or Applied Mathematics (Modeling) E40)                                      | <input type="checkbox"/>   | <input type="checkbox"/>             | <input type="checkbox"/>                   | <input type="checkbox"/>                        |                                                                                                                                         |
| E44. Codes and Cryptography                                                                                                             | <input type="checkbox"/>   | <input type="checkbox"/>             | <input type="checkbox"/>                   | <input type="checkbox"/>                        |                                                                                                                                         |
| E45. Biomathematics                                                                                                                     | <input type="checkbox"/>   | <input type="checkbox"/>             | <input type="checkbox"/>                   | <input type="checkbox"/>                        |                                                                                                                                         |
| E46. Operations Research (all courses)                                                                                                  | <input type="checkbox"/>   | <input type="checkbox"/>             | <input type="checkbox"/>                   | <input type="checkbox"/>                        |                                                                                                                                         |
| E47. Senior Seminar/ Independent Study in Mathematics                                                                                   | <input type="checkbox"/>   | <input type="checkbox"/>             | <input type="checkbox"/>                   | <input type="checkbox"/>                        |                                                                                                                                         |
| E48. All other advanced level mathematics (excluding Math for Secondary School Teachers, Probability or Statistics courses)             | <input type="checkbox"/>   | <input type="checkbox"/>             | <input type="checkbox"/>                   | <input type="checkbox"/>                        |                                                                                                                                         |
| E49. Mathematics for Secondary School Teachers (all such courses not counted above)                                                     | <input type="checkbox"/>   | <input type="checkbox"/>             | <input type="checkbox"/>                   | <input type="checkbox"/>                        |                                                                                                                                         |

**E. Mathematics Courses (Fall 2021) (cont.)**

Four-Year Mathematics Questionnaire

In the next several pages you will enter data about courses your department is teaching. For each course that is taught, you will be asked to enter the fall 2021 enrollment and the number of sections of the course. Depending upon what you indicated above, you will be asked about distance/remote learning enrollment.

**The following instructions apply throughout Sections E (mathematics courses), F (statistics courses), and G (computer science courses) (pages 12-23).**

- Do NOT include any **dual enrollment** sections or enrollments in these tables. (In this questionnaire, a *dual enrollment* section is one that is conducted in a high school, taught by a high school teacher, and allows students to receive high school credit and, simultaneously, college credit from your institution for the course. These courses were reported in Section B. Include courses taught at high schools by college faculty)
- Column (a): Report **distance/remote learning enrollments** separately from other enrollments. Distance/remote learning courses are those courses offered by your institution for credit, in which more than half of the instruction occurs with the instructor and the students separated by time and /or place, facilitated by technology (e.g. courses in which more than half of the course is taught online synchronously or asynchronously, or by computer software, or by other technologies).
- Columns (c) and (d) for **Calculus and Introductory Statistics classes**. You will be asked to list separately classes taught in a **large lecture format** (with recitation/problem/laboratory sections) and sections that meet as a class with an instructor at a regularly scheduled time (and are not divided into recitation sections). For example, for Mainstream Calculus I, you will be asked for both the number of large lecture courses (E13-1 column (c)) and the total number of recitation sections for all the large lectures (E13-1 column (d)). Please treat any large class that is sometimes broken up into smaller units as a “lecture/recitation” class (even if there is no lecture); if neither the lecture/recitation or individual class format seems an appropriate description of the enrollment, enter the enrollment under “other”.
- **Courses, sections, and sessions.** In this questionnaire, “course” is used to refer to the topic area (e.g., Calculus 1 or Number Theory). You may have multiple faculty teaching the same course in the same term but at different times or locations; these divisions of the topic area into separate instances of teaching are called sections. Within a section, you may have times when the students are divided into laboratory or recitation sessions; these are counted as recitation sessions, not as separate sections.
- For all courses except as marked in E13, E14, E15, E16, F1, and F2, please do not treat **recitation sessions** as separate sections. Instead, please treat both the lecture component and any associated recitation sessions as a single section.
- Do not fill in any shaded boxes.
- Any **unshaded box that is left blank** will be interpreted as reporting a count of zero.
- Except where specifically stated to the contrary, the tables in Sections E, F, and G deal with **enrollments in fall term 2021**.
- If an undergraduate course contains a mixture of graduate and undergraduate students, report them all in column (b).



**E. Mathematics Courses (Fall 2021) (cont.)**

Four-Year Mathematics Questionnaire

E51. You reported a total of [#] sections in fall 2021, distributed by course type as shown below. For each course type, please provide the number of sections taught by tenured or tenure-eligible faculty, other full-time faculty, part-time faculty, and graduate teaching assistants.

◆ Cells left blank will be interpreted as zeros

| Type of course and your applicable catalog course codes                              | Total number of sections (a) | Of the number in column (a), how many sections are taught by: |                             |                   |                                           |
|--------------------------------------------------------------------------------------|------------------------------|---------------------------------------------------------------|-----------------------------|-------------------|-------------------------------------------|
|                                                                                      |                              | Full-time faculty <sup>1</sup>                                |                             | Part-time Faculty | Graduate Teaching Assistants <sup>2</sup> |
|                                                                                      |                              | Tenured or Tenure-eligible, Faculty (b)                       | Other Full-time Faculty (c) | (d)               | (e)                                       |
| E51a. Precollege level<br>(course code list)                                         |                              |                                                               |                             |                   |                                           |
| E51b. Introductory level, including precalculus<br>(course code list)                |                              |                                                               |                             |                   |                                           |
| E51c. Mainstream Calculus I) —Lecture with separate recitation<br>(course code list) |                              |                                                               |                             |                   |                                           |
| E51d. Mainstream Calculus I) — Sections that meet as a class<br>(course code list)   |                              |                                                               |                             |                   |                                           |
| E51e. Mainstream Calculus I) —Other sections<br>(course code list)                   |                              |                                                               |                             |                   |                                           |
| E51f. Mainstream Calculus II —Lecture with separate recitation<br>(course code list) |                              |                                                               |                             |                   |                                           |

<sup>1</sup>If your institution does not recognize tenure, report sections taught by your permanent full-time faculty in column (b) and sections taught by other full-time faculty in column (c). If your institution does recognize tenure but has **faculty with renewable contracts**, report these faculty as **other full-time faculty** (column c).

**Full-time faculty teaching in your department and holding joint appointments with other departments** should be counted in column (b) if they are tenured, tenure-eligible, or permanent (if your institution does not recognize tenure) in your department. Faculty who are not tenured, tenure-eligible, or permanent in your department should be counted in column (d) if their fall 2021 teaching in your department is less than or equal to 50% of their total fall teaching assignment, and they should be reported in column (c) otherwise. (Example: If a tenured physics professor with a joint appointment in your department teaches a total of two courses in fall 2021, with exactly one being in your department and hence mathematics comprised 50% of the fall teaching assignment, then that person would be counted as parttime in your department.)

<sup>2</sup> Report a section of a course as being taught by a **graduate teaching assistant (GTA)** if and only if that section is taught *independently* by the GTA, i.e., when it is the GTA's own course and the GTA is the instructor of record.



**E. Mathematics Courses (Fall 2021) (cont.)**

Four-Year Mathematics Questionnaire

◆ Cells left blank will be interpreted as zeros

| Type of course and your applicable catalog course codes                                              | Total number of sections (a) | Of the number in column (a), how many sections are taught by: |                             |                   |                                           |
|------------------------------------------------------------------------------------------------------|------------------------------|---------------------------------------------------------------|-----------------------------|-------------------|-------------------------------------------|
|                                                                                                      |                              | Full-time faculty <sup>1</sup>                                |                             | Part-time Faculty | Graduate Teaching Assistants <sup>2</sup> |
|                                                                                                      |                              | Tenured or Tenure-eligible, Faculty (b)                       | Other Full-time Faculty (c) | (d)               | (e)                                       |
| E51g. Mainstream Calculus II — Sections that meet as a class<br>(course code list)                   |                              |                                                               |                             |                   |                                           |
| E51h. Mainstream Calculus II —Other sections<br>(course code list)                                   |                              |                                                               |                             |                   |                                           |
| E51i. Mainstream Calculus III (and IV, etc.) —Lecture with separate recitation<br>(course code list) |                              |                                                               |                             |                   |                                           |
| E51j. Mainstream Calculus III (and IV, etc.) —Sections that meet as a class<br>(course code list)    |                              |                                                               |                             |                   |                                           |
| E51k. Mainstream Calculus III (and IV, etc.) —Other sections<br>(course code list)                   |                              |                                                               |                             |                   |                                           |
| E51l. Non-mainstream Calculus —Lecture with separate recitation<br>(course code list)                |                              |                                                               |                             |                   |                                           |
| E51m. Non-mainstream Calculus — Sections that meet as a class<br>(course code list) —Other sections  |                              |                                                               |                             |                   |                                           |
| E51n. Non-mainstream Calculus —Other sections not listed above<br>(course code list)                 |                              |                                                               |                             |                   |                                           |
| E51o. Non-mainstream Calculus II, III, etc.<br>(course code list)                                    |                              |                                                               |                             |                   |                                           |
| E51p. Other Calculus Level Courses<br>(course code list)                                             |                              |                                                               |                             |                   |                                           |
| E51q. Advanced Undergraduate Level<br>(course code list)                                             |                              |                                                               |                             |                   |                                           |

**F. Probability and Statistics Courses (Fall 2021)**

Four-Year Mathematics Questionnaire

F. Does your department offer any Probability and/or Statistics Courses?

Yes..... ☐ —————> If Yes, go to F1 below.No ..... ☐ —————> If No, go to Section G.

Which of the following courses are taught in your department in Fall 2021, will be taught in Spring 2022, or were taught at any time in 2020-21? You may use different titles for these courses, and may have multiple courses that match a particular course name.

- Enter an X in each applicable box.
- Do **not** include courses taught in other departments, learning centers, or developmental/remedial programs separate from your mathematics program or department.
- Include courses taught through distance/remote education.
- Please also indicate which catalog codes are used to identify those courses. This information will be used to generate a reduced course list suitable for your department so reporting on enrollments will be easier.
- Make sure that no course is reported in more than one row.
- **Calculus and Introductory Statistics classes.** You will be asked to list separately classes taught in a **large lecture format** (with recitation/problem/laboratory sections) and, sections that meet as a class with an instructor at a regularly scheduled time (and are not divided into recitation sections). Please treat any large class that is sometimes broken up into smaller units as a "lecture/recitation" class (even if there is no lecture); if neither the lecture/recitation or individual class format seems an appropriate description of the enrollment, enter the enrollment under "other."

| Name of Course (or equivalent)                                                                                                                         | Taught in fall 2021      | Catalog code(s) used for Fall 2021 (use comma to separate codes) <sup>1</sup> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|-------------------------------------------------------------------------------|
| <b>Statistics</b>                                                                                                                                      |                          |                                                                               |
| <b>INTRODUCTORY STATISTICS (no calculus prerequisite; designed for non-majors/minors [General Education Courses] but may be taken by major/minors)</b> |                          |                                                                               |
| F1-1. Lecture with separately scheduled recitation/problem/laboratory sessions                                                                         | <input type="checkbox"/> |                                                                               |
| F1-2. Individual sections not in F1-1, that meet as a class with an instructor at a regularly scheduled time                                           | <input type="checkbox"/> |                                                                               |
| F1-3. Other sections not listed above                                                                                                                  | <input type="checkbox"/> |                                                                               |
| <b>INTRODUCTORY STATISTICS (calculus prerequisite)</b>                                                                                                 |                          |                                                                               |
| F2-1. Lecture with separately scheduled recitation/problem/laboratory sessions                                                                         | <input type="checkbox"/> |                                                                               |
| F2-2. Individual sections not in F2-1, that meet as a class with an instructor at a regularly scheduled time                                           | <input type="checkbox"/> |                                                                               |
| F2-3. Sections not listed above                                                                                                                        | <input type="checkbox"/> |                                                                               |
| <b>OTHER INTRODUCTORY STATISTICS courses</b>                                                                                                           |                          |                                                                               |

|                                                                                               |                          |  |
|-----------------------------------------------------------------------------------------------|--------------------------|--|
| F3. Statistics for pre-service elementary and/or middle grade teachers                        | <input type="checkbox"/> |  |
| F4. Statistics for pre-service secondary school teachers                                      | <input type="checkbox"/> |  |
| F5. Intermediate statistics (non-calculus)                                                    | <input type="checkbox"/> |  |
| F6. Other introductory level Probability or Statistics courses designed for non-majors/minors | <input type="checkbox"/> |  |

<sup>1</sup> For F-1 and F-2, enter course identifiers that are sufficiently distinct to separate courses with recitation sessions, courses that meet as a class, and other sections.

**F. Probability and Statistics Courses (Fall 2021) (cont.)**

Four-Year Mathematics Questionnaire

Which of the following courses are taught in your department in Fall 2021, will be taught in Spring 2022, or were taught at any time in 2020-21? You may use different titles for these courses, and may have multiple courses that match a particular course name.

- Enter an X in each applicable box.
- Do **not** include courses taught in other departments, learning centers, or developmental/remedial programs separate from your mathematics program or department.
- Include courses taught through distance/remote education.
- Please also indicate which catalog codes are used to identify those courses. This information will be used to generate a reduced course list suitable for your department so reporting on enrollments will be easier.
- Make sure that no course is reported in more than one row.
- **Calculus and Introductory Statistics classes.** You will be asked to list separately classes taught in a **large lecture format** (with recitation/problem/laboratory sections) and, sections that meet as a class with an instructor at a regularly scheduled time (and are not divided into recitation sections). Please treat any large class that is sometimes broken up into smaller units as a "lecture/recitation" class (even if there is no lecture); if neither the lecture/recitation or individual class format seems an appropriate description of the enrollment, enter the enrollment under "other."

| Name of Course (or equivalent)                                | Taught in fall 2021<br>(a) | Will be taught in spring 2022<br>(b) | Taught during academic year 2020-21<br>(c) | Offer as distance/remote learning course <sup>1</sup><br>(d) | Catalog code(s) used for courses offered Fall 2021 (not for courses offered Spring 2022 or previous academic year A) (use comma to separate codes)<br>(e) |
|---------------------------------------------------------------|----------------------------|--------------------------------------|--------------------------------------------|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>INTERMEDIATE AND ADVANCED LEVEL</b>                        |                            |                                      |                                            |                                                              |                                                                                                                                                           |
| F7. Combined Probability & Statistics (calculus prerequisite) | <input type="checkbox"/>   | <input type="checkbox"/>             | <input type="checkbox"/>                   | <input type="checkbox"/>                                     |                                                                                                                                                           |
| F8. Probability (calculus prerequisite)                       | <input type="checkbox"/>   | <input type="checkbox"/>             | <input type="checkbox"/>                   | <input type="checkbox"/>                                     |                                                                                                                                                           |
| F9. Mathematical Statistics (calculus prerequisite)           | <input type="checkbox"/>   | <input type="checkbox"/>             | <input type="checkbox"/>                   | <input type="checkbox"/>                                     |                                                                                                                                                           |
| F10. Stochastic Processes                                     | <input type="checkbox"/>   | <input type="checkbox"/>             | <input type="checkbox"/>                   | <input type="checkbox"/>                                     |                                                                                                                                                           |
| F11. Data Science/Analytics/Statistical Learning              | <input type="checkbox"/>   | <input type="checkbox"/>             | <input type="checkbox"/>                   | <input type="checkbox"/>                                     |                                                                                                                                                           |
| F12. Design & Analysis of Experiments (ANOVA)                 | <input type="checkbox"/>   | <input type="checkbox"/>             | <input type="checkbox"/>                   | <input type="checkbox"/>                                     |                                                                                                                                                           |
| F13. Applied Regression                                       | <input type="checkbox"/>   | <input type="checkbox"/>             | <input type="checkbox"/>                   | <input type="checkbox"/>                                     |                                                                                                                                                           |
| F14. Linear Models I                                          | <input type="checkbox"/>   | <input type="checkbox"/>             | <input type="checkbox"/>                   | <input type="checkbox"/>                                     |                                                                                                                                                           |
| F15. Linear Models II                                         | <input type="checkbox"/>   | <input type="checkbox"/>             | <input type="checkbox"/>                   | <input type="checkbox"/>                                     |                                                                                                                                                           |
| F16. Biostatistics                                            | <input type="checkbox"/>   | <input type="checkbox"/>             | <input type="checkbox"/>                   | <input type="checkbox"/>                                     |                                                                                                                                                           |

|                                                                     |                          |                          |                          |                          |  |
|---------------------------------------------------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--|
| F17. Nonparametric Statistics                                       | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |  |
| F18. Categorical Data Analysis                                      | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |  |
| F19. Sample Survey Design & Analysis                                | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |  |
| F20. Statistical Computing or Software                              | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |  |
| F21. Bayesian Statistics                                            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |  |
| F22. Statistical Consulting                                         | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |  |
| F23. Senior Seminar/Capstone                                        | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |  |
| F24. All other upper level Probability & Statistics related courses | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |  |

<sup>1</sup>Distance/remote learning courses are those courses offered by your institution for credit, in which half or more of the instruction occurs with the instructor and the students separated by time and /or place, and facilitated by technology (e.g. courses in which the majority of the course is taught online synchronously or asynchronously, or by computer software, or by other technologies).

**F. Probability and Statistics Courses (Fall 2021) (cont.)**

Four-Year Mathematics Questionnaire

F25. Please enter the total fall 2021 enrollments, number of sections, and recitation sections below, as indicated.

**Instructions**

- Do NOT include any **dual enrollment** sections or enrollments in these tables. (In this questionnaire, a *dual enrollment* section is one that is conducted in a high school, taught by a high school teacher, and allows students to receive high school credit and, simultaneously, college credit from your institution for the course. These courses were reported in Section B. Include courses taught at high schools by college faculty)
- Column (a): Report **distance/remote learning enrollments** separately from other enrollments. Distance/remote learning courses are those courses offered by your institution for credit, in which more than half of the instruction occurs with the instructor and the students separated by time and /or place, facilitated by technology (e.g. courses in which more than half of the course is taught online synchronously or asynchronously, or by computer software, or by other technologies).
- Columns (c) and (d) for **Calculus and Introductory Statistics classes**. You will be asked to list separately classes taught in a **large lecture format** (with recitation/problem/laboratory sections) and sections that meet as a class with an instructor at a regularly scheduled time (and are not divided into recitation sections). For example, for Introductory Statistics, you will be asked for both the number of large lecture courses (F1-1 column (c)) and the total number of recitation sections for all the large lectures (F1-1 column (d)). Please treat any large class that is sometimes broken up into smaller units as a “lecture/recitation” class (even if there is no lecture); if neither the lecture/recitation or individual class format seems an appropriate description of the enrollment, enter the enrollment under “other”.
- Courses, sections, and sessions**. In this questionnaire, “course” is used to refer to the topic area (e.g., Calculus 1 or Number Theory). You may have multiple faculty teaching the same course in the same term but at different times or locations; these divisions of the topic area into separate instances of teaching are called sections. Within a section, you may have times when the students are divided into laboratory or recitation sessions; these are counted as recitation sessions, not as separate sections.
- For all courses except as marked in F1 and F2, please do not treat **recitation sessions** as separate sections. Instead, please treat both the lecture component and any associated recitation sessions as a single section.
- Do not fill in any shaded boxes.
- Any **unshaded box that is left blank** will be interpreted as reporting a count of zero.
- Except where specifically stated to the contrary, the tables in Sections E, F, and G deal with **enrollments in fall term 2021**.
- If an undergraduate course contains a mixture of graduate and undergraduate students, report them all in column (b).

◆ **Cells left blank will be interpreted as zeros**

| Your catalog course codes | Total fall 2021 distance/remote education | Total fall 2021 enrollment NOT in distance/ | Number of sections corresponding | Total number of recitation/problem/ |
|---------------------------|-------------------------------------------|---------------------------------------------|----------------------------------|-------------------------------------|
|---------------------------|-------------------------------------------|---------------------------------------------|----------------------------------|-------------------------------------|



**F. Probability and Statistics Courses (Fall 2021) (cont.)**

Four-Year Mathematics Questionnaire

F26. You reported a total of # sections in fall 2021, distributed by course type as shown below. For each course type, please provide the number of sections taught by tenured or tenure-eligible faculty, other full-time faculty, part-time faculty, and graduate teaching assistants.

## Instructions

- Any **unshaded box that is left blank** will be interpreted as reporting a count of zero.
- Except where specifically stated to the contrary, the tables in Sections E, F, and G deal with **enrollments in fall term 2021**.

| ◆ Cells left blank will be interpreted as zeros                                                            |                              | Of the number in column (a), how many sections are taught by: |                             |                   |                                           |
|------------------------------------------------------------------------------------------------------------|------------------------------|---------------------------------------------------------------|-----------------------------|-------------------|-------------------------------------------|
|                                                                                                            |                              | Full-time faculty <sup>1</sup>                                |                             | Part-time Faculty | Graduate Teaching Assistants <sup>2</sup> |
|                                                                                                            |                              | Tenured or Tenure-eligible, Faculty (b)                       | Other Full-time Faculty (c) |                   |                                           |
| Type of course and your applicable catalog course codes                                                    | Total number of sections (a) |                                                               |                             | (d)               | (e)                                       |
| a. Introductory Statistics (no calculus prerequisite) —Lecture with separate recitation (course code list) |                              |                                                               |                             |                   |                                           |
| b. Introductory Statistics (no calculus prerequisite) —Sections that meet as a class (course code list)    |                              |                                                               |                             |                   |                                           |
| c. Introductory Statistics (no calculus prerequisite) —Other sections (course code list)                   |                              |                                                               |                             |                   |                                           |

<sup>1</sup>If your institution does not recognize tenure, report sections taught by your permanent full-time faculty in column (b) and sections taught by other full-time faculty in column (c). If your institution does recognize tenure but has **faculty with renewable contracts**, report these faculty as **other full-time faculty** (column c).

**Full-time faculty teaching in your department and holding joint appointments with other departments** should be counted in column (b) if they are tenured, tenure-eligible, or permanent (if your institution does not recognize tenure) in your department. Faculty who are not tenured, tenure-eligible, or permanent in your department should be counted in column (d) if their fall 2021 teaching in your department is less than or equal to 50% of their total fall teaching assignment, and they should be reported in column (c) otherwise. (Example: If a tenured physics professor with a joint appointment in your department teaches a total of two courses in fall 2021, with exactly one being in your department and hence mathematics comprised 50% of the fall teaching assignment, then that person would be counted as parttime in your department.)

<sup>2</sup> Report a section of a course as being taught by a **graduate teaching assistant (GTA)** if and only if that section is taught *independently* by the GTA, i.e., when it is the GTA's own course and the GTA is the instructor of record.



**F. Probability and Statistics Courses (Fall 2021) (cont.)**

Four-Year Mathematics Questionnaire

◆ Cells left blank will be interpreted as zeros

| Type of course and your applicable catalog course codes                                                                   | Total number of sections<br>(a) | Of the number in column (a), how many sections are taught by: |                             |                   |                                           |
|---------------------------------------------------------------------------------------------------------------------------|---------------------------------|---------------------------------------------------------------|-----------------------------|-------------------|-------------------------------------------|
|                                                                                                                           |                                 | Full-time faculty <sup>1</sup>                                |                             | Part-time Faculty | Graduate Teaching Assistants <sup>2</sup> |
|                                                                                                                           |                                 | Tenured or Tenure-eligible, Faculty (b)                       | Other Full-time Faculty (c) | (d)               | (e)                                       |
| d. Introductory Statistics (calculus prerequisite) (not for majors) — Lecture with separate recitation (course code list) |                                 |                                                               |                             |                   |                                           |
| e. Introductory Statistics (calculus prerequisite) (not for majors) — Sections that meet as a class (course code list)    |                                 |                                                               |                             |                   |                                           |
| f. Introductory Statistics (calculus prerequisite) (not for majors) — Other sections (course code list)                   |                                 |                                                               |                             |                   |                                           |
| g. Statistics for Pre-service Teachers (course code list)                                                                 |                                 |                                                               |                             |                   |                                           |
| h. Other introductory level Statistics (course code list)                                                                 |                                 |                                                               |                             |                   |                                           |
| i. Advanced Undergraduate Level (course code list)                                                                        |                                 |                                                               |                             |                   |                                           |

**G. Computer Science Courses (Fall 2021)**

Four-Year Mathematics Questionnaire

G. Does your department offer any Computer Science courses?

Yes ..... ☐ —————→ If Yes, go to G1 below.

No ..... ☐ —————→ If No, go to Section H.

- Please refer to the course reporting instructions at the beginning of Section E.

In December 2013, a joint IEEE Computer Society/ACM Task Force issued its recommendations on “Computer Science Curricula 2013.”

That report, which listed 18 Knowledge Areas, is available by clicking  
<http://www.acm.org/education/CS2013-final-report.pdf>

Which of the following courses are taught in your department in Fall 2021? You may use different titles for these courses, and may have multiple courses that match a particular course name.

- See the complete instructions for this table in Section E.

| Name of Course (or equivalent)                                                                | Taught in fall 2021      | Catalog code(s) used for courses offered Fall 2021 (not for courses offered Spring 2022 or previous academic year A) (use comma to separate codes) |
|-----------------------------------------------------------------------------------------------|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                               | (a)                      | (c)                                                                                                                                                |
| <b>Computer Science</b>                                                                       |                          |                                                                                                                                                    |
| <b>GENERAL EDUCATION COURSES</b>                                                              |                          |                                                                                                                                                    |
| G1. Computers and Society, Issues in CS                                                       | <input type="checkbox"/> |                                                                                                                                                    |
| G2. Intro. to Software Packages                                                               | <input type="checkbox"/> |                                                                                                                                                    |
| G3. Other CS General Education Courses                                                        | <input type="checkbox"/> |                                                                                                                                                    |
| <b>INTRODUCTORY CS COURSES</b>                                                                |                          |                                                                                                                                                    |
| G4. Computer Programming I                                                                    | <input type="checkbox"/> |                                                                                                                                                    |
| G5. Computer Programming II                                                                   | <input type="checkbox"/> |                                                                                                                                                    |
| G6. Discrete Structures DS) <sup>4</sup> , but not math courses E21 or E30 in Section E above | <input type="checkbox"/> |                                                                                                                                                    |
| G7. All other introductory level CS courses                                                   | <input type="checkbox"/> |                                                                                                                                                    |

**G. Computer Science Courses (Fall 2021) (cont.)**

Four-Year Mathematics Questionnaire

| Name of Course (or equivalent)                                                                                | Taught in fall 2021      | Catalog code(s) used for courses offered Fall 2021 (not for courses offered Spring 2022 or previous academic year A) (use comma to separate codes) |
|---------------------------------------------------------------------------------------------------------------|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                               | (a)                      | (e)                                                                                                                                                |
| <b>INTERMEDIATE LEVEL</b>                                                                                     |                          |                                                                                                                                                    |
| G8. Algorithms and Complexity (AL) <sup>2</sup>                                                               | <input type="checkbox"/> |                                                                                                                                                    |
| G9. Architecture and Organization (AR) <sup>2</sup>                                                           | <input type="checkbox"/> |                                                                                                                                                    |
| G10. Operating Systems (OS) <sup>2</sup>                                                                      | <input type="checkbox"/> |                                                                                                                                                    |
| G11. Networking and Communication (NC) <sup>2</sup>                                                           | <input type="checkbox"/> |                                                                                                                                                    |
| G12. Programming Languages (PL) <sup>2</sup>                                                                  | <input type="checkbox"/> |                                                                                                                                                    |
| G13. Human-Computer Interaction (HCI) <sup>2</sup>                                                            | <input type="checkbox"/> |                                                                                                                                                    |
| G14. Intelligent Systems (IS) <sup>2</sup>                                                                    | <input type="checkbox"/> |                                                                                                                                                    |
| G15. Information Management (IM) <sup>2</sup>                                                                 | <input type="checkbox"/> |                                                                                                                                                    |
| G16. Social Issues and Professional Practice (SP) <sup>2</sup>                                                | <input type="checkbox"/> |                                                                                                                                                    |
| G17. Software Development Fundamentals (SDF) <sup>2</sup>                                                     | <input type="checkbox"/> |                                                                                                                                                    |
| G18. Computational Science (CN) <sup>2</sup>                                                                  | <input type="checkbox"/> |                                                                                                                                                    |
| <b>UPPER LEVEL</b>                                                                                            |                          |                                                                                                                                                    |
| G19. Graphics and Visualization (GV) <sup>2</sup>                                                             | <input type="checkbox"/> |                                                                                                                                                    |
| G20. Information Assurance and Security (IAS) <sup>2</sup>                                                    | <input type="checkbox"/> |                                                                                                                                                    |
| G21. Parallel and Distributed Computing (PD) <sup>2</sup>                                                     | <input type="checkbox"/> |                                                                                                                                                    |
| G22. All other intermediate or advanced level CS Courses (including knowledge areas PBD, SE, SF) <sup>2</sup> | <input type="checkbox"/> |                                                                                                                                                    |

<sup>1</sup>Distance/remote learning courses are those courses offered by your institution for credit, in which half or more of the instruction occurs with the instructor and the students separated by time and /or place, and facilitated by technology (e.g. courses in which the majority of the course is taught online synchronously or asynchronously, or by computer software, or by other technologies).

<sup>2</sup> Knowledge areas from Computer Science Curricula 2013

**G. Computer Science Courses (Fall 2021) (cont.)**

Four-Year Mathematics Questionnaire

G23. Please enter the total fall 2021 enrollments and number of sections below, as indicated.

**Instructions**

- Do NOT include any **dual enrollment** sections or enrollments in these tables. (In this questionnaire, a *dual enrollment* section is one that is conducted in a high school, taught by a high school teacher, and allows students to receive high school credit and, simultaneously, college credit from your institution for the course. These courses were reported in Section B. Include courses taught at high schools by college faculty)
- Column (a): Report **distance/remote learning enrollments** separately from other enrollments. Distance/remote learning courses are those courses offered by your institution for credit, in which more than half of the instruction occurs with the instructor and the students separated by time and /or place, facilitated by technology (e.g. courses in which more than half of the course is taught online synchronously or asynchronously, or by computer software, or by other technologies).
- Courses, sections, and sessions.** In this questionnaire, “course” is used to refer to the topic area (e.g., Calculus 1 or Number Theory). You may have multiple faculty teaching the same course in the same term but at different times or locations; these divisions of the topic area into separate instances of teaching are called sections. Within a section, you may have times when the students are divided into laboratory or recitation sessions; these are counted as recitation sessions, not as separate sections.
- Do not fill in any shaded boxes.
- Any **unshaded box that is left blank** will be interpreted as reporting a count of zero.
- Except where specifically stated to the contrary, the tables in Sections E, F, and G deal with **enrollments in fall term 2021**.
- If an undergraduate course contains a mixture of graduate and undergraduate students, report them all in column (b).

◆ **Cells left blank will be interpreted as zeros**

| Your catalog course codes | Total fall 2021 distance/remote education enrollments <sup>1</sup><br>(a) | Total fall 2021 enrollment NOT in distance/remote education and NOT dual enrollments <sup>2</sup><br>(b) | Number of sections corresponding to column (b) <sup>3</sup><br>(c) |
|---------------------------|---------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
|                           |                                                                           |                                                                                                          |                                                                    |
|                           |                                                                           |                                                                                                          |                                                                    |
|                           |                                                                           |                                                                                                          |                                                                    |
|                           |                                                                           |                                                                                                          |                                                                    |
|                           |                                                                           |                                                                                                          |                                                                    |
|                           |                                                                           |                                                                                                          |                                                                    |
|                           |                                                                           |                                                                                                          |                                                                    |
|                           |                                                                           |                                                                                                          |                                                                    |

[illegible]

<sup>2</sup>Do not include any dual enrollments (see Section B)

<sup>4</sup>Example: suppose your department offers four 100-student sections of a course and that each is divided into five student discussion sessions that meet separately from the lectures. Report 4\*5=20 recitation/problem/laboratory sessions associated with the course, even if each discussion meets several times per week.

**G. Computer Science Courses (Fall 2021) (cont.)**

Four-Year Mathematics Questionnaire

G24. You reported a total of # sections in fall 2021, distributed by course type as shown below. For each course type, please provide the number of sections taught by tenured or tenure-eligible faculty, other full-time faculty, part-time faculty, and graduate teaching assistants.

**Instructions**

- Any **unshaded box that is left blank** will be interpreted as reporting a count of zero.
- Except where specifically stated to the contrary, the tables in Sections E, F, and G deal with **enrollments in fall term 2021**.

| ◆ Cells left blank will be interpreted as zeros         |                              | Of the number in column (a), how many sections are taught by: |                             |                   |                                           |
|---------------------------------------------------------|------------------------------|---------------------------------------------------------------|-----------------------------|-------------------|-------------------------------------------|
|                                                         |                              | Full-time faculty <sup>1</sup>                                |                             | Part-time Faculty | Graduate Teaching Assistants <sup>2</sup> |
|                                                         |                              | Tenured or Tenure-eligible, Faculty (b)                       | Other Full-time Faculty (c) |                   |                                           |
| Type of course and your applicable catalog course codes | Total number of sections (a) |                                                               |                             | (d)               | (e)                                       |
| a. General Education Course<br>(course code list)       |                              |                                                               |                             |                   |                                           |
| b. Introductory CS<br>(course code list)                |                              |                                                               |                             |                   |                                           |
| c. Intermediate Level<br>(course code list)             |                              |                                                               |                             |                   |                                           |
| d. Upper Level<br>(course code list)                    |                              |                                                               |                             |                   |                                           |

<sup>1</sup>If your institution does not recognize tenure, report sections taught by your permanent full-time faculty in column (b) and sections taught by other full-time faculty in column (c). If your institution does recognize tenure but has **faculty with renewable contracts, report these faculty as other full-time faculty** (column c).

**Full-time faculty teaching in your department and holding joint appointments with other departments** should be counted in column (b) if they are tenured, tenure-eligible, or permanent (if your institution does not recognize tenure) in your department. Faculty who are not tenured, tenure-eligible, or permanent in your department should be counted in column (d) if their fall 2021 teaching in your department is less than or equal to 50% of their total fall teaching assignment, and they should be reported in column (c) otherwise. (Example: If a tenured physics professor with a joint appointment in your department teaches a total of two courses in fall 2021, with exactly one being in your department and hence mathematics comprised 50% of the fall teaching assignment, then that person would be counted as parttime in your department.)

<sup>2</sup> Report a section of a course as being taught by a **graduate teaching assistant (GTA)** if and only if that section is taught *independently* by the GTA, i.e., when it is the GTA's own course and the GTA is the instructor of record.

**H. Precalculus/Introductory Mathematics and Statistics Instruction****Introductory Mathematics**

H1. Has your department offered a non-traditional “pathways” course sequence within the last five years?

Yes ..... ☐

No ..... ☐

(“Pathways” is defined to be a single course or course sequence that enables students to complete a college-level gateway mathematics or statistics course that is aligned to students' academic and/or career goals within one academic year.)

H2. Does your department offer any Precollege or Introductory Level (including Precalculus) mathematics courses for credit (courses E1-E12) in Fall 2021?

Yes..... ☐ —————→ If Yes, go to H3

No ..... ☐ —————→ If No, go to H4.

H3. How often are each of the following instructional strategies used in the Precollege and Introductory Level mathematics courses (courses E1-E12) you offered for credit in Fall 2021?

|                                                                                         | At least<br>once a<br>week | Occasionally             | Almost<br>never          |
|-----------------------------------------------------------------------------------------|----------------------------|--------------------------|--------------------------|
| a. Focusing on conceptual understanding over formulas and procedures                    | <input type="checkbox"/>   | <input type="checkbox"/> | <input type="checkbox"/> |
| b. Integrating real-world applications                                                  | <input type="checkbox"/>   | <input type="checkbox"/> | <input type="checkbox"/> |
| c. Using student-centered active learning strategies                                    | <input type="checkbox"/>   | <input type="checkbox"/> | <input type="checkbox"/> |
| d. Using assessments such as regular graded homework or quizzes used to inform teaching | <input type="checkbox"/>   | <input type="checkbox"/> | <input type="checkbox"/> |

**H. Precalculus/Introductory Mathematics and Statistics Instruction (cont.)****Introductory Statistics**

H4. Does your department offer an Introductory Statistics (no calculus prerequisite) course (course F1) in Fall 2021?

Yes..... ☐ —————→ If Yes, go to H5

No ..... ☐ —————→ If No, go to H8.

H5. How many different kinds of introductory statistics courses designed for non-majors that have no calculus prerequisite does your department offer? (e.g. statistics for social scientists, for life scientists, or as general education courses for a broad audience)

1 ..... ☐

2 ..... ☐

3 ..... ☐

More than 3 ..... ☐

The following questions are about instruction in the course F1: Introductory Statistics (no calculus prerequisite) on page 15. If you offer more than one such course, choose the course that is aimed at the most general audience.

H6. How often are each of these instructional strategies used in the Introductory Statistics courses taught in your department in Fall 2021?

|                                                                                         | At least<br>once a<br>week | Occasionally             | Almost<br>never          |
|-----------------------------------------------------------------------------------------|----------------------------|--------------------------|--------------------------|
| a. Focusing on conceptual understanding over formulas and procedures                    | <input type="checkbox"/>   | <input type="checkbox"/> | <input type="checkbox"/> |
| b. Integrating real world applications                                                  | <input type="checkbox"/>   | <input type="checkbox"/> | <input type="checkbox"/> |
| c. Students collect, organize, and analyze real data                                    | <input type="checkbox"/>   | <input type="checkbox"/> | <input type="checkbox"/> |
| d. Using student-centered active learning strategies                                    | <input type="checkbox"/>   | <input type="checkbox"/> | <input type="checkbox"/> |
| e. Using assessments such as regular graded homework or quizzes used to inform teaching | <input type="checkbox"/>   | <input type="checkbox"/> | <input type="checkbox"/> |



Four-Year Mathematics Questionnaire

**H. Precalculus/Introductory Mathematics and Statistics Instruction (cont.)**

- H7. Technology used in teaching Introductory Statistics could include graphing calculators, statistical software, or online applets.

How successful is your program in adopting each of the following use of technology in your Introductory Statistics (no calculus prerequisite) courses taught in Fall 2021?

|                                                                      | Very Successful          | Somewhat Successful      | Not Successful           |
|----------------------------------------------------------------------|--------------------------|--------------------------|--------------------------|
| a. Students use technology to explore concepts                       | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| b. Instructors use technology to demonstrate concepts                | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| c. Students use technology to analyze data                           | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| d. Students' ability to use technology to solve problems is assessed | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

- H8. Are there other introductory statistics courses at your institution, offered by departments outside of the mathematical sciences?

Yes..... ☐ —————> If Yes, go to H9

No ..... ☐ —————> If No, go to H10.

- H9. Enter the Fall 2021 total enrollment in all such introductory statistics courses, offered outside of the mathematical sciences, at your institution. \_\_\_\_\_

- H10. Are there introductory data science courses at your institution offered by other departments?

Yes..... ☐ —————> If Yes, go to H11

No ..... ☐ —————> If No, go to Section I.

- H11. Enter the Fall 2021 total enrollment in all such introductory data science courses, offered outside of your department at your institution. \_\_\_\_\_

**I. Undergraduate Program (Fall 2021)**

Four-Year Mathematics Questionnaire

If you do not offer a major in a mathematical science, check here ☐ and go to I5. Otherwise go to I1.

- I1. Report the total number of your departmental majors who received their bachelor's degrees in the mathematical sciences or computer sciences between July 1, 2020 and June 30, 2021. Include joint majors and double majors<sup>1</sup>

Number of majors receiving degrees.....

- I2. Of the undergraduate degrees described in I1, please report the number who majored in each of the following categories. Each student should be reported only once. Include all double and joint majors<sup>1</sup> in your totals. Use the Other Mathematics Major category for a major in your department who does not fit into one of the listed categories.

| Area of Major                                  | Men | Women | Nonbinary |
|------------------------------------------------|-----|-------|-----------|
| a. Mathematics (including applied) .....       |     |       |           |
| b. Mathematics Education .....                 |     |       |           |
| c. Statistics .....                            |     |       |           |
| d. Computer Science .....                      |     |       |           |
| e. Actuarial Mathematics .....                 |     |       |           |
| f. Joint <sup>1</sup> Mathematics Majors ..... |     |       |           |
| g. Other Mathematics Majors .....              |     |       |           |

<sup>1</sup> A "double major" is a student who completes the degree requirements of two separate majors, one in mathematics and one in another program or department. A "joint major" is a student who completes a single major in your department that integrates courses from mathematics and some other program or department and typically requires fewer credit hours than the sum of the credit hours required by the separate majors.

**I. Undergraduate Program (Fall 2021) (cont.)**

Four-Year Mathematics Questionnaire

13. To what extent must majors in your department complete the following? Check one box in each row.

|                                                                                                                    | Required of<br>all majors | Required of some<br>but not all majors | Not required<br>of any major |
|--------------------------------------------------------------------------------------------------------------------|---------------------------|----------------------------------------|------------------------------|
| a. Modern Algebra I.....                                                                                           | <input type="checkbox"/>  | <input type="checkbox"/>               | <input type="checkbox"/>     |
| b. Real Analysis I.....                                                                                            | <input type="checkbox"/>  | <input type="checkbox"/>               | <input type="checkbox"/>     |
| c. Modern Algebra I or Real Analysis I<br>(majors may choose either to fulfill this<br>requirement) .....          | <input type="checkbox"/>  | <input type="checkbox"/>               | <input type="checkbox"/>     |
| d. A one-year upper level sequence .....                                                                           | <input type="checkbox"/>  | <input type="checkbox"/>               | <input type="checkbox"/>     |
| e. At least one computer science course .....                                                                      | <input type="checkbox"/>  | <input type="checkbox"/>               | <input type="checkbox"/>     |
| f. At least one statistics course .....                                                                            | <input type="checkbox"/>  | <input type="checkbox"/>               | <input type="checkbox"/>     |
| g. At least one applied mathematics course<br>beyond calculus level (in Section E) .....                           | <input type="checkbox"/>  | <input type="checkbox"/>               | <input type="checkbox"/>     |
| h. A capstone experience (e.g., a senior<br>project, a senior thesis, a senior seminar,<br>or an internship) ..... | <input type="checkbox"/>  | <input type="checkbox"/>               | <input type="checkbox"/>     |
| i. An exit exam (written or oral) .....                                                                            | <input type="checkbox"/>  | <input type="checkbox"/>               | <input type="checkbox"/>     |

14. Give your best estimate of the percentage of your department's graduating majors from the previous academic year 2020-21 (reported in I1) in each of the following categories. Please make the totals add to 100 percent.

|                                                                                                       |                        |
|-------------------------------------------------------------------------------------------------------|------------------------|
| a. Who went into pre-college teaching .....                                                           | <input type="text"/> % |
| b. Who went to graduate school in the mathematical sciences .....                                     | <input type="text"/> % |
| c. Who went to professional school or to graduate school outside of the<br>mathematical sciences..... | <input type="text"/> % |
| d. Who took jobs in business, industry, government, etc.....                                          | <input type="text"/> % |
| e. Who had other post-graduation plans known to the department .....                                  | <input type="text"/> % |
| f. Whose plans are not known to the department .....                                                  | <input type="text"/> % |

**I. Undergraduate Program (Fall 2021) (cont.)**

Four-Year Mathematics Questionnaire

15. Many departments today use a spectrum of program-assessment methods. Please indicate whether each of the following apply to your department's undergraduate program-assessment efforts during the last six years.

|                                                                                                                                      | Yes                      | No                       |
|--------------------------------------------------------------------------------------------------------------------------------------|--------------------------|--------------------------|
| a. We conducted a review of our undergraduate program that included one or more reviewers from outside of our institution .....      | <input type="checkbox"/> | <input type="checkbox"/> |
| b. We asked graduates of our undergraduate program to comment on and suggest changes in our undergraduate program .....              | <input type="checkbox"/> | <input type="checkbox"/> |
| c. Other departments at our institution were invited to comment on the preparation that their students received in our courses ..... | <input type="checkbox"/> | <input type="checkbox"/> |
| d. Data on our students' progress in subsequent mathematics courses were gathered and analyzed .....                                 | <input type="checkbox"/> | <input type="checkbox"/> |
| e. We have developed a set of student learning outcomes for our program .....                                                        | <input type="checkbox"/> | <input type="checkbox"/> |
| f. We have assessed student learning objectives in courses required in our major .....                                               | <input type="checkbox"/> | <input type="checkbox"/> |
| g. We have a placement system for first-year students and we gathered and analyzed data on its effectiveness .....                   | <input type="checkbox"/> | <input type="checkbox"/> |
| h. Our department's program assessment activities led to changes in our undergraduate program .....                                  | <input type="checkbox"/> | <input type="checkbox"/> |

16. Please indicate the extent to which the following activities have taken place in the past year in your department in response to increased national attention to equity, diversity, and inclusion issues (here the term “demographic” includes race, ethnicity, gender, disability status, and other characteristics of individuals).

|                                                                                                                                      | None                     | Some                     | A lot                    | Not applicable           |
|--------------------------------------------------------------------------------------------------------------------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| a. Faculty discussion designed to increase awareness of equity, diversity, and inclusion issues                                      | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |                          |
| b. Student discussion designed to increase awareness of equity, diversity, and inclusion issues                                      | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |                          |
| c. Program or policy changes to affect the demographic balance of faculty in the mathematical sciences.                              | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |                          |
| d. Program or policy changes to affect the demographic balance of undergraduate students in mathematical sciences.                   | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |                          |
| e. Program or policy changes intended to affect the demographic balance of graduate students in mathematical sciences.               | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| f. Consideration of existing or new programs to assist underrepresented groups and/or at-risk students in the mathematical sciences. | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |                          |

g. Please describe any other steps your department or institution has taken in the past year that deal with equity, diversity, and inclusion as they pertain to the study of and employment in the mathematical sciences.

**I. Undergraduate Program (Fall 2021) (cont.)**

Four-Year Mathematics Questionnaire

17. For each of the following opportunities, indicate whether or not it is available to your undergraduate mathematical sciences students through your department or institutions.

|                                                                                                       | Yes                      | No                       |
|-------------------------------------------------------------------------------------------------------|--------------------------|--------------------------|
| a. Honors sections of departmental courses .....                                                      | <input type="checkbox"/> | <input type="checkbox"/> |
| b. An undergraduate Mathematical Science Club .....                                                   | <input type="checkbox"/> | <input type="checkbox"/> |
| c. Special mathematics programs to encourage women .....                                              | <input type="checkbox"/> | <input type="checkbox"/> |
| d. Special mathematics programs to encourage minorities .....                                         | <input type="checkbox"/> | <input type="checkbox"/> |
| e. Opportunities to participate in mathematical science contests .....                                | <input type="checkbox"/> | <input type="checkbox"/> |
| f. Special mathematics statistics lectures/colloquia not part of a<br>mathematical science club ..... | <input type="checkbox"/> | <input type="checkbox"/> |
| g. Mathematical sciences outreach opportunities in local K–12<br>schools .....                        | <input type="checkbox"/> | <input type="checkbox"/> |
| h. Undergraduate research opportunities in mathematical sciences .....                                | <input type="checkbox"/> | <input type="checkbox"/> |
| i. Independent study opportunities in mathematical sciences.....                                      | <input type="checkbox"/> | <input type="checkbox"/> |
| j. Assigned faculty advisers in mathematical sciences .....                                           | <input type="checkbox"/> | <input type="checkbox"/> |
| k. Opportunity to write a senior thesis in mathematical sciences.....                                 | <input type="checkbox"/> | <input type="checkbox"/> |
| l. A career day for mathematical sciences majors .....                                                | <input type="checkbox"/> | <input type="checkbox"/> |
| m. Special advising about graduate school opportunities in<br>mathematical sciences.....              | <input type="checkbox"/> | <input type="checkbox"/> |
| n. Opportunity for an internship experience.....                                                      | <input type="checkbox"/> | <input type="checkbox"/> |
| o. Opportunity to participate in a senior seminar .....                                               | <input type="checkbox"/> | <input type="checkbox"/> |
| p. Opportunity to tutor, grade papers, or TA in the department .....                                  | <input type="checkbox"/> | <input type="checkbox"/> |
| q. Opportunity to provide mathematical or statistical consulting to<br>client.....                    | <input type="checkbox"/> | <input type="checkbox"/> |

18. Give your best estimate of the number of all of your majors who have participated in each of the following activities over the past year September 1, 2020 – August 31, 2021.

- Undergraduate research project in the mathematical sciences \_\_\_\_\_
- Internship in mathematical sciences \_\_\_\_\_
- Mathematical or statistical consulting for client \_\_\_\_\_

I. Undergraduate Program (Fall 2021) (cont.)

Four-Year Mathematics Questionnaire

19. Does your department offer interdisciplinary course(s) in any of the following areas below: (Check all that apply.) An interdisciplinary course is one in which mathematics is taught with relation to another field such as mathematics and economics, or mathematics and education; do not include calculus courses.

- a. Mathematics and finance or business ..... ☐
- b. Mathematics and biology ..... ☐
- c. Mathematics and the study of the environment ..... ☐
- d. Mathematics and engineering or the physical sciences..... ☐
- e. Mathematics and economics..... ☐
- f. Mathematics and social sciences other than economics ..... ☐
- g. Mathematics and education ..... ☐
- h. Mathematics and the humanities..... ☐
- i. Mathematics and computer science ..... ☐
- j. Mathematics and social justice..... ☐
- k. Other ..... ☐

110a. Does your department offer a minor in statistics?

- Yes..... ☐ —————> If Yes, go to I10b
- No ..... ☐ —————> If No, go to I11.

110b. How many students graduated with a minor in statistics from your department between July 1, 2020 and June 30, 2021? \_\_\_\_\_

**I. Undergraduate Program (Fall 2021) (cont.)**

Four-Year Mathematics Questionnaire

I11. Does your department offer a major in statistics?

Yes..... ☐ —————→ If Yes, go to I12No ..... ☐ —————→ If No, go to I13.

I12. To what extent must statistics majors in your department complete the following? Check one box in each row.

|                                                                                                                    | Required of<br>all majors | Required of some<br>but not all majors | Not required<br>of any major |
|--------------------------------------------------------------------------------------------------------------------|---------------------------|----------------------------------------|------------------------------|
| a. Calculus I .....                                                                                                | <input type="checkbox"/>  | <input type="checkbox"/>               | <input type="checkbox"/>     |
| b. Calculus II .....                                                                                               | <input type="checkbox"/>  | <input type="checkbox"/>               | <input type="checkbox"/>     |
| c. Multivariable Calculus.....                                                                                     | <input type="checkbox"/>  | <input type="checkbox"/>               | <input type="checkbox"/>     |
| d. Linear Algebra/Matrix Theory .....                                                                              | <input type="checkbox"/>  | <input type="checkbox"/>               | <input type="checkbox"/>     |
| e. At least one computer science course .....                                                                      | <input type="checkbox"/>  | <input type="checkbox"/>               | <input type="checkbox"/>     |
| f. At least one applied mathematics course<br>(not including a, b, c, d above) .....                               | <input type="checkbox"/>  | <input type="checkbox"/>               | <input type="checkbox"/>     |
| g. A capstone experience (e.g., a senior<br>project, a senior thesis, a senior<br>seminar, or an internship) ..... | <input type="checkbox"/>  | <input type="checkbox"/>               | <input type="checkbox"/>     |
| h. An exit exam (written or oral) .....                                                                            | <input type="checkbox"/>  | <input type="checkbox"/>               | <input type="checkbox"/>     |
| i. At least one upper level Probability course                                                                     | <input type="checkbox"/>  | <input type="checkbox"/>               | <input type="checkbox"/>     |
| j. At least one upper-level Mathematical<br>Statistics course .....                                                | <input type="checkbox"/>  | <input type="checkbox"/>               | <input type="checkbox"/>     |
| k. At least one applied statistics course .....                                                                    | <input type="checkbox"/>  | <input type="checkbox"/>               | <input type="checkbox"/>     |
| l. At least one upper-level Linear Models<br>course .....                                                          | <input type="checkbox"/>  | <input type="checkbox"/>               | <input type="checkbox"/>     |
| m. One Bayesian Inference course.....                                                                              | <input type="checkbox"/>  | <input type="checkbox"/>               | <input type="checkbox"/>     |

I13. Does your institution allow a student to meet an institutional or divisional graduation requirement in the mathematical sciences using an Advanced Placement course (taken while the student was in high school)?

Yes ..... ☐No ..... ☐



**J. Pre-service Teacher Education in Mathematics**

Four-Year Mathematics Questionnaire

**Questions regarding the preparation of pre-service grades 9-12 mathematics teachers:**

- J1. Does your institution offer a program of study that leads to obtaining credentials to teach mathematics in public high schools (any license that includes grades 9-12) in your state?
- Yes..... ☐ —————> If Yes, go to J2
- No ..... ☐ —————> If No, skip to J5.
- J2. How many semester hours of mathematics or statistics courses from your department are required by your institution's program of certification for pre-service secondary mathematics teachers (grades 9-12)? \_\_\_\_\_
- J3. How many semester hours of mathematics or statistics courses from your department with a primary focus on high school mathematics from an advanced viewpoint are required in your institution's program of certification for pre-service secondary mathematics teachers (grades 9-12)? \_\_\_\_\_

**J. Pre-service Teacher Education in Mathematics (cont.)**

Four-Year Mathematics Questionnaire

- J4. Considering the teacher preparation program at your institution, in each of the following core areas indicate whether the core area is required of all students seeking mathematics certification, if the course is generally taken by those seeking certification (if it is not required), and if in that core area your department offers a special course that is specifically designed for pre-service secondary mathematics teachers.

| Course                                                                                                                                    | Required                 |                          | If required is "No" Generally Taken |                          | Special Course Offered   |                          |
|-------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|--------------------------|-------------------------------------|--------------------------|--------------------------|--------------------------|
|                                                                                                                                           | Yes                      | No                       | Yes                                 | No                       | Yes                      | No                       |
| a. Advanced Calculus/Analysis                                                                                                             | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| b. Modern Algebra                                                                                                                         | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| c. Number Theory                                                                                                                          | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| d. Geometry                                                                                                                               | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| e. Discrete Mathematics                                                                                                                   | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| f. History of Mathematics                                                                                                                 | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| g. Introductory Statistics that includes a simulation-based approach to inference (whether or not accompanied by a normal-based approach) | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| h. Introductory Statistics that only includes a normal-based (non-simulation-based) approach to inference                                 | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| i. Statistical Methods with an introductory course as a prerequisite                                                                      | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| j. Statistical Modeling                                                                                                                   | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| k. Probability and/or statistics with calculus prerequisite                                                                               | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| l. Other (specify)<br>_____                                                                                                               | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

**J. Pre-service Teacher Education in Mathematics (cont.)**

Four-Year Mathematics Questionnaire

**Questions regarding the preparation of pre-service teachers of mathematics in grades 6-8:**

- J5. Does your institution offer a program of study that leads to obtaining credentials to teach mathematics in public middle schools (any license that includes grades 6-8) in your state?

Yes, and the program has different requirements from those for teaching mathematics at the elementary or high school levels .....

☐

Go to J6.

Yes, but the program is identical to that for either the high school mathematics teacher preparation program or the elementary mathematics teacher preparation program. ....

☐

Skip to J9.

No .....

☐

Skip to J9.

- J6. How many semester hours of courses in mathematics from your department are required by your institution's program of certification for pre-service middle grades (6-8) teachers of mathematics?

\_\_\_\_\_

- J7. How many semester hours of courses in statistics from your department are required by your institution's program of certification for pre-service middle grades (6-8) teachers of mathematics?

\_\_\_\_\_

- J8. How many semester hours of courses from your department on fundamental ideas of mathematics appropriate for middle grade teachers are required by your institution's program of certification for pre-service middle grades (6-8) teachers of mathematics? \_\_\_\_\_

**J. Pre-service Teacher Education in Mathematics (cont.)** Four-Year Mathematics Questionnaire**Questions regarding the preparation of pre-service teachers of mathematics in grades K-5.**

- J9. Does your institution offer a program of study that leads to obtaining credentials to teach mathematics in public elementary schools (any license that includes grades K-5) in your state?
- Yes..... ☐ —————> If Yes, go to J10
- No ..... ☐ —————> If No, skip to Section K.
- J10. How many semester hours of courses in mathematics from your department are required by your institution's program of certification for pre-service elementary grades (K-5) teachers of mathematics? \_\_\_\_\_
- J11. How many semester hours of courses in statistics from your department are required by your institution's program of certification for pre-service elementary grades (K-5) teachers of mathematics? \_\_\_\_\_
- J12. How many semester hours of courses from your department on fundamental ideas of mathematics appropriate for elementary teachers are required by your institution's program of certification for prospective elementary grades (K-5) teachers of mathematics? \_\_\_\_\_

**K. Comments and Suggestions**

Four-Year Mathematics Questionnaire

If you found some question(s) difficult to interpret or answer, please let us know. We welcome suggestions to improve future surveys (e.g., CBMS 2025).

Comments:

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*Thank you for completing this questionnaire. We know it was a time-consuming process and we hope that the resulting survey report, which we hope to publish in spring 2023, will be of use to you and your department.*

*Please keep a copy of your responses to this questionnaire in case questions arise.*

## Appendix VI

|                                                                                                                                                                                                                                                                                                           |                                                                                      |                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------|
|                                                                                                                                                                                                                          | <b>Annual Survey</b><br>of the Mathematical Sciences<br>AMS • ASA • IMS • MAA • SIAM | <b>Other Full-time Faculty Survey</b> |
| <a href="http://www.ams.org/annual-survey/forms">www.ams.org/annual-survey/forms</a>                                                                                                                                                                                                                      |                                                                                      |                                       |
| Return to: American Mathematical Society • P.O. Box 6248 • Providence, RI 02940-6248<br>Email: <a href="mailto:ams-survey@ams.org">ams-survey@ams.org</a> • Fax: 401-331-3842 (Attn: Surveys) • Tel: 800-321-4267, ext. 4189<br>Web: <a href="http://www.ams.org/deptprofile">www.ams.org/deptprofile</a> |                                                                                      |                                       |

Institution name: \_\_\_\_\_

Department name: \_\_\_\_\_

City/State/Zipcode: \_\_\_\_\_ Date: \_\_\_\_\_

Name of person completing form: \_\_\_\_\_ Title: \_\_\_\_\_

Email: \_\_\_\_\_ Phone: \_\_\_\_\_

Please complete this form by February 25, 2022, keeping a copy for your records, and return it to the above address.

The purpose of this brief questionnaire is to obtain detailed information about the early career arcs of individuals with PhDs in mathematical sciences. The results of this survey will be reported in the next 2021 CBMS Statistical Abstract of Undergraduate Programs in the Mathematical Sciences in the United States.

1. Indicate the number of individuals in your department in 2020–21 who were postdoctoral faculty (those in a temporary position primarily intended to provide an opportunity to extend graduate training or to further research experience) \_\_\_\_\_ [1.a]; of these how many are not classifiable as postdoctoral research faculty in your department in 2021–22 \_\_\_\_\_ [1.b] (include postdocs who remain in your department in a different appointment).

Of those reported in [1.b], give the number whose employment status in 2021–22 (at your institution or elsewhere) is

- |                                                       |                                        |
|-------------------------------------------------------|----------------------------------------|
| _____ [1.c] A tenure-track appointment                | _____ [1.g] A non-academic appointment |
| _____ [1.d] Another postdoctoral research appointment | _____ [1.h] Unemployed                 |
| _____ [1.e] A renewable appointment                   | _____ [1.i] Unknown                    |
| _____ [1.f] A non-renewable appointment               |                                        |

2. Indicate the number of faculty in your department during 2020–21 not counted in [1.a] who were in renewable non-tenure-track positions (e.g. lecturer, teaching professional, professor of the practice) \_\_\_\_\_ [2.a]; of these how many are not in your department in 2021–22 \_\_\_\_\_ [2.b].

Indicate the number of faculty in your department in 2021–22 in renewable non-tenure-track, non-postdoc positions \_\_\_\_\_ [2.c]

Of those reported in [2.c], indicate the number who are typically engaged in each of the following activities (note that the sum of the values entered here may be larger than that in [2.c]):

- |                                                                |                                                         |
|----------------------------------------------------------------|---------------------------------------------------------|
| _____ [2.d] Teaching                                           | _____ [2.i] Advise undergraduate research projects      |
| _____ [2.e] Research                                           | _____ [2.j] Serve as academic advisor to undergraduates |
| _____ [2.f] Attend research conferences with financial support | _____ [2.k] Serve on university/college committees      |
| _____ [2.g] Attend teaching conferences with financial support | _____ [2.l] Serve as department course coordinators     |
| _____ [2.h] Serve on department committees                     |                                                         |

3. Indicate the number of faculty in your department during 2020–21 not counted in [1.a] who were in fixed-term (not renewable) non-tenure-track positions \_\_\_\_\_ [3.a]; of these how many are not in your department in 2021–22 \_\_\_\_\_ [3.b].

Indicate the number of faculty in your department in 2021–22 in fixed-term (not renewable) non-tenure-track, non-postdoc positions \_\_\_\_\_ [3.c]

Of those reported in [3.c], indicate the number who are typically engaged in each of the following activities (note that the sum of the values entered here may be larger than that in [3.c]):

- |                                                                |                                                         |
|----------------------------------------------------------------|---------------------------------------------------------|
| _____ [3.d] Teaching                                           | _____ [3.i] Advise undergraduate research projects      |
| _____ [3.e] Research                                           | _____ [3.j] Serve as academic advisor to undergraduates |
| _____ [3.f] Attend research conferences with financial support | _____ [3.k] Serve on university/college committees      |
| _____ [3.g] Attend teaching conferences with financial support | _____ [3.l] Serve as department course coordinators     |
| _____ [3.h] Serve on department committees                     |                                                         |



## **Two-Year Mathematics Questionnaire**



**SURVEY OF UNDERGRADUATE PROGRAMS IN THE MATHEMATICAL SCIENCES**



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**General Information**Two-Year College Mathematics Questionnaire

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As part of a random sample, your department has been selected to participate in the CBMS 2021 National Survey, the importance of which has been endorsed by all of our major professional societies. Please read the instructions in each section carefully and complete all of the pertinent items as indicated.

If your college does not have a departmental or divisional structure, consider the group of all mathematics instructors to be the "mathematics department" for the purpose of this survey.

If you belong to a multi-campus system, please report for the entire multi-campus system to which you belong. If it is not appropriate to report for the entire system (i.e., because at least one campus directly administers its own program), please call Westat at 1-855-770-0558.

This questionnaire should be completed by the person who is directly in charge of the mathematics program or department on your campus. It may be that a different person, such as the college Registrar or a person in Institutional Research, is best equipped to answer Section F on course enrollment; we provide a method for delegating that section to another person in case that will help you. Please contact Westat at 855-770-0558 or send an email to [cbms2021@westat.com](mailto:cbms2021@westat.com) if you would like to delegate a section to another person.

Report on all of your courses and instructors that fall under the general heading of the mathematics program or department. Include all mathematics and statistics courses taught within your mathematics program or department.

We have classified your department as belonging to a two-year college, to a college or campus within a two-year system, or to a two-year branch of a university system. If this is not correct, please contact Richelle (Rikki) Blair at the email address or telephone number given below.

We recommend completing this questionnaire online because the online system will automatically skip those questions that are not applicable to you (based on the responses you give). However, this survey may be completed using a hard-copy questionnaire.

If you have any questions, please contact Richelle (Rikki) Blair, Associate Director for Two-Year Colleges, by email at [richelle.blair@sbcglobal.net](mailto:richelle.blair@sbcglobal.net) or by phone at 440-212-5965. For help with the online questionnaire, call Westat at 855-770-0558 or send an email to [cbms2021@westat.com](mailto:cbms2021@westat.com).

**Please return your completed questionnaire by October 29, 2021, either online or by mailing a hard copy to:**

CBMS Survey  
Westat  
1600 Research Boulevard  
Rockville, MD 20850-3129

**Please retain a copy of your responses to this questionnaire in case questions arise.**

**A. General Information**

Two-Year College Mathematics Questionnaire

**PLEASE PRINT CLEARLY**

A1. Name of your campus: \_\_\_\_\_

A2. Name of your department: \_\_\_\_\_

A3. Mailing address of the multi-campus organization to which your campus belongs (if any). (Write NA if your campus does not belong to a multi-campus organization.)  
\_\_\_\_\_  
\_\_\_\_\_

A4. We have classified your department as belonging to a two-year college or to a college campus within a two-year college system, or to a two-year branch of a university system. Do you agree?

Yes..... ☐ → go to the next question.No ..... ☐ → please contact Richelle (Rikki) Blair,  
Survey Associate Director, by email  
([richelle.blair@sbcglobal.net](mailto:richelle.blair@sbcglobal.net)) or by phone  
(440-212-5965) before proceeding any further.

A5. What is the unit (= academic discipline group) that most directly administers the mathematics program on your campus? (Check one box.)

Mathematics Department (department does not offer Computer Science) ..... ☐Mathematics and Computer Science Department or Division (department also offers Computer Science, whether or not it is part of the title) ..... ☐Mathematics and Science Department or Division ..... ☐Other Departments or Division ..... ☐

**A. General Information (cont.)**

Two-Year College Mathematics Questionnaire

A6. To help us project enrollment for the current academic year (2021–2022), please give the following enrollment figures for the previous academic year (2020–2021) not counting summer enrollment.

- a. Fall 2020 total student enrollment in your mathematics courses .....
- b. Entire academic year 2020–2021 enrollment in your mathematics courses .....
- c. Calculus II total enrollment in winter/spring 2021.....
- d. Calculus II total number of sections in winter/spring 2021 .....

A7. Does your college organize its **developmental education**, including mathematics, in a separately administered department or division?

Yes ..... ☐

No ..... ☐

A8. Your name or contact person  
in your department:

A9. Your email address or contact  
person's e-mail address:

A10. Your phone number or contact  
person's phone number including  
area code:

A11. Campus mailing address:

**B. Mathematics Faculty in Mathematics Department/Program (Fall 2021)**

- If you belong to a multi-campus system, please report for the entire system.
- Bold faculty categories defined in this section will be used in later sections.

B1. For fall 2021, what is the **total number of full-time mathematics faculty in your department/program**, both permanent and temporary, including those on leave, on sabbatical, or have otherwise temporarily left the department?

Number of full-time mathematics faculty .....

B2. Of the number in B1, how many are tenured, tenure-eligible, on a tenure-track, or otherwise considered to be permanent faculty (including appointive and probationary faculty, and those who are on leave or sabbatical)? We will refer to these as "**permanent full-time faculty**."

Number of permanent full-time faculty .....

B3. Of the number in B1 and excluding those in B2, how many are continuing full-time faculty? We will refer to these as "**non-tenure-track full-time faculty**."

Number of non-tenure track full-time faculty .....

B4. Give the number of "**other full-time faculty**" by computing B1 minus (B2 and B3)

B5. For the **permanent full-time faculty** reported in B2,

a. give the required teaching assignment in weekly contact hours .....

b. give the number of office hours required weekly in association with the teaching assignment in B5a (count all office hours, including those offered online).....

B6. For the **permanent full-time faculty** reported in B2, how many teach extra credit hours for extra pay at your campus or within your organization?

Number who teach extra credit hours for extra pay at your campus or within your organization .....

Two-Year College Mathematics Questionnaire

**B. Mathematics Faculty in Mathematics Department/Program (Fall 2021) (cont.)**

B7. For the **permanent full-time faculty** reported in B2, how many permanent faculty teach extra credit hours per week in the following categories?

a. Number who teach 1–3 credit hours extra weekly.....

b. Number who teach 4–6 credit hours extra weekly .....

c. Number who teach 7 or more credit hours extra weekly .....

B8. For fall 2021, how many **part-time mathematics faculty** are teaching in your department? (Note: none of these were reported above.)

a. Number of part-time mathematics faculty **paid by your college** .....

b. Number of part-time faculty paid only by a third party, such as a school district paying faculty who teach dual-enrollment courses (= courses taught in high school by high school teachers for which students may obtain high school credit and simultaneous college credit through your institution) ....

c. **Total number of part-time faculty** (add B8a and B8b) .....

B9. How many **part-time faculty paid by your college** (reported in B8a) teach 6 or more hours per week?

Number in B8a teaching 6 or more hours/week .....

B10. Are office hours required by college policy for the **part-time faculty paid by your college** (reported in B8a)?

Yes ..... ☐

No ..... ☐

**C. Distance/Remote Learning**

Two-Year College Mathematics Questionnaire

**Definition:** Distance/remote learning courses are courses offered by your institution for credit, in which half or more of the instruction occurs with the instructor and the students separated by time and/or place and facilitated by technology (e.g. courses in which the majority of the course is taught online, or by computer software, or by other technologies). Exclude distance/remote courses that temporarily moved to remote due to the pandemic.

C1. Overall, how have attitudes towards online learning changed as a result of the COVID-19 experience?

|                                        | More favorable           | No change                | Less favorable           |
|----------------------------------------|--------------------------|--------------------------|--------------------------|
| a. Faculty interest in online teaching | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| b. Faculty use of online tools         | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| c. Student interest in online teaching | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| d. Student use of online tools         | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

- C2. Many colleges have conducted online instruction as a way of addressing the COVID-19 pandemic, offering online instruction either as an alternative to face-to-face instruction or as a supplement to face-to-face instruction. How is that instruction coordinated with what has normally been called distance education? Please indicate which of the following applies to your department for each listed time period, using the following definitions.

- On-campus (face-to-face) learners—students who would be expected to attend most classes in person.
- Remote learners—students who would be expected to attend most classes remotely.

| Policy                                                                                                                                                                                    | Prior to pandemic<br>(prior to spring 2020) |                          | During pandemic<br>(spring 2020-summer 2021) |                          | Fall 2021                |                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|--------------------------|----------------------------------------------|--------------------------|--------------------------|--------------------------|
|                                                                                                                                                                                           | Yes                                         | No                       | Yes                                          | No                       | Yes                      | No                       |
| a. We offer some course sections only to on-campus (face-to-face) learners; remote learners might be offered the same course, but the students would not normally be in the same section. | <input type="checkbox"/>                    | <input type="checkbox"/> | <input type="checkbox"/>                     | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| b. We offer some course sections only to remote learners; on-campus (face-to-face) learners might be offered the same course, but the students would not normally be in the same section. | <input type="checkbox"/>                    | <input type="checkbox"/> | <input type="checkbox"/>                     | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| c. We offer some "hybrid" courses to on-campus (face-to-face) learners and remote learners in the same section.                                                                           | <input type="checkbox"/>                    | <input type="checkbox"/> | <input type="checkbox"/>                     | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

- C3. Has your department taught any distance/remote learning courses **(other than courses moved online in response to the COVID-19 pandemic)** in 2018-2021?

Yes..... ☐ —————> go to C4.

No ..... ☐ —————> skip to D1.

- C4. Which best characterizes the delivery format of the majority of your distance/remote learning courses (not including courses moved online in response to the COVID-19 pandemic)? (Check one box.)

Sections are taught only online, and only asynchronously ..... ☐

Sections are taught only online, with an opportunity to meet synchronously online..... ☐

Sections use a mixture of online and face-to-face sessions ..... ☐

Other ..... ☐

**C. Distance/Remote Learning (cont.)**

Two-Year College Mathematics Questionnaire

- C5. In most of your distance/remote learning courses, how and where do students take the majority of their tests (not including courses moved online in response to the COVID-19 pandemic)? (Check one box.)

Online, not monitored ..... ☐

Online, but using some kind of monitoring technology ..... ☐

At a monitored testing site ..... ☐

Combination of the above ..... ☐

- C6. Give the maximum number of hours of the weekly teaching assignment that can be met by teaching distance/remote learning classes (prior to the COVID-19 pandemic) as defined above.

No limit ☐

- C7. Rate the following challenges that your department faces when creating and/or offering distance/remote learning mathematics courses. (Please check one box in each line.)

| Challenge                                                                | Not a challenge          | Somewhat of a challenge  | Very significant challenge |
|--------------------------------------------------------------------------|--------------------------|--------------------------|----------------------------|
| a. Designing appropriate assessments of student learning.                | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>   |
| b. Maintaining academic integrity on assessments.                        | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>   |
| c. Grade inflation.                                                      | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>   |
| d. Building/maintaining community among faculty and students.            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>   |
| e. Maintaining academic quality instruction.                             | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>   |
| f. Engaging students online.                                             | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>   |
| g. Replicating active learning in a virtual environment.                 | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>   |
| h. Availability of equipment and technical support for faculty/students. | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>   |



**D. Pathways and Instructional Strategies**

Two-Year College Mathematics Questionnaire

D1. Has your mathematics department or developmental education department offered a "pathways" course sequence within the last five years?

("Pathways" is defined to be a single course or course sequence that enables students to complete a college-level gateway mathematics or statistics course that is aligned to students' academic and/or career goals within one academic year.)

Yes ..... ☐ Go to D2

No..... ☐ Go to D3

D2. Which of the following pathways courses or course sequences have you offered within the last five years?

|                                                 | Implemented?             |                          | Intermediate Algebra a pre-requisite? |                          |
|-------------------------------------------------|--------------------------|--------------------------|---------------------------------------|--------------------------|
|                                                 | Yes                      | No                       | Yes                                   | No                       |
| a. Quantitative Reasoning/Quantitative Literacy | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>              | <input type="checkbox"/> |
| b. Statistics                                   | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>              | <input type="checkbox"/> |
| c. STEM                                         | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>              | <input type="checkbox"/> |
| d. Other (specify)                              | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>              | <input type="checkbox"/> |

D3. How often are each of the following instructional strategies used in courses below the level of Calculus (courses F1-F10 in Section F) you offered for credit in Fall 2021?

|                                                                                                                 | At least once a week     | Occasionally             | Almost never             |
|-----------------------------------------------------------------------------------------------------------------|--------------------------|--------------------------|--------------------------|
| a. Focusing on conceptual understanding to support the acquisition and understanding of formulas and procedures | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| b. Integrating real-world applications                                                                          | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| c. Using student-centered active learning strategies                                                            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| d. Using assessments such as regular graded homework or quizzes used to inform teaching                         | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

- D4. How often are each of these instructional strategies used in the Introductory Statistics courses taught in your department in Fall 2021?

|                                                                                                                 | At least<br>once a<br>week | Occasionally             | Almost<br>never          |
|-----------------------------------------------------------------------------------------------------------------|----------------------------|--------------------------|--------------------------|
| a. Focusing on conceptual understanding to support the acquisition and understanding of formulas and procedures | <input type="checkbox"/>   | <input type="checkbox"/> | <input type="checkbox"/> |
| b. Integrating real-world applications                                                                          | <input type="checkbox"/>   | <input type="checkbox"/> | <input type="checkbox"/> |
| c. Using student-centered active learning strategies                                                            | <input type="checkbox"/>   | <input type="checkbox"/> | <input type="checkbox"/> |
| d. Using assessments such as regular graded homework or quizzes used to inform teaching                         | <input type="checkbox"/>   | <input type="checkbox"/> | <input type="checkbox"/> |

- D5. Technology used in teaching Introductory Statistics could include graphing calculators, statistical software, or online applets. How successful is your program in adopting each of the following use of technology in your Introductory Statistics (no calculus prerequisite) courses taught Fall 2021?

|                                                                      | Very<br>Successful       | Somewhat<br>Successful   | Not<br>Successful        |
|----------------------------------------------------------------------|--------------------------|--------------------------|--------------------------|
| a. Students use technology to explore concepts                       | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| b. Instructors use technology to demonstrate concepts                | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| c. Students use technology to analyze data                           | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| e. Students' ability to use technology to solve problems is assessed | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

**E. Dual Enrollment Courses**

Two-Year College Mathematics Questionnaire

- If you belong to a multi-campus system, please report for the entire system.
- **Definition:** We use the term “dual/concurrent-enrollment courses” to refer to courses conducted on a high school campus **by high school teachers**, for which students may obtain high school credit and, simultaneously, college credit through your institution.

E1. Does your department participate in any dual-enrollment program of the type defined above?

Yes..... ☐ —————> go to E2.

No ..... ☐ —————> go to Section F.

E2. Please provide the head-count enrollment for your dual-enrollment program (as defined above) for the spring term of 2021 and for the current fall term of 2021.

| Course             | Total Dual<br>Enrollments<br>Last Term =<br>Spring 2021 | Total Dual<br>Enrollments<br>This Term =<br>Fall 2021 |
|--------------------|---------------------------------------------------------|-------------------------------------------------------|
| a. College Algebra |                                                         |                                                       |
| b. Precalculus     |                                                         |                                                       |
| c. Calculus I      |                                                         |                                                       |
| d. Statistics      |                                                         |                                                       |
| e. Other           |                                                         |                                                       |

E3. Does your department assign any of its own full-time or part-time faculty to teach courses in a high school for which high school students may receive both high school and college credit through your institution?

Yes..... ☐ —————> go to E4.

No ..... ☐ —————> go to Section F.

E4. In fall 2021, how many students are enrolled in the courses conducted on a high school campus and taught by your full-time or part-time faculty (who are not employees of the high school) and through which high school students may receive both high school and college credit through your institution?

Number of students.....

**F. Mathematics Courses (Fall 2021)**

Two-Year College Mathematics Questionnaire

Which of the following courses are taught in your department in Fall 2021? You may use different titles for these courses and may have multiple courses that match a particular course name. Choose the course in the list that best matches the mathematical content of your course(s). If you belong to a multi-campus system, please report for the entire system.

- Do **not** include courses taught in other departments, learning centers, or developmental/remedial programs separate from your mathematics program or department.
- For each course that you offer, please indicate whether co-requisites are available (supports for underprepared students so they may be placed directly into college-level courses).
- Please also indicate which catalog codes are used to identify those courses. This information will be used to generate a reduced course list suitable for your department so reporting on enrollments will be easier.

| Name of Course (or equivalent)                                                    | Offer course in Fall 2021 |                          | Co-requisite available   |                          | Catalog code(s) used (use comma to separate codes) |
|-----------------------------------------------------------------------------------|---------------------------|--------------------------|--------------------------|--------------------------|----------------------------------------------------|
|                                                                                   | Yes                       | No                       | Yes                      | No                       |                                                    |
| <b>Developmental/Pre-College Courses</b>                                          |                           |                          |                          |                          |                                                    |
| F1. Course(s) that are primarily Arithmetic/Basic Mathematics content             | <input type="checkbox"/>  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |                                                    |
| F2. Course(s) that are primarily Pre-Algebra content                              | <input type="checkbox"/>  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |                                                    |
| F3. Course(s) that are primarily Elementary Algebra (high school level) content   | <input type="checkbox"/>  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |                                                    |
| F4. Course(s) that are primarily Intermediate Algebra (high school level) content | <input type="checkbox"/>  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |                                                    |
| F5. Geometry (high school level)                                                  | <input type="checkbox"/>  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |                                                    |
| F6. Other developmental mathematics courses                                       | <input type="checkbox"/>  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |                                                    |
| <b>College Algebra/Trigonometry/Precalculus</b>                                   |                           |                          |                          |                          |                                                    |
| F7. College Algebra (level beyond intermediate Algebra)                           | <input type="checkbox"/>  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |                                                    |
| F8. Trigonometry                                                                  | <input type="checkbox"/>  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |                                                    |
| F9. College Algebra and Trigonometry, combined                                    | <input type="checkbox"/>  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |                                                    |
| F10. Introduction to Mathematical Modeling                                        | <input type="checkbox"/>  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |                                                    |
| F11. Precalculus/Elementary Functions/Analytic Geometry                           | <input type="checkbox"/>  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |                                                    |
| <b>Calculus/Differential Equations/Linear Algebra</b>                             |                           |                          |                          |                          |                                                    |
| F12. Mainstream Calculus I <sup>a</sup>                                           | <input type="checkbox"/>  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |                                                    |
| F13. Mainstream Calculus II <sup>a</sup>                                          | <input type="checkbox"/>  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |                                                    |
| F14. Mainstream Calculus III <sup>a</sup>                                         | <input type="checkbox"/>  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |                                                    |
| F15. Non-Mainstream Calculus I <sup>b</sup>                                       | <input type="checkbox"/>  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |                                                    |
| F16. Non-Mainstream Calculus II <sup>b</sup>                                      | <input type="checkbox"/>  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |                                                    |
| F17. Differential Equations                                                       | <input type="checkbox"/>  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |                                                    |
| F18. Linear Algebra                                                               | <input type="checkbox"/>  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |                                                    |
| <b>Discrete Math/Statistics/Probability/Finite Math</b>                           |                           |                          |                          |                          |                                                    |
| F19. Discrete Mathematics                                                         | <input type="checkbox"/>  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |                                                    |
| F20. Elementary Statistics (with or without probability) <sup>c</sup>             | <input type="checkbox"/>  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |                                                    |
| F21. Probability (with or without statistics) <sup>c</sup>                        | <input type="checkbox"/>  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |                                                    |
| F22. Finite Mathematics                                                           | <input type="checkbox"/>  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |                                                    |

<sup>a</sup> Typically a trigonometry-based course for mathematics, physical sciences, and engineering majors.

<sup>b</sup> Typically non-trigonometry-based, for business, life sciences, and social science majors.

<sup>c</sup> Do **not** count the same course in both lines F20 and F21.

**F. Mathematics Courses (Fall 2021) (cont.)**

Two-Year College Mathematics Questionnaire

| Name of Course (or equivalent)                              | Offer course in Fall 2021 |                          | Co-requisite available   |                          | Catalog code(s) used (use comma to separate codes) |
|-------------------------------------------------------------|---------------------------|--------------------------|--------------------------|--------------------------|----------------------------------------------------|
|                                                             | Yes                       | No                       | Yes                      | No                       |                                                    |
| <b>Mathematics for Liberal Arts/Quantitative Literacy</b>   |                           |                          |                          |                          |                                                    |
| F23. Mathematics for Liberal Arts/ Math Appreciation        | <input type="checkbox"/>  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |                                                    |
| F24. Quantitative Literacy/Quantitative Reasoning           | <input type="checkbox"/>  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |                                                    |
| <b>Mathematics Courses for Teacher Preparation</b>          |                           |                          |                          |                          |                                                    |
| F25. Mathematics for Elementary School Teachers I           | <input type="checkbox"/>  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |                                                    |
| F26. Mathematics for Elementary School Teachers II          | <input type="checkbox"/>  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |                                                    |
| F27. Other Mathematics Courses for Teacher Preparation      | <input type="checkbox"/>  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |                                                    |
| <b>Business/Technical Mathematics</b>                       |                           |                          |                          |                          |                                                    |
| F28. Business Mathematics                                   | <input type="checkbox"/>  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |                                                    |
| F29. Non-Calculus-Based Technical Mathematics               | <input type="checkbox"/>  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |                                                    |
| F30. Calculus-Based Technical Mathematics (transfer course) | <input type="checkbox"/>  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |                                                    |
| <b>Other Mathematics Courses</b>                            |                           |                          |                          |                          |                                                    |
| F31. Other Mathematics Courses (non-transfer)               | <input type="checkbox"/>  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |                                                    |
| F32. Other Mathematics Courses (transfer)                   | <input type="checkbox"/>  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |                                                    |

## Two-Year College Mathematics Questionnaire

- ◆ Cells left blank will be interpreted as zeros

[illegible]

<sup>a</sup> Do not include distance/remote learning students, who are counted in column d.

b Distance/remote learning courses are courses offered by your institution for credit, in which half or more of the instruction occurs with the instructor and the students separated by time and/or place and facilitated by technology (e.g., courses in which the majority of the course is taught online either synchronously or asynchronously, or by computer software or other technologies).

**F. Mathematics Courses (Fall 2021) (cont.)**

Two-Year College Mathematics Questionnaire

You reported a total of # sections in fall 2021, distributed by course type as shown below. For each course type, please provide the number of sections taught by part-time faculty.

◆ Cells left blank will be interpreted as zeros

| Type of course and your applicable catalog course codes           | Total number of sections<br>(a) | Number of sections taught by part-time faculty<br>(b) |
|-------------------------------------------------------------------|---------------------------------|-------------------------------------------------------|
| Precollege level (courses F1-F6)<br>(course code list)            |                                 |                                                       |
| Precalculus level (courses F7-F11)<br>(course code list)          |                                 |                                                       |
| Mainstream calculus (courses F12-F14)<br>(course code list)       |                                 |                                                       |
| Non-mainstream calculus (courses F15-F16)<br>(course code list)   |                                 |                                                       |
| Advanced level (courses F17-F19)<br>(course code list)            |                                 |                                                       |
| Statistics, Probability (courses F20-F21)<br>(course code list)   |                                 |                                                       |
| Service courses (courses F22-F28)<br>(course code list)           |                                 |                                                       |
| Technical mathematics (courses F29-F30)<br>(course code list)     |                                 |                                                       |
| Other mathematics courses (courses F31-F32)<br>(course code list) |                                 |                                                       |

G. Faculty Educational Level, by Subject Field

Two-Year College Mathematics Questionnaire

G1. For the **permanent full-time faculty** (including those on leave or sabbatical) reported in B2 ( ), complete the following table showing the area of each faculty member's highest earned degree. The total of all faculty listed in this table should equal , the number reported in B2 (on page 3).

- If you belong to a multi-campus system, please report for the entire system.

| Highest Degree | Number of Full-Time Faculty by Major Field of Highest Degree |            |                       |
|----------------|--------------------------------------------------------------|------------|-----------------------|
|                | Mathematics                                                  | Statistics | Mathematics Education |
| a. Doctorate   |                                                              |            |                       |
| b. Master's    |                                                              |            |                       |
| c. Bachelor's  |                                                              |            |                       |



G. Faculty Educational Level, by Subject Field (cont.)

Two-Year College Mathematics Questionnaire

- G2. For the **part-time faculty** reported in B8c ( ) (including those paid by your college and those paid by a third party), complete the following table showing the area of each faculty member's highest earned degree. The total of all faculty listed in this table should equal , the number reported in B8c (on page 4).
- If you belong to a multi-campus system, please report for the entire system.

| Highest Degree | Number of Part-Time Faculty by Major Field of Highest Degree |            |                       |       |
|----------------|--------------------------------------------------------------|------------|-----------------------|-------|
|                | Mathematics                                                  | Statistics | Mathematics Education | Other |
| a. Doctorate   |                                                              |            |                       |       |
| b. Master's    |                                                              |            |                       |       |
| c. Bachelor's  |                                                              |            |                       |       |

## H. Faculty by Gender and Ethnicity/Race

Two-Year College Mathematics Questionnaire

### Instructions:

- If you belong to a multi-campus system, please report for the entire system.
- Include the data for the mathematics courses that are considered to be within YOUR mathematics department.
- For the **permanent full-time faculty** (including those on leave) reported in B2 and for the **part-time faculty** reported in B8a (those paid by your college), complete the following table giving data about gender and ethnicity/race.
- The total of full-time faculty should equal the figure given in B2. The total of part-time faculty should equal   , the figure reported in B8a.

| Ethnic/Racial Status and Gender Identity             |            |  | Number of Faculty                   |                            |
|------------------------------------------------------|------------|--|-------------------------------------|----------------------------|
|                                                      |            |  | Permanent Full-Time Faculty From B2 | Part-Time Faculty From B8a |
| 1. American Indian, Alaskan Native                   | Men        |  | Age < 40                            | Age > 40                   |
|                                                      | Women      |  |                                     |                            |
|                                                      | Non-binary |  |                                     |                            |
| 2. Asian                                             | Men        |  |                                     |                            |
|                                                      | Women      |  |                                     |                            |
|                                                      | Non-binary |  |                                     |                            |
| 3. Black or African American (non- Hispanic)         | Men        |  |                                     |                            |
|                                                      | Women      |  |                                     |                            |
|                                                      | Non-binary |  |                                     |                            |
| 4. Mexican-American, Puerto Rican, or other Hispanic | Men        |  |                                     |                            |
|                                                      | Women      |  |                                     |                            |
|                                                      | Non-binary |  |                                     |                            |
| 5. White (non-Hispanic)                              | Men        |  |                                     |                            |
|                                                      | Women      |  |                                     |                            |

|                                      |            |  |  |  |  |
|--------------------------------------|------------|--|--|--|--|
|                                      | Non-binary |  |  |  |  |
| 6. Native Hawaiian, Pacific Islander | Men        |  |  |  |  |
|                                      | Women      |  |  |  |  |
|                                      | Non-binary |  |  |  |  |
| 7. Status not known or other         | Men        |  |  |  |  |
|                                      | Women      |  |  |  |  |
|                                      |            |  |  |  |  |
|                                      | Non-binary |  |  |  |  |

I. Faculty Age Profile

- I1. Complete the following table showing the number of faculty who belong in each of the age categories below.
- Consider only the permanent full-time faculty (including those on leave) reported in B2 (on page 3).
  - If you belong to a multi-campus system, please report for the entire system.
  - The total faculty listed should equal the number reported in B2.

| Age                 | Number of Faculty |       |            |
|---------------------|-------------------|-------|------------|
|                     | Men               | Women | Non-binary |
| a. Under 30 .....   |                   |       |            |
| b. 30-34 .....      |                   |       |            |
| c. 35-39 .....      |                   |       |            |
| d. 40-44 .....      |                   |       |            |
| e. 45-49 .....      |                   |       |            |
| f. 50-54 .....      |                   |       |            |
| g. 55-59 .....      |                   |       |            |
| h. 60-64 .....      |                   |       |            |
| i. 65-69 .....      |                   |       |            |
| j. 70 and over..... |                   |       |            |

**J. Faculty Employment and Mobility**

Two-Year College Mathematics Questionnaire

- If you belong to a multi-campus system, please report for the entire system.

J1. How many of the        **permanent full-time faculty** members you reported in B2 (on page 3) were newly appointed to a permanent full-time position this year (2021–2022)?

Number of faculty newly appointed on a permanent full-time position this year (2021-2022)?

if 0                      → go to J3.

if 1 or more → go to J2.

J2. Of the faculty members counted in J1, how many had the following as their main activity in the academic year preceding their appointment? Report only **one** main activity per person. The total in J2 should equal       , the number reported in J1 above.

- |                                                                      |                      |
|----------------------------------------------------------------------|----------------------|
| a. Attending graduate school.....                                    | <input type="text"/> |
| b. Teaching in a four-year college or university .....               | <input type="text"/> |
| c. Teaching in another two-year college .....                        | <input type="text"/> |
| d. Teaching in a secondary school .....                              | <input type="text"/> |
| e. Part-time or full-time temporary employment by your college ..... | <input type="text"/> |
| f. Nonacademic employment .....                                      | <input type="text"/> |
| g. Unemployed .....                                                  | <input type="text"/> |
| h. Status unknown .....                                              | <input type="text"/> |

J3. How many of your faculty who were **permanent full-time faculty** in the previous year (2020-2021) are no longer part of your **permanent full-time faculty**?.....

**J. Faculty Employment and Mobility (cont.)**

Two-Year College Mathematics Questionnaire

- J4. For each newly appointed **permanent full-time faculty** member reported in J1, give the following data. Copy this page to add more faculty if necessary. For each new hire, check one box in each column.

|             | Gender Identity                                                                                                    | Ethnicity/Race                                                                                                                                                                                                                                     | Highest Degree Earned                                                                                                            |
|-------------|--------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| New Hire #1 | Man ..... <input type="checkbox"/><br>Woman ..... <input type="checkbox"/><br>Other ..... <input type="checkbox"/> | Am Indian ..... <input type="checkbox"/><br>Asian ..... <input type="checkbox"/><br>Black..... <input type="checkbox"/><br>Hispanic ..... <input type="checkbox"/><br>White ..... <input type="checkbox"/><br>Other ..... <input type="checkbox"/> | Bachelor's ..... <input type="checkbox"/><br>Master's ..... <input type="checkbox"/><br>Doctorate ..... <input type="checkbox"/> |
| New Hire #2 | Man ..... <input type="checkbox"/><br>Woman ..... <input type="checkbox"/><br>Other ..... <input type="checkbox"/> | Am Indian ..... <input type="checkbox"/><br>Asian ..... <input type="checkbox"/><br>Black..... <input type="checkbox"/><br>Hispanic ..... <input type="checkbox"/><br>White ..... <input type="checkbox"/><br>Other ..... <input type="checkbox"/> | Bachelor's ..... <input type="checkbox"/><br>Master's ..... <input type="checkbox"/><br>Doctorate ..... <input type="checkbox"/> |
| New Hire #3 | Man ..... <input type="checkbox"/><br>Woman ..... <input type="checkbox"/><br>Other ..... <input type="checkbox"/> | Am Indian ..... <input type="checkbox"/><br>Asian ..... <input type="checkbox"/><br>Black..... <input type="checkbox"/><br>Hispanic ..... <input type="checkbox"/><br>White ..... <input type="checkbox"/><br>Other ..... <input type="checkbox"/> | Bachelor's ..... <input type="checkbox"/><br>Master's ..... <input type="checkbox"/><br>Doctorate ..... <input type="checkbox"/> |
| New Hire #4 | Man ..... <input type="checkbox"/><br>Woman ..... <input type="checkbox"/><br>Other ..... <input type="checkbox"/> | Am Indian ..... <input type="checkbox"/><br>Asian ..... <input type="checkbox"/><br>Black..... <input type="checkbox"/><br>Hispanic ..... <input type="checkbox"/><br>White ..... <input type="checkbox"/><br>Other ..... <input type="checkbox"/> | Bachelor's ..... <input type="checkbox"/><br>Master's ..... <input type="checkbox"/><br>Doctorate ..... <input type="checkbox"/> |
| New Hire #5 | Man ..... <input type="checkbox"/><br>Woman ..... <input type="checkbox"/><br>Other ..... <input type="checkbox"/> | Am Indian ..... <input type="checkbox"/><br>Asian ..... <input type="checkbox"/><br>Black..... <input type="checkbox"/><br>Hispanic ..... <input type="checkbox"/><br>White ..... <input type="checkbox"/><br>Other ..... <input type="checkbox"/> | Bachelor's ..... <input type="checkbox"/><br>Master's ..... <input type="checkbox"/><br>Doctorate ..... <input type="checkbox"/> |
| New Hire #6 | Man ..... <input type="checkbox"/><br>Woman ..... <input type="checkbox"/><br>Other ..... <input type="checkbox"/> | Am Indian ..... <input type="checkbox"/><br>Asian ..... <input type="checkbox"/><br>Black..... <input type="checkbox"/><br>Hispanic ..... <input type="checkbox"/><br>White ..... <input type="checkbox"/><br>Other ..... <input type="checkbox"/> | Bachelor's ..... <input type="checkbox"/><br>Master's ..... <input type="checkbox"/><br>Doctorate ..... <input type="checkbox"/> |

**K Professional Activities and Evaluation of Faculty**

Two-Year College Mathematics Questionnaire

K1. Is professional development required of your faculty?

|                              | Yes                      | No                       |
|------------------------------|--------------------------|--------------------------|
| a. Permanent full-time ..... | <input type="checkbox"/> | <input type="checkbox"/> |
| b. Part-time .....           | <input type="checkbox"/> | <input type="checkbox"/> |

K2. In general, how frequently are mathematics faculty evaluated? (Check one in each row.)

|                              | At least<br>once a year  | At least<br>once every<br>other year | Occasionally             | Never                    | Not<br>applicable        |
|------------------------------|--------------------------|--------------------------------------|--------------------------|--------------------------|--------------------------|
| a. Full-time (tenured) ..... | <input type="checkbox"/> | <input type="checkbox"/>             | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| b. Full-time (non-tenured)   | <input type="checkbox"/> | <input type="checkbox"/>             | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| c. Part-time .....           | <input type="checkbox"/> | <input type="checkbox"/>             | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

K3. Check all evaluation methods that are used for **part-time faculty** paid by your college (reported in B8(a)) and for **permanent full-time faculty** (reported in B2). (Check yes or no for both part-time and full-time faculty on each line.)

| Evaluation Mode                                                                                   | Full-Time<br>Faculty in<br>B2 |                          | Part-Time<br>Faculty in<br>B8a |                          |
|---------------------------------------------------------------------------------------------------|-------------------------------|--------------------------|--------------------------------|--------------------------|
|                                                                                                   | Yes                           | No                       | Yes                            | No                       |
| a. Observation of classes by other faculty members or department chair .....                      | <input type="checkbox"/>      | <input type="checkbox"/> | <input type="checkbox"/>       | <input type="checkbox"/> |
| b. Observation of classes by division head (if different from chair) or other administrator ..... | <input type="checkbox"/>      | <input type="checkbox"/> | <input type="checkbox"/>       | <input type="checkbox"/> |
| c. Evaluation forms completed by students .....                                                   | <input type="checkbox"/>      | <input type="checkbox"/> | <input type="checkbox"/>       | <input type="checkbox"/> |
| d. Evaluation of written course material such as lesson plans, syllabi, or exams.....             | <input type="checkbox"/>      | <input type="checkbox"/> | <input type="checkbox"/>       | <input type="checkbox"/> |
| e. Self-evaluation such as teaching portfolios.....                                               | <input type="checkbox"/>      | <input type="checkbox"/> | <input type="checkbox"/>       | <input type="checkbox"/> |
| f. Written peer evaluations .....                                                                 | <input type="checkbox"/>      | <input type="checkbox"/> | <input type="checkbox"/>       | <input type="checkbox"/> |
| g. Other (specify) .....                                                                          | <input type="checkbox"/>      | <input type="checkbox"/> | <input type="checkbox"/>       | <input type="checkbox"/> |

**L. Academic Support and Enrichment Opportunities for Students**

- If you belong to a multi-campus system, please report for the entire system.

L1. Is some form of placement process in mathematics required for first-time enrollees?

Yes..... ☐ → go to L2

No ..... ☐ → go to L3.

L2. If placement is required, check all the components/criteria that would be considered.

- |                                  |                          |
|----------------------------------|--------------------------|
| a. Placement Test alone          | <input type="checkbox"/> |
| b. Self-placement                | <input type="checkbox"/> |
| c. Counseling/advisement         | <input type="checkbox"/> |
| d. Scores on State test          | <input type="checkbox"/> |
| e. High School record            | <input type="checkbox"/> |
| f. ACT/SAT                       | <input type="checkbox"/> |
| g. Scores on national assessment | <input type="checkbox"/> |

L3. Does your college/department periodically assess the effectiveness of the mathematics placement program?

Yes..... ☐ →

No ..... ☐ →

L4. Check all opportunities available to your mathematics students. (Please check one box in each line.)

|                                                                                | Yes                      | No                       |
|--------------------------------------------------------------------------------|--------------------------|--------------------------|
| a. Mathematics club .....                                                      | <input type="checkbox"/> | <input type="checkbox"/> |
| b. Special mathematics programs to encourage women .....                       | <input type="checkbox"/> | <input type="checkbox"/> |
| c. Special mathematics programs to encourage minorities .....                  | <input type="checkbox"/> | <input type="checkbox"/> |
| d. Special mathematics lectures/colloquia not part of a mathematics club ..... | <input type="checkbox"/> | <input type="checkbox"/> |
| e. Mathematics outreach opportunities in local K–12 schools .....              | <input type="checkbox"/> | <input type="checkbox"/> |
| f. Opportunities to participate in undergraduate research in mathematics ..... | <input type="checkbox"/> | <input type="checkbox"/> |
| g. Independent study opportunities in mathematics .....                        | <input type="checkbox"/> | <input type="checkbox"/> |
| h. Assigned faculty advisors in mathematics .....                              | <input type="checkbox"/> | <input type="checkbox"/> |
| i. Other (specify) .....                                                       | <input type="checkbox"/> | <input type="checkbox"/> |



**M. Mathematics Preparation of K–12 Teachers**

Two-Year College Mathematics Questionnaire

- If you belong to a multi-campus system, please report for the entire system.

M1. Does your department have any mathematics courses or programs that lead to obtaining credentials to teach mathematics in public elementary schools (any license that includes grades K-5) in your state?

Yes ..... ☐ If yes, go to M2

No ..... ☐ If no, go to M3.

M2. Does your department have a faculty member assigned to coordinate mathematics program courses for pre-service elementary school teachers?

Yes ..... ☐

No ..... ☐

M3. Does your department have any mathematics courses or programs that lead to obtaining credentials to teach mathematics in public middle schools (any license that includes grades 6-8) in your state?

Yes, and the program has different requirements from those for teaching mathematics at the elementary or high school levels..... ☐ If yes, go to M4.

Yes, but the program is identical to that for either the high school mathematics teacher preparation program or the elementary mathematics teacher preparation program. .... ☐ If yes, go to M4.

No ..... ☐ If no, go to M5.

M4. Does your department have a faculty member assigned to coordinate mathematics courses or programs for pre-service middle school teachers?

Yes ..... ☐

No ..... ☐

M5. Does your department have any mathematics courses or programs that leads to obtaining credentials to teach mathematics in public high schools (any license that includes grades 9-12) in your state?

Yes..... ☐ If yes, go to M6

No ..... ☐ If no, go to M7.

M6. Does your department have a faculty member assigned to coordinate mathematics courses or programs for pre-service secondary school teachers?

Yes ..... ☐

No ..... ☐

**M. Mathematics Preparation of K–12 Teachers (cont.)**

Two-Year College Mathematics Questionnaire

- M7. Other than the courses “Mathematics for Elementary School Teachers I, II, and other Mathematics courses for Teacher Preparation” reported on lines F23, F24, and F25, do you designate any sections of your other mathematics program courses as “especially designed for pre-service elementary school teachers”?

Yes ..... ☐

No ..... ☐

- M8. Which of the following groups can meet their entire mathematics course or licensure requirement for teaching via an organized program in your department? Consider “pre- service” and “career switchers” as distinct categories. “Career switchers” usually are post- baccalaureate older adults returning for teaching licensure after a non-teaching career and often under state-approved special licensure rules. (Check one on each row.)

|                                                                | Yes                      | No                       |
|----------------------------------------------------------------|--------------------------|--------------------------|
| a. Pre-service elementary school teachers .....                | <input type="checkbox"/> | <input type="checkbox"/> |
| b. Pre-service middle school teachers .....                    | <input type="checkbox"/> | <input type="checkbox"/> |
| c. Pre-service secondary school teachers .....                 | <input type="checkbox"/> | <input type="checkbox"/> |
| d. In-service elementary school teachers .....                 | <input type="checkbox"/> | <input type="checkbox"/> |
| e. In-service middle school teachers .....                     | <input type="checkbox"/> | <input type="checkbox"/> |
| f. In-service secondary school teachers .....                  | <input type="checkbox"/> | <input type="checkbox"/> |
| g. Career switchers moving to elementary school teaching ..... | <input type="checkbox"/> | <input type="checkbox"/> |
| h. Career switchers moving to middle school teaching.....      | <input type="checkbox"/> | <input type="checkbox"/> |
| i. Career switchers moving to secondary school teaching.....   | <input type="checkbox"/> | <input type="checkbox"/> |

- M9. Does your institution offer pedagogical courses in mathematics for teacher licensure for any of the three grade levels listed below? (Check all that apply.)

Grades PK-5 ..... ☐

Grades 6-8 ..... ☐

Grades 9-12 ..... ☐

If any or all are checked, Go to M10

No ..... ☐ go to N1

- M10. Where are the pedagogical courses in mathematics for teacher licensure taught?

|                                        | Yes                      | No                       |
|----------------------------------------|--------------------------|--------------------------|
| a. In the mathematics department ..... | <input type="checkbox"/> | <input type="checkbox"/> |
| b. Elsewhere in the institution .....  | <input type="checkbox"/> | <input type="checkbox"/> |

**N. Issues of Professional Concern**

Two-Year College Mathematics Questionnaire

N1. Below are concerns often cited by two-year college mathematics departments. Please read each item carefully and check the box in each row that best reflects your view.

|                                                                                                                        | Minor or<br>no<br>concern<br>for us | Somewhat<br>of a<br>concern<br>for us | Major<br>concern<br>for us |
|------------------------------------------------------------------------------------------------------------------------|-------------------------------------|---------------------------------------|----------------------------|
| a. Faculty staying current regarding effective teaching strategies .....                                               | <input type="checkbox"/>            | <input type="checkbox"/>              | <input type="checkbox"/>   |
| b. Courses taken for both high school and college credit).....                                                         | <input type="checkbox"/>            | <input type="checkbox"/>              | <input type="checkbox"/>   |
| c. Staffing statistics courses .....                                                                                   | <input type="checkbox"/>            | <input type="checkbox"/>              | <input type="checkbox"/>   |
| d. Unrealistic student understanding of the demands of college work.....                                               | <input type="checkbox"/>            | <input type="checkbox"/>              | <input type="checkbox"/>   |
| e. Part-time faculty teach too many courses .....                                                                      | <input type="checkbox"/>            | <input type="checkbox"/>              | <input type="checkbox"/>   |
| f. Faculty salaries too low .....                                                                                      | <input type="checkbox"/>            | <input type="checkbox"/>              | <input type="checkbox"/>   |
| g. Class sizes too large .....                                                                                         | <input type="checkbox"/>            | <input type="checkbox"/>              | <input type="checkbox"/>   |
| h. Low student motivation .....                                                                                        | <input type="checkbox"/>            | <input type="checkbox"/>              | <input type="checkbox"/>   |
| i. Too many students needing remediation .....                                                                         | <input type="checkbox"/>            | <input type="checkbox"/>              | <input type="checkbox"/>   |
| j. Successful progress of students through developmental courses to more advanced mathematics courses is too low ..... | <input type="checkbox"/>            | <input type="checkbox"/>              | <input type="checkbox"/>   |
| k. Student success rate in transfer-level math courses is too low .....                                                | <input type="checkbox"/>            | <input type="checkbox"/>              | <input type="checkbox"/>   |
| l. Too few students who intend to transfer actually do transfer .....                                                  | <input type="checkbox"/>            | <input type="checkbox"/>              | <input type="checkbox"/>   |
| m. Inadequate travel funds for faculty ..                                                                              | <input type="checkbox"/>            | <input type="checkbox"/>              | <input type="checkbox"/>   |
| n. Learning disabilities are not adequately supported .....                                                            | <input type="checkbox"/>            | <input type="checkbox"/>              | <input type="checkbox"/>   |
| o. Inadequate classroom facilities for teaching with technology .....                                                  | <input type="checkbox"/>            | <input type="checkbox"/>              | <input type="checkbox"/>   |
| p. Inadequate classroom facilities for teaching with active learning .....                                             | <input type="checkbox"/>            | <input type="checkbox"/>              | <input type="checkbox"/>   |
| q. Inadequate computer facilities for part-time faculty use .....                                                      | <input type="checkbox"/>            | <input type="checkbox"/>              | <input type="checkbox"/>   |
| r. Inadequate computer facilities for student use .....                                                                | <input type="checkbox"/>            | <input type="checkbox"/>              | <input type="checkbox"/>   |

**N. Issues of Professional Concern (cont.)**

Two-Year College Mathematics Questionnaire

N1. Continued

|                                                                                                                | Minor or<br>No<br>problem<br>for us | Somewhat<br>of a<br>problem<br>for us | Major<br>problem<br>for us |
|----------------------------------------------------------------------------------------------------------------|-------------------------------------|---------------------------------------|----------------------------|
| s. Classroom and other duties make it difficult for faculty to engage in professional development .....        | <input type="checkbox"/>            | <input type="checkbox"/>              | <input type="checkbox"/>   |
| t. Curriculum alignment between high schools and college .....                                                 | <input type="checkbox"/>            | <input type="checkbox"/>              | <input type="checkbox"/>   |
| u. Lack of curricular flexibility because of transfer requirements .....                                       | <input type="checkbox"/>            | <input type="checkbox"/>              | <input type="checkbox"/>   |
| v. Other barriers that inhibit curricular changes .....                                                        | <input type="checkbox"/>            | <input type="checkbox"/>              | <input type="checkbox"/>   |
| w. Maintaining high and consistent expectations of students across different sections of the same course ..... | <input type="checkbox"/>            | <input type="checkbox"/>              | <input type="checkbox"/>   |
| x. High cost of textbooks .....                                                                                | <input type="checkbox"/>            | <input type="checkbox"/>              | <input type="checkbox"/>   |
| y. Lack of flexibility in curricular redesign .....                                                            | <input type="checkbox"/>            | <input type="checkbox"/>              | <input type="checkbox"/>   |
| z. Maintaining common standards between distance/remote learning courses and related courses .....             | <input type="checkbox"/>            | <input type="checkbox"/>              | <input type="checkbox"/>   |

**N. Issues of Professional Concern (cont.)**

Two-Year College Mathematics Questionnaire

N2. Many departments today use a spectrum of program assessment methods. Please check all that apply to your department's program assessment efforts during the last six years.

|                                                                                                                                     | Yes                      | No                       |
|-------------------------------------------------------------------------------------------------------------------------------------|--------------------------|--------------------------|
| a. A review of our mathematics program was conducted that included one or more reviewers from outside our institution .....         | <input type="checkbox"/> | <input type="checkbox"/> |
| b. Students in our mathematics program were asked to comment on and suggest changes in the program.....                             | <input type="checkbox"/> | <input type="checkbox"/> |
| c. Other departments at our institution were invited to comment on the preparation that their students received in our courses..... | <input type="checkbox"/> | <input type="checkbox"/> |
| d. Data on students' progress in subsequent mathematics courses were gathered and analyzed .....                                    | <input type="checkbox"/> | <input type="checkbox"/> |
| e. Data on the placement system for first-year students were analyzed for effectiveness .....                                       | <input type="checkbox"/> | <input type="checkbox"/> |
| f. Department's program assessment activities led to changes in the mathematics program .....                                       | <input type="checkbox"/> | <input type="checkbox"/> |
| g. A process is in place to analyze data and assess achievement of student learning outcomes.....                                   | <input type="checkbox"/> | <input type="checkbox"/> |

- N3. Please indicate the extent to which the following activities have taken place in the past year in your department in response to increased national attention to equity, diversity, and inclusion issues (here the term “demographic” includes race, ethnicity, gender, disability status, and other characteristics of individuals).

|                                                                                                                                      | None                     | Some                     | A lot                    |
|--------------------------------------------------------------------------------------------------------------------------------------|--------------------------|--------------------------|--------------------------|
| a. Faculty discussion designed to increase awareness of equity, diversity, and inclusion issues                                      | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| b. Student discussion designed to increase awareness of equity, diversity, and inclusion issues                                      | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| c. Program or policy changes to affect the demographic balance of faculty in the mathematical sciences.                              | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| d. Program or policy changes to affect the demographic balance of undergraduate students in mathematical sciences.                   | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| e. Consideration of existing or new programs to assist underrepresented groups and/or at-risk students in the mathematical sciences. | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

- g. Please describe any other steps your department or institution has taken in the past year that deal with equity, diversity, and inclusion as they pertain to the study of and employment in the mathematical sciences.
-

**O. Comments and Suggestions**

Two-Year College Mathematics Questionnaire

If you found some question(s) difficult to interpret or answer, please let us know. We welcome suggestions to improve future surveys (e.g., CBMS 2025).

Comments: \_\_\_\_\_

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*Thank you for completing this questionnaire. We know it was a time-consuming process and we hope that the resulting survey report, which we hope to publish in spring 2023, will be of use to you and your department.*

*Please keep a copy of your responses to this questionnaire in case questions arise.*

## **Four-Year Statistics Questionnaire**



**SURVEY OF UNDERGRADUATE PROGRAMS IN THE MATHEMATICAL SCIENCES**



**General Information**

Statistics Questionnaire-

As part of a random sample, your department has been chosen to participate in the NSF-funded CBMS2021 National Survey of Undergraduate Mathematical Sciences Programs. The presidents of all U.S. mathematical sciences organizations have endorsed it and ask for your cooperation.

We assure you that no individual departmental data, except the names of responding departments, will be released.

This survey provides data about the nation's undergraduate mathematical and statistical effort that is available from no other source. You can see the results of a similar survey fielded six years ago by going to [www.ams.org/cbms](http://www.ams.org/cbms), where the CBMS 2015 report is available online.

All departments in this survey are in universities and colleges that offer at least a bachelor's degree. They may or may not offer an undergraduate major in statistics. Most of the statistics departments in our random sample also offer higher degrees in statistical sciences.

We have classified your department as belonging to a university or four-year college. If this is not correct, please contact Ellen Kirkman, Survey Director, at 336-758-5351 or at [Kirkman@wfu.edu](mailto:Kirkman@wfu.edu).

Please report on undergraduate programs in the statistical sciences (including probability) that are under the direction of your department. Do not include data for other departments or for branches or campuses of your institution that are budgetarily separate from your own. Also, if your department is broader than just statistics (e.g., Department of Statistics and Computer Science or Statistics and Operations Research), please report on all the courses offered by your department.

This survey may be completed either online or using a hard-copy questionnaire. We recommend using the online system because it will do some of the work for you; e.g., it will automatically skip those questions that are not applicable (based on the response you give), gray out portions of questions that do not apply, remind you of previous responses, and provide definitions when you let your cursor hover over certain highlighted words.

If you have any questions while filling out this survey form, please call the Survey Director, Ellen Kirkman, at 336-758-5351 or contact her by e-mail at [Kirkman@wfu.edu](mailto:Kirkman@wfu.edu). For help with the online questionnaire, call Westat at 855-770-0558 or send an email to [cbms2021@westat.com](mailto:cbms2021@westat.com).

**Please complete the questionnaire by October 29, 2021 online or by mailing a hard copy to:**

**CBMS Survey**

**Westat**

**1600 Research Boulevard, RB 3103**

**Rockville, MD 20850-312**

**Please retain a copy of your responses to this questionnaire in case questions arise.**

**A. General Information**

A1. Name of your Institution: \_\_\_\_\_

A2. Name of your Department: \_\_\_\_\_

A3. We have classified your department as being part of a university or four-year college. Do you agree?

Yes..... ☐ → If Yes, go to A4 below.

No ..... ☐ → If No, please call Ellen Kirkman, Survey Director,  
at 336-758-5351

A4. If your college or university does not recognize tenure, check this box. ☐

A5. Contact person in your department:

A6. Contact person's e-mail address:

A7. Contact person's phone number  
including area code:

A8. Contact person's mailing address:

a. Street .....

b. Street2.....

c. City .....

d. State .....

e. Zip code .....

**B. Dual-Enrollment Courses**

Statistics Questionnaire

**Definition:** We use the term dual (or concurrent) enrollment courses to refer to courses taught in a high school by high school teachers, for which high school students may obtain high school credit and, simultaneously, college credit through your institution.

B1. Does your department participate in any dual enrollment programs of this type?

Yes..... ☐ —————> If Yes, go to B2.

No ..... ☐ —————> If No, go to B3.

B2. Please complete the following table giving the number of students enrolled in your dual enrollment program (as defined above) for the previous term (spring 2021) and the current fall term of 2021. These enrollments should NOT be included in subsequent sections.

| Course              | Total Dual Enrollments    |                         |
|---------------------|---------------------------|-------------------------|
|                     | Last Term=<br>Spring 2021 | This Term=<br>Fall 2021 |
| a. Statistics ..... |                           |                         |

B3. Does your department assign any of its own full-time or part-time faculty to teach courses conducted on a high school campus for which high school students may receive both high school and college credit (through your institution)?

Yes..... ☐ —————> If Yes, go to B4.

No ..... ☐ —————> If No, go to C1.

B4. In fall 2021, how many students are enrolled in the courses taught in a high school by your full-time or part-time faculty and through which high school students may receive both high school and college credit (through your institution)? (Include these enrollments in subsequent sections).

Number of students.....

**C. Distance/ remote learning**

Statistics Questionnaire

**Definition:** Distance/remote learning courses are those courses offered by your institution for credit, in which half or more of the instruction occurs with the instructor and the students separated by time and /or place, and facilitated by technology (e.g. courses in which half or more of the course is taught online, either synchronously or asynchronously, or by computer software, or by other technologies). Include only distance/remote learning courses offered in normal practice, not courses that became distance/remote due to the COVID-19 pandemic.

- C1. Overall, how have attitudes towards online learning changed as a result of the COVID-19 experience?

|                                        | More favorable           | No change                | Less favorable           |
|----------------------------------------|--------------------------|--------------------------|--------------------------|
| a. Faculty interest in online teaching | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| b. Faculty use of online tools         | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| c. Student interest in online teaching | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| d. Student use of online tools         | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

C2. Many colleges have conducted online instruction as a way of addressing the COVID-19 pandemic, offering online instruction either as an alternative to face-to-face instruction or as a supplement to face-to-face instruction. How is that instruction coordinated with what has normally been called distance education? Please indicate which of the following applies to your department for each listed time period, using the following definitions.

- On-campus (face-to-face) learners—students who would be expected to attend most classes in person.
- Remote learners—students who would be expected to attend most classes remotely.

| Policy                                                                                                                                                                                    | Prior to pandemic (prior to spring 2020) |                          | During pandemic (spring 2020–summer 2021) |                          | Fall 2021                |                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|--------------------------|-------------------------------------------|--------------------------|--------------------------|--------------------------|
|                                                                                                                                                                                           | Yes                                      | No                       | Yes                                       | No                       | Yes                      | No                       |
| a. We offer some course sections only to on-campus (face-to-face) learners; remote learners might be offered the same course, but the students would not normally be in the same section. | <input type="checkbox"/>                 | <input type="checkbox"/> | <input type="checkbox"/>                  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| b. We offer some course sections only to remote learners; on-campus (face-to-face) learners might be offered the same course, but the students would not normally be in the same section. | <input type="checkbox"/>                 | <input type="checkbox"/> | <input type="checkbox"/>                  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| c. We offer some “hybrid” courses to on-campus (face-to-face) learners and remote learners in the same section.                                                                           | <input type="checkbox"/>                 | <input type="checkbox"/> | <input type="checkbox"/>                  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

C3. Has your department taught any distance/remote learning course **(other than courses moved online in response to the COVID-19 pandemic)** within the calendar years 2018-2021?

Yes..... ☐ —————> If Yes, go to C4.

No ..... ☐ —————> If No, skip to section D.

- C4. Which best characterizes the format/structure of the majority of your distance/remote learning courses (not including courses moved online in response to the COVID-19 pandemic)? (*Choose one response.*)

Sections are taught only online, and only asynchronously ..... ☐

Sections are taught only online, with an opportunity to meet synchronously online..... ☐

Sections use a mixture of online and face-to-face sessions..... ☐

Other ..... ☐

- C5. In most of your distance/remote learning courses, how are the majority of the tests administered (not including courses moved online in response to the COVID-19 pandemic)? (*Choose one response.*)

Online, not monitored ..... ☐

Online, but using some kind of monitoring technology ..... ☐

At a monitored testing site ..... ☐

Combination of monitoring methods ..... ☐

- C6. Rate the following challenges that your department faces when creating and/or offering distance/remote learning statistics courses. (Please check one box in each line.)

| Challenge                                                                | Not a challenge          | Somewhat of a challenge  | Very significant challenge |
|--------------------------------------------------------------------------|--------------------------|--------------------------|----------------------------|
| a. Designing appropriate assessments of student learning.                | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>   |
| b. Maintaining academic integrity on assessments.                        | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>   |
| c. Grade inflation.                                                      | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>   |
| d. Building/maintaining community among faculty and students.            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>   |
| e. Maintaining academic quality instruction.                             | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>   |
| f. Engaging students online.                                             | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>   |
| g. Replicating active learning in a virtual environment.                 | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>   |
| h. Availability of equipment and technical support for faculty/students. | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>   |

**D. Faculty Profile (Fall 2021)**

Statistics Questionnaire

Please indicate whether the following types of faculty are actively teaching one or more courses in fall 2021.

## Definitions

- **Full-time faculty.** Faculty who are full-time employees in the institution and more than half-time in the department. For example, if a tenured computer science professor with a joint appointment in your department teaches a total of two courses in fall 2021, with exactly one being in your department (i.e., statistics is 50% of the fall teaching assignment), then that person would be counted as part-time in your department.

| Faculty Type |                                                                                                                                                                                                                                          | Teach in Fall 2021       |                          |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|--------------------------|
|              |                                                                                                                                                                                                                                          | Yes                      | No                       |
| D1.          | Full-time faculty who are more than half-time in your department                                                                                                                                                                         |                          |                          |
| a.           | Tenured or tenure-eligible, or permanent (if your institution does not recognize tenure) faculty.....                                                                                                                                    | <input type="checkbox"/> | <input type="checkbox"/> |
| b.           | Other full-time faculty (e.g., permanent faculty only if your institution also has tenured or tenure-eligible faculty, faculty with renewable positions such as teaching professionals, postdoctoral faculty, and visiting faculty)..... | <input type="checkbox"/> | <input type="checkbox"/> |
| D2.          | Part-time faculty (those who are half-time or less in your department) ....                                                                                                                                                              | <input type="checkbox"/> | <input type="checkbox"/> |
| D3.          | Graduate teaching assistant(s) who teach courses independently (not counting the teaching of recitation sessions) .....                                                                                                                  | <input type="checkbox"/> | <input type="checkbox"/> |

**E. Probability and Statistics Courses (Fall 2021)**

Statistics Questionnaire

Which of the following courses are taught in your department in Fall 2021? You may use different titles for these courses, and may have multiple courses that match a particular course name.

- Enter an X in each applicable box.
- Do **not** include courses taught in other departments, learning centers, or developmental/remedial programs separate from your statistics program or department.
- Please also indicate which catalog codes are used to identify those courses taught in Fall 2021. This information will be used to generate a reduced course list suitable for your department so reporting on enrollments will be easier.
- Make sure that no course is reported in more than one row.
- If your department is broader than just statistics (e.g., Department of Statistics and Computer Science or Statistics and Operations Research), please use E24 to report on the courses outside of probability and statistics.
- **Introductory Statistics classes.** You will be asked to list separately Introductory Statistics classes taught in a large lecture format (with recitation/problem/laboratory sections) and, sections that meet as a class with an instructor at a regularly scheduled time (and are not divided into recitation sections). For example, you will be asked about large lecture courses (E1-1 column). Please treat any large class that is sometimes broken up into smaller units as a "lecture/recitation" class (even if there is no lecture); if neither the lecture/recitation or individual class format seems an appropriate description of the enrollment, enter the enrollment under "other."

| Name of Course (or equivalent)                                                                                                                         | Taught in Fall 2021      | Catalog code(s) used for courses offered Fall 2021 (not for courses offered Spring 2022 or previous academic year) (use comma to separate codes) |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Statistics</b>                                                                                                                                      |                          |                                                                                                                                                  |
| <b>INTRODUCTORY STATISTICS</b> (no calculus prerequisite; designed for non-majors/minors [General Education Courses] but may be taken by major/minors) |                          |                                                                                                                                                  |
| E1-1. Lecture with separately scheduled recitation/problem/laboratory sessions                                                                         | <input type="checkbox"/> |                                                                                                                                                  |
| E1-2. Individual sections not in E1-1, that meet as a class with an instructor at a regularly scheduled time                                           | <input type="checkbox"/> |                                                                                                                                                  |
| E1-3. Other sections not listed above                                                                                                                  | <input type="checkbox"/> |                                                                                                                                                  |
| <b>INTRODUCTORY STATISTICS</b> (calculus prerequisite)                                                                                                 |                          |                                                                                                                                                  |
| E2-1. Lecture with separately scheduled recitation/problem/laboratory sessions                                                                         | <input type="checkbox"/> |                                                                                                                                                  |
| E2-2. Individual sections not in E2-1, that meet as a class with an instructor at a regularly scheduled time                                           | <input type="checkbox"/> |                                                                                                                                                  |
| E2-3. Sections not listed above                                                                                                                        | <input type="checkbox"/> |                                                                                                                                                  |

<sup>1</sup>Distance/remote learning courses are those courses offered by your institution for credit, in which half or more of the instruction occurs with the instructor and the students separated by time and /or place, and facilitated by technology (e.g. courses in which the majority of the course is taught online either synchronously or asynchronously, or by computer software, or by other technologies).

<sup>2</sup>For E1 and E2, enter course identifiers that are sufficiently distinct to separate courses with recitation sessions, courses that meet as a class, and other sections.



**E. Probability and Statistics Courses (Fall 2021) (cont.)**

Statistics Questionnaire

| Name of Course (or equivalent)                                                                | Taught in<br>Fall 2021   | Catalog code(s)<br>used for courses<br>offered Fall 2021<br>(not for courses<br>offered Spring<br>2022 or previous<br>academic year)<br>(use comma to<br>separate codes) |
|-----------------------------------------------------------------------------------------------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>OTHER INTRODUCTORY STATISTICS courses</b>                                                  |                          |                                                                                                                                                                          |
| E3. Statistics for pre-service elementary and/or middle grade teachers                        | <input type="checkbox"/> |                                                                                                                                                                          |
| E4. Statistics for pre-service secondary school teachers                                      | <input type="checkbox"/> |                                                                                                                                                                          |
| E5. Intermediate statistics (non-calculus)                                                    | <input type="checkbox"/> |                                                                                                                                                                          |
| E6. Other introductory level Probability or Statistics courses designed for non-majors/minors | <input type="checkbox"/> |                                                                                                                                                                          |

Which of the following courses are taught in your department in Fall 2021, will be taught in Spring 2022, or were taught at any time in 2020-21? You may use different titles for these courses, and may have multiple courses that match a particular course name.

- Enter an X in each applicable box.
- Do **not** include courses taught in other departments, learning centers, or developmental/remedial programs separate from your statistics program or department.
- Please also indicate which catalog codes are used to identify those courses taught in Fall 2021. This information will be used to generate a reduced course list suitable for your department so reporting on enrollments will be easier.
- Make sure that no course is reported in more than one row.
- If your department is broader than just statistics (e.g., Department of Statistics and Computer Science or Statistics and Operations Research), please use E24 to report on the courses outside of probability and statistics.

| Name of Course (or equivalent)                                | Taught in<br>Fall 2021<br>(a) | Will be<br>taught in<br>Spring<br>2022<br>(b) | Taught<br>during<br>academic<br>year 2020-<br>21<br>(c) | Offer as<br>distance/<br>remote<br>learning<br>course <sup>1</sup><br>(d) | Catalog<br>code(s) used<br>for courses<br>offered Fall<br>2021 (not for<br>courses<br>offered<br>Spring 2022<br>or previous<br>academic<br>year) (use<br>comma to<br>separate<br>codes)<br>(e) |
|---------------------------------------------------------------|-------------------------------|-----------------------------------------------|---------------------------------------------------------|---------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>INTERMEDIATE AND ADVANCED LEVEL</b>                        |                               |                                               |                                                         |                                                                           |                                                                                                                                                                                                |
| E7. Combined Probability & Statistics (calculus prerequisite) | <input type="checkbox"/>      | <input type="checkbox"/>                      | <input type="checkbox"/>                                | <input type="checkbox"/>                                                  |                                                                                                                                                                                                |
| E8. Probability (calculus prerequisite)                       | <input type="checkbox"/>      | <input type="checkbox"/>                      | <input type="checkbox"/>                                | <input type="checkbox"/>                                                  |                                                                                                                                                                                                |
| E9. Mathematical Statistics (calculus prerequisite)           | <input type="checkbox"/>      | <input type="checkbox"/>                      | <input type="checkbox"/>                                | <input type="checkbox"/>                                                  |                                                                                                                                                                                                |
| E10. Stochastic Processes                                     | <input type="checkbox"/>      | <input type="checkbox"/>                      | <input type="checkbox"/>                                | <input type="checkbox"/>                                                  |                                                                                                                                                                                                |
| E11. Data Science/Analytics/Statistical Learning              | <input type="checkbox"/>      | <input type="checkbox"/>                      | <input type="checkbox"/>                                | <input type="checkbox"/>                                                  |                                                                                                                                                                                                |
| E12. Design & Analysis of Experiments (ANOVA)                 | <input type="checkbox"/>      | <input type="checkbox"/>                      | <input type="checkbox"/>                                | <input type="checkbox"/>                                                  |                                                                                                                                                                                                |
| E13. Applied Regression                                       | <input type="checkbox"/>      | <input type="checkbox"/>                      | <input type="checkbox"/>                                | <input type="checkbox"/>                                                  |                                                                                                                                                                                                |
| E14. Linear Models I                                          | <input type="checkbox"/>      | <input type="checkbox"/>                      | <input type="checkbox"/>                                | <input type="checkbox"/>                                                  |                                                                                                                                                                                                |

|                                                                     |                          |                          |                          |                          |  |
|---------------------------------------------------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--|
| E15. Linear Models II                                               | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |  |
| E16. Biostatistics                                                  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |  |
| E17. Nonparametric Statistics                                       | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |  |
| E18. Categorical Data Analysis                                      | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |  |
| E19. Sample Survey Design & Analysis                                | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |  |
| E20. Statistical Computing or Software                              | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |  |
| E21. Bayesian Statistics                                            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |  |
| E22. Statistical Consulting                                         | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |  |
| E23. Senior Seminar/Capstone                                        | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |  |
| E24. All other upper level Probability & Statistics related courses | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |  |

<sup>1</sup>Distance/ remote learning courses are those courses offered by your institution for credit, in which half or more of the instruction occurs with the instructor and the students separated by time and /or place, and facilitated by technology (e.g. courses in which the majority of the course is taught online, or by computer software, or by other technologies).

**E. Probability and Statistics Courses (Fall 2021)**

Statistics Questionnaire

In the next several pages you will enter data about courses your department is teaching. For each course that is taught, you will be asked to enter the fall 2021 enrollment and the number of sections of the course. Depending upon what you indicated above, you will be asked about distance/ remote learning enrollment.

**The following instructions apply throughout Section E (pages 10-12).**

- **Courses, sections, and sessions.** In this questionnaire, “course” is used to refer to the topic area (e.g., Stochastic Processes or Biostatistics). You may have multiple faculty teaching the same course in the same term but at different times or locations; these divisions of the topic area into separate instances of teaching are called sections. Within a section, you may have times when the students are divided into laboratory or recitation sessions; these are counted as recitation sessions, not as separate sections.
- Below please do not treat recitation sessions as separate sections. Instead, please treat both the lecture component and any associated recitation sessions as a single section.
- If a course is cross-listed in both statistics and another department (such as mathematics, psychology, or engineering), count all students, regardless of how the course is listed on the students’ transcripts.
- Do not fill in any shaded boxes.
- Any unshaded box that is left blank will be interpreted as reporting a count of zero.
- Except where specifically stated to the contrary, the tables in Section E deal with enrollments in fall term 2021.



**E. Probability and Statistics Courses (Fall 2021) (cont.)**

Statistics Questionnaire

E26. You reported a total of # sections in fall 2021, distributed by course type as shown below. For each course type, please provide the number of sections taught by tenured or tenure-eligible faculty, other full-time faculty, part-time faculty, and graduate teaching assistants.

◆ Cells left blank will be interpreted as zeros

| Type of course and your applicable catalog course codes                                                                        | Total number of sections<br>(a) | Of the number in column (a), how many sections are taught by: |                                |                          |                                                  |
|--------------------------------------------------------------------------------------------------------------------------------|---------------------------------|---------------------------------------------------------------|--------------------------------|--------------------------|--------------------------------------------------|
|                                                                                                                                |                                 | Full-time faculty <sup>1</sup>                                |                                | Part-time Faculty<br>(d) | Graduate Teaching Assistants <sup>2</sup><br>(e) |
|                                                                                                                                |                                 | Tenured or Tenure-eligible, Faculty<br>(b)                    | Other Full-time Faculty<br>(c) |                          |                                                  |
| E26a. Introductory Statistics (no calculus prerequisite) —Lecture with separate recitation<br>(course code list)               |                                 |                                                               |                                |                          |                                                  |
| E26b. Introductory Statistics (no calculus prerequisite) —Sections that meet as a class<br>(course code list)                  |                                 |                                                               |                                |                          |                                                  |
| E26c. Introductory Statistics (no calculus prerequisite) —Other sections<br>(course code list)                                 |                                 |                                                               |                                |                          |                                                  |
| E26d. Introductory Statistics (calculus prerequisite) (not for majors) —Lecture with separate recitation<br>(course code list) |                                 |                                                               |                                |                          |                                                  |
| E26e. Introductory Statistics (calculus prerequisite) (not for majors) —Sections that meet as a class<br>(course code list)    |                                 |                                                               |                                |                          |                                                  |

<sup>1</sup>If your institution does not recognize tenure, report sections taught by your permanent full-time faculty in column (b) and sections taught by other full-time faculty in column (c). If your institution does recognize tenure but has **faculty with renewable contracts**, report these faculty as **other full-time faculty** (column c).

**Full-time faculty teaching in your department and holding joint appointments with other departments** should be counted in column (b) if they are tenured, tenure-eligible, or permanent (if your institution does not recognize tenure) in your department. Faculty who are not tenured, tenure-eligible, or permanent in your department should be counted in column (d) if their fall 2021 teaching in your department is less than or equal to 50% of their total fall teaching assignment, and they should be reported in column (c) otherwise. (Example: If a tenured computer science professor with a joint appointment in your department teaches a total of two courses in fall 2021, with exactly one being in your department and hence statistics comprised 50% of the fall teaching assignment, then that person would be counted as parttime in your department.)

<sup>2</sup> Report a section of a course as being taught by a **graduate teaching assistant (GTA)** if and only if that section is taught independently by the GTA, i.e., when it is the GTA's own course and the GTA is the instructor of record.

**E. Probability and Statistics Courses (Fall 2021) (cont.)**

Statistics Questionnaire

| <b>◆ Cells left blank will be interpreted as zeros</b><br><br>Type of course and your applicable catalog course codes |                                 | Of the number in column (a), how many sections are taught by: |                                |                   |                                           |
|-----------------------------------------------------------------------------------------------------------------------|---------------------------------|---------------------------------------------------------------|--------------------------------|-------------------|-------------------------------------------|
|                                                                                                                       |                                 | Full-time faculty <sup>1</sup>                                |                                | Part-time Faculty | Graduate Teaching Assistants <sup>2</sup> |
|                                                                                                                       | Total number of sections<br>(a) | Tenured or Tenure-eligible, Faculty<br>(b)                    | Other Full-time Faculty<br>(c) | (d)               | (e)                                       |
| E26f. Introductory Statistics (calculus prerequisite) (not for majors) — Other sections<br>(course code list)         |                                 |                                                               |                                |                   |                                           |
| E26g. Statistics for Pre-service Teachers<br>(course code list)                                                       |                                 |                                                               |                                |                   |                                           |
| E26h. Intermediate statistics (non-calculus)<br>(course code list)                                                    |                                 |                                                               |                                |                   |                                           |
| E26i. Other introductory level Statistics<br>(course code list)                                                       |                                 |                                                               |                                |                   |                                           |
| E26j. Advanced Undergraduate Level<br>(course code list)                                                              |                                 |                                                               |                                |                   |                                           |

**F. Undergraduate Program (Fall 2021)**

Statistics Questionnaire

If you do not offer a major in statistics, check here ☐ and go to F5. Otherwise go to F1.

- F1. Report the total number of your departmental majors who received their bachelor's degrees between July 1, 2020 and June 30, 2021. Include joint majors and double majors.<sup>1</sup>

Number of majors receiving degrees.....

- F2. Of the undergraduate degrees described in F1, please report the number who majored in each of the following categories. Each student should be reported only once. Include all double and joint majors<sup>1</sup> in your totals. Use the Other Major category for a major in your department who does not fit into one of the listed categories.

| Area of Major                                                    | Men | Women | Nonbinary |
|------------------------------------------------------------------|-----|-------|-----------|
| a. Statistics.....                                               |     |       |           |
| b. Biostatistics .....                                           |     |       |           |
| c. Actuarial Science .....                                       |     |       |           |
| d. Data Science .....                                            |     |       |           |
| e. Joint <sup>1</sup> Statistics and with Computer Science ..... |     |       |           |
| f. Joint <sup>1</sup> Statistics and with Mathematics .....      |     |       |           |
| g. Other Mathematics Majors.....                                 |     |       |           |
| h. Statistics Education                                          |     |       |           |
| i. Other Majors                                                  |     |       |           |

<sup>1</sup> A "double major" is a student who completes the degree requirements of two separate majors, one in statistics and one in another program or department. A "joint major" is a student who completes a single major in your department that integrates courses from statistics and some other program or department and typically requires fewer credit hours than the sum of the credit hours required by the separate majors.



**F. Undergraduate Program (Fall 2021) (cont.)**

Statistics Questionnaire

F3. To what extent must majors in your department complete the following? Check one box in each row.

|                                                                                                                    | Required of<br>all majors | Required of some<br>but not all majors | Not required<br>of any major |
|--------------------------------------------------------------------------------------------------------------------|---------------------------|----------------------------------------|------------------------------|
| a. Calculus I .....                                                                                                | <input type="checkbox"/>  | <input type="checkbox"/>               | <input type="checkbox"/>     |
| b. Calculus II .....                                                                                               | <input type="checkbox"/>  | <input type="checkbox"/>               | <input type="checkbox"/>     |
| c. Multivariable Calculus .....                                                                                    | <input type="checkbox"/>  | <input type="checkbox"/>               | <input type="checkbox"/>     |
| d. Linear Algebra/Matrix Theory .....                                                                              | <input type="checkbox"/>  | <input type="checkbox"/>               | <input type="checkbox"/>     |
| e. At least one computer science course .....                                                                      | <input type="checkbox"/>  | <input type="checkbox"/>               | <input type="checkbox"/>     |
| f. At least one applied mathematics course<br>(not including a, b, c, d above).....                                | <input type="checkbox"/>  | <input type="checkbox"/>               | <input type="checkbox"/>     |
| g. A capstone experience (e.g., a senior<br>project, a senior thesis, a senior seminar,<br>or an internship) ..... | <input type="checkbox"/>  | <input type="checkbox"/>               | <input type="checkbox"/>     |
| h. An exit exam (written or oral) .....                                                                            | <input type="checkbox"/>  | <input type="checkbox"/>               | <input type="checkbox"/>     |
| i. At least one upper level Probability course                                                                     | <input type="checkbox"/>  | <input type="checkbox"/>               | <input type="checkbox"/>     |
| j. At least one upper-level Mathematical<br>Statistics course .....                                                | <input type="checkbox"/>  | <input type="checkbox"/>               | <input type="checkbox"/>     |
| k. At least one applied statistics course .....                                                                    | <input type="checkbox"/>  | <input type="checkbox"/>               | <input type="checkbox"/>     |
| l. At least one upper-level Linear Models<br>course .....                                                          | <input type="checkbox"/>  | <input type="checkbox"/>               | <input type="checkbox"/>     |
| m. A Bayesian Inference course.....                                                                                | <input type="checkbox"/>  | <input type="checkbox"/>               | <input type="checkbox"/>     |

F4. Give your best estimate of the percentage of your department's graduating majors from the previous academic year 2020-2021 (reported in I1) in each of the following categories. Please make the totals add to 100 percent.

|                                                                                                       |                        |
|-------------------------------------------------------------------------------------------------------|------------------------|
| a. Who went into pre-college teaching .....                                                           | <input type="text"/> % |
| b. Who went to graduate school in the statistical sciences.....                                       | <input type="text"/> % |
| c. Who went to professional school or to graduate school outside of the<br>statistical sciences ..... | <input type="text"/> % |
| d. Who took jobs in business, industry, government, etc.....                                          | <input type="text"/> % |
| e. Who had other post-graduation plans known to the department .....                                  | <input type="text"/> % |
| f. Whose plans are not known to the department .....                                                  | <input type="text"/> % |

**F. Undergraduate Program (Fall 2021) (cont.)**

Statistics Questionnaire

- F5. Many departments today use a spectrum of program-assessment methods. Please indicate whether each of the following apply to your department's undergraduate program-assessment efforts during the last six years.

|                                                                                                                                      | Yes                      | No                       |
|--------------------------------------------------------------------------------------------------------------------------------------|--------------------------|--------------------------|
| a. We conducted a review of our undergraduate program that included one or more reviewers from outside of our institution.....       | <input type="checkbox"/> | <input type="checkbox"/> |
| b. We asked graduates of our undergraduate program to comment on and suggest changes in our undergraduate program .....              | <input type="checkbox"/> | <input type="checkbox"/> |
| c. Other departments at our institution were invited to comment on the preparation that their students received in our courses ..... | <input type="checkbox"/> | <input type="checkbox"/> |
| d. Data on our students' progress in subsequent statistics courses were gathered and analyzed .....                                  | <input type="checkbox"/> | <input type="checkbox"/> |
| e. We have developed a set of student learning outcomes for our program .....                                                        | <input type="checkbox"/> | <input type="checkbox"/> |
| f. We have assessed student learning objectives in courses required in our major.....                                                | <input type="checkbox"/> | <input type="checkbox"/> |
| g. We have a placement system for first-year students and we gathered and analyzed data on its effectiveness .....                   | <input type="checkbox"/> | <input type="checkbox"/> |
| h. Our department's program assessment activities led to changes in our undergraduate program .....                                  | <input type="checkbox"/> | <input type="checkbox"/> |

**F. Undergraduate Program (Fall 2021) (cont.)**

Statistics Questionnaire

F6. Please indicate the extent to which the following activities have taken place in the past year in your department in response to increased national attention to equity, diversity, and inclusion issues (here the term "demographic" includes race, ethnicity, gender, disability status, and other characteristics of individuals).

|                                                                                                                                      | None                     | Some                     | A lot                    | Not applicable           |
|--------------------------------------------------------------------------------------------------------------------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| a. Faculty discussion designed to increase awareness of equity, diversity, and inclusion issues                                      | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |                          |
| b. Student discussion designed to increase awareness of equity, diversity, and inclusion issues                                      | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |                          |
| c. Program or policy changes to affect the demographic balance of faculty in the mathematical sciences.                              | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |                          |
| d. Program or policy changes to affect the demographic balance of undergraduate students in mathematical sciences.                   | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |                          |
| e. Program or policy changes intended to affect the demographic balance of graduate students in mathematical sciences.               | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| f. Consideration of existing or new programs to assist underrepresented groups and/or at-risk students in the mathematical sciences. | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |                          |

g. Please describe any other steps your department or institution has taken in the past year that deal with equity, diversity, and inclusion as they pertain to the study of and employment in the mathematical sciences.

---

- F7. For each of the following opportunities, indicate whether or not it is available to your undergraduate mathematical sciences students through your department or institutions.

|                                                                                                                     | Available                | Not available            |
|---------------------------------------------------------------------------------------------------------------------|--------------------------|--------------------------|
| a. Honors sections of departmental courses .....                                                                    | <input type="checkbox"/> | <input type="checkbox"/> |
| b. An undergraduate statistics club .....                                                                           | <input type="checkbox"/> | <input type="checkbox"/> |
| c. Special statistics programs to encourage women .....                                                             | <input type="checkbox"/> | <input type="checkbox"/> |
| d. Special statistics programs to encourage minorities .....                                                        | <input type="checkbox"/> | <input type="checkbox"/> |
| e. Opportunities to participate in statistics or data science competitions .....                                    | <input type="checkbox"/> | <input type="checkbox"/> |
| f. Special statistics lectures/colloquia not part of a statistics club .....                                        | <input type="checkbox"/> | <input type="checkbox"/> |
| g. Statistics outreach opportunities in local K–12 schools .....                                                    | <input type="checkbox"/> | <input type="checkbox"/> |
| h. Undergraduate research opportunities in statistics .....                                                         | <input type="checkbox"/> | <input type="checkbox"/> |
| i. Independent study opportunities in statistics .....                                                              | <input type="checkbox"/> | <input type="checkbox"/> |
| j. Assigned faculty advisers in statistics .....                                                                    | <input type="checkbox"/> | <input type="checkbox"/> |
| k. Opportunity to write a senior thesis in statistics .....                                                         | <input type="checkbox"/> | <input type="checkbox"/> |
| l. A career day for statistics majors .....                                                                         | <input type="checkbox"/> | <input type="checkbox"/> |
| m. Special advising about graduate school opportunities in statistical sciences .....                               | <input type="checkbox"/> | <input type="checkbox"/> |
| n. Opportunity for an internship experience or part-time employment in a professional statistical opportunity ..... | <input type="checkbox"/> | <input type="checkbox"/> |
| o. Opportunity to participate in a senior seminar .....                                                             | <input type="checkbox"/> | <input type="checkbox"/> |
| p. Supervised consultation working in a consulting lab with clients .....                                           | <input type="checkbox"/> | <input type="checkbox"/> |
| q. Opportunity to tutor, grade papers, or TA in the department .....                                                | <input type="checkbox"/> | <input type="checkbox"/> |
| r. A defined gathering space for majors .....                                                                       | <input type="checkbox"/> | <input type="checkbox"/> |

- F8. Give your best estimate of the number of all of your majors who have participated in each of the following activities over the past year September 1, 2020 – August 31, 2021.

- Undergraduate research project in statistics \_\_\_\_\_
- Internship in statistics \_\_\_\_\_
- Statistical consulting to client \_\_\_\_\_

**F. Undergraduate Program (Fall 2021) (cont.)**

Statistics Questionnaire

F9a. Does your department offer a minor in statistics?

Yes..... ☐ → If Yes, go to F9b.No ..... ☐ → If No, go to F10.

F9b. How many students graduated with a minor in statistics from your department between July 1, 2020 and June 30, 2021?

Number of students..... 

F10. Does your institution allow a student to meet an institutional, divisional, or general education graduation requirement in the mathematical sciences using an Advanced Placement course (taken while the student was in high school)?

Yes ..... ☐No ..... ☐

**G. Introductory Statistics Instruction**

Statistics Questionnaire

G1. Has your department offered a non-traditional “pathways” course sequence within the last five years?

Yes ..... ☐

No ..... ☐

(“Pathways” is defined to be a single course or course sequence that enables students to complete a college-level gateway mathematics or statistics course that is aligned to students’ academic and/or career goals within one academic year.)

**The following questions are about instruction in course E1: Introductory Statistics for non-majors/minors (no calculus prerequisite) on page 7.**

G2. How many different kinds of introductory statistics courses designed for non-majors (general education courses) that have no calculus prerequisite does your department offer? (e.g. statistics for social scientists, for life scientists, etc.)

1 ..... ☐

2 ..... ☐

3 ..... ☐

More than 3 ..... ☐

The following questions are about instruction in course E1: Introductory Statistics (no calculus prerequisite) on page 7. If you offer more than one such course, choose the course that is aimed at the most general audience.

**G. Introductory Statistics Instruction (cont.)**

Statistics Questionnaire

- G3. How often are each of these instructional strategies used in the Introductory Statistics courses taught in your department in Fall 2021?

|                                                                                   | At least<br>once a<br>week | Occasionally             | Almost<br>never          |
|-----------------------------------------------------------------------------------|----------------------------|--------------------------|--------------------------|
| a. Focusing on conceptual understanding over formulas and procedures              | <input type="checkbox"/>   | <input type="checkbox"/> | <input type="checkbox"/> |
| b. Integrating real world applications                                            | <input type="checkbox"/>   | <input type="checkbox"/> | <input type="checkbox"/> |
| c. Students collect, organize, and analyze real data                              | <input type="checkbox"/>   | <input type="checkbox"/> | <input type="checkbox"/> |
| d. Using student-centered active learning strategies                              | <input type="checkbox"/>   | <input type="checkbox"/> | <input type="checkbox"/> |
| e. Using assessment such as regular graded homework or quizzes to inform teaching | <input type="checkbox"/>   | <input type="checkbox"/> | <input type="checkbox"/> |

- G4. Technology used in teaching Introductory Statistics could include graphing calculators, statistical software, or online applets.

How successful is your program in adopting each of the following use of technology in your Introductory Statistics (no calculus prerequisite) courses taught in Fall 2021?

|                                                                      | Very<br>Successful       | Somewhat<br>Successful   | Not<br>Successful        |
|----------------------------------------------------------------------|--------------------------|--------------------------|--------------------------|
| a. Students use technology to explore concepts                       | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| b. Instructors use technology to demonstrate concepts                | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| c. Students use technology to analyze data                           | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| d. Students' ability to use technology to solve problems is assessed | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

**G. Introductory Statistics Instruction (cont.)**

Statistics Questionnaire

- G5. Are there other introductory statistics courses at your institution, offered by departments outside of the mathematical sciences?

Yes..... ☐ —————→ If Yes, go to G6

No ..... ☐ —————→ If No, go to G7

- G6. Enter the Fall 2021 total enrollment in all such introductory statistics courses, offered outside of the mathematical sciences, at your institution. \_\_\_\_\_

- G7. Are there introductory data science courses at your institution offered by other departments?

Yes..... ☐ —————→ If Yes, go to G8

No ..... ☐ —————→ If No, go to Section H

- G8. Enter the Fall 2021 total enrollment in all such introductory data science courses, offered outside of your department at your institution. \_\_\_\_\_



**H. Pre-service Teacher Education in Statistics**

Statistics Questionnaire

**Questions regarding the statistical preparation of pre-service teachers:**

- H1. Does your institution offer a program of study that leads to obtaining credentials to teach mathematics in public high schools (any license that includes grades 9-12) in your state?

Yes..... ☐ —————> If Yes, go to H2

No ..... ☐ —————> If No, skip to H3

- H2. If yes, for each of the following core areas indicate whether the core area is required of all students seeking certification that leads to obtaining credentials as a grades 9-12 mathematics teacher in public high schools in your state; if the course is generally taken by those seeking certification but not required; and if in that core area your department offers a special course that is specifically designed for pre-service grades 9-12 mathematics teachers.

| Course                                                                                                                                    | Required                 |                          | If<br>"Required"<br>is "No":<br>Generally<br>Taken |                          | Special<br>Course<br>Offered |                          |
|-------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|--------------------------|----------------------------------------------------|--------------------------|------------------------------|--------------------------|
|                                                                                                                                           | Yes                      | No                       | Yes                                                | No                       | Yes                          | No                       |
| a. Introductory Statistics that includes a simulation-based approach to inference (whether or not accompanied by a normal-based approach) | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>                           | <input type="checkbox"/> | <input type="checkbox"/>     | <input type="checkbox"/> |
| b. Introductory Statistics that only includes a normal-based (non-simulation-based) approach to inference                                 | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>                           | <input type="checkbox"/> | <input type="checkbox"/>     | <input type="checkbox"/> |
| c. Statistical Methods with an introductory course as a prerequisite                                                                      | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>                           | <input type="checkbox"/> | <input type="checkbox"/>     | <input type="checkbox"/> |
| d. Statistical modeling                                                                                                                   | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>                           | <input type="checkbox"/> | <input type="checkbox"/>     | <input type="checkbox"/> |
| e. Probability and/or statistics with calculus prerequisite                                                                               | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>                           | <input type="checkbox"/> | <input type="checkbox"/>     | <input type="checkbox"/> |
| f. Other (please specify)                                                                                                                 | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/>                           | <input type="checkbox"/> | <input type="checkbox"/>     | <input type="checkbox"/> |

**H. Pre-service Teacher Education in Statistics (cont.)**

Statistics Questionnaire

- H3. Does your institution offer a program of study that leads to obtaining credentials to teach mathematics at public middle schools (any license that includes grades 6-8) that is distinct from either the elementary or high school levels?

Yes..... ☐ —————> If Yes, go to H4

No ..... ☐ —————> If No, skip to H5

- H4. If yes, how many semester hours of courses in statistics from your department are required by your institution's program of certification for pre-service middle grades teachers?

Number of semester hours.....

- H5. Does your institution offer a program of study that leads to obtaining credentials to teach mathematics in public elementary schools (any license that includes grades K-5) in your state?

Yes..... ☐ —————> If Yes, go to H6

No ..... ☐ —————> If No, skip to Section I

- H6. If yes, how many semester hours of courses in statistics from your department are required by your institution's program of certification for pre-service elementary grades teachers?

Number of semester hours.....

**I. Comments and Suggestions**

Statistics Questionnaire

If you found some question(s) difficult to interpret or answer, please let us know. We welcome suggestions to improve future surveys (e.g., CBMS 2025).

Comments: \_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

*Thank you for completing this questionnaire. We know it was a time-consuming process and we hope that the resulting survey report, which we hope to publish in spring 2023, will be of use to you and your department.*

*Please keep a copy of your responses to this questionnaire in case questions arise.*

## Appendix IX

# Standard Errors

**TABLE S.1**

| SEs              |                                                                        |              |      |                                             |
|------------------|------------------------------------------------------------------------|--------------|------|---------------------------------------------|
|                  | Four-Year College & University<br>Mathematics & Statistics Departments |              |      | Two-Year College<br>Mathematics Departments |
|                  | 2021                                                                   | 2021 by Dept |      | 2021                                        |
|                  |                                                                        | Math         | Stat |                                             |
| Mathematics      | 121.7                                                                  | 121.7        | na   | 120.1                                       |
| Statistics       | 22.4                                                                   | 24.7         | 14.2 | 33.8                                        |
| Computer Science | 18.8                                                                   | 18.8         | na   | na                                          |
| Total            | 114.6094534                                                            | 133.5        | 14.2 | 144.7                                       |

**TABLE S.2**

| SEs                     |                        |                                             |
|-------------------------|------------------------|---------------------------------------------|
| Mathematics Departments | Statistics Departments | Two-Year College<br>Mathematics Departments |
| 2021                    | 2021                   | 2021                                        |
| 16.2                    |                        | 47.1                                        |
| 66.3                    |                        | 42.0                                        |
| 62.7                    |                        | 26.3                                        |
| 17.1                    |                        | 31.2                                        |
| 121.7                   |                        | 120.1                                       |
| 19.5                    | 12.0                   | 33.8                                        |
| 8.4                     | 5.7                    |                                             |
| 24.7                    | 14.2                   | 33.8                                        |
| 3.3                     |                        |                                             |
| 7.2                     |                        |                                             |
| 7.8                     |                        |                                             |
| 2.9                     |                        |                                             |
| 18.8                    |                        |                                             |
| 133.5                   | 14.2                   | 144.7                                       |

**TABLE S.3**

| SEs    |
|--------|
| 20–21  |
| 1581.0 |
| 725.6  |
| 0.0    |
| 591.4  |
| 0.0    |
| 237.5  |
| 105.8  |
| 415.2  |
| 545.2  |
| 9.0    |
| na     |
| 217.6  |
| na     |
| 266.5  |
| 1828.7 |
| 671.4  |
| 392.7  |
| 118.2  |
| 1885.6 |
| 689.0  |

TABLE S.6

| SEs                                              |                                      |                         |                |                                         |              |                        |                            |
|--------------------------------------------------|--------------------------------------|-------------------------|----------------|-----------------------------------------|--------------|------------------------|----------------------------|
|                                                  | Percentage of Sections Taught by     |                         |                |                                         |              |                        |                            |
|                                                  | Tenured/<br>Tenure-<br>Eligible<br>% | Other<br>Full-Time<br>% | Part-Time<br>% | Graduate<br>Teaching<br>Assistants<br>% | Unknown<br>% | Enrollment<br>in 1000s | Average<br>Section<br>Size |
| <b>Four-Year Colleges &amp; Universities</b>     |                                      |                         |                |                                         |              |                        |                            |
| Non-Mainstream Calculus I                        |                                      |                         |                |                                         |              |                        |                            |
| Lecture with separate recitation                 | 6.6                                  | 10.1                    | 7.5            | 8.4                                     | 0.0          | 5.0                    | 14.5                       |
| Sections that meet as a class                    | 6.7                                  | 8.4                     | 5.0            | 6.1                                     | 0.0          | 11.7                   | 4.9                        |
| Other sections                                   | 0.0                                  | 0.0                     | 0.0            | 0.0                                     | 0.0          | 0.2                    | 0.0                        |
| Course total 2021<br>(2010, 2015)                | 6.1                                  | 7.8                     | 4.6            | 5.4                                     | 0.0          | 12.2                   | 4.5                        |
| Non-Mainstream Calculus II, III, etc.            |                                      |                         |                |                                         |              |                        |                            |
| Course total 2021<br>(2010, 2015)                | 23.2                                 | 26.1                    | 13.8           | 0.0                                     | 0.0          | 3.7                    | 19.8                       |
| Total Non-Mnstrm Calculus I & II, III,<br>etc.   | 5.8                                  | 7.8                     | 4.2            | 5.0                                     | 0.0          | 13.8                   | 4.8                        |
| (2010, 2015)                                     |                                      |                         |                |                                         |              |                        |                            |
| <b>Two-Year Colleges</b>                         |                                      |                         |                |                                         |              |                        |                            |
|                                                  | Full-Time<br>%                       |                         | Part-Time<br>% |                                         |              |                        |                            |
| Total Non-Mnstrm Calculus I & II<br>(2010, 2015) | 5.9                                  |                         | 5.9            |                                         |              | 2.0                    | 2.1                        |

TABLE S.9

| SEs                                  |                      |                                      |                      |                                      |                      |
|--------------------------------------|----------------------|--------------------------------------|----------------------|--------------------------------------|----------------------|
| Four-Year Mathematics<br>Departments |                      | Two-Year Mathematics<br>Departments  |                      | Statistics Departments               |                      |
| Distance-<br>Learning<br>Enrollments | Other<br>Enrollments | Distance-<br>Learning<br>Enrollments | Other<br>Enrollments | Distance-<br>Learning<br>Enrollments | Other<br>Enrollments |
| 4246                                 | 14747                | 35753                                | 19237                |                                      |                      |
| 25277                                | 55630                | 29460                                | 18613                |                                      |                      |
| 13188                                | 29512                | 11829                                | 5635                 |                                      |                      |
| 2953                                 | 9996                 | 4070                                 | 2705                 |                                      |                      |
| 3925                                 | 12441                | 2780                                 | 1466                 |                                      |                      |
| 7352                                 | 16091                | 25613                                | 15524                | 7257                                 | 9477                 |

TABLE S.10

| SEs                     |           |              |       |                   |
|-------------------------|-----------|--------------|-------|-------------------|
| Mathematics Departments |           |              |       | Two-Year Colleges |
| Univ (PhD)              | Univ (MA) | College (BA) | Total | Total             |
| 4.5                     | 11.5      | 5.5          | 4.5   | 6.5               |
| 9.1                     | 8.2       | 4.9          | 4.0   |                   |
|                         |           |              |       |                   |
| 11.1                    | 11.1      | 7.2          | 5.9   | 4.5               |
| 11.1                    | 9.3       | 7.2          | 5.9   | 4.5               |
| 0.0                     | 7.1       | 0.0          | 1.0   |                   |
|                         |           |              |       |                   |
| 12.0                    | 12.2      | 7.3          | 6.0   | 6.2               |
| 12.0                    | 10.9      | 7.3          | 5.9   | 6.1               |
| 0.0                     | 7.1       | 0.0          | 1.0   | 1.0               |
|                         |           |              |       |                   |
| 10.9                    | 13.2      | 8.0          | 6.5   | 6.6               |
| 10.9                    | 11.9      | 7.9          | 6.5   | 6.6               |
| 0.0                     | 8.5       | 4.1          | 3.4   | 1.6               |
|                         |           |              |       |                   |
| 10.4                    | 6.5       | 4.0          | 3.3   | 3.6               |
| 10.4                    | 4.1       | 4.0          | 3.3   | 3.6               |
| 0.0                     | 5.1       | 0.0          | 0.7   |                   |

TABLE S.11

| SEs                     |           |              |       |                        |                   |
|-------------------------|-----------|--------------|-------|------------------------|-------------------|
| Mathematics Departments |           |              |       | Statistics Departments | Two-Year Colleges |
| Univ (PhD)              | Univ (MA) | College (BA) | Total | Total                  | Total             |
| 9.7                     | 8.8       | 5.8          | 4.6   | 5.7                    | 3.8               |
|                         |           |              |       |                        |                   |
| 17.4                    | 12.5      | 6.8          | 5.8   | 6.1                    |                   |
| 14.0                    | 12.5      | 6.3          | 5.3   | 7.5                    |                   |
| 14.0                    | 0.0       | 2.2          | 1.9   | 6.7                    |                   |
| 0.0                     | 0.0       | 3.1          | 2.5   | 7.2                    |                   |
|                         |           |              |       |                        |                   |
|                         |           |              |       |                        |                   |
| 11.6                    | 7.3       | 6.9          | 5.8   | 5.5                    | 5.2               |
| 11.6                    | 7.3       | 6.8          | 5.7   | 5.5                    | 5.2               |
| 0.0                     | 0.0       | 1.6          | 1.2   |                        |                   |
|                         |           |              |       |                        |                   |
| 14.0                    | 8.8       | 5.8          | 4.7   | 5.5                    | 4.1               |
| 14.0                    | 8.8       | 5.6          | 4.6   | 5.5                    | 4.1               |
| 0.0                     | 0.0       | 1.6          | 1.2   |                        |                   |
|                         |           |              |       |                        |                   |
| 14.7                    | 12.6      | 5.9          | 5.0   | 7.1                    |                   |
| 16.3                    | 12.9      | 8.6          | 7.2   | 7.8                    |                   |
| 9.1                     | 5.3       | 6.2          | 5.0   | 5.0                    |                   |
|                         |           |              |       |                        |                   |
| 17.6                    | 13.3      | 7.8          | 6.7   | 8.2                    | 6.6               |
| 18.0                    | 13.3      | 8.1          | 6.9   | 8.2                    |                   |
| 9.1                     | 0.0       | 5.3          |       | 2.6                    | 2.2               |
|                         |           |              |       |                        |                   |
| 16.5                    | 5.3       | 4.6          | 3.8   | 5.9                    | 3.6               |
| 16.5                    | 0.0       | 4.4          | 3.6   | 5.9                    | 3.6               |
| 0.0                     | 5.3       | 1.6          | 1.4   |                        |                   |



TABLE S.12

| SEs                                                               |                     |                         |           |              |       |                        |                   |
|-------------------------------------------------------------------|---------------------|-------------------------|-----------|--------------|-------|------------------------|-------------------|
|                                                                   |                     | Mathematics Departments |           |              |       | Statistics Departments | Two-Year Colleges |
|                                                                   |                     | Univ (PhD)              | Univ (MA) | College (BA) | Total | Total                  | Total             |
| Students use technology to explore concepts                       |                     |                         |           |              |       |                        |                   |
|                                                                   | Very successful     | 18.0                    | 13.7      | 8.9          | 7.5   | 7.8                    | 6.9               |
|                                                                   | Somewhat successful | 18.0                    | 13.7      | 8.8          | 7.3   | 7.5                    | 6.9               |
|                                                                   | Not successful      | 0.0                     | 0.0       | 1.6          | 1.2   | 3.6                    | 1.6               |
| Instructors use technology to demonstrate concepts                |                     |                         |           |              |       |                        |                   |
|                                                                   | Very successful     | 18.1                    | 11.9      | 6.2          | 5.2   | 7.1                    | 6.3               |
|                                                                   | Somewhat successful | 18.1                    | 11.9      | 6.1          | 5.2   | 7.1                    | 6.3               |
|                                                                   | Not successful      | 0.0                     | 0.0       | 1.6          | 1.2   |                        | 1.2               |
| Students use technology to analyze data                           |                     |                         |           |              |       |                        |                   |
|                                                                   | Very successful     | 18.1                    | 13.6      | 7.5          | 6.5   | 7.7                    | 6.4               |
|                                                                   | Somewhat successful | 18.1                    | 13.6      | 7.3          | 6.3   | 7.3                    | 6.1               |
|                                                                   | Not successful      | 0.0                     | 0.0       | 2.2          | 1.7   | 3.6                    | 3.2               |
| Students' ability to use technology to solve problems is assessed |                     |                         |           |              |       |                        |                   |
|                                                                   | Very successful     | 18.0                    | 13.6      | 8.3          | 7.1   | 8.2                    | 6.8               |
|                                                                   | Somewhat successful | 18.0                    | 13.6      | 8.2          | 6.9   | 8.1                    | 6.9               |
|                                                                   | Not successful      | 0.0                     | 7.3       | 1.6          | 1.6   | 2.6                    | 3.7               |

TABLE S.13

| SEs                                          |        |
|----------------------------------------------|--------|
|                                              | 2021   |
| <b>Four-Year Colleges &amp; Universities</b> |        |
| Mathematics Departments                      |        |
| Full-time faculty                            | 756    |
| Part-time faculty                            | 245    |
| Statistics Departments (PhD)                 |        |
| Full-time faculty                            | 100    |
| Part-time faculty                            | 74     |
| <b>Two-Year College Mathematics Programs</b> |        |
| Full-time faculty                            | 1117.1 |
| Part-time faculty                            | 1630.9 |

TABLE S.14

| SEs                     |                     |                 |         |  |
|-------------------------|---------------------|-----------------|---------|--|
| Fall 2021               |                     |                 |         |  |
| Total                   | TTE                 | Other Full-Time | Postdoc |  |
| 756                     | 337                 | 480             | 140     |  |
| 673                     | 335                 | 396             | 140     |  |
| 232                     | 27                  | 229             |         |  |
|                         |                     |                 |         |  |
| 100                     | 47                  | 87              | 25      |  |
| 83                      | 47                  | 66              | 25      |  |
| 38                      | 2                   | 38              |         |  |
| 769                     | 346                 | 489             | 143     |  |
| Total Full-Time Faculty | Full-Time Permanent | Other Full-Time |         |  |
| 1117                    | 919                 | 470             |         |  |

TABLE S.15

| SEs                          |                    |                 |                 |         |
|------------------------------|--------------------|-----------------|-----------------|---------|
| Fall 2021                    |                    |                 |                 |         |
| Total                        | Tenured            | Tenure-Eligible | Other Full-Time | Postdoc |
| 756                          | 285                | 69              | 479             | 140     |
| 215                          | 61                 | 25              | 164             | 34      |
| 100                          | 41                 | 21              | 87              | 25      |
| 60                           | 20                 | 12              | 49              | 14      |
| July 1, 2015 – June 30, 2021 |                    |                 |                 |         |
| Total Full-Time              | Full-Time Age < 40 |                 |                 |         |
| 918.7                        | 224.5              |                 |                 |         |
| 590.8                        | 121.9              |                 |                 |         |
| 2.8%                         | 4%                 |                 |                 |         |

TABLE S.16

| SEs                                                    |                                               |       |       |       |       |       |       |       |       |     |     |                  |
|--------------------------------------------------------|-----------------------------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-----|-----|------------------|
| Four-Year College & University Mathematics Departments | Percentage of Tenured/Tenure-Eligible Faculty |       |       |       |       |       |       |       |       |     |     | Average Age 2021 |
|                                                        | <30                                           | 30–34 | 35–39 | 40–44 | 45–49 | 50–54 | 55–59 | 60–64 | 65–69 | >69 | %   |                  |
|                                                        | 0.0                                           | 0.1   | 0.1   | 0.1   | 0.1   | 0.1   | 0.1   | 0.1   | 0.1   | 0.1 | 0.3 |                  |
|                                                        | 0.0                                           | 0.1   | 0.0   | 0.1   | 0.0   | 0.1   | 0.1   | 0.1   | 0.0   | 0.0 | 0.1 |                  |
|                                                        | 0.1                                           | 0.1   | 0.1   | 0.1   | 0.1   | 0.1   | 0.1   | 0.0   | 0.0   | 0.0 | 0.2 |                  |
|                                                        | 0.1                                           | 0.1   | 0.1   | 0.1   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0 | 0.1 |                  |
| Total tenured & tenure-eligible faculty                | 0.1                                           | 0.2   | 0.2   | 0.2   | 0.2   | 0.2   | 0.1   | 0.2   | 0.1   | 0.2 |     |                  |
| Two-Year College Mathematics Programs                  | Percentage of Permanent Full-Time Faculty     |       |       |       |       |       |       |       |       |     |     | 0.8              |
|                                                        | <30                                           | 30–34 | 35–39 | 40–44 | 45–49 | 50–54 | 55–59 | >59   |       |     |     |                  |
|                                                        |                                               |       |       |       |       |       |       |       |       |     |     |                  |
|                                                        |                                               |       |       |       |       |       |       |       |       |     |     |                  |
| Full-time permanent faculty                            | 1.0                                           | 1.0   | 1.0   | 1.0   | 1.0   | 2.0   | 1.0   | 2.0   |       |     |     |                  |
| Note: 0 means less than one half of 1%.                |                                               |       |       |       |       |       |       |       |       |     |     |                  |

Note: 0 means less than one half of 1%.

TABLE S.17

| SEs                                     |                                               |       |       |       |       |       |       |       |       |     |                        |
|-----------------------------------------|-----------------------------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-----|------------------------|
| All Statistics<br>Departments           | Percentage of Tenured/Tenure-Eligible Faculty |       |       |       |       |       |       |       |       |     | Average<br>Age<br>2021 |
|                                         | <30                                           | 30–34 | 35–39 | 40–44 | 45–49 | 50–54 | 55–59 | 60–64 | 65–69 | >69 |                        |
|                                         | %                                             | %     | %     | %     | %     | %     | %     | %     | %     | %   |                        |
| Tenured men                             | 0.2                                           | 0.2   | 0.3   | 0.4   | 0.3   | 0.3   | 0.4   | 0.3   | 0.4   | 0.5 | 13.1                   |
| Tenured women                           | 0.0                                           | 0.4   | 0.3   | 0.3   | 0.2   | 0.3   | 0.3   | 0.2   | 0.2   | 0.2 | 4.2                    |
| Tenure-eligible men                     | 0.4                                           | 0.4   | 0.3   | 0.3   | 0.4   | 0.3   | 0.2   | 0.2   | 0.2   | 0.2 | 4.3                    |
| Tenure-eligible women                   | 0.2                                           | 0.5   | 0.3   | 0.4   | 0.2   | 0.2   | 0.2   | 0.2   | 0.1   | 0.1 | 2.5                    |
| Total tenured & tenure-eligible faculty | 0.8                                           | 1.6   | 1.2   | 1.4   | 1.1   | 1.1   | 1.1   | 1.0   | 0.9   | 1.0 |                        |

TABLE S.18

| SEs        |                                |                                                              |                                |                  |              |
|------------|--------------------------------|--------------------------------------------------------------|--------------------------------|------------------|--------------|
| Asian<br>% | Black,<br>Not<br>Hispanic<br>% | Mexican<br>American/<br>Puerto Rican/<br>Other Hispanic<br>% | White,<br>Not<br>Hispanic<br>% | AIAN & NHPI<br>& | Unknown<br>% |
| 0.2        | 0.1                            | 0.0                                                          | 0.6                            | 0.0              | 0.1          |
| 0.1        | 0.0                            | 0.0                                                          | 0.1                            | 0.0              | 0.0          |
| 0.1        | 0.1                            | 0.0                                                          | 0.1                            | 0.0              | 0.0          |
| 0.0        | 0.0                            | 0.0                                                          | 0.1                            | 0.0              | 0.0          |
| 0.1        | 0.0                            | 0.0                                                          | 0.2                            | 0.0              | 0.1          |
| 0.1        | 0.0                            | 0.0                                                          | 0.1                            | 0.0              | 0.0          |
| 0.1        | 0.0                            | 0.0                                                          | 0.2                            | 0.0              | 0.1          |
| 0.1        | 0.0                            | 0.0                                                          | 0.2                            | 0.0              | 0.1          |
| 0.3        | 0.1                            | 0.1                                                          | 1.0                            | 0.0              | 0.3          |
| 0.1        | 0.1                            | 0.1                                                          | 0.4                            | 0.0              | 0.1          |

TABLE S.19

| SEs        |                                |                                                              |                                |                  |              |
|------------|--------------------------------|--------------------------------------------------------------|--------------------------------|------------------|--------------|
| Asian<br>% | Black,<br>Not<br>Hispanic<br>% | Mexican<br>American/<br>Puerto Rican/<br>Other Hispanic<br>% | White,<br>Not<br>Hispanic<br>% | AIAN & NHPI<br>& | Unknown<br>% |
| 0.5        | 0.1                            | 0.1                                                          | 0.7                            | 0.0              | 0.3          |
| 0.3        | 0.1                            | 0.1                                                          | 0.4                            | 0.0              | 0.1          |
| 0.3        | 0.1                            | 0.1                                                          | 0.3                            | 0.0              | 0.1          |
| 0.2        | 0.1                            | 0.1                                                          | 0.2                            | 0.0              | 0.1          |
| 0.3        | 0.0                            | 0.0                                                          | 0.4                            | 0.0              | 0.2          |
| 0.2        | 0.0                            | 0.0                                                          | 0.1                            | 0.0              | 0.1          |
| 0.2        | 0.1                            | 0.1                                                          | 0.4                            | 0.0              | 0.1          |
| 0.3        | 0.1                            | 0.1                                                          | 0.5                            | 0.0              | 0.1          |
| 0.8        | 0.2                            | 0.2                                                          | 1.0                            | 0.0              | 0.6          |
| 0.4        | 0.1                            | 0.2                                                          | 0.8                            | 0.0              | 0.2          |

TABLE S.20

| SEs       |
|-----------|
| 2020–2021 |
| 16.1      |
| 8.9       |
| 14.9      |
| 27.8      |
| 7         |

**TABLE SP.1**

| SEs                     |                                                                 |            |                  |
|-------------------------|-----------------------------------------------------------------|------------|------------------|
|                         | Percentage Whose Institutions Have a Certification Program for: |            |                  |
|                         | Grades K–5                                                      | Grades 6–8 | Secondary (9–12) |
| Mathematics Departments | Fall 2021                                                       | Fall 2021  | Fall 2021        |
| Univ (PhD)              | 9.9                                                             | 9.6        | 8.1              |
| Univ (MA)               | 10.0                                                            | 4.8        | 0.0              |
| Coll (BA)               | 7.7                                                             | 6.1        | 4.3              |
| Total Math Depts        | 6.0                                                             | 4.9        | 3.3              |

**TABLE SP.2**

| SEs               |
|-------------------|
| 5.7<br>2.3<br>1.7 |
| 4.6<br>1.8<br>1.6 |
| 4.6<br>2.0<br>1.6 |

**TABLE SP.3**

| SEs |
|-----|
| 9.0 |
| 5.7 |
| 1.7 |
| 3.4 |
| 1.9 |

TABLE SP.4

| SEs          |             |             |            |
|--------------|-------------|-------------|------------|
| Univ (PhD) % | Univ (MA) % | Coll (BA) % | All Math % |
| 8.8          | 8.7         | 9.0         | 6.4        |
|              | 4.5         | 3.2         | 2.4        |
| 10.9         | 11.6        | 9.2         | 6.7        |
| 9.3          | 14.1        | 7.2         | 5.6        |
| 10.4         | 5.6         | 5.3         | 4.0        |
| 10.2         | 8.8         | 8.5         | 6.2        |
| Univ (PhD) % | Univ (MA) % | Coll (BA) % | All Math % |
| 12.0         | 8.7         | 9.4         | 6.7        |
| 6.1          | 9.7         | 7.6         | 5.6        |
| 11.7         | 13.8        | 10.6        | 7.7        |
| 10.9         | 8.8         | 7.6         | 5.5        |
| 6.0          |             | 5.0         | 3.5        |
| 5.6          | 4.5         | 5.0         | 3.6        |
| Univ (PhD) % | Univ (MA) % | Coll (BA) % | All Math % |
| 11.3         | 14.0        | 9.6         | 7.2        |
| 10.5         | 7.8         | 7.4         | 5.5        |
| 1.9          | 5.6         | 6.7         | 4.8        |
|              | 13.4        |             | 2.4        |
|              |             | 2.7         | 1.9        |
| 5.3          | 4.5         | 4.8         | 3.5        |



TABLE SP.5

| SEs          |             |             |            |
|--------------|-------------|-------------|------------|
| Univ (PhD) % | Univ (MA) % | Coll (BA) % | All Math % |
|              |             | 5.6         | 4.0        |
|              | 8.3         |             | 1.8        |
|              |             | 5.6         | 4.0        |
| 0.0          | 8.3         | 7.9         | 6.0        |
| Univ (PhD) % | Univ (MA) % | Coll (BA) % | All Math % |
|              |             | 16.1        | 11.8       |
|              |             | 12.7        | 9.1        |
| 31.4         | 12.2        | 14.2        | 10.6       |
| 10.9         | 13.6        | 10.2        | 7.9        |
| 32.6         | 12.2        | 4.1         | 4.5        |
| 10.9         | 18.5        |             | 4.7        |
| Univ (PhD) % | Univ (MA) % | Coll (BA) % | All Math % |
|              | 8.3         | 10.0        | 8.6        |
| 31.7         | 16.4        | 10.5        | 9.3        |
| 32.6         | 15.2        | 5.6         | 5.5        |
| 10.9         |             |             | 0.4        |

TABLE SP.6

| SEs                                                                   |                   |                   |                  |                                                |                   |                   |                  |                                                                                         |                   |                   |                  |
|-----------------------------------------------------------------------|-------------------|-------------------|------------------|------------------------------------------------|-------------------|-------------------|------------------|-----------------------------------------------------------------------------------------|-------------------|-------------------|------------------|
| Percentage of Departments with Secondary Certification Program Where: |                   |                   |                  |                                                |                   |                   |                  |                                                                                         |                   |                   |                  |
| Course Is Required                                                    |                   |                   |                  | Course Is Generally Taken,<br>but Not Required |                   |                   |                  | Math Dept Offers Special Course<br>in the Subject for Secondary<br>Pre-Service Teachers |                   |                   |                  |
| Univ<br>(PhD)<br>%                                                    | Univ<br>(MA)<br>% | Coll<br>(BA)<br>% | All<br>Math<br>% | Univ<br>(PhD)<br>%                             | Univ<br>(MA)<br>% | Coll<br>(BA)<br>% | All<br>Math<br>% | Univ<br>(PhD)<br>%                                                                      | Univ<br>(MA)<br>% | Coll<br>(BA)<br>% | All<br>Math<br>% |
| 10.6                                                                  | 9.7               | 8.6               | 6.7              | 15.0                                           | 14.2              | 9.8               | 7.6              | 5.7                                                                                     | 11.5              | 4.9               | 4.1              |
| 7.2                                                                   | 7.8               | 8.5               | 6.5              | 14.8                                           | 0.0               | 13.0              | 10.3             | 6.0                                                                                     |                   | 0.5               | 0.7              |
| 10.2                                                                  | 10.4              | 4.2               | 3.8              | 12.1                                           | 14.0              | 6.1               | 5.1              | 5.7                                                                                     |                   | 4.9               | 3.8              |
| 8.8                                                                   | 7.5               | 6.4               | 4.9              | 15.2                                           | 0.0               | 8.5               | 6.6              | 12.1                                                                                    | 9.5               | 7.2               | 5.9              |
| 11.2                                                                  | 12.7              | 7.8               | 6.1              | 13.8                                           | 19.1              | 10.7              | 8.6              | 6.3                                                                                     |                   | 4.2               | 3.3              |
| 7.4                                                                   | 10.2              | 7.1               | 5.5              | 6.7                                            | 9.8               | 8.2               | 6.2              |                                                                                         |                   | 5.2               | 4.1              |
| 8.4                                                                   | 13.1              | 8.5               | 6.6              | 8.2                                            | 14.4              | 7.6               | 5.9              |                                                                                         |                   | 3.6               | 2.8              |
| 4.5                                                                   | 6.8               | 3.9               | 3.1              | 4.9                                            | 8.5               | 2.7               | 2.4              |                                                                                         |                   | 3.2               | 2.5              |
| 6.5                                                                   | 3.9               | 0.4               | 1.0              | 4.7                                            | 6.8               | 3.1               | 2.6              |                                                                                         |                   | 1.6               | 1.3              |
| 10.7                                                                  | 13.1              | 8.1               | 6.3              | 13.7                                           | 21.1              | 10.4              | 8.6              |                                                                                         |                   | 5.1               | 3.8              |
| 10.8                                                                  | 12.3              | 7.8               | 6.0              | 13.8                                           | 14.1              | 6.7               | 5.9              | 8.0                                                                                     | 13.0              | 5.5               | 4.7              |
| 13.2                                                                  | 14.7              | 8.6               | 6.9              | 11.2                                           | 18.5              | 9.2               | 7.4              | 9.4                                                                                     | 11.8              |                   | 1.9              |

TABLE SP.7

| SEs                                                                              |                   |                  |                                                |                   |                  |                                                                                         |                   |                  |
|----------------------------------------------------------------------------------|-------------------|------------------|------------------------------------------------|-------------------|------------------|-----------------------------------------------------------------------------------------|-------------------|------------------|
| Percentage of Statistics Departments with Secondary Certification Program Where: |                   |                  |                                                |                   |                  |                                                                                         |                   |                  |
| Course Is Required                                                               |                   |                  | Course Is Generally Taken,<br>but Not Required |                   |                  | Stat Dept Offers<br>Special Course in the Subject for<br>Secondary Pre-Service Teachers |                   |                  |
| Univ<br>(PhD)<br>%                                                               | Univ<br>(MA)<br>% | All<br>Stat<br>% | Univ<br>(PhD)<br>%                             | Univ<br>(MA)<br>% | All<br>Stat<br>% | Univ<br>(PhD)<br>%                                                                      | Univ<br>(MA)<br>% | All<br>Stat<br>% |
| 9.3                                                                              | 28.7              | 8.8              |                                                |                   |                  | 5.7                                                                                     |                   | 4.6              |
| 13.7                                                                             | 28.7              | 12.0             |                                                |                   |                  |                                                                                         |                   |                  |
| 11.0                                                                             |                   | 8.8              | 8.6                                            |                   | 6.4              |                                                                                         |                   |                  |
| 6.9                                                                              |                   | 5.4              | 7.4                                            |                   | 5.7              |                                                                                         |                   |                  |
| 13.5                                                                             | 33.1              | 11.9             | 17.6                                           |                   | 14.7             |                                                                                         |                   |                  |
| 10.0                                                                             | 28.7              | 9.2              | 8.6                                            |                   | 6.9              |                                                                                         | 28.7              | 4.6              |
| 14.7                                                                             | 0.0               | 11.7             |                                                |                   |                  |                                                                                         |                   |                  |
| 14.7                                                                             |                   | 11.7             |                                                |                   |                  |                                                                                         |                   |                  |
| 28.7                                                                             | 40.6              | 21.7             |                                                |                   |                  |                                                                                         |                   |                  |
| 33.1                                                                             | 40.6              | 21.7             |                                                |                   |                  |                                                                                         |                   |                  |
| 28.7                                                                             |                   | 15.3             |                                                |                   |                  |                                                                                         |                   |                  |

TABLE SP.8.A

| SEs               |           |              |       |                  |                   |
|-------------------|-----------|--------------|-------|------------------|-------------------|
| Mathematics Depts |           |              |       | Statistics Depts | Two-Year Colleges |
| Univ (PhD)        | Univ (MA) | College (BA) | Total | Total            | Total             |
| 9.1               | 11.6      | 6.8          | 5.2   | 6.8              | 6.4               |
| 8.8               | 10.9      | 7.1          | 5.5   | 5.7              | 5.7               |
| 5.6               | 9.0       | 6.6          | 5.0   | 4.5              | 4.6               |
|                   |           |              |       |                  |                   |
| 8.0               | 9.1       | 6.2          | 4.7   | 5.4              | 4.0               |
| 8.0               | 9.1       | 6.2          | 4.7   | 5.4              | 4.0               |
|                   |           |              |       |                  | 0.6               |
|                   |           |              |       |                  |                   |
| 9.1               | 10.7      | 6.0          | 4.7   | 7.5              | 6.4               |
| 8.5               | 10.0      | 7.5          | 5.6   | 6.8              | 5.8               |
| 7.5               | 11.6      | 7.0          | 5.4   | 5.0              | 3.2               |
|                   |           |              |       |                  |                   |
| 8.5               | 9.6       | 7.4          | 5.4   | 5.7              | 6.2               |
| 8.3               | 9.6       | 7.3          | 5.4   | 5.4              | 6.0               |
| 3.0               | 0.0       | 2.9          | 2.1   | 2.3              | 2.1               |

TABLE SP.8.B(a)

| SEs               |              |                 |       |                 |              |                 |       |               |              |                 |       |
|-------------------|--------------|-----------------|-------|-----------------|--------------|-----------------|-------|---------------|--------------|-----------------|-------|
| Prior to Pandemic |              |                 |       | During Pandemic |              |                 |       | Fall 2021     |              |                 |       |
| Univ<br>(PhD)     | Univ<br>(MA) | College<br>(BA) | Total | Univ<br>(PhD)   | Univ<br>(MA) | College<br>(BA) | Total | Univ<br>(PhD) | Univ<br>(MA) | College<br>(BA) | Total |
| 8.5               | 6.3          | 7.6             | 5.8   | 9.1             | 11.4         | 6.7             | 5.2   | 8.1           | 7.3          | 7.0             | 5.4   |
| 9.1               | 10.0         | 7.0             | 5.4   | 8.0             | 10.8         | 7.3             | 5.7   | 9.1           | 9.3          | 6.8             | 5.3   |
| 4.7               | 6.6          | 3.9             | 3.1   | 8.6             | 11.5         | 6.5             | 5.3   | 9.0           | 11.1         | 6.6             | 5.1   |

TABLE SP.8.B(b)

| SEs               |                 |           |
|-------------------|-----------------|-----------|
| Prior to Pandemic | During Pandemic | Fall 2021 |
| 6.8               | 7.7             | 6.0       |
| 7.7               | 6.6             | 7.4       |
| 6.4               | 7.7             | 7.6       |

TABLE SP.8.B(c)

| SEs               |                 |           |
|-------------------|-----------------|-----------|
| Prior to Pandemic | During Pandemic | Fall 2021 |
| 3.7               | 6.4             | 5.7       |
| 5.9               | 5.7             | 6.2       |
| 6.0               | 6.8             | 6.6       |

TABLE SP.9

| SEs               |           |              |       |                  |           |       |                |
|-------------------|-----------|--------------|-------|------------------|-----------|-------|----------------|
| Mathematics Depts |           |              |       | Statistics Depts |           |       | Two-Year Depts |
| Univ (PhD)        | Univ (MA) | College (BA) | Total | Univ (PhD)       | Univ (MA) | Total |                |
| 8.7               | 8.5       | 6.3          | 4.8   | 7.3              | 13.3      | 6.3   | 4.8            |
| 8.7               | 8.5       | 6.3          | 4.8   | 7.3              | 13.3      | 6.3   | 4.8            |
| 10.6              | 13.2      | 8.7          | 6.8   | 9.9              | 21.7      | 8.7   | 6.4            |
| 8.8               | 8.6       | 6.8          | 5.2   | 9.6              | 15.3      | 8.2   | 6.3            |
| 7.7               | 10.4      | 5.9          | 4.7   | 5.2              | 19.8      | 5.7   | 2.6            |
| 4.4               | 8.0       | 3.9          | 3.1   |                  |           |       |                |
| 7.1               | 7.2       | 7.5          | 5.6   | 7.1              | 21.7      | 7.1   | 6.3            |
| 9.9               | 13.8      | 9.1          | 7.0   | 9.8              | 21.7      | 8.6   | 5.6            |
| 8.2               | 6.5       | 4.1          | 3.3   |                  |           |       | 5.3            |
| 10.6              | 13.2      | 7.3          | 5.9   | 9.9              | 15.3      | 8.5   | 6.6            |

TABLE SP.10

| SEs               |           |              |       |                  |                   |
|-------------------|-----------|--------------|-------|------------------|-------------------|
| Mathematics Depts |           |              |       | Statistics Depts | Two-Year Colleges |
| Univ (PhD)        | Univ (MA) | College (BA) | Total | Total            | Total             |
| 5.9               | 8.8       | 5.4          | 4.2   | 6.7              | 5.7               |
| 10.8              | 11.3      | 7.4          | 5.7   | 8.6              | 7.1               |
| 10.5              | 8.5       | 6.6          | 5.1   | 7.4              | 6.4               |
|                   |           |              |       |                  |                   |
| 4.3               | 8.8       | 3.8          | 3.1   | 5.0              | 4.3               |
| 9.1               | 12.4      | 6.4          | 5.1   | 8.5              | 7.2               |
| 9.6               | 13.3      | 7.1          | 5.7   | 8.8              | 6.9               |
|                   |           |              |       |                  |                   |
| 8.9               | 13.3      | 7.7          | 6.0   | 8.7              | 6.3               |
| 10.3              | 13.7      | 8.0          | 6.3   | 8.8              | 6.9               |
| 7.3               | 6.5       | 6.2          | 4.7   | 5.0              | 5.4               |
|                   |           |              |       |                  |                   |
| 7.2               | 0.0       | 1.8          | 1.6   |                  |                   |
| 10.2              | 13.1      | 8.6          | 6.7   | 8.8              | 6.9               |
| 10.8              | 13.1      | 8.7          | 6.8   | 8.8              | 6.9               |
|                   |           |              |       |                  |                   |
| 9.3               | 8.1       | 5.4          | 4.3   | 8.4              | 6.3               |
| 10.7              | 10.8      | 8.3          | 6.4   | 8.6              | 7.1               |
| 8.5               | 8.1       | 7.0          | 5.3   | 4.1              | 5.4               |
|                   |           |              |       |                  |                   |
| 6.7               | 0.0       | 1.8          | 1.6   | 3.0              |                   |
| 10.6              | 13.5      | 7.6          |       | 8.5              | 6.8               |
| 10.8              | 13.5      | 7.6          | 6.1   | 8.4              | 6.8               |
|                   |           |              |       |                  |                   |
| 4.3               | 0.0       | 1.8          | 1.4   | 5.7              | 1.3               |
| 10.2              | 11.8      | 7.4          | 5.8   | 8.8              | 6.9               |
| 10.5              | 11.8      | 7.5          | 5.8   | 8.5              | 6.9               |
|                   |           |              |       |                  |                   |
| 10.7              | 12.0      | 7.4          | 5.9   | 8.2              | 6.2               |
| 10.7              | 13.5      | 8.2          | 6.5   | 8.6              | 6.9               |
| 0.0               | 12.7      | 6.0          | 4.8   | 5.0              | 5.0               |

TABLE SP.11.A

| SEs                     |              |                 |       |
|-------------------------|--------------|-----------------|-------|
| Mathematics Departments |              |                 |       |
| Univ<br>(PhD)           | Univ<br>(MA) | College<br>(BA) | Total |
| 1.0                     | 6.3          | 3.4             | 2.6   |
| 3.0                     | 3.2          | 3.6             | 2.7   |
| 4.7                     |              | 0.3             | 0.6   |
|                         | 5.1          |                 | 0.7   |
|                         |              | 2.3             | 1.7   |
| 3.0                     | 4.0          |                 | 0.7   |
|                         |              |                 |       |
| 1.0                     |              |                 | 0.1   |
| 4.1                     | 4.0          | 1.6             | 1.4   |
| 4.2                     | 3.2          | 2.5             | 1.9   |
| 3.8                     | 4.0          | 1.6             | 1.4   |
| 3.8                     |              | 2.3             | 1.8   |
| 3.0                     | 4.0          |                 | 0.7   |
| 4.7                     |              |                 | 0.6   |
| 3.0                     |              |                 | 0.4   |
|                         |              |                 |       |
| 3.8                     |              |                 | 0.5   |
| 3.8                     | 4.0          | 0.9             | 1.0   |
|                         |              | 2.8             | 2.1   |
| 3.8                     |              |                 | 0.5   |
| 3.8                     |              |                 | 0.5   |
|                         |              |                 |       |
|                         | 3.2          |                 | 0.4   |
|                         |              |                 |       |
|                         |              |                 |       |
| 3.0                     | 4.0          | 2.8             | 2.2   |
|                         |              | 3.5             | 2.6   |
|                         | 4.0          |                 | 0.5   |



TABLE SP.11.B

| SEs                     |           |              |       |                        |           |       |
|-------------------------|-----------|--------------|-------|------------------------|-----------|-------|
| Mathematics Departments |           |              |       | Statistics Departments |           |       |
| Univ (PhD)              | Univ (MA) | College (BA) | Total | Univ (PhD)             | Univ (MA) | Total |
| 9.3                     | 11.3      | 6.1          | 5.0   | 8.6                    | 19.4      | 7.8   |
| 9.1                     | 11.8      | 5.5          | 4.6   | 8.6                    | 13.3      | 7.7   |
| 9.9                     | 8.7       | 5.5          | 4.4   | 8.6                    | 19.4      | 7.8   |
| 6.1                     | 3.5       |              | 0.8   | 6.9                    |           | 5.7   |
| 6.7                     | 9.6       | 4.5          | 3.7   | 8.6                    | 13.3      | 7.6   |
| 4.6                     |           |              | 0.5   | 4.0                    |           | 3.3   |
| 9.3                     | 7.8       | 3.6          | 3.1   | 8.5                    | 17.4      | 7.7   |
|                         | 5.0       |              | 0.7   | 7.3                    | 13.3      | 6.3   |
|                         |           |              |       | 4.0                    |           | 3.3   |
|                         | 3.5       | 2.6          | 2.1   | 4.0                    | 19.4      | 5.0   |
| 3.6                     |           | 1.0          | 0.9   |                        | 13.3      | 6.0   |
|                         | 5.5       |              | 1.4   | 7.6                    | 20.1      | 7.1   |
| 7.0                     |           | 0.3          | 0.8   | 7.3                    | 20.1      | 7.0   |
| 5.7                     | 6.7       | 1.3          | 1.5   | 8.6                    | 17.4      | 7.7   |
|                         |           |              |       | 8.0                    | 19.4      | 7.3   |
| 4.6                     |           | 1.0          | 0.9   | 5.5                    | 13.3      | 5.0   |
| 3.6                     |           | 3.9          | 3.0   | 7.6                    | 13.3      | 6.6   |
| 7.3                     | 8.9       | 4.5          | 3.7   | 8.6                    | 19.4      | 7.7   |

TABLE SP.12

| SEs                                              |                           |                                       |                                            |                               |                                                          |                                  |
|--------------------------------------------------|---------------------------|---------------------------------------|--------------------------------------------|-------------------------------|----------------------------------------------------------|----------------------------------|
| Honors Sections<br>of Courses<br>for Majors<br>% | Math or Stat<br>Club<br>% | Special<br>Programs<br>for Women<br>% | Special<br>Programs<br>for Minorities<br>% | Math or Stat<br>Contests<br>% | Special Math<br>or Stat Colloquia<br>for Undergrads<br>& | Outreach in<br>K–12 Schools<br>% |
| 9.7                                              | 0.0                       | 9.0                                   | 8.7                                        | 5.3                           | 9.7                                                      | 9.6                              |
| 12.2                                             | 0.0                       | 9.9                                   | 9.5                                        | 11.4                          | 9.2                                                      | 12.3                             |
| 6.2                                              | 7.6                       | 5.7                                   | 5.1                                        | 6.8                           | 7.1                                                      | 7.8                              |
| 5.1                                              | 5.6                       | 4.8                                   | 4.3                                        | 5.2                           | 5.5                                                      | 6.0                              |
| 9.0                                              | 8.1                       | 8.8                                   | 8.1                                        | 7.7                           | 8.6                                                      | 7.7                              |
| 21.7                                             | 20.1                      | 21.7                                  | 19.8                                       | 19.8                          | 19.8                                                     |                                  |
| 8.1                                              | 7.5                       | 8.0                                   | 7.3                                        | 7.0                           | 7.7                                                      | 6.6                              |
| na                                               | 4.9                       | 3.7                                   | 4.5                                        | na                            | 5.3                                                      | 7.1                              |



**TABLE SP.14**

| Activity                                                    | SEs               |             |            |            |                   |             |            |
|-------------------------------------------------------------|-------------------|-------------|------------|------------|-------------------|-------------|------------|
|                                                             | All Math<br>Depts | PhD<br>Math | MA<br>Math | BA<br>Math | All Stat<br>Depts | PhD<br>Stat | MA<br>Stat |
| Undergraduate research project in the mathematical sciences | 2910.3            | 835.7       | 380.6      | 2715.1     | 124.1             | 122.4       | 45.0       |
| Internship in mathematical sciences                         | 873.7             | 220.1       | 101.9      | 820.5      | 196.7             | 115.7       | 175.4      |
| Mathematical or statistical consulting to client            | 1034.7            | 67.5        | 93.9       | 1029.3     | 114.8             | 63.0        | 102.9      |

**TABLE SP.15**

| SEs                    |                        |                        |                     |
|------------------------|------------------------|------------------------|---------------------|
| Univ<br>(PhD)          | Univ<br>(MA)           | Coll<br>(BA)           | All Departments     |
| Offered<br>Course<br>% | Offered<br>Course<br>% | Offered<br>Course<br>% | Offered Course<br>% |
| 9.5                    | 11.3                   | 6.3                    | 5.0                 |
| 9.5                    | 7.6                    | 6.1                    | 4.8                 |
| 4.5                    |                        | 2.6                    | 2.0                 |
| 9.1                    | 9.7                    | 5.2                    | 4.1                 |
| 8.2                    | 5.8                    | 2.5                    | 2.3                 |
| 5.5                    | 4.6                    | 3.9                    | 3.0                 |
| 8.9                    | 12.4                   | 7.1                    | 5.5                 |
|                        | 4.6                    | 1.9                    | 1.5                 |
| 9.1                    | 11.7                   | 7.9                    | 6.1                 |
| 4.5                    |                        | 3.8                    | 2.9                 |
| 5.1                    | 9.9                    | 4.7                    | 3.7                 |

TABLE SP.16

| SEs                   |           |                   |           |                      |           |                   |           |                      |           |                   |
|-----------------------|-----------|-------------------|-----------|----------------------|-----------|-------------------|-----------|----------------------|-----------|-------------------|
| Four-Year Mathematics |           |                   |           | Two-Year Mathematics |           |                   |           | Four-Year Statistics |           |                   |
| 5                     |           |                   |           | 6                    |           |                   |           | 5                    |           |                   |
| Dual Enrollments      |           | Other Enrollments |           | Dual Enrollments     |           | Other Enrollments |           | Dual Enrollments     |           | Other Enrollments |
| Spring 2021           | Fall 2021 | Fall 2021         | Fall 2021 | Spring 2021          | Fall 2021 | Fall 2021         | Fall 2021 | Spring 2021          | Fall 2021 | Fall 2021         |
| 5741.0                | 6326.6    | 27723.0           |           | 15882.4              | 20634.2   |                   | 30625.2   | na                   | na        | na                |
| 2924.3                | 2082.0    | 18464.0           |           | 11375.8              | 12508.8   |                   | 15887.6   | na                   | na        | na                |
| 1888.2                | 3426.0    | 32361.0           |           | 6540.7               | 6783.2    |                   | 15537.8   | na                   | na        | na                |
| 2728.3                | 1504.3    | 17773.2           |           | 6125.2               | 6386.2    |                   | 33760.7   | 26.7                 | 1032.2    | 11865.0           |
| 1969.5                | 898.0     | na                |           | 6206.3               | 6646.8    |                   | na        | na                   | na        | na                |

**TABLE SP.17**

| SEs                                     |                                        |                           |
|-----------------------------------------|----------------------------------------|---------------------------|
| Four-Year<br>Mathematics<br>Departments | Two-Year<br>Mathematics<br>Departments | Statistics<br>Departments |
| 2.3                                     | 7.0                                    | 0                         |
| 1590.5                                  | 2869.8                                 |                           |

TABLE SP.18

| SEs                                                                                          |                        |                   |                      |                                     |                   |                      |                           |                   |                      |
|----------------------------------------------------------------------------------------------|------------------------|-------------------|----------------------|-------------------------------------|-------------------|----------------------|---------------------------|-------------------|----------------------|
|                                                                                              | Required in All Majors |                   |                      | Required in Some but Not All Majors |                   |                      | Not Required in Any Major |                   |                      |
| Mathematics Department Requirements                                                          | Univ<br>(PhD)<br>%     | Univ<br>(MA)<br>% | College<br>(BA)<br>% | Univ<br>(PhD)<br>%                  | Univ<br>(MA)<br>% | College<br>(BA)<br>% | Univ<br>(PhD)<br>%        | Univ<br>(MA)<br>% | College<br>(BA)<br>% |
| Modern Algebra I                                                                             | 10.2                   | 13.9              | 8.6                  | 10.2                                | 12.8              | 8.4                  | 3.9                       | 13.8              | 6.6                  |
| Real Analysis I                                                                              | 10.2                   | 12.8              | 7.5                  | 10.1                                | 13.3              | 8.4                  | 3.9                       | 14.2              | 7.8                  |
| Modern Algebra I or Real Analysis I (major<br>may choose either to fulfill this requirement) | 9.2                    | 11.6              | 7.6                  | 8.3                                 | 13.3              | 6.2                  | 10.4                      | 15.1              | 8.5                  |
| A one-year upper-level sequence                                                              | 9.4                    | 10.7              | 8.3                  | 9.2                                 | 4.6               | 9.0                  | 10.0                      | 11.5              | 8.5                  |
| At least one computer science course                                                         | 10.2                   | 14.3              | 8.7                  | 8.8                                 | 9.1               | 5.2                  | 9.0                       | 14.3              | 6.8                  |
| At least one statistics course                                                               | 9.8                    | 9.9               | 7.6                  | 9.5                                 | 8.4               | 7.3                  | 10.2                      | 5.7               | 5.7                  |
| At least one applied mathematics course beyond<br>course E21                                 | 7.5                    | 12.8              | 8.5                  | 10.3                                | 13.1              | 8.7                  | 10.3                      | 14.4              | 7.3                  |
| A capstone experience (senior project, thesis,<br>seminar, internship)                       | 10.0                   | 12.8              | 6.3                  | 8.5                                 | 9.0               | 2.9                  | 10.0                      | 10.3              | 5.9                  |
| An exit exam (written or oral)                                                               | 3.9                    | 4.6               | 7.2                  | 7.1                                 | 7.9               | 2.8                  | 7.8                       | 9.0               | 7.6                  |

TABLE SP.19.A

|                                                                                      | SEs                    |             |                |  |                                     |             |                |  |                           |             |                |  |
|--------------------------------------------------------------------------------------|------------------------|-------------|----------------|--|-------------------------------------|-------------|----------------|--|---------------------------|-------------|----------------|--|
|                                                                                      | Required in All Majors |             |                |  | Required in Some but Not All Majors |             |                |  | Not Required in Any Major |             |                |  |
| Percentage of Mathematics Departments that Require:                                  | Univ (PhD) %           | Univ (MA) % | College (BA) % |  | Univ (PhD) %                        | Univ (MA) % | College (BA) % |  | Univ (PhD) %              | Univ (MA) % | College (BA) % |  |
| (a) Calculus I                                                                       | 0.0                    | 0.0         | 0.0            |  |                                     |             |                |  |                           |             |                |  |
| (b) Calculus II                                                                      | 0.0                    | 0.0         | 10.1           |  |                                     |             | 10.1           |  |                           |             |                |  |
| (c) Multivariable Calculus                                                           | 0.0                    | 43.3        | 24.6           |  |                                     |             | 10.1           |  |                           | 43.3        | 23.8           |  |
| (d) Linear Algebra/Matrix Theory                                                     | 0.0                    | 43.3        | 10.1           |  |                                     | 43.3        | 10.1           |  |                           |             |                |  |
| (e) At least one computer science course                                             | 32.0                   | 43.0        | 28.3           |  | 32.0                                | 43.3        |                |  | 36.3                      |             | 28.3           |  |
| (f) At least one applied mathematics course, not incl. (a), (b), (c), (d)            | 25.8                   | 43.3        | 27.2           |  |                                     |             |                |  | 25.8                      | 43.3        | 27.2           |  |
| (g) A capstone experience (e.g., a senior thesis or project, seminar, or internship) | 32.0                   | 43.3        | 11.0           |  | 32.0                                |             |                |  | 32.9                      | 43.3        | 11.0           |  |
| (h) An exit exam (oral or written)                                                   |                        |             | 23.8           |  |                                     |             |                |  | 0.0                       | 0.0         | 23.8           |  |
| (i) One Probability course                                                           | 32.0                   | 36.3        | 23.8           |  | 32.0                                | 36.3        | 3.3            |  |                           |             | 23.8           |  |
| (j) One Mathematical Statistics course                                               | 32.0                   | 0.0         | 24.6           |  | 32.0                                |             | 10.1           |  |                           |             | 23.8           |  |
| (k) One applied statistics course                                                    | 32.0                   | 0.0         | 0.0            |  | 32.0                                |             |                |  |                           |             |                |  |
| (l) One Linear Models course                                                         | 32.0                   | 0.0         | 25.6           |  | 25.8                                |             | 3.3            |  | 33.9                      |             | 25.9           |  |
| (m) One Bayesian Inference course                                                    | 33.9                   |             |                |  |                                     | 36.3        | 11.0           |  | 33.9                      | 36.3        | 11.0           |  |



TABLE SP.19.B

|                                                                                      | SEs                    |             |                                     |             |                           |             |
|--------------------------------------------------------------------------------------|------------------------|-------------|-------------------------------------|-------------|---------------------------|-------------|
|                                                                                      | Required in All Majors |             | Required in Some but Not All Majors |             | Not Required in Any Major |             |
| Percentage of Statistics Departments that Require:                                   | Univ (PhD) %           | Univ (MA) % | Univ (PhD) %                        | Univ (MA) % | Univ (PhD) %              | Univ (MA) % |
| (a) Calculus I                                                                       | 5.1                    | 22.9        | 3.1                                 | 22.9        | 4.3                       |             |
| (b) Calculus II                                                                      | 5.8                    | 22.9        | 4.3                                 | 22.9        | 4.3                       |             |
| (c) Multivariable Calculus                                                           | 8.4                    | 27.2        | 6.0                                 | 22.2        | 7.1                       | 27.2        |
| (d) Linear Algebra/Matrix Theory                                                     | 7.3                    | 27.2        | 5.1                                 |             | 5.8                       | 27.2        |
| (e) At least one computer science course                                             | 7.1                    | 22.2        | 5.3                                 | 22.2        | 5.3                       |             |
| (f) At least one applied mathematics course, not incl. (a), (b), (c), (d)            | 6.6                    | 22.2        | 7.5                                 |             | 8.8                       | 22.2        |
| (g) A capstone experience (e.g., a senior thesis or project, seminar, or internship) | 8.8                    | 27.2        | 5.8                                 | 22.2        | 8.9                       | 27.2        |
| (h) An exit exam (oral or written)                                                   |                        | 22.2        |                                     | 22.2        | 0.0                       | 27.2        |
| (i) One Probability course                                                           | 7.6                    | 28.7        | 5.1                                 |             | 6.4                       | 28.7        |
| (j) One Mathematical Statistics course                                               | 8.6                    | 22.2        | 7.3                                 |             | 6.4                       | 22.2        |
| (k) One applied statistics course                                                    | 7.6                    | 0.0         | 5.8                                 |             | 5.8                       |             |
| (l) One Linear Models course                                                         | 8.8                    | 28.7        | 6.4                                 |             | 7.9                       | 28.7        |
| (m) One Bayesian Inference course                                                    | 5.5                    | 22.2        | 6.2                                 |             | 7.7                       | 22.2        |

TABLE SP.20

| SEs                                  |               |              |              |
|--------------------------------------|---------------|--------------|--------------|
| Academic Years 2020–2021 & 2021–2022 |               |              |              |
| All Math Depts<br>2020–2022<br>%     | PhD Math<br>% | MA Math<br>% | BA Math<br>% |
| 5.6                                  | 7.6           | 11.5         | 7.4          |
| 3.8                                  | 9.0           | 9.3          | 4.6          |
| 4.8                                  | 8.2           | 11.1         | 6.0          |
| 3.6                                  | 8.9           | 8.8          | 4.4          |
| 4.3                                  | 8.5           | 8.6          | 5.5          |
| 1.1                                  | 6.7           | 5.5          |              |
| 2.4                                  | 7.0           | 5.1          | 2.9          |
| 4.6                                  | 9.1           | 11.1         | 5.9          |
| 5.5                                  | 8.8           | 9.4          | 7.1          |
| 4.1                                  | 7.5           | 10.6         | 5.2          |
| 5.4                                  | 8.1           | 11.1         | 6.8          |
| 4.0                                  | 8.7           | 9.8          | 4.9          |
| 2.6                                  | 7.5           | 9.4          | 2.8          |
| 4.1                                  | 9.1           | 10.4         | 5.0          |
| 6.0                                  | 8.9           | 9.0          | 7.7          |

TABLE SP.20 (continued)

| SEs                                  |               |              |              |
|--------------------------------------|---------------|--------------|--------------|
| Academic Years 2020–2021 & 2021–2022 |               |              |              |
| All Math Depts<br>2020–2022<br>%     | PhD Math<br>% | MA Math<br>% | BA Math<br>% |
| 1.9                                  | 6.3           | 5.9          | 1.6          |
| 3.3                                  | 7.6           | 4.5          | 4.2          |
| 2.6                                  | 9.0           | 10.6         | 2.5          |
| 4.6                                  | 8.6           | 11.4         | 5.6          |
| 3.9                                  | 9.0           | 10.8         | 4.5          |
| 5.4                                  | 7.6           | 10.5         | 6.8          |
| 2.9                                  | 8.8           | 8.3          | 3.4          |
| 2.3                                  | 8.0           | 7.0          | 2.4          |
| 3.3                                  | 8.1           | 10.5         | 3.6          |
| 1.4                                  | 8.4           | 6.6          | 0.3          |
| 3.0                                  | 8.0           | 7.8          | 3.5          |
| 5.5                                  | 9.0           | 11.0         | 6.9          |
| 5.5                                  | 9.0           | 10.4         | 7.1          |

TABLE SP.21

| SEs                                 |                  |                 |                 |  |                                     |                  |                 |
|-------------------------------------|------------------|-----------------|-----------------|--|-------------------------------------|------------------|-----------------|
| Academic Years<br>2020–21 & 2021–22 |                  |                 |                 |  | Academic Years<br>2020–21 & 2021–22 |                  |                 |
| All Math<br>Depts<br>%              | PhD<br>Math<br>% | MA<br>Math<br>% | BA<br>Math<br>% |  | All Stat<br>Depts<br>%              | PhD<br>Stat<br>% | MA<br>Stat<br>% |
| 5.6                                 | 10.0             | 11.8            | 7.0             |  | 7.5                                 | 8.0              | 19.4            |
| 4.8                                 | 10.0             | 11.2            | 5.7             |  | 7.6                                 | 8.6              | 13.3            |
| 4.5                                 | 9.3              | 10.3            | 5.6             |  | 7.7                                 | 8.6              | 20.1            |
| 1.5                                 | 9.7              | 7.4             | 0.3             |  | 6.3                                 | 6.9              | 17.4            |
| 1.7                                 | 9.1              | 8.1             | 1.1             |  | 7.8                                 | 8.6              | 17.4            |
| 2.9                                 | 7.9              | 7.1             | 3.4             |  | 7.7                                 | 8.6              | 13.3            |
| 0.6                                 | 3.6              | 3.5             |                 |  | 6.3                                 | 7.3              | 13.3            |
| 0.7                                 |                  | 5.0             |                 |  | 5.7                                 | 6.5              | 13.3            |
| 2.0                                 | 3.6              |                 | 2.6             |  | 5.0                                 | 4.8              | 17.4            |
| 0.9                                 | 3.6              |                 | 1.0             |  | 6.6                                 | 7.6              | 13.3            |
| 1.7                                 | 3.6              | 6.0             | 1.9             |  | 6.6                                 | 6.9              | 19.4            |
| 1.0                                 | 4.6              | 3.5             | 1.0             |  | 6.6                                 | 7.3              | 17.4            |
| 1.5                                 | 7.9              | 7.4             | 0.9             |  | 7.6                                 | 8.4              | 20.1            |
| 3.8                                 | 8.6              | 11.6            | 4.5             |  | 7.7                                 | 8.6              | 19.4            |
| 1.8                                 |                  | 4.3             | 2.3             |  | 7.5                                 | 8.5              | 17.4            |
| 0.8                                 |                  |                 | 1.0             |  | 6.3                                 | 6.9              | 17.4            |
| 3.8                                 | 6.5              | 6.0             | 4.8             |  | 7.3                                 | 8.2              | 17.4            |

**TABLE SP.22**

| SEs                     |                |                   |                        |                |
|-------------------------|----------------|-------------------|------------------------|----------------|
| Mathematics Departments |                |                   | Statistics Departments |                |
| Univ (PhD)<br>%         | Univ (MA)<br>% | College (BA)<br>% | Univ (PhD)<br>%        | Univ (MA)<br>% |
| 3.0                     | 4.7            | 3.3               | 0.4                    | 1.1            |
| 2.8                     | 5.0            | 3.0               | 1.5                    | 5.4            |
| 2.4                     | 0.8            | 2.7               | 2.1                    | 6.7            |
| 4.2                     | 5.3            | 4.0               | 4.8                    | 14.3           |
| 0.7                     | 0.0            | 1.0               | 0.6                    | 2.2            |
| 7.4                     | 9.0            | 5.1               | 6.5                    | 18.0           |

**TABLE SP.23**

| SEs                               |                |                   |                        |                |
|-----------------------------------|----------------|-------------------|------------------------|----------------|
| Four-Year Mathematics Departments |                |                   | Statistics Departments |                |
| Univ (PhD)<br>%                   | Univ (MA)<br>% | College (BA)<br>% | Univ (PhD)<br>%        | Univ (MA)<br>% |
| 9.7                               | 11.0           | 6.8               | 8.4                    | 21.7           |
| 8.8                               | 12.4           | 6.2               | 8.4                    | 21.7           |
| 8.9                               | 9.0            | 8.1               | 9.3                    | 19.8           |
| 9.5                               | 11.7           | 5.8               | 9.1                    | 24.3           |
| 8.9                               | 0.0            | 3.6               | 5.3                    | 15.3           |
| 7.8                               | 0.0            | 4.5               | 7.3                    | 19.8           |
| 9.7                               | 8.7            | 7.0               | 6.6                    | 18.1           |
| 8.3                               | 6.8            | 5.0               | 7.7                    | 21.7           |

**TABLE SP.24**

| SEs               |                   |             |            |            |                   |             |            |
|-------------------|-------------------|-------------|------------|------------|-------------------|-------------|------------|
| Meets Requirement | All Math<br>Depts | PhD<br>Math | MA<br>Math | BA<br>Math | All Stat<br>Depts | PhD<br>Stat | MA<br>Stat |
| Yes (%)           | 2.5               | 6.0         | 3.7        | 3.2        | 4.6               | 3.0         | 19.4       |
| No (%)            | 2.5               | 6.0         | 3.7        | 3.2        | 4.6               | 3.0         | 19.4       |

**TABLE SP.25**

| SEs                     |           |              |        |
|-------------------------|-----------|--------------|--------|
| Mathematics Departments |           |              |        |
| Univ (PhD)              | Univ (MA) | College (BA) | Total  |
| 8.9                     | 12.4      | 5.8          | 4.7    |
| 345.2                   | 2394.7    | 184.0        | 2409.3 |
| 7.1                     | 7.2       | 4.8          | 3.8    |

TABLE SP.26

| SEs                                                                                                                              |                   |           |              |                  |       |                   |
|----------------------------------------------------------------------------------------------------------------------------------|-------------------|-----------|--------------|------------------|-------|-------------------|
|                                                                                                                                  | Mathematics Depts |           |              | Statistics Depts |       | Two-Year Colleges |
|                                                                                                                                  | Univ (PhD)        | Univ (MA) | College (BA) | Total            | Total | Total             |
| Faculty discussion designed to increase awareness of DEI issues                                                                  |                   |           |              |                  |       |                   |
| None                                                                                                                             | 5.3               | 5.8       | 6.0          | 4.5              | 6.2   | 5.2               |
| Some                                                                                                                             | 9.3               | 10.0      | 7.4          | 5.8              | 8.4   | 7.4               |
| A lot                                                                                                                            | 8.7               | 8.6       | 6.9          | 5.4              | 7.9   | 6.7               |
| Student discussion designed to increase awareness of DEI issues                                                                  |                   |           |              |                  |       |                   |
| None                                                                                                                             | 9.1               | 9.9       | 7.3          | 5.8              | 7.6   | 7.2               |
| Some                                                                                                                             | 9.4               | 11.3      | 7.4          | 5.9              | 8.5   | 7.0               |
| A lot                                                                                                                            | 4.5               | 7.2       | 2.3          | 2.1              | 7.6   | 1.9               |
| Program or policy changes to affect the demographic balance of faculty in the mathematical sciences                              |                   |           |              |                  |       |                   |
| None                                                                                                                             | 9.7               | 10.8      | 7.3          | 5.7              | 8.4   | 7.0               |
| Some                                                                                                                             | 9.7               | 10.8      | 7.3          | 5.7              | 8.5   | 7.0               |
| A lot                                                                                                                            | 3.6               | 0.0       | 2.0          | 1.5              | 5.9   | 2.5               |
| Program or policy changes to affect the demographic balance of undergraduate students in the mathematical sciences               |                   |           |              |                  |       |                   |
| None                                                                                                                             | 9.7               | 11.3      | 7.8          | 6.2              | 8.4   | 6.9               |
| Some                                                                                                                             | 9.7               | 11.3      | 7.2          | 5.7              | 8.5   | 6.9               |
| A lot                                                                                                                            | 0.0               | 0.0       | 5.0          | 3.7              | 5.9   | 1.7               |
| Program or policy changes intended to affect the demographic balance of graduate students in the mathematical sciences           |                   |           |              |                  |       |                   |
| None                                                                                                                             | 9.6               | 11.7      | 6.3          | 5.2              | 7.6   |                   |
| Some                                                                                                                             | 9.7               | 9.7       | 3.7          | 3.3              | 8.6   |                   |
| A lot                                                                                                                            | 3.5               | 0.0       | 0.0          | 0.5              | 7.6   |                   |
| Not applicable                                                                                                                   | 0.0               | 8.6       | 6.6          | 5.1              | 4.0   |                   |
| Consideration of existing or new programs to assist underrepresented groups and/or at-risk students in the mathematical sciences |                   |           |              |                  |       |                   |
| None                                                                                                                             | 8.2               | 10.6      |              | 6.3              | 6.2   | 7.2               |
| Some                                                                                                                             | 9.1               | 11.1      | 7.6          | 6.0              | 8.4   | 7.2               |
| A lot                                                                                                                            | 5.8               | 4.6       | 5.0          | 3.8              | 7.9   | 5.1               |



TABLE SP.27

| Section B                                                   | SEs           |               |                 |          | Doctoral Stat |
|-------------------------------------------------------------|---------------|---------------|-----------------|----------|---------------|
|                                                             | Doctoral Math | Master's Math | Bachelor's Math | All Math |               |
| Renewable positions filled for 2020–2021                    | 99.7          | 108.4         | 63.3            | 193.6    | 33.0          |
| Number that left renewable position for 2021                | 15.2          | 5.7           | 8.9             | 19.7     | 2.5           |
| Percent that left renewable position for 2021               | 0.7           | 0.3           | 0.3             | 0.3      | 0.4           |
| Renewable positions filled for 2021–2022                    | 99.9          | 109.1         | 59.1            | 195.0    | 35.0          |
| Number active in teaching                                   | 111.8         | 107.9         | 58.2            | 200.9    | 26.3          |
| Percent active in teaching                                  | 5.0           | 6.3           | 2.0             | 2.9      | 3.8           |
| Number active in research                                   | 63.6          | 20.4          | 21.7            | 75.9     | 35.8          |
| Percent active in research                                  | 2.8           | 1.2           | 0.7             | 1.1      | 5.1           |
| Number that will attend a research conference with support  | 55.4          | 12.9          | 20.7            | 61.8     | 33.2          |
| Percent that will attend a research conference with support | 2.5           | 0.7           | 0.7             | 0.9      | 4.7           |
| Number that will attend a teaching conference with support  | 49.9          | 14.1          | 25.7            | 60.4     | 21.3          |
| Percent that will attend a teaching conference with support | 2.2           | 0.8           | 0.9             | 0.9      | 3.0           |
| Number that will serve in department committees             | 50.4          | 35.0          | 37.5            | 88.5     | 19.4          |
| Percent that will serve in department committees            | 2.2           | 2.0           | 1.3             | 1.3      | 2.8           |
| Number that advise undergrad. research projects             | 29.9          | 10.5          | 17.8            | 37.9     | 13.5          |
| Percent that advise undergrad. research projects            | 1.3           | 0.6           | 0.6             | 0.6      | 1.9           |
| Number that serve as academic advisor                       | 37.2          | 30.1          | 27.6            | 57.6     | 22.9          |
| Percent that serve as academic advisor                      | 1.7           | 1.7           | 0.9             |          | 3.3           |
| Number that serve on univ. committees                       | 33.7          | 18.4          | 26.6            | 49.2     | 13.6          |
| Percent that serve on univ. committees                      | 1.5           | 1.1           | 0.9             | 0.7      | 1.9           |
| Number that serve as course coordinator                     | 33.7          | 15.0          | 23.7            | 59.0     | 7.7           |
| Percent that serve as course coordinator                    | 1.5           | 0.9           | 0.8             | 0.9      | 1.1           |

TABLE SP.28

|                                                             | SEs           |               |                 |          |               |
|-------------------------------------------------------------|---------------|---------------|-----------------|----------|---------------|
|                                                             | Doctoral Math | Master's Math | Bachelor's Math | All Math | Doctoral Stat |
| Number of Fixed-term positions filled for 2019–2020         | 115.4         | 31.6          | 31.7            | 133.5    | 25.5          |
| Number that left renewable position for 2020                | 14.5          | 4.6           | 10.9            | 19.7     | 1.9           |
| Percent that left renewable position for 2020               | 1.5           | 0.9           | 1.4             | 0.9      | 0.9           |
| Number of Fixed-term positions filled for 2020–2021         | 114.6         | 35.5          | 31.8            | 134.1    | 25.1          |
| Number active in teaching                                   | 47.3          | 41.4          | 27.6            | 74.5     | 15.2          |
| Percent active in teaching                                  | 4.8           | 7.5           | 3.3             | 3.1      | 8.0           |
| Number active in research                                   | 32.6          | 11.1          | 12.8            | 39.3     | 12.0          |
| Percent active in research                                  | 3.3           | 2.0           | 1.5             | 1.7      | 6.3           |
| Number that will attend a research conference with support  | 22.2          | 9.5           | 11.8            | 28.1     | 10.6          |
| Percent that will attend a research conference with support | 2.3           | 1.7           | 1.4             | 1.2      | 5.6           |
| Number that will attend a teaching conference with support  | 14.7          | 10.9          | 11.8            | 22.5     | 9.3           |
| Percent that will attend a teaching conference with support | 1.5           | 2.0           | 1.4             | 0.9      | 4.9           |
| Number that will serve in department committees             | 18.0          | 17.2          | 17.8            | 30.9     | 9.7           |
| Percent that will serve in department committees            | 1.8           | 3.1           | 2.1             | 1.3      | 5.1           |
| Number that advise undergrad. research projects             | 14.5          | 2.2           | 7.9             | 16.9     | 3.5           |
| Percent that advise undergrad. research projects            | 1.5           | 0.4           | 0.9             | 0.7      | 1.9           |
| Number that serve as academic advisor                       | 17.2          | 6.3           | 14.8            | 23.9     | 6.1           |
| Percent that serve as academic advisor                      | 1.7           | 1.1           | 1.7             |          | 3.2           |
| Number that serve on univ. committees                       | 12.1          | 6.3           | 10.9            | 16.9     | 4.5           |
| Percent that serve on univ. committees                      | 1.2           | 1.1           | 1.3             | 0.7      | 2.4           |
| Number that serve as course coordinator                     | 13.0          | 6.1           | 8.9             | 18.3     | 2.8           |
| Percent that serve as course coordinator                    | 1.3           | 1.1           | 1.0             | 0.8      | 1.5           |

TABLE E.2

| SEs                              |              |              |                     |                        |              |                     |
|----------------------------------|--------------|--------------|---------------------|------------------------|--------------|---------------------|
| Fall 2021 Enrollments (in 1000s) |              |              |                     |                        |              |                     |
| Mathematics Departments          |              |              |                     | Statistics Departments |              |                     |
| Univ<br>(PhD)                    | Univ<br>(MA) | Coll<br>(BA) | Total Math<br>Depts | Univ<br>(PhD)          | Univ<br>(MA) | Total Stat<br>Depts |
| 7                                | 6            | 13           | 16                  |                        |              |                     |
| 46                               | 27           | 40           | 66                  |                        |              |                     |
| 55                               | 23           | 24           | 63                  |                        |              |                     |
| 16                               | 4            | 5            | 17                  |                        |              |                     |
| 94                               | 51           | 63           | 122                 |                        |              |                     |
| 8                                | 11           | 15           | 20                  | 12                     | 9            | 12                  |
| 3                                | 2            | 7            | 8                   | 6                      | 2            | 6                   |
| 11                               | 12           | 20           | 25                  | 14                     | 11           | 14                  |
| 0                                | 0            | 3            | 3                   |                        |              |                     |
| 3                                | 1            |              | 7                   |                        |              |                     |
| 4                                | 2            | 7            | 8                   |                        |              |                     |
| 0                                | 0            | 3            | 3                   |                        |              |                     |
| 8                                | 5            | 16           | 19                  |                        |              |                     |
| 93                               | 61           | 80           | 133                 | 14                     | 11           | 14                  |

TABLE E.5

| SEs                     |                                             |                    |           |                                   |         |                   |
|-------------------------|---------------------------------------------|--------------------|-----------|-----------------------------------|---------|-------------------|
|                         | Number of Calculus-Level Sections Taught by |                    |           |                                   |         |                   |
|                         | Tenured/<br>Tenure-Eligible                 | Other<br>Full-Time | Part-Time | Graduate<br>Teaching<br>Assistant | Unknown | Total<br>Sections |
| Mathematics Departments |                                             |                    |           |                                   |         |                   |
| Univ (PhD)              | 326                                         | 429                | 163       | 370                               | 0       | 742               |
| Univ (MA)               | 335                                         | 209                | 150       | 16                                | 271     | 593               |
| Coll (BA)               | 678                                         | 369                | 198       | 0                                 | 10      | 1000              |
|                         | 829                                         | 619                | 299       | 370                               | 271     | 1375              |

TABLE E.7

| SEs                          |                              |                   |                        |                              |                   |
|------------------------------|------------------------------|-------------------|------------------------|------------------------------|-------------------|
|                              | Sections<br>Taught by<br>TTE | Total<br>Sections |                        | Sections<br>Taught by<br>TTE | Total<br>Sections |
| Mathematics Departments      |                              |                   | Statistics Departments |                              |                   |
| Advanced Mathematics courses |                              |                   |                        |                              |                   |
| Univ (PhD)                   | 278                          | 358               |                        |                              |                   |
| Univ (MA)                    | 208                          | 259               |                        |                              |                   |
| Coll (BA)                    | 402                          | 410               |                        |                              |                   |
| Total advanced mathematics   | 557                          | 626               |                        |                              |                   |
| Advanced Statistics courses  |                              |                   |                        |                              |                   |
| Univ (PhD)                   | 86                           | 102               |                        | 45                           | 93                |
| Univ (MA)                    | 83                           | 90                |                        | 39                           | 57                |
| Coll (BA)                    | 264                          | 298               |                        |                              |                   |
| Total advanced statistics    | 292                          | 330               |                        | 48                           | 94                |
| Total all advanced courses   | 756                          | 837               |                        | 48                           | 94                |

**TABLE E.8**

| SEs                                                       |                    |           |                                   |         |                   |
|-----------------------------------------------------------|--------------------|-----------|-----------------------------------|---------|-------------------|
| Number of Lower-Level Computer Science Sections Taught by |                    |           |                                   |         |                   |
| Tenured/<br>Tenure-Eligible/<br>Permanent                 | Other<br>Full-Time | Part-Time | Graduate<br>Teaching<br>Assistant | Unknown | Total<br>Sections |
| 59                                                        | 72                 | 42        | 0                                 | 0       | 141               |
| 24                                                        | 5                  | 18        | 0                                 | 0       | 50                |
| 247                                                       | 80                 | 43        | 0                                 | 0       | 279               |
| 252                                                       | 109                | 63        | 0                                 | 0       | 309               |

**TABLE E.9**

| SEs                                                        |                    |           |                                   |         |                   |
|------------------------------------------------------------|--------------------|-----------|-----------------------------------|---------|-------------------|
| Number of Middle-Level Computer Science Sections Taught by |                    |           |                                   |         |                   |
| Tenured/<br>Tenure-Eligible                                | Other<br>Full-Time | Part-Time | Graduate<br>Teaching<br>Assistant | Unknown | Total<br>Sections |
| 39                                                         | 113                | 28        | 0                                 | 0       | 147               |
| 64                                                         | 2                  | 18        | 0                                 | 0       | 81                |
| 293                                                        | 82                 | 30        | 0                                 | 0       | 344               |
| 301                                                        | 139                | 45        | 0                                 | 0       | 380               |

**TABLE E.10**

| SEs                            |              |              |                 |                  |              |                 |                   |
|--------------------------------|--------------|--------------|-----------------|------------------|--------------|-----------------|-------------------|
| Average Section Size Fall 2021 |              |              |                 |                  |              |                 | All Depts<br>2021 |
| Mathematics Depts              |              |              |                 | Statistics Depts |              |                 |                   |
| Univ<br>(PhD)                  | Univ<br>(MA) | Coll<br>(BA) | Overall<br>Math | Univ<br>(PhD)    | Univ<br>(MA) | Overall<br>Stat |                   |
| 17                             | 4            | 1            | 2               |                  |              |                 | 2                 |
| 6                              | 2            | 1            | 1               |                  |              |                 | 1                 |
| 6                              | 2            | 1            | 2               |                  |              |                 | 2                 |
| 4                              | 1            | 1            | 2               |                  |              |                 | 2                 |
| 5                              | 2            | 1            | 1               | 8                | 12           | 7               | 1                 |
| 2                              | 2            | 2            | 1               | 4                | 4            | 4               | 1                 |
| 0                              |              | 1            | 0               |                  |              |                 | 0                 |
| 1                              | 13           | 3            | 2               |                  |              |                 | 2                 |
| 5                              | 17           | 3            | 2               |                  |              |                 | 2                 |
| 0                              | 0            | 6            | 6               |                  |              |                 | 6                 |

**TABLE E.11**

| SEs                             |              |                 |
|---------------------------------|--------------|-----------------|
| Average Recitation Section Size |              |                 |
| Univ<br>(PhD)                   | Univ<br>(MA) | College<br>(BA) |
| 17                              | 18           | 6               |
| 9                               | 0            | 0               |
| 13                              | 9            | 3               |
| 23                              | 0            |                 |
| 11                              | 0            | 4               |
| 23                              | 47           |                 |

TABLE F.1

| SEs                   |            |                 |     |           |           |           |                 |     |           |           |           |                 |
|-----------------------|------------|-----------------|-----|-----------|-----------|-----------|-----------------|-----|-----------|-----------|-----------|-----------------|
|                       | Univ (PhD) |                 |     |           |           | Univ (MA) |                 |     |           |           | Coll (BA) |                 |
|                       | Tenured    | Tenure-Eligible | OFT | Post-Docs | Part-Time | Tenured   | Tenure-Eligible | OFT | Post-Docs | Part-Time | Tenured   | Tenure-Eligible |
| Mathematics Depts     |            |                 |     |           |           |           |                 |     |           |           |           |                 |
| Doctoral faculty      | 138        | 40              | 306 | 116       | 58        | 77        | 22              | 50  | 5         | 66        | 93        | 28              |
| Doctoral (F)          | 30         | 14              | 68  | 28        | 29        | 21        | 11              | 16  | 2         | 40        | 33        | 15              |
| Non-doctoral faculty  | 15         | 0               | 80  | 0         | 124       | 12        | 6               | 172 | 0         | 80        | 14        | 5               |
| Non-doctoral (F)      | 7          | 0               | 48  | 0         | 60        | 4         | 3               | 80  | 0         | 42        | 8         | 2               |
| Total Mathematics     | 140        | 40              | 320 | 116       | 159       | 82        | 24              | 199 | 5         | 115       | 98        | 27              |
| Total Mathematics (F) | 33         | 14              | 90  | 28        | 73        | 23        | 11              | 85  | 2         | 57        | 36        | 15              |
| Statistics            |            |                 |     |           |           |           |                 |     |           |           |           |                 |
| Doctoral faculty      | 41         | 19              | 66  | 25        | 59        |           |                 |     |           |           |           |                 |
| Doctoral (F)          | 20         | 10              | 33  | 14        | 27        |           |                 |     |           |           |           |                 |
| Non-doctoral faculty  | 1          | 2               | 38  | 0         | 47        |           |                 |     |           |           |           |                 |
| Non-doctoral (F)      | 0          | 2               | 25  | 0         | 27        |           |                 |     |           |           |           |                 |
| Total Statistics      | 41         | 21              | 87  | 25        | 74        |           |                 |     |           |           |           |                 |
| Total Statistics (F)  | 20         | 12              | 49  | 14        | 33        |           |                 |     |           |           |           |                 |

TABLE F.1.1

| SEs                   |                                    |                 |     |           |           |
|-----------------------|------------------------------------|-----------------|-----|-----------|-----------|
|                       | Tenured                            | Tenure-Eligible | OFT | Post-Docs | Part-Time |
| Mathematics Depts     | Univ (PhD) + Univ (MA) + Coll (BA) |                 |     |           |           |
| Doctoral faculty      | 282                                | 69              | 396 | 140       | 104       |
| Doctoral (F)          | 58                                 | 25              | 95  | 34        | 57        |
| Non-doctoral faculty  | 24                                 | 8               | 229 | 0         | 183       |
| Non-doctoral (F)      | 12                                 | 4               | 118 | 0         | 92        |
| Total Mathematics     | 285                                | 69              | 479 | 140       | 245       |
| Total Mathematics (F) | 61                                 | 25              | 164 | 34        | 117       |
| Statistics Depts      | Univ (PhD)                         |                 |     |           |           |
| Doctoral faculty      | 41                                 | 19              | 66  | 25        | 59        |
| Doctoral (F)          | 20                                 | 10              | 33  | 14        | 27        |
| Non-doctoral faculty  | 1                                  | 2               | 38  | 0         | 47        |
| Non-doctoral (F)      | 0                                  | 2               | 25  | 0         | 27        |
| Total Statistics      | 41                                 | 21              | 87  | 25        | 74        |
| Total Statistics (F)  | 20                                 | 12              | 49  | 14        | 33        |



TABLE F.2

|                 | SEs        |                 |     |           |           |                 |     |           |           |                 |     |           |
|-----------------|------------|-----------------|-----|-----------|-----------|-----------------|-----|-----------|-----------|-----------------|-----|-----------|
|                 | Univ (PhD) |                 |     |           | Univ (MA) |                 |     |           | Coll (BA) |                 |     |           |
|                 | Tenured    | Tenure-Eligible | OFT | Post-Docs | Tenured   | Tenure-Eligible | OFT | Post-Docs | Tenured   | Tenure-Eligible | OFT | Post-Docs |
| Men, 2021       | 118        | 31              | 247 | 89        | 66        | 15              | 119 | 3         | 70        | 18              | 58  | 9         |
| Women, 2021     | 33         | 14              | 90  | 28        | 23        | 11              | 85  | 2         | 36        | 15              | 52  | 4         |
| Nonbinary, 2021 | 2          | 2               | 3   | 2         | 1         | 1               | 1   | 0         | 3         | 2               | 2   | 0         |
| Total, 2021     | 140        | 40              | 320 | 116       | 82        | 24              | 199 | 5         | 98        | 27              | 98  | 11        |
|                 |            |                 |     |           |           |                 |     |           | 285       | 69              | 479 | 140       |

TABLE F.3

| SEs             |                                 |                 |                 |           |
|-----------------|---------------------------------|-----------------|-----------------|-----------|
|                 | Doctoral Statistics Departments |                 |                 |           |
|                 | Tenured                         | Tenure-Eligible | Other Full-Time | Post-Docs |
| Men, 2021       | 34                              | 15              | 49              | 14        |
| Women, 2021     | 20                              | 12              | 49              | 14        |
| Nonbinary, 2021 | 1                               | 2               | 0               | 0         |
| Total, 2021     | 41                              | 21              | 87              | 25        |

TABLE F.4

|                           |  | SEs      |            |            |            |            |            |            |            |            |          |
|---------------------------|--|----------|------------|------------|------------|------------|------------|------------|------------|------------|----------|
|                           |  | <30<br>% | 30–34<br>% | 35–39<br>% | 40–44<br>% | 45–49<br>% | 50–54<br>% | 55–59<br>% | 60–64<br>% | 65–69<br>% | >69<br>% |
| <b>Mathematics Depts.</b> |  |          |            |            |            |            |            |            |            |            |          |
| Univ (PhD)                |  |          |            |            |            |            |            |            |            |            |          |
| Tenured men               |  | 0.1      | 0.2        | 0.2        | 0.3        | 0.3        | 0.3        | 0.3        | 0.3        | 0.3        | 0.4      |
| Tenured women             |  | 0.0      | 0.3        | 0.1        | 0.2        | 0.1        | 0.1        | 0.2        | 0.1        | 0.1        | 0.2      |
| Tenure-eligible men       |  | 0.2      | 0.3        | 0.3        | 0.2        | 0.2        | 0.1        | 0.1        | 0.2        | 0.1        | 0.1      |
| Tenure-eligible women     |  | 0.2      | 0.3        | 0.2        | 0.2        | 0.1        | 0.1        | 0.2        | 0.1        | 0.1        | 0.1      |
| Total Univ (PhD)          |  | 0.5      | 1.0        | 0.7        | 0.9        | 0.8        | 0.7        | 0.7        | 0.8        | 0.6        | 0.8      |
| Univ (MA)                 |  |          |            |            |            |            |            |            |            |            |          |
| Tenured men               |  | 0.1      | 0.2        | 0.1        | 0.2        | 0.2        | 0.2        | 0.2        | 0.3        | 0.3        | 0.3      |
| Tenured women             |  | 0.0      | 0.2        | 0.1        | 0.2        | 0.1        | 0.2        | 0.1        | 0.1        | 0.1        | 0.1      |
| Tenure-eligible men       |  | 0.3      | 0.2        | 0.3        | 0.2        | 0.2        | 0.2        | 0.2        | 0.1        | 0.1        | 0.1      |
| Tenure-eligible women     |  | 0.1      | 0.3        | 0.1        | 0.2        | 0.2        | 0.1        | 0.1        | 0.1        | 0.1        | 0.1      |
| Total Univ (MA)           |  | 0.5      | 0.9        | 0.6        | 0.8        | 0.7        | 0.7        | 0.6        | 0.7        | 0.6        | 0.5      |
| Coll (BA)                 |  |          |            |            |            |            |            |            |            |            |          |
| Tenured men               |  | 0.0      | 0.0        | 0.0        | 0.1        | 0.1        | 0.1        | 0.1        | 0.1        | 0.1        | 0.1      |
| Tenured women             |  | 0.0      | 0.1        | 0.0        | 0.1        | 0.0        | 0.0        | 0.0        | 0.1        | 0.0        | 0.0      |
| Tenure-eligible men       |  | 0.1      | 0.1        | 0.1        | 0.0        | 0.0        | 0.0        | 0.0        | 0.0        | 0.0        | 0.0      |
| Tenure-eligible women     |  | 0.1      | 0.1        | 0.0        | 0.1        | 0.0        | 0.0        | 0.0        | 0.0        | 0.0        | 0.0      |
| Total Coll (BA)           |  | 0.2      | 0.2        | 0.2        | 0.2        | 0.2        | 0.2        | 0.2        | 0.2        | 0.2        | 0.2      |
| <b>Statistics Depts.</b>  |  |          |            |            |            |            |            |            |            |            |          |
| Univ (PhD)                |  |          |            |            |            |            |            |            |            |            |          |
| Tenured men               |  | 0.2      | 0.2        | 0.3        | 0.4        | 0.3        |            | 0.4        | 0.3        | 0.4        | 0.5      |
| Tenured women             |  | 0.0      | 0.4        | 0.3        |            | 0.2        | 0.3        | 0.3        | 0.2        | 0.2        | 0.2      |
| Tenure-eligible men       |  | 0.4      | 0.4        | 0.3        | 0.3        | 0.4        | 0.3        | 0.2        | 0.2        | 0.2        | 0.2      |
| Tenure-eligible women     |  | 0.2      | 0.5        | 0.3        | 0.4        | 0.2        | 0.2        | 0.2        | 0.2        | 0.2        | 0.1      |
| Total Univ (PhD)          |  | 0.8      | 1.6        | 1.2        | 1.4        | 1.1        | 1.1        | 1.1        | 1.0        | 0.9        | 1.0      |

TABLE F.5

|                             | SEs                             |                             |                                                              |                             |                      |              |
|-----------------------------|---------------------------------|-----------------------------|--------------------------------------------------------------|-----------------------------|----------------------|--------------|
|                             | Percentage of Full-Time Faculty |                             |                                                              |                             |                      |              |
|                             | Asian<br>%                      | Black, Not<br>Hispanic<br>% | Mexican<br>American/<br>Puerto Rican/<br>Other Hispanic<br>% | White, Not<br>Hispanic<br>% | AIAN or<br>NHPI<br>% | Unknown<br>% |
| PhD Mathematics Departments |                                 |                             |                                                              |                             |                      |              |
| All full-time men           | 0.4                             | 0.2                         | 0.2                                                          | 1.6                         | 0.0                  | 0.6          |
| All full-time women         | 0.3                             | 0.1                         | 0.1                                                          | 0.6                         | 0.0                  | 0.2          |
| MA Mathematics Departments  |                                 |                             |                                                              |                             |                      |              |
| All full-time men           | 0.5                             | 0.2                         | 0.3                                                          | 1.1                         | 0.0                  | 0.8          |
| All full-time women         | 0.3                             | 0.2                         | 0.2                                                          | 0.7                         | 0.0                  | 0.3          |
| BA Mathematics Departments  |                                 |                             |                                                              |                             |                      |              |
| All full-time men           | 0.2                             | 0.1                         | 0.1                                                          | 0.5                         | 0.0                  | 0.2          |
| All full-time women         | 0.1                             | 0.1                         | 0.1                                                          | 0.4                         | 0.0                  | 0.1          |
| PhD Statistics Departments  |                                 |                             |                                                              |                             |                      |              |
| All full-time men           | 0.8                             | 0.2                         | 0.2                                                          | 1.0                         | 0.0                  | 0.6          |
| All full-time women         | 0.4                             | 0.1                         | 0.2                                                          | 0.8                         | 0.0                  | 0.2          |

TABLE F.6

|                             | SEs                             |                             |                                                              |                             |                      |              |
|-----------------------------|---------------------------------|-----------------------------|--------------------------------------------------------------|-----------------------------|----------------------|--------------|
|                             | Percentage of Full-time Faculty |                             |                                                              |                             |                      |              |
|                             | Asian<br>%                      | Black, Not<br>Hispanic<br>% | Mexican<br>American/<br>Puerto Rican/<br>Other Hispanic<br>% | White, Not<br>Hispanic<br>% | AIAN or<br>NHPI<br>% | Unknown<br>% |
| PhD Mathematics Departments |                                 |                             |                                                              |                             |                      |              |
| All part-time men           | 0.8                             | 0.7                         | 0.8                                                          | 3.0                         | 0.2                  | 1.2          |
| All part-time women         | 0.6                             | 0.8                         | 0.3                                                          | 3.0                         | 0.1                  | 0.5          |
| MA Mathematics Departments  |                                 |                             |                                                              |                             |                      |              |
| All part-time men           | 0.3                             | 0.4                         | 0.5                                                          | 1.7                         | 0.2                  | 0.5          |
| All part-time women         | 0.2                             | 0.4                         | 0.3                                                          | 1.3                         | 0.1                  | 0.3          |
| BA Mathematics Departments  |                                 |                             |                                                              |                             |                      |              |
| All part-time men           | 0.2                             | 0.1                         | 0.1                                                          | 0.7                         | 0.0                  | 0.3          |
| All part-time women         | 0.1                             | 0.1                         | 0.0                                                          | 0.7                         | 0.0                  | 0.2          |
| PhD Statistics Departments  |                                 |                             |                                                              |                             |                      |              |
| All part-time men           | 2.9                             | 1.3                         | 2.6                                                          | 7.8                         | 0.3                  | 3.4          |
| All part-time women         | 1.8                             | 0.7                         | 0.5                                                          | 8.0                         | 0.1                  | 1.2          |

TABLE FY.5

|                                                                                                           | SEs                     |              |                 |                        |                        |              |                        |  |
|-----------------------------------------------------------------------------------------------------------|-------------------------|--------------|-----------------|------------------------|------------------------|--------------|------------------------|--|
|                                                                                                           | Mathematics Departments |              |                 |                        | Statistics Departments |              |                        |  |
|                                                                                                           | Univ<br>(PhD)           | Univ<br>(MA) | College<br>(BA) | All Depts.<br>Combined | Univ<br>(PhD)          | Univ<br>(MA) | All Depts.<br>Combined |  |
| Percentage of departments that offer Introductory Statistics with no calculus prerequisite                | 10                      | 9            | 6               | 5                      | 7                      | 13           | 6                      |  |
| Number of different kinds of Introductory Statistics courses for non-majors with no calculus prerequisite |                         |              |                 |                        |                        |              |                        |  |
| 1                                                                                                         | 17                      | 12           | 7               | 6                      | 6                      | 19           | 6                      |  |
| 2                                                                                                         | 14                      | 12           | 6               | 5                      | 8                      | 19           | 8                      |  |
| 3                                                                                                         | 14                      |              | 2               | 2                      | 8                      |              | 7                      |  |
| More than 3                                                                                               |                         |              | 3               | 2                      | 8                      | 17           | 7                      |  |

TABLE FY.6

|                                                                                         | SEs                        |              |                 |                            |              |                 |
|-----------------------------------------------------------------------------------------|----------------------------|--------------|-----------------|----------------------------|--------------|-----------------|
|                                                                                         | Mathematics Departments    |              |                 | Statistics Departments     |              |                 |
|                                                                                         | At Least<br>Once a<br>Week | Occasionally | Almost<br>Never | At Least<br>Once a<br>Week | Occasionally | Almost<br>Never |
| a. Focusing on conceptual understanding over formulas and procedures                    | 6                          | 6            | 1               | 5                          | 5            |                 |
| b. Integrating real world applications                                                  | 5                          | 5            | 1               | 5                          | 5            |                 |
| c. Students collect, organize, and analyze real data                                    | 5                          | 7            | 5               | 7                          | 8            | 5               |
| d. Using student-centered active learning strategies                                    | 7                          | 7            | 4               | 8                          | 8            | 3               |
| e. Using assessments such as regular graded homework or quizzes used to inform teaching | 4                          | 4            | 1               | 6                          | 6            |                 |

TABLE FY.7

|                                                                      | SEs                     |                        |                   |                        |                        |                   |
|----------------------------------------------------------------------|-------------------------|------------------------|-------------------|------------------------|------------------------|-------------------|
|                                                                      | Mathematics Departments |                        |                   | Statistics Departments |                        |                   |
|                                                                      | Very<br>Successful      | Somewhat<br>Successful | Not<br>Successful | Very<br>Successful     | Somewhat<br>Successful | Not<br>Successful |
| a. Students use technology to explore concepts                       | 7                       | 7                      | 1                 | 8                      | 8                      | 4                 |
| b. Instructors use technology to demonstrate concepts                | 5                       | 5                      | 1                 | 7                      | 7                      |                   |
| c. Students use technology to analyze data                           | 6                       | 6                      | 2                 | 8                      | 7                      | 4                 |
| d. Students' ability to use technology to solve problems is assessed | 7                       | 7                      | 2                 | 8                      | 8                      | 3                 |

**TABLE FY.8**

| SEs                                                               |                   |              |                 |       |                  |              |       |
|-------------------------------------------------------------------|-------------------|--------------|-----------------|-------|------------------|--------------|-------|
|                                                                   | Mathematics Depts |              |                 |       | Statistics Depts |              |       |
|                                                                   | Univ<br>(PhD)     | Univ<br>(MA) | College<br>(BA) | Total | Univ<br>(PhD)    | Univ<br>(MA) | Total |
| Average estimated outside enrollment in Introductory Statistics   | 166               | 46           | 14              | 23    | 47               | 0            | 43    |
| Estimated outside national enrollment in Introductory Statistics  | 17626             | 5459         | 10154           | 21094 | 1079             | 49           | 1107  |
| Average estimated outside enrollment in Introductory Data Science | 67                | 0            | 12              | 26    | 28               |              | 28    |
| Estimated outside national enrollment in Data Science             | 6137              | 0            | 2211            | 6527  | 896              |              | 896   |

TABLE TYE.2

| SEs     |
|---------|
| 2021    |
| 157,543 |

TABLE TYE.3

| SEs   |
|-------|
| 2021  |
| 8.4   |
| 8.3   |
| 12.9  |
| 28.6  |
| 0.8   |
| 7.7   |
| 30.6  |
| 7.9   |
| 3.9   |
| 2.4   |
| 15.9  |
| 14.1  |
| 5.7   |
| 3.5   |
| 3.4   |
| 1.8   |
| 1.9   |
| 2.1   |
| 1.7   |
| 32.9  |
| 10.1  |
| 4.6   |
| 18.1  |
| 14.6  |
| 2.2   |
| 0.9   |
| 0.3   |
| 3.6   |
|       |
| 3.7   |
| 2.4   |
| 5.3   |
| 1.2   |
| 144.7 |



TABLE TYE.4

| SEs              |                         |      |
|------------------|-------------------------|------|
| Course Numbers   | Type of Course          | 2021 |
| 1–5a             | Pre-college level       | 47   |
| 6–10             | Precalculus level       | 42   |
| 11–16            | Calculus level          | 26   |
| 19–20            | Statistics, Probability | 34   |
| 17–18 &<br>21–31 | Remaining courses       | 31   |
| 1–31             | Total, All Courses      | 145  |

TABLE TYE.5

| SEs |
|-----|
| 6.9 |
| 6.8 |
| 6.4 |
| 4.7 |
| 1.9 |
| 6.1 |
| 5.0 |
| 6.9 |
| 4.6 |
| 3.5 |
| 6.9 |
| 4.3 |
| 6.4 |
| 6.8 |
| 6.4 |
| 2.0 |
| 6.4 |
| 6.1 |
| 5.1 |
| 3.8 |
| 4.6 |
| 5.5 |
| 6.7 |
| 6.6 |
| 6.9 |
| 6.8 |
| 4.6 |
| 6.3 |
| na  |
| 6.1 |
| 3.7 |
| 5.9 |
| 3.9 |

**TABLE TYE.6**

| SEs |
|-----|
| 4.3 |
| 6.4 |
| 6.1 |
| 5.1 |
| 3.8 |
| 5.5 |
| 6.7 |
| 6.9 |
| 6.1 |
| 3.7 |

**TABLE TYE.7**

| SEs                     |                                             |
|-------------------------|---------------------------------------------|
| Average<br>Section Size | Percentage of<br>Sections with<br>Size > 30 |
| 1.04                    | 0.00                                        |
| 0.85                    | 0.00                                        |
| 1.09                    | 0.01                                        |
| 0.96                    | 0.00                                        |
| 0.64                    | 0.00                                        |

**TABLE TYE.7.1**

| SEs                          |                                                             |
|------------------------------|-------------------------------------------------------------|
| 2021 Average<br>Section Size | Percentage of 2021<br>Departments with Average<br>Size > 30 |
| 1.69                         | 0.00                                                        |
| 0.85                         | 0.01                                                        |
| 1.41                         | 0.01                                                        |
| 0.99                         | 0.01                                                        |
| 0.89                         | 0.01                                                        |

TABLE TYE.8

| SEs                            |                                        |                                |                                        |
|--------------------------------|----------------------------------------|--------------------------------|----------------------------------------|
| Average On-Campus Section Size | Average Distance Learning Section Size | Average On-Campus Section Size | Average Distance Learning Section Size |
| 1.4                            | 1.8                                    | 2.6                            | 2.3                                    |
| 1.7                            | 1.6                                    | 1.9                            | 2.8                                    |
| 1.4                            | 1.2                                    | 1.0                            | 1.0                                    |
| 1.4                            | 3.5                                    | 0.9                            | 3.8                                    |
| 3.9                            | 0.0                                    | 6.3                            | 4.3                                    |
| 1.7                            | 1.8                                    | 1.1                            | 1.4                                    |
| 1.3                            | 1.1                                    | 1.3                            | 1.1                                    |
| 1.4                            | 1.2                                    | 1.5                            | 1.7                                    |
| 1.6                            | 4.1                                    | 3.4                            | 2.1                                    |
| 4.0                            | 6.0                                    |                                | 5.3                                    |
| 1.1                            | 1.4                                    | 2.3                            | 2.0                                    |
| 1.2                            | 1.5                                    |                                |                                        |
| 1.4                            | 1.8                                    | 1.4                            | 0.8                                    |
| 1.5                            | 1.9                                    | 2.3                            | 2.2                                    |
| 1.7                            | 1.7                                    | 1.3                            | 3.0                                    |
| 5.6                            | 1.5                                    | 1.8                            | 2.3                                    |
| 2.2                            | 2.1                                    |                                |                                        |

TABLE TYE.9

| SEs                |                                                |                                                    |
|--------------------|------------------------------------------------|----------------------------------------------------|
| Number of Sections | Number of Sections Taught by Part-Time Faculty | Percentage of Sections Taught by Part-Time Faculty |
| 1436.6             | 1556.3                                         | 0.1                                                |
| 903.0              | 718.9                                          | 0.1                                                |
| 404.8              | 131.4                                          | 0.0                                                |
| 114.3              | 71.1                                           | 0.2                                                |
| 103.7              | 55.5                                           | 0.1                                                |
| 829.1              | 716.4                                          | 0.2                                                |
| 622.4              | 542.9                                          | 0.1                                                |
| 123.3              | 170.1                                          | 0.3                                                |
| 131.3              | 98.0                                           | 0.2                                                |
| 3284.4             | 2755.8                                         | 0.1                                                |

**TABLE TYE.10**

| SEs                  |                                              |
|----------------------|----------------------------------------------|
| Implemented in 2021? | Intermediate Algebra a Prerequisite in 2021? |
| Yes                  | Yes                                          |
| 6.5                  |                                              |
| 4.6                  | 6.8                                          |
| 4.5                  | 5.8                                          |
| 6.6                  | 7.9                                          |
| 12.9                 | 13.7                                         |

**TABLE TYE.11**

| SEs                       |                              |                                 |
|---------------------------|------------------------------|---------------------------------|
| 2021                      | 2021                         | 2021                            |
| Total Enrollments (1000s) | Distance Enrollments (1000s) | Percentage Distance Enrollments |
| 8.4                       | 6.6                          | 0.1                             |
| 8.3                       | 6.3                          | 0.1                             |
| 12.9                      | 8.1                          | 0.1                             |
| 28.6                      | 22.1                         | 0.1                             |
| 0.8                       | 0.2                          | 0.7                             |
| 7.7                       | 5.8                          | 0.1                             |
| 30.6                      | 20.5                         | 0.0                             |
| 7.9                       | 5.9                          | 0.0                             |
| 3.9                       | 2.1                          | 0.1                             |
| 2.4                       | 1.1                          | 0.1                             |
| 15.9                      | 10.9                         | 0.1                             |
| 14.1                      | 10.7                         | 0.0                             |
| 5.7                       | 3.9                          | 0.0                             |
| 3.5                       | 2.6                          | 0.1                             |
| 3.4                       | 2.5                          | 0.0                             |
| 1.8                       | 0.9                          | 0.3                             |
| 1.9                       | 1.5                          | 0.1                             |
| 2.1                       | 1.6                          | 0.1                             |
| 1.7                       | 1.5                          | 0.1                             |

TABLE TYE.11 (continued)

| SEs                             |                                    |                                       |
|---------------------------------|------------------------------------|---------------------------------------|
| 2021                            | 2021                               | 2021                                  |
| Total<br>Enrollments<br>(1000s) | Distance<br>Enrollments<br>(1000s) | Percentage<br>Distance<br>Enrollments |
| 32.9                            | 25.2                               | 0.0                                   |
| 10.1                            | 4.1                                | 0.4                                   |
| 4.6                             | 3.2                                | 0.2                                   |
| 18.1                            | 11.9                               | 0.1                                   |
| 14.6                            | 9.8                                | 0.1                                   |
| 2.2                             | 1.5                                | 0.1                                   |
| 0.9                             | 0.8                                | 0.1                                   |
| 0.3                             | 0.3                                | 0.0                                   |
| 3.6                             | 2.9                                | 0.1                                   |
|                                 |                                    |                                       |
| 3.7                             | 2.5                                | 0.1                                   |
| 2.4                             | 1.8                                | 0.1                                   |
| 5.3                             | 4.7                                | 0.1                                   |
| 1.2                             | 0.9                                | 0.1                                   |
| 144.7                           | 107.7                              | 0.0                                   |

TABLE TYE.12

| SEs |
|-----|
| 4.8 |
| 4.8 |
| 6.4 |
| 6.3 |
| 2.6 |
| 6.3 |
| 5.6 |
| 5.3 |
| 6.6 |

TABLE TYE.13

| SEs  |
|------|
| 2021 |
| na   |
| 6    |
| 8    |
| 8    |
| 8    |
| 8    |
| 7    |
| 6    |
| 8    |
| 6    |
| 6    |
| na   |
| na   |
| 5    |
| 4    |
| 5    |
| 4    |
| 7    |
| 5    |
| 6    |
| 8    |

**TABLE TYF.1**

| SEs                            |        |
|--------------------------------|--------|
| Two-Year Colleges              | 2021   |
| Full-time permanent faculty    | 918.7  |
| Full-time continuing faculty   | 261.6  |
| Other full-time faculty        | 326.9  |
| Part-time faculty paid by TYC  | 1630.9 |
| Part-time, paid by third party | 969.1  |

**TABLE TYF.2**

| SEs                                                                                      |                                             |          |          |          |          |     |
|------------------------------------------------------------------------------------------|---------------------------------------------|----------|----------|----------|----------|-----|
|                                                                                          | Teaching Assignment in Weekly Contact Hours |          |          |          |          |     |
|                                                                                          | <10                                         | 10 to 12 | 13 to 15 | 16 to 18 | 19 to 21 | >21 |
| Percentage of Two-Year Colleges                                                          | 2.3                                         | 4.4      | 6.7      | 4.2      | 4.2      | 2.0 |
| <b>Full-Time Permanent Faculty</b>                                                       |                                             |          |          |          |          |     |
| A. Average weekly contact hours: 0.5                                                     |                                             |          |          |          |          |     |
| B. Percentage who teach extra hours for extra pay at their own two-year college: 4%      |                                             |          |          |          |          |     |
| C. Percentage teaching 1–3 extra hours for extra pay: 4%                                 |                                             |          |          |          |          |     |
| D. Percentage teaching 4–6 extra hours for extra pay: 4%                                 |                                             |          |          |          |          |     |
| E. Percentage teaching 7 or more extra hours for extra pay: 5%                           |                                             |          |          |          |          |     |
| <b>Part-Time Faculty</b>                                                                 |                                             |          |          |          |          |     |
| F. Percentage who teach 6 or more hours weekly: 4%                                       |                                             |          |          |          |          |     |
| G. Percentage of two-year colleges requiring part-time faculty to hold office hours: 6.5 |                                             |          |          |          |          |     |

**TABLE TYF.3**

| SEs                                          |       |
|----------------------------------------------|-------|
| Number no longer part of 2021–2022 faculty   | 63.4  |
| Total full-time permanent faculty, fall 2021 | 918.7 |

**TABLE TYF.4**

| SEs                                   |      |                                 |
|---------------------------------------|------|---------------------------------|
| Highest Degree                        | 2021 | Percentage of Full-Time Faculty |
| Doctorate                             | 3    | 0.03                            |
| Master's                              | 3    | 0.03                            |
| Bachelor's                            | 0    | 0.00                            |
| Number of full-time permanent faculty | 919  |                                 |

**TABLE TYF.5**

| SEs                                |                                     |          |            |                        |
|------------------------------------|-------------------------------------|----------|------------|------------------------|
|                                    | Percentage Having as Highest Degree |          |            |                        |
| Field of Degree                    | Doctorate                           | Master's | Bachelor's | Total Percent in Field |
| Mathematics                        | 2.2                                 | 3.5      | 0.0        | 2.7                    |
| Statistics                         | 0.6                                 | 1.5      | 0.0        | 1.6                    |
| Mathematics Education              | 1.6                                 | 1.6      | 0.3        | 2.0                    |
| Other fields                       | 0.5                                 | 1.5      | 0.3        | 1.7                    |
| Total percentage by highest degree | 3.3                                 | 3.3      | 0.4        |                        |

**TABLE TYF.6**

| SEs                         |        |
|-----------------------------|--------|
| Highest Degree              | 2021   |
| Doctorate                   | 1.1    |
| Master's                    | 3.0    |
| Bachelor's                  | 2.7    |
| Number of part-time faculty | 2066.4 |



**TABLE TYF.7**

| SEs                                |                                     |          |            |                        |
|------------------------------------|-------------------------------------|----------|------------|------------------------|
|                                    | Percentage Having as Highest Degree |          |            |                        |
| Field of Degree                    | Doctorate                           | Master's | Bachelor's | Total Percent in Field |
| Mathematics                        | 0.9                                 | 5.4      | 1.9        | 5.9                    |
| Statistics                         | 0.1                                 | 0.6      | 0.0        | 0.6                    |
| Mathematics Education              | 0.4                                 | 5.0      | 1.6        | 5.2                    |
| Other fields                       | 0.5                                 | 2.7      | 1.4        | 3.4                    |
| Total percentage by highest degree | 1.1                                 | 3.0      | 2.7        |                        |

**TABLE TYF.8**

| SEs   |               |
|-------|---------------|
|       | 2021          |
| Men   | 632.2<br>2.8% |
| Women | 590.8<br>2.8% |
| Total | 796           |

**TABLE TYF.9**

| SEs          |                             |                   |
|--------------|-----------------------------|-------------------|
|              | Percentage of               |                   |
|              | Full-Time Permanent Faculty | Part-Time Faculty |
| Men          | 2.8                         | 3.1               |
| Women        | 2.8                         | 3.1               |
| Total number | 796                         | 1313.2            |

**TABLE TYF.10**

| SEs                                                               |       |
|-------------------------------------------------------------------|-------|
|                                                                   | 2021  |
| Percentage of ethnic minorities among full-time permanent faculty | 2.6%  |
| Number of full-time permanent ethnic minority faculty             | 262.4 |
| Number of full-time permanent faculty                             | 918.7 |

**TABLE TYF.11**

| SEs                                              |       |
|--------------------------------------------------|-------|
| Ethnic Group                                     | 2021  |
| American Indian / Eskimo / Aleut                 | 0.3   |
| Asian / Pacific Islander                         | 1.2   |
| Black (non-Hispanic)                             | 1.3   |
| Mexican American / Puerto Rican / other Hispanic | 1.4   |
| White (non-Hispanic)                             | 5.4   |
| Status unknown                                   | 1.6   |
| Not reported                                     |       |
| Number of full-time permanent faculty            | 918.7 |

**TABLE TYF.12**

| SEs                                               |                                       |                                                           |                                     |
|---------------------------------------------------|---------------------------------------|-----------------------------------------------------------|-------------------------------------|
| Ethnic Group                                      | Number of Full-Time Permanent Faculty | Percentage of Ethnic Group in Full-Time Permanent Faculty | Percentage of Women in Ethnic Group |
| American Indian, Alaskan Native                   | 30.2                                  | 0.3                                                       | 78.5                                |
| Asian / Pacific Islander                          | 113.2                                 | 1.2                                                       | 9.9                                 |
| Black or African American (non-Hispanic)          | 124.5                                 | 1.3                                                       | 7.4                                 |
| Mexican American, Puerto Rican, or other Hispanic | 129.1                                 | 1.4                                                       | 7.1                                 |
| White (non-Hispanic)                              | 613.2                                 | 5.4                                                       | 2.9                                 |
| Status not known or other                         | 148.5                                 | 1.6                                                       | 6.3                                 |
| Not reported                                      |                                       |                                                           |                                     |
| Total                                             | 918.7                                 |                                                           | 3.8                                 |

**TABLE TYF.13**

| SEs                  |                                 |                                          |
|----------------------|---------------------------------|------------------------------------------|
|                      | Percentage Among                |                                          |
| Ethnic Group         | All Full-Time Permanent Faculty | Full-Time Permanent Faculty Under Age 40 |
| Ethnic minorities    | 2.6                             | 8.8                                      |
| White (non-Hispanic) | 5.4                             | 9.8                                      |
| Unknown              | 1.6                             | 2.0                                      |
| Not reported         |                                 |                                          |
| Number               | 918.7                           | 66.6                                     |

**TABLE TYF.14**

| SEs                                                     |        |
|---------------------------------------------------------|--------|
|                                                         | 2021   |
| Percentage of ethnic minorities among part-time faculty | 4.0    |
| Number of part-time faculty                             | 1630.9 |

**TABLE TYF.15**

| SEs                                               |                                |                                             |                              |
|---------------------------------------------------|--------------------------------|---------------------------------------------|------------------------------|
|                                                   |                                | Percentage of                               |                              |
| Ethnic Group                                      | Number of<br>Part-Time Faculty | Ethnic Group Among<br>All Part-Time Faculty | Women Within<br>Ethnic Group |
| American Indian, Alaskan Native                   | 18.6                           | 0.0                                         | 22.4                         |
| Asian / Pacific Islander                          | 191.7                          | 0.0                                         | 5.6                          |
| Black or African American (non-Hispanic)          | 171.9                          | 0.0                                         | 9.4                          |
| Mexican American, Puerto Rican, or other Hispanic | 171.6                          | 0.0                                         | 12.8                         |
| White (non-Hispanic)                              | 1031.2                         | 0.1                                         | 4.1                          |
| Status not known or other                         | 352.4                          | 0.0                                         | 8.1                          |
| Total                                             |                                |                                             | 3.1                          |

**TABLE TYF.16**

| SEs          |                                           |                                       |
|--------------|-------------------------------------------|---------------------------------------|
|              | Percentage of Full-Time Permanent Faculty | Number of Full-Time Permanent Faculty |
| Age          | 2021                                      | 2021                                  |
| <30          | 0.9                                       | 82.2                                  |
| 30–34        | 1.1                                       | 97.1                                  |
| 35–39        | 1.3                                       | 130.0                                 |
| 40–44        | 1.3                                       | 131.7                                 |
| 45–49        | 1.4                                       | 149.0                                 |
| 50–54        | 1.8                                       | 190.5                                 |
| 55–59        | 1.5                                       | 152.4                                 |
| >59          | 1.9                                       | 185.4                                 |
| Not reported |                                           |                                       |
| Total        |                                           | 918.7                                 |

**TABLE TYF.17**

| SEs          |                                           |     |            |                                     |
|--------------|-------------------------------------------|-----|------------|-------------------------------------|
|              | Percentage of Full-Time Permanent Faculty |     |            | Percentage of Women<br>in Age Group |
| Age          | Women                                     | Men | Non-Binary |                                     |
| <35          | 0.8                                       | 1.1 | 0.0        | 5.2                                 |
| 35–44        | 1.2                                       | 1.3 | 0.0        | 4.5                                 |
| 45–54        | 1.7                                       | 1.5 | 0.0        | 4.5                                 |
| >54          | 1.6                                       | 1.9 | 0.0        | 4.6                                 |
| Not reported | 3.2                                       | 3.2 | 0.0        | 5.1                                 |
| Total        | 5.3                                       | 5.4 | 0          |                                     |

**TABLE TYF.18**

| SEs                                                                |      |
|--------------------------------------------------------------------|------|
| Percentage of New Faculty from:                                    | 2021 |
| A. Graduate School                                                 | 7.7  |
| B. Teaching in a four-year college or university                   | 2.2  |
| C. Teaching in another two-year college                            | 8.5  |
| D. Teaching in a secondary school                                  | 6.2  |
| E. Part-time or full-time temporary employment at the same college | 16.2 |
| F. Nonacademic employment                                          | 0.0  |
| G. Unemployed                                                      | 0.0  |
| H. Unknown                                                         | 11.0 |
| Total number appointed                                             | 69.6 |

**TABLE TYF.19**

| SEs            |           |
|----------------|-----------|
| Highest Degree | 2021–2022 |
| Doctorate      | 12.0      |
| Master's       | 13.3      |
| Bachelor's     | 0.0       |
| Unknown        | 7.4       |

**TABLE TYF.20**

| SEs                                               |           |                                                |
|---------------------------------------------------|-----------|------------------------------------------------|
| Ethnic Group                                      | 2021–2022 | Percentage of Women in                         |
|                                                   |           | Ethnic Group for 2021–2022<br>New Appointments |
| American Indian                                   | 0.0       | 0.0                                            |
| Asian / Pacific Islander                          | 5.6       | 4.5                                            |
| Black or African American (non-Hispanic)          | 1.9       | 1.9                                            |
| Mexican American, Puerto Rican, or other Hispanic | 5.1       | 0.0                                            |
| White (non-Hispanic)                              | 11.7      | 11.9                                           |
| Other                                             | 4.5       | 4.0                                            |
| Unknown                                           | 7.4       | 0.0                                            |
| Percentage of women among all new hires           | 15.2      |                                                |

**TABLE TYF.21**

| SEs                                                                     |                                                 |
|-------------------------------------------------------------------------|-------------------------------------------------|
|                                                                         | Percentage of Two-Year<br>Colleges in Fall 2021 |
| Colleges that require teaching<br>evaluations for all full-time faculty | 0                                               |
| Colleges that require teaching<br>evaluations for all part-time faculty | 4.1                                             |

**TABLE TYF.22**

| SEs                                                                                         |                                                          |                   |
|---------------------------------------------------------------------------------------------|----------------------------------------------------------|-------------------|
|                                                                                             | Percentage of Departments<br>Using Evaluation Method for |                   |
| Method of Evaluating Teaching                                                               | Part-Time Faculty                                        | Full-Time Faculty |
| A. Observation of classes by other faculty                                                  | 6.5                                                      | 7.0               |
| B. Observation of classes by division head (if different from chair) or other administrator | 6.7                                                      | 7.0               |
| C. Evaluation forms completed by students                                                   | 2.6                                                      | 1.6               |
| D. Evaluation of written course material such as lesson plans, syllabus, or exams           | 7.3                                                      | 7.0               |
| E. Self-evaluation such as teaching portfolios                                              | 5.5                                                      | 7.0               |
| F. Written peer evaluations                                                                 | 5.2                                                      | 6.6               |
| G. Other methods                                                                            | 7.5                                                      | 7.3               |

**TABLE TYF.23**

| SEs                                                                                                                   |           |
|-----------------------------------------------------------------------------------------------------------------------|-----------|
| Faculty Development                                                                                                   | Fall 2021 |
| Percentage of institutions requiring continuing education or professional development for full-time permanent faculty | 4.8       |

TABLE TYF.24

| SEs                                                                           |      |
|-------------------------------------------------------------------------------|------|
| Problem                                                                       | 2021 |
| A. Maintaining vitality of faculty                                            | 4.7  |
| B. Dual-enrollment courses                                                    | 2.9  |
| C. Staffing statistics courses                                                | 3.3  |
| D. Students don't understand demands of college work                          | 6.9  |
| E. Need to use part-time faculty for too many courses                         | 3.1  |
| F. Faculty salaries too low                                                   | 6.3  |
| G. Class sizes too large                                                      | 3.1  |
| H. Low student motivation                                                     | 7.1  |
| I. Too many students needing remediation                                      | 7.0  |
| J. Lack of student progress from developmental to advanced courses            | 6.1  |
| K. Low success rate in transfer-level courses                                 | 5.1  |
| L. Too few students who intend to transfer actually do                        | 1.9  |
| M. Inadequate travel funds for faculty                                        | 3.8  |
| N. Learning disabilities are not adequately supported                         | 3.7  |
| O. Inadequate classroom facilities for use of technology                      | 3.7  |
| P. Inadequate classroom facilities for teaching with active learning          | 3.9  |
| Q. Inadequate computer facilities for part-time faculty use                   | 4.3  |
| R. Inadequate computer facilities for student services                        | 1.9  |
| S. Heavy classroom duties prevent personal & teaching enrichment by faculty   | 4.1  |
| T. Coordinating mathematics courses with high schools                         | 4.3  |
| U. Lack of curricular flexibility because of transfer rules                   | 2.1  |
| V. Other barriers that inhibit curricular changes                             | 2.7  |
| W. Maintaining high and consistent expectations across sections               | 5.1  |
| X. High cost of textbooks                                                     | 4.3  |
| Y. Lack of flexibility in curricular redesign                                 | 1.6  |
| Z. Maintaining common standards between distance learning and related courses | 5.6  |
| AA. Use of distance education                                                 | na   |



TABLE TYF.25

| SEs                                                                           |                                                           |                          |                  |
|-------------------------------------------------------------------------------|-----------------------------------------------------------|--------------------------|------------------|
|                                                                               | Percentage of Department Heads<br>Classifying Problems as |                          |                  |
| Problem                                                                       | Minor or<br>No Problem                                    | Somewhat of<br>a Problem | Major<br>Problem |
| A. Maintaining vitality of faculty                                            | 7.1                                                       | 6.9                      | 4.7              |
| B. Dual-enrollment courses                                                    | 7.2                                                       | 7.1                      | 2.9              |
| C. Staffing statistics courses                                                | 7.2                                                       | 7.0                      | 3.3              |
| D. Students don't understand demands of college work                          | 3.2                                                       | 7.0                      | 6.9              |
| E. Need to use part-time faculty for too many courses                         | 6.5                                                       | 6.1                      | 3.1              |
| F. Faculty salaries too low                                                   | 4.6                                                       | 7.0                      | 6.3              |
| G. Class sizes too large                                                      | 6.7                                                       | 6.4                      | 3.1              |
| H. Low student motivation                                                     | 1.4                                                       | 7.1                      | 7.1              |
| I. Too many students needing remediation                                      | 4.1                                                       | 7.1                      | 7.0              |
| J. Lack of student progress from developmental to advanced courses            | 5.5                                                       | 6.9                      | 6.1              |
| K. Low success rate in transfer-level courses                                 | 6.2                                                       | 7.0                      | 5.1              |
| L. Too few students who intend to transfer actually do                        | 7.1                                                       | 7.3                      | 1.9              |
| M. Inadequate travel funds for faculty                                        | 6.2                                                       | 5.5                      | 3.8              |
| N. Learning disabilities are not adequately supported                         | 7.0                                                       | 6.6                      | 3.7              |
| O. Inadequate classroom facilities for use of technology                      | 6.6                                                       | 6.0                      | 3.7              |
| P. Inadequate classroom facilities for teaching with active learning          | 6.6                                                       | 6.1                      | 3.9              |
| Q. Inadequate computer facilities for part-time faculty use                   | 6.7                                                       | 5.9                      | 4.3              |
| R. Inadequate computer facilities for student services                        | 6.3                                                       | 6.2                      | 1.9              |
| S. Heavy classroom duties prevent personal & teaching enrichment by faculty   | 7.3                                                       | 7.2                      | 4.1              |
| T. Coordinating mathematics courses with high schools                         | 7.0                                                       | 7.2                      | 4.3              |
| U. Lack of curricular flexibility because of transfer rules                   | 6.9                                                       | 6.8                      | 2.1              |
| V. Other barriers that inhibit curricular changes                             | 7.2                                                       | 6.8                      | 2.7              |
| W. Maintaining high and consistent expectations across sections               | 6.9                                                       | 7.2                      | 5.1              |
| X. High cost of textbooks                                                     | 7.3                                                       | 7.1                      | 4.3              |
| Y. Lack of flexibility in curricular redesign                                 | 6.4                                                       | 6.3                      | 1.6              |
| Z. Maintaining common standards between distance learning and related courses | 6.9                                                       | 7.1                      | 5.6              |

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## Reference

This report presents the findings of an NSF-supported, statistically designed survey of the undergraduate mathematical sciences in the United States that was sponsored by the Conference Board of the Mathematical Sciences (CBMS) in fall 2021. It includes detailed studies of undergraduate mathematical sciences departments and programs in two- and four-year colleges and universities, and of statistics departments at all degree levels. Continuing the work of previous CBMS survey reports, this report gives data on enrollments, undergraduate degrees granted, faculty demographics, teaching formats, and pedagogical methods used; and presents information on a family of special topics of current interest. Special topics for fall 2021 include: the mathematical education of teachers of pre-college mathematics/statistics, percentage of departments offering distance learning courses and practices in distance learning courses, dual enrollment courses in mathematics and statistics, estimates of post-graduation plans of graduates of four-year mathematics departments and statistics departments, and profiles of other full-time faculty at four-year mathematics and statistics departments. Chapter 1 presents integrated two- and four-year findings. Chapter 2 presents data from the special projects of CBMS 2021 for two- and four-year mathematics departments, while Chapters 3 and 4 concentrate on four-year mathematics and statistics departments, showing findings by type of department (bachelor's, master's, or doctoral). Chapter 5 explores the mathematics and statistics courses of four-year colleges and universities. Chapter 6 details enrollment, course offerings, and instructional practices in mathematics departments at two-year colleges, while Chapter 7 describes faculty, administration, and special topics in mathematics departments at two-year colleges.

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